THL Series

Industrial Robot

Robot controller TSL3000
Robot controller TSL3000E
Robot controller TS3000
Robot controller TS3000E

INSTRUCTION MANUAL

INDUSTRIAL ROBOT

TRANSPORTATION AND INSTALLATION MANUAL

Notice

- Make sure that this instruction manual is delivered to the final user of SHIBAURA MACHINE's industrial robot.
- Before operating the industrial robot, read through and completely understand this manual.
- After reading through this manual, keep it nearby for future reference.

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Shibaura Machine

This Instruction Manual applies to the following robots:

THL Series: THL300, THL400, THL500, THL600, THL700, THL800, THL900, THL1000, THL1200

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The information contained in this manual is subject to change without prior notice to effect improvements.

Preface

This manual describes the basic specifications of the industrial robot and controller, and how to unpack and install them. Specifically, it describes how to unpack the shipment containing the equipment, how to install the equipment, how to connect wiring and air piping, and how to attach tools. Be sure to look through this manual before unpacking the shipment.
Before beginning the work according to this manual, read through the Safety Manual so that you can understand the safety measures.

This manual is divided into the following five parts:

Section 1-4	Specifications This section describes the basic specifications and names of respective units of the robot and controller.
Section 5-8	Transportation This section describes how to remove the robot and controller from their boxes and how to transport them to the installation site. This section also discusses how to temporarily store the equipment after unpacking the shipment.
Section 9-15	Installation This section discusses the equipment installation environment, space requirements, and how to install the equipment.
Section 16-18	System Connections This section describes how to connect the robot, controller and peripheral equipment.
Section 19-25	Tool Interface This section describes how to connect the tool to the robot arm and how to connect pipes and wires to the tool. This section also discusses maximum permissible loads of the tool.

Precautions on Safety

Important information on the robot and controller is noted in the instruction manual to prevent injury to the user and persons nearby, prevent damage to assets and to ensure correct use.
Make sure that the following details (indications and symbols) are well understood before reading this manual. Always observe the information that is noted.

[Explanation of indications]

Indication	Meaning of indication
DANGER	This means that "incorrect handling will lead to fatalities or major injuries".
WARNING	This means that "incorrect handling will lead to fatalities or serious injuries."
CAUTION	This means that "incorrect handling may lead to personal injuries *1) or physical damage *2)".

*1) Injuries refer to injuries, burns and electric shocks, etc., which do not require hospitalization or long term treatment.

*2) Physical damage refers to major fires due to destruction of assets or resources.

[Explanation of symbols]

Symbol	Meaning of symbol
	This means that the action is prohibited (must not be done). The details of the actions actually prohibited are indicated with pictures or words in or near the symbol.
	This means that the action is mandatory (must be done). The details of the actions that must be done are indicated with pictures or words in or near the symbol.
	This means danger and caution. The details are indicated with pictures or words in or near the symbol.

CAUTION

- Always read through the Safety Manual provided separately before starting actual work to ensure safety work covering from the robot installation to operation.

[Installation and transportation]

Always observe the following items to safely use the robot.

DANGER	
Prohibited	• DO NOT install or operate if any parts are damaged or missing. Doing so could lead to electric shocks, fires or faults. • DO NOT install the robot where it may be subject to fluids such as water. Doing so could lead to electric shocks, fires or faults. • Do not place the robot near combustible matters. Doing so could lead to fires if the matter ignites due to a fault, etc.
Mandatory	• Always secure the robot with the attached clamps before transporting it. Failure to do so could lead to injuries if the arm moves when the robot is suspended (for THL500 to THL1200). • Wire the robot after installation. Wiring the robot before installation could lead to electric shocks or injuries. • Always use the power voltage and power capacity designated by SHIBAURA MACHINE. Failure to do so could lead to device faults or fires. • Always use the designated power cable. Using a cable other than that designated could lead to fires or faults.
Always ground	• Completely connect the grounding cable. Failure to do so could lead to electric shocks or fires if a fault or fault current occurs. Noise could lead to malfunction. Also, it could cause misoperation by noise.

CAUTION	
Prohibited	• NEVER lift the robot by the arm 2 cover. Doing so will apply an excessive force on the robot's mechanism section and could lead to faults. • For the controller, secure the ample space for air vent. Heating of controller could lead to malfunction.
Mandatory	• When lifting the robot (for THL500 to THL1200), lift it up slowly. The robot will tilt slightly, so lifting it up suddenly could be hazardous. • When storing the robot, secure it to the base. The robot will be unstable if just set down, and it could tilt over.

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1. Specifications (THL300 and THL400)

1.1 Correspondence between the Robots and the Robot Controllers

Table 1.1 shows the correspondence between the robots and the robot controllers. There are several models of robots and controllers. The THL series robots are compatible with the THL3000, THL3000E, TH3000 and TH3000E robot controllers.

Table 1.1 Correspondence between the Robots and the Robot Controllers

		Robot			
		THL300, THL400	THL500, THL600, THL700	THL800, THL900, THL1000	THL1200
Robot controller	TSL3000	○	○	○	○
	TSL3000E	○	○	○	○
	TS3000	○	○	○	○
	TS3000E	○	○	○	○

○: Supported

1.2 Robot Configuration Diagram (TSL3000)

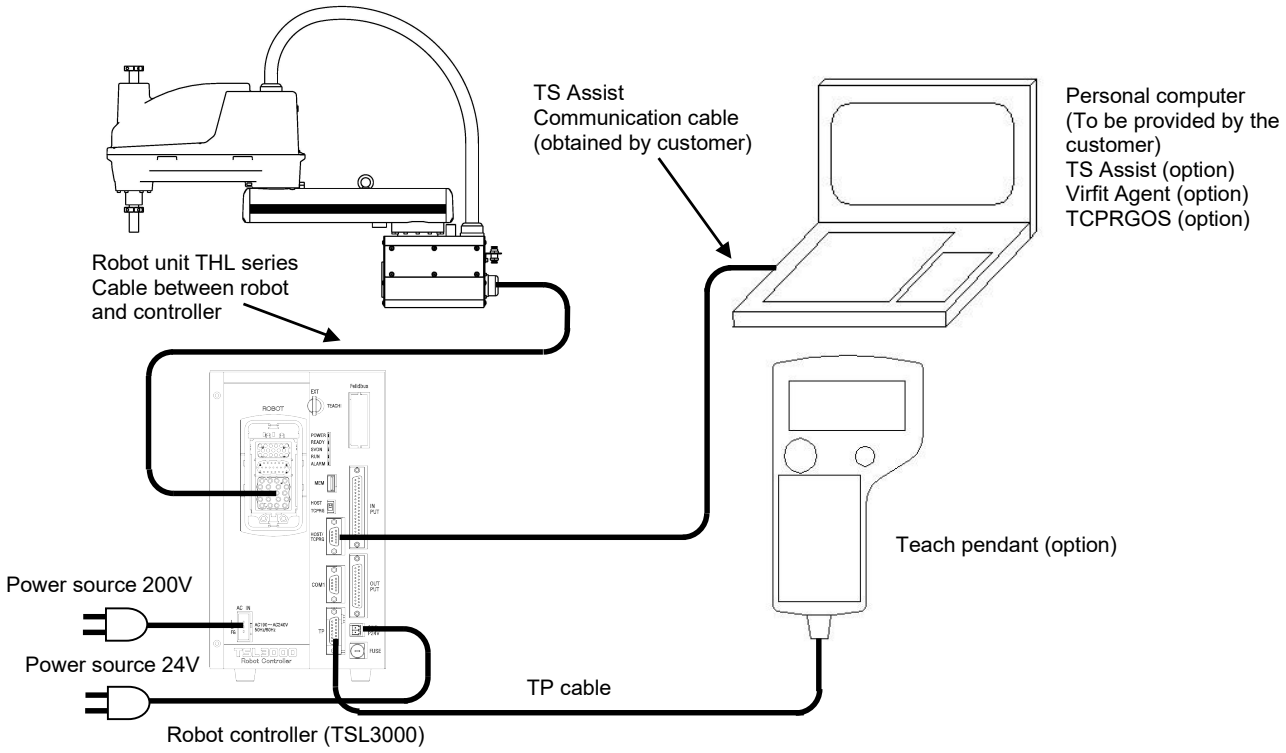


Fig. 1.1 Robot Configuration (TSL3000)

For connection of the 200-volt power source, see "16.1.2 Connecting the Power Cable "ACIN"". For connection of 24-volt power source, see "16.1.7 Connecting the External Input/Output Power Cable "GNDP24V"".

1.3 Robot Configuration Diagram (TSL3000E)

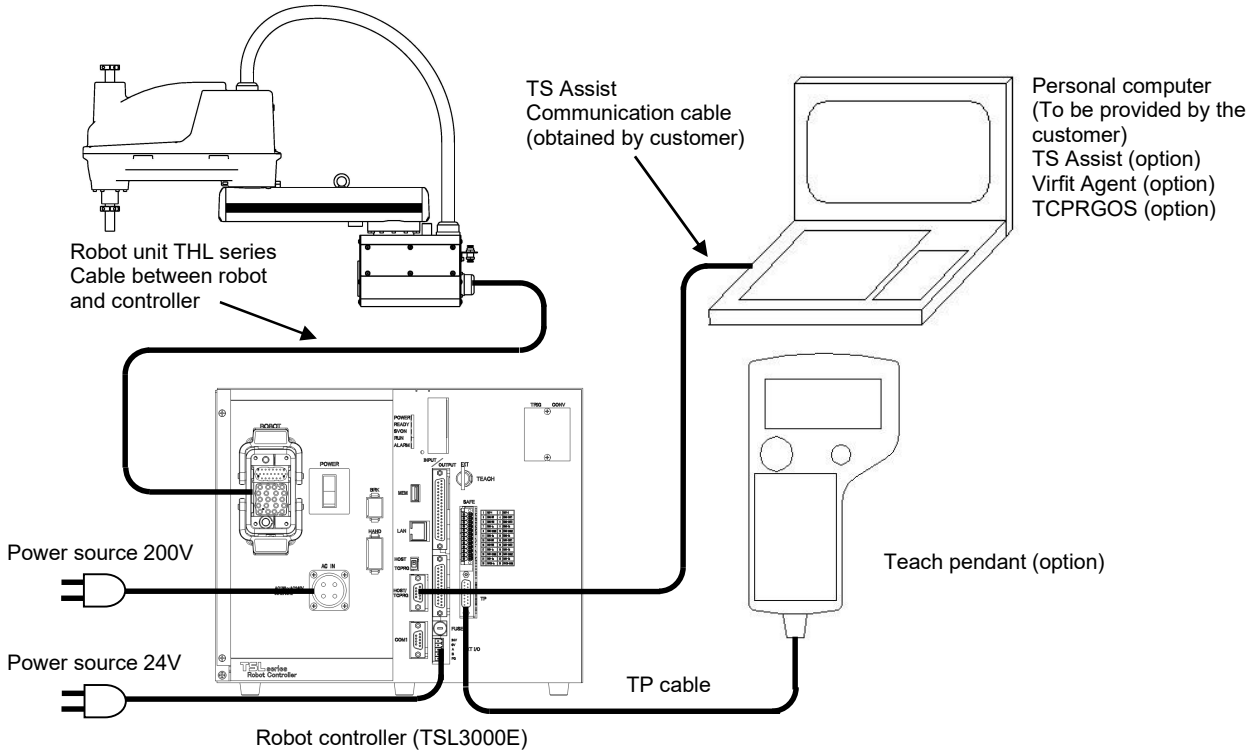


Fig. 1.2 Robot Configuration (TSL3000E)

For connection of the 200-volt power source, see "17.1.2 Connecting the Power Cable". For connection of 24-volt power source, see "17.1.5 Connecting the External Input/Output Power Cable".

1.4 Robot Configuration Diagram (TS3000 and TS3000E)

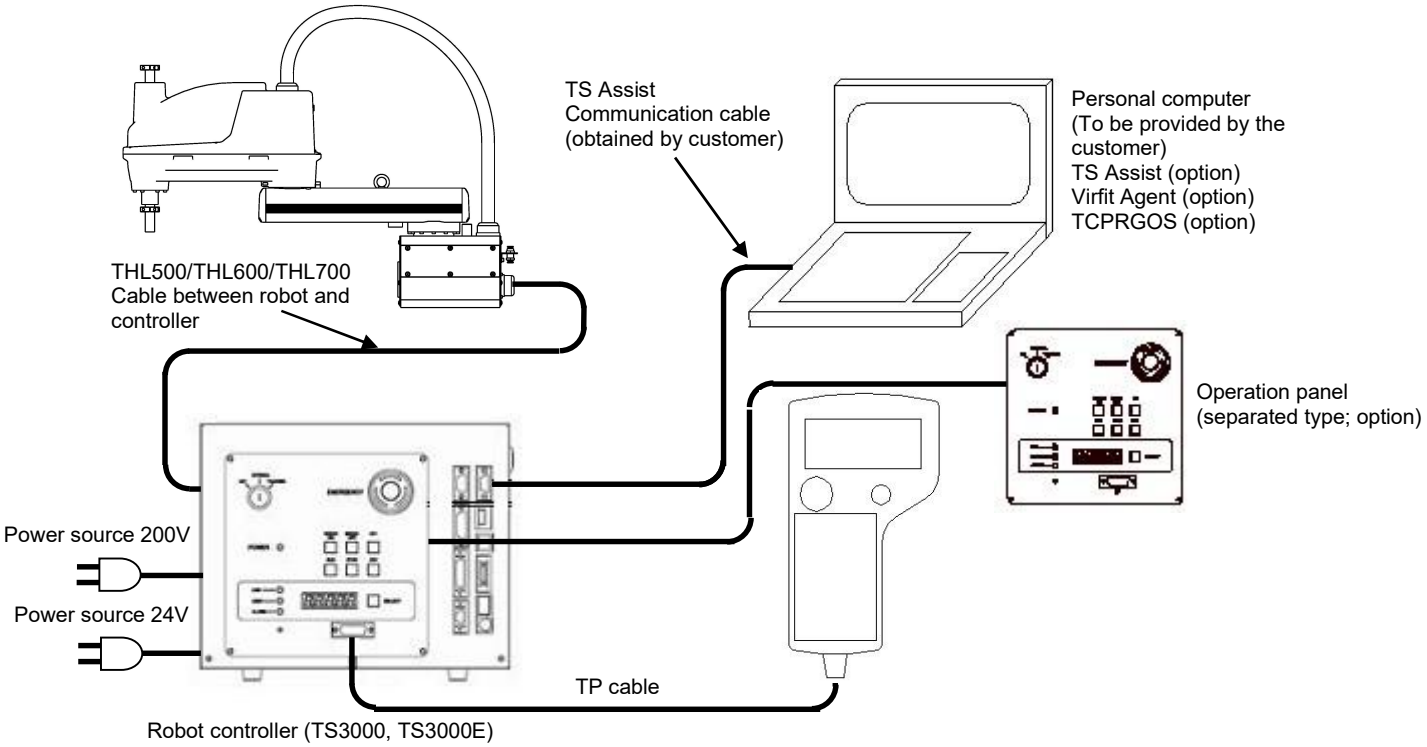


Fig. 1.3 Robot Configuration (TS3000 and TS3000E)

For connection of the 200-volt power source, see "18.1.2 Connecting the Power Cable "ACIN"". For connection of 24-volt power source, see "18.1.7 Connecting the External Input/Output Power Cable "EMS"".

1.5 Name of Each Part

Fig. 1.4 shows the name of each part of the robot (THL300 and THL400).

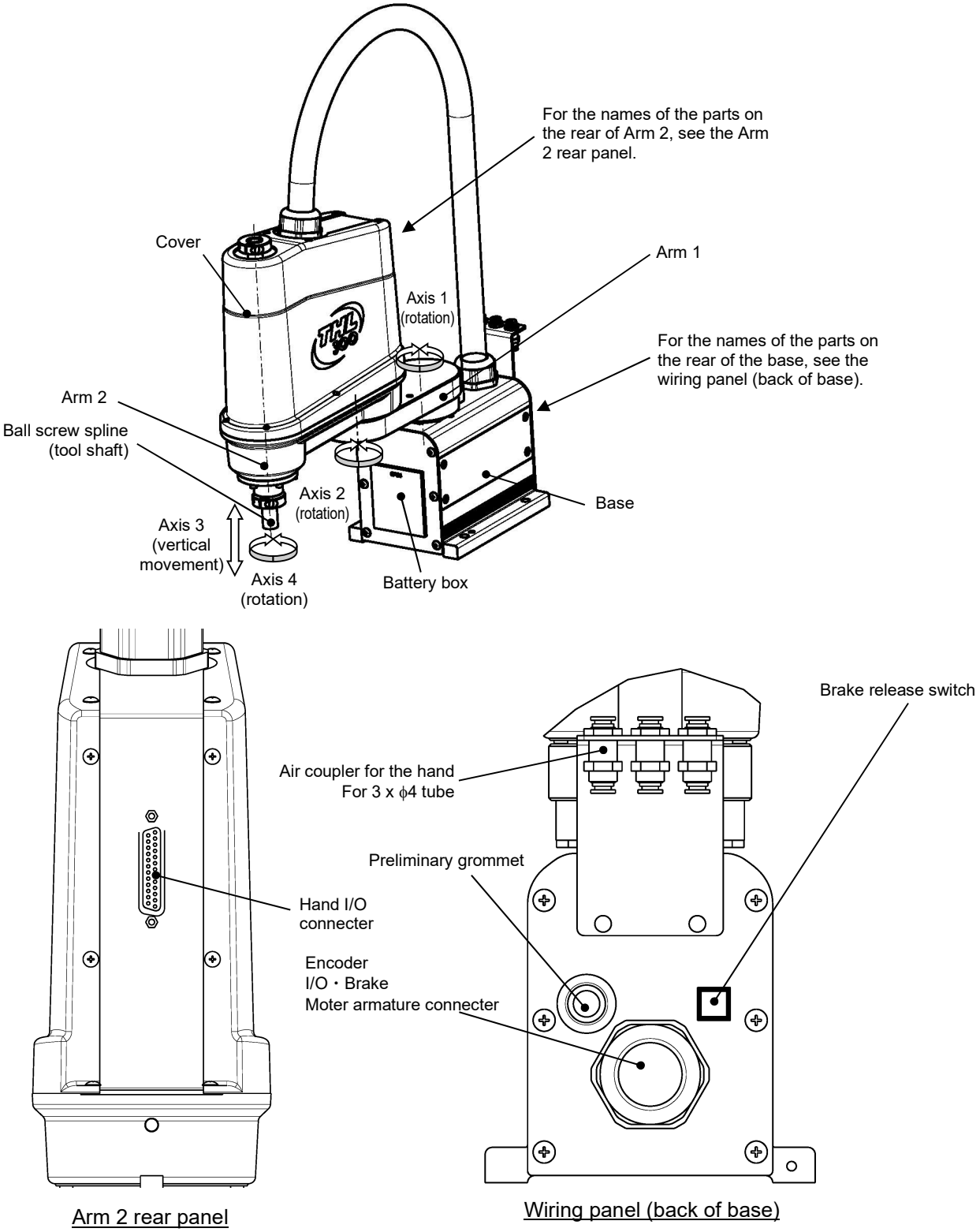


Fig. 1.4 Name of each part (THL300 and THL400)

1.6 External Dimensions

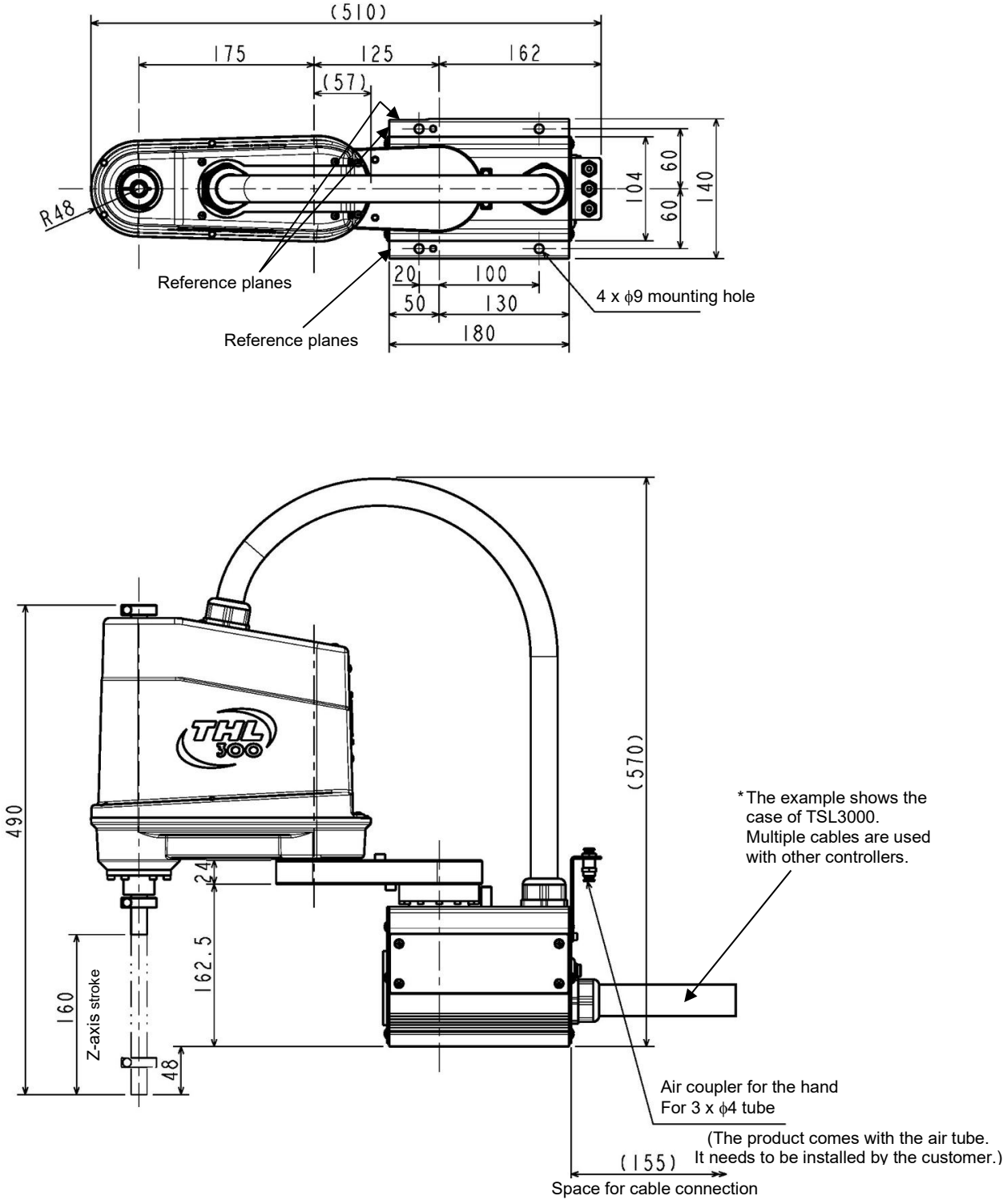


Fig. 1.5 External dimensions of robot (THL300)

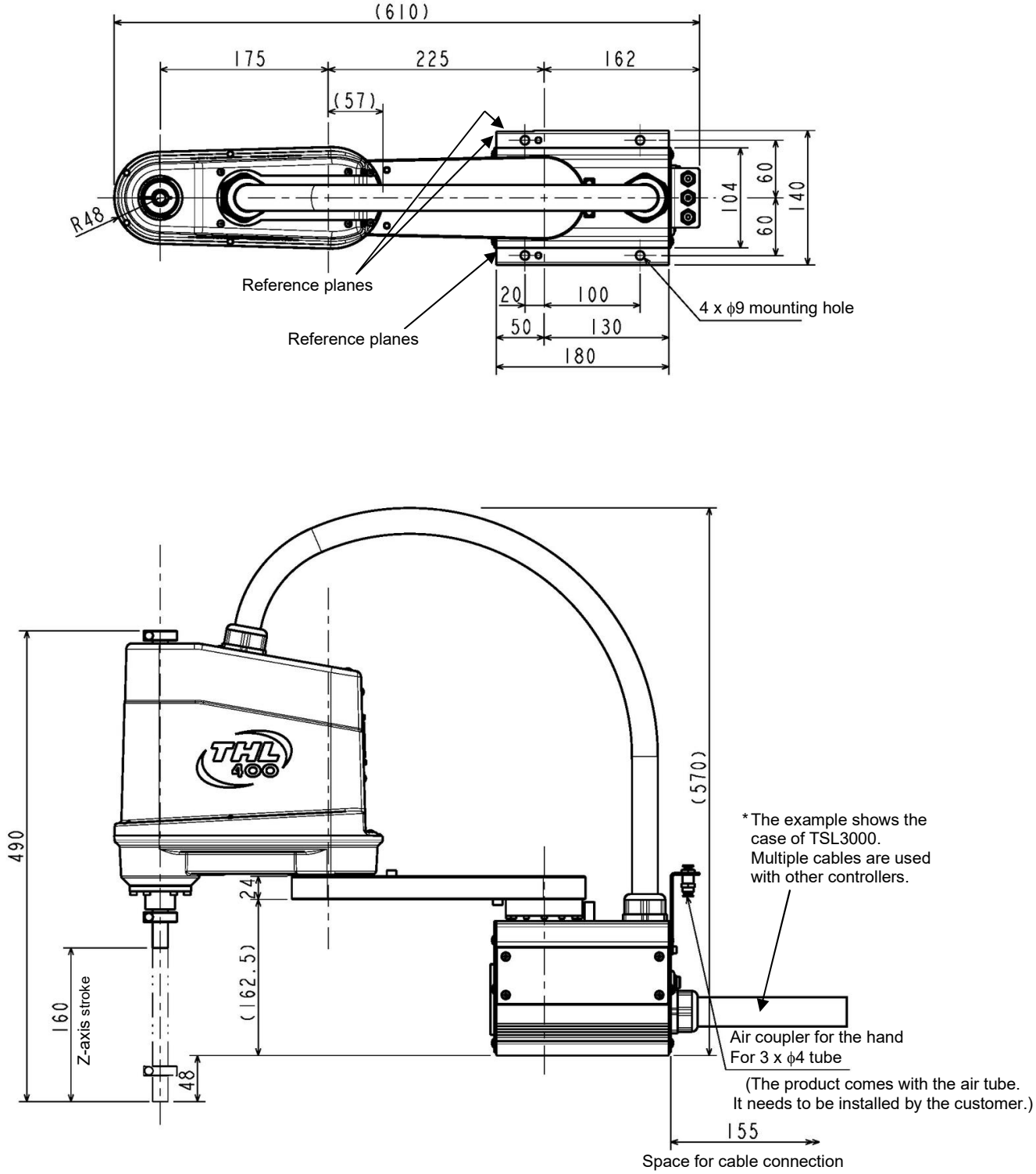


Fig. 1.6 External dimensions of robot (THL400)

1.7 Specifications Table

Item		Specifications	
Structure		Horizontal multi-joint type SCARA robot	
Model		THL300	THL400
Applicable controller		TSL3000, TSL3000E, TS3000, TS3000E	
Mass of robot body		12 kg	13 kg
No. of controlled axes		4	
Arm length		300 mm (125 mm + 175 mm)	400 mm (225 mm + 175 mm)
Motor capacity / Current limit	Axis 1	200 W / 9.84 A_{0-P}	
	Axis 2	100 W / 4.89 A_{0-P}	
	Axis 3	100 W / 4.70 A_{0-P}	
	Axis 4	100 W / 4.89 A_{0-P}	
Operating range	Axis 1	±125 deg	
	Axis 2	±145 deg	
	Axis 3	0 to 160 mm	
	Axis 4	±360 deg	
Maximum speed (*1)	Axis 1	660 deg/s	
	Axis 2	660 deg/s	
	Axis 3	1120 mm/s	
	Axis 4	1500 deg/s	
	Composite speed of axes 1 and 2	5.1 m/s	6.3 m/s
Rated payload mass (*1)		2 kg	
Maximum payload mass (*1)		5 kg	
Permissible load inertia (*1)		0.05 kg·m^2	
Repeatability (*2)	X,Y	±0.01 mm	
	Z	±0.015 mm	
	C	±0.007 deg	
Cycle time (*3) (When payload mass is 2 kg)		0.48 s	0.47 s
Drive system		By means of AC servo motors	
Position detection method		Absolute	
Paint color (*4)		Arm 2: Equivalent to PANTONE 293C, acrylic paint Body: White alumite processing Arm cover: White	
Power supply capacity		0.7 kVA	

*1: There are restrictions on speed and acceleration speed depending on the operation pattern, load, and offset amount.

*2: This is unidirectional position repeatability when the ambient temperature is constant at 20°C. It is not absolute positioning accuracy. The values of X, Y and C are what Z value is in upper limit. It does not guarantee the trajectory accuracy.

*3: The standard cycle operation pattern cannot achieve continuous operation exceeding the effective load rate. With horizontal direction 300 mm, vertical direction 25 mm round trip, and rough positioning.

*4: The tone may be different among production lots, but it does not affect the product quality.

! CAUTION

- Micro vibration may occur depending on the robot posture. If micro vibration occurs, reduce the acceleration speed to use the robot.
- Put the Z-axis (axis 3) in the raised position as much as possible, when moving Axes 1, 2, and 4.

  Moving Axis 1, 2, or 4 when the Z-axis is in low positions can lead to premature damage to the ball screw spline (Z-axis shaft).

  If Axis 1, 2, or 4 must be moved while the Z-axis is in low positions due to unavoidable circumstances, prevent the ball screw spline vibrations by using the SPEED, ACCEL/DECEL, and PAYLOAD commands to adjust the operation speed and acceleration.
- If Axis 1, 2, or 4 is moved while the Z-axis is in low positions, be extremely careful not to collide with any objects. Even if Axis 1, 2, or 4 is moved at low speed, a collision or other impact with an object can damage the ball screw spline (Z-axis shaft) before alarm occurs.

2. Specifications (THL500, THL600 and THL700)

2.1 Correspondence between robot and controller, and equipment configuration diagram

For the correspondence between robot and controller, and equipment configuration diagram, see the description of 1.1 through 1.4.

2.2 Name of Each Parts

Fig. 2.1 shows the name of each part of the robot (THL500, THL600 and THL700).

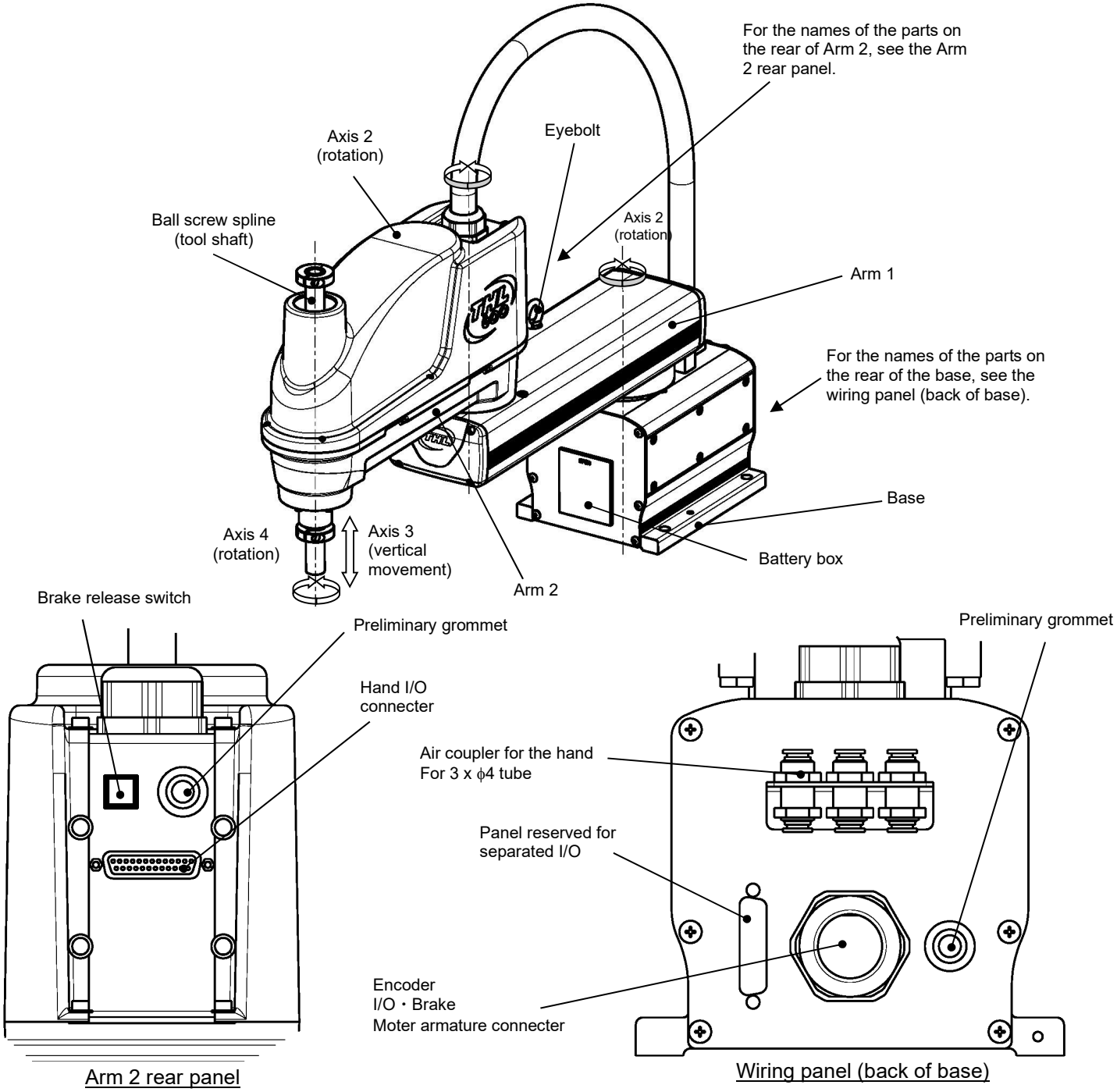


Fig. 2.1 Name of each part (THL500, THL600 and THL700)

2.3 External Dimensions

Figs. 2.2 to 2.4 refer to the external dimensions of the robot (THL500, THL600 and THL700).

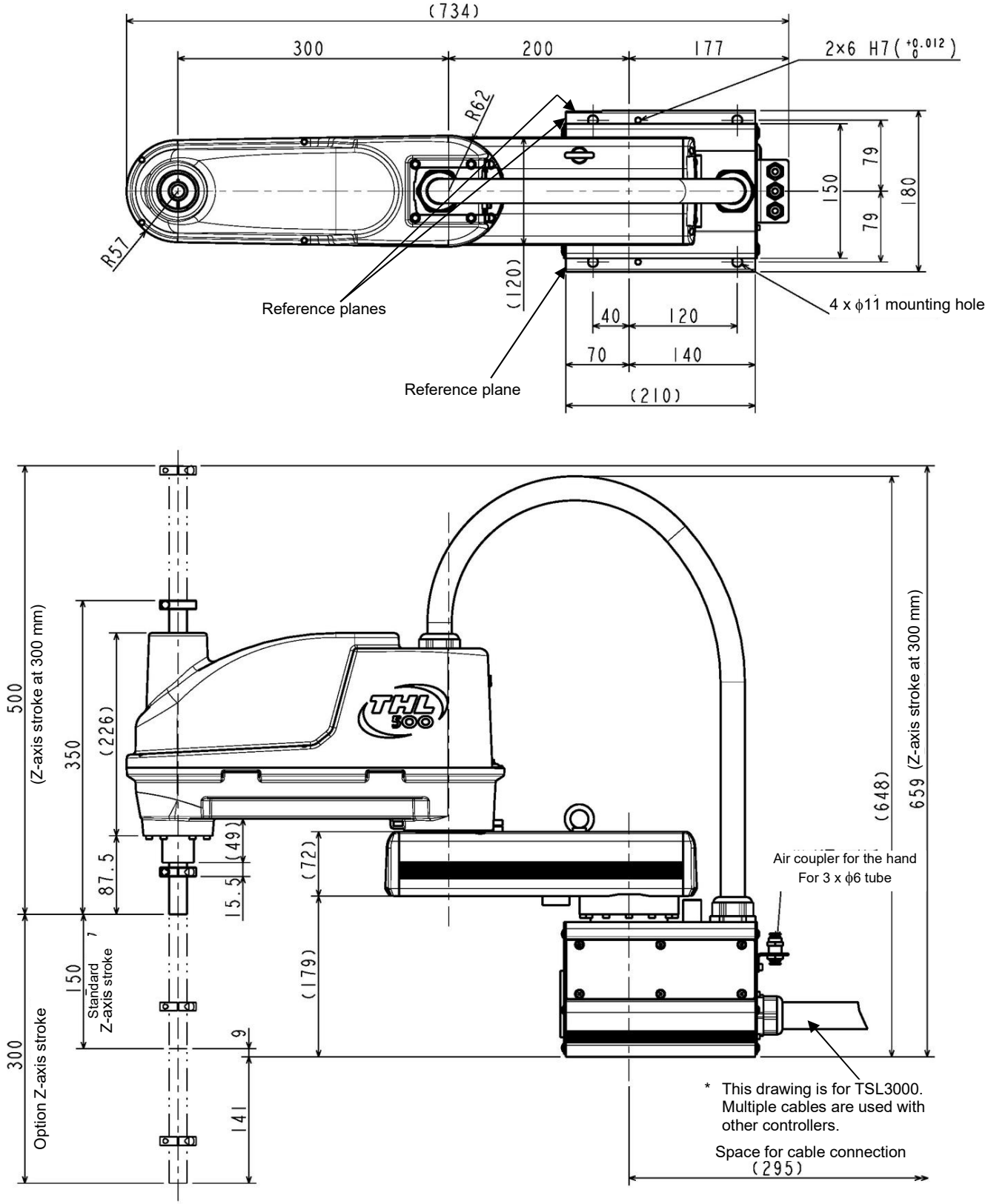


Fig. 2.2 External dimensions of robot (THL500)

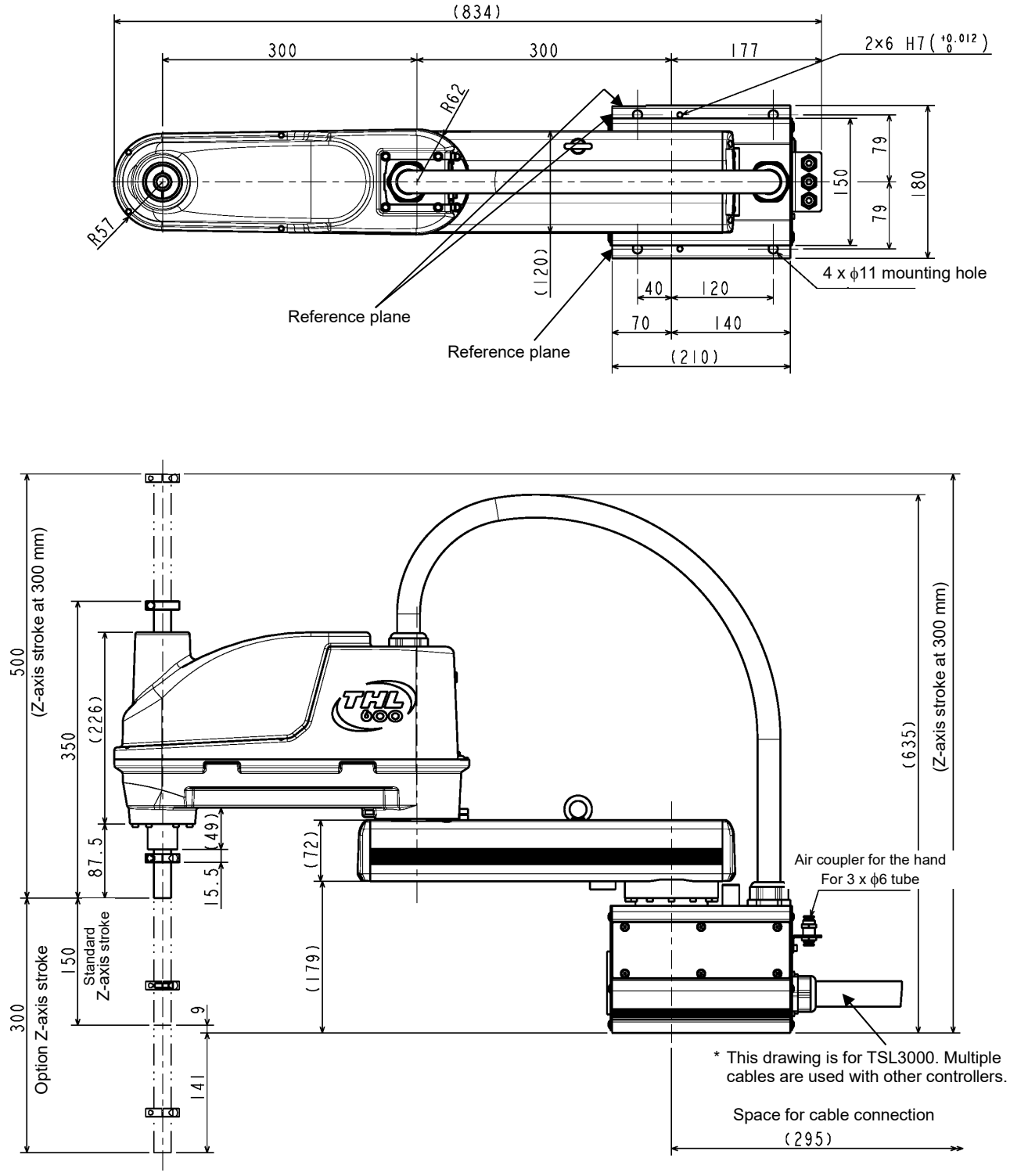


Fig. 2.3 External dimensions of robot (THL600)

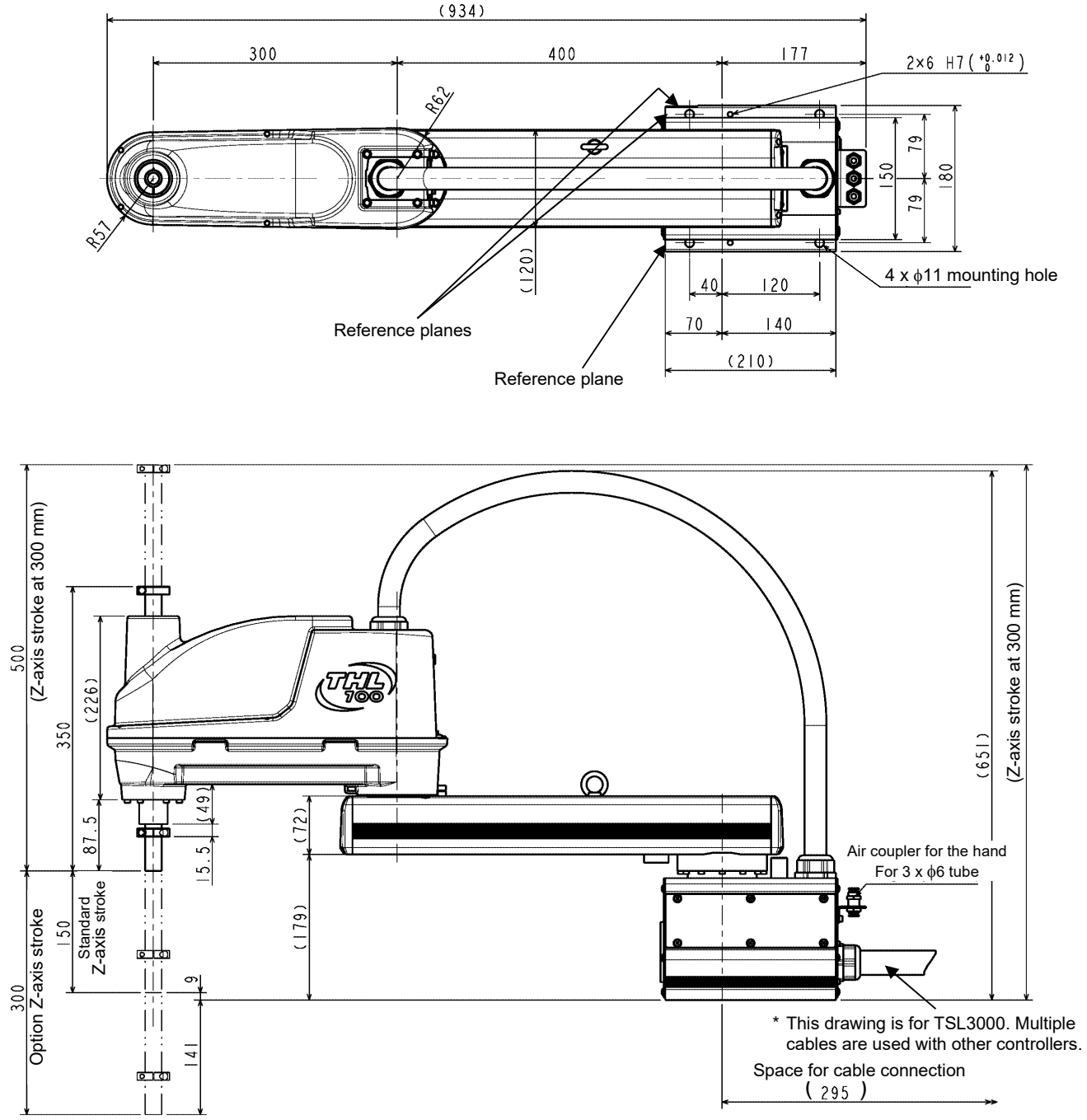


Fig. 2.4 External dimensions of robot (THL700)

2.4 Specifications Table

Item		Specifications		
Structure		Horizontal multi-joint type SCARA robot		
Model		THL500	THL600	THL700
Applicable controller		TSL3000, TSL3000E, TS3000, TS3000E		
Mass of robot body		22 kg	23 kg	24 kg
No. of controlled axes		4 axis		
Arm length		500 mm (200 mm + 300 mm)	600 mm (300 mm + 300 mm)	700 mm (400 mm + 300 mm)
Motor capacity / Current limit	Axis 1	400 W / 14.7 A_{0-P}		
	Axis 2	200 W / 9.37 A_{0-P}		
	Axis 3	200 W / 6.90 A_{0-P}		
	Axis 4	200 W / 9.37 A_{0-P}		
Operating range	Axis 1	±125 deg		
	Axis 2	±145 deg		
	Axis 3	0 to 150 mm [option: 0 to 300 mm]		
	Axis 4	±360 deg		
Maximum speed (*1)	Axis 1	450 deg/s		
	Axis 2	450 deg/s		
	Axis 3	2000 mm/s		
	Axis 4	1700 deg/s		
	Composite speed of axes 1 and 2	6.3 m/s	7.1 m/s	7.9 m/s
Rated payload mass (*1)		2 kg		
Maximum payload mass (*1)		10 kg		
Permissible load inertia (*1)		0.2 kg·m^2		
Repeatability (*2)	X,Y	±0.01 mm		
	Z	±0.015 mm		
	C	±0.007 deg		
Cycle time (*3) (When payload mass is 2 kg)		0.45 s		0.50 s
Drive system		By means of AC servo motors		
Position detection method		Absolute		
Paint color (*4)		Arm 2: Equivalent to PANTONE 293C, acrylic paint Body: White alumite processing Arm cover: White		
Power supply capacity		1.4 kVA		

*1: There are restrictions on speed and acceleration speed depending on the operation pattern, load, and offset amount.

*2: This is unidirectional position repeatability when the ambient temperature is constant at 20°C. It is not absolute positioning accuracy. The values of X, Y and C are what Z value is in upper limit. It does not guarantee the trajectory accuracy.

*3: The standard cycle operation pattern cannot achieve continuous operation exceeding the effective load rate. With horizontal direction 300 mm, vertical direction 25 mm round trip, and rough positioning.

*4: The tone may be different among production lots, but It does not affect the product quality.

! CAUTION

- Micro vibration may occur depending on the robot posture. If micro vibration occurs, reduce the acceleration speed to use the robot.
- Put the Z-axis (axis 3) in the raised position as much as possible, when moving Axes 1, 2, and 4.

  Moving Axis 1, 2, or 4 when the Z-axis is in low positions can lead to premature damage to the ball screw spline (Z-axis shaft).

  If Axis 1, 2, or 4 must be moved while the Z-axis is in low positions due to unavoidable circumstances, prevent the ball screw spline vibrations by using the SPEED, ACCEL/DECEL, and PAYLOAD commands to adjust the operation speed and acceleration.
- If Axis 1, 2, or 4 is moved while the Z-axis is in low positions, be extremely careful not to collide with any objects. Even if Axis 1, 2, or 4 is moved at low speed, a collision or other impact with an object can damage the ball screw spline (Z-axis shaft) before alarm occurs.

3. Specifications (THL800, THL900 and THL1000)

3.1 Correspondence between robot and controller, and equipment configuration diagram

For the correspondence between robot and controller, and equipment configuration diagram, see the description of 1.1 through 1.4.

3.2 Name of Each Parts

Fig. 3.1 shows the name of each part of the robot (THL800, THL900 and THL1000).

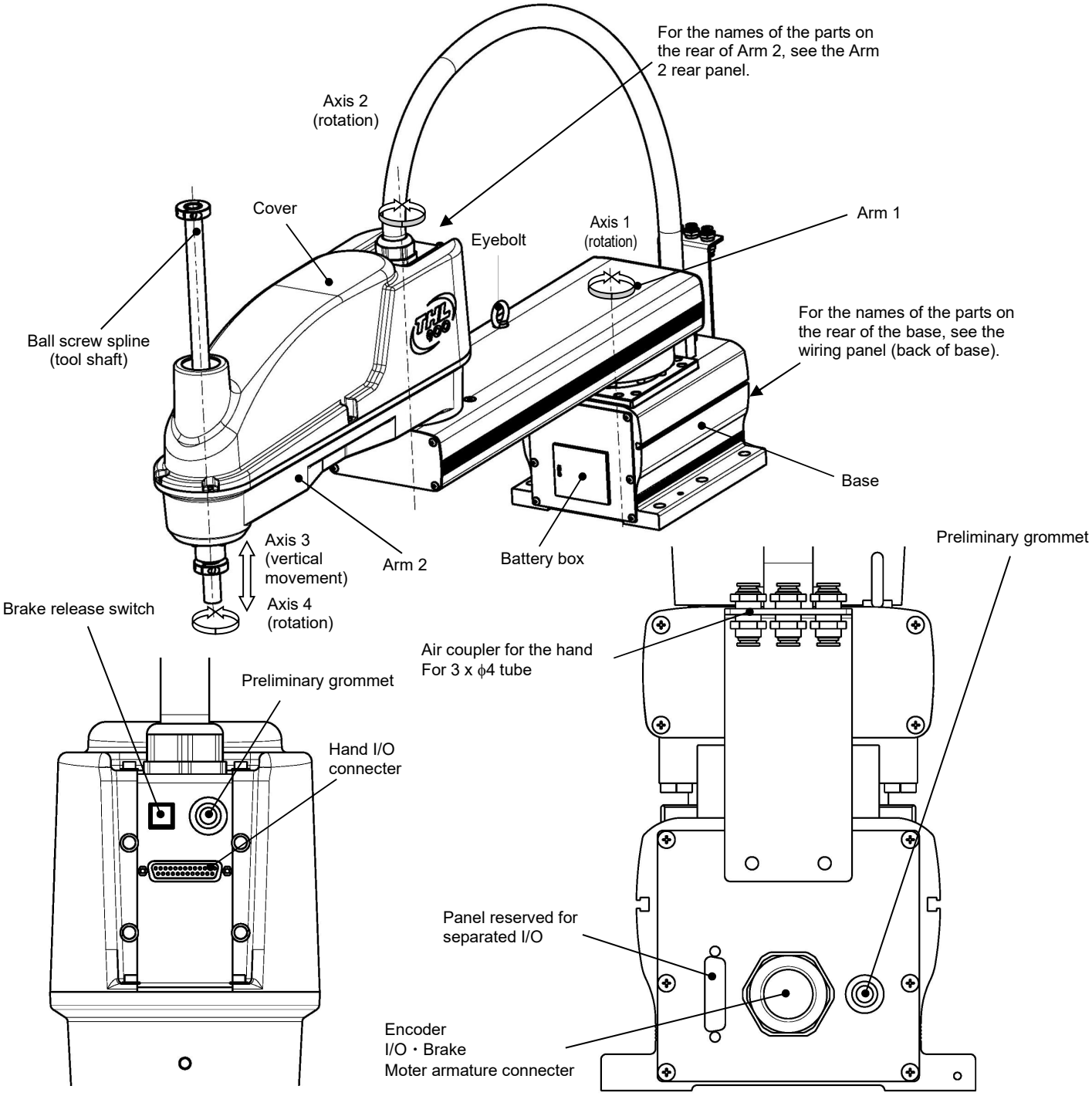


Fig. 3.1 Name of each part (THL800, THL900 and THL1000)

3.3 External Dimensions

Figs. 3.2 to 3.4 refer to the external dimensions of the robot (THL800, THL900 and THL1000).

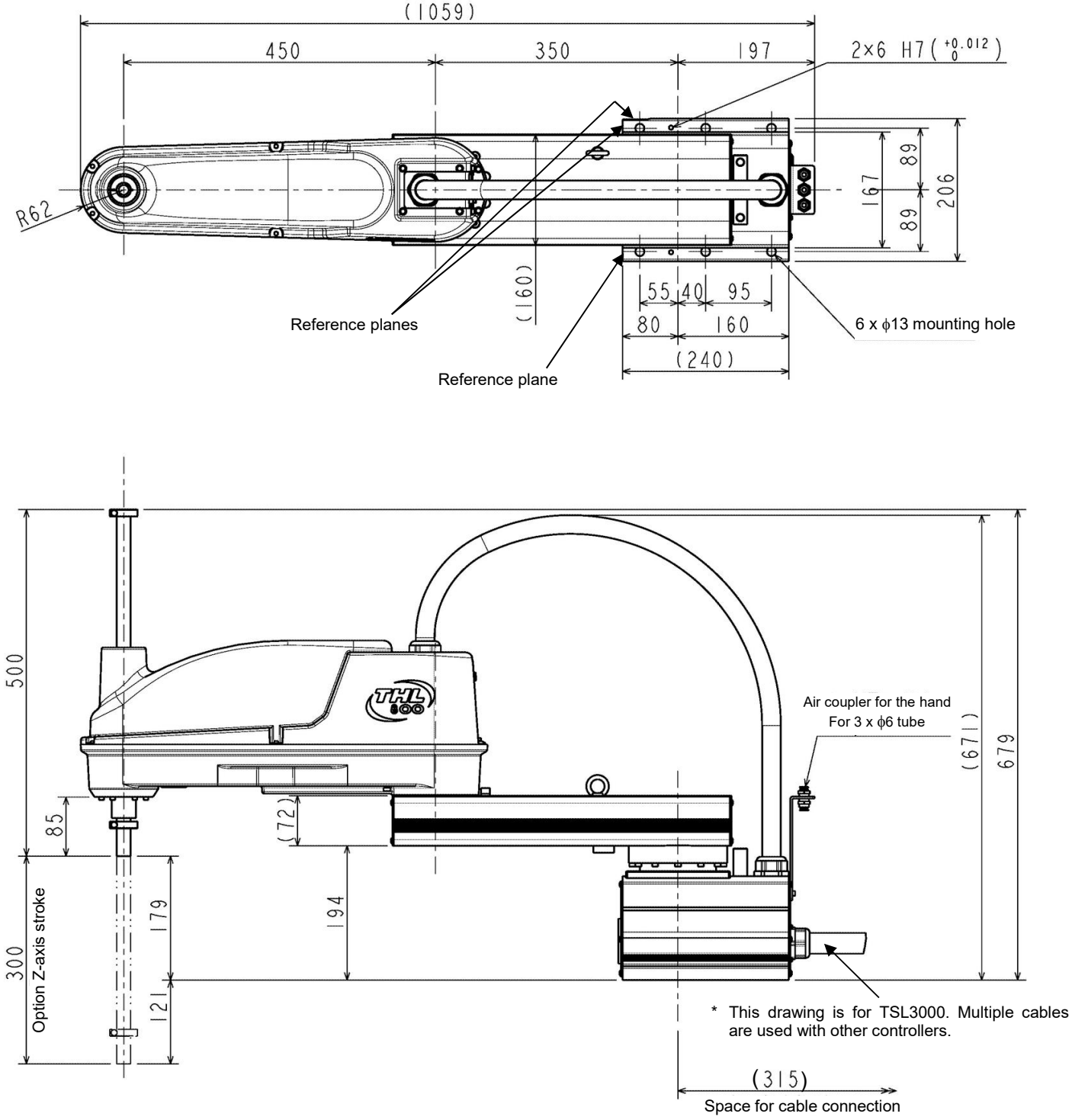


Fig. 3.2 External dimensions of robot (THL800)

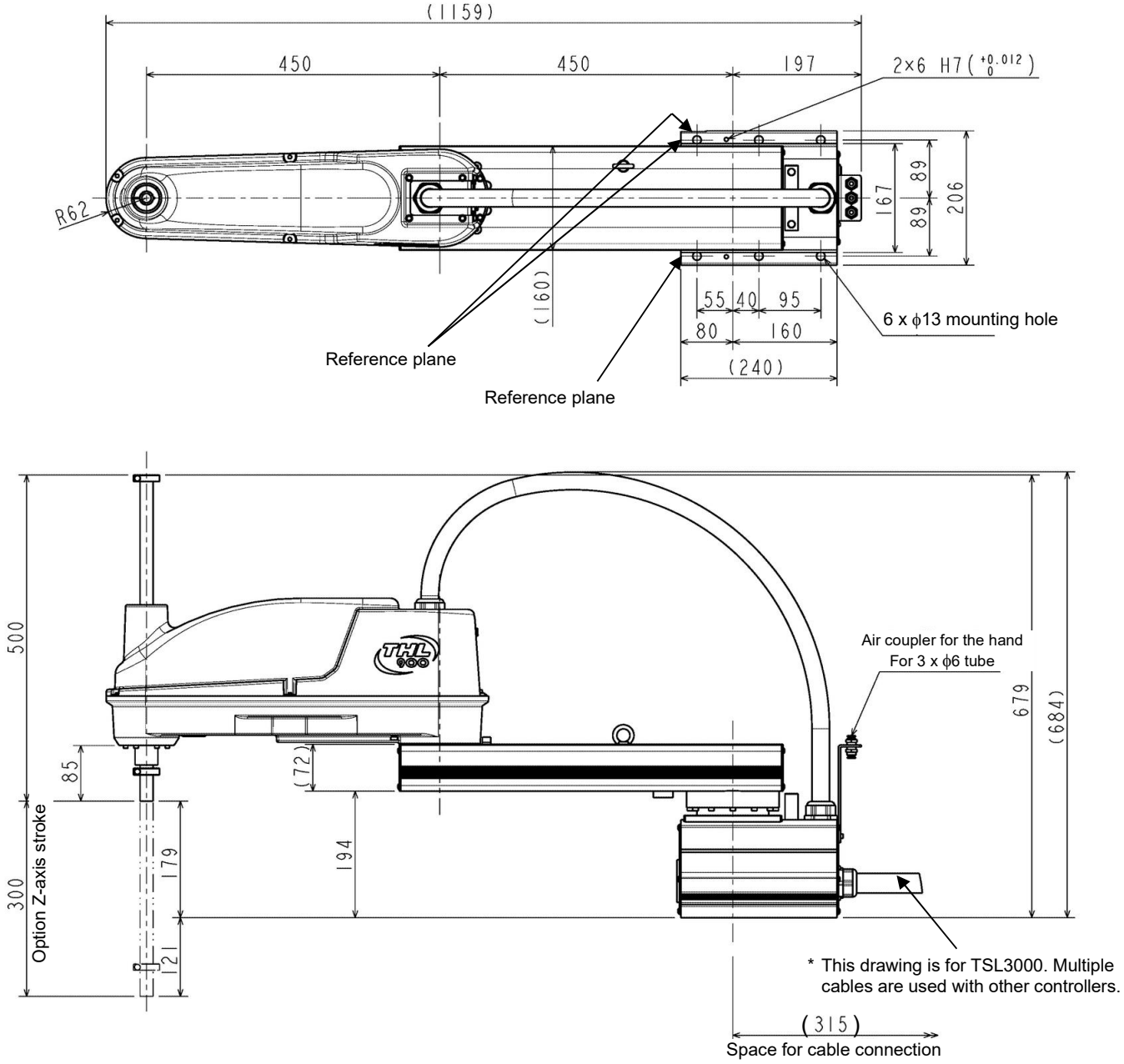


Fig. 3.3 External dimensions of robot (THL900)

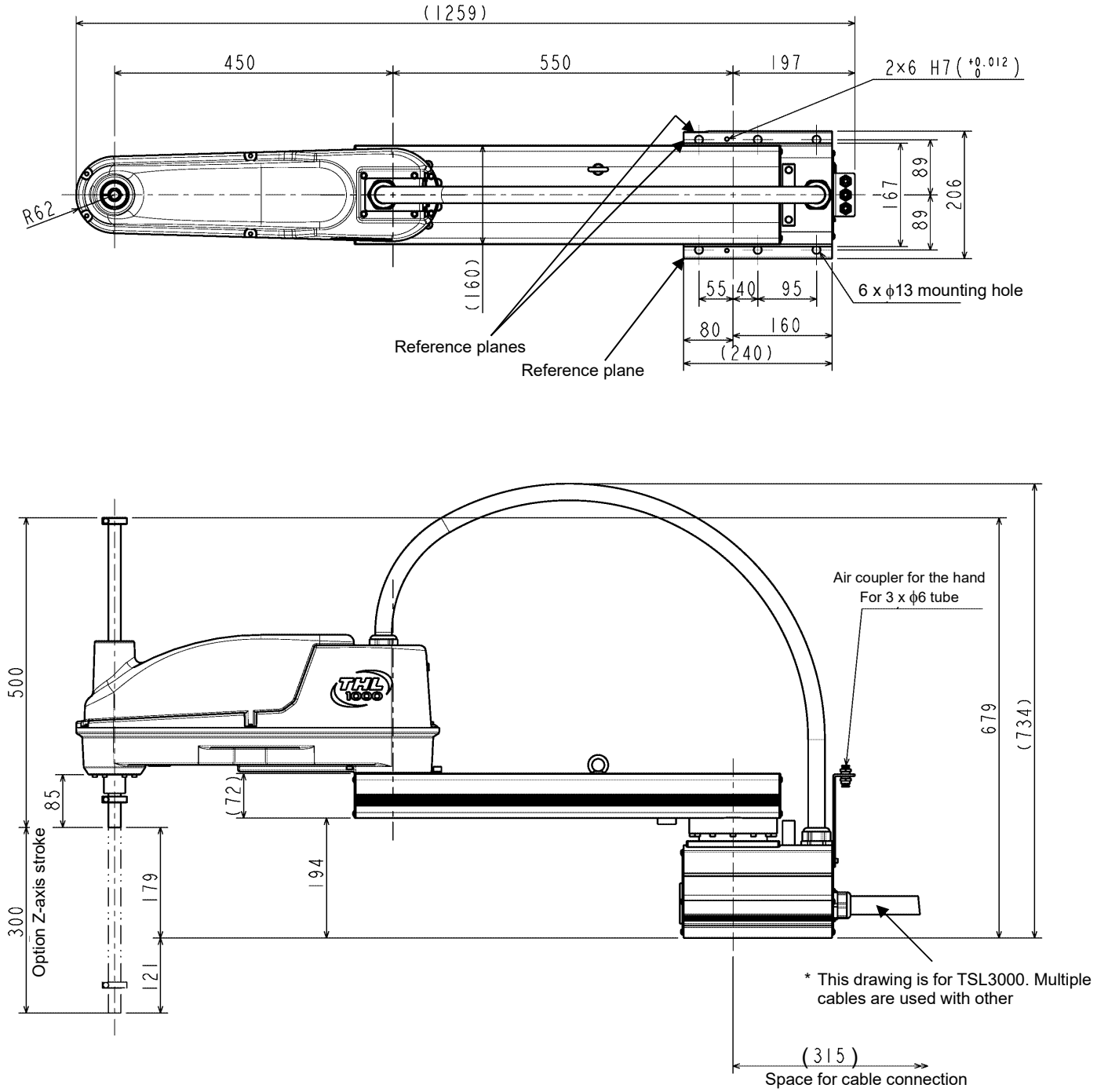


Fig. 3.4 External dimensions of robot (THL1000)

3.4 Specifications Table

Item		Specifications		
Structure		Horizontal multi-joint type SCARA robot		
Model		THL800	THL900	THL1000
Applicable controller		TSL3000, TSL3000E, TS3000, TS3000E		
Mass of robot body		33 kg	35 kg	37 kg
No. of controlled axes		4 axis		
Arm length		800 mm (350 mm + 450 mm)	900 mm (450 mm + 450 mm)	1000 mm (550 mm + 450 mm)
Motor capacity / Current limit	Axis 1	400 W / 14.7 A_{0-P}		
	Axis 2	200 W / 9.37 A_{0-P}		
	Axis 3	200 W / 6.90 A_{0-P}		
	Axis 4	200 W / 9.37 A_{0-P}		
Operating range	Axis 1	±125 deg		
	Axis 2	±145 deg		
	Axis 3	0 to 300 mm		
	Axis 4	±360 deg		
Maximum speed (*1)	Axis 1	187.5 deg/s		
	Axis 2	217.5 deg/s		
	Axis 3	2000 mm/s		
	Axis 4	1700 deg/s		
	Composite speed of axes 1 and 2	4.3 m/s	4.6 m/s	5.0 m/s
Rated payload mass (*1)		2 kg		
Maximum payload mass (*1)		10 kg		
Permissible load inertia (*1)		0.2 kg·m^2		
Repeatability (*2)	X,Y	±0.02 mm		
	Z	±0.015 mm		
	C	±0.007 deg		
Cycle time (*3) (When payload mass is 2 kg)		0.47 s	0.48 s	0.48 s
Drive system		By means of AC servo motors		
Position detection method		Absolute		
Paint color (*4)		Arm 2: Equivalent to PANTONE 293C, acrylic paint Body: White alumite processing Arm cover: White		
Power supply capacity		1.4 kVA		

*1: There are restrictions on speed and acceleration speed depending on the operation pattern, load, and offset amount.

*2: This is unidirectional position repeatability when the ambient temperature is constant at 20°C. It is not absolute positioning accuracy. The values of X, Y and C are what Z value is in upper limit. It does not guarantee the trajectory accuracy.

*3: The standard cycle operation pattern cannot achieve continuous operation exceeding the effective load rate. With horizontal direction 300 mm, vertical direction 25 mm round trip, and rough positioning.

*4: The tone may be different among production lots, but It does not affect the product quality.

! CAUTION

- Micro vibration may occur depending on the robot posture. If micro vibration occurs, reduce the acceleration speed to use the robot.
- Put the Z-axis (axis 3) in the raised position as much as possible, when moving Axes 1, 2, and 4.

  Moving Axis 1, 2, or 4 when the Z-axis is in low positions can lead to premature damage to the ball screw spline (Z-axis shaft).

  If Axis 1, 2, or 4 must be moved while the Z-axis is in low positions due to unavoidable circumstances, prevent the ball screw spline vibrations by using the SPEED, ACCEL/DECEL, and PAYLOAD commands to adjust the operation speed and acceleration.
- If Axis 1, 2, or 4 is moved while the Z-axis is in low positions, be extremely careful not to collide with any objects. Even if Axis 1, 2, or 4 is moved at low speed, a collision or other impact with an object can damage the ball screw spline (Z-axis shaft) before alarm occurs.

4. Specifications (THL1200)

4.1 Correspondence between robot and controller, and equipment configuration diagram

For the correspondence between robot and controller, and equipment configuration diagram, see the description of 1.1 through 1.4.

4.2 Name of Each Parts

Fig. 4.1 shows the name of each part of the robot (THL1200).

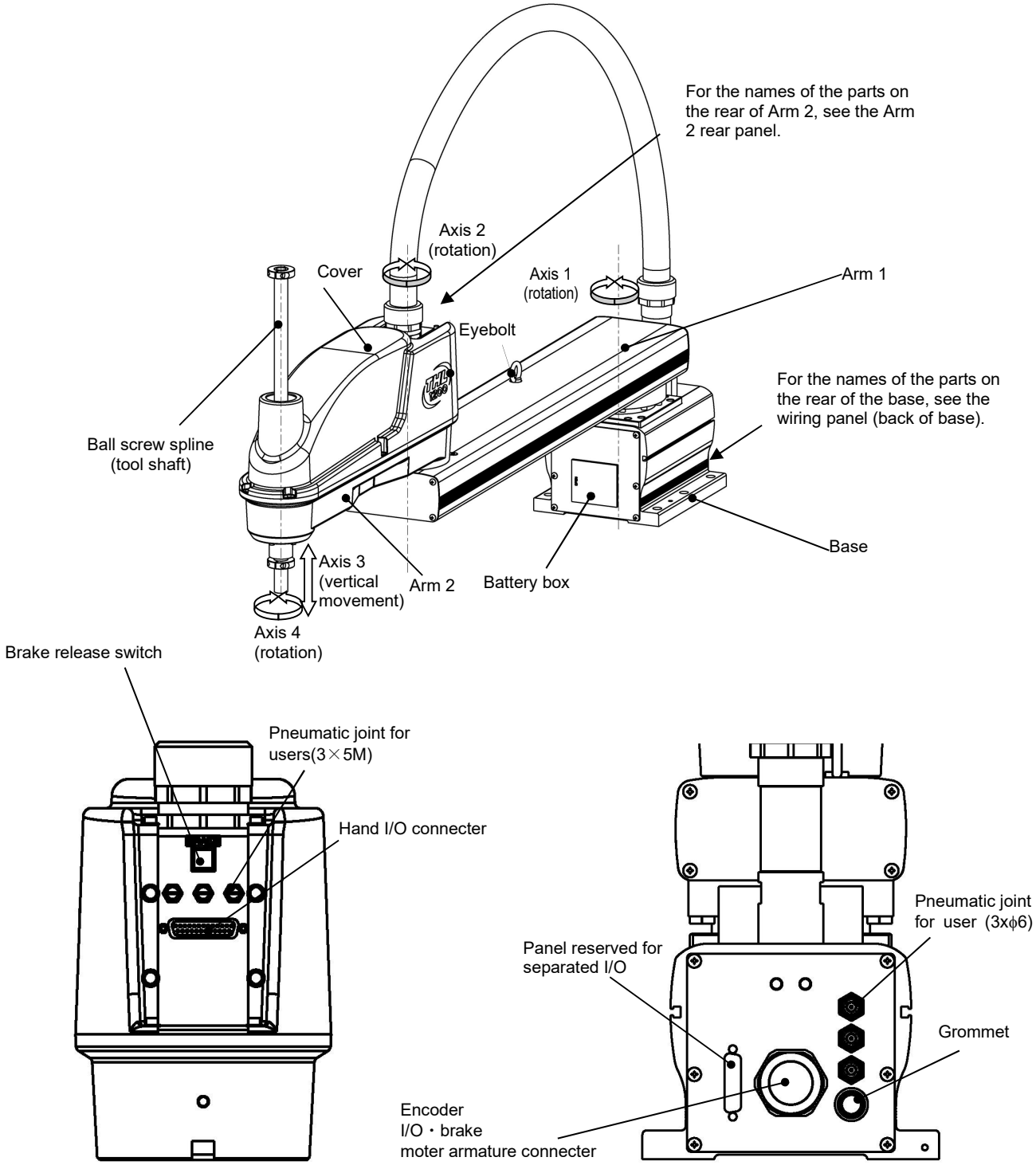


Fig. 4.1 Name of each part (THL1200)

4.3 External Dimensions

Figs. 4.2 refer to the external dimensions of the robot (THL1200).

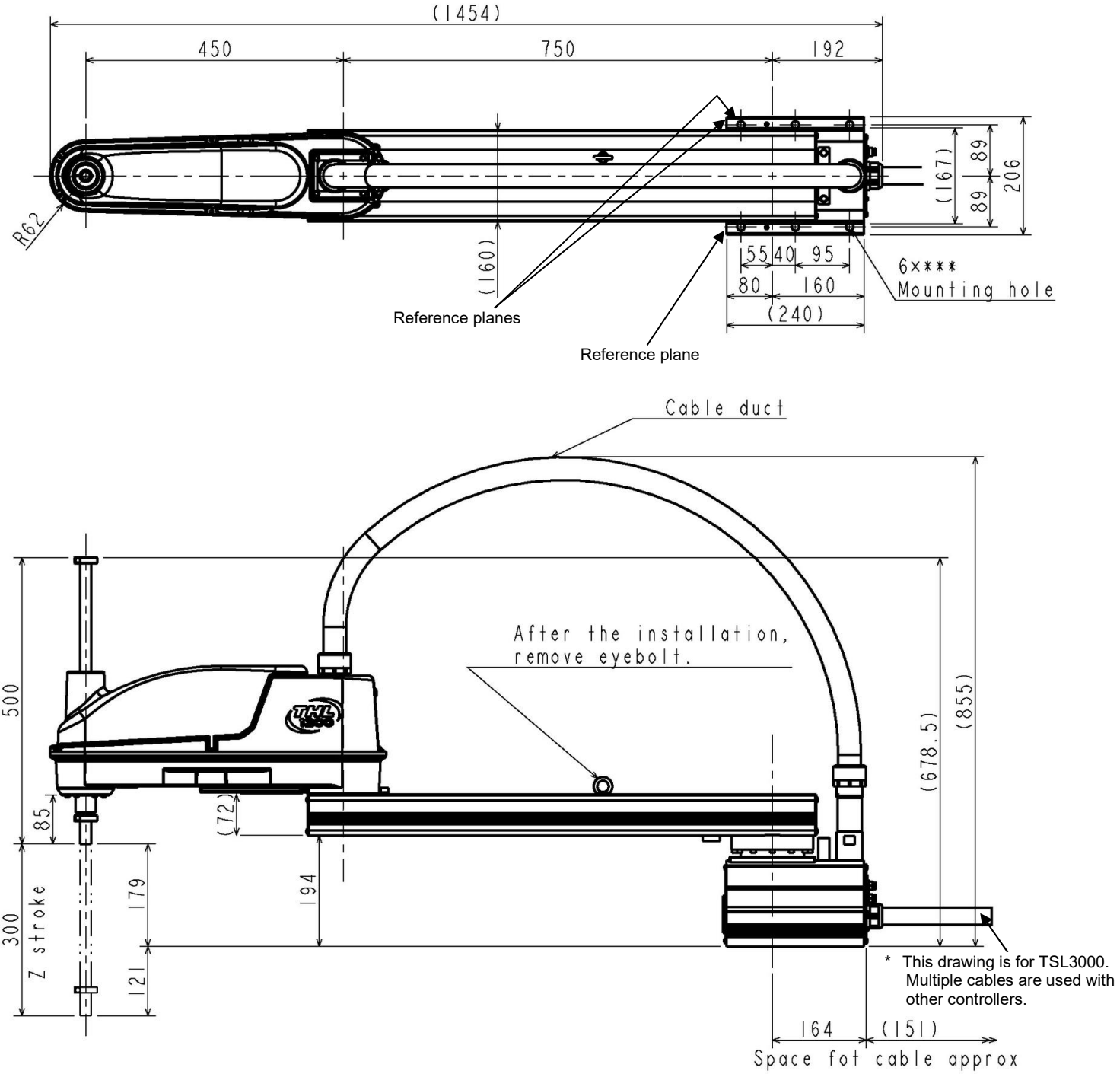


Fig. 4.2 External dimensions of robot (THL1200)

4.4 Specifications Table

Item		Specifications
Structure		Horizontal multi-joint type SCARA robot
Model		THL1200
Applicable controller		TSL3000, TSL3000E, TS3000, TS3000E
Mass of robot body		40 kg
No. of controlled axes		4 axis
Arm length		1200 mm (750 mm + 450 mm)
Motor capacity / Current limit	Axis 1	400 W / 14.7 A_{0-P}
	Axis 2	200 W / 9.37 A_{0-P}
	Axis 3	200 W / 6.90 A_{0-P}
	Axis 4	200 W / 9.37 A_{0-P}
Operating range	Axis 1	±125 deg
	Axis 2	±145 deg
	Axis 3	0 to 300 mm
	Axis 4	±360 deg
Maximum speed (*1)	Axis 1	187.5 deg/s
	Axis 2	217.5 deg/s
	Axis 3	2000 mm/s
	Axis 4	1700 deg/s
	Composite speed of axes 1 and 2	5.7 m/s
Rated payload mass (*1)		2 kg
Maximum payload mass (*1)		10 kg
Permissible load inertia (*1)		0.2 kg·m^2
Repeatability (*2)	X,Y	±0.05 mm
	Z	±0.03 mm
	C	±0.014 deg
Cycle time (*3) (When payload mass is 2 kg)		0.58 s
Drive system		By means of AC servo motors
Position detection method		Absolute
Paint color (*4)		Arm 2: Equivalent to PANTONE 293C, acrylic paint Body: White alumite processing Arm cover: White
Power supply capacity		1.4 kVA

*1: There are restrictions on speed and acceleration speed depending on the operation pattern, load, and offset amount.

*2: This is unidirectional position repeatability when the ambient temperature is constant at 20°C. It is not absolute positioning accuracy. The values of X, Y and C are what Z value is in upper limit. It does not guarantee the trajectory accuracy.

*3: The standard cycle operation pattern cannot achieve continuous operation exceeding the effective load rate. With horizontal direction 300 mm, vertical direction 25 mm round trip, and rough positioning.

*4: The tone may be different among production lots, but It does not affect the product quality.

! CAUTION

- Micro vibration may occur depending on the robot posture. If micro vibration occurs, reduce the acceleration speed to use the robot.
- Put the Z-axis (axis 3) in the raised position as much as possible, when moving Axes 1, 2, and 4.

  Moving Axis 1, 2, or 4 when the Z-axis is in low positions can lead to premature damage to the ball screw spline (Z-axis shaft).

  If Axis 1, 2, or 4 must be moved while the Z-axis is in low positions due to unavoidable circumstances, prevent the ball screw spline vibrations by using the SPEED, ACCEL/DECEL, and PAYLOAD commands to adjust the operation speed and acceleration.
- If Axis 1, 2, or 4 is moved while the Z-axis is in low positions, be extremely careful not to collide with any objects. Even if Axis 1, 2, or 4 is moved at low speed, a collision or other impact with an object can damage the ball screw spline (Z-axis shaft) before alarm occurs.

5. Transportation (THL300 and THL400)

5.1 Unpacking (TSL3000)

For the TSL3000E, see "5.2 Unpacking (TSL3000E) ".
For the TS3000 and TS3000E, see "5.3 Unpacking (TS3000 and TS3000E)".
The robot and controller are shipped separately in corrugated cardboards. Fig. 5.1 shows each packaging state.
Open the packages in a location easily accessible, where the equipment is to be installed. Take careful precautions not to damage the robot and controller.
After opening the packages, make sure that all the accessories are present and that nothing has been damaged during transport.
For the controller accessories, see the accessory list packed with the controller.

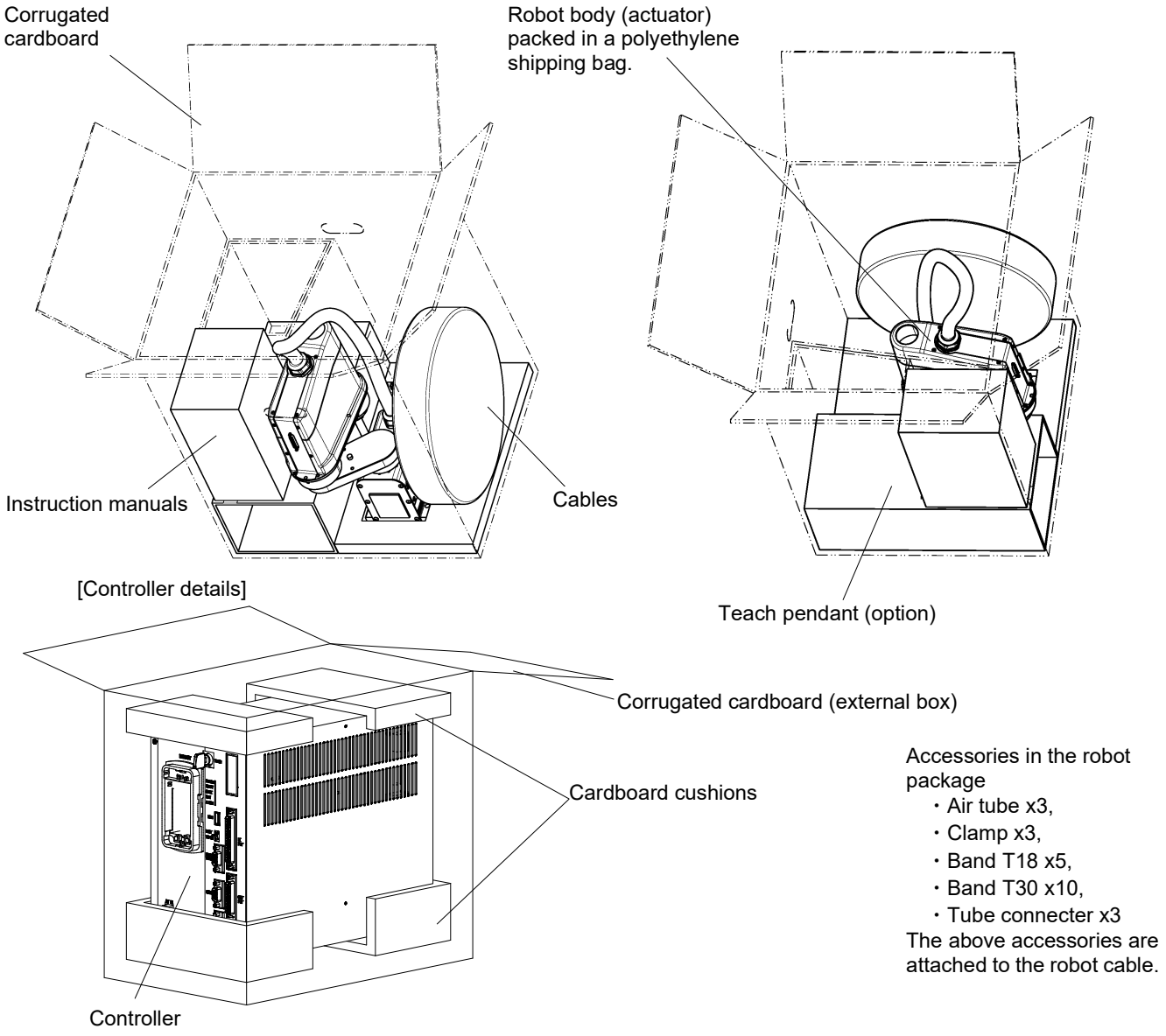


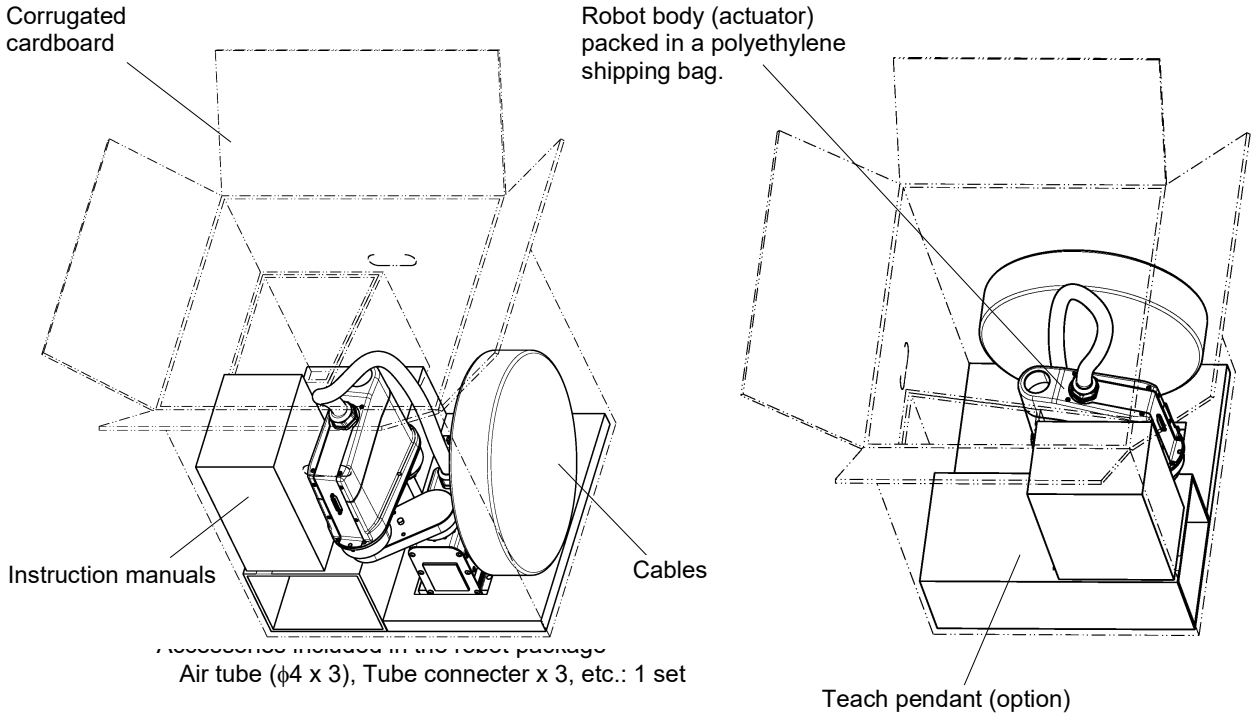
Fig. 5.1 Packaging state (THL300/THL400 and TSL3000)

DANGER

- If any parts of the equipment are found damaged or any accessories are missing after the shipment containing the robot and controller have reached your office, DO NOT install and operate them. Otherwise, the equipment will malfunction. Contact SHIBAURA MACHINE immediately.
- Dispose of the wooden pallet, corrugated cardboards, polyethylene shipping bags and cushion material according to the customer's in-house regulations.

5.2 Unpacking (TSL3000E)

For the TSL3000, see "5.1 Unpacking (TSL3000) ".
For the TS3000 and TS3000E, see "5.3 Unpacking (TS3000 and TS3000E)".
The robot and controller are shipped separately in corrugated cardboards.
Fig. 5.2 shows each packaging state.
Open the packages in a location easily accessible, where the equipment is to be installed. Take careful precautions not to damage the robot and controller.
After opening the packages, make sure that all the accessories are present and that nothing has been damaged during transport.
For the controller accessories, see the accessory list packed with the controller.



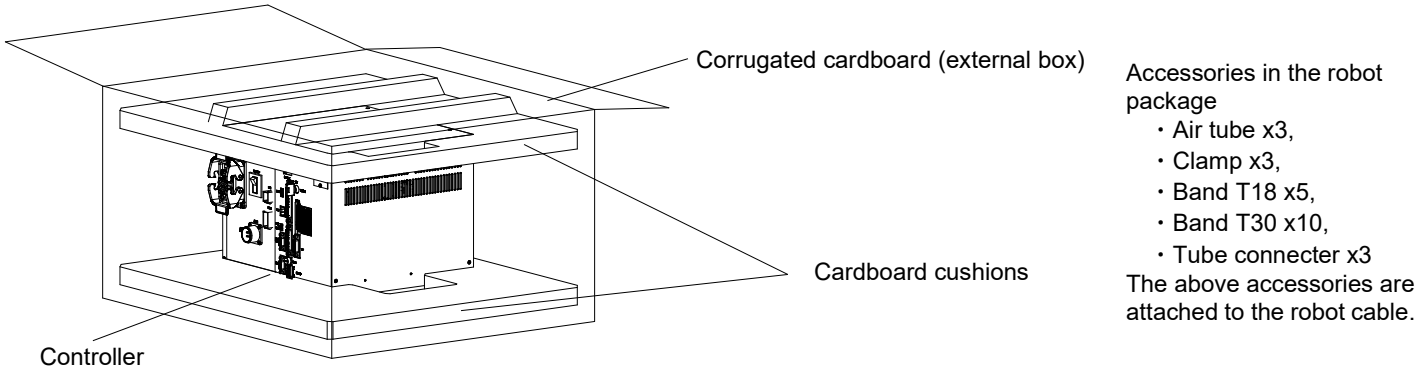


Fig. 5.2 Packaging state (THL300/THL400 and TSL3000E)

DANGER

- If any parts of the equipment are found damaged or any accessories are missing after the shipment containing the robot and controller have reached your office, DO NOT install and operate them. Otherwise, the equipment will malfunction. Contact SHIBAURA MACHINE immediately.
- Dispose of the wooden pallet, corrugated cardboards, polyethylene shipping bags and cushion material according to the customer's in-house regulations.

5.3 Unpacking (TS3000 and TS3000E)

For the TSL3000, see "5.1 Unpacking (TSL3000)".
For the TSL3000, see "5.2 Unpacking (TSL3000E)".
The robot and controller are shipped separately in corrugated cardboards.
Fig. 5.3 shows each packaging state.
Open the packages in a location easily accessible, where the equipment is to be installed. Take careful precautions not to damage the robot and controller.
After opening the packages, make sure that all the accessories are present and that nothing has been damaged during transport.
For the controller accessories, see the accessory list packed with the controller.

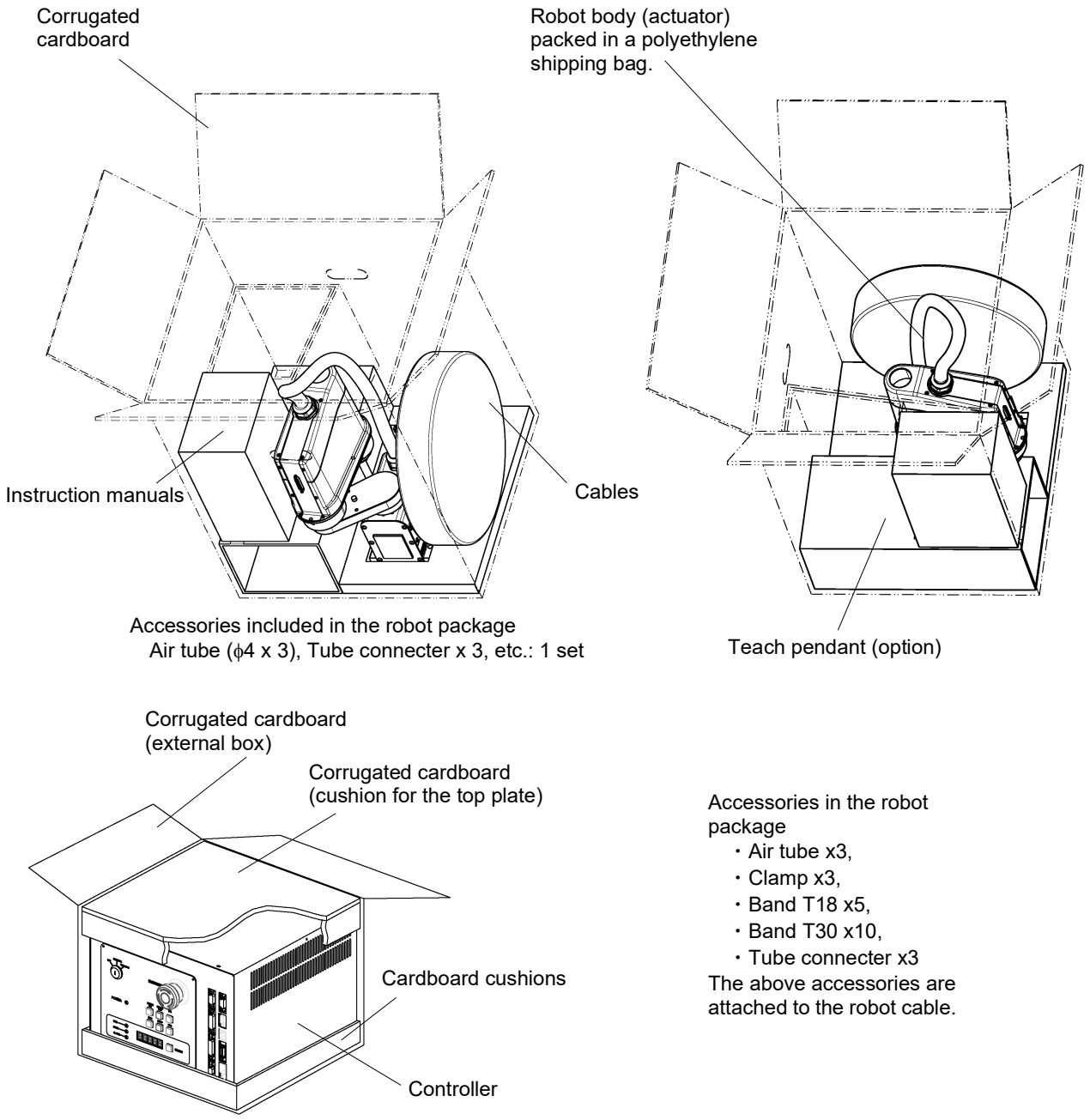


Fig. 5.3 Packaging state (THL300/THL400 and TS3000/TS3000E)

DANGER

- If any parts of the equipment are found damaged or any accessories are missing after the shipment containing the robot and controller have reached your office, DO NOT install and operate them. Otherwise, the equipment will malfunction. Contact SHIBAURA MACHINE immediately.
- Dispose of the wooden pallet, corrugated cardboards, polyethylene shipping bags and cushion material according to the customer's in-house regulations.

5.4 Transportation

Move the robot and controller very carefully. Make sure that no excessive impact or vibration is exerted on the equipment. If the equipment is to be subject to vibration over a long period, be sure to tighten all the clamp and base set bolts completely.
If the equipment is to be moved to a location some distance from where it was unpacked, reposition the cushions as they were and put the equipment back into the corrugated cardboards.

5.4.1 Mass and Dimensions

The mass and outer dimentions of the robot are shown in Figs. 5.4 to 5.5.
For the controller mass and external dimensions, see "13.1 External Dimensions", "14.1 External Dimensions" or "15.1 External Dimensions".

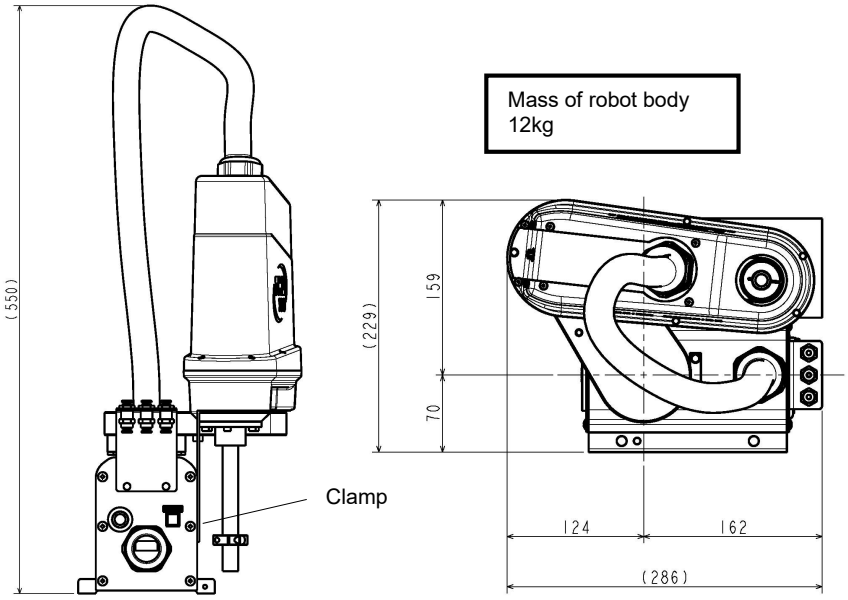


Fig. 5.4 Packaging state (THL300)

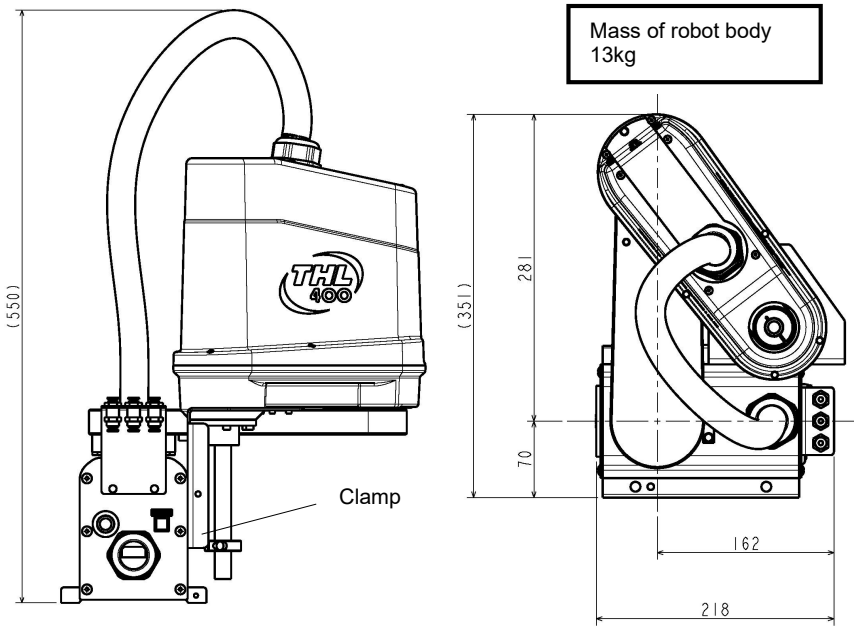


Fig. 5.5 Packaging state (THL400)

5.4.2 Transporting the Robot

In principle, the robot should be transported in the state shown in Figs. 5.4 to 5.5 above. Fold back and secure the arm with the attached clamp in Figs. 5.6 above. Then, fix a tip of the ball screw spline in the binding band in Figs. 5.7 above. (The robot is shipped in this posture. After you have unpacked the shipment, you should move it as it is.) At this time, take careful precautions not to impose a large force on the tool shaft. Especially, the THL300 and THL400 are shipped with their ball screw spline shaft drawn out long, so be careful. Further, the THL300 and the THL400 cannot be lifted for transportation.

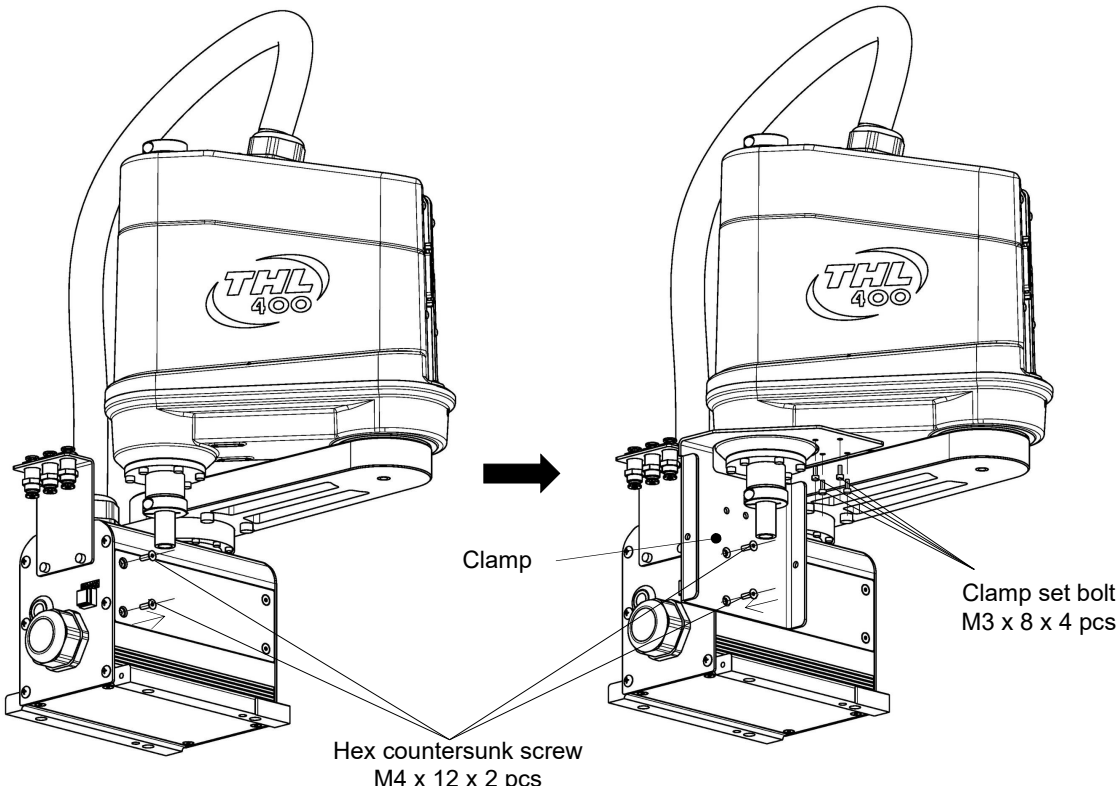


Fig. 5.6 Attaching the clamp (THL300/THL400)

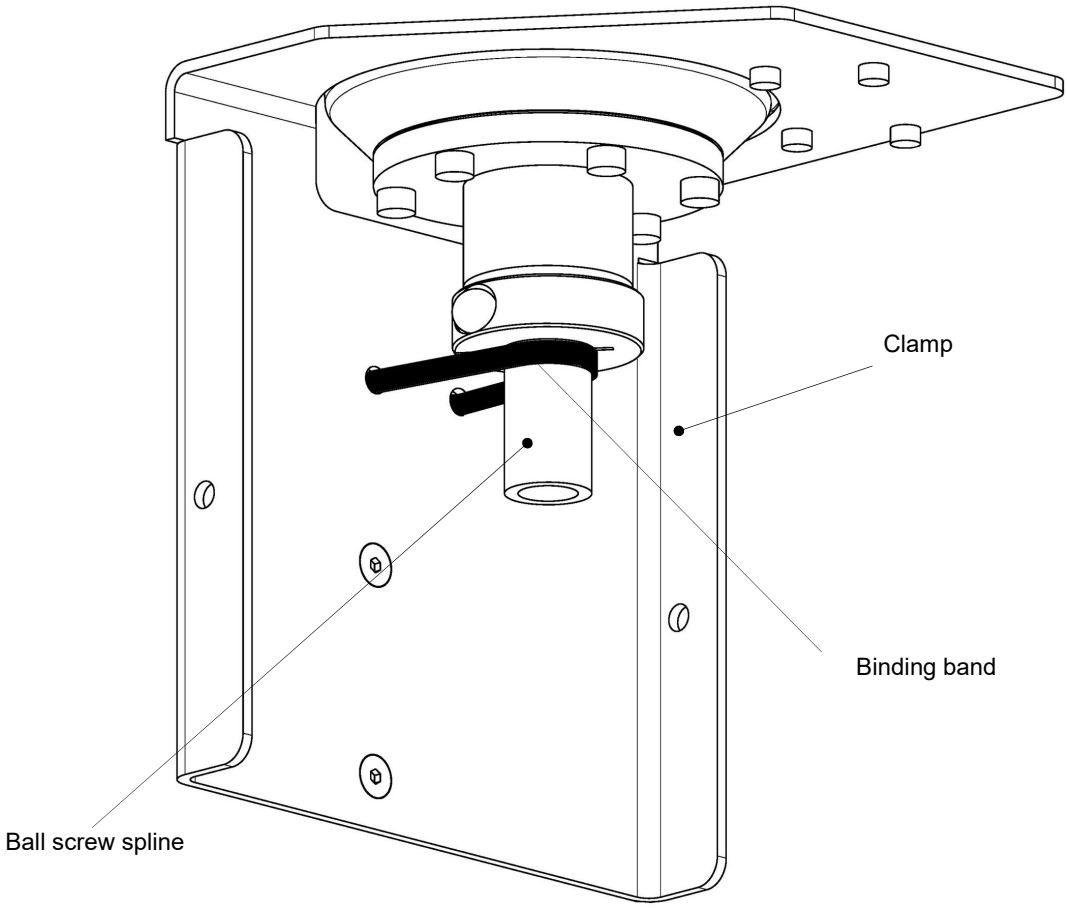


Fig. 5.7 Fixing the ball screw spline (THL300/THL400)

DANGER

- Be sure to secure the arm with the attached clamp before transporting the robot. Failure to do so could cause a hazardous situation as the arm will move when the robot is lifted.

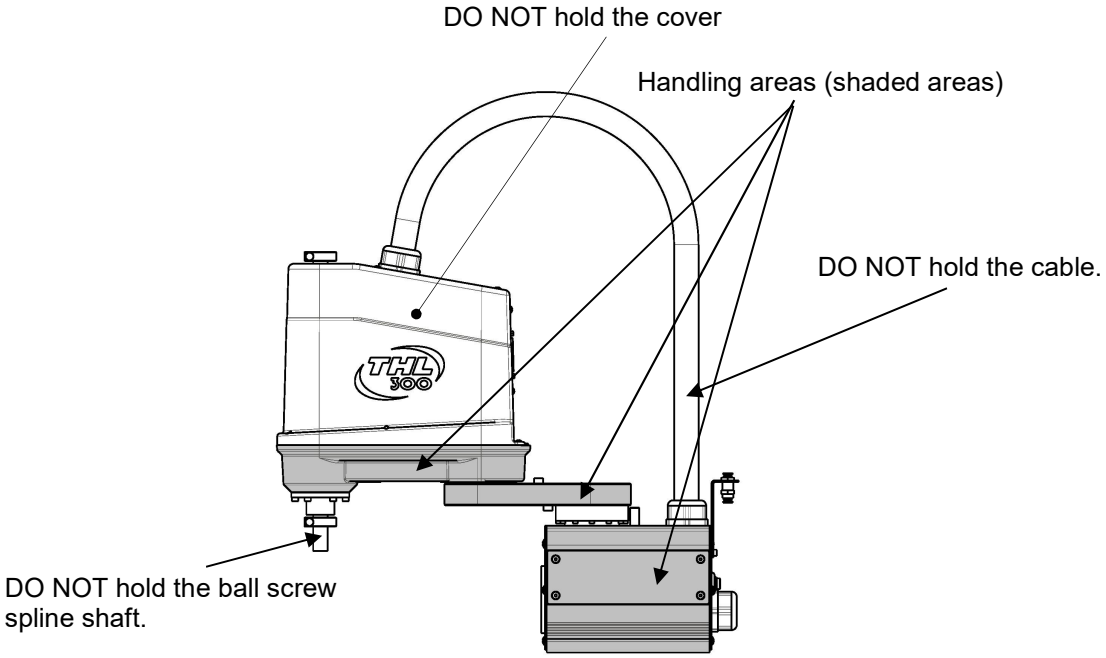


Fig. 5.8 Robot Handling Locations (THL300 and THL400)

After the installation, remove the clamp used for transport.

CAUTION

- When lifting up the robot by workers, hold the locations (shaded areas) by hands as shown in Fig. 5.8. Holding the arm 2 cover, the cable and the ball screw spline shaft will apply excess force, possibly causing failure.
- When carrying the robot by workers, take careful precautions to prevent their hand or leg from being caught in the robot.
- The work should be performed by two (2) or more workers.
- Never touch the ball screw spline shaft with a bare hand. If you touch it with a bare hand, earlier rust formation may result. Be sure to use gloves.

5.4.3 Transporting the Controller

Disconnect all cables and teach pendant before transporting the controller.

DANGER

- When placing the controller on the floor, etc., make sure not to have your hands or feet caught.

5.5 Storage

Avoid storing the robot and controller for long periods of time after unpacking them. If this is unavoidable, however, strictly observe the following precautions for storage.

5.5.1 Storage Precautions for the Robot

CAUTION

- Secure the base completely to prevent the robot from falling over. When placed directly on the floor, the robot is unstable and will fall over.
- Avoid direct sunshine and high temperature/humidity when the robot is stored. The timing belts and resin covers may deteriorate.
- Seal the robot in a vinyl bag to prevent rust development and contaminant. Put a desiccant in the bag to absorb moisture. As the tool shaft is susceptible to rust development, coat it with rust-preventive agent or grease the entire tool shaft beforehand.
  For the grease application procedure, see the maintenance manual "Greasing Ball Screw Spline Unit and Applying Anticorrosive".
- Before the use, apply the grease to the tool shaft.
- Before starting an operation, perform running completely.

5.5.2 Storage Precautions for the Controller

CAUTION

- Keep the controller out of direct sunlight. Otherwise, the controller interior will be excessively heated up, causing a trouble.
- Seal the controller in a vinyl bag to prevent rust development and contaminant. Put a desiccant in the bag to absorb moisture.

6. Transportation (THL500, THL600 and THL700)

6.1 Unpacking (TSL3000)

For information on TSL3000E, refer to Section 6.2. For information on TS3000 and TS3000E, refer to Section 6.3. The robot and controller are shipped separately in corrugated cardboards. Fig. 6.1 shows each packaging state.
Open the packages in a location easily accessible, where the equipment is to be installed. Take careful precautions not to damage the robot and controller.
After opening the packages, make sure that all the accessories are present and that nothing has been damaged during transport.
For the controller accessories, see the accessory list packed with the controller.

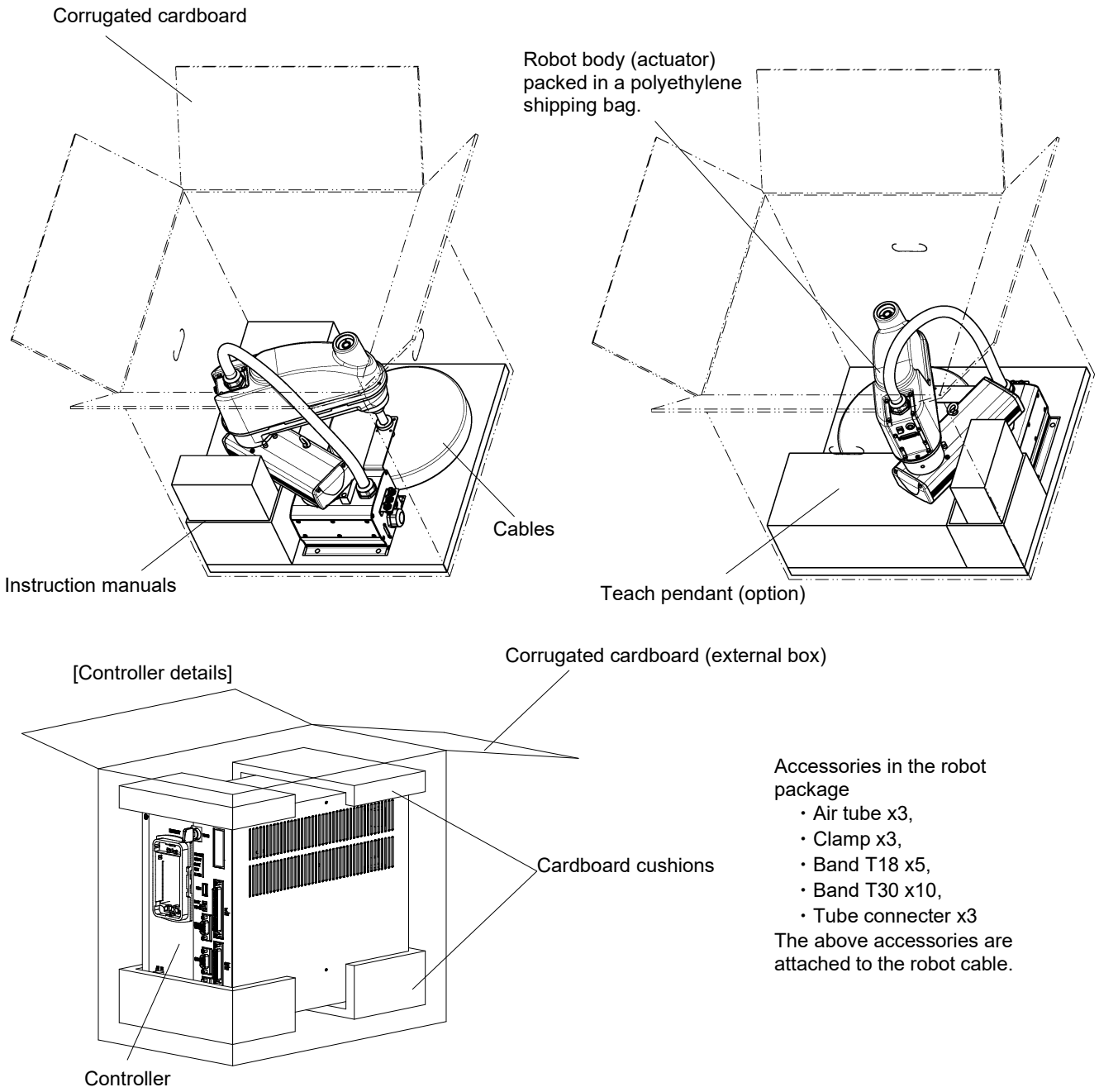


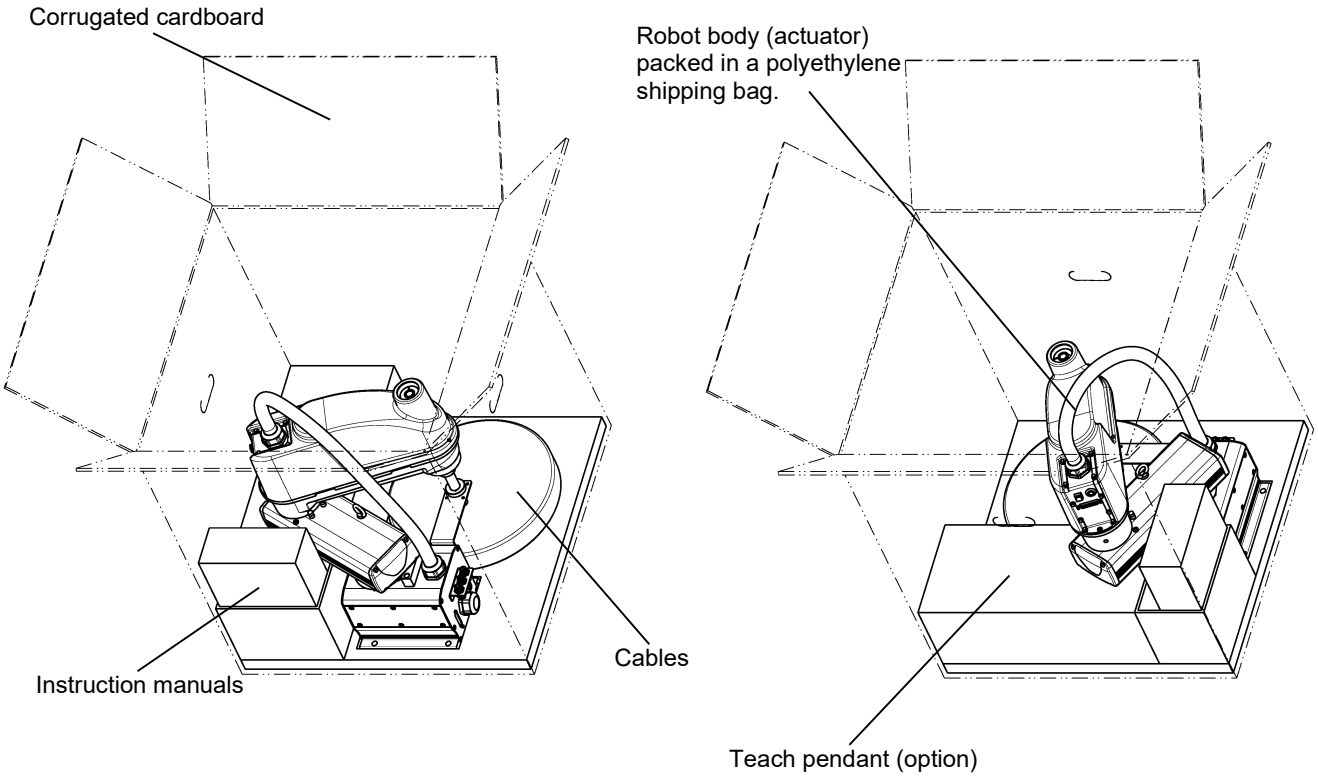
Fig. 6.1 Packaging state (THL500/THL600/THL700 and TSL3000)

DANGER

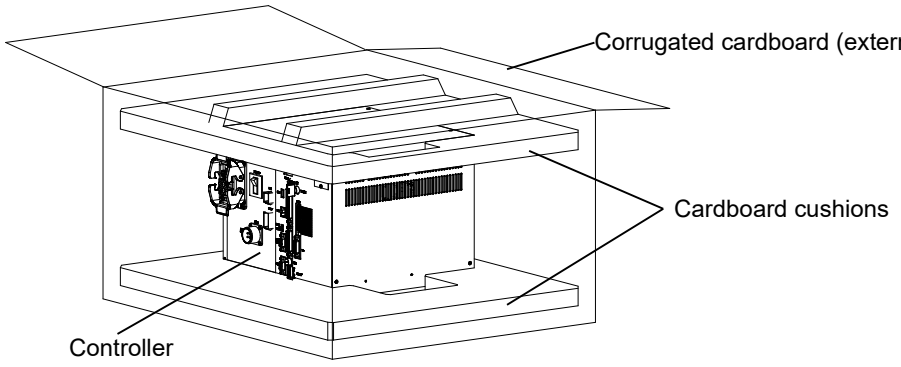
- If any parts of the equipment are found damaged or any accessories are missing after the shipment containing the robot and controller have reached your office, DO NOT install and operate them. Otherwise, the equipment will malfunction. Contact SHIBAURA MACHINE immediately.
- Dispose of the wooden pallet, corrugated cardboards, polyethylene shipping bags and cushion material according to the customer's in-house regulations.

6.2 Unpacking (TSL3000E)

For information on TSL3000, refer to Section 6.1. For information on TS3000 and TS3000E, refer to Section 6.3. The robot and controller are shipped separately in corrugated cardboards. Fig. 6.2 shows each packaging state.
Open the packages in a location easily accessible, where the equipment is to be installed. Take careful precautions not to damage the robot and controller.
After opening the packages, make sure that all the accessories are present and that nothing has been damaged during transport.
For the controller accessories, see the accessory list packed with the controller.



[Controller details]



Accessories in the robot package
- Air tube x3,
- Clamp x3,
- Band T18 x5,
- Band T30 x10,
- Tube connecter x3

The above accessories are attached to the robot cable.

Fig. 6.2 Packaging state (THL500/THL600/THL700 and TSL3000E)

DANGER

- If any parts of the equipment are found damaged or any accessories are missing after the shipment containing the robot and controller have reached your office, DO NOT install and operate them. Otherwise, the equipment will malfunction. Contact SHIBAURA MACHINE immediately.
- Dispose of the wooden pallet, corrugated cardboards, polyethylene shipping bags and cushion material according to the customer's in-house regulations.

6.3 Unpacking (TS3000 and TS3000E)

For information on TSL3000, refer to Section 6.1. For information on TSL3000E, refer to Section 6.2. The robot and controller are shipped separately in corrugated cardboards. Fig. 6.3 shows each packaging state.

Open the packages in a location easily accessible, where the equipment is to be installed. Take careful precautions not to damage the robot and controller.

After opening the packages, make sure that all the accessories are present and that nothing has been damaged during transport.

For the controller accessories, see the accessory list packed with the controller.

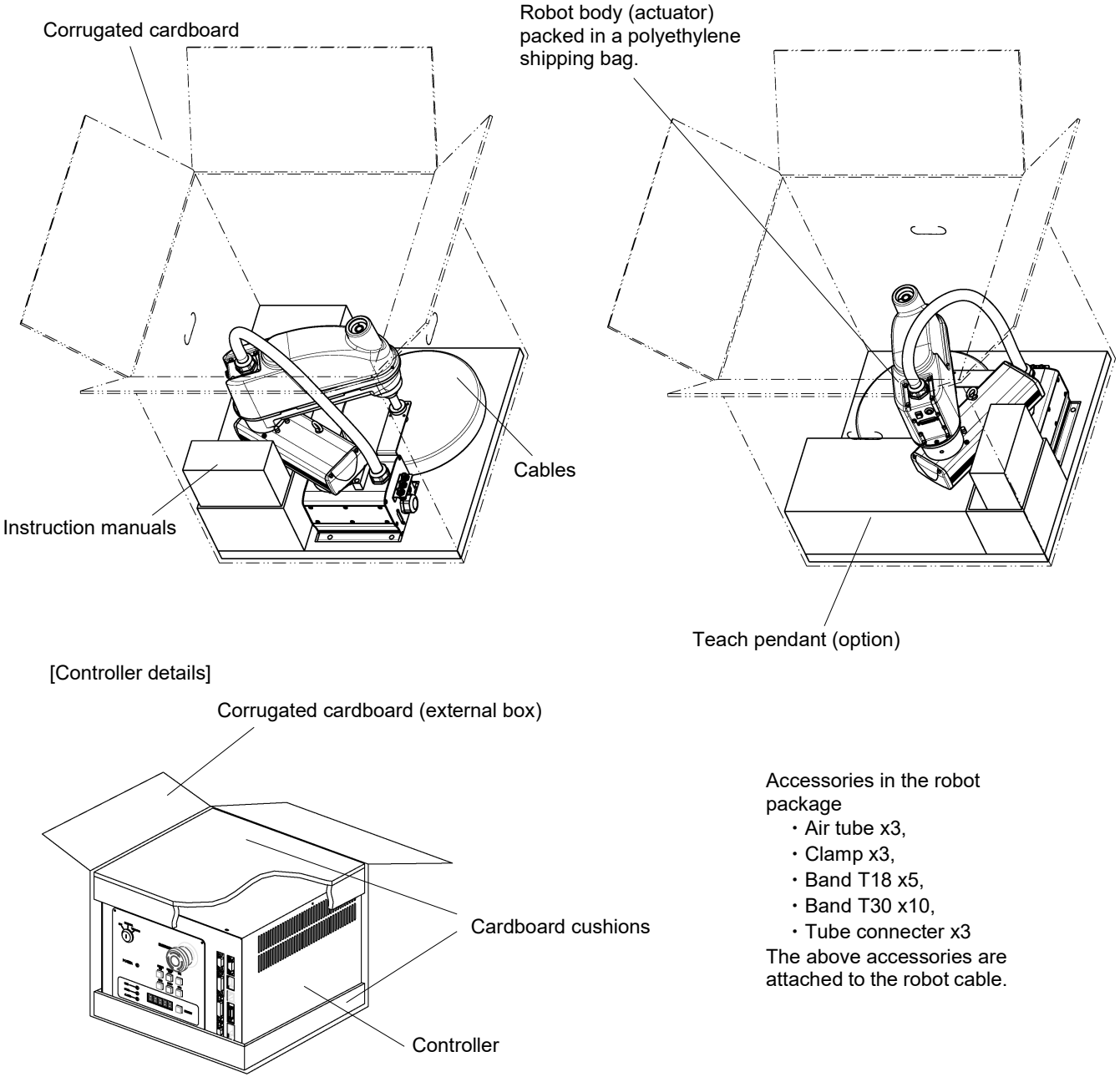


Fig. 6.3 Packaging state (THL500/THL600/THL700 and TS3000/TS3000E)

DANGER

- If any parts of the equipment are found damaged or any accessories are missing after the shipment containing the robot and controller have reached your office, DO NOT install and operate them. Otherwise, the equipment will malfunction. Contact SHIBAURA MACHINE immediately.
- Dispose of the wooden pallet, corrugated cardboards, polyethylene shipping bags and cushion material according to the customer's in-house regulations.

6.4 Transportation

Move the robot and controller very carefully. Make sure that no excessive impact or vibration is exerted on the equipment. If the equipment is to be subject to vibration over a long period, be sure to tighten all the clamp and base set bolts completely.
If the equipment is to be moved to a location some distance from where it was unpacked, reposition the cushions as they were and put the equipment back into the corrugated cardboards.

6.4.1 Mass and Dimensions

The mass and outer dimensions of the robot are shown in Figs. 6.4 to 6.6.
For the controller mass and external dimensions, see "13.1 External Dimensions", "14.1 External Dimensions" or "15.1 External Dimensions".

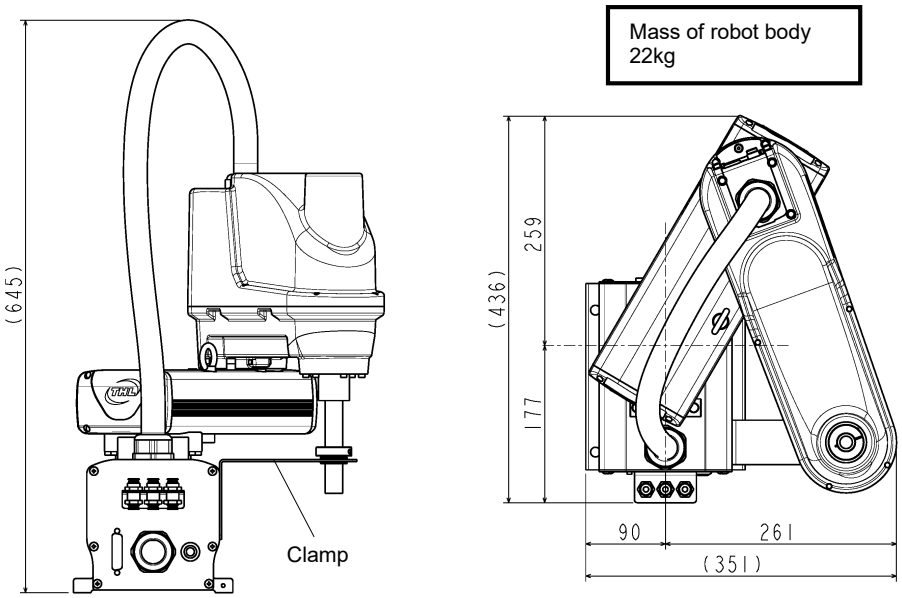


Fig. 6.4 Packaging state (THL500)

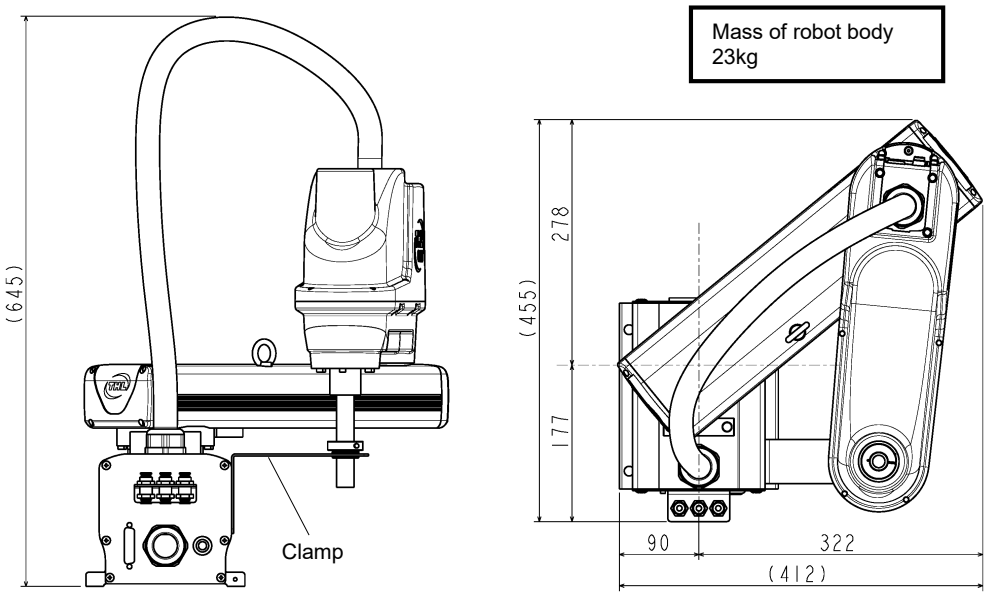


Fig. 6.5 Packaging state (THL600)

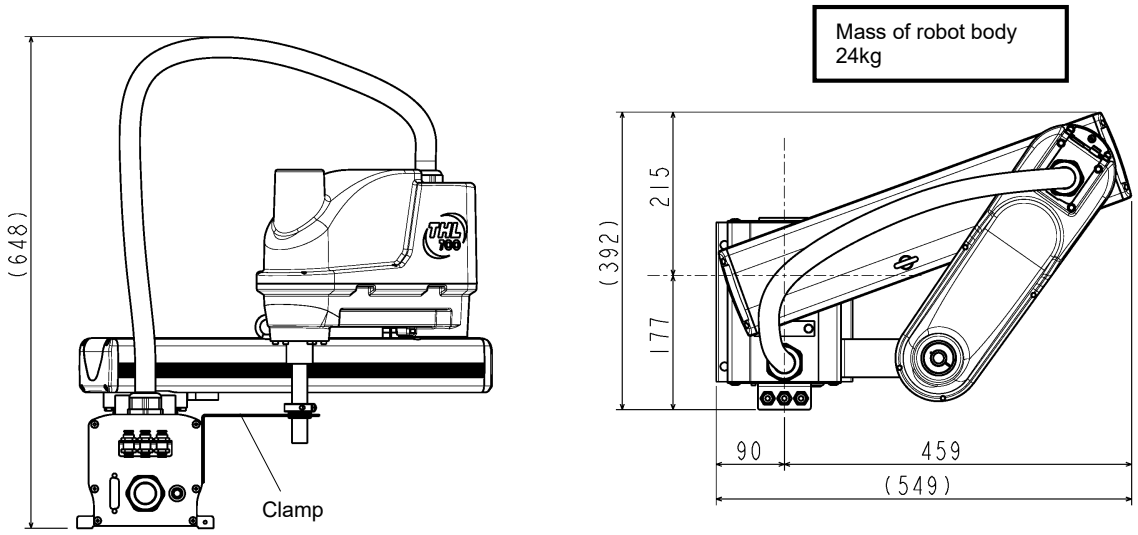


Fig. 6.6 Packaging state (THL700)

6.4.2 Transporting the Robot

In principle, the robot should be transported in the state shown in Figs. 6.4 to 6.6 above. Fold back and secure the arm with the attached clamp in Figs. 6.7 above. (The robot is shipped in this posture. After you have unpacked the shipment, you should move it as it is.) At this time, take careful precautions not to impose a large force on the tool shaft.

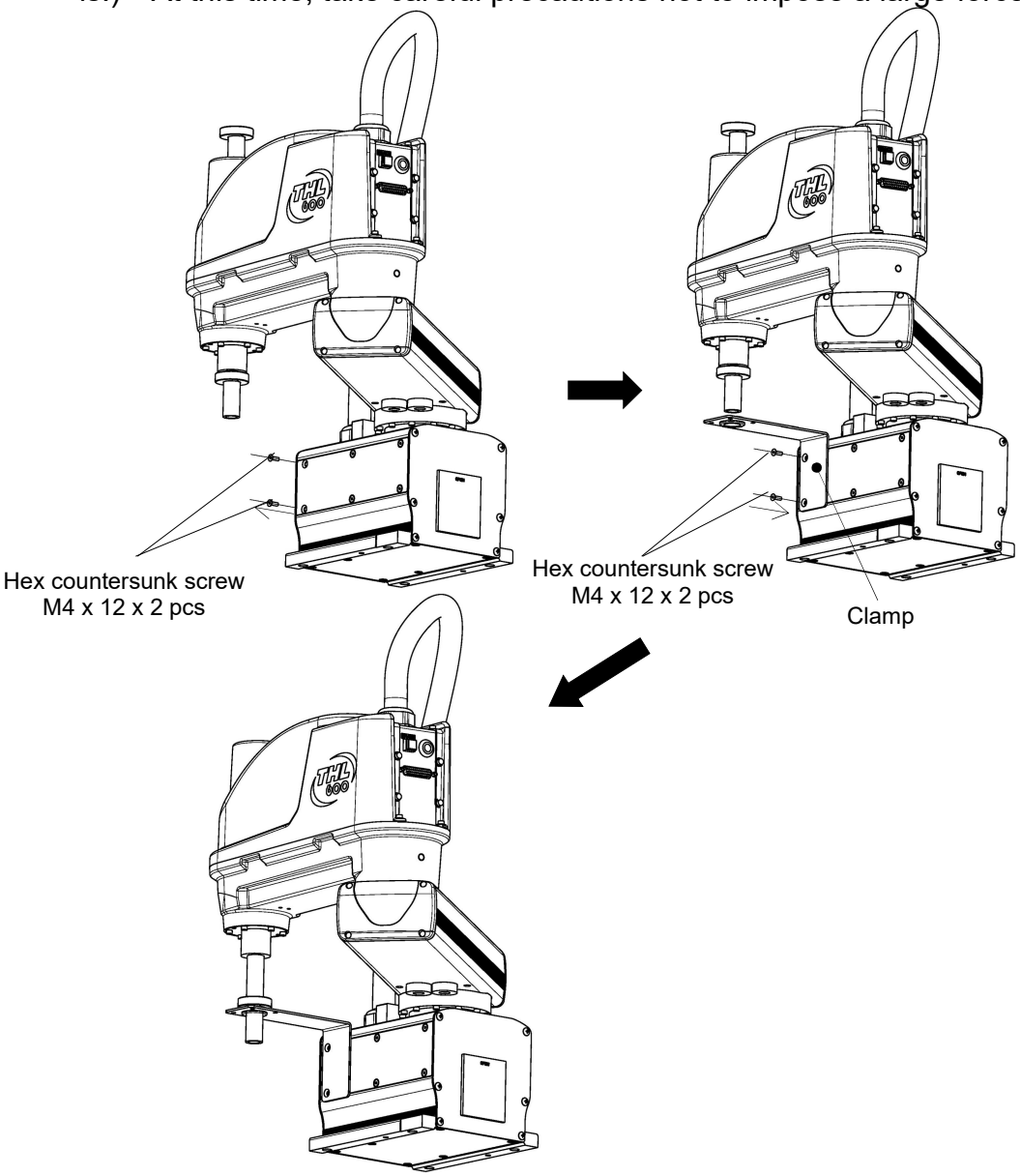


Fig. 6.7 Attaching the clamp (THL500/THL600/THL700)

DANGER

- Be sure to secure the arm with the attached clamp before transporting the robot. Failure to do so could cause a hazardous situation as the arm will move when the robot is lifted.

It is possible to lift up and transport THL500, THL600 and THL700. Pass the wire through the attached eyebolt, then lift up the robot carefully, as shown in Fig. 6.8.

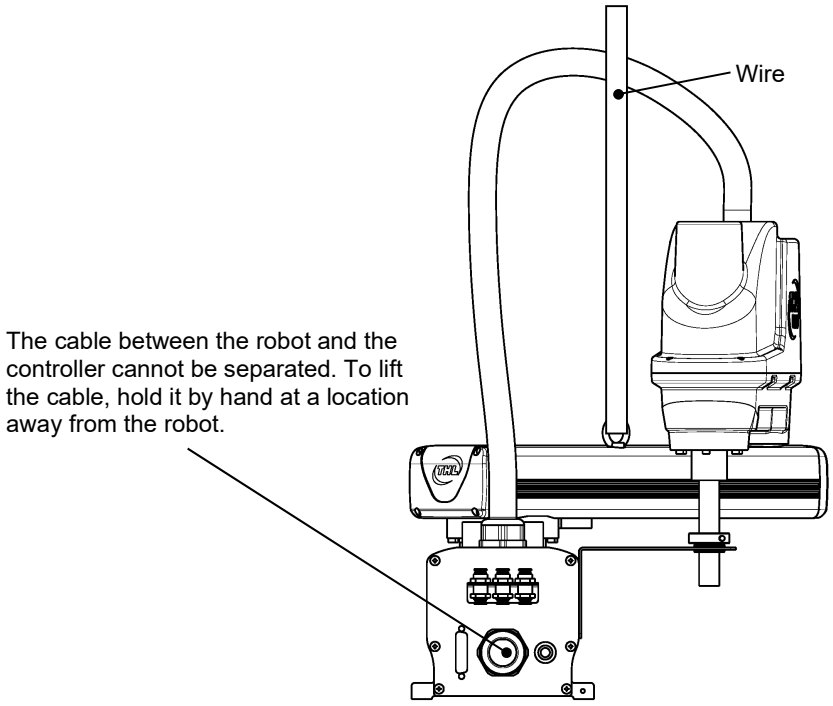


Fig. 6.8 Lifting up the robot

! CAUTION

- The wire to be used should be such that can well withstand the mass of the robot.
- When lifting up the robot, it may tilt a little. Lift it up slowly.
- Lifting up and down should be performed carefully so that any impact cannot be exerted on the robot.
- When carrying the robot by workers, take careful precautions to prevent their hand or leg from being caught in the robot.

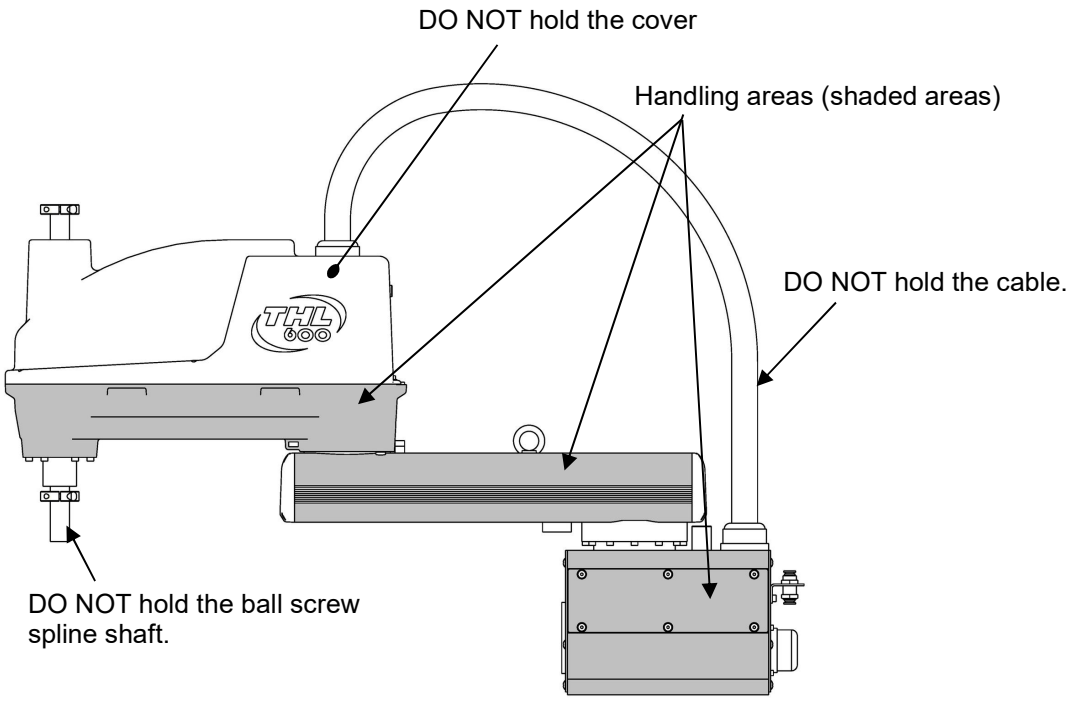


Fig. 6.9 Robot Handling Locations (THL500, THL600 and THL700)

After the installation, remove the clamp and eyebolt used for transport.

! CAUTION

- When lifting up the robot by workers, hold the locations (shaded areas) by hands as shown in Fig. 6.9. Holding the arm 2 cover, the cable and the ball screw spline shaft will apply excess force, possibly causing failure.
- When carrying the robot by workers, take careful precautions to prevent their hand or leg from being caught in the robot.
- Never touch the ball screw spline shaft with a bare hand. If you touch it with a bare hand, earlier rust formation may result. Be sure to use gloves.
- The work should be performed by two (2) or more workers.
- Plug up the threaded bore which took off an eyebolt by a bolt attached to the robot.

6.4.3 Transporting the Controller

Disconnect all cables and teach pendant before transporting the controller.

DANGER

- When placing the controller on the floor, etc., make sure not to have your hands or feet caught.

6.5 Storage

Avoid storing the robot and controller for long periods of time after unpacking them. If this is unavoidable, however, strictly observe the following precautions for storage.

6.5.1 Storage Precautions for the Robot

CAUTION

- Secure the base completely to prevent the robot from falling over. When placed directly on the floor, the robot is unstable and will fall over.
- Avoid direct sunshine and high temperature/humidity when the robot is stored. The timing belts and resin covers may deteriorate.
- Seal the robot in a vinyl bag to prevent rust development and contaminant. Put a desiccant in the bag to absorb moisture. As the tool shaft is susceptible to rust development, coat it with rust-preventive agent or grease the entire tool shaft beforehand.
  For the grease application procedure, see the maintenance manual "Greasing Ball Screw Spline Unit and Applying Anticorrosive".
- Before the use, apply the grease to the tool shaft.
- Before starting an operation, perform running completely.
- During storage, the life of the backup batteries will shorten. It is recommended to replace the batteries at the time of operation.

6.5.2 Storage Precautions for the Controller

CAUTION

- Keep the controller out of direct sunlight. Otherwise, the controller interior will be excessively heated up, causing a trouble.
- Seal the controller in a vinyl bag to prevent rust development and contaminant. Put a desiccant in the bag to absorb moisture.

7. Transportation (THL800, THL900 and THL1000)

7.1 Unpacking (TSL3000)

For information on TSL3000E, refer to Section 7.2. For information on TS3000 and TS3000E, refer to Section 7.3. The robot and controller are shipped separately in corrugated cardboards. Fig. 7.1 shows each packaging state.
Open the packages in a location easily accessible, where the equipment is to be installed. Take careful precautions not to damage the robot and controller.
After opening the packages, make sure that all the accessories are present and that nothing has been damaged during transport.
For the controller accessories, see the accessory list packed with the controller.

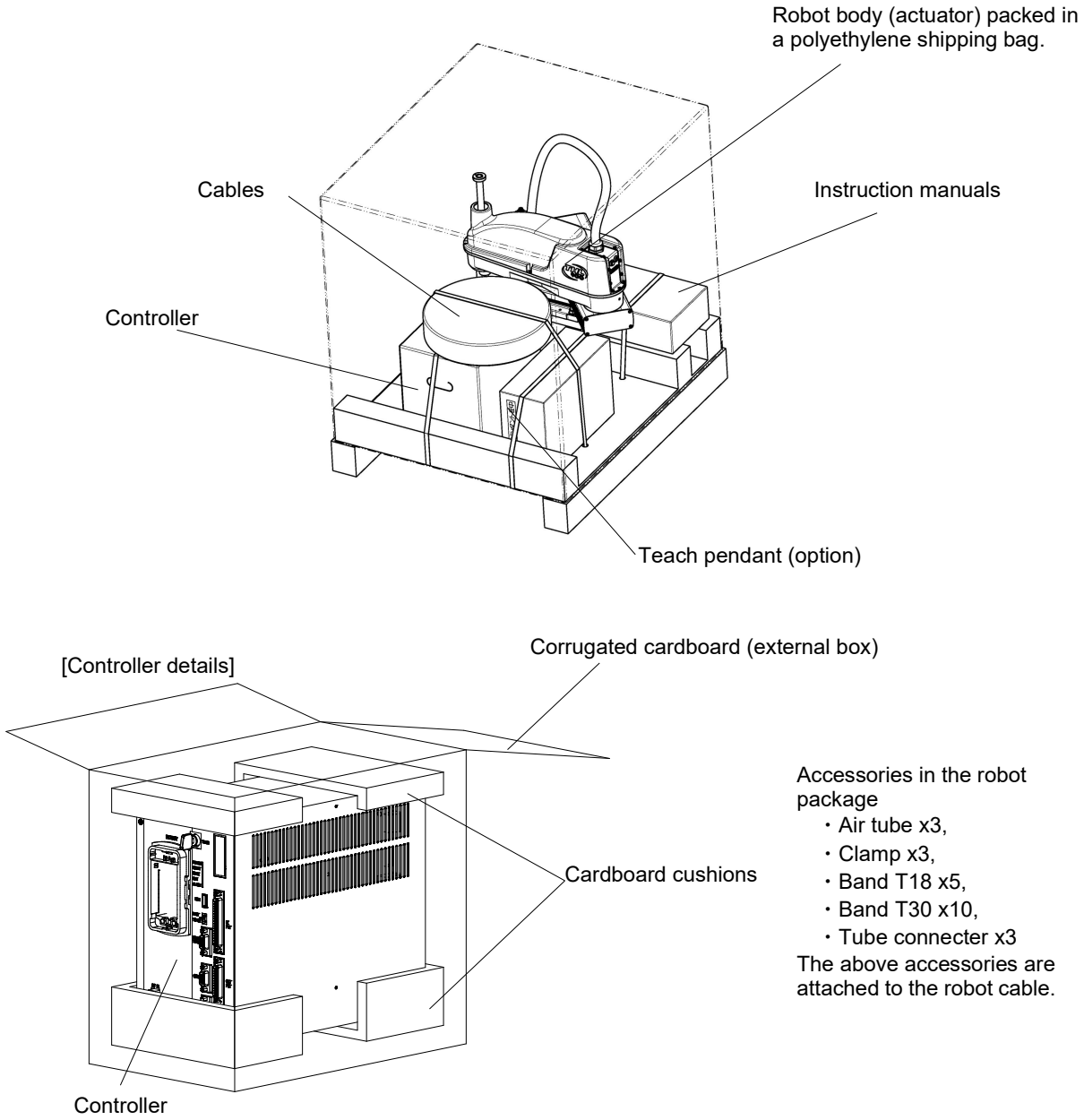


Fig. 7.1 Packaging state (THL800/THL900/THL1000 and TSL3000)

DANGER

- If any parts of the equipment are found damaged or any accessories are missing after the shipment containing the robot and controller have reached your office, DO NOT install and operate them. Otherwise, the equipment will malfunction. Contact SHIBAURA MACHINE immediately.
- Dispose of the wooden pallet, corrugated cardboards, polyethylene shipping bags and cushion material according to the customer's in-house regulations.

7.2 Unpacking (TS3000E)

For information on TSL3000, refer to Section 7.1. For information on TS3000 and TS3000E, refer to Section 7.3. The robot and controller are shipped separately in corrugated cardboards. Fig. 7.2 shows each packaging state.

Open the packages in a location easily accessible, where the equipment is to be installed. Take careful precautions not to damage the robot and controller.

After opening the packages, make sure that all the accessories are present and that nothing has been damaged during transport.

For the controller accessories, see the accessory list packed with the controller.

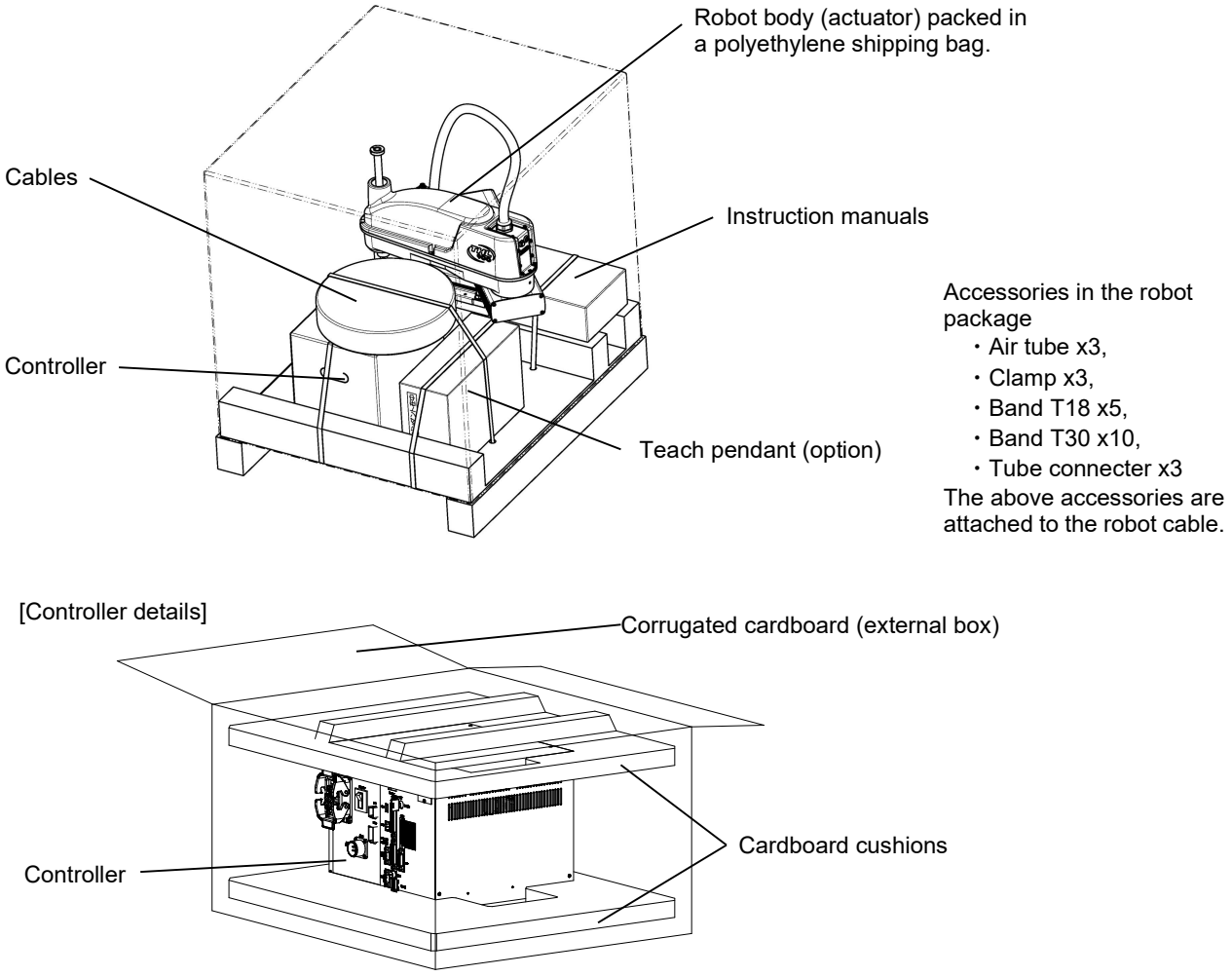


Fig. 7.2 Packaging state (THL800/THL900/THL1000 and TSL3000E)

! DANGER

- If any parts of the equipment are found damaged or any accessories are missing after the shipment containing the robot and controller have reached your office, DO NOT install and operate them. Otherwise, the equipment will malfunction. Contact SHIBAURA MACHINE immediately.
- Dispose of the wooden pallet, corrugated cardboards, polyethylene shipping bags and cushion material according to the customer's in-house regulations.

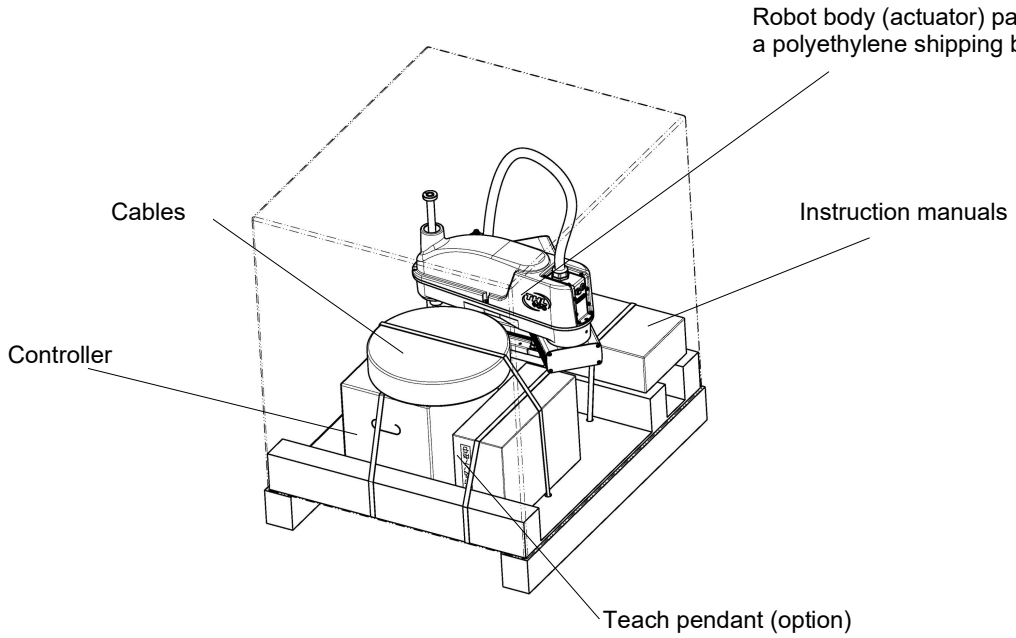
7.3 Unpacking (TS3000 and TS3000E)

For information on TSL3000, refer to Section 7.1. For information on TSL3000E, refer to Section 7.2. The robot and controller are shipped separately in corrugated cardboards. Fig. 7.3 shows each packaging state.

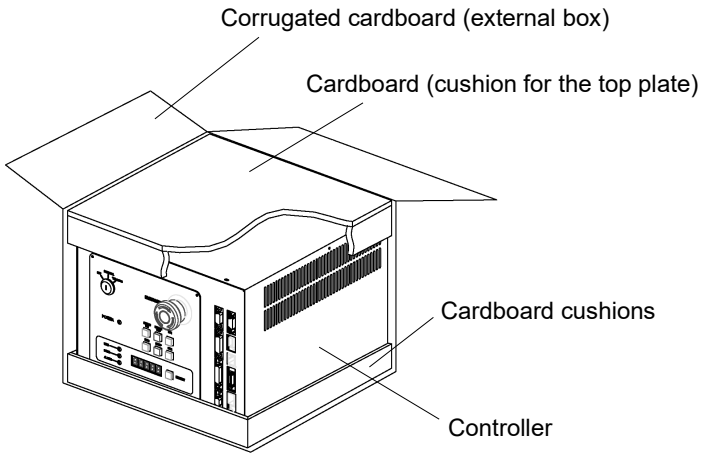
Open the packages in a location easily accessible, where the equipment is to be installed. Take careful precautions not to damage the robot and controller.

After opening the packages, make sure that all the accessories are present and that nothing has been damaged during transport.

For the controller accessories, see the accessory list packed with the controller.



[Controller details]



Accessories in the robot package
- Air tube x3,
- Clamp x3,
- Band T18 x5,
- Band T30 x10,
- Tube connecter x3

The above accessories are attached to the robot cable.

Fig. 7.3 Packaging state (THL800/THL900/THL1000 and TS3000/TS3000E)

DANGER

- If any parts of the equipment are found damaged or any accessories are missing after the shipment containing the robot and controller have reached your office, DO NOT install and operate them. Otherwise, the equipment will malfunction. Contact SHIBAURA MACHINE immediately.
- Dispose of the wooden pallet, corrugated cardboards, polyethylene shipping bags and cushion material according to the customer's in-house regulations.

7.4 Transportation

Move the robot and controller very carefully. Make sure that no excessive impact or vibration is exerted on the equipment. If the equipment is to be subject to vibration over a long period, be sure to tighten all the clamp and base set bolts completely.
If the equipment is to be moved to a location some distance from where it was unpacked, reposition the cushions as they were and put the equipment back into the corrugated cardboards.

7.4.1 Mass and Dimensions

The mass and outer dimensions of the robot are shown in Figs. 7.4 to 7.6.
For the controller mass and external dimensions, see "13.1 External Dimensions", "14.1 External Dimensions" or "15.1 External Dimensions".

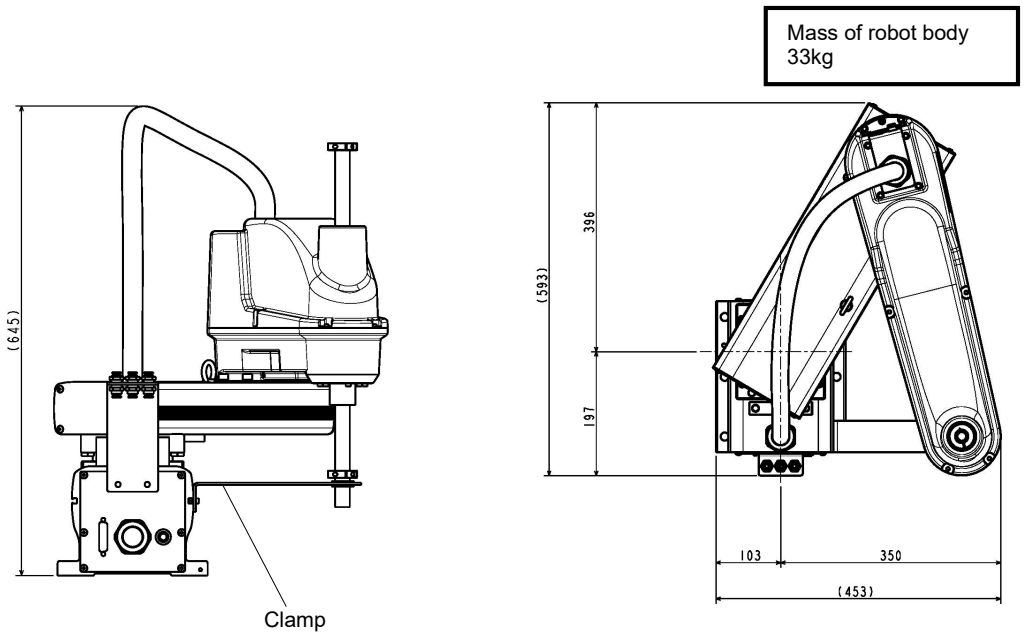


Fig. 7.4 Packaging state (THL800)

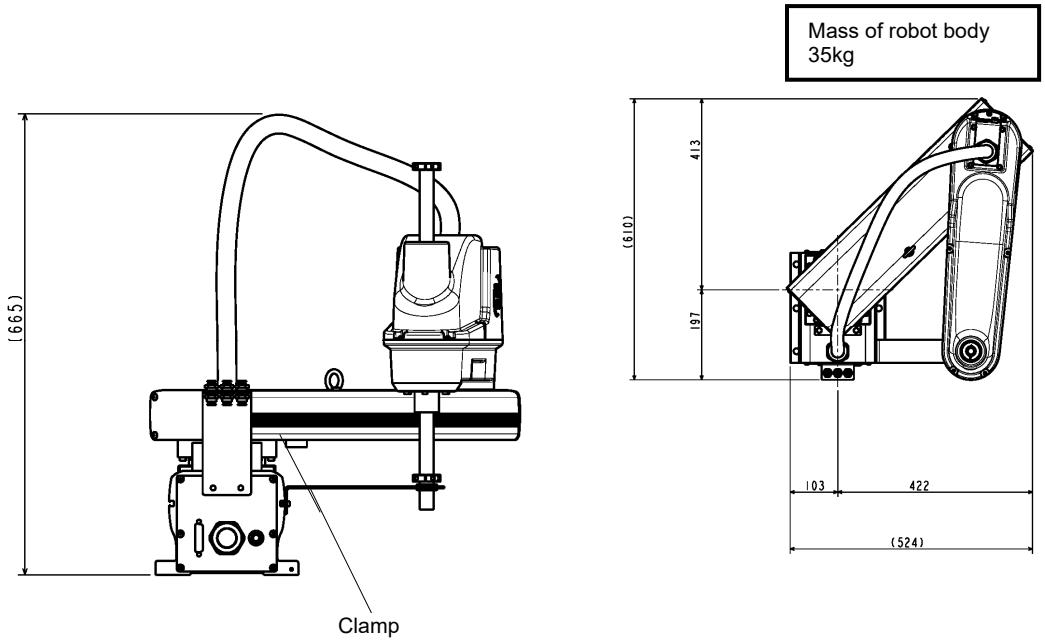


Fig. 7.5 Packaging state (THL900)

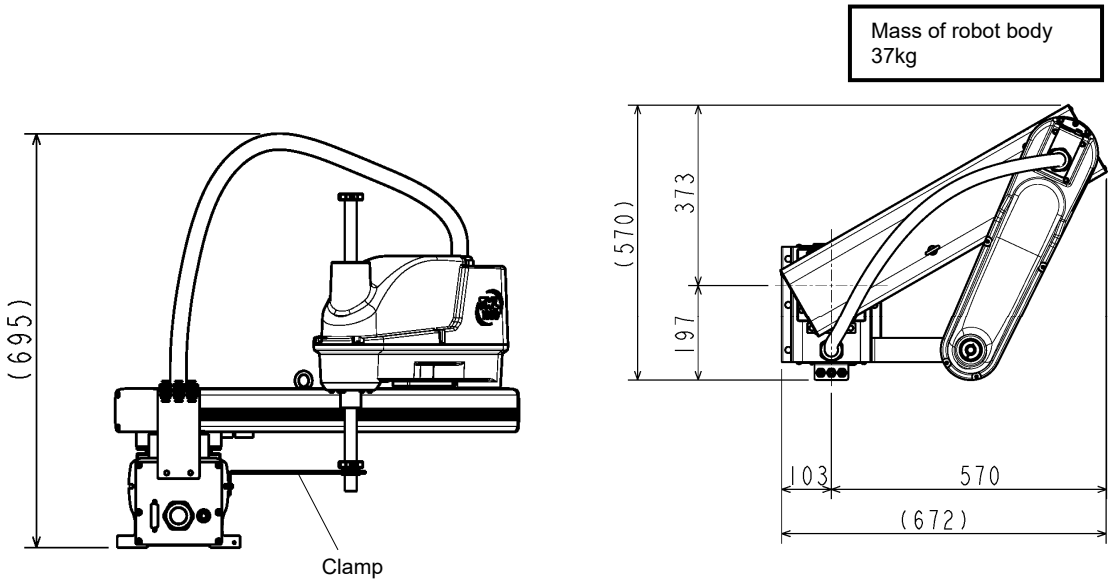


Fig. 7.6 Packaging state (THL1000)

7.4.2 Transporting the Robot

In principle, the robot should be transported in the state shown in Figs. 7.4 to 7.6 above. Fold back and secure the arm with the attached clamp in Figs. 7.7 above. (The robot is shipped in this posture. After you have unpacked the shipment, you should move it as it is.) At this time, take careful precautions not to impose a large force on the tool shaft.

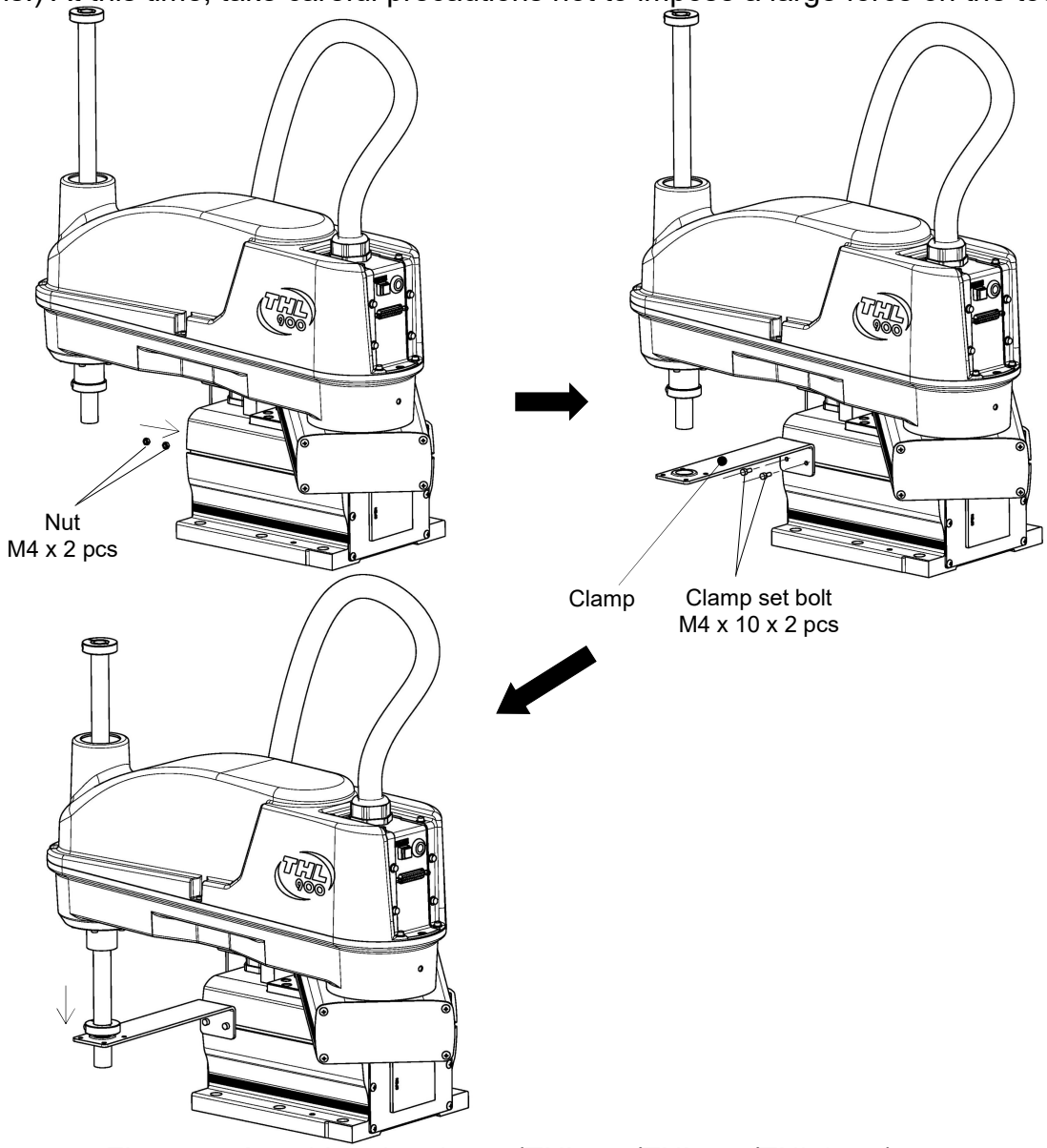


Fig. 7.7 Attaching the clamp (THL800/THL900/THL1000)

DANGER

- Be sure to secure the arm with the attached clamp before transporting the robot. Failure to do so could cause a hazardous situation as the arm will move when the robot is lifted.

It is possible to lift up and transport THL800, THL900 and THL1000. Pass the wire through the attached eyebolt, then lift up the robot carefully, as shown in Fig. 7.8.

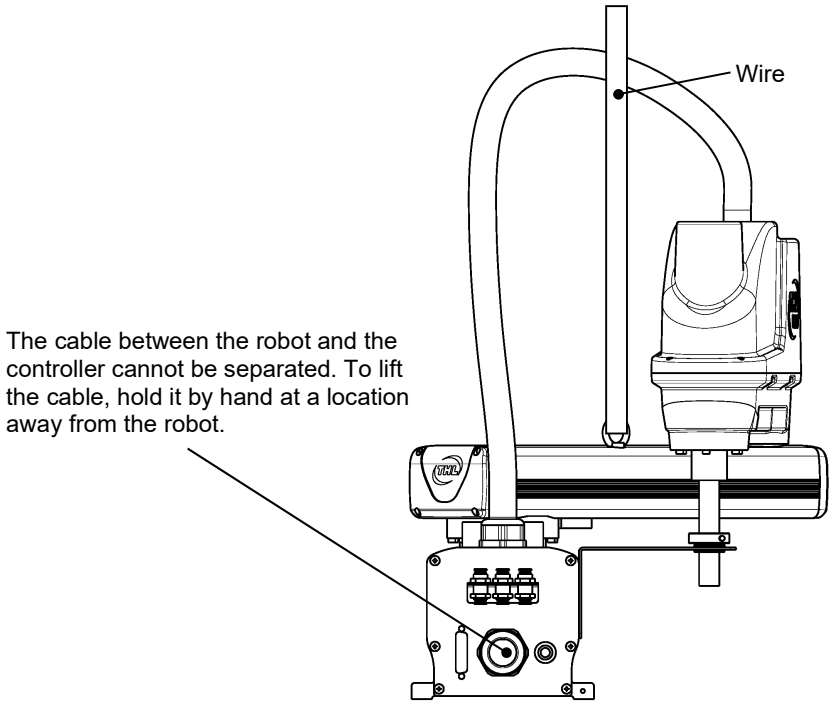


Fig. 7.8 Lifting up the robot

! CAUTION

- The wire to be used should be such that can well withstand the mass of the robot.
- When lifting up the robot, it may tilt a little. Lift it up slowly.
- Lifting up and down should be performed carefully so that any impact cannot be exerted on the robot.
- When carrying the robot by workers, take careful precautions to prevent their hand or leg from being caught in the robot.

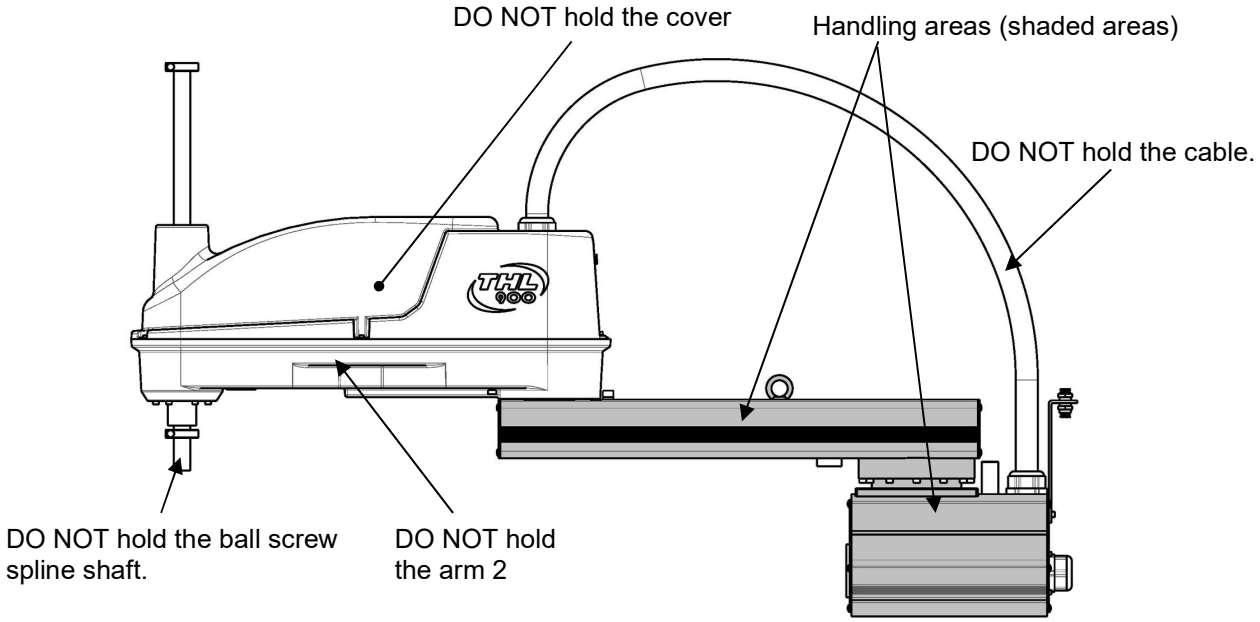


Fig. 7.8 Robot Handling Locations

After the installation, remove the clamp and eyebolt used for transport.

CAUTION

- When lifting up the robot by workers, hold the locations (shaded areas) by hands as shown in Fig. 7.8. Holding the arm 2, the cable and the ball screw spline shaft will apply excess force, possibly causing failure.
- When carrying the robot by workers, take careful precautions to prevent their hand or leg from being caught in the robot.
- Never touch the ball screw spline shaft with a bare hand. If you touch it with a bare hand, earlier rust formation may result. Be sure to use gloves.
- The work should be performed by two (2) or more workers.
- Plug up the threaded bore which took off an eyebolt by a bolt attached to the robot.

7.4.3 Transporting the Controller

Disconnect all cables and teach pendant before transporting the controller.

DANGER

- When placing the controller on the floor, etc., make sure not to have your hands or feet caught.

7.5 Storage

Avoid storing the robot and controller for long periods of time after unpacking them. If this is unavoidable, however, strictly observe the following precautions for storage.

7.5.1 Storage Precautions for the Robot

CAUTION

- Secure the base completely to prevent the robot from falling over. When placed directly on the floor, the robot is unstable and will fall over.
- Avoid direct sunshine and high temperature/humidity when the robot is stored. The timing belts and resin covers may deteriorate.
- Seal the robot in a vinyl bag to prevent rust development and contaminant. Put a desiccant in the bag to absorb moisture. As the tool shaft is susceptible to rust development, coat it with rust-preventive agent or grease the entire tool shaft beforehand.
  For the grease application procedure, see the maintenance manual "Greasing Ball Screw Spline Unit and Applying Anticorrosive".
- Before the use, apply the grease to the tool shaft.
- Before starting an operation, perform running completely.
- During storage, the life of the backup batteries will shorten. It is recommended to replace the batteries at the time of operation.

7.5.2 Storage Precautions for the Controller

CAUTION

- Keep the controller out of direct sunlight. Otherwise, the controller interior will be excessively heated up, causing a trouble.
- Seal the controller in a vinyl bag to prevent rust development and contaminant. Put a desiccant in the bag to absorb moisture.

8. Transportation (THL1200)

8.1 Unpacking (TSL3000)

For information on TSL3000E, refer to Section 8.2. For information on TS3000 and TS3000E, refer to Section 8.3. The robot is shipped on the wooden palette and the controller is shipped separately in corrugated cardboards. Fig. 8.1 shows packaging state.

Open the packages in a location easily accessible, where the equipment is to be installed. Take careful precautions not to damage the robot and controller.

After opening the packages, make sure that all the accessories are present and that nothing has been damaged during transport.

For the controller accessories, see the accessory list packed with the controller.

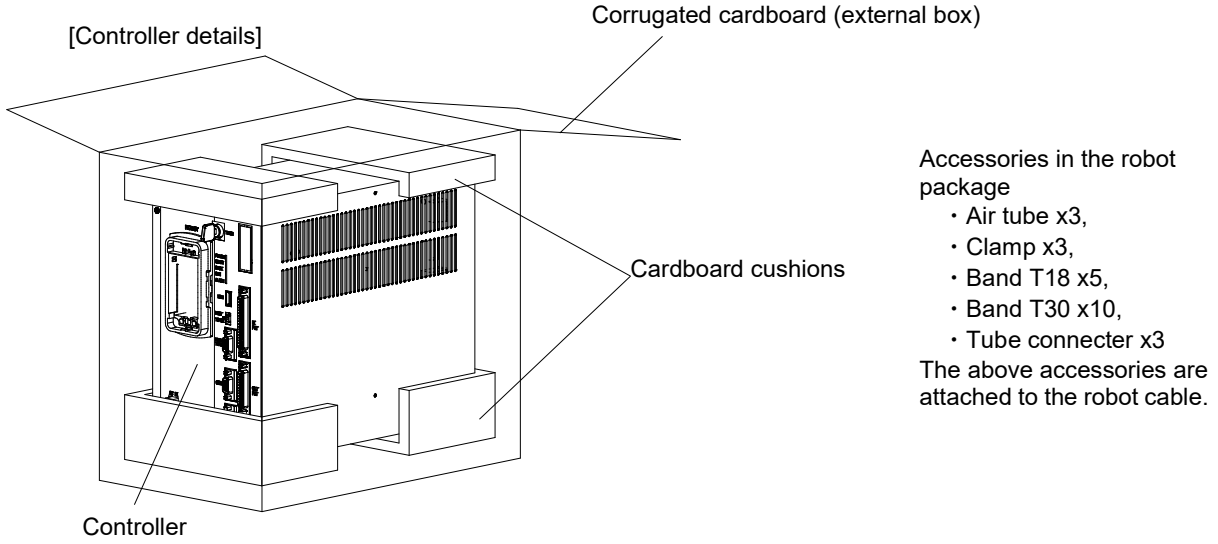


Fig. 8.1 Packaging state (TSL3000)

! DANGER

- If any parts of the equipment are found damaged or any accessories are missing after the shipment containing the robot and controller have reached your office, DO NOT install and operate them. Otherwise, the equipment will malfunction. Contact SHIBAURA MACHINE immediately.
- Dispose of the wooden pallet, corrugated cardboards, polyethylene shipping bags and cushion material according to the customer's in-house regulations.

8.2 Unpacking (TS3000E)

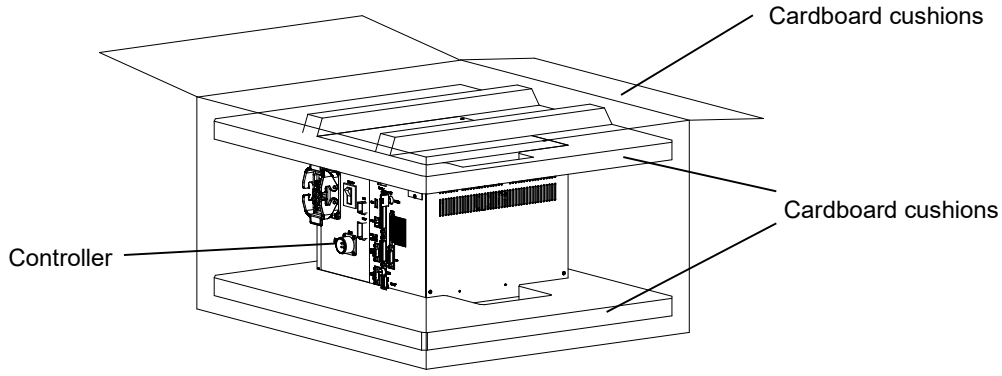
For information on TSL3000, refer to Section 8.1. For information on TS3000 and TS3000E, refer to Section 8.3. The robot is shipped on the wooden palette and the controller is shipped separately in corrugated cardboards. Fig. 8.2 shows packaging state.

Open the packages in a location easily accessible, where the equipment is to be installed. Take careful precautions not to damage the robot and controller.

After opening the packages, make sure that all the accessories are present and that nothing has been damaged during transport.

For the controller accessories, see the accessory list packed with the controller.

[Controller details]



Accessories in the robot package
- Air tube x3,
- Clamp x3,
- Band T18 x5,
- Band T30 x10,
- Tube connecter x3

The above accessories are attached to the robot cable.

Fig. 8.2 Packaging state (TSL3000E)

DANGER

- If any parts of the equipment are found damaged or any accessories are missing after the shipment containing the robot and controller have reached your office, DO NOT install and operate them. Otherwise, the equipment will malfunction. Contact SHIBAURA MACHINE immediately.
- Dispose of the wooden pallet, corrugated cardboards, polyethylene shipping bags and cushion material according to the customer's in-house regulations.

8.3 Unpacking (TS3000 and TS3000E)

For information on TSL3000, refer to Section 8.1. For information on TSL3000E, refer to Section 8.2. The robot is shipped on the wooden palette and the controller is shipped separately in corrugated cardboards. Fig. 8.3 shows packaging state.
Open the packages in a location easily accessible, where the equipment is to be installed. Take careful precautions not to damage the robot and controller.
After opening the packages, make sure that all the accessories are present and that nothing has been damaged during transport.
For the controller accessories, see the accessory list packed with the controller.

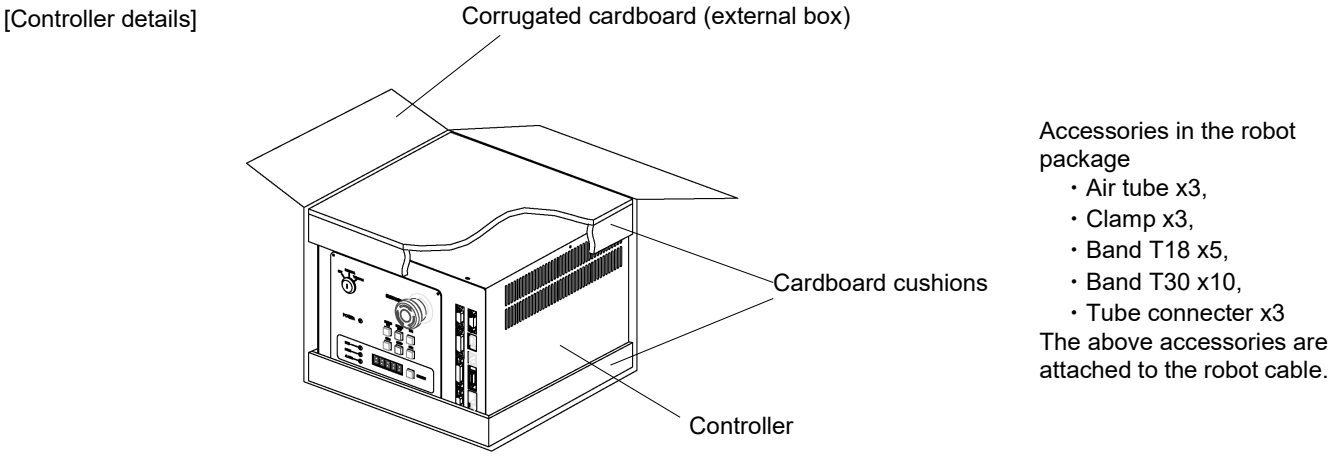


Fig. 8.3 Packaging state (TS3000/TS3000E)

! DANGER

- If any parts of the equipment are found damaged or any accessories are missing after the shipment containing the robot and controller have reached your office, DO NOT install and operate them. Otherwise, the equipment will malfunction. Contact SHIBAURA MACHINE immediately.
- Dispose of the wooden pallet, corrugated cardboards, polyethylene shipping bags and cushion material according to the customer's in-house regulations.

8.4 Transportation

Move the robot and controller very carefully. Make sure that no excessive impact or vibration is exerted on the equipment. If the equipment is to be subject to vibration over a long period, be sure to tighten all the clamp and base set bolts completely.
If the equipment is to be moved to a location some distance from where it was unpacked, reposition the cushions as they were and put the equipment back into the corrugated cardboards.

8.4.1 Mass and Dimensions

The mass and outer dimensions of the robot are shown in Figs. 8.4.
For the controller mass and external dimensions, see "13.1 External Dimensions", "14.1 External Dimensions" or "15.1 External Dimensions".

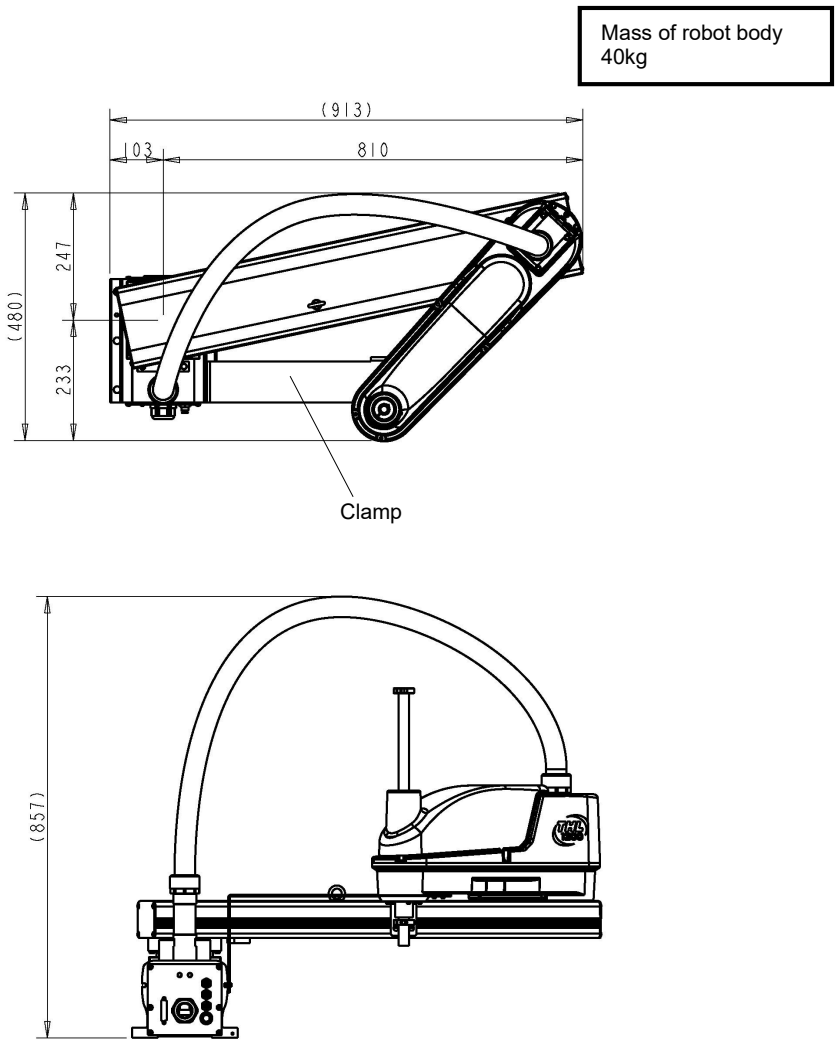


Fig. 8.4 Packaging state (THL1200)

8.4.2 Transporting the Robot

In principle, the robot should be transported in the state shown in Figs. 8.4 above. Fold back and secure the arm with the attached clamp in Figs. 8.5 above. Then, fix a tip of the ball screw spline in the binding band in Figs. 8.6 above. (The robot is shipped in this posture. After you have unpacked the shipment, you should move it as it is.) At this time, take careful precautions not to impose a large force on the tool shaft.

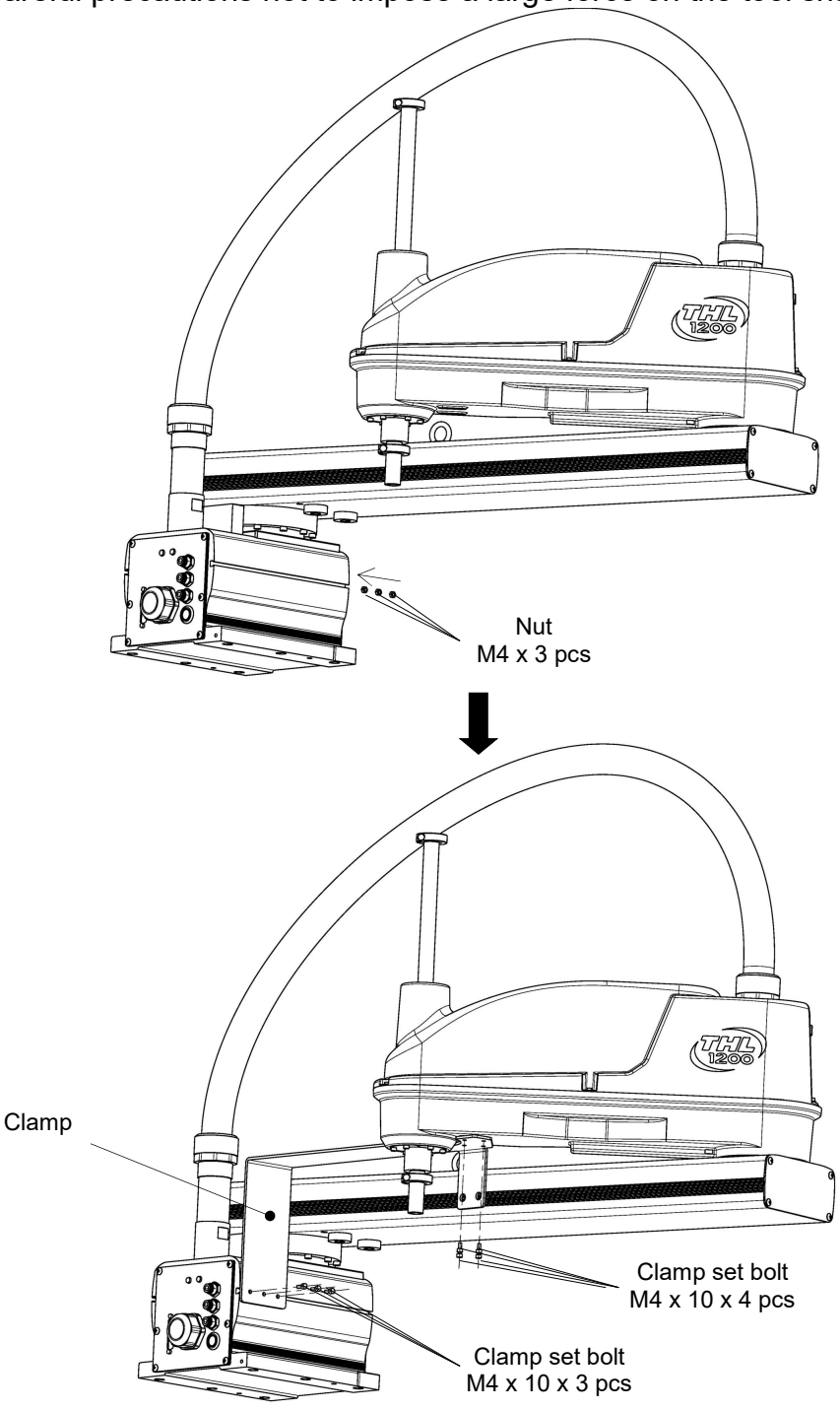


Fig. 8.5 Attaching the clamp (THL1200)

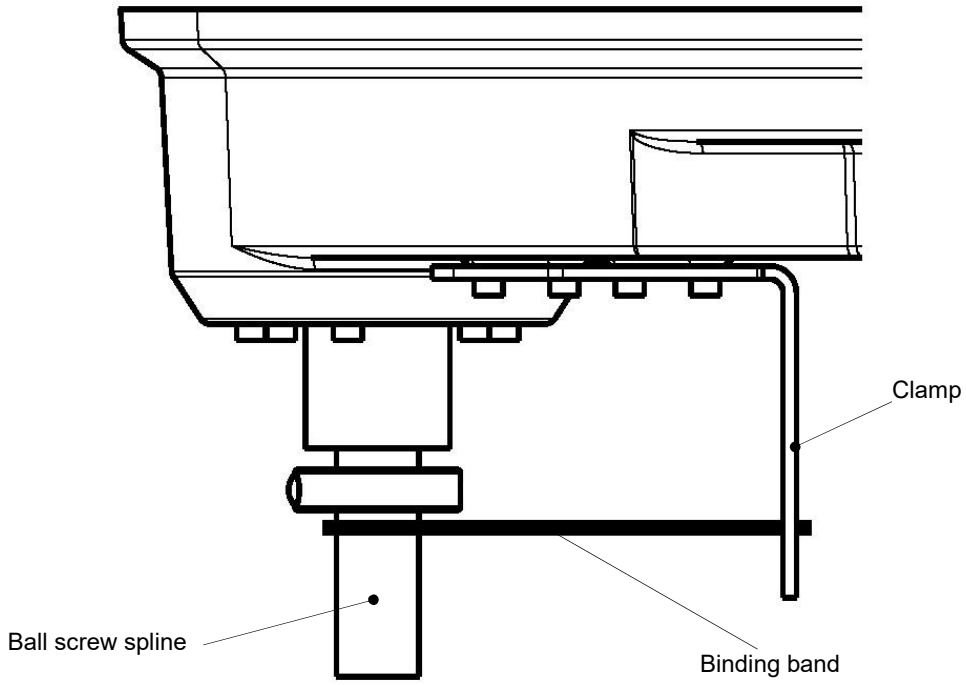


Fig. 8.6 Fixing the ball screw spline (THL1200)

DANGER

- Be sure to secure the arm with the attached clamp before transporting the robot. Failure to do so could cause a hazardous situation as the arm will move when the robot is lifted.

It is possible to lift up and transport THL1200. Pass the wire through the attached eyebolt, then lift up the robot carefully, as shown in Fig. 8.7.

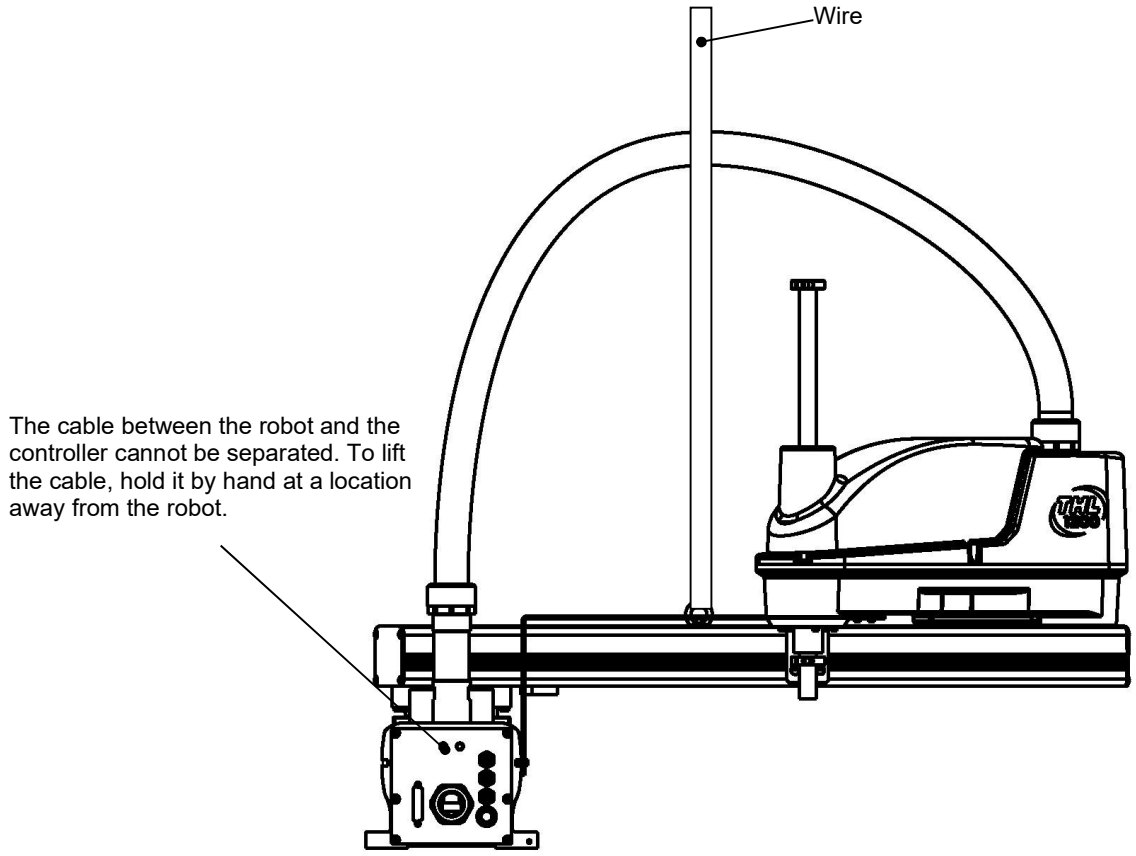


Fig. 8.7 Lifting up the robot

! CAUTION

- The wire to be used should be such that can well withstand the mass of the robot.
- When lifting up the robot, it may tilt a little. Lift it up slowly.
- Lifting up and down should be performed carefully so that any impact cannot be exerted on the robot.
- When carrying the robot by workers, take careful precautions to prevent their hand or leg from being caught in the robot.

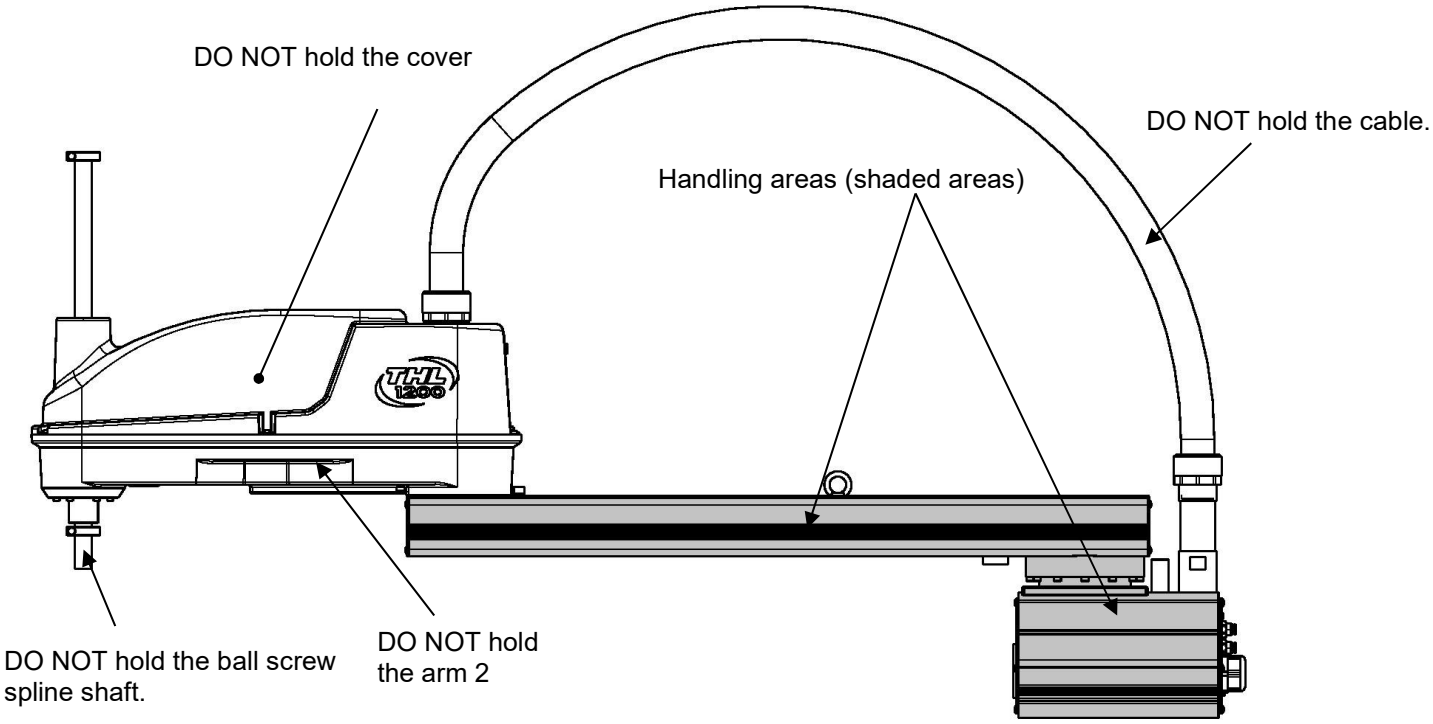


Fig. 8.8 Robot Handling Locations

After the installation, remove the clamp and eyebolt used for transport.

! CAUTION

- When lifting up the robot by workers, hold the locations (shaded areas) by hands as shown in Fig. 8.8. Holding the arm 2, the cable and the ball screw spline shaft will apply excess force, possibly causing failure.
- When carrying the robot by workers, take careful precautions to prevent their hand or leg from being caught in the robot.
- Never touch the ball screw spline shaft with a bare hand. If you touch it with a bare hand, earlier rust formation may result. Be sure to use gloves.
- The work should be performed by two (2) or more workers.
- Plug up the threaded bore which took off an eyebolt by a bolt attached to the robot.

8.4.3 Transporting the Controller

Disconnect all cables and teach pendant before transporting the controller.

DANGER

- When placing the controller on the floor, etc., make sure not to have your hands or feet caught.

8.5 Storage

Avoid storing the robot and controller for long periods of time after unpacking them. If this is unavoidable, however, strictly observe the following precautions for storage.

8.5.1 Storage Precautions for the Robot

CAUTION

- Secure the base completely to prevent the robot from falling over. When placed directly on the floor, the robot is unstable and will fall over.
- Avoid direct sunshine and high temperature/humidity when the robot is stored. The timing belts and resin covers may deteriorate.
- Seal the robot in a vinyl bag to prevent rust development and contaminant. Put a desiccant in the bag to absorb moisture. As the tool shaft is susceptible to rust development, coat it with rust-preventive agent or grease the entire tool shaft beforehand.
  For the grease application procedure, see the maintenance manual "Greasing Ball Screw Spline Unit and Applying Anticorrosive".
- Before the use, apply the grease to the tool shaft.
- Before starting an operation, perform running completely.
- During storage, the life of the backup batteries will shorten. It is recommended to replace the batteries at the time of operation.

8.5.2 Storage Precautions for the Controller

CAUTION

- Keep the controller out of direct sunlight. Otherwise, the controller interior will be excessively heated up, causing a trouble.
- Seal the controller in a vinyl bag to prevent rust development and contaminant. Put a desiccant in the bag to absorb moisture.

9. Robot Installation (THL300 and THL400)

9.1 Installation Environment

Table 9.1 shows the environmental conditions for the location in which the robot and controller are to be installed.

Table 9.1 Environmental conditions for robot and controller

Item	Specifications
Temperature	In operation: 0 to 40°C In storage: –10 to 50°C
Humidity	20 to 80% (Non-condensing) DO NOT install the robot where it may be subject to fluids such as water.
Altitude	1000 m or less
Vibration	In operation: 0.98 m/s^2 or less
Dust	No inductive dust should exist. Consult with SHIBAURA MACHINE first if you wish to use the robot and controller in a dusty environment.
Gas	No corrosive or combustible gas should exist.
Degrees of protection	IEC60529 IP10(robot only) IP20(controller only)
Overvoltage category	IEC60664-1 ClassⅢ(controller only)
Protection against electric shock	IEC61140 Class Ⅰ (controller only)
Pollution Degree	IEC60664-1 Pollution Degree 3(controller only)
Sunlight	The robot and controller should not be exposed to direct sunlight.
Power noise	A heavy noise source should not exist nearby.
Magnetic field	A heavy magnetic field source should not exist nearby.
Other ambient environment	Must be free from iron powder, oil, salt, and organic solvents. Must not be exposed to water.

DANGER

- Do not place the robot or controller near combustible.
  Doing so could lead to fires if it ignites due to a fault, etc.

CAUTION

- In the case where batteries for detecting the motor position are of alkaline type (standard type), the batteries can overheat, leak battery fluid, or rupture when used under high temperatures. Also, high temperatures can reduce the performance and lifespan of the battery. If using the robot under high temperatures, please consult with the SHIBAURA MACHINE sales office.
- When starting fast-movement operation in low-temperature environment, errors may occur because of the increased torque. In operating the robot in low-temperature environment, start by making low-speed continuous operation for a several minutes to soften the grease before starting high-speed operations.

9.2 Installation

Before installing the robot, you should plan a layout, fully considering the working envelope (or operating range), coordinate system and space for maintenance.

9.2.1 External Dimensions and Working Envelope

Figs. 9.1 and 9.2 show the external view drawing and working envelope of the robot. Each axis can operate within the working envelope. To prevent the robot from moving out of the working envelope by misoperation, the robot is equipped with mechanical stoppers outside the working envelope. Additionally, soft limits that can be set by the user are provided. For further information, see the “Instruction Manual: User Parameters” provided separately.

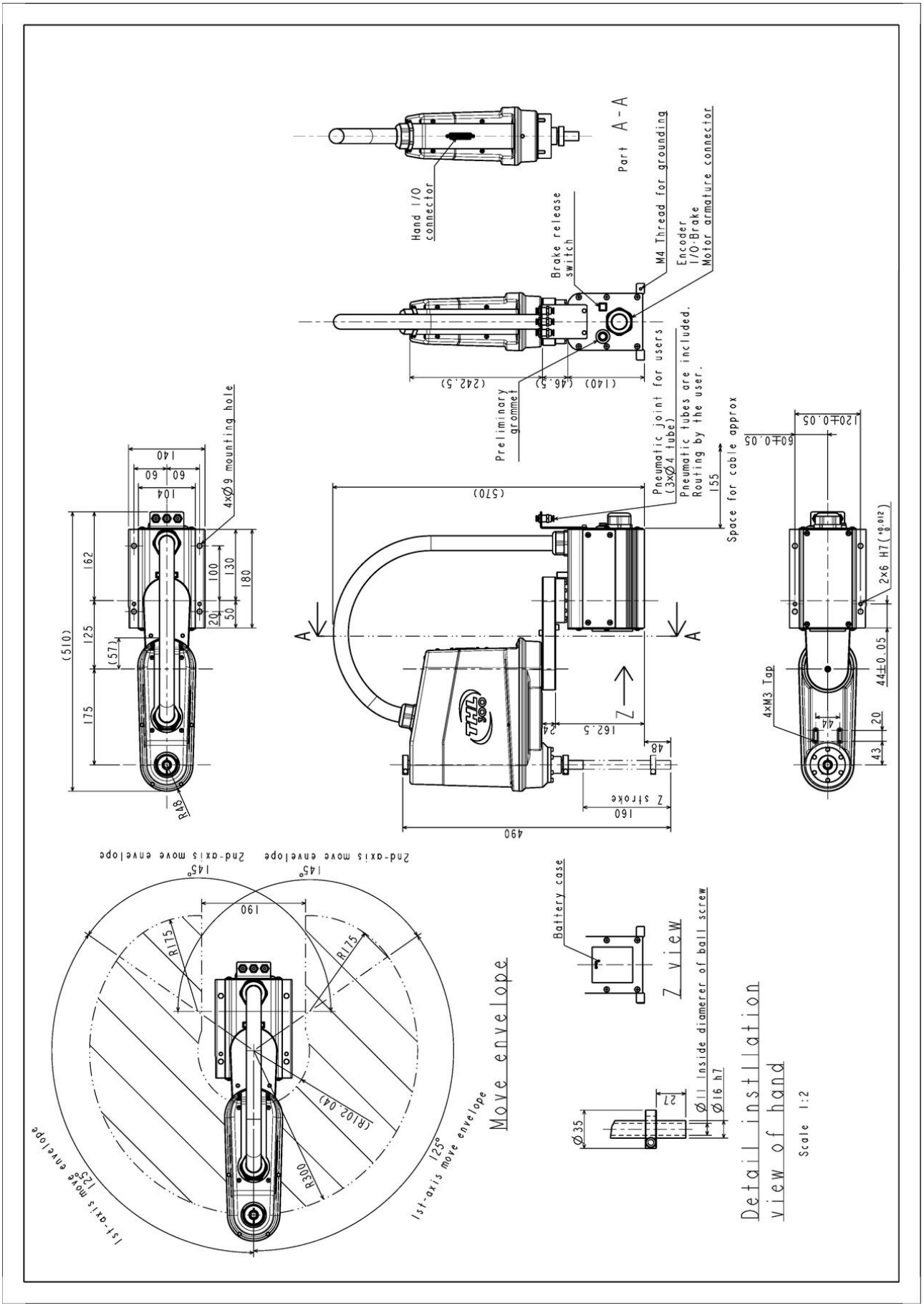


Fig. 9.1 External view and working envelope (THL300)

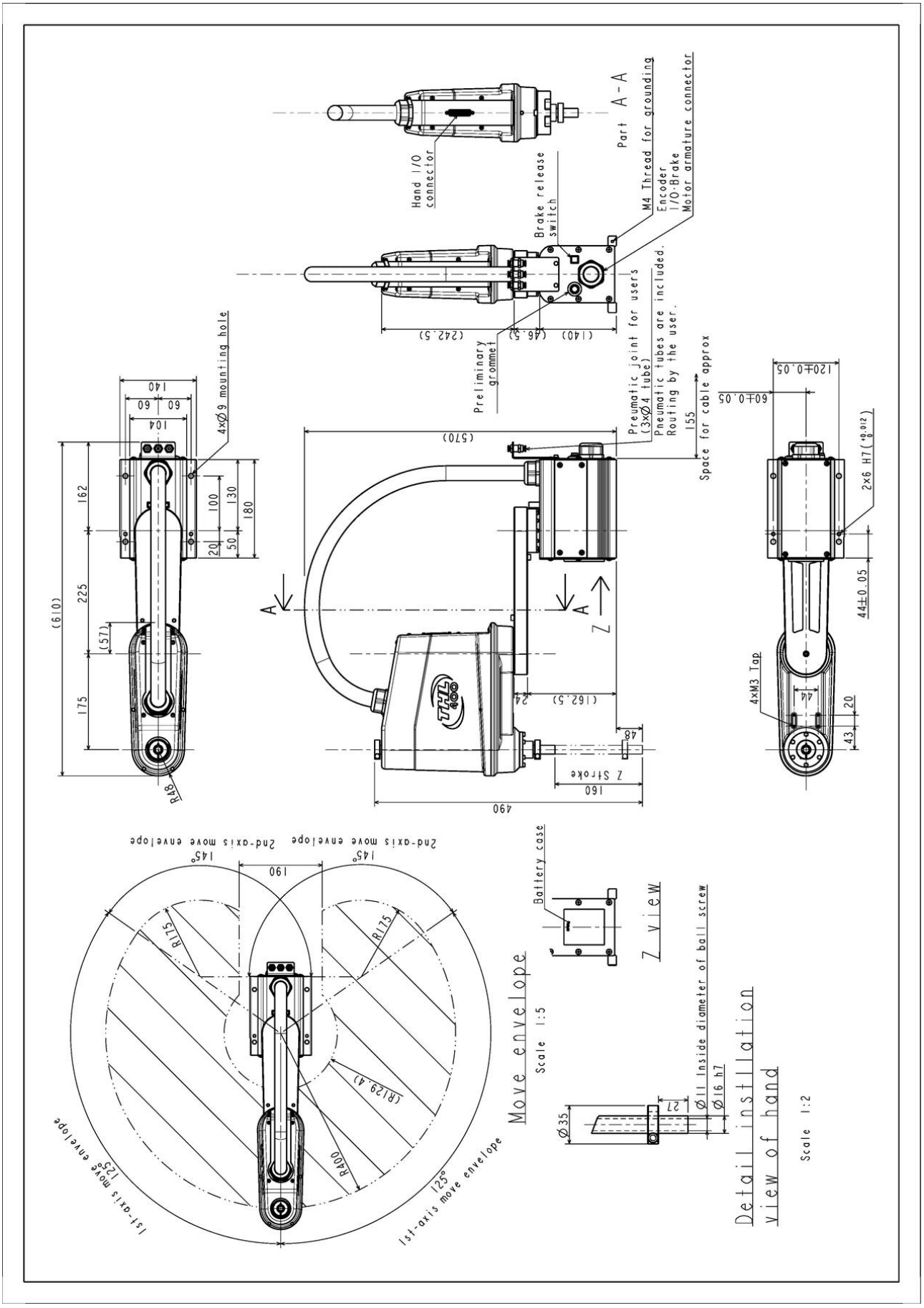


Fig. 9.2 External view and working envelope (THL400)

9.2.2 Changing the Operation Ranges of Axes 1 and 2

Mechanical stoppers are installed to each robot to mechanically control the operation range of each axis. Changing the mechanical operation range of a robot by modifying these mechanical stoppers is referred to as "change in operating range."
Here, how to change the operating ranges of Axes 1 and 2 of a robot is explained.

Regarding how to change the operating range of Axis 3, see Para. 9.2.3.
Note that Axis 4 is different from other operating axes; it restricts the operating range with software limits only, not with mechanical stoppers. Therefore, regarding how to change the operating range of Axis 4, see Para. 9.2.4.

! CAUTION

- To change operating ranges, design and produce mechanical stoppers by referring to this document according to your usage.
- When mechanical stoppers have been changed and then operating ranges have been changed, be sure to also change software limits to prevent contact with the mechanical stoppers while operating a robot.
- Mechanical stoppers do not securely restrict the movable ranges of robots.

  When the power of a robot is turned on, never enter the operating range of the robot.
- If a robot collides with a mechanical stopper, the robot detects the collision and stops, but the mechanical stopper may be damaged. Avoid reusing the mechanical stopper.
- The mechanical stopper reference drawings shown in this document do not fully satisfy the customer's usage.

  Design, produce and install mechanical stoppers according to your usage, such as the operating range.
- The failures of a robot caused by mechanical stoppers will be excluded from the warranty coverage.

As shown in Fig. 9.3, the operation range can be altered by changing the position of the mechanical stopper.

Table 9.2 Operation ranges before and after change (THL300 and THL400)

		Before change	After change
Axis 1 operation range	+ direction	125°	105°
	- direction	125°	105°
Axis 2 operation range	+ direction	145°	120°
	- direction	145°	120°

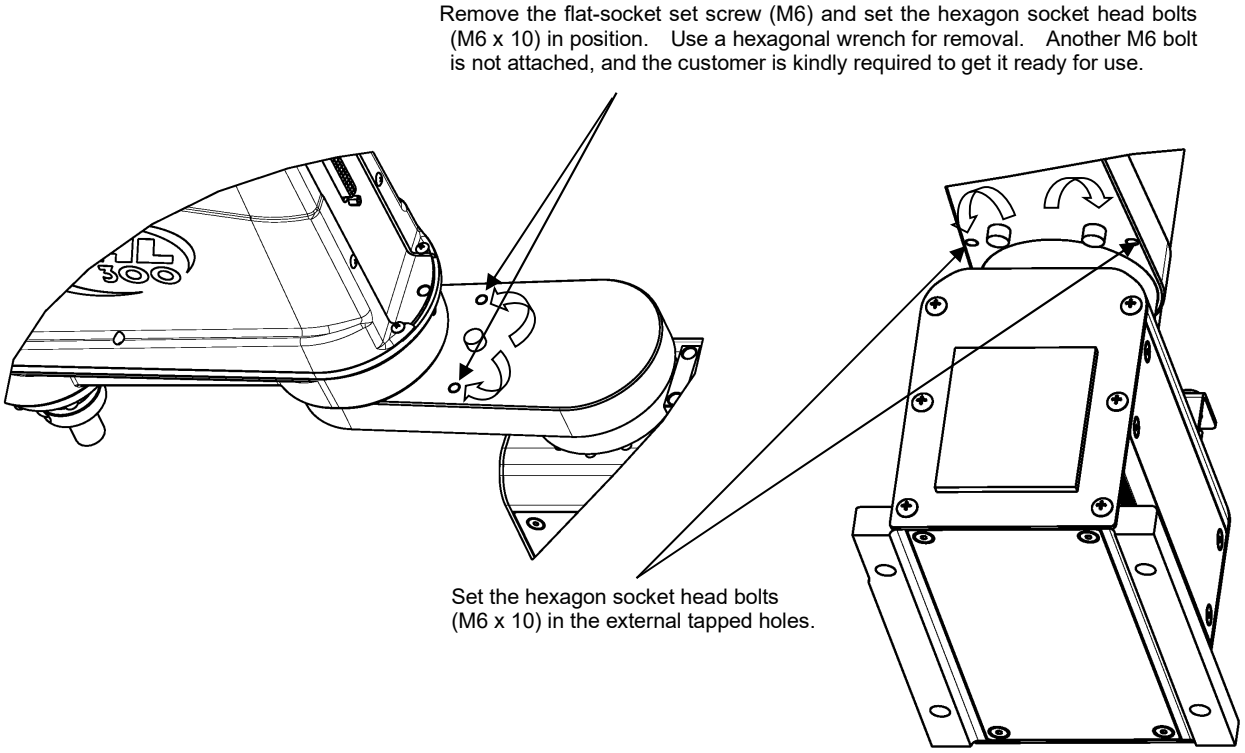


Fig. 9.3 Changing the Operating Ranges of Axes 1 and 2 (THL300 and THL400)

If the operation range is changed, it is necessary to change the user parameter. Regarding how to change software limits, see Para. 9.2.4.

9.2.3 Changing the Operating Range of Axis 3

At the factory shipment of a robot, software limits and mechanical stoppers are preset so that the Z stroke of Axis 3 is between 0 and 160 mm. Fig. 9.4 shows the settings of operating range at the factory shipment.

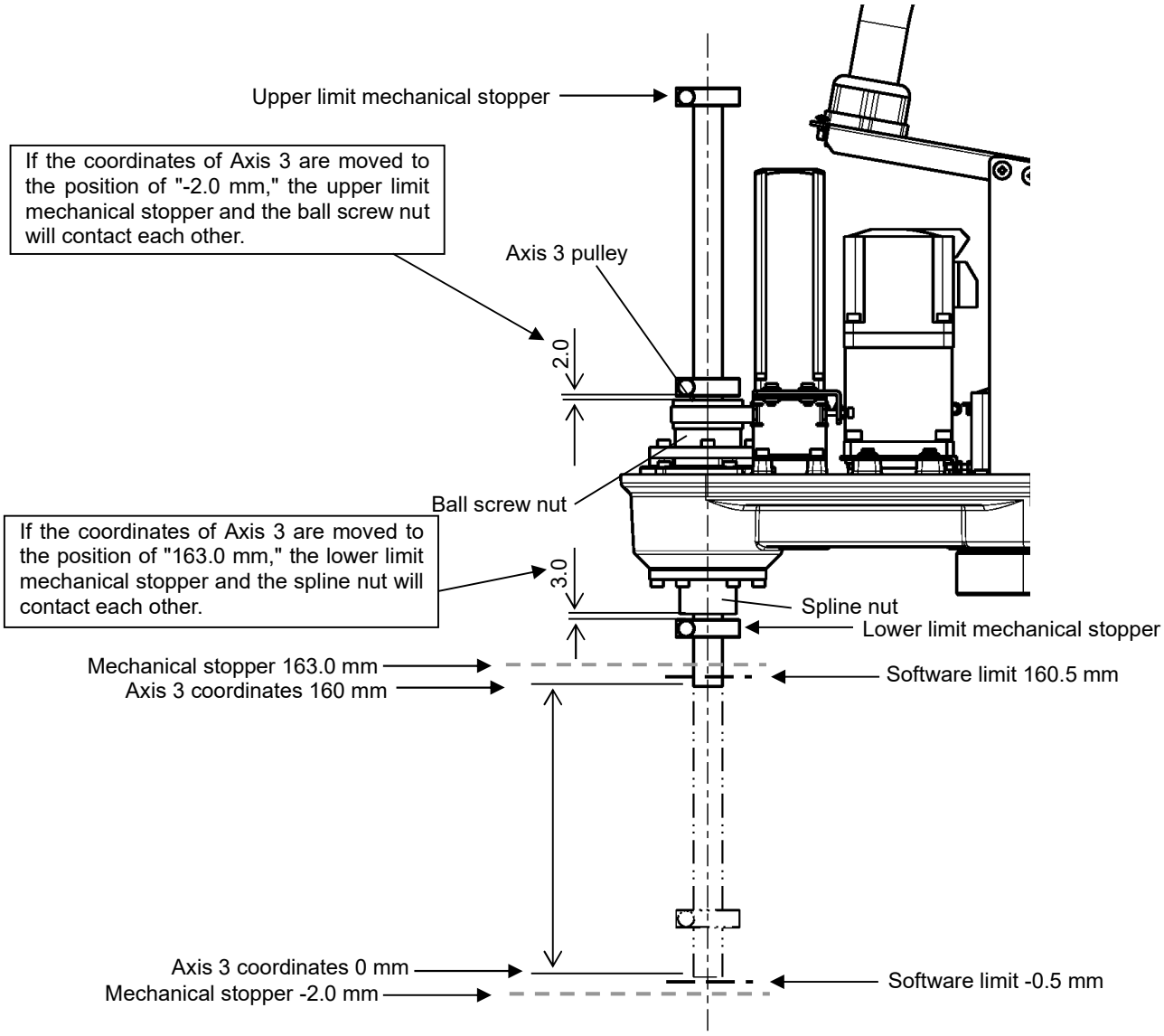


Fig. 9.4 Changing the Operating Range of Axis 3 (THL300 and THL400)

1) Remove the Arm 2 cover. The cover is fastened to Arm 2 and the harness guide with four (4) hexagon socket head bolts (M3 x 16) and ten (10) cross recessed truss head screw (M3 x 10 x 2 pcs. and M3 x 6 x 8 pcs.). It is recommended to lower Axis 3 in advance until the upper limit stopper of the ball screw enters the cover.

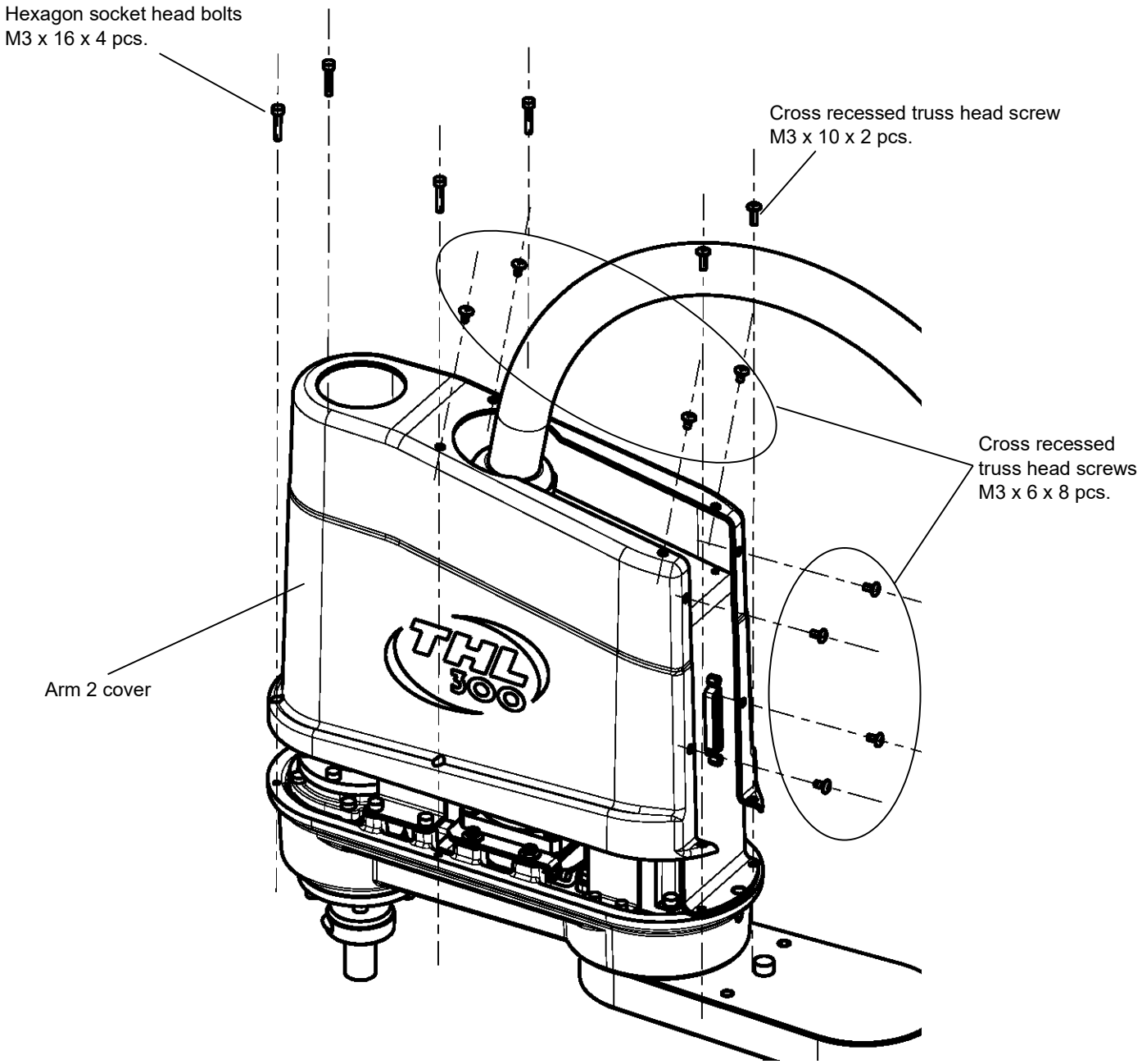
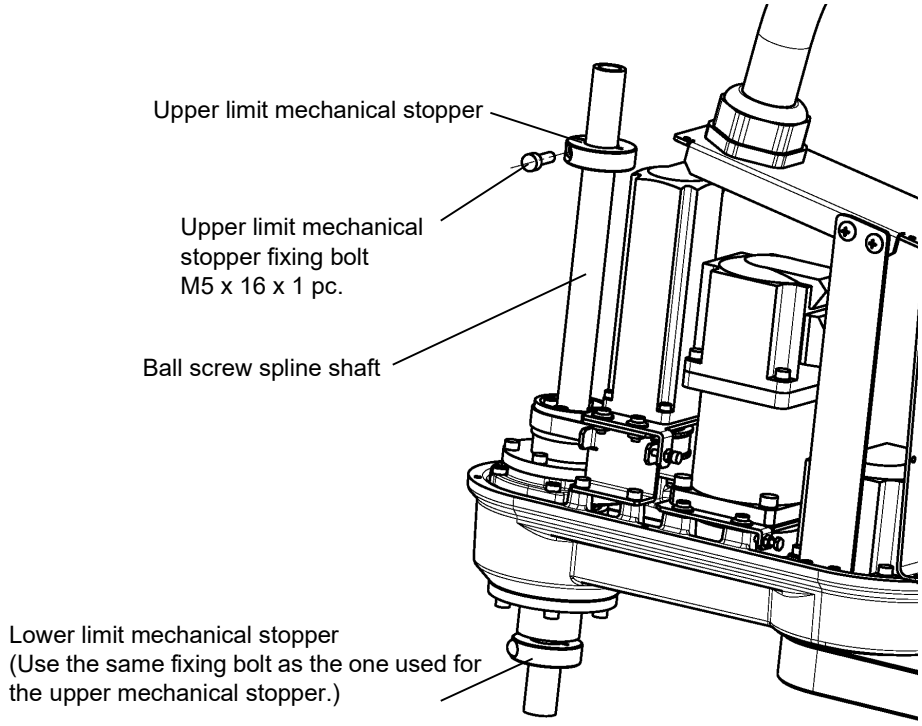


Fig. 9.5 Removing the Arm 2 Cover (THL300 and THL400)

2) Loosen the fixing bolts of the mechanical stoppers, move the mechanical stoppers to a desired position, and fix them again. When fixing the mechanical stoppers, be sure to apply Loctite to the fixing bolts.



Change the fixing position of the mechanical stopper.
Example: When the upper limit mechanical stopper is moved 75 mm downward.

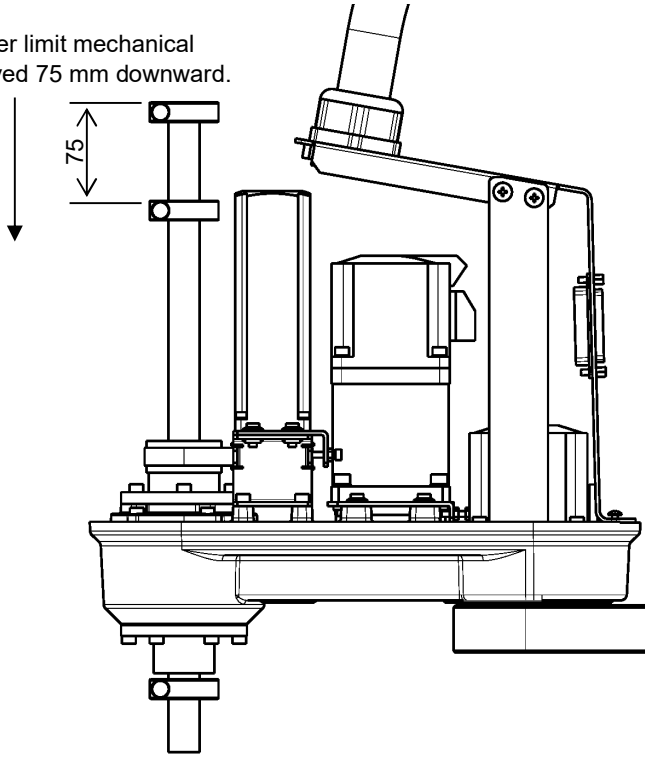


Fig. 9.6 Changing the Operating Range (THL300 and THL400)

3) When the mechanical stoppers are changed, be sure to also change the software limits. Regarding how to change software limits, see Para. 9.2.4 and Fig. 9.8. After changing the software limit, while pressing the axis 3 brake release switch, move Axis 3 manually, and make sure that the software limit is correctly set.

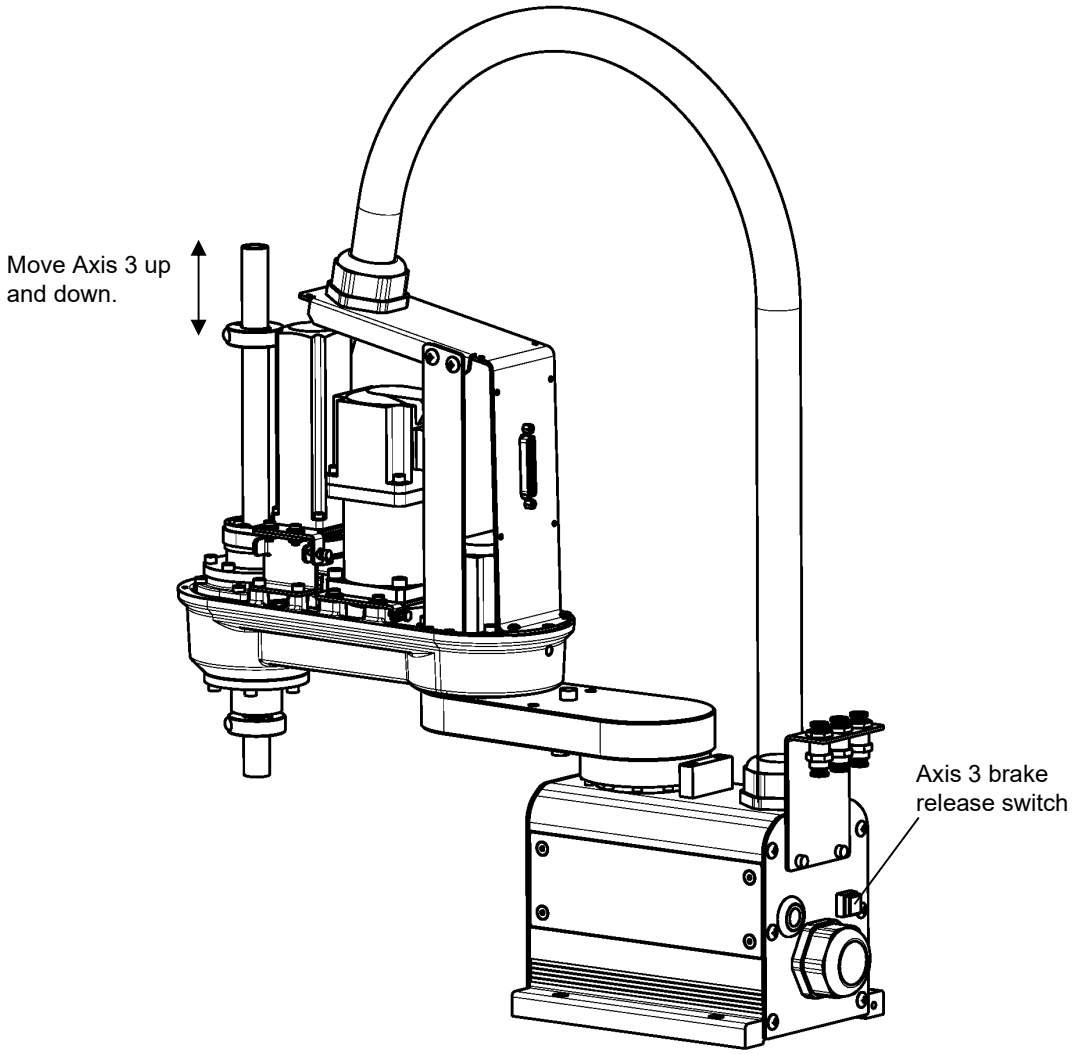


Fig. 9.7 Checking Software Limit Change (THL300 and THL400)

Fig. 9.8 shows the settings of the operating range when the upper limit mechanical stopper is moved 75 mm downward.

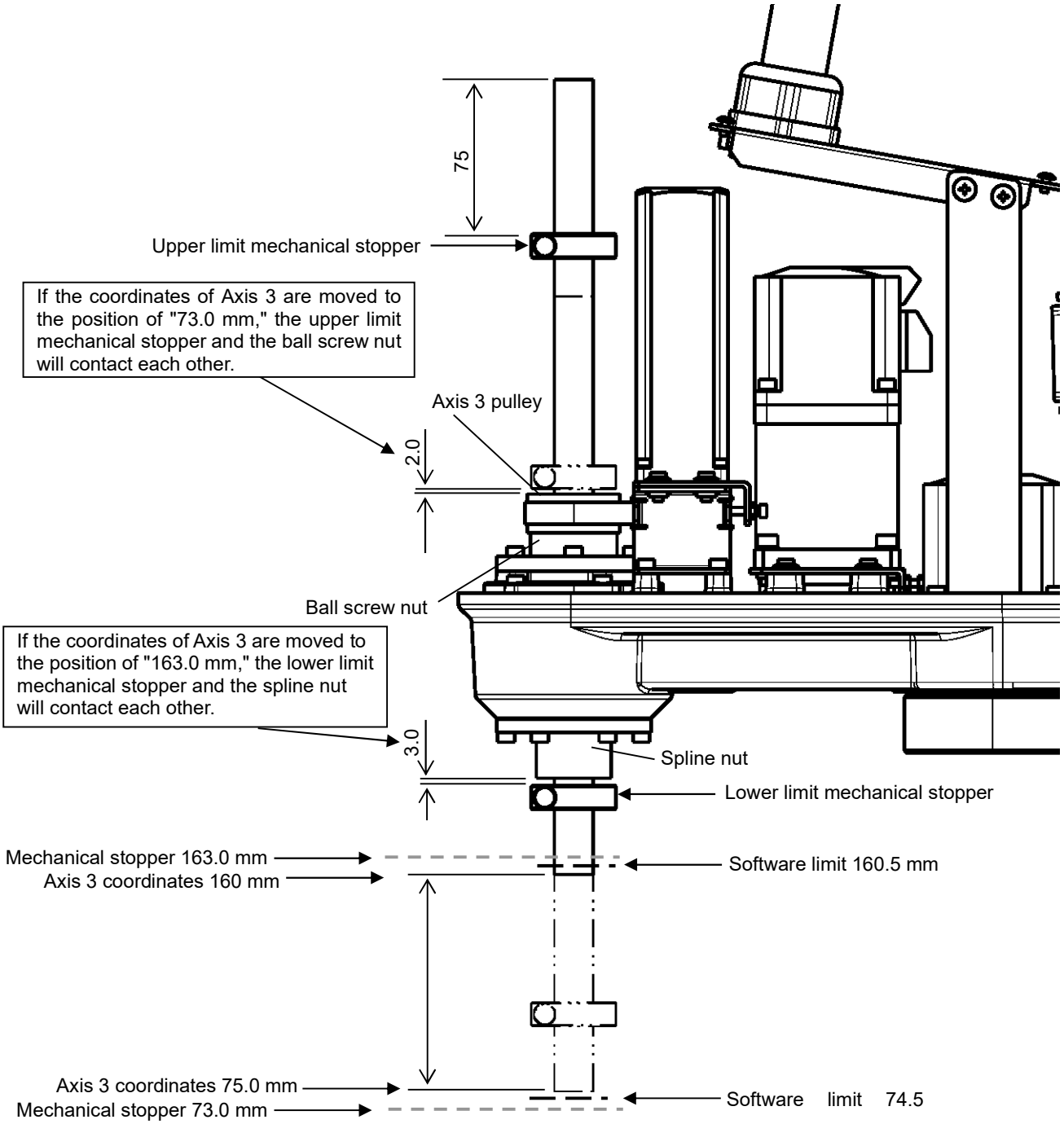


Fig. 9.8 Changing the Operating Range of Axis 3 (THL300 and THL400)

4) Remove the Arm 2 cover. The cover is fastened to Arm 2 and the harness guide with four (4) hexagon socket head bolts (M3 x 16) and ten (10) cross recessed truss head screw (M3 x 10 x 2 pcs. and M3 x 6 x 8 pcs.). It is recommended to lower Axis 3 in advance until the upper limit stopper of the ball screw enters the cover.

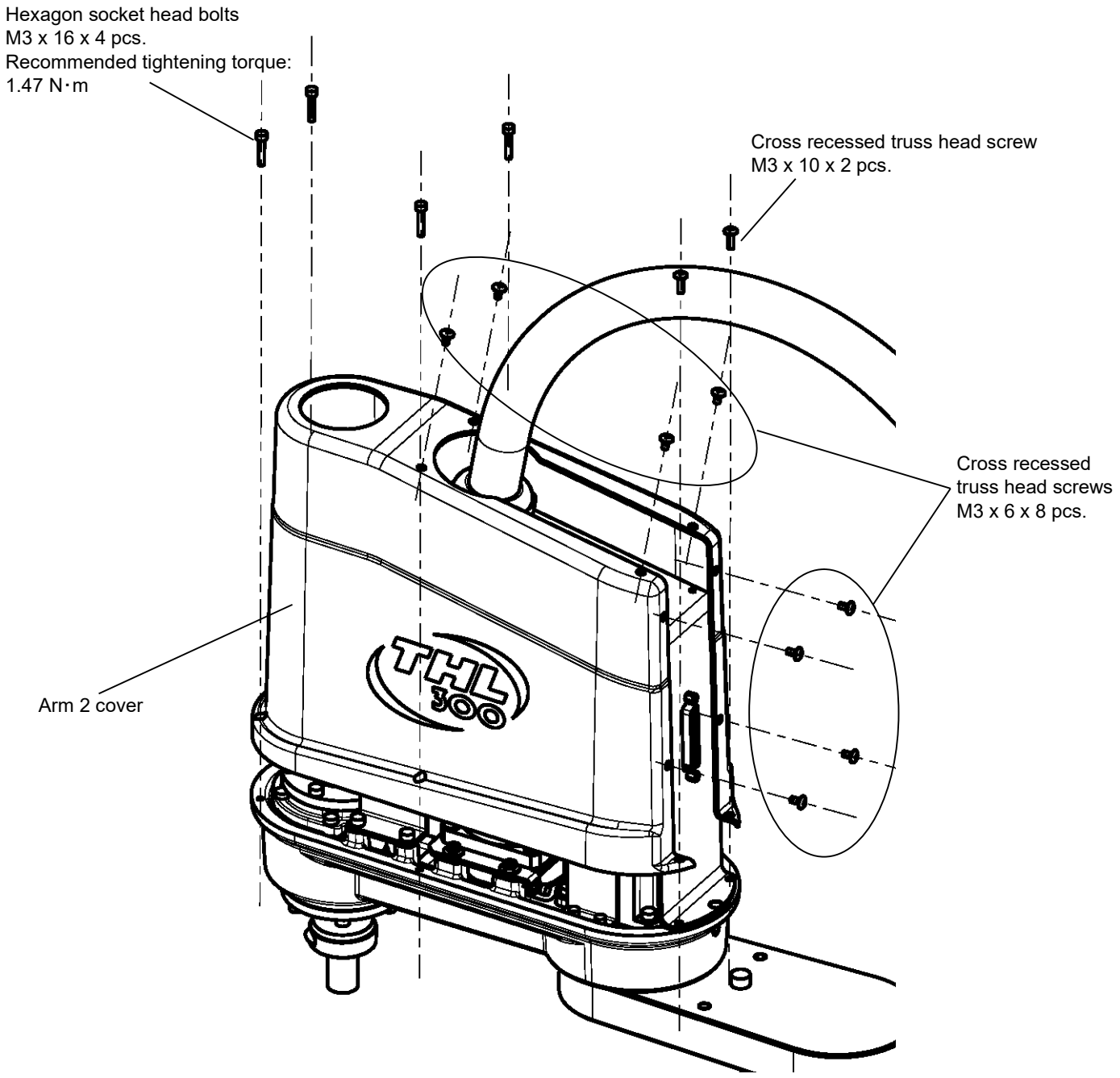


Fig. 9.9 Removing the Arm 2 Cover (THL300 and THL400)

9.2.4 Changing Software Limits

When the mechanical stoppers are changed, be sure to also change the software limits. There are the following two types of methods to change software limits.

[1] Change the setting values of the "User Parameter File (file name: USER.PAR)." There are software limits that can be set by the customer. For more details, refer to the item of [U14] SOFTWARE LIMIT in the "Instruction Manual: User Parameters" separately available.

[2] By manipulating the teaching pendant, change software limits in the utility mode "J-LIM."
For more details, refer to "10.8 Joint Limit Setting [J-LIM]" in Chapter 10, "Utilities" in the "Instruction Manual: Operations" separately available.

Note that if software limits are changed using the above two types of methods, the factory preset values of the software limits set up in the "User Parameter File (file name: USER.PAR)" will be overwritten and saved.

Thus, **be sure to create a backup of the "User Parameter File" before changing software limits** so as to get a grasp of the factory preset values of the software limits in the "User Parameter File."

Also, when software limits were changed, be sure to turn OFF and then ON the power.
If the power is not turned OFF and then ON again, changes to the user parameters will not take effect.

9.2.5 Coordinate System

The robot's joint angle origin (0° or 0 mm position) is factory-calibrated according to the base reference planes. Fig. 9.10 shows the base coordinate system and origin of each axis joint angle. The coordinate system is common between the THL300 and THL400. Figure shows the THL300.

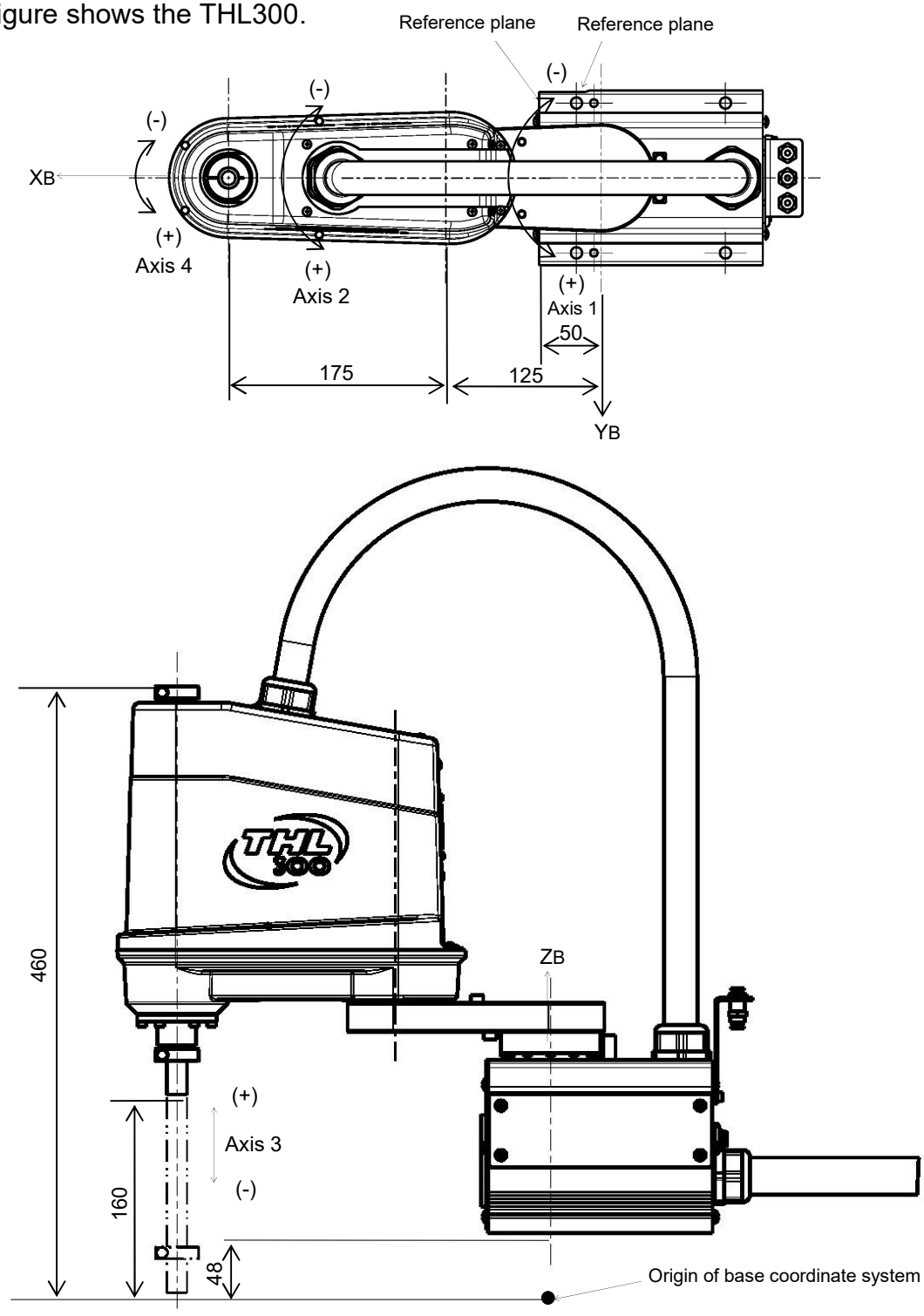


Fig. 9.10 Base coordinate system and joint angle origin (THL300)

9.2.6 Installing the Robot

The robot is secured, using the mounting holes on the base (four places). Use M8 hexagon socket head cap screws.
Table 9.3 lists the loads applied to the frame during horizontal operation, and Fig. 9.11 shows how to install the robot. Reference planes are provided on the base unit.
To align the robot position in the base coordinate system, or to replace the robot, provide adequate reference planes. Then, contact such reference planes to the base reference planes and secure the robot. Further, a pin hole is provided. This hole can be used for positioning.

! CAUTION

- The robot will suddenly accelerate and decelerate during operation. When installing it on a frame, make sure that the frame has sufficient strength and rigidity.
  If the robot is installed on a frame that does not have sufficient rigidity, vibration will occur while the robot is operating, and could lead to faults.
  When installing the robot on the floor, secure the robot with anchor bolts, etc.
- Install the robot on a level place. Failure to do so could lead to a drop in performance or faults.

Table 9.3 Load applied to the frame during horizontal operation (THL300 and THL400)

Model	Loads applied to the frame during horizontal operation [Nm]	Robot main body mass [kg]
THL300	110	12
THL400	110	13

* These values are reference values. When designing a frame, take the safety factor into consideration.

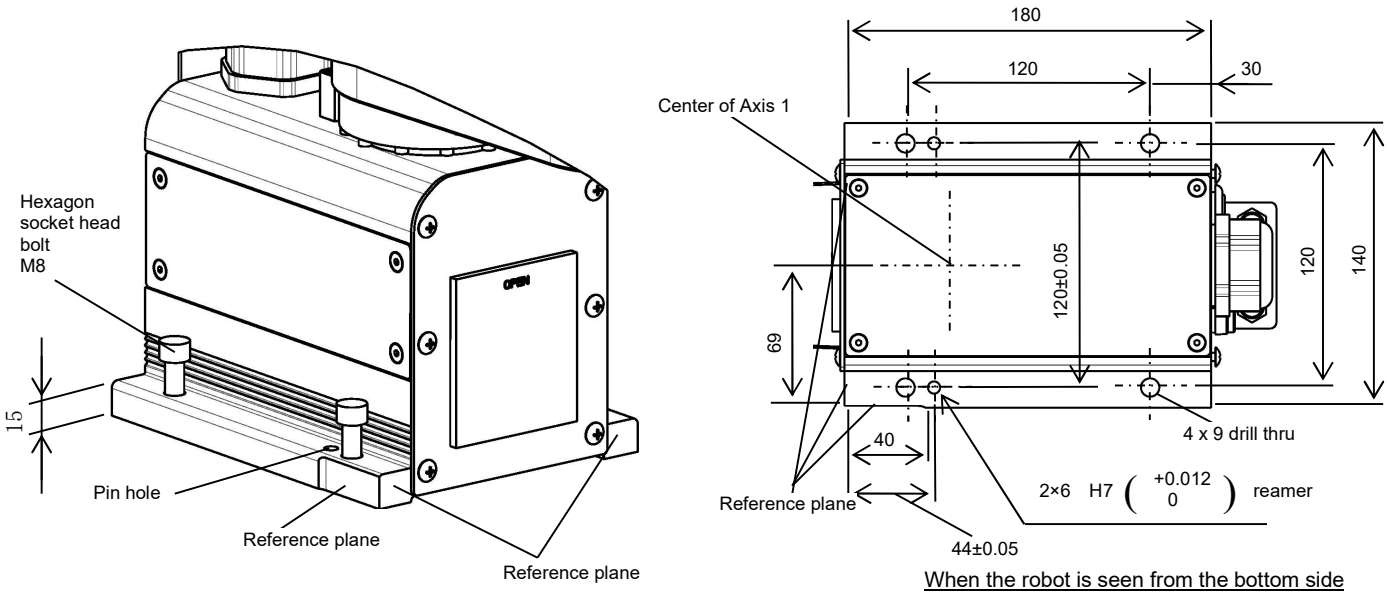


Fig. 9.11 Installation method (THL300 and THL400)

9.3 Precautions for Handling the Teach Pendant

Be careful of the following matters when handling the teach pendant.

! CAUTION

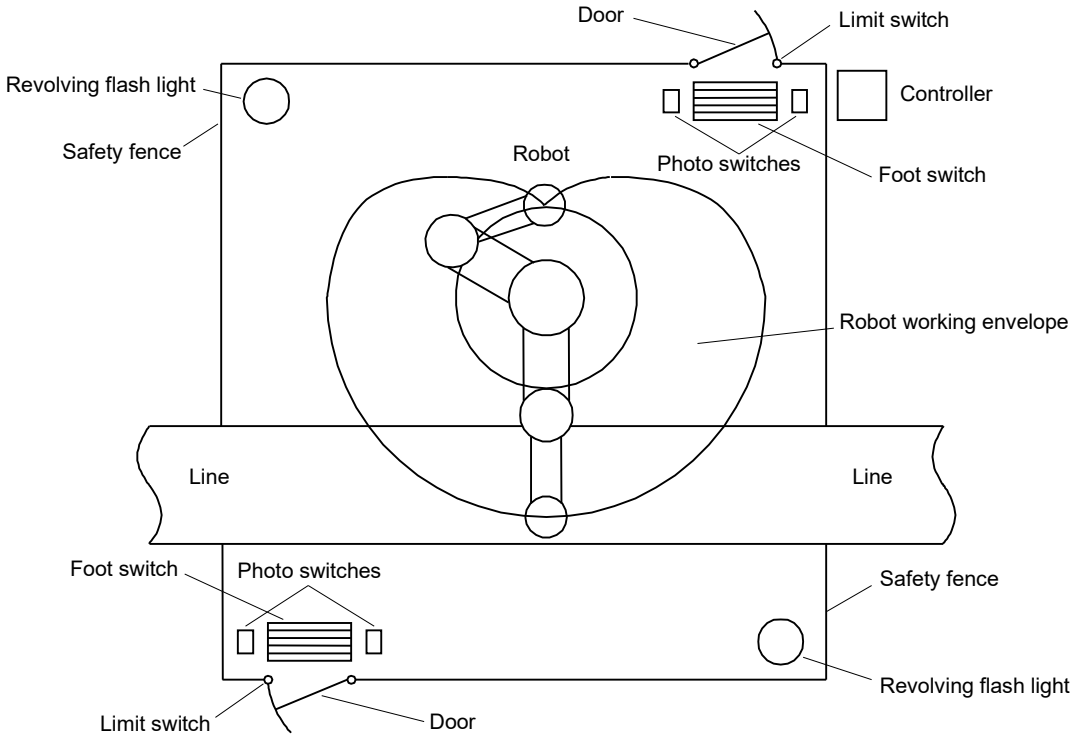
- DO NOT drop the teach pendant or hit it against anything.
- DO NOT pull the cable running from the teach pendant.
- DO NOT press the switches on the teach pendant with anything sharp (like the tip of a knife, pencil, ball-point pen, etc.).
- DO NOT place or use the teach pendant near open flames.
- DO NOT leave the teach pendant in direct sunlight for a long period of time.

9.4 Safety Measures

a) When installing the robot, provide sufficient space to carry out the work safely.

b) Clarify the hazard zone, and provide a safety fence so that other persons cannot enter the zone easily. The hazard zone is the zone near the robot's working space where a hazardous state could occur if a person enters.

c) Provide limit switches, photo switches, foot switch, etc., at the entrance of the safety fence to provide an emergency stop function that will stop the robot if a person enters the hazard zone. The emergency stop function should be an electrically independent normal close contact b (closed in normal operation) with compulsive opening function and must not be automatically recovered.



d) The controller should be installed at a place outside the hazard zone where the operator can view the robot movement.

10. Robot Installation (THL500, THL600 and THL700)

10.1 Installation Environment

For mounting environment, see "9.1 Installation Environment".

10.2 Installation

Before installing the robot, you should plan a layout, fully considering the working envelope (or operating range), coordinate system and space for maintenance.

10.2.1 External Dimensions and Working Envelope

Figs. 10.1 to 10.3 show the external view drawing and working envelope of the robot. Each axis can operate within the working envelope. To prevent the robot from moving out of the working envelope by misoperation, the robot is equipped with mechanical stoppers outside the working envelope. Additionally, soft limits that can be set by the user are provided. For further information, see the “Instruction Manual: User Parameters” provided separately.

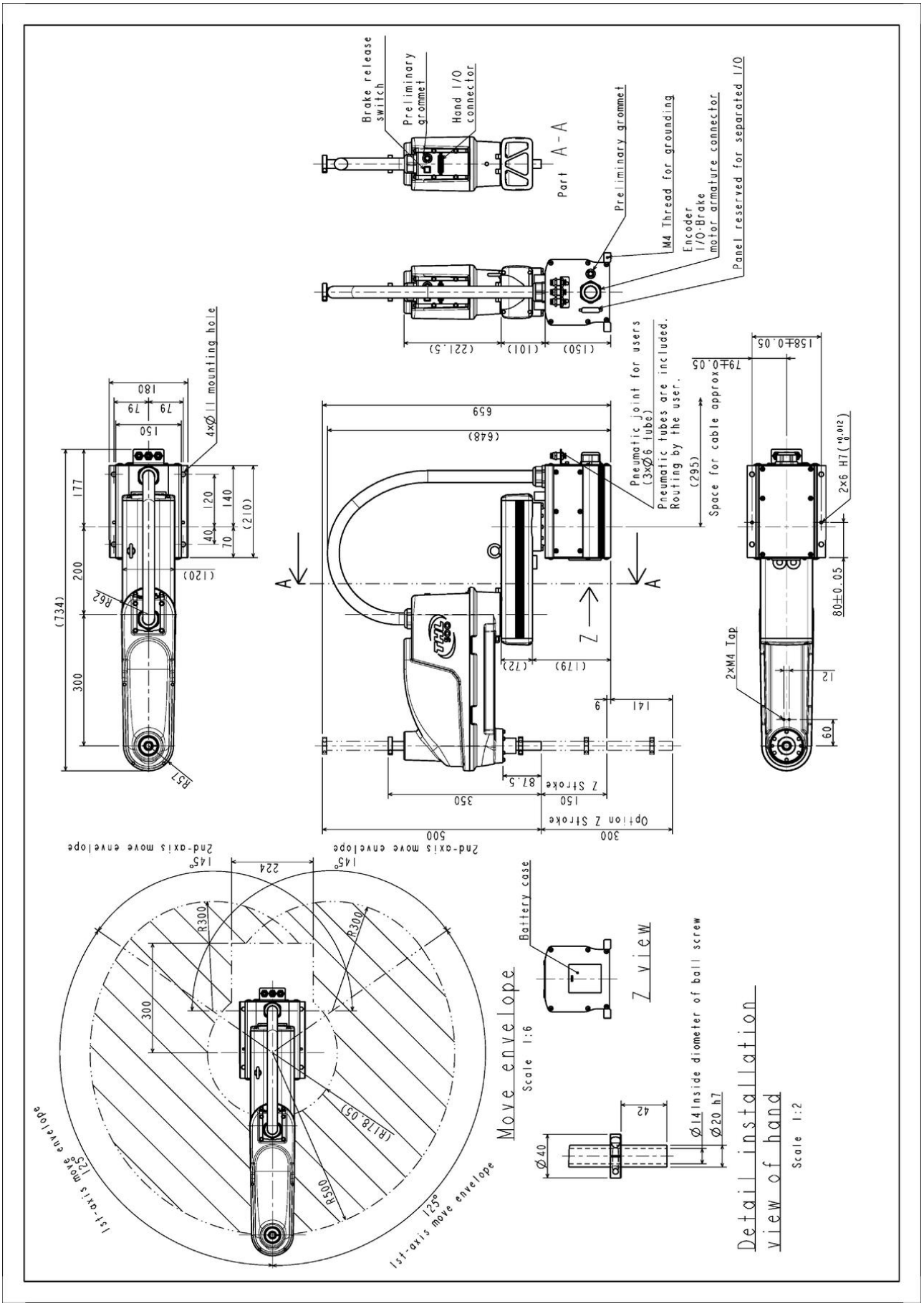


Fig. 10.1 External view and working envelope (THL500)

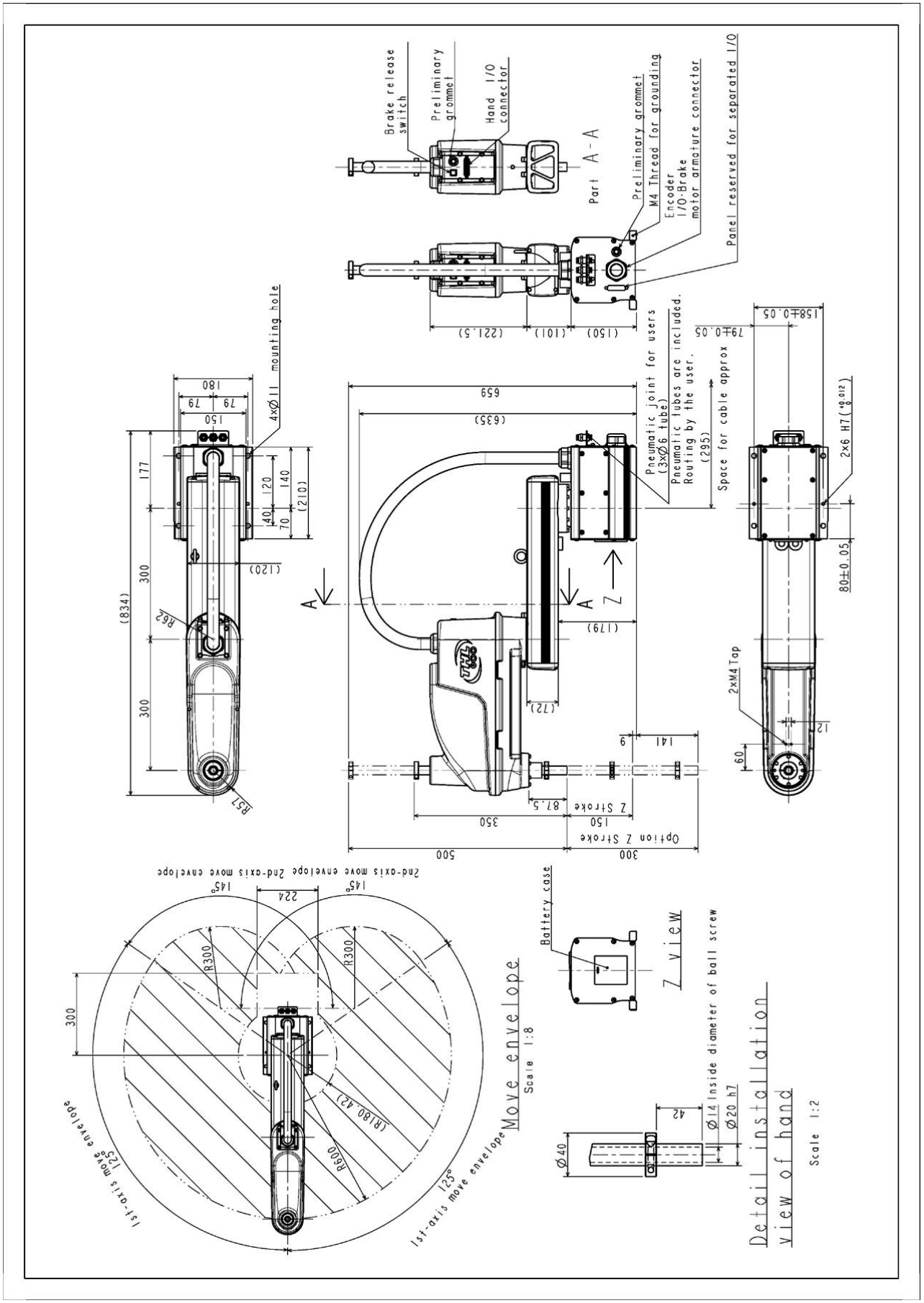


Fig. 10.2 External view and working envelope (THL600)

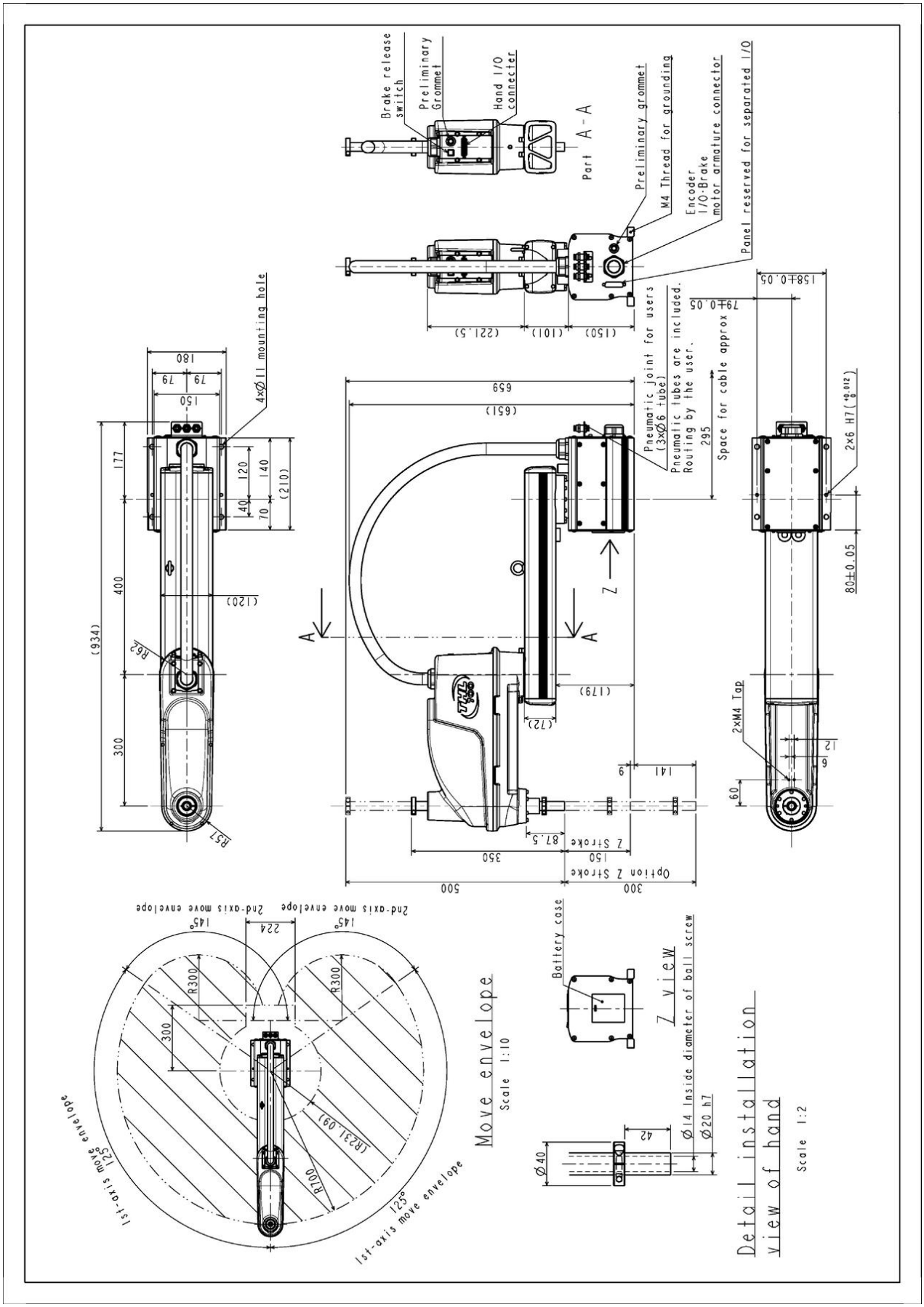


Fig. 10.3 External view and working envelope (THL700)

10.2.2 Changing the Operating Ranges of Axes 1 and 2

Mechanical stoppers are installed to each robot to mechanically control the operation range of each axis. Changing the mechanical operation range of a robot by modifying these mechanical stoppers is referred to as "change in operating range."
Here, how to change the operating ranges of Axes 1 and 2 of a robot is explained.

Regarding how to change the operating range of Axis 3, Para. 10.2.3.
Note that Axis 4 is different from other operating axes; it restricts the operating range with software limits only, not with mechanical stoppers. Therefore, regarding how to change the operating range of Axis 4, see Para. 9.2.4.

! CAUTION

- To change operating ranges, design and produce mechanical stoppers by referring to this document according to your usage.
- When mechanical stoppers have been changed and then operating ranges have been changed, be sure to also change software limits to prevent contact with the mechanical stoppers while operating a robot.
- Mechanical stoppers do not securely restrict the movable ranges of robots.

  When the power of a robot is turned on, never enter the operating range of the robot.
- If a robot collides with a mechanical stopper, the robot detects the collision and stops, but the mechanical stopper may be damaged. Avoid reusing the mechanical stopper.
- The mechanical stopper reference drawings shown in this document do not fully satisfy the customer's usage.

  Design, produce and install mechanical stoppers according to your usage, such as the operating range.
- The failures of a robot caused by mechanical stoppers will be excluded from the warranty coverage.

As shown in Fig. 10.4, the operation range can be altered by changing the position of the mechanical stopper.

Table 10.1 Operation ranges before and after change (THL500, THL600 and THL700)

		Before change	After change
Axis 1 operation range	+ direction	125°	105°
	- direction	125°	105°
Axis 2 operation range	+ direction	145°	110°
	- direction	145°	110°

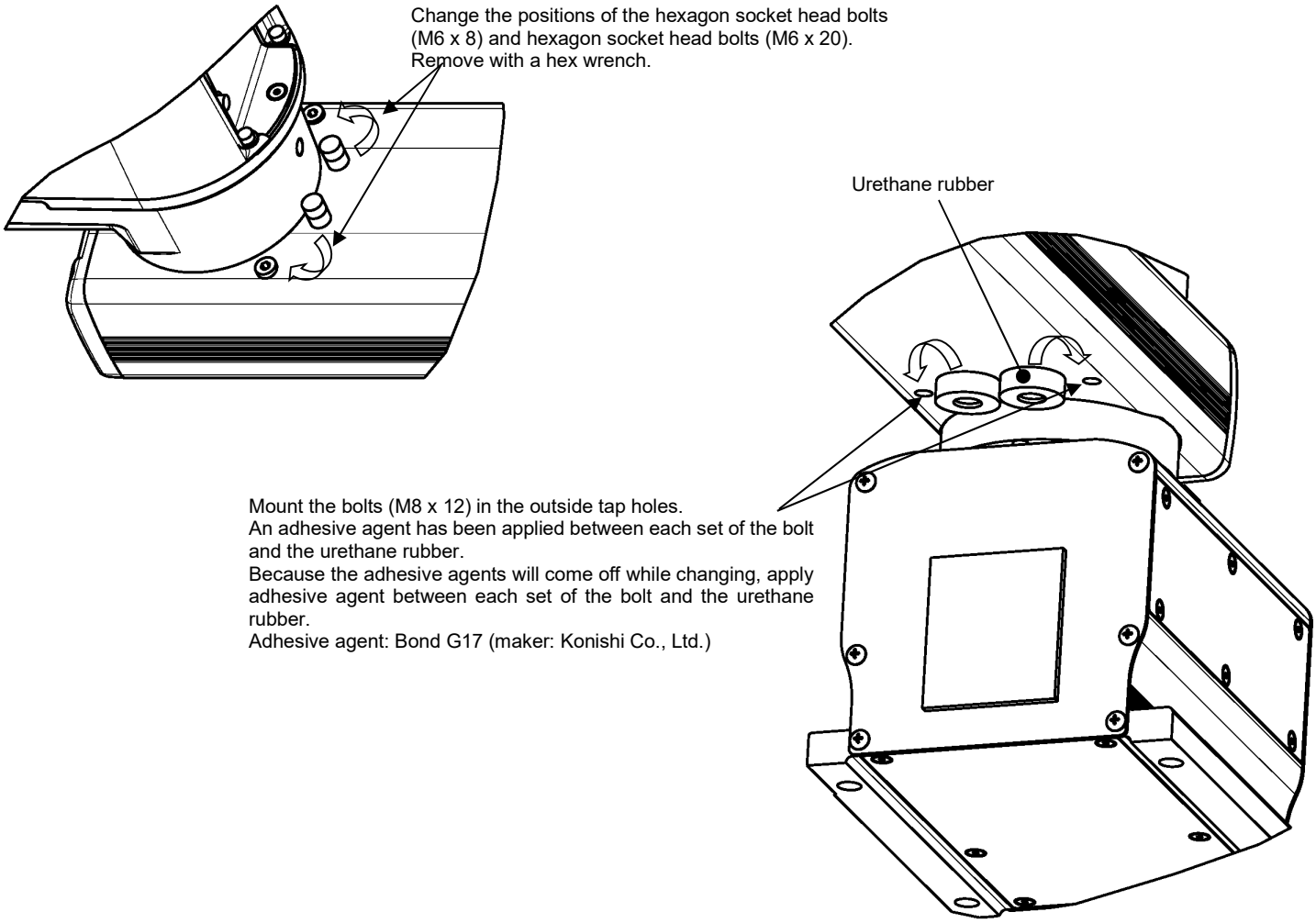


Fig. 10.4 Changing the Operating Ranges of Axes 1 and 2 (THL500, THL600 and THL700)

If the operation range is changed, it is necessary to change the user parameter. For the software limit change procedure, see "9.2.4 Changing Software Limits".

10.2.3 Changing the Operating Range of Axis 3

At the factory shipment of a robot, software limits and mechanical stoppers are preset so that the Z stroke of Axis 3 is between 0 and 150 mm (0 through 300 mm for Z-long specifications). Fig. 10.5 shows the settings of operating range at the factory shipment.

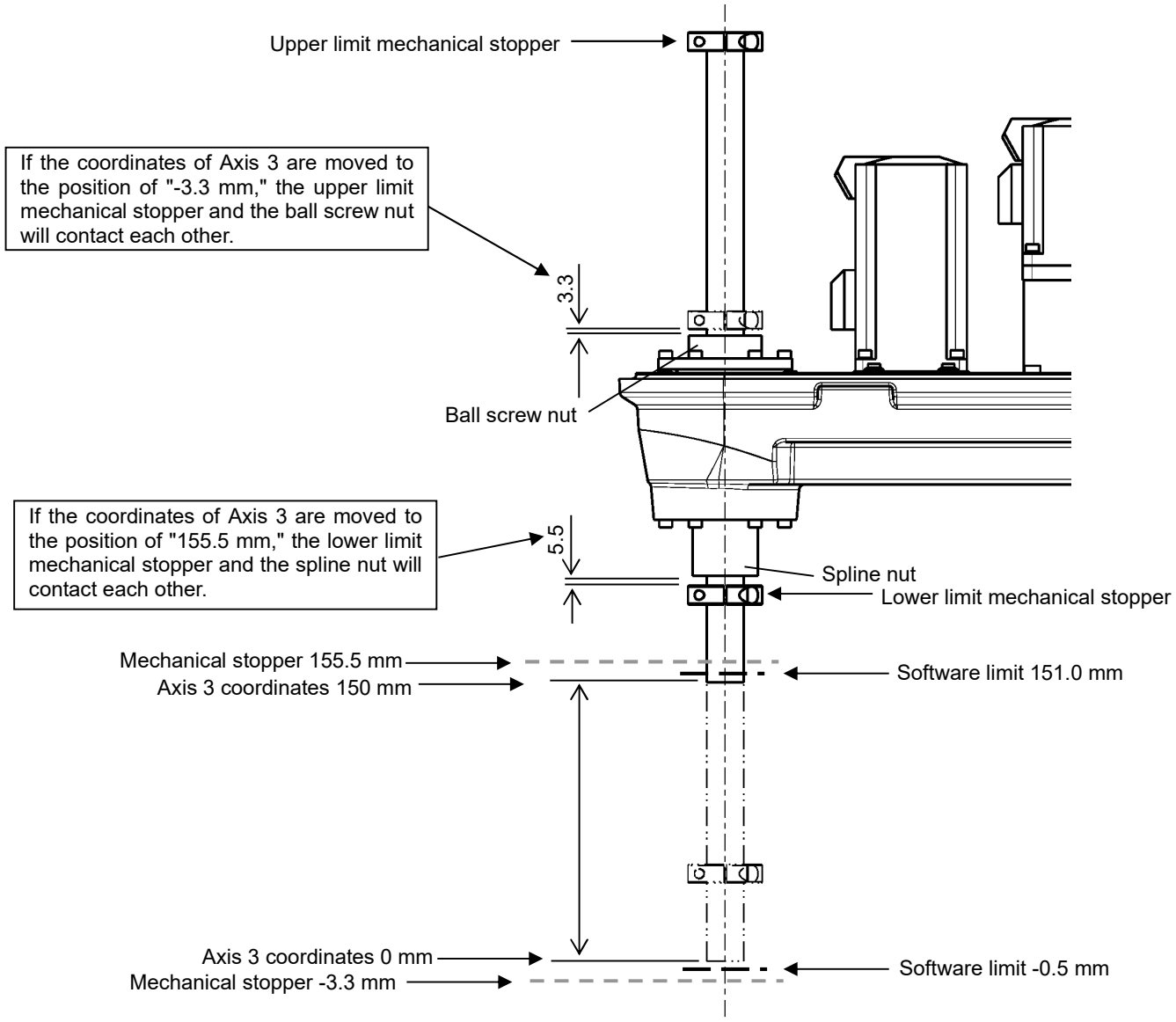


Fig. 10.5 Changing the Operating Range of Axis 3 (THL500, THL600 and THL700)

1) Remove the Arm 2 cover. The cover is fastened to Arm 2 and the harness guide with 14 hexagon socket head bolts (M3 x 16 x 4 pcs., M4 x 6 x 8 pcs. and M4 x 10 x 2 pcs.)

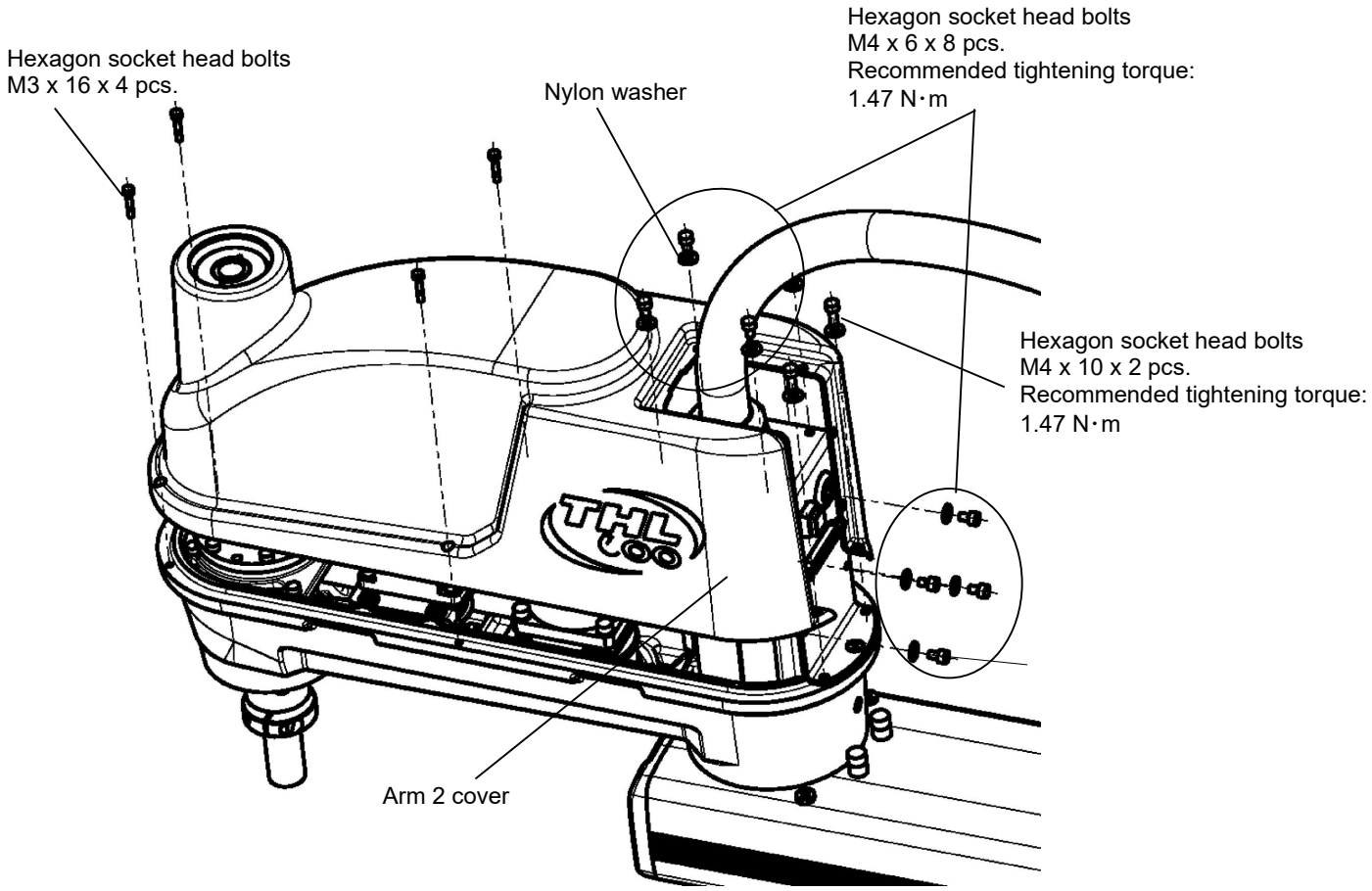


Fig. 10.6 Removing the Arm 2 Cover (THL500, THL600 and THL700)

2) Loosen the fixing bolts of the mechanical stoppers, move the mechanical stoppers to a desired position, and fix them again. When fixing the mechanical stoppers, be sure to apply Loctite to the fixing bolts.

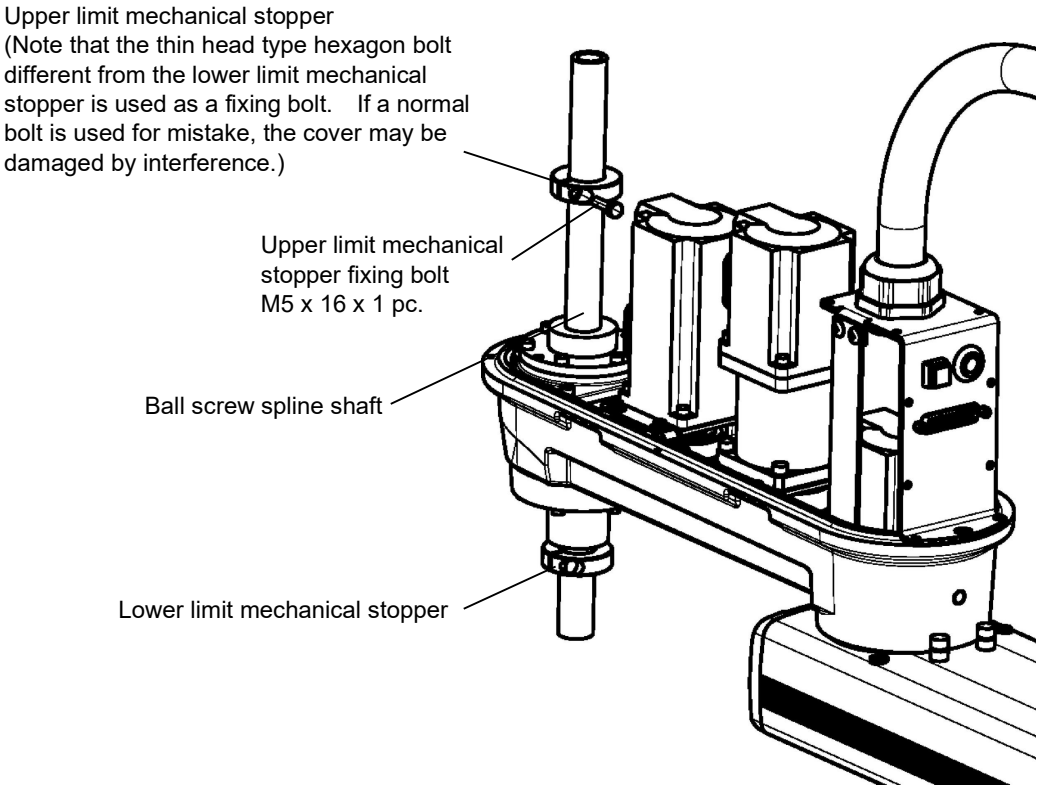


Fig. 10.7 Checking Software Limit Change (THL500, THL600 and THL700)

3) When the mechanical stoppers are changed, be sure to also change the software limits. Regarding how to change software limits, see Para. 9.2.4 and Fig. 10.10. After changing the software limit, while pressing the axis 3 brake release switch, move Axis 3 manually, and make sure that the software limit is correctly set.

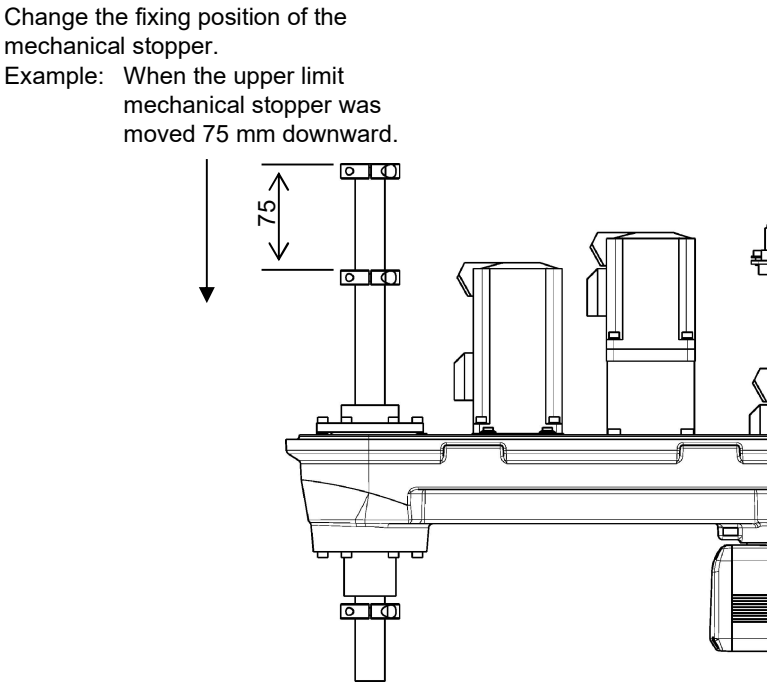


Fig. 10.8 Changing the Operating Range (THL500, THL600 and THL700)

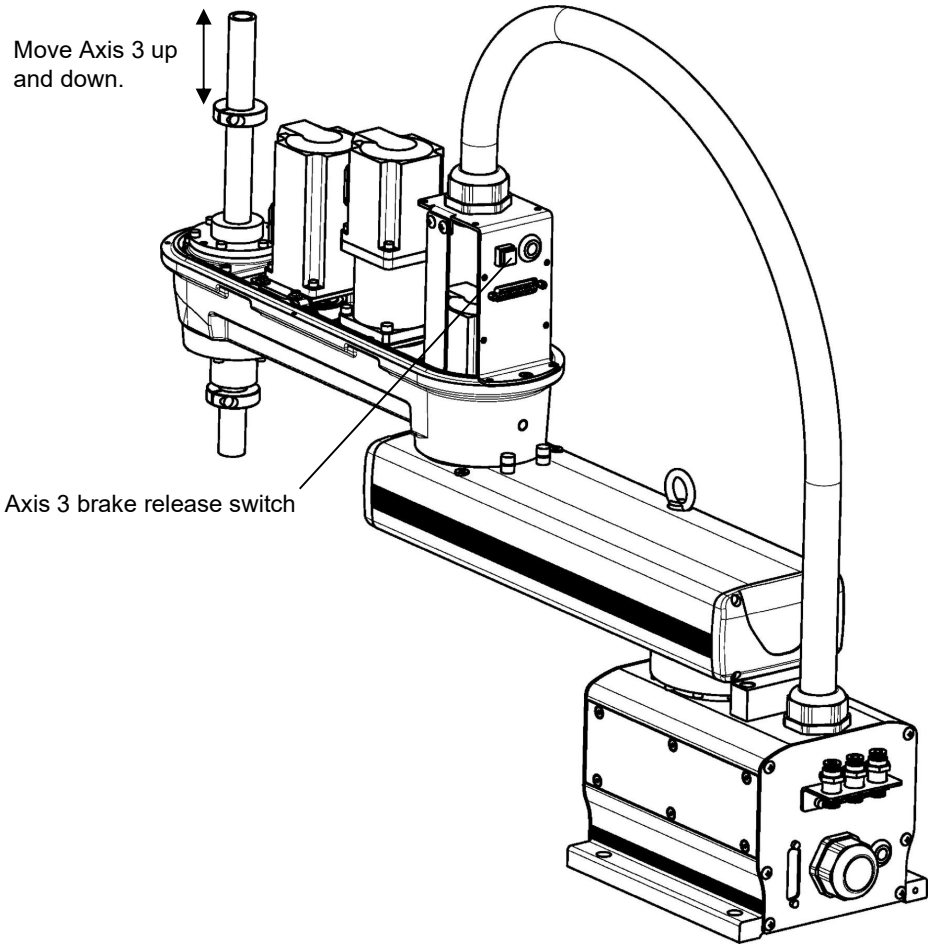


Fig. 10.9 Checking Software Limit Change (THL500, THL600 and THL700)

Fig. 10.10 shows the settings of the operating range when the upper limit mechanical stopper was moved 75 mm downward.

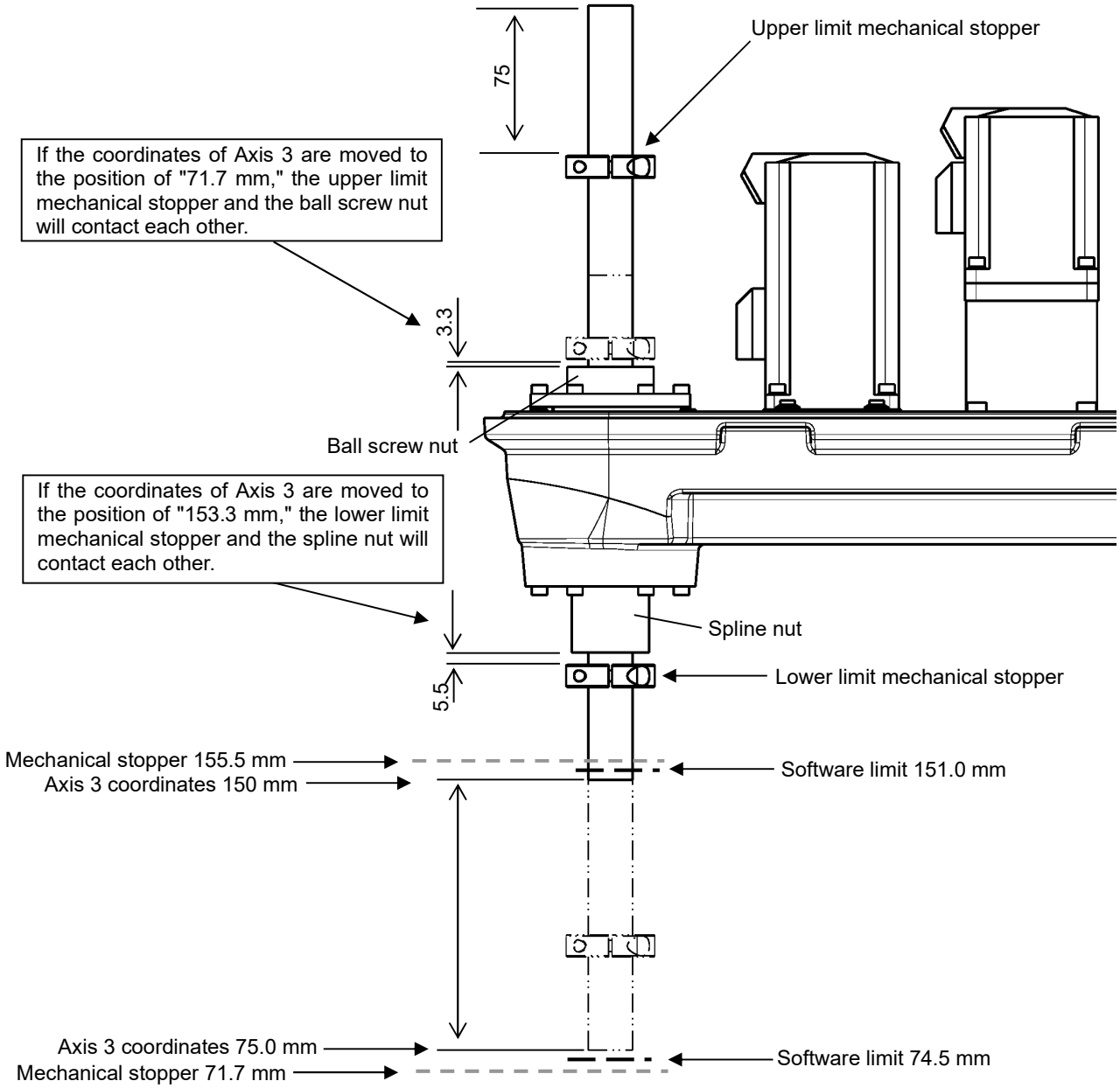


Fig. 10.10 Changing the Operating Range of Axis 3 (THL500, THL600 and THL700)

4) Fasten the Arm 2 cover to Arm 2 and the harness guide with 14 hexagon socket head bolts (M3 x 16 x 4 pcs., M4 x 6 x 8 pcs. and M4 x 10 x 2 pcs.) and nylon washers. (Application of Loctite during installation is not necessary.)

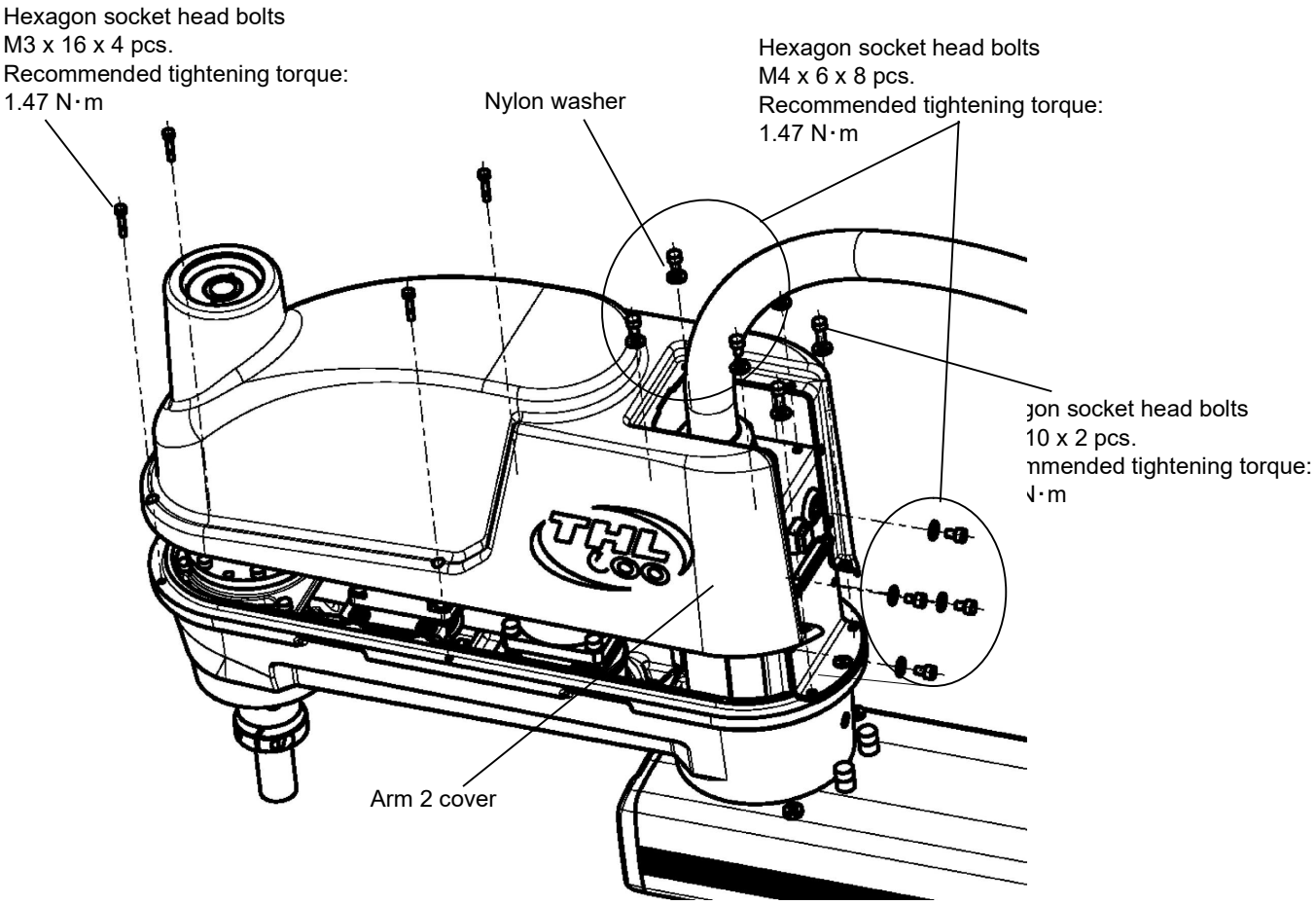


Fig. 10.11 Removing the Arm 2 Cover (THL500, THL600 and THL700)

After installation, manually move the ball screw spline shaft up and down while holding down the brake release switch, and check that the holes for the ball screws of the Arm 2 cover and the mechanical stoppers are not interfering with each other.

10.2.4 Changing Software Limits

For the software limit change procedure, see "9.2.4 Changing Software Limits".

10.2.5 Coordinate System

The robot's joint angle origin (0° or 0 mm position) is factory-calibrated according to the base reference planes. Fig. 10.12 shows the base coordinate system and origin of each axis joint angle. The coordinate system is common among the THL500, THL600, and THL700.

Figure shows the THL600.

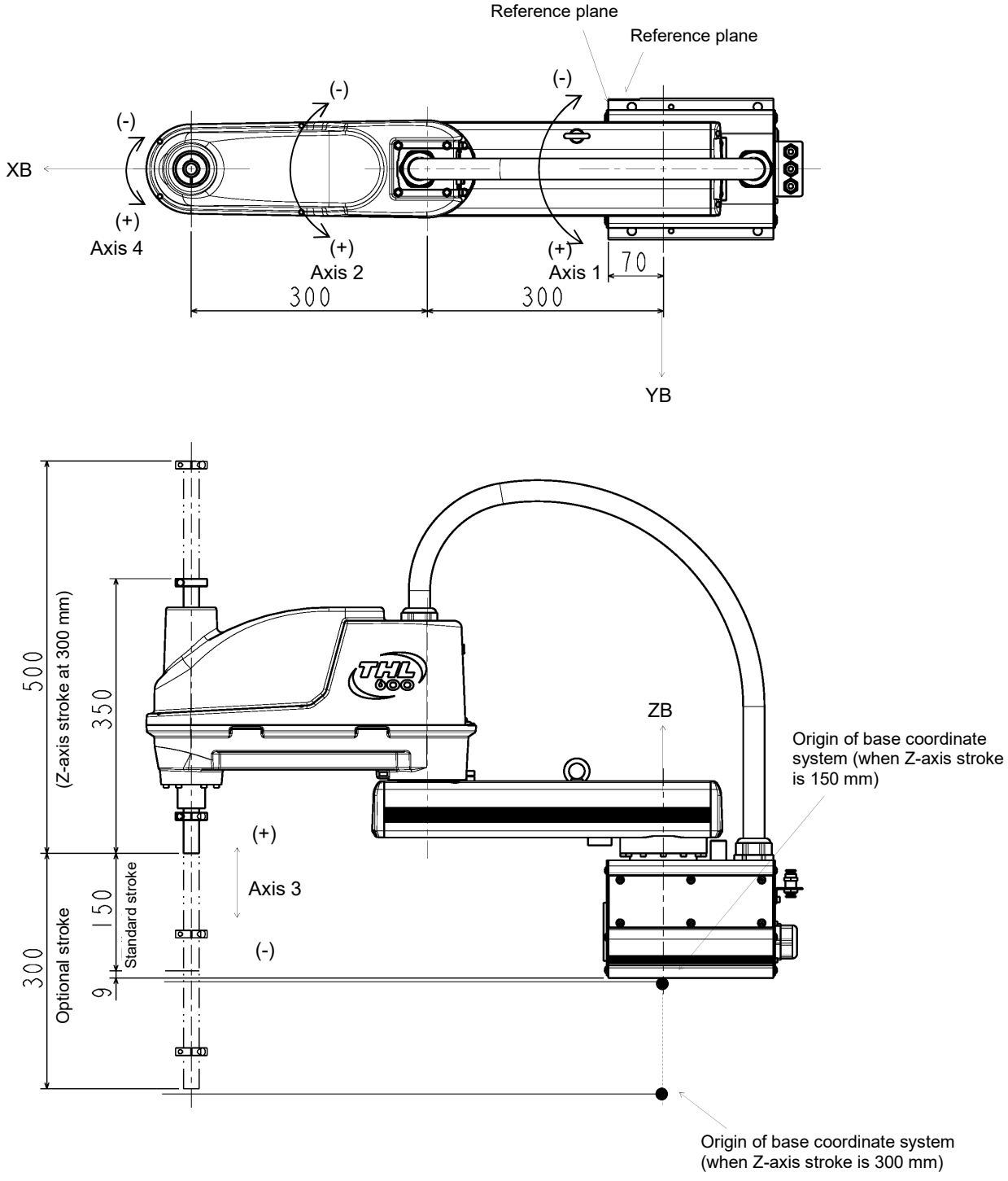


Fig. 10.12 Base coordinate system and joint angle origin (THL600)

10.2.6 Installing the Robot

The robot is secured, using the mounting holes on the base (four places). Use M10 hexagon socket head cap screws.
Table 10.2 lists the loads applied to the frame during horizontal operation, and Fig. 10.13 shows how to install the robot. Reference planes are provided on the base unit. To align the robot position in the base coordinate system, or to replace the robot, provide adequate reference planes. Then, contact such reference planes to the base reference planes and secure the robot. Further, a pin hole is provided. This hole can be used for positioning.

! CAUTION

- The robot will suddenly accelerate and decelerate during operation. When installing it on a frame, make sure that the frame has sufficient strength and rigidity.
  If the robot is installed on a frame that does not have sufficient rigidity, vibration will occur while the robot is operating, and could lead to faults.
  When installing the robot on the floor, secure the robot with anchor bolts, etc.
- Install the robot on a level place. Failure to do so could lead to a drop in performance or faults.

Table 10.2 Load applied to the frame during horizontal operation (THL500, THL600 and THL700)

Model	Loads applied to the frame during horizontal operation [Nm]	Robot main body mass [kg]
THL500	350	22
THL600	350	23
THL700	350	24

* These values are reference values. When designing a frame, take the safety factor into consideration.

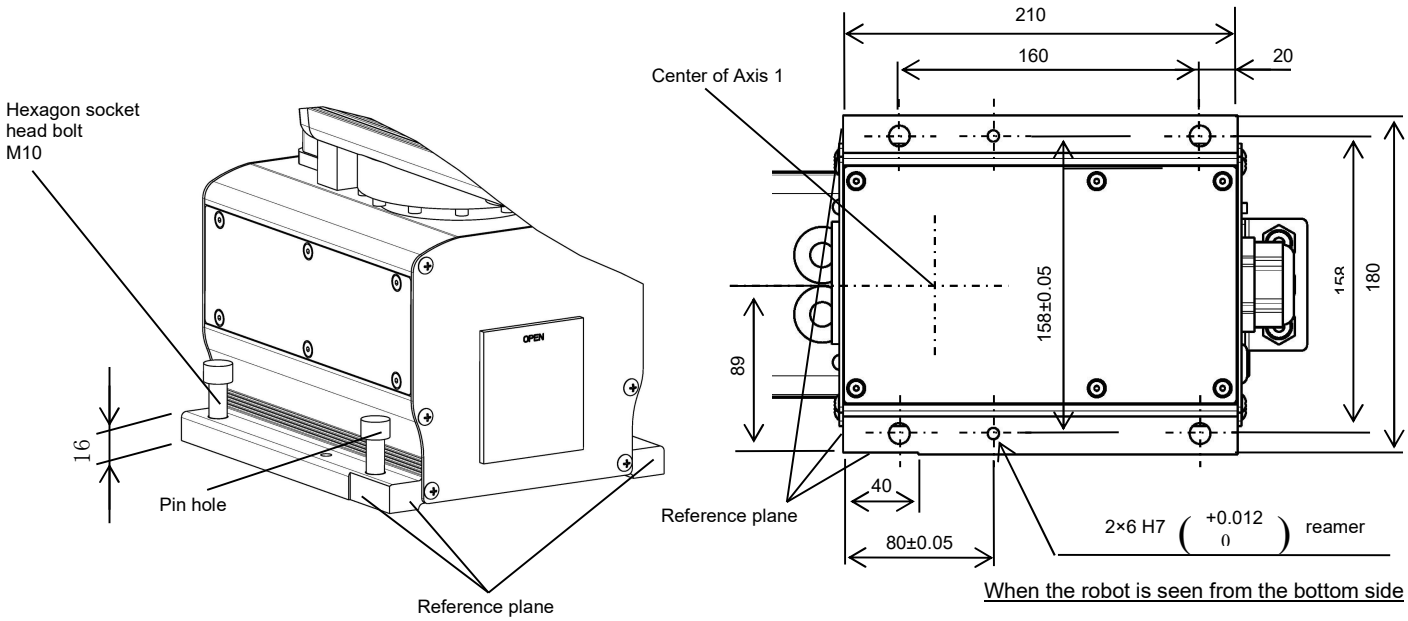


Fig. 10.13 Installation method (THL500, THL600 and THL700)

10.3 Precautions for Handling the Teach Pendant

For the teach pendant handing procedure, see "9.3 Precautions for Handling the Teach Pendant".

10.4 Safety Measures

For safety measures, see "9.4 Safety Measures".

11. Robot Installation (THL800, THL900 and THL1000)

11.1 Installation Environment

For mounting environment, see "9.1 Installation Environment".

11.2 Installation

Before installing the robot, you should plan a layout, fully considering the working envelope (or operating range), coordinate system and space for maintenance.

11.2.1 External Dimensions and Working Envelope

Figs. 11.1 to 11.3 show the external view drawing and working envelope of the robot. Each axis can operate within the working envelope. To prevent the robot from moving out of the working envelope by misoperation, the robot is equipped with mechanical stoppers outside the working envelope. Additionally, soft limits that can be set by the user are provided. For further information, see the “Instruction Manual: User Parameters” provided separately.

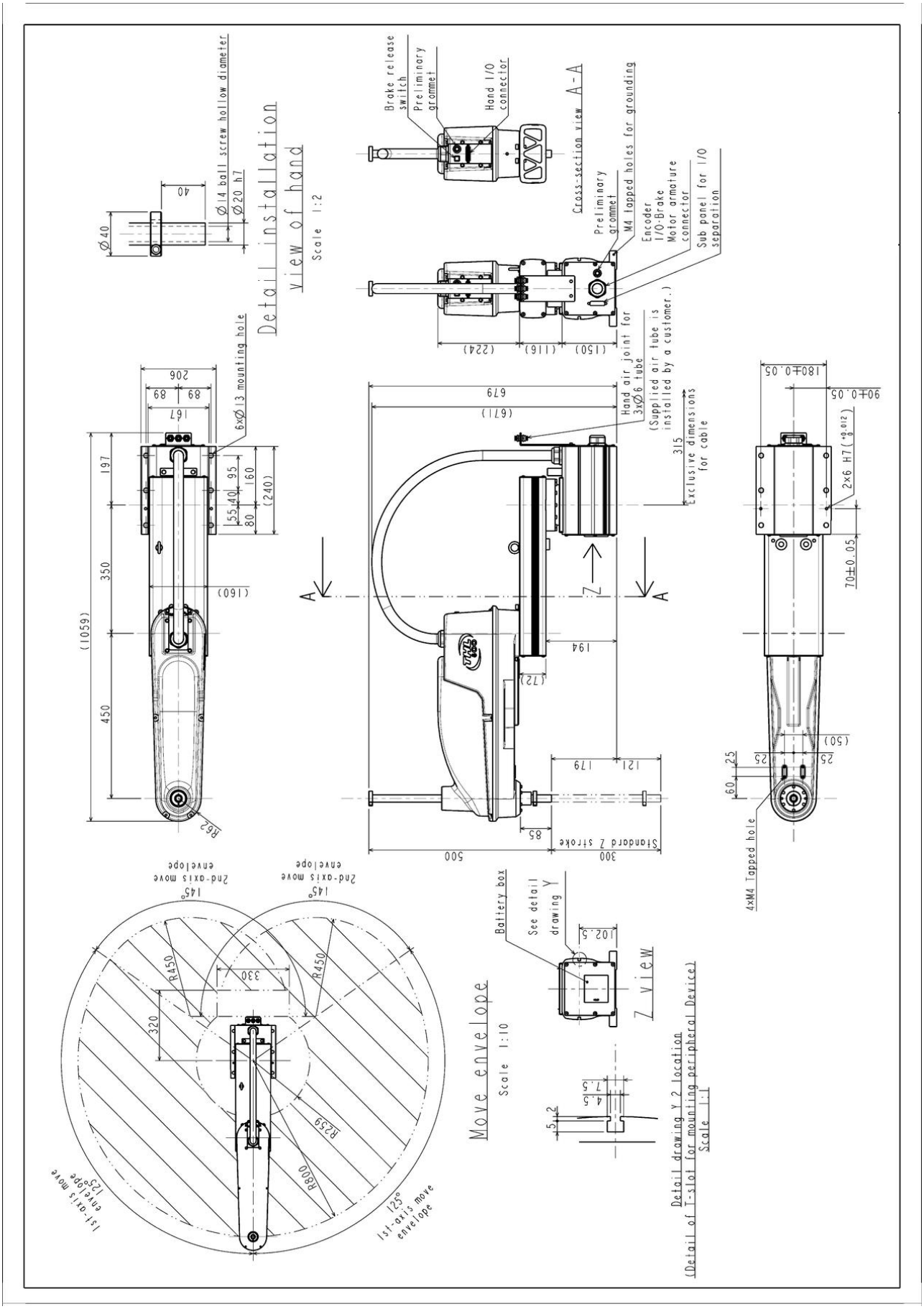


Fig. 11.1 External view and working envelope (THL800)

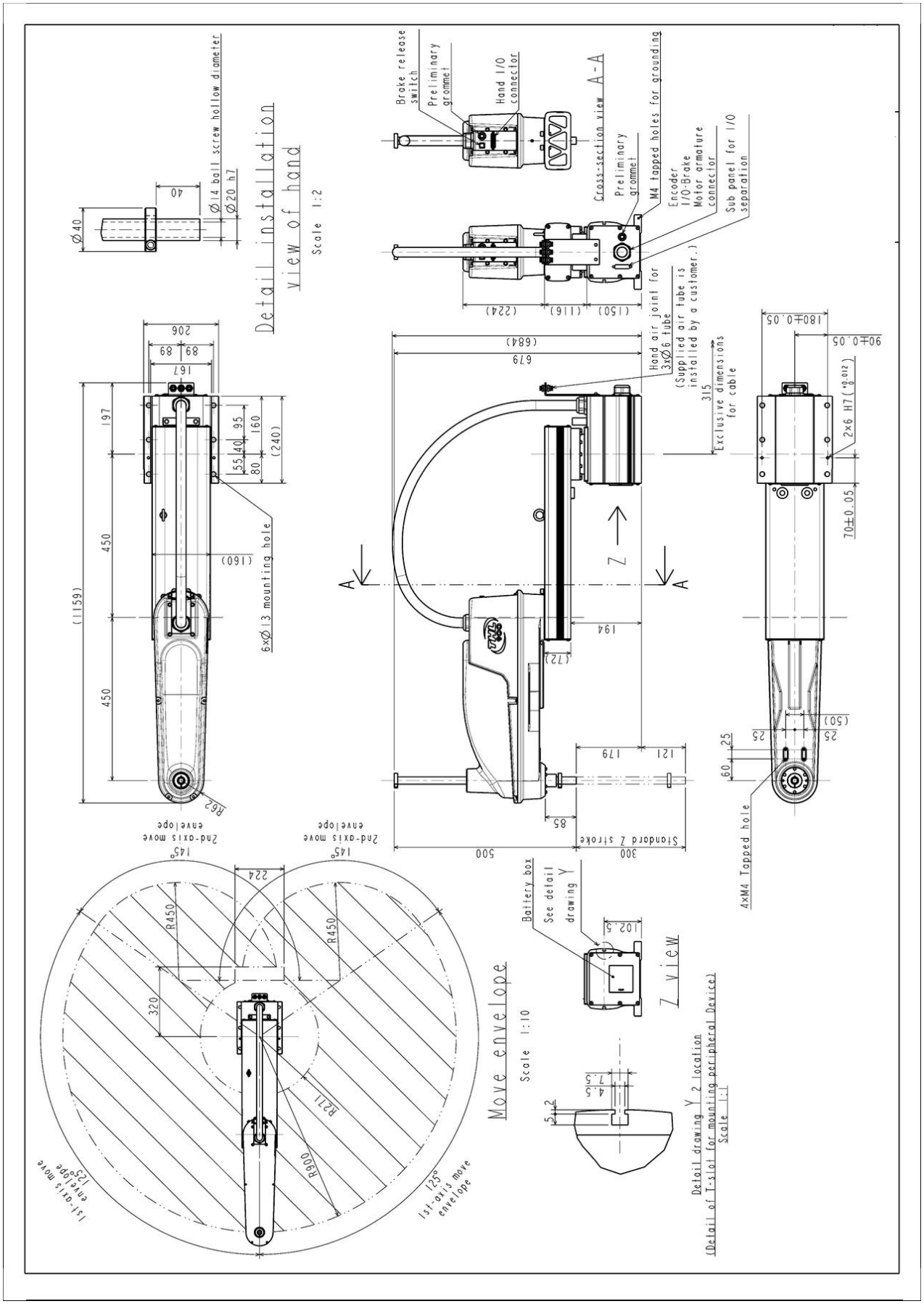


Fig. 11.2 External view and working envelope (THL900)

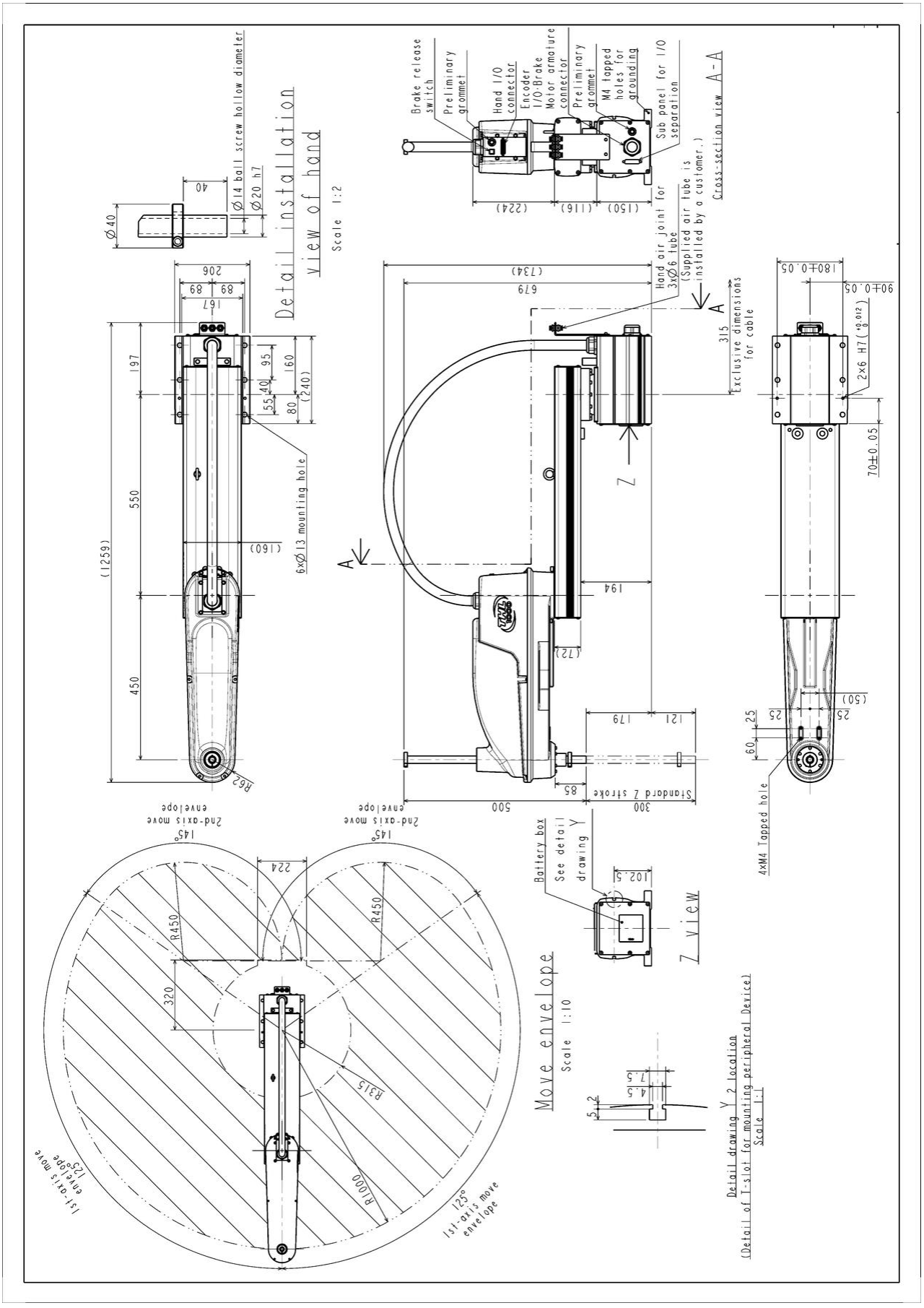


Fig. 11.3 External view and working envelope (THL1000)

11.2.2 Changing the Operating Ranges of Axes 1 and 2

Mechanical stoppers are installed to each robot to mechanically control the operation range of each axis. Changing the mechanical operation range of a robot by modifying these mechanical stoppers is referred to as "change in operating range."
Here, how to change the operating ranges of Axes 1 and 2 of a robot is explained.

Regarding how to change the operating range of Axis 3, Para. 11.2.3.
Note that Axis 4 is different from other operating axes; it restricts the operating range with software limits only, not with mechanical stoppers. Therefore, regarding how to change the operating range of Axis 4, see Para. 9.2.4.

! CAUTION

- To change operating ranges, design and produce mechanical stoppers by referring to this document according to your usage.
- When mechanical stoppers have been changed and then operating ranges have been changed, be sure to also change software limits to prevent contact with the mechanical stoppers while operating a robot.
- Mechanical stoppers do not securely restrict the movable ranges of robots.

  When the power of a robot is turned on, never enter the operating range of the robot.
- If a robot collides with a mechanical stopper, the robot detects the collision and stops, but the mechanical stopper may be damaged. Avoid reusing the mechanical stopper.
- The mechanical stopper reference drawings shown in this document do not fully satisfy the customer's usage.

  Design, produce and install mechanical stoppers according to your usage, such as the operating range.
- The failures of a robot caused by mechanical stoppers will be excluded from the warranty coverage.

As shown in Fig. 11.4, the operation range can be altered by changing the position of the mechanical stopper.

Table 11.1 Operation ranges before and after change (THL800, THL900 and THL1000)

		Before change	After change
Axis 1 operation range	+ direction	125°	110°
	- direction	125°	110°
Axis 2 operation range	+ direction	145°	120°
	- direction	145°	120°

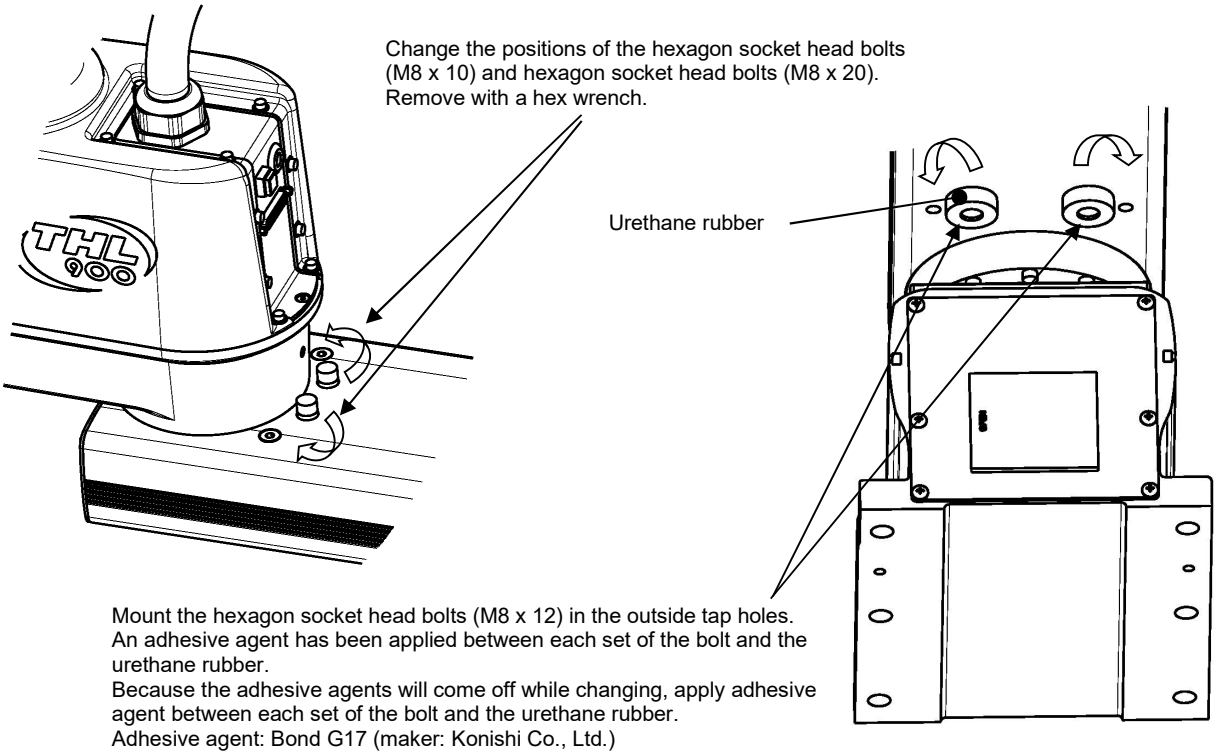


Fig. 11.4 Changing the Operating Ranges of Axes 1 and 2 (THL800, THL900 and THL1000)

If the operation range is changed, it is necessary to change the user parameter. For the software limit change procedure, see "9.2.4 Changing Software Limits".

11.2.3 Changing the Operating Range of Axis 3

At the factory shipment of a robot, software limits and mechanical stoppers are preset so that the Z stroke of Axis 3 is between 0 and 300 mm. Fig. 11.5 shows the settings of operating range at the factory shipment.

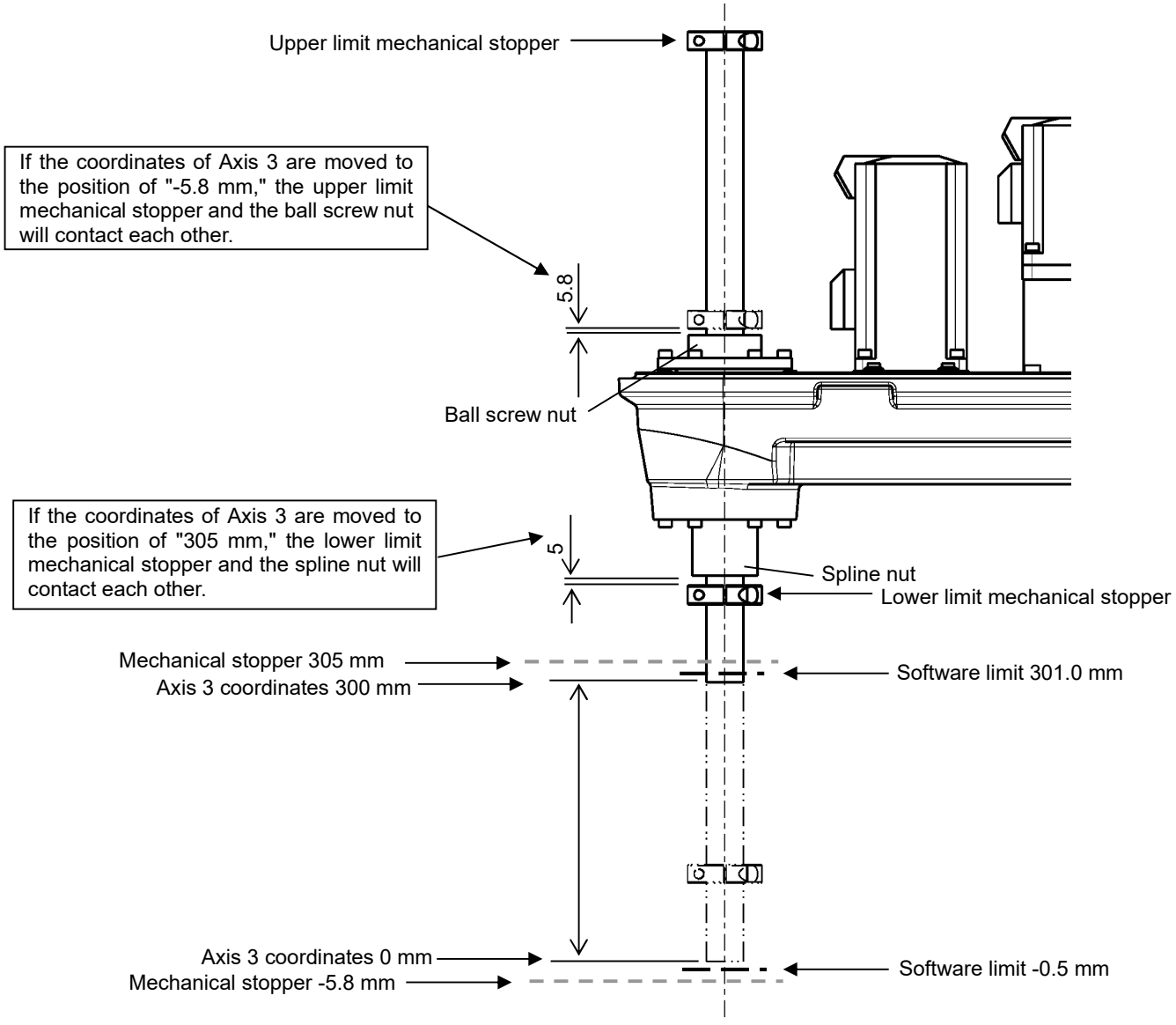


Fig. 11.5 Changing the Operating Range of Axis 3 (THL800, THL900 and THL1000)

1) Remove the Arm 2 cover. The cover is fastened to Arm 2 and the harness guide with 14 hexagon socket head bolts (M3 x 6 x 4 pcs., M4 x 6 x 8 pcs. and M4 x 10 x 2 pcs.)

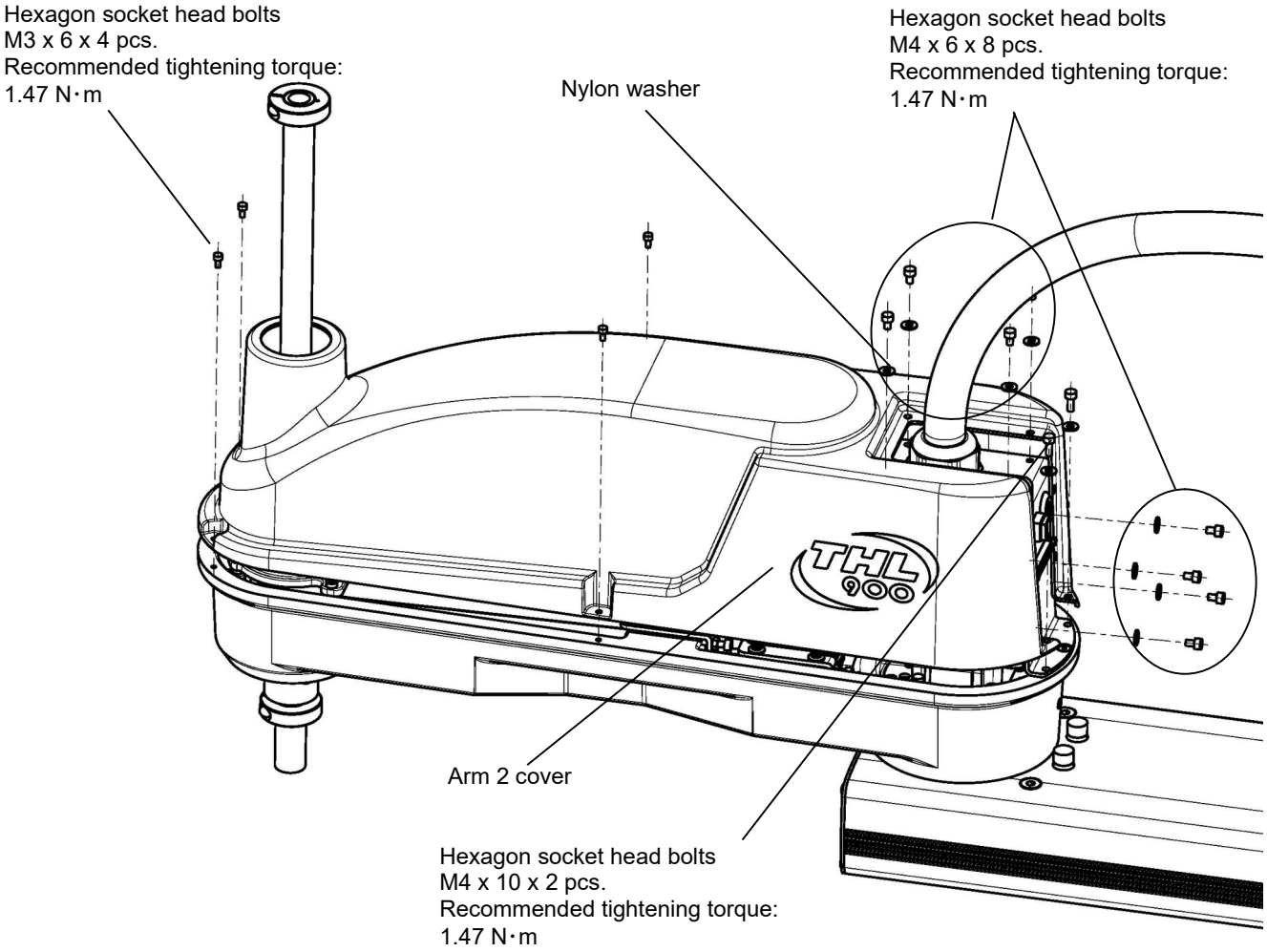


Fig. 11.6 Removing the Arm 2 Cover (THL800, THL900 and THL1000)

2) Loosen the fixing bolts of the mechanical stoppers, move the mechanical stoppers to a desired position, and fix them again. When fixing the mechanical stoppers, be sure to apply Loctite to the fixing bolts.

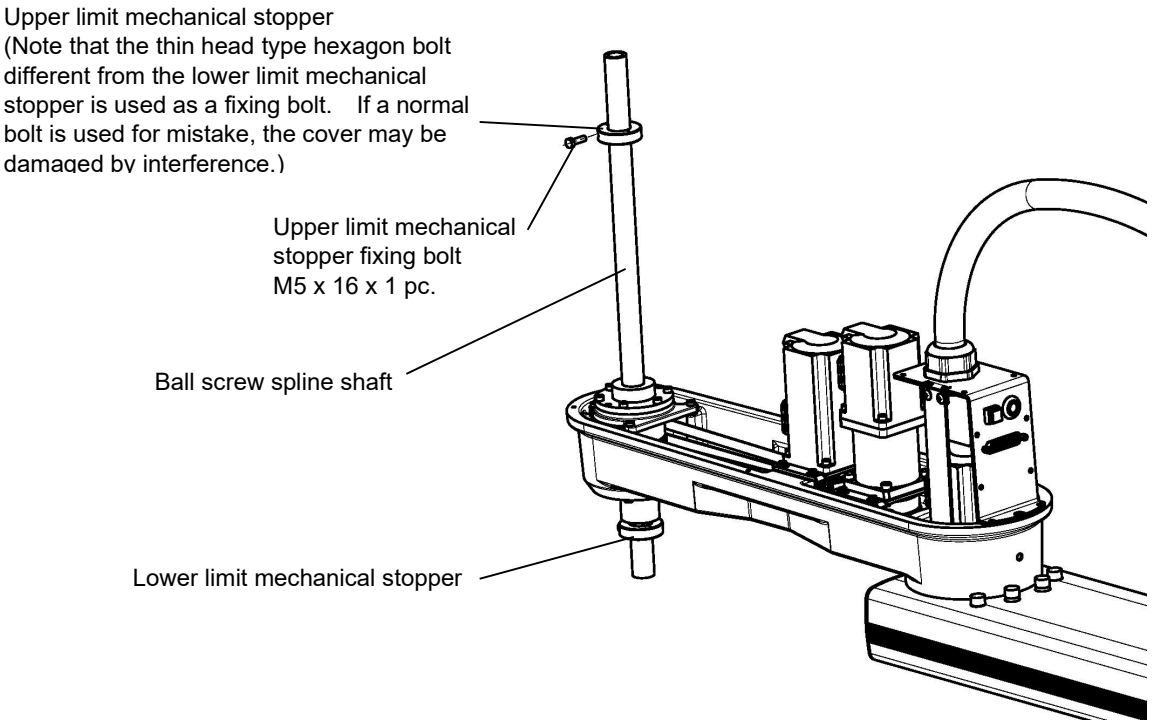


Fig. 11.7 Checking Software Limit Change (THL800, THL900 and THL1000)

3) When the mechanical stoppers are changed, be sure to also change the software limits. Regarding how to change software limits, see Para. 9.2.4 and Fig. 11.10. After changing the software limit, while pressing the axis 3 brake release switch, move Axis 3 manually, and make sure that the software limit is correctly set.

Change the fixing position of the mechanical stopper.
Example: When the upper limit mechanical stopper was moved 75 mm downward.

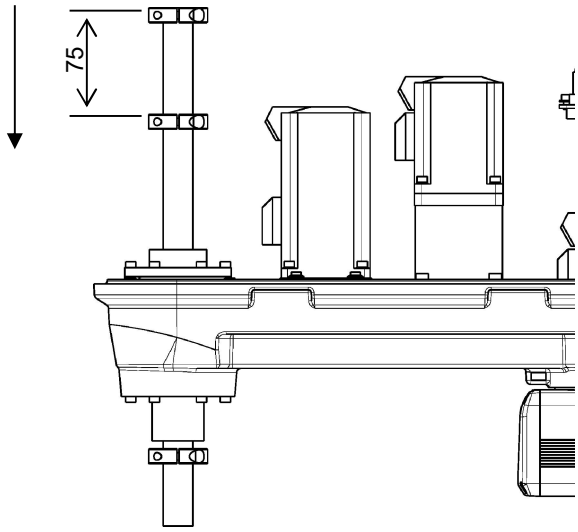


Fig. 11.8 Changing the Operating Range (THL800, THL900 and THL1000)

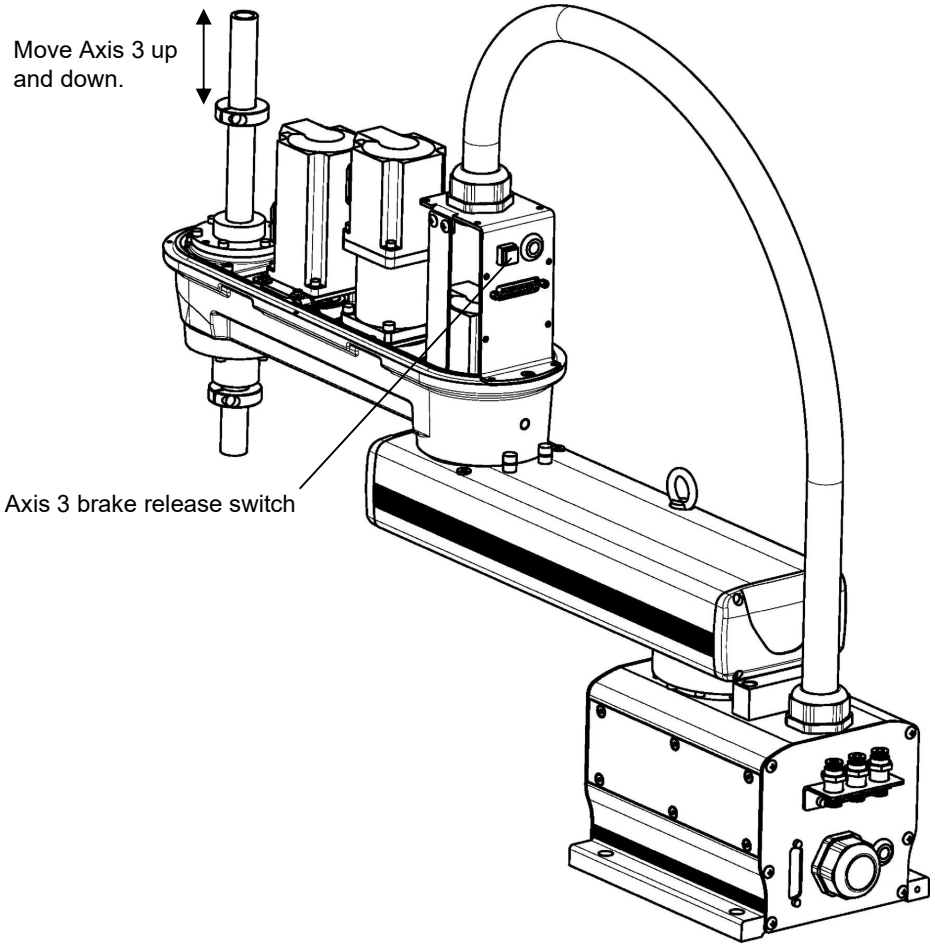


Fig. 11.9 Checking Software Limit Change (THL800, THL900 and THL1000)

Fig. 11.10 shows the settings of the operating range when the upper limit mechanical stopper was moved 75 mm downward.

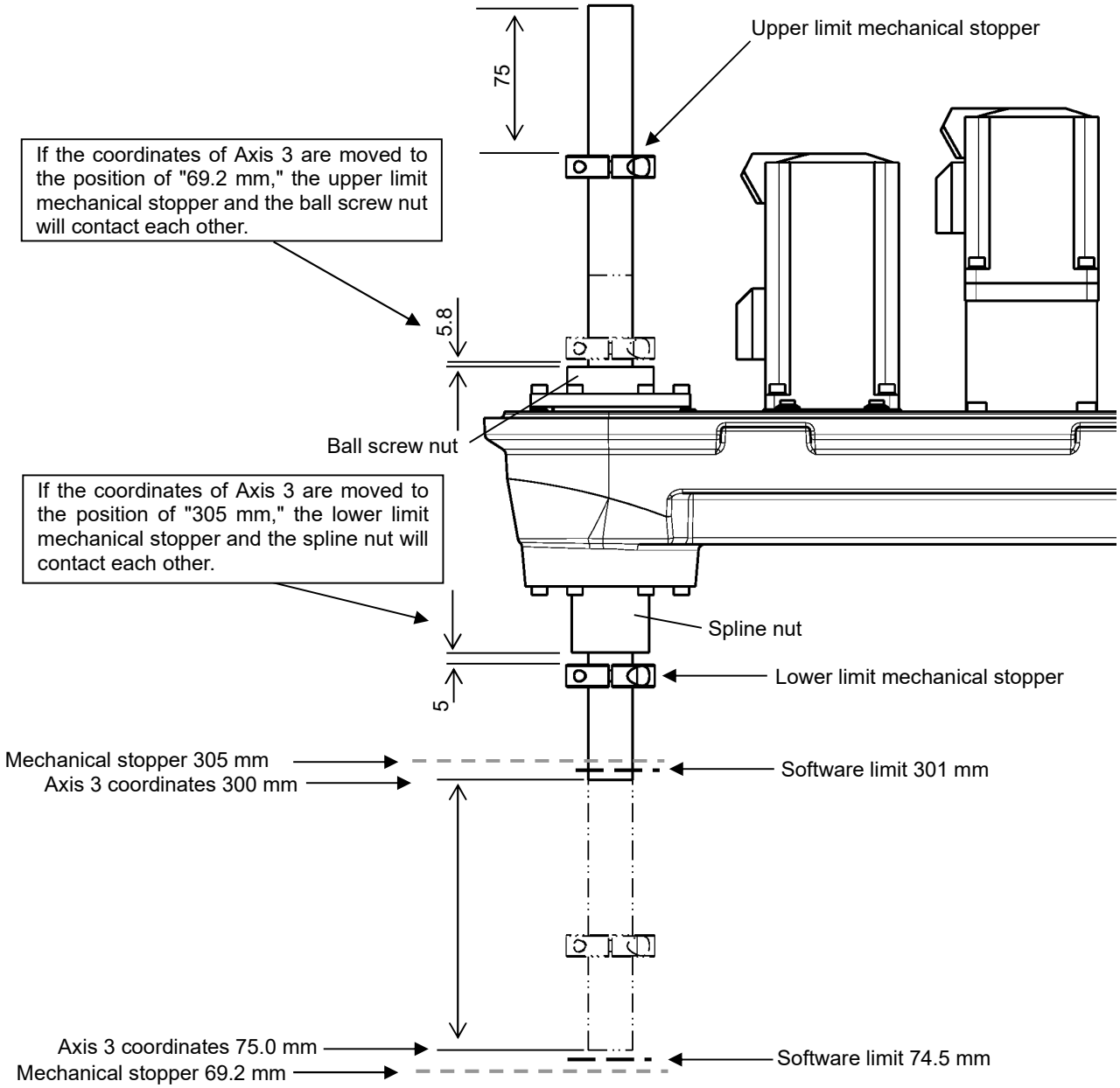


Fig. 11.10 Changing the Operating Range of Axis 3 (THL800, THL900 and THL1000)

4) Fasten the Arm 2 cover to Arm 2 and the harness guide with 14 hexagon socket head bolts (M3 x 6 x 4 pcs., M4 x 6 x 8 pcs. and M4 x 10 x 2 pcs.) and nylon washers. (Application of Loctite during installation is not necessary.)

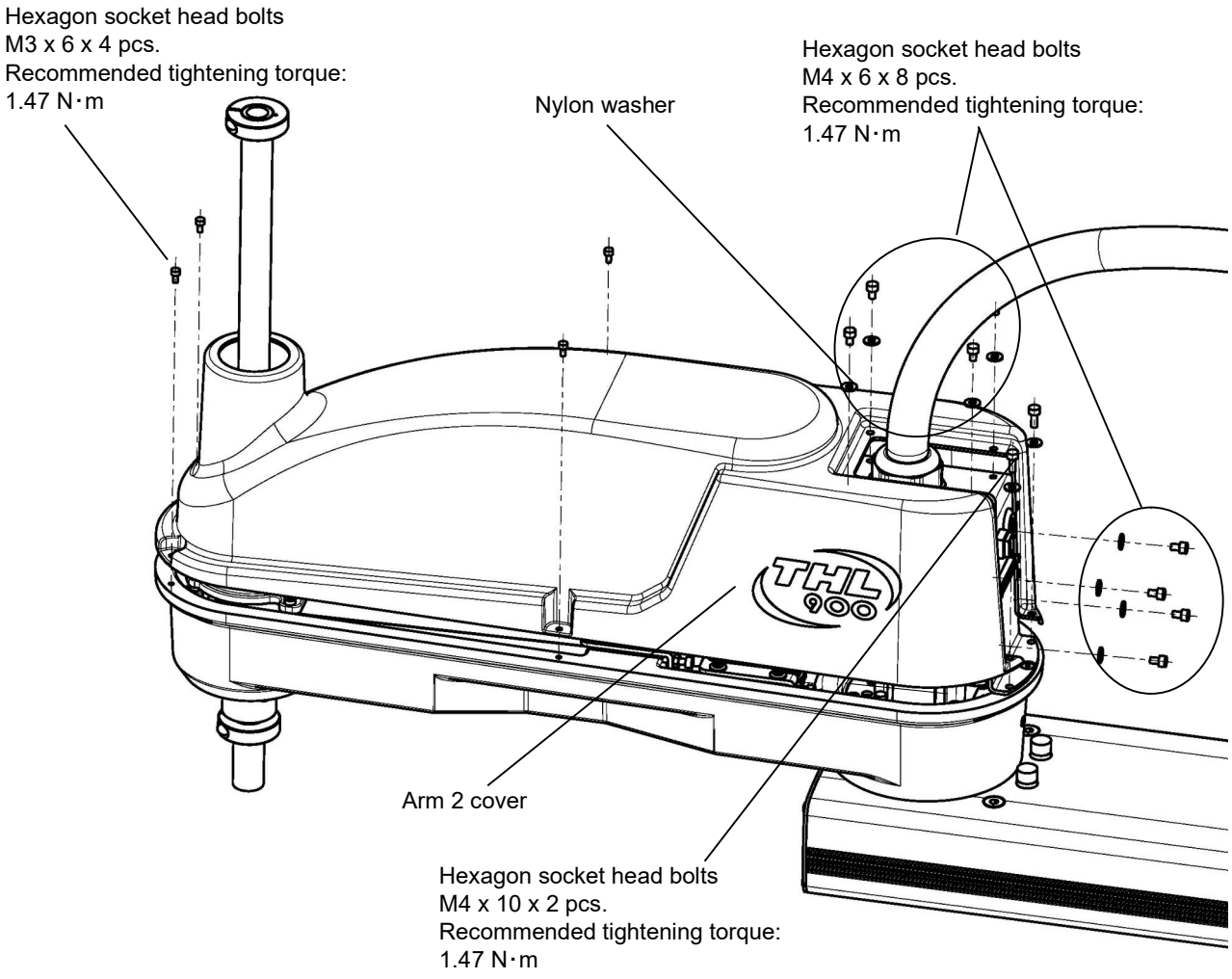


Fig. 11.11 Removing the Arm 2 Cover (THL800, THL900 and THL1000)

After installation, manually move the ball screw spline shaft up and down while holding down the brake release switch, and check that the holes for the ball screws of the Arm 2 cover and the mechanical stoppers are not interfering with each other.

11.2.4 Changing Software Limits

For the software limit change procedure, see "9.2.4 Changing Software Limits".

11.2.5 Coordinate System

The robot's joint angle origin (0° or 0 mm position) is factory-calibrated according to the base reference planes. Fig. 11.12 shows the base coordinate system and origin of each axis joint angle. The coordinate system is common among the THL800, THL900, and THL1000.

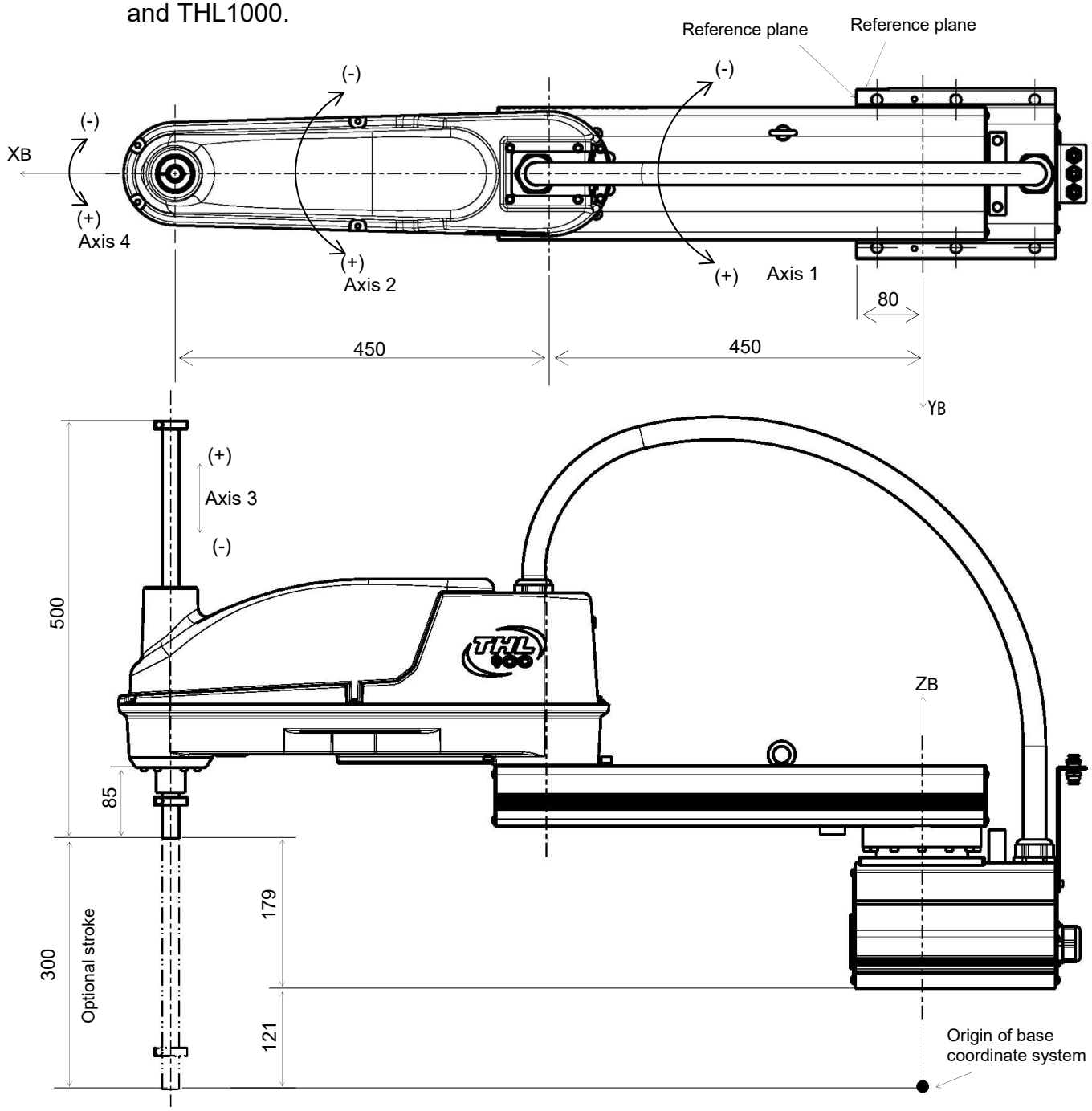


Fig. 11.12 Base coordinate system and joint angle origin (THL900)

11.2.6 Installing the Robot

The robot is secured, using the mounting holes on the base (six places). Use M12 hexagon socket head cap screws.
Table 11.2 lists the loads applied to the frame during horizontal operation, and Fig. 11.13 shows how to install the robot. Reference planes are provided on the base unit. To align the robot position in the base coordinate system, or to replace the robot, provide adequate reference planes. Then, contact such reference planes to the base reference planes and secure the robot. Further, a pin hole is provided. This hole can be used for positioning.

! CAUTION

- The robot will suddenly accelerate and decelerate during operation. When installing it on a frame, make sure that the frame has sufficient strength and rigidity.
  If the robot is installed on a frame that does not have sufficient rigidity, vibration will occur while the robot is operating, and could lead to faults.
  When installing the robot on the floor, secure the robot with anchor bolts, etc.
- Install the robot on a level place. Failure to do so could lead to a drop in performance or faults.

Table 11.2 Load applied to the frame during horizontal operation (THL800, THL900 and THL1000)

Model	Loads applied to the frame during horizontal operation [Nm]	Robot main body mass [kg]
THL800	550	33
THL900	550	35
THL1000	550	37

* These values are reference values. When designing a frame, take the safety factor into consideration.

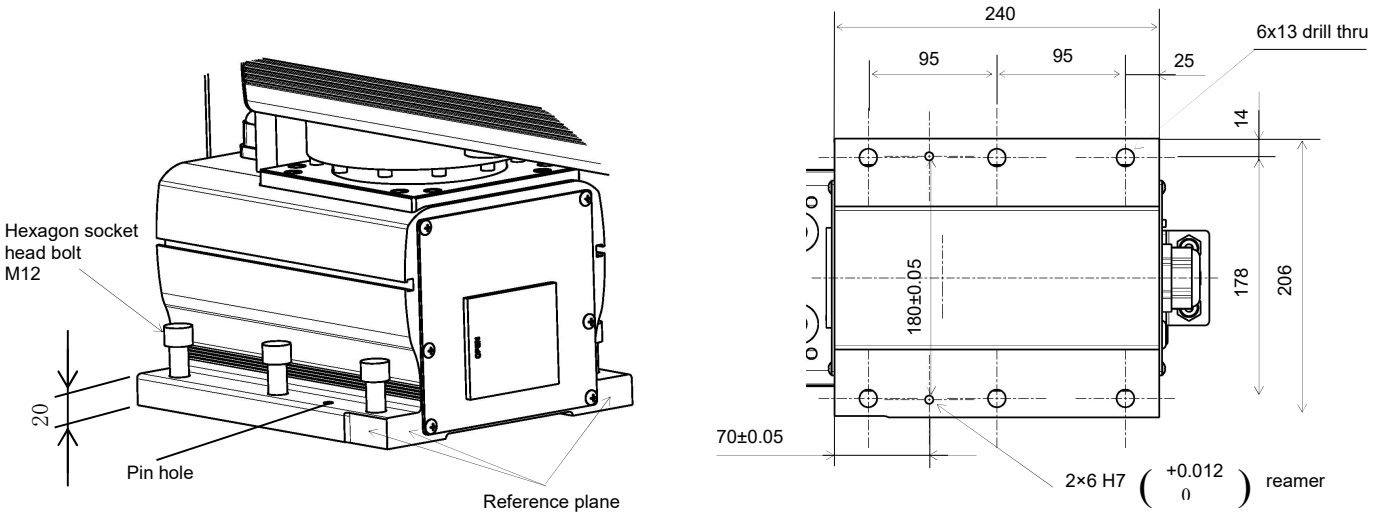


Fig. 11.13 Installation method (THL800, THL900 and THL1000)

11.3 Precautions for Handling the Teach Pendant

For the teach pendant handing procedure, see "9.3 Precautions for Handling the Teach Pendant".

11.4 Safety Measures

For safety measures, see "9.4 Safety Measures".

12. Robot Installation (THL1200)

12.1 Installation Environment

For mounting environment, see "9.1 Installation Environment".

12.2 Installation

Before installing the robot, you should plan a layout, fully considering the working envelope (or operating range), coordinate system and space for maintenance.

12.2.1 External Dimensions and Working Envelope

Figs. 12.1 show the external view drawing and working envelope of the robot.
Each axis can operate within the working envelope. To prevent the robot from moving out of the working envelope by misoperation, the robot is equipped with mechanical stoppers outside the working envelope. Additionally, soft limits that can be set by the user are provided. For further information, see the “Instruction Manual: User Parameters” provided separately.

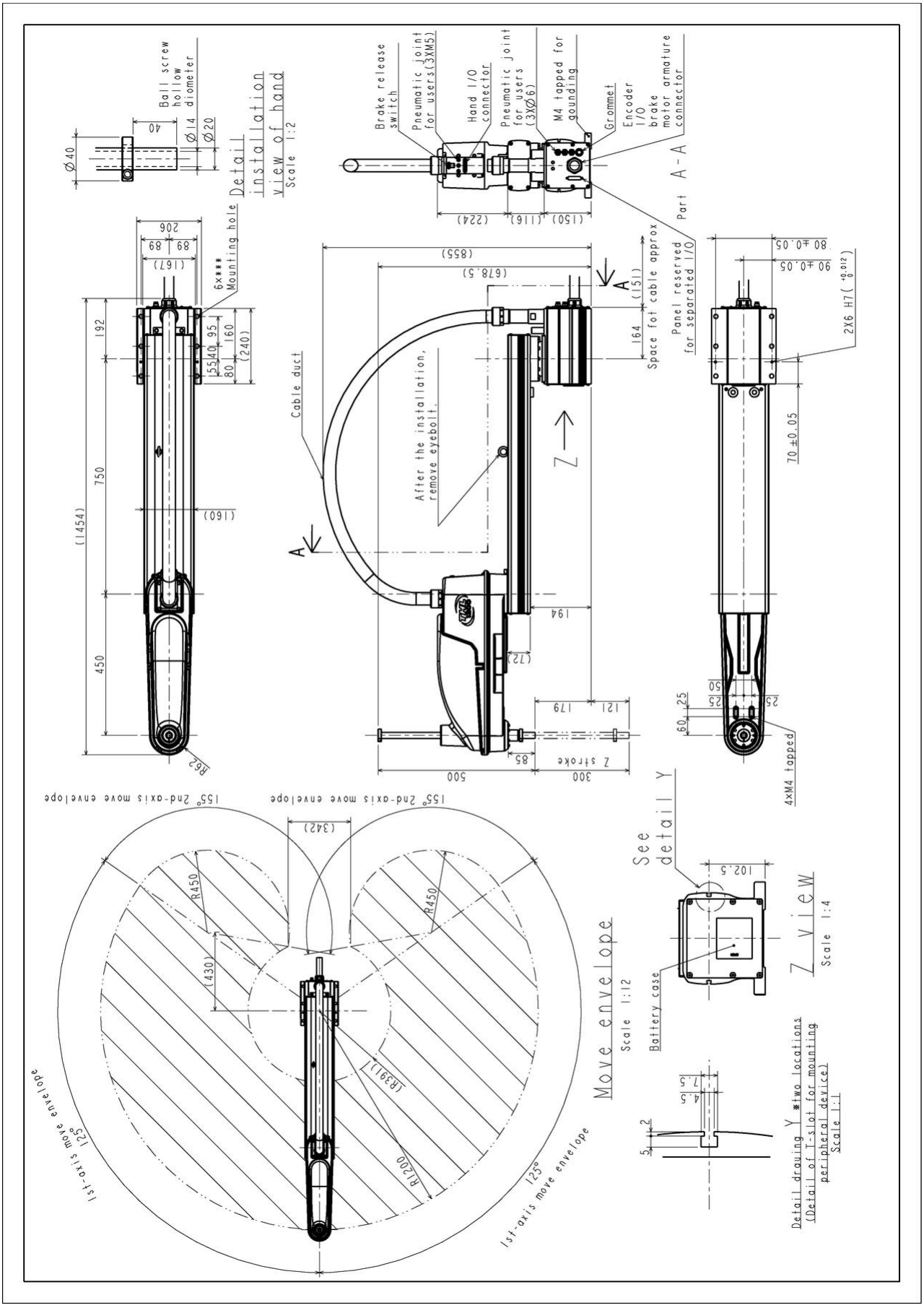


Fig. 12.3 External view and working envelope (THL1200)

12.2.2 Changing the Operating Ranges of Axes 1 and 2

Mechanical stoppers are installed to each robot to mechanically control the operation range of each axis. Changing the mechanical operation range of a robot by modifying these mechanical stoppers is referred to as "change in operating range."
Here, how to change the operating ranges of Axes 1 and 2 of a robot is explained.

Regarding how to change the operating range of Axis 3, Para. 12.2.3.
Note that Axis 4 is different from other operating axes; it restricts the operating range with software limits only, not with mechanical stoppers. Therefore, regarding how to change the operating range of Axis 4, see Para. 9.2.4.

! CAUTION

- To change operating ranges, design and produce mechanical stoppers by referring to this document according to your usage.
- When mechanical stoppers have been changed and then operating ranges have been changed, be sure to also change software limits to prevent contact with the mechanical stoppers while operating a robot.
- Mechanical stoppers do not securely restrict the movable ranges of robots.

  When the power of a robot is turned on, never enter the operating range of the robot.
- If a robot collides with a mechanical stopper, the robot detects the collision and stops, but the mechanical stopper may be damaged. Avoid reusing the mechanical stopper.
- The mechanical stopper reference drawings shown in this document do not fully satisfy the customer's usage.

  Design, produce and install mechanical stoppers according to your usage, such as the operating range.
- The failures of a robot caused by mechanical stoppers will be excluded from the warranty coverage.

As shown in Fig. 12.2, the operation range can be altered by changing the position of the mechanical stopper.

Table 12.1 Operation ranges before and after change (THL1200)

		Before change	After change(1)	After change(2)
Axis 1 operation range	+ direction	125°	110°	
	- direction	125°	110°	
Axis 2 operation range	+ direction	155°	145°	120°
	- direction	155°	145°	120°

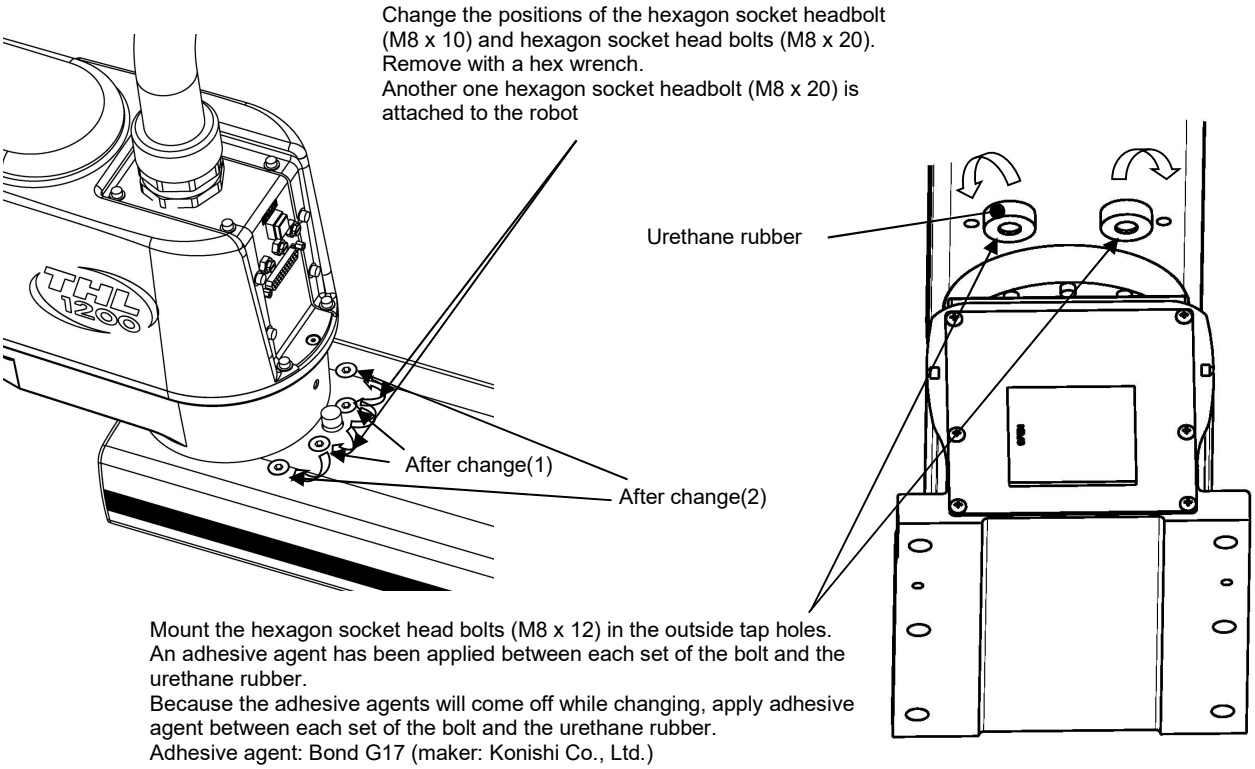


Fig. 12.2 Changing the Operating Ranges of Axes 1 and 2 (THL1200)

If the operation range is changed, it is necessary to change the user parameter. For the software limit change procedure, see "9.2.4 Changing Software Limits".

12.2.3 Changing the Operating Range of Axis 3

At the factory shipment of a robot, software limits and mechanical stoppers are preset so that the Z stroke of Axis 3 is between 0 and 300 mm. Fig. 11.5 shows the settings of operating range at the factory shipment.

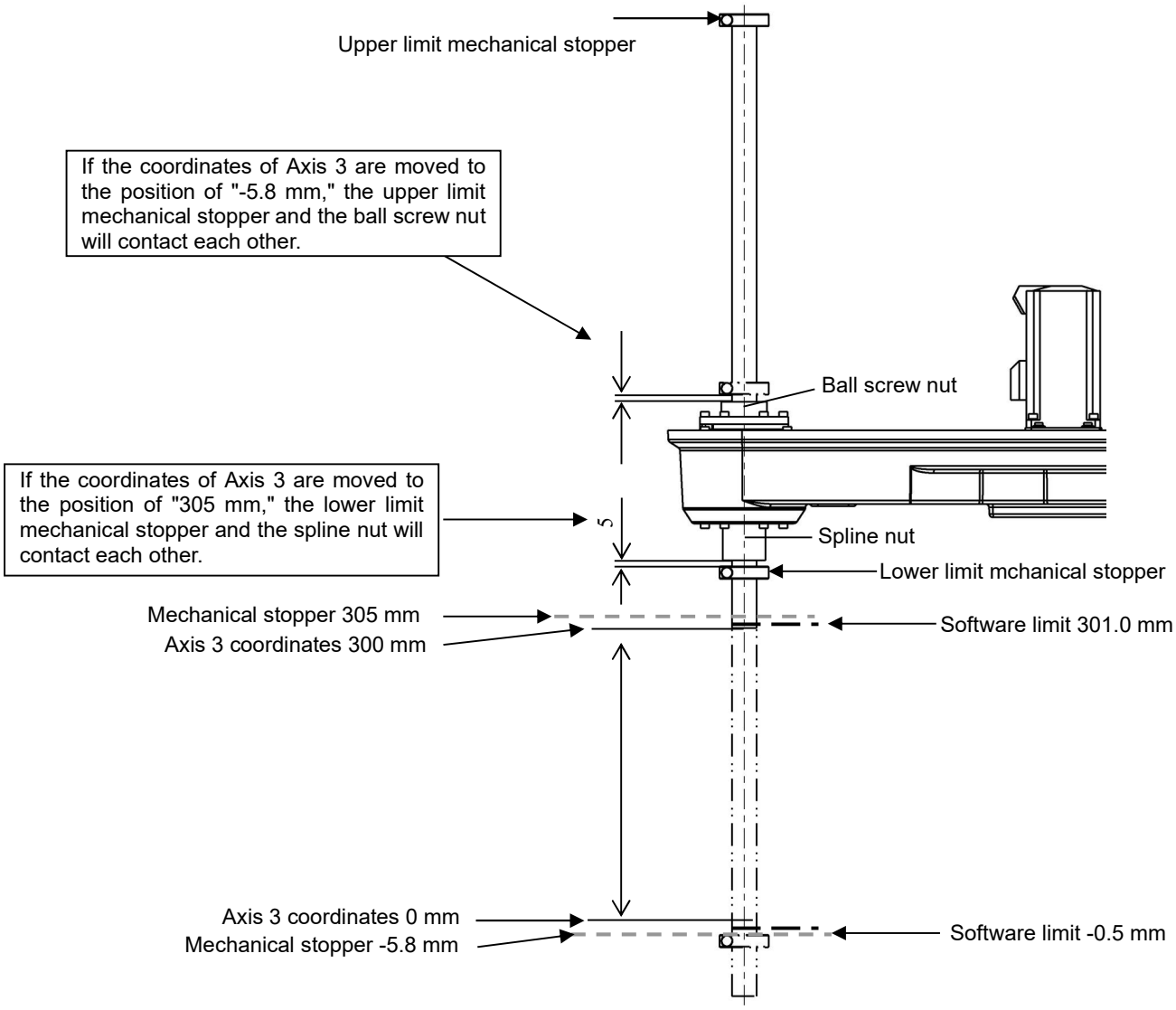


Fig. 12.3 Changing the Operating Range of Axis 3 (THL1200)

1) Remove the Arm 2 cover. The cover is fastened to Arm 2 and the harness guide with 14 hexagon socket head bolts (M3 x 6 x 4 pcs., M4 x 6 x 8 pcs. and M4 x 10 x 2 pcs.)

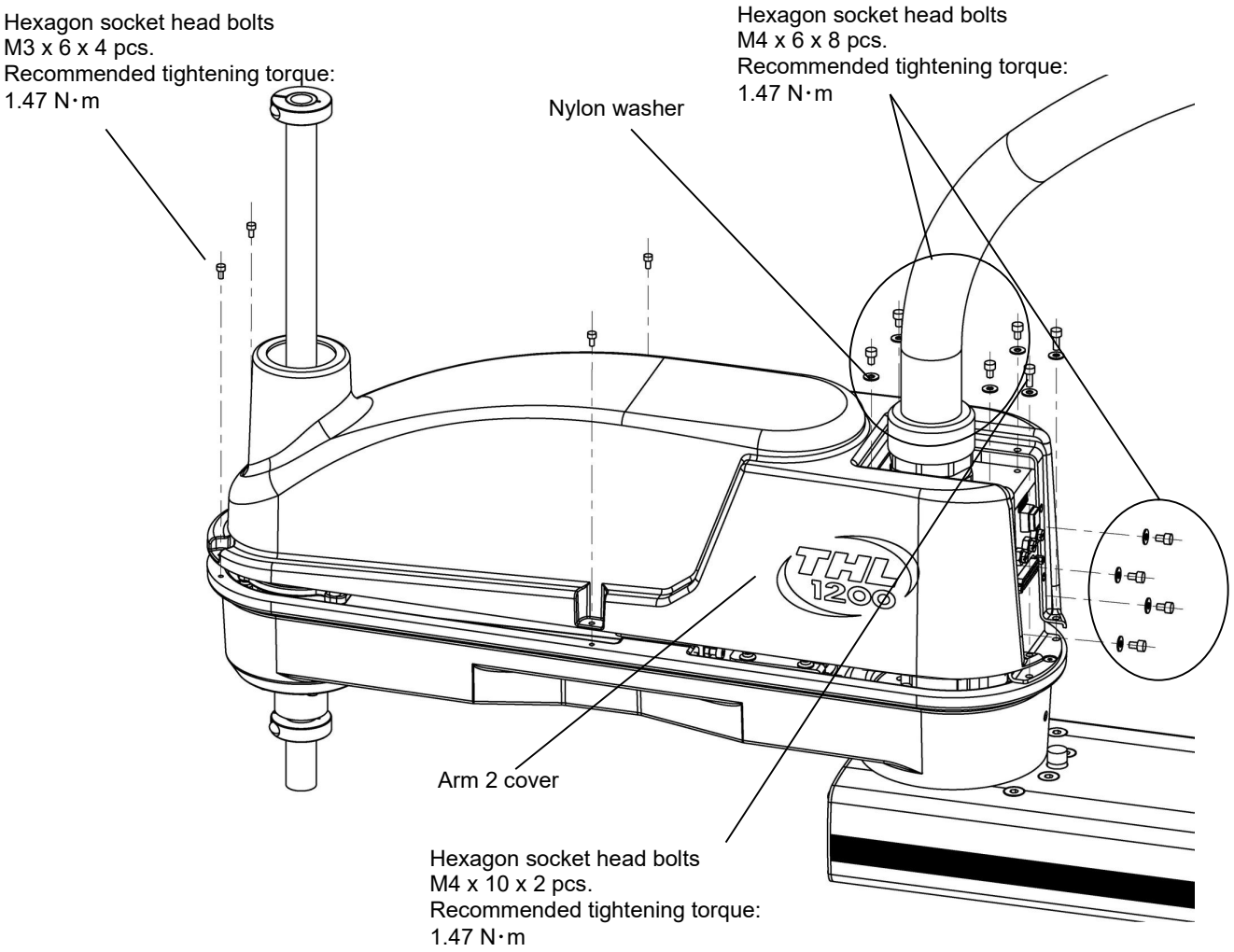


Fig. 12.4 Removing the Arm 2 Cover (THL1200)

2) Loosen the fixing bolts of the mechanical stoppers, move the mechanical stoppers to a desired position, and fix them again. When fixing the mechanical stoppers, be sure to apply Loctite to the fixing bolts.

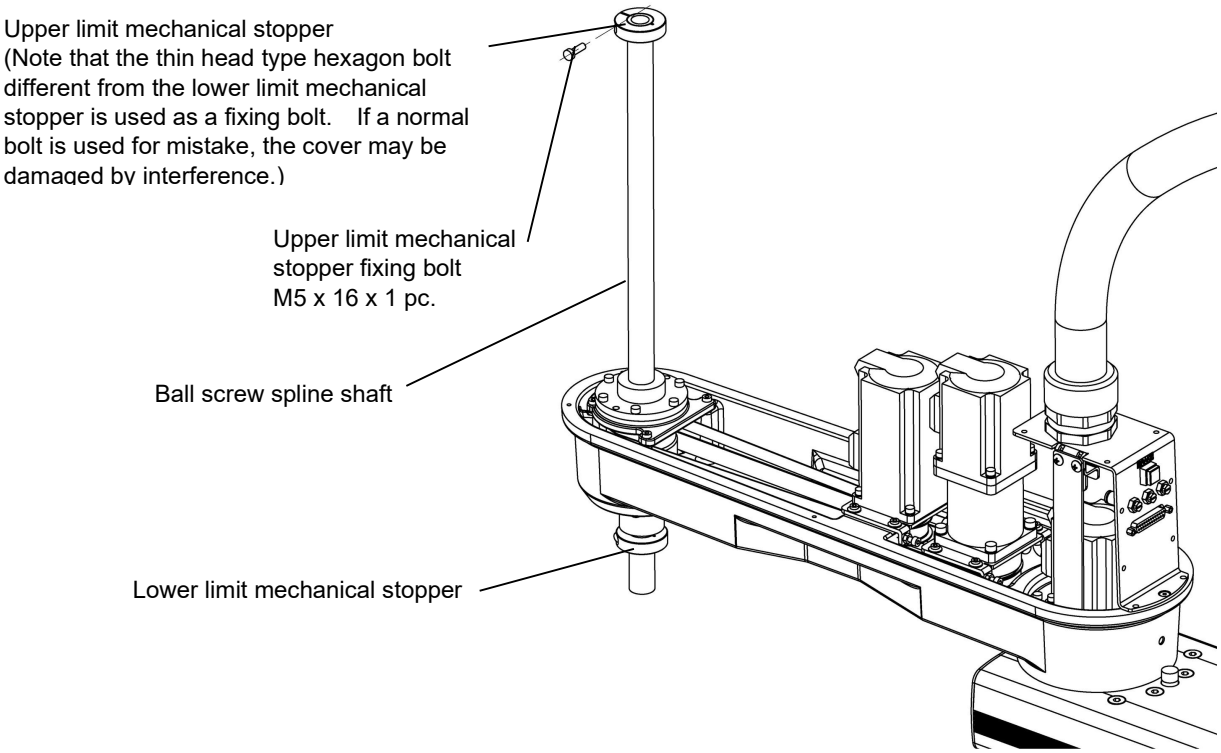


Fig. 12.5 Checking Software Limit Change (THL1200)

3) When the mechanical stoppers are changed, be sure to also change the software limits. Regarding how to change software limits, see Para. 9.2.4 and Fig. 12.10. After changing the software limit, while pressing the axis 3 brake release switch, move Axis 3 manually, and make sure that the software limit is correctly set.

Change the fixing position of the mechanical stopper.
Example: When the upper limit mechanical stopper was moved 75 mm downward.

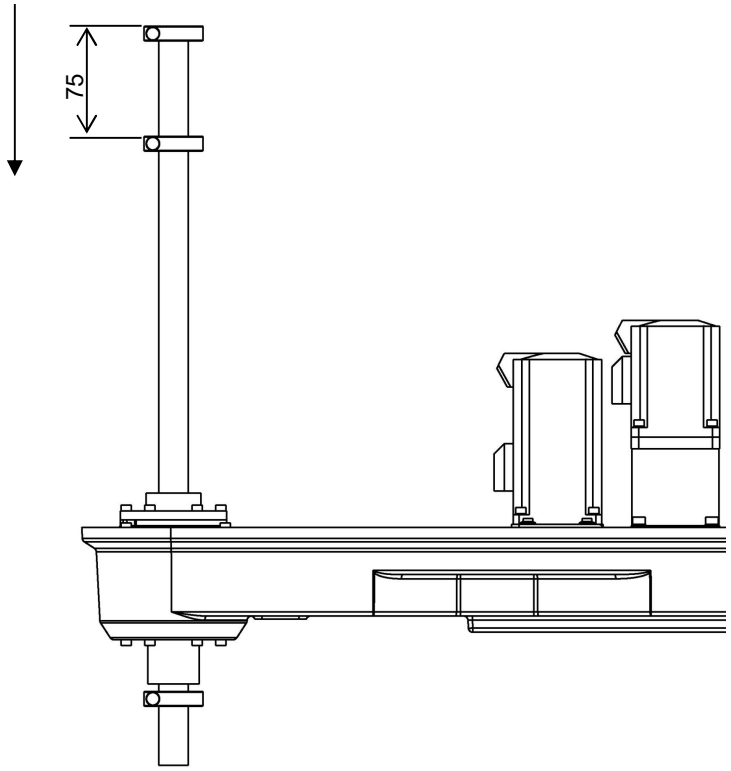


Fig. 12.6 Changing the Operating Range (THL1200)

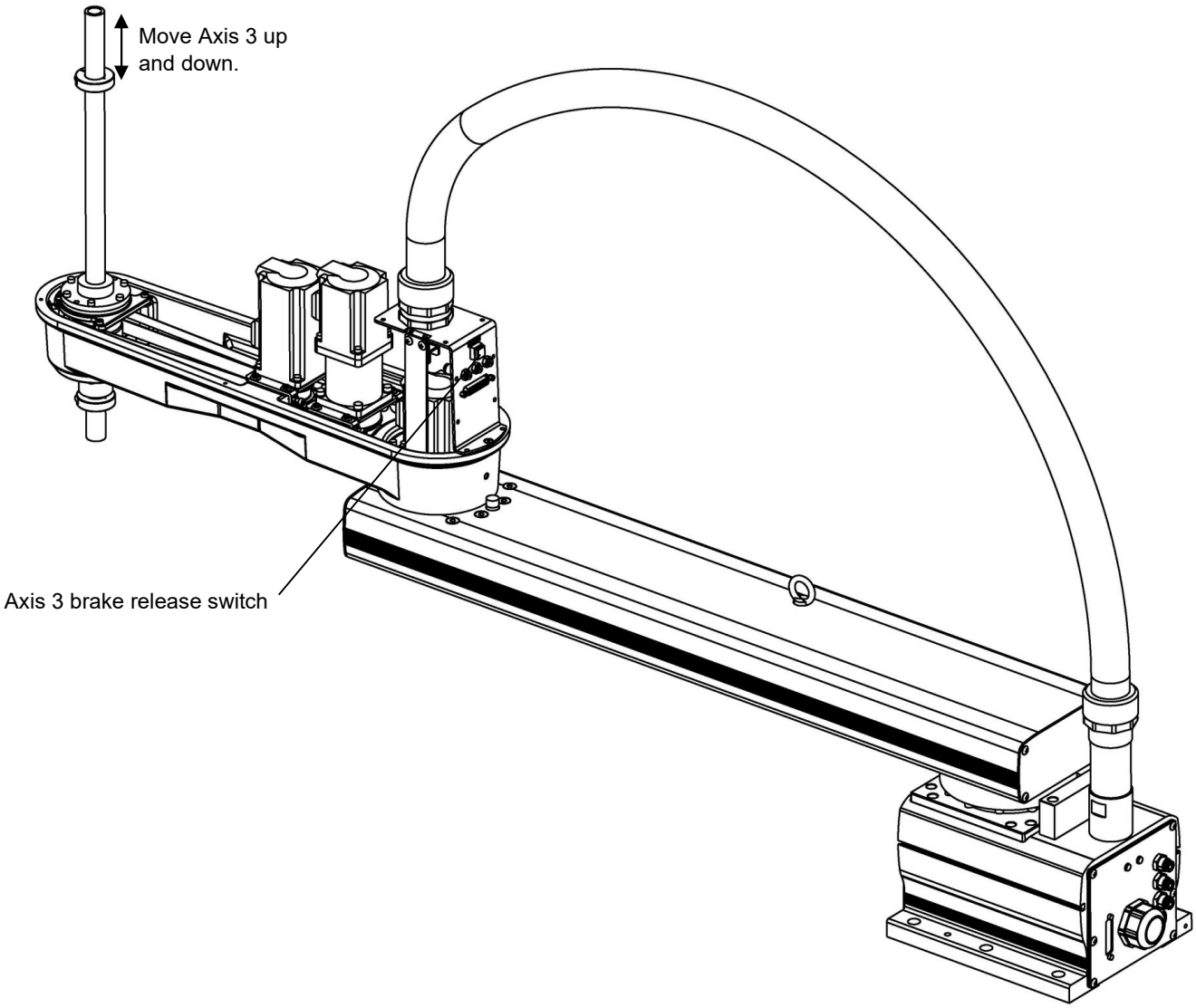


Fig. 12.7 Checking Software Limit Change (THL1200)

Fig. 12.10 shows the settings of the operating range when the upper limit mechanical stopper was moved 75 mm downward.

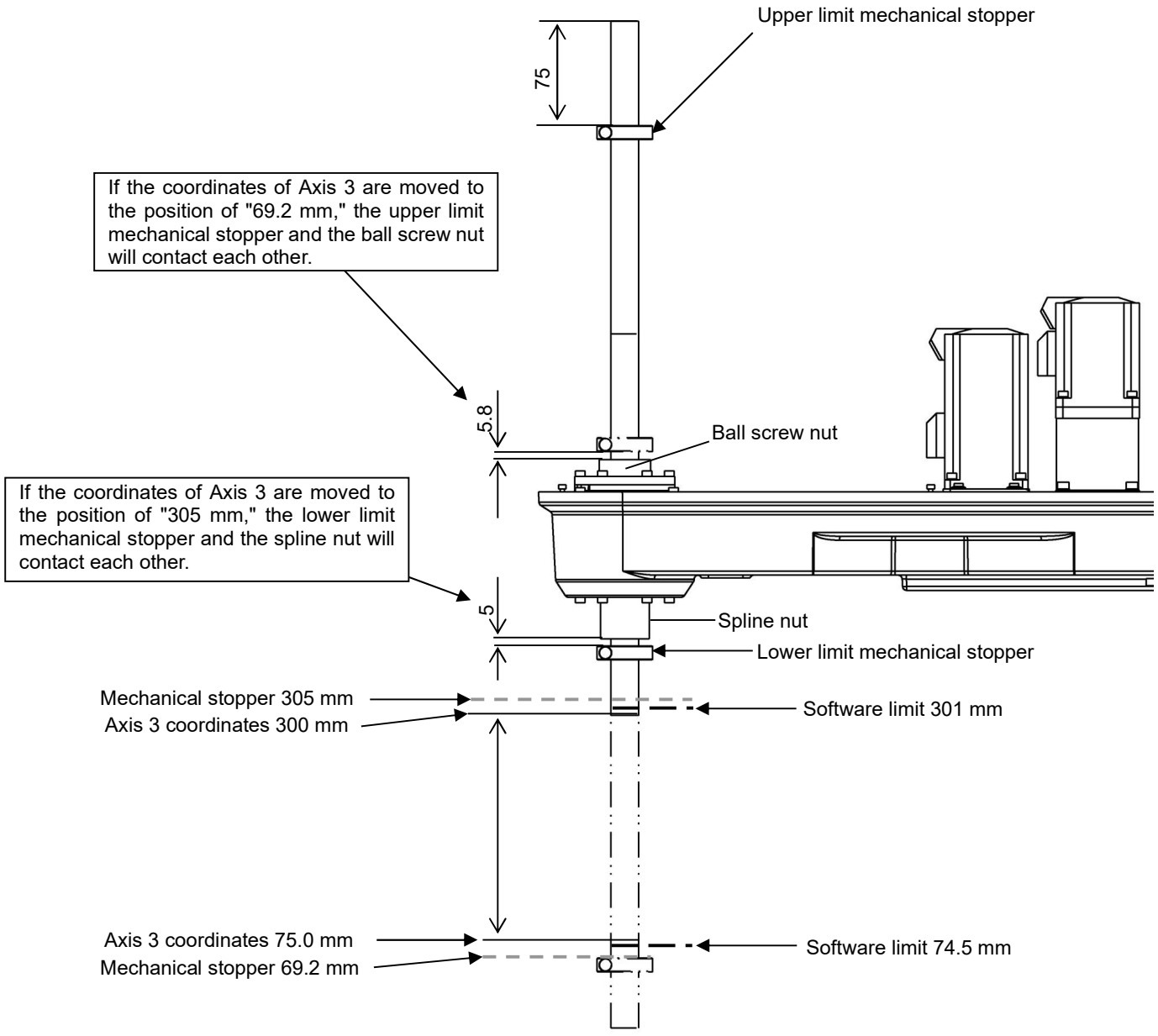


Fig. 12.8 Changing the Operating Range of Axis 3 (THL1200)

4) Fasten the Arm 2 cover to Arm 2 and the harness guide with 14 hexagon socket head bolts (M3 x 6 x 4 pcs., M4 x 6 x 8 pcs. and M4 x 10 x 2 pcs.) and nylon washers. (Application of Loctite during installation is not necessary.)

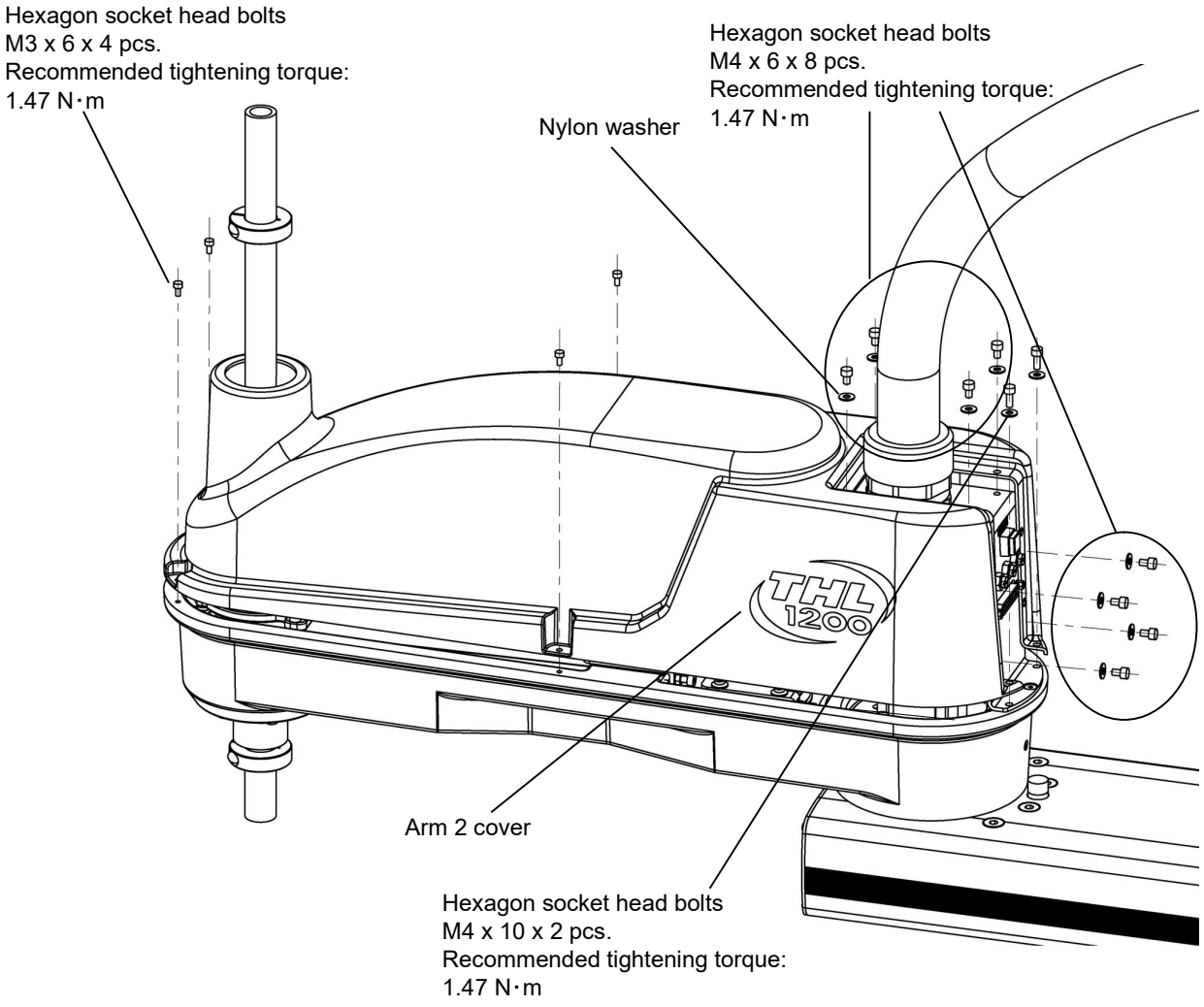


Fig. 12.9 Removing the Arm 2 Cover (THL1200)

After installation, manually move the ball screw spline shaft up and down while holding down the brake release switch, and check that the holes for the ball screws of the Arm 2 cover and the mechanical stoppers are not interfering with each other.

12.2.4 Changing Software Limits

For the software limit change procedure, see "9.2.4 Changing Software Limits".

12.2.5 Coordinate System

The robot's joint angle origin (0° or 0 mm position) is factory-calibrated according to the base reference planes. Fig. 12.12 shows the base coordinate system and origin of each axis joint angle. The coordinate system is common among the THL1200. Figure shows the THL1200.

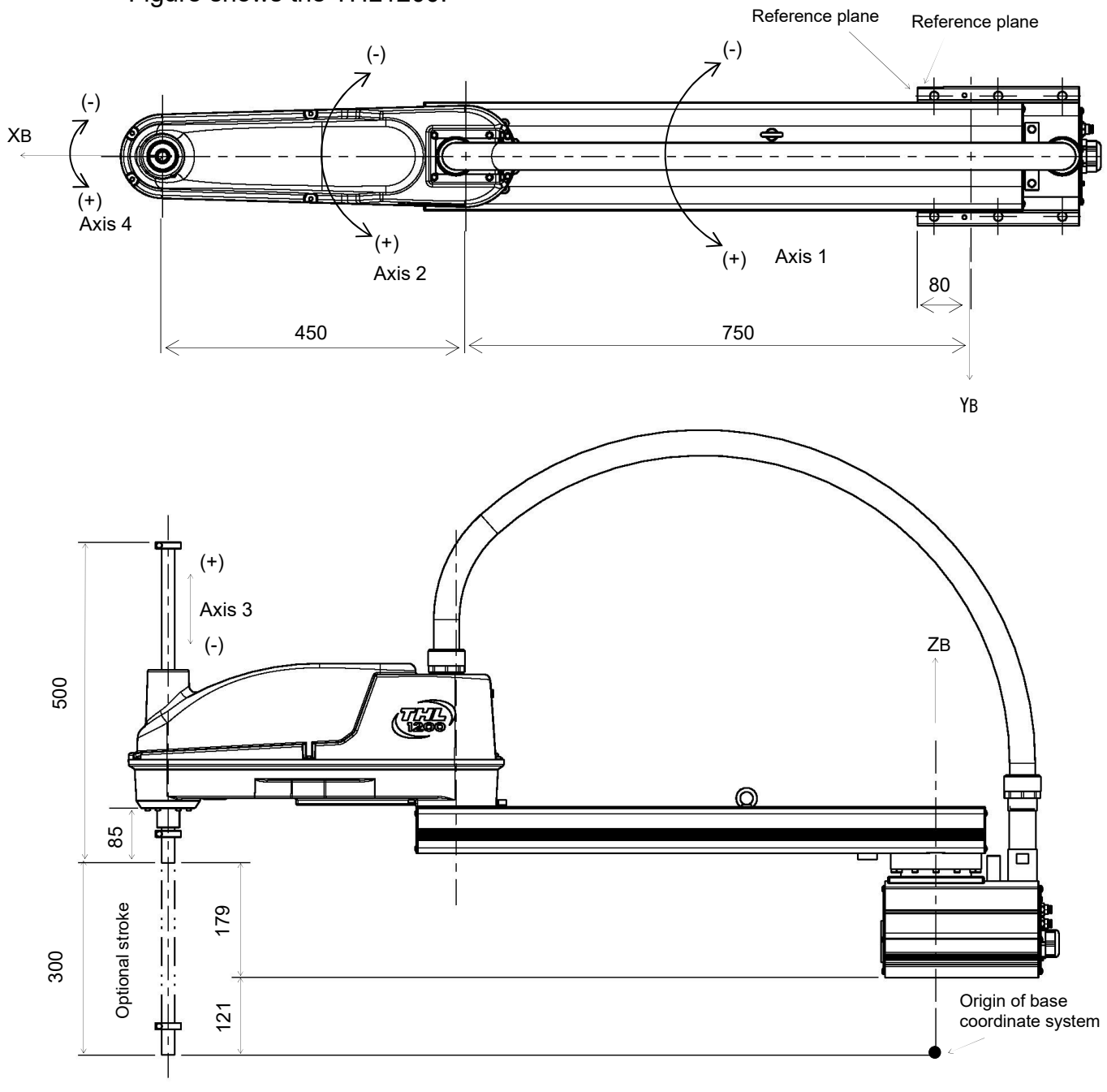


Fig. 12.10 Base coordinate system and joint angle origin (THL1200)

12.2.6 Installing the Robot

The robot is secured, using the mounting holes on the base (six places). Use M12 hexagon socket head cap screws.
Table 12.2 lists the loads applied to the frame during horizontal operation, and Fig. 12.11 shows how to install the robot. Reference planes are provided on the base unit. To align the robot position in the base coordinate system, or to replace the robot, provide adequate reference planes. Then, contact such reference planes to the base reference planes and secure the robot. Further, a pin hole is provided. This hole can be used for positioning.

! CAUTION

- The robot will suddenly accelerate and decelerate during operation. When installing it on a frame, make sure that the frame has sufficient strength and rigidity.
  If the robot is installed on a frame that does not have sufficient rigidity, vibration will occur while the robot is operating, and could lead to faults.
  When installing the robot on the floor, secure the robot with anchor bolts, etc.
- Install the robot on a level place. Failure to do so could lead to a drop in performance or faults.

Table 12.2 Load applied to the frame during horizontal operation (THL1200)

Model	Loads applied to the frame during horizontal operation [Nm]	Robot main body mass [kg]
THL1000	550	40

* These values are reference values. When designing a frame, take the safety factor into consideration.

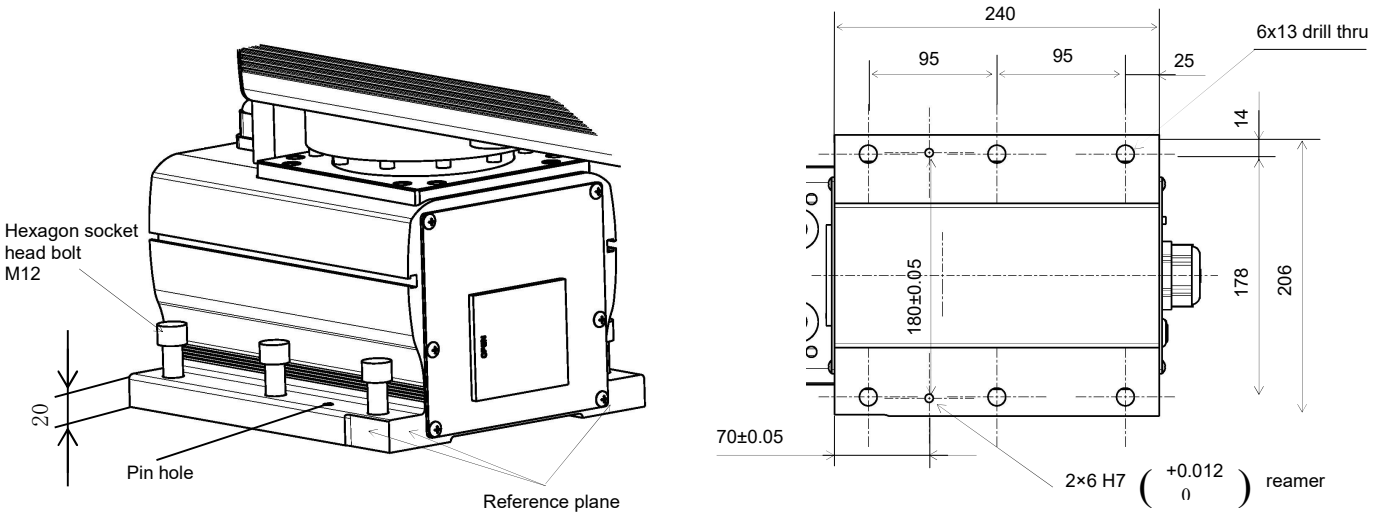


Fig. 12.11 Installation method (THL1200)

12.3 Precautions for Handling the Teach Pendant

For the teach pendant handing procedure, see "9.3 Precautions for Handling the Teach Pendant".

12.4 Safety Measures

For safety measures, see "9.4 Safety Measures".

13. Installing the Controller (TSL3000)

13.1 External Dimensions

For information on TSL3000E, refer to Section 14. For information on TS3000 and TS3000E, refer to Section 15. External view of the controller TSL3000 is shown in Fig. 13.1.

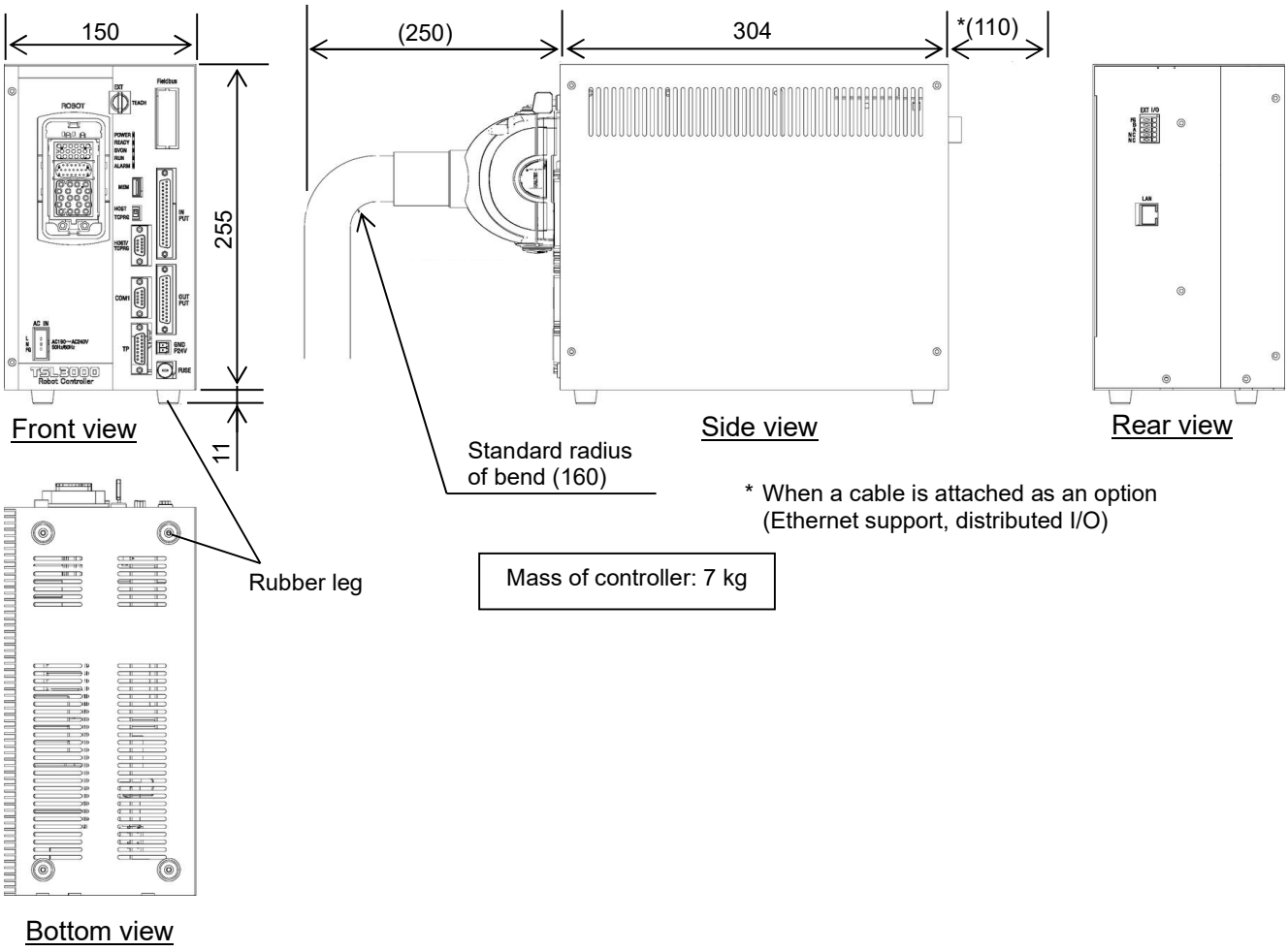


Fig. 13.1 External view of controller (TSL3000)

13.2 Precautions for Direct Installation

It is necessary to provide a clearance of 50 mm or more in the horizontal direction and a clearance of 100 mm or more in the upward direction near the controller.

CAUTION

- Provide a ventilation space at the side of the controller so that the air vent holes are not blocked. The space equal to the length of the rubber legs should be kept below the lower surface. Otherwise, the cooling capacity will drop, causing controller failure.
- DO NOT stack the controllers.
- DO NOT place any object on top of the controller.

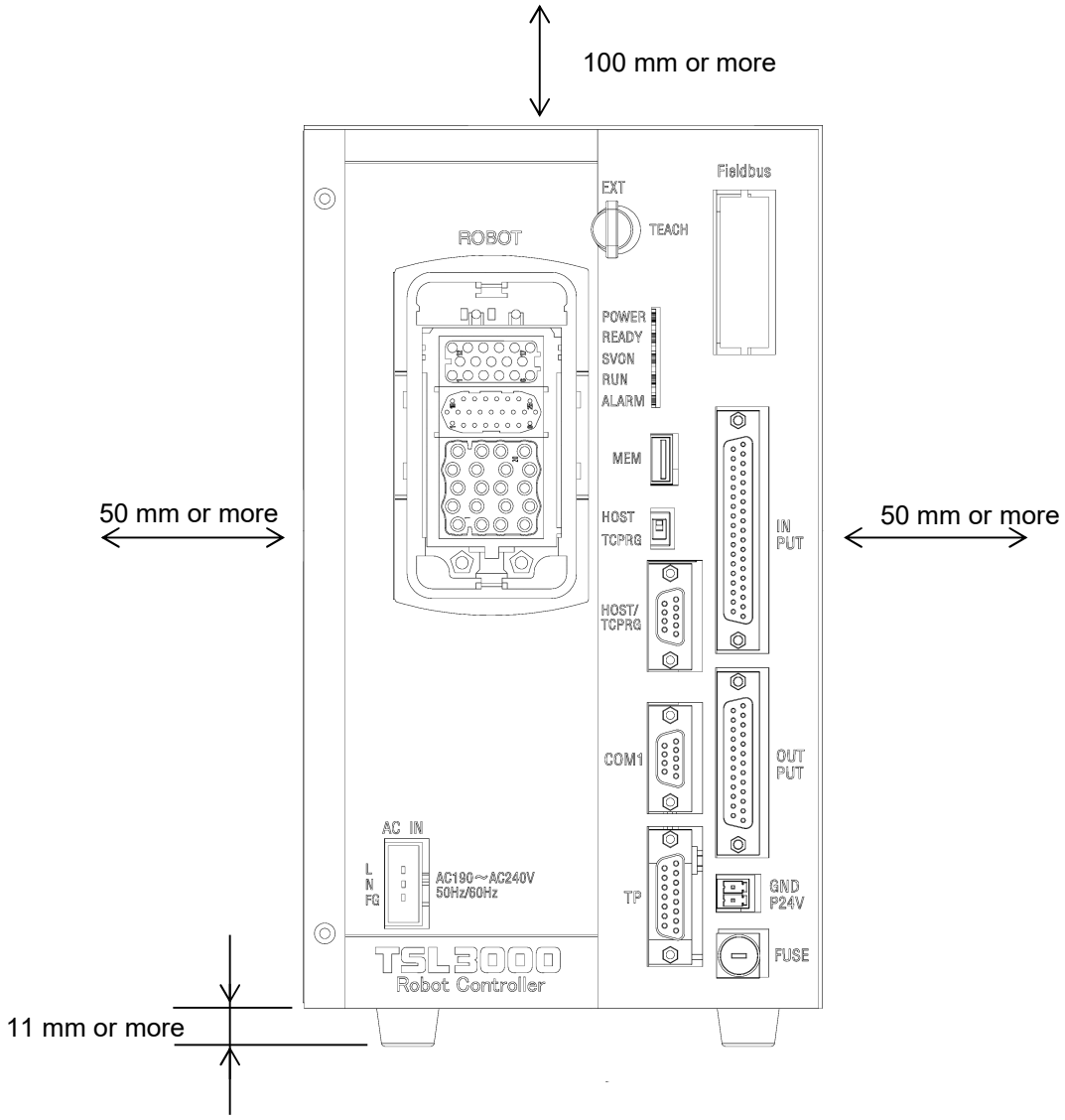


Fig. 13.2 Controller ventilation space (TSL3000)

13.3 Rack Mounting Dimensions

When installing the robot controller TSL3000 to the control panel, attach the mounting fixtures using the screw holes for mounting the rubber legs on the bottom surface, and then fasten the controller to the control panel.

Note) The mounting fixtures (2 piece set) are optional. The customer is requested to do assembly.

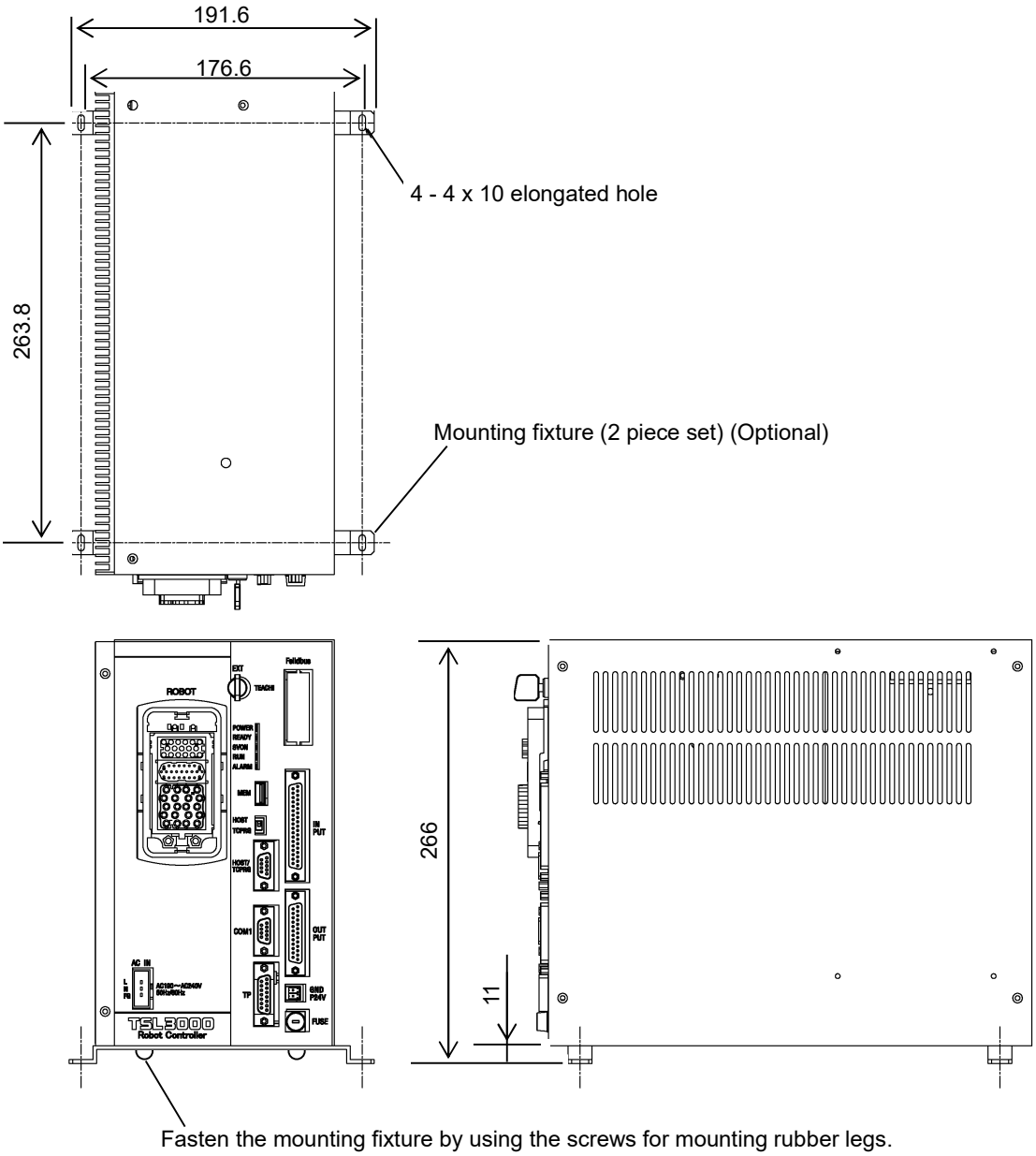


Fig. 13.3 Screw hole dimensions for securing controller (TSL3000)

13.4 Precautions for Assembling the Control Panel

When installing the robot controller TSL3000 to the control panel, be careful with the following items.

a) When installing the robot controller to the control panel, remove the rubber legs on the bottom surface and use the holes for installing the rubber legs to fix the controller in position.

b) If there are options such as Ethernet support and distributed I/O, it is necessary to connect a cable to the rear of the controller, and a space of at least 110 mm is necessary in the rear direction when doing so.

c) To do maintenance of the controller, do it with the upper cover removed. (See Fig. 13.4.)

d) When installing the controller, make sure that the maintenance of the controller can be done without any problem. Especially when storing the controller in the control panel, it will be necessary to take out the controller from the control panel for maintenance.

   Specifically, be careful of the following points.

   1) Arrange the cables around the rear side of the controller (so that the controller can be removed).
   2) Connect all cables in such a manner that the robot can be operated even if the controller is removed from the rack.

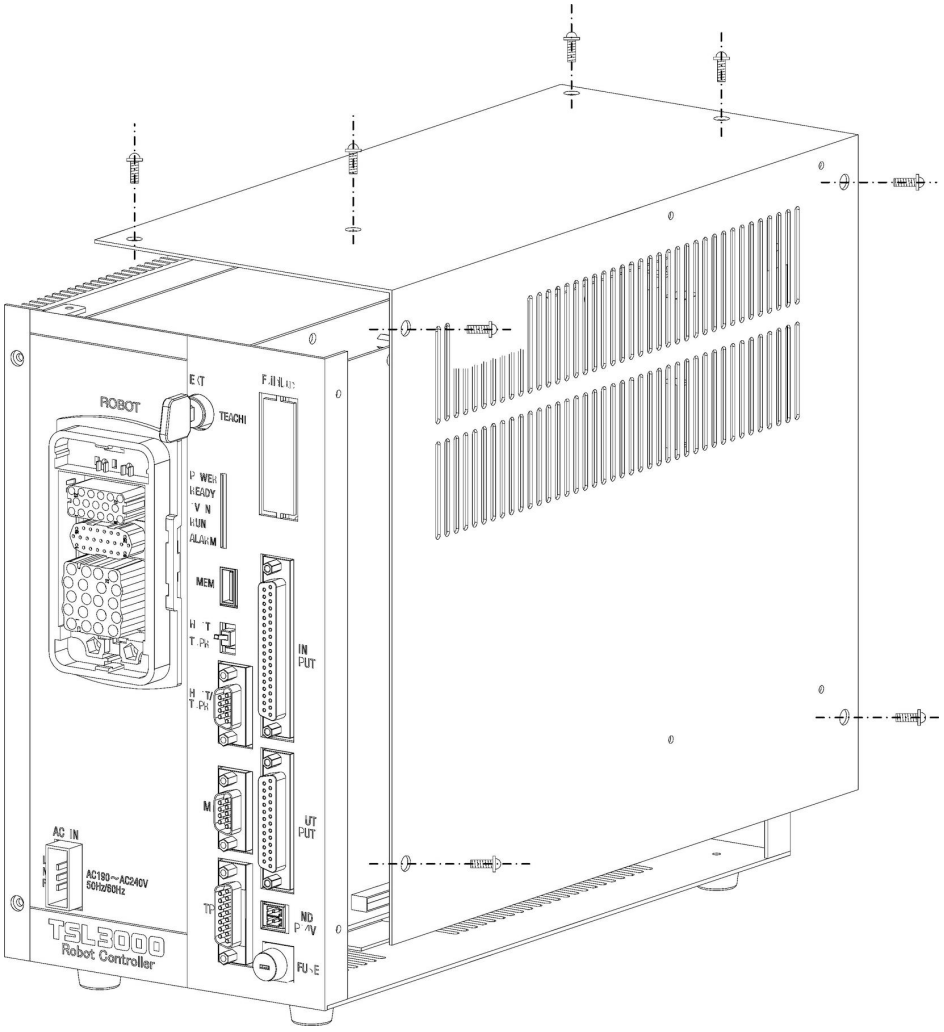


Fig. 13.4 Removing upper cover (TSL3000)

e) Allocate a clearance of about 250 mm in front of the controller for connecting the robot cable connector.

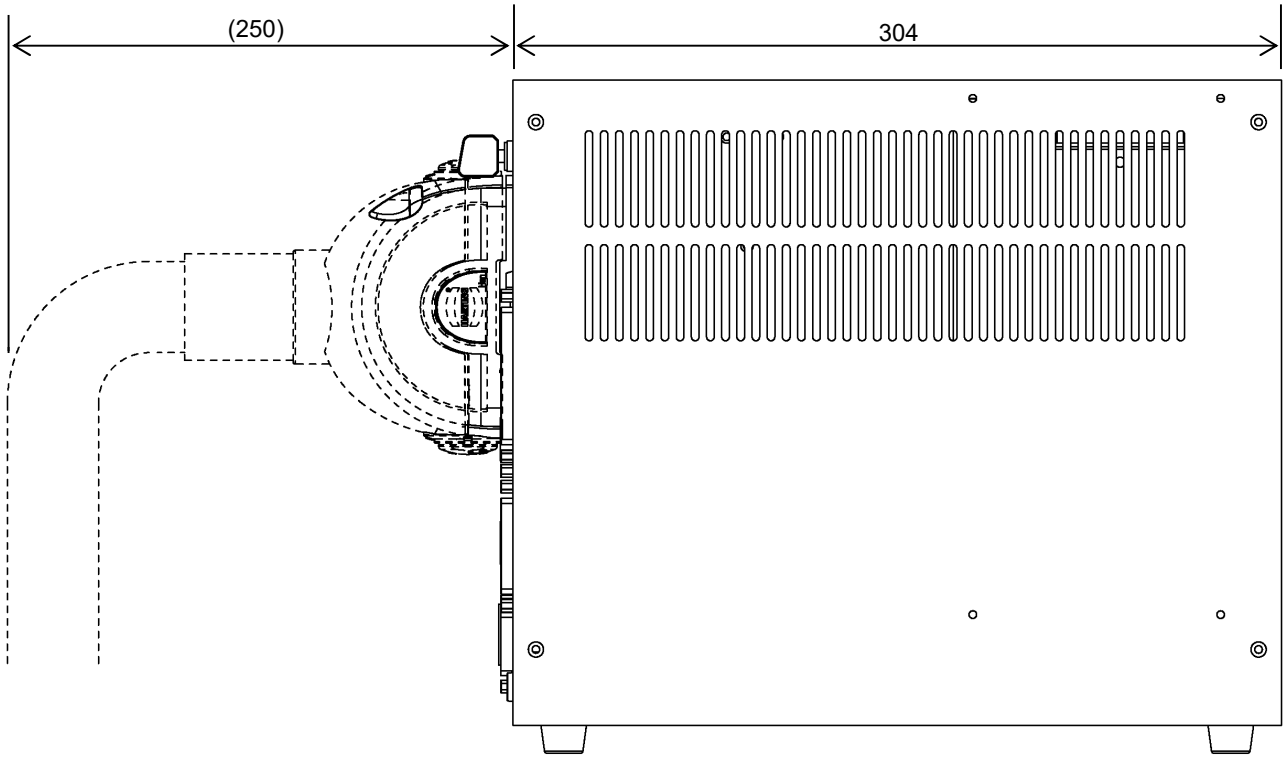


Fig. 13.5 Clearance of controller front side (TSL3000)

CAUTION

- If the control panel is completely sealed up, open air release holes, ventilate the air forcibly with a fan, or conduct indirect cooling to release heat from the control panel. Otherwise, heat will be confined in the control panel and the controller, causing failure.

14. Installing the Controller (TSL3000E)

14.1 External Dimensions

For information on TSL3000, refer to Section 13. For information on TS3000 and TS3000E, refer to Section 15. External view of the controller TSL3000E is shown in Fig. 14.1.

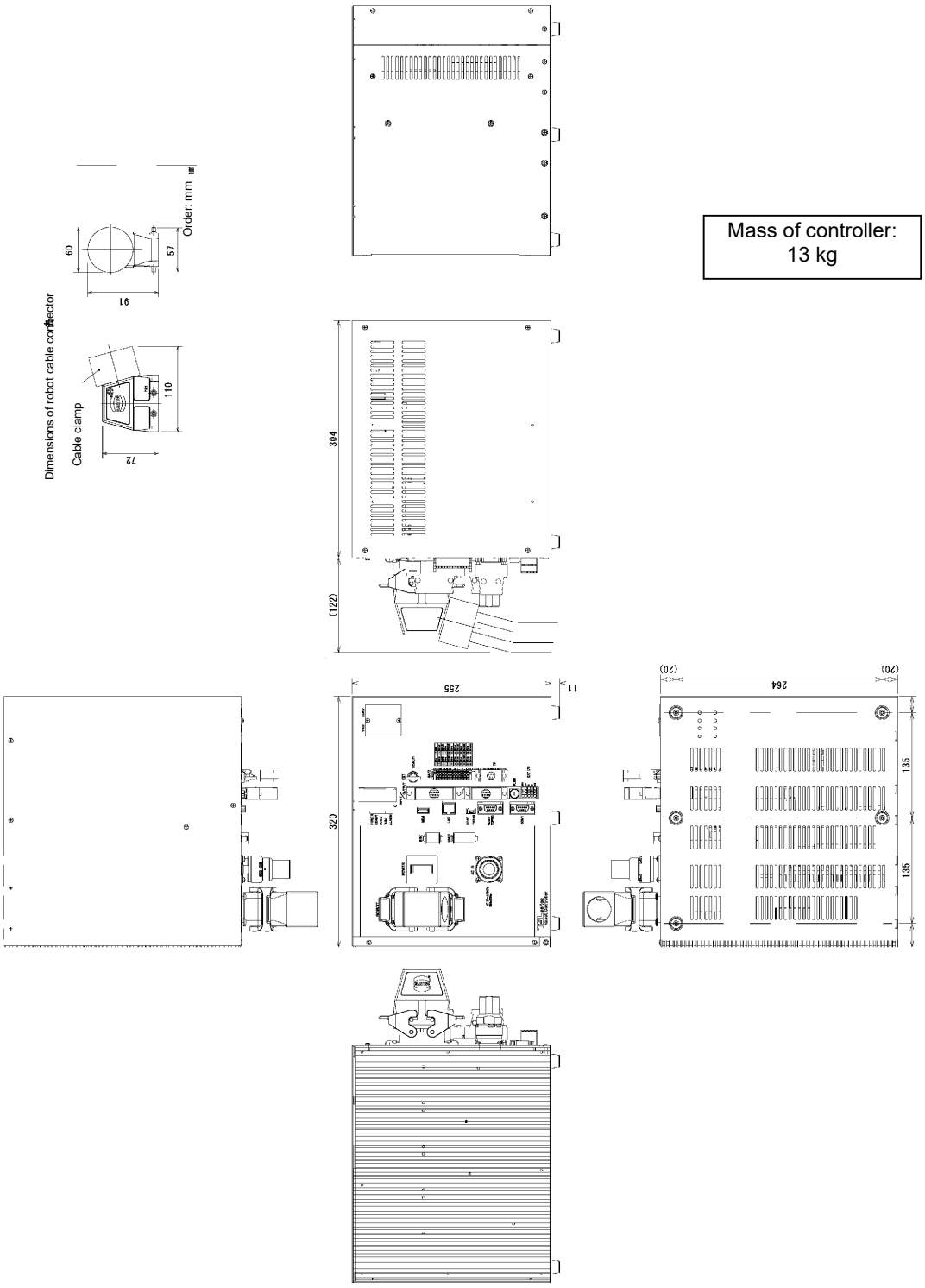


Fig. 14.1 External view of controller (TSL3000E)

14.2 Precautions for Direct Installation

It is necessary to provide a clearance of 50 mm or more in the horizontal direction and a clearance of 100 mm or more in the upward direction near the controller.

CAUTION

- Provide a ventilation space at the side of the controller so that the air vent holes are not blocked. The space equal to the length of the rubber legs should be kept below the lower surface. Otherwise, the cooling capacity will drop, causing controller failure.
- DO NOT stack the controllers.
- DO NOT place any object on top of the controller.

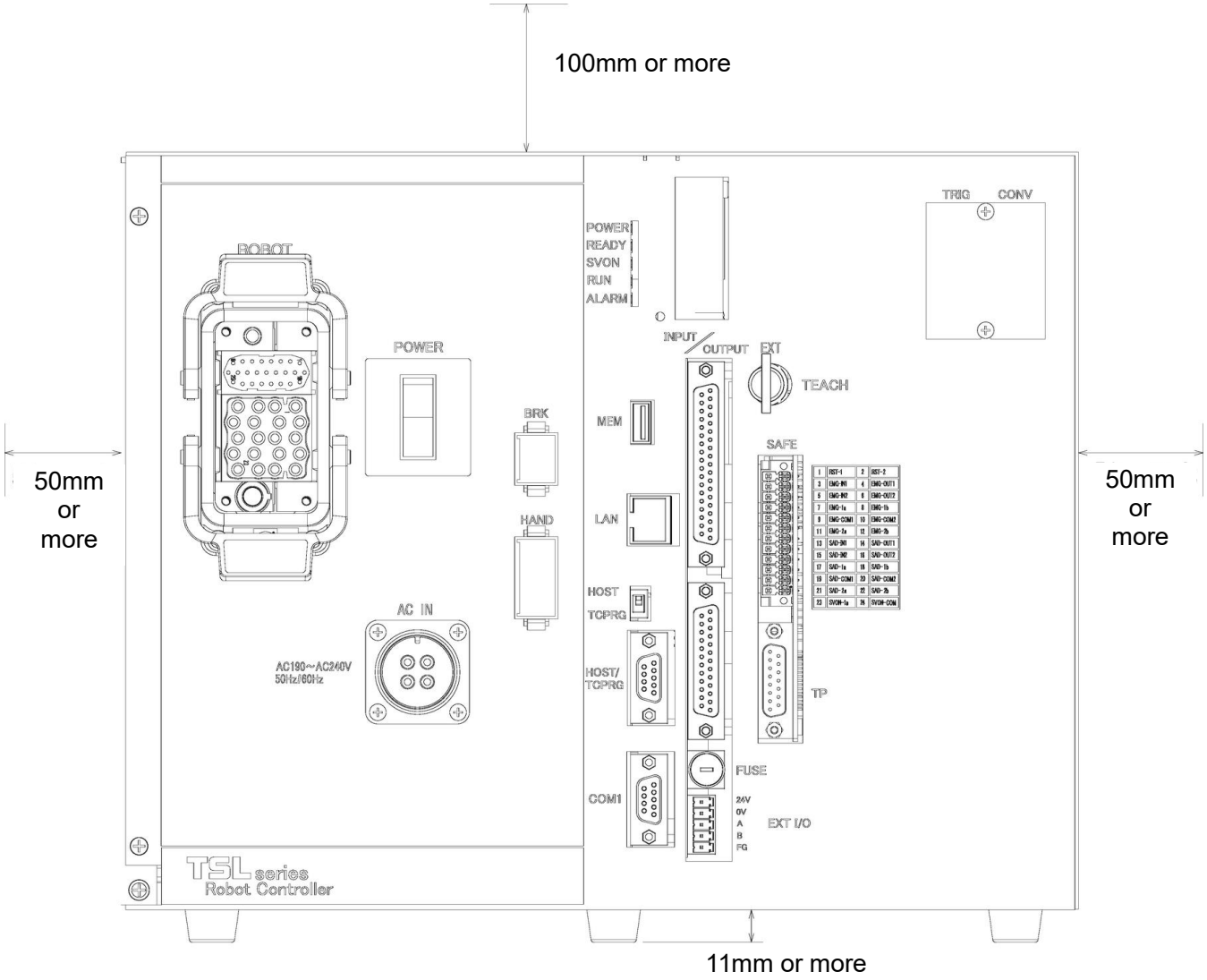


Fig. 14.2 Controller ventilation space (TSL3000E)

14.3 Rack Mounting Dimensions

When installing the robot controller TSL3000E to the control panel, attach the mounting fixtures using the screw holes for mounting the rubber legs on the bottom surface, and then fasten the controller to the control panel.

Note) The mounting fixtures (2 piece set) are optional. The customer is requested to do assembly.

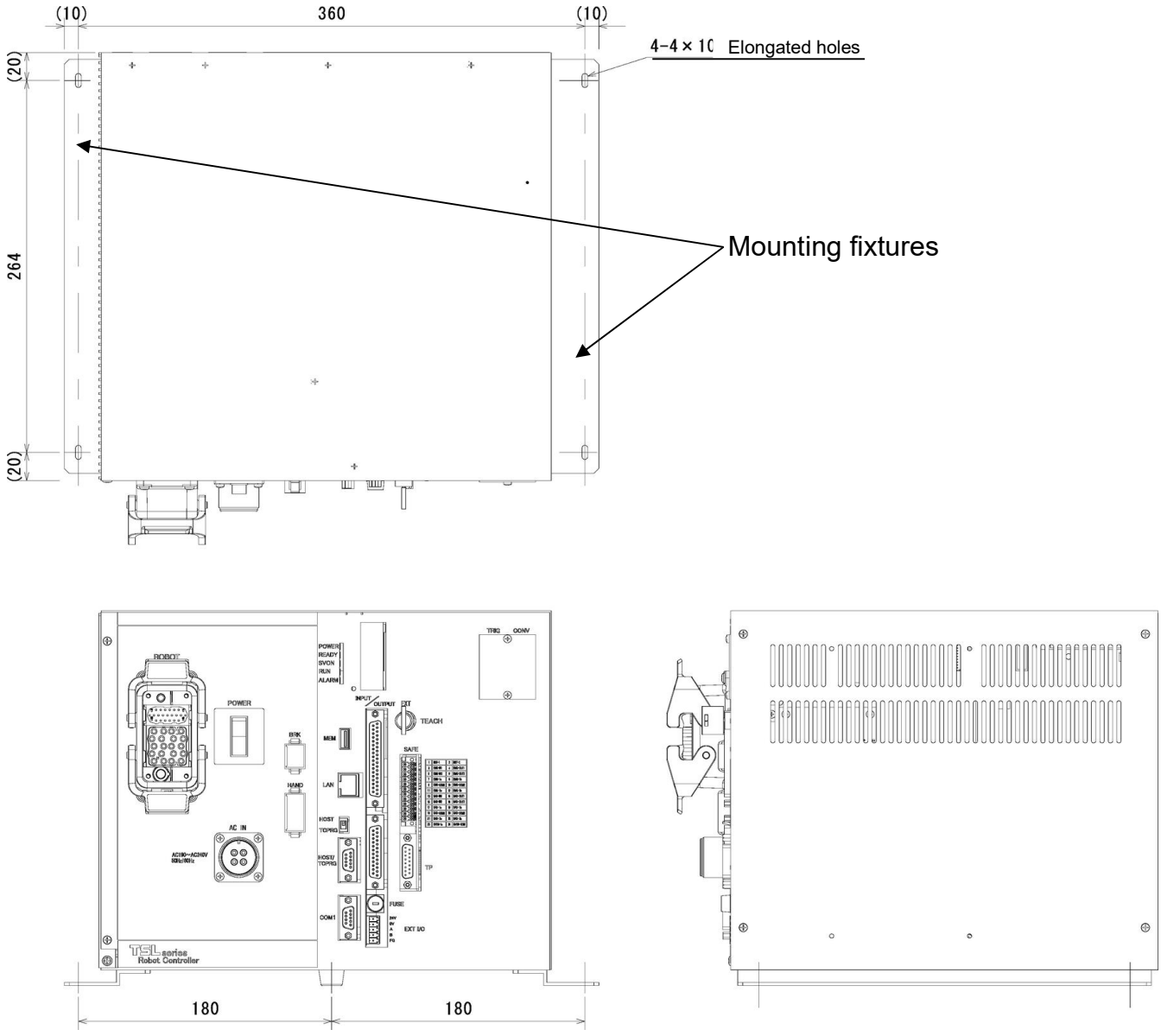


Fig. 14.3 Screw hole dimensions for securing controller (TSL3000E)

14.4 Precautions for Assembling the Control Panel

When installing the robot controller TSL3000E to the control panel, be careful with the following items.

a) When installing the robot controller to the control panel, remove the rubber legs on the bottom surface and use the holes for installing the rubber legs to fix the controller in position.

b) To do maintenance of the controller, do it with the upper cover removed. (See Fig. 14.4.)

c) When installing the controller, make sure that the maintenance of the controller can be done without any problem. Especially when storing the controller in the control panel, it will be necessary to take out the controller from the control panel for maintenance.

Specifically, be careful of the following points.

1) Arrange the cables around the rear side of the controller (so that the controller can be removed).
2) Connect all cables in such a manner that the robot can be operated even if the controller is removed from the rack.

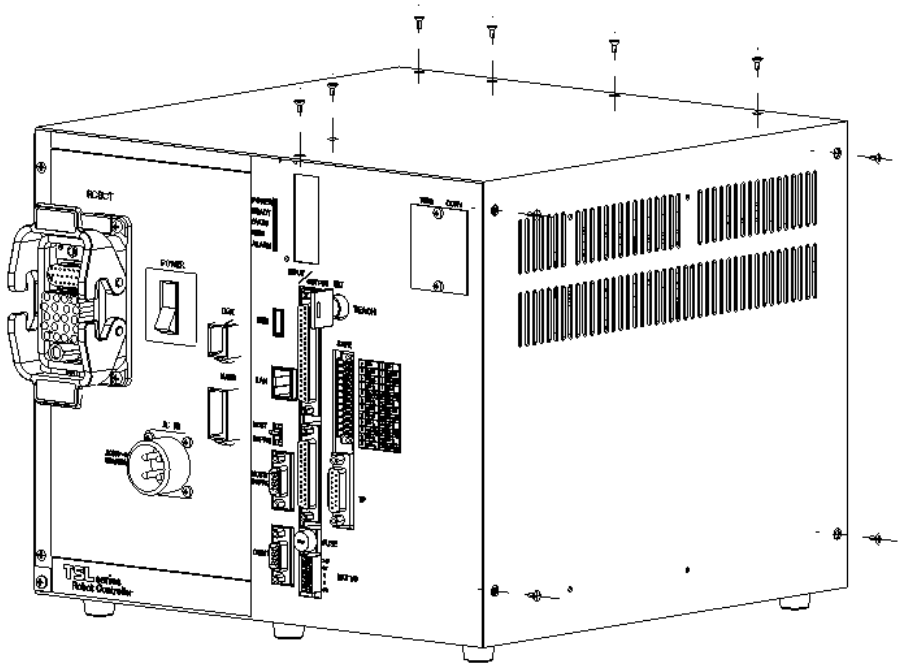


Fig. 14.4 Removing upper cover (TSL3000E)

d) Allocate a clearance of about 122 mm in front of the controller for connecting the robot cable connector.

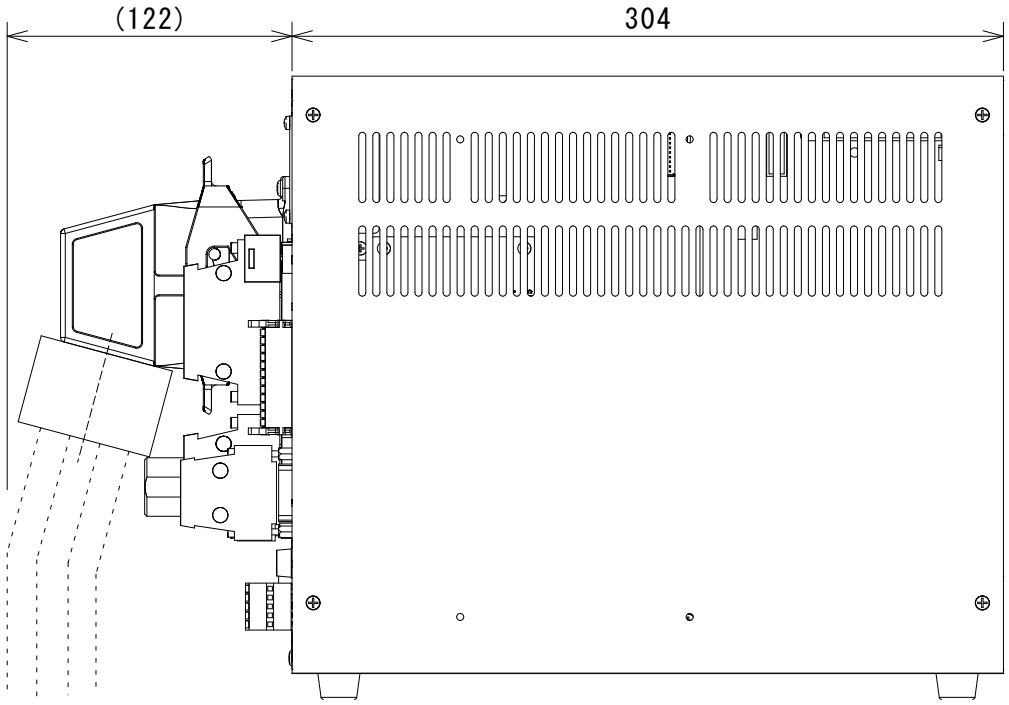


Fig. 14.5 Clearance of controller front side (TSL3000E)

CAUTION

- If the control panel is completely sealed up, open air release holes, ventilate the air forcibly with a fan, or conduct indirect cooling to release heat from the control panel. Otherwise, heat will be confined in the control panel and the controller, causing failure.

15. Installing the Controller (TS3000 and TS3000E)

15.1 External Dimensions

For information on TSL3000, refer to Section 13.

For information on TSL3000E, refer to Section 14.

External view of the controller TS3000 and TS3000E are shown in Fig. 12.1.

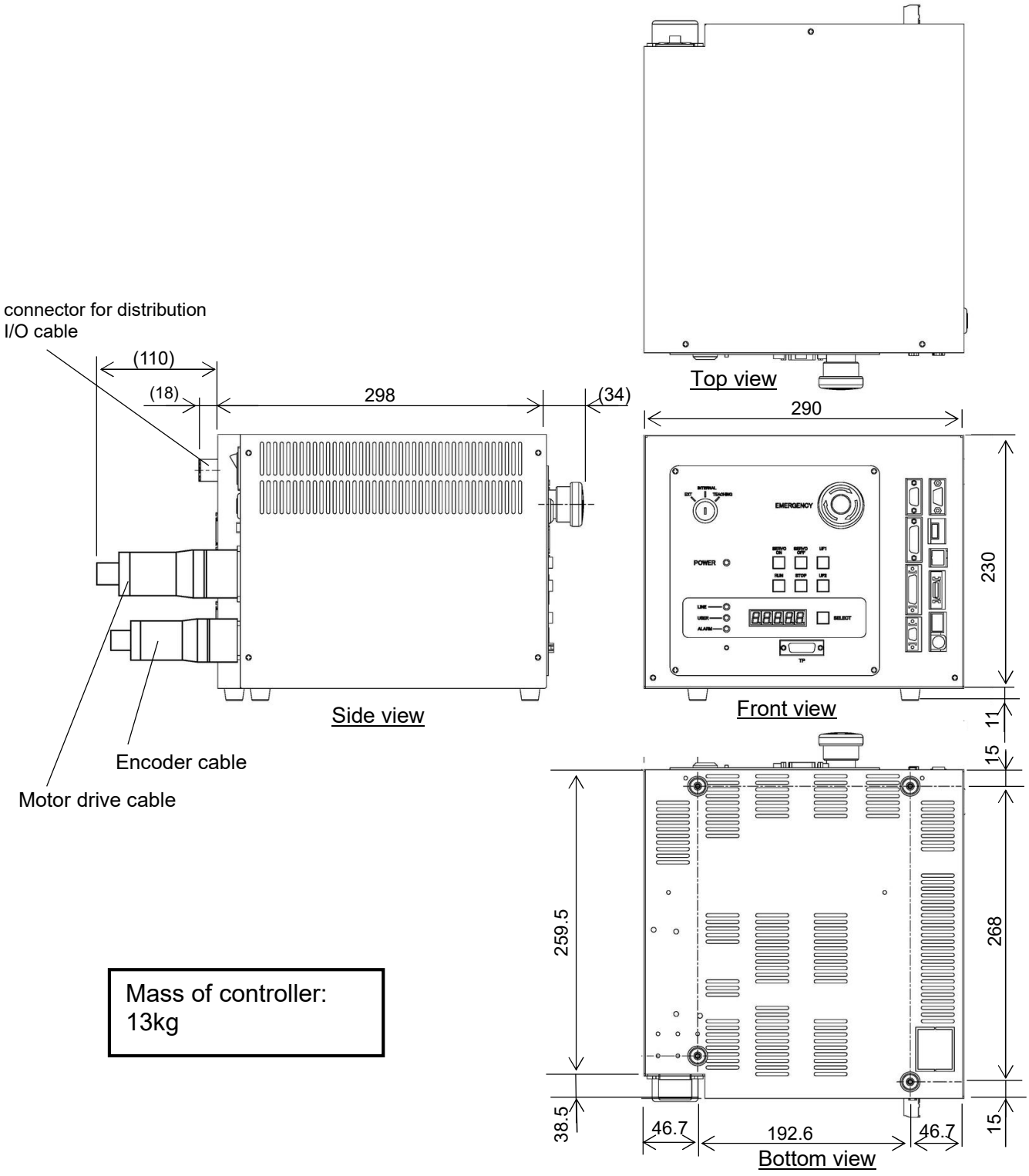


Fig. 15.1 External view of controller (TS3000 and TS3000E)

15.2 Precautions for Direct Installation

It is necessary to provide a clearance of 50 mm or more in the horizontal direction and a clearance of 100 mm or more in the upward direction near the controller.

CAUTION

- Provide a ventilation space at the side of the controller so that the air vent holes are not blocked. The space equal to the length of the legs should be kept below the lower surface.
  Failure to do so could cause the cooling performance to drop or to faults.
- DO NOT stack the controllers.
- DO NOT place any object on top of the controller.

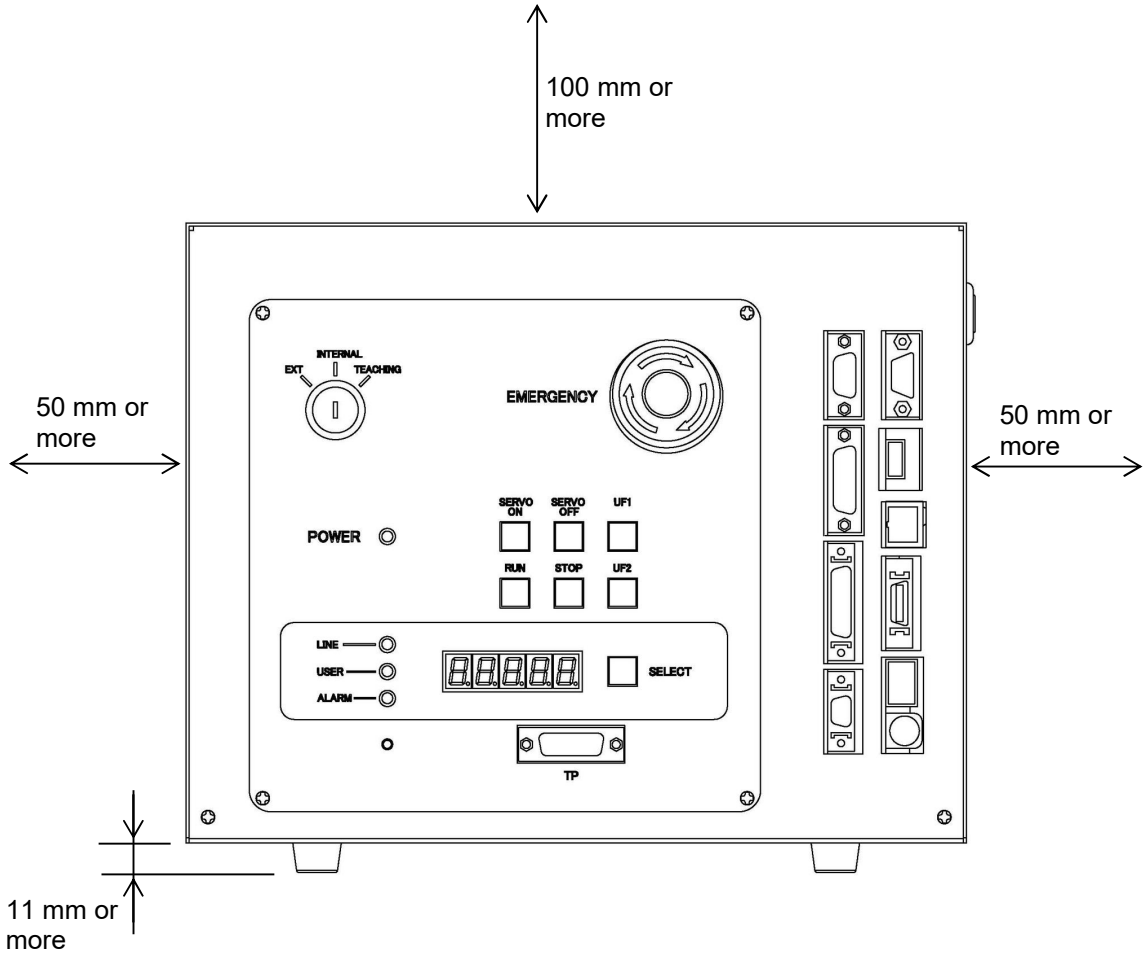


Fig. 15.2 Controller ventilation space (TS3000 and TS3000E)

15.3 Rack Mounting Dimensions

When mounting the robot controller TS3000 and TS3000E in a rack, mount the side brackets using the screw holes provided on both ends of the front panel, and secure the controller to the rack. The side brackets ① in Fig. 15.3 are optional.

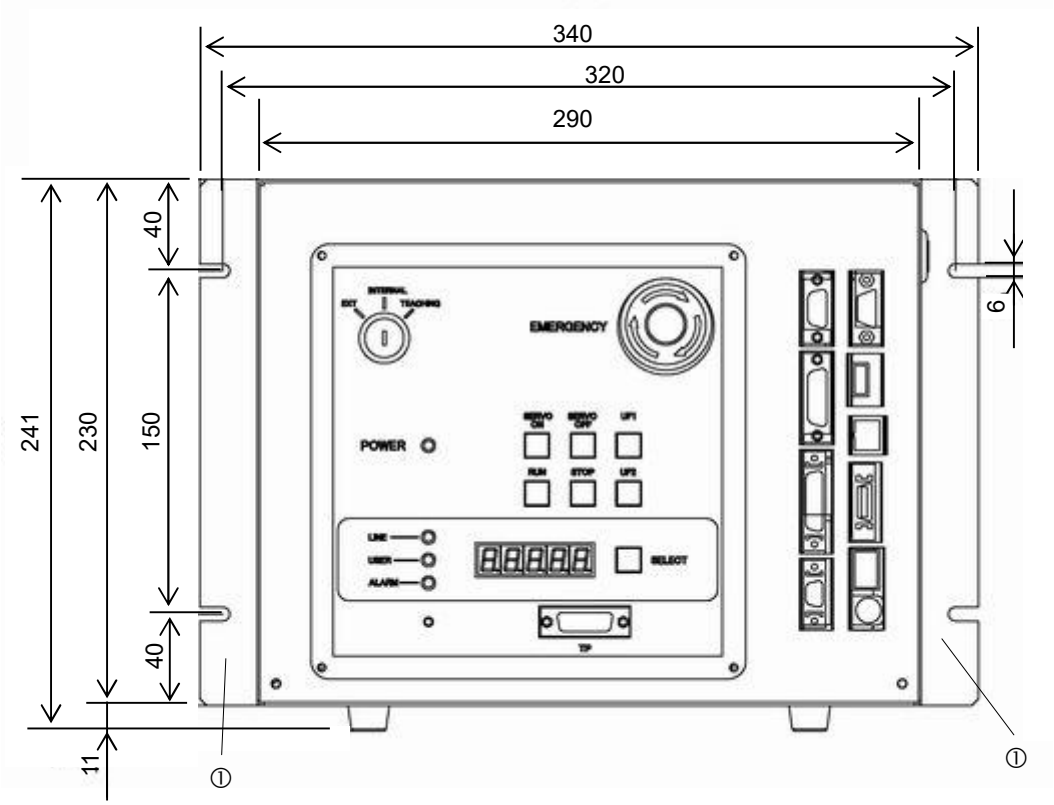


Fig. 15.3 Screw hole dimensions for securing controller (TS3000 and TS3000E)

15.4 Precautions for Rack Mounting

Pay attention to the following matters when rack-mounting the robot controller TS3000 and TS3000E.

a) When mounting the robot controller in a rack, use the screw holes provided on both ends of the front panel, and secure the controller. (Optional side brackets are required.)

CAUTION

- If the rack is completely sealed, provide holes to allow the air to be let out, force-ventilate the rack with a fan, or cool it indirectly so that the heat will not be trapped in the rack.
  If the heat is trapped in the rack or controller, faults could occur.

As the cable connectors are connected to the rear side of the controller, provide a space of 110 mm or more on the rear side.

For maintenance, the upper cover should be removed. (See Fig. 15.4.)

Install the controller for enabling easy access to the controller when conducting maintenance. In particular, when the controller is mounted in a rack, the controller must be removed from the rack during maintenance. Specifically, be careful of the following points.

1) Arrange the cables around the rear side of the controller (so that the controller can be removed).
2) Arrange the cables between the controller and control panel when the control panel is separated.
3) Connect all cables in such a manner that the robot can be operated even if the controller is removed from the rack.

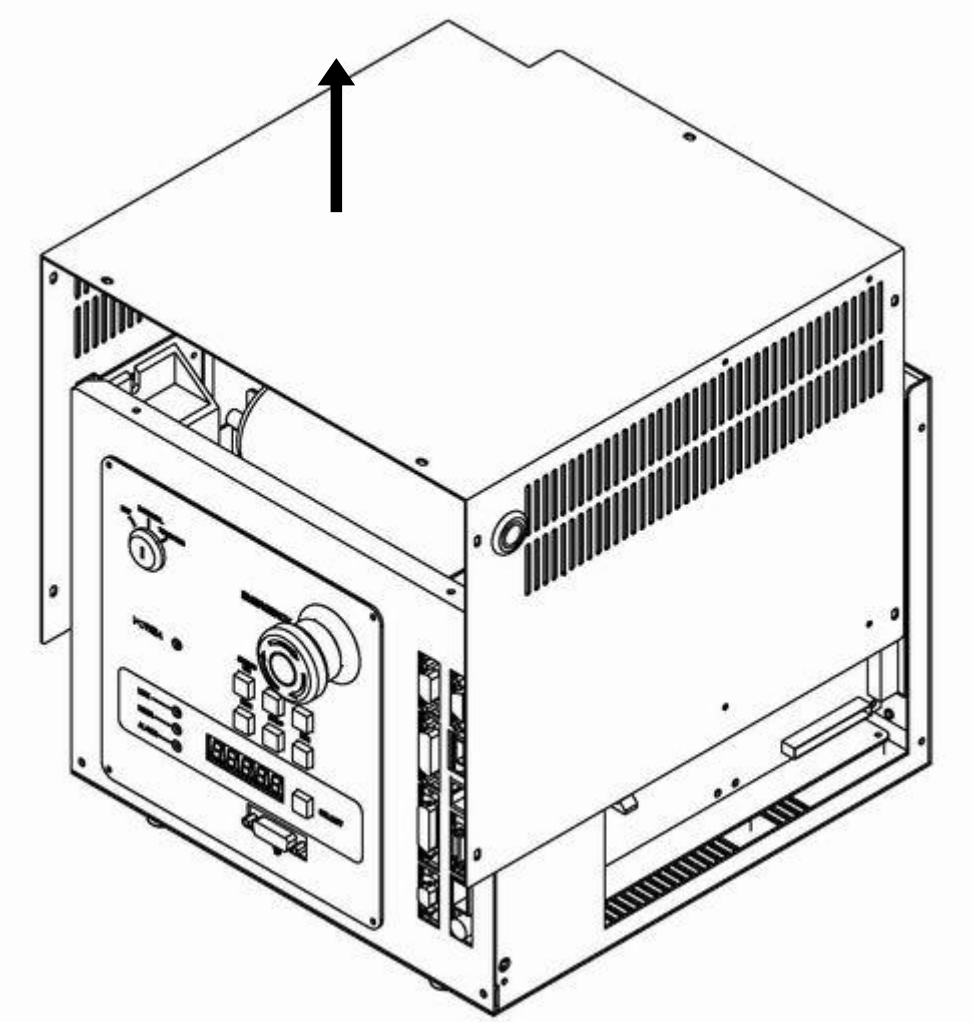


Fig. 15.4 Removing upper cover (TS3000 and TS3000E)

b) When the controller is mounted on the rack, the weight of the controller should be supported with the legs of the controller. Screw holes for rack-mounting the controller are used for securing the controller panel, and the weight of the controller cannot be supported only with these screws.

c) On the front of the controller, a clearance of approx. 90 mm should be provided for connecting the connector of the teach pendant. Even if the teach pendant is not used, a clearance of approx. 60 mm is required for connecting a dummy plug.

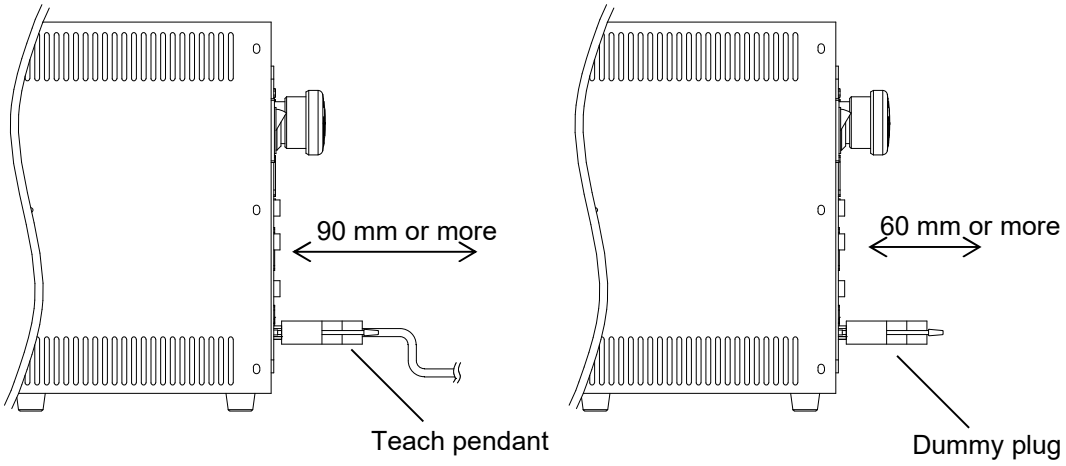


Fig. 15.5 Clearance of controller front side (TS3000 and TS3000E)

d) On the back of the controller, a clearance of approx. 110 mm should be provided for connecting the connector of the teach pendant.

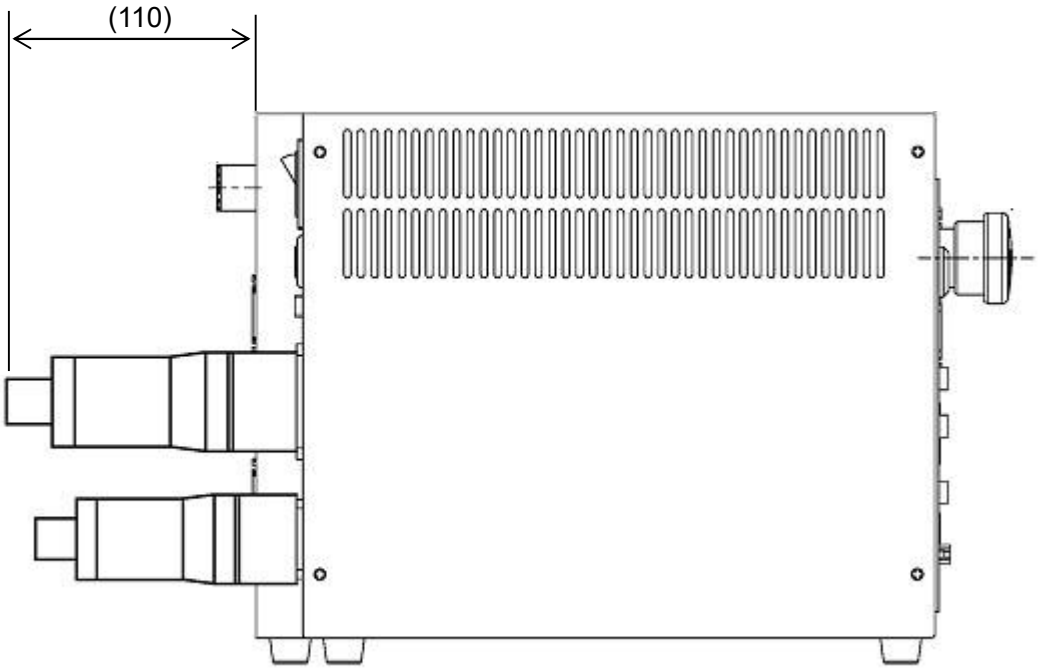


Fig. 15.6 Clearance of controller back side (TS3000 and TS3000E)

16. System Connections (TSL3000)

16.1 Cable Wiring

For information on TSL3000E, refer to Section 17. For information on TS3000 and TS3000E, refer to Section 18. This section describes the various types of cables and connectors and explains how these are to be connected.

16.1.1 Connector Arrangement on the Controller

The cables connected to the robot controller are shown in Fig. 16.1.

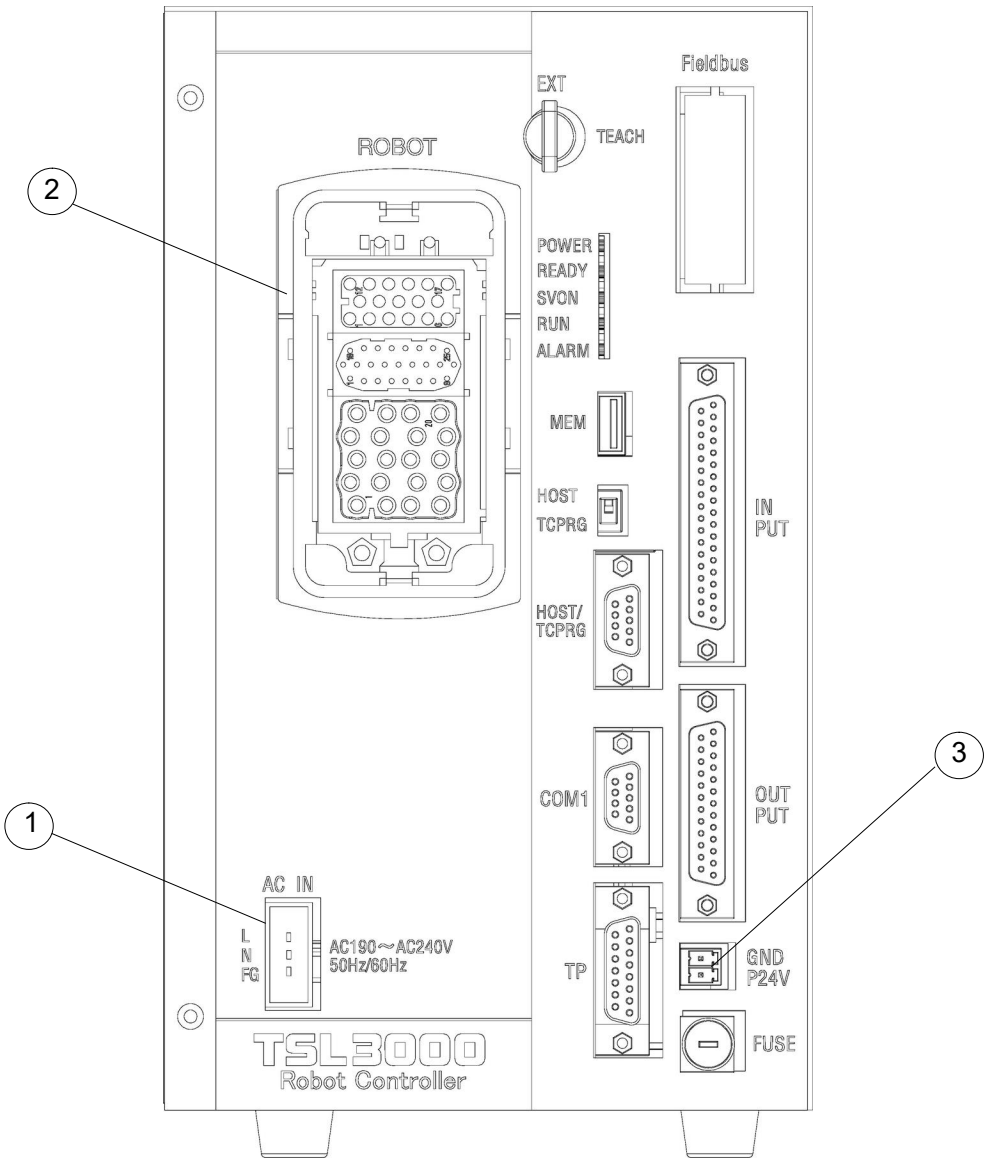


Fig. 16.1 Robot controller connector arrangement (TSL3000)

① Power cable (ACIN)
② Motor cable, encoder cable, robot control signal cable, brake signal cable (ROBOT)
③ External input/output power cable (GNDP24V)

In the subsequent paragraphs, we explain how to connect cables ①, ② and ③. For information on how to connect other cables, refer to the TSL3000 Interface Manual STE85364.

16.1.2 Connecting the Power Cable "ACIN" (① of Fig. 16.1) (Plug connector attached)

The power cable is used to supply the main AC power to the controller. The plug connector is attached.

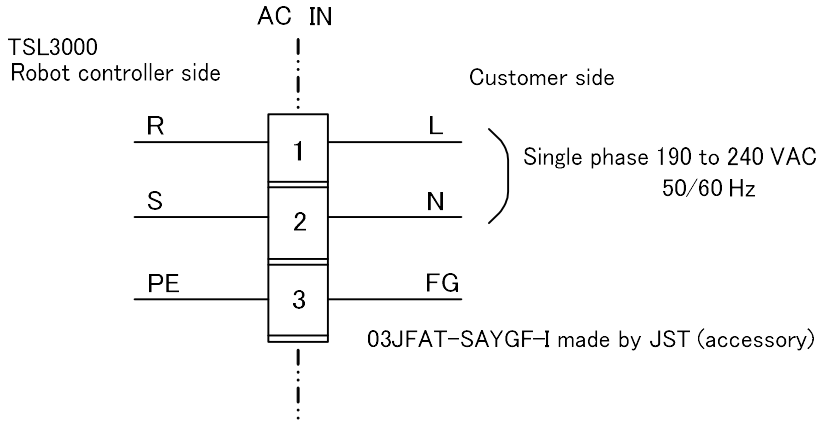


Table 16.1 Power supply specifications (TSL3000)

Power supply	Single phase, 190 - 240 V AC, 50/60 Hz ±1 Hz
Instantaneous power failure	Within 40 msec
Grounding	JIS class D

The connector is ACIN (① of Fig. 16.1).

ACIN plug connector: Type: 03JFAT-SAYGF-I Maker: J.S.T. Mfg. Co., Ltd.
Wire: 0.8 mm^2 to 2.0 mm^2 (AWG#18 to AWG#14)

As the cable is not an accessory, use the attached plug connector connected to ACIN on the controller side to manufacture a cable.

! DANGER

- Be sure to use the designated wire. Failure to do so could lead to fires or faults.
- When connecting the connector and wires, make sure not to mistake the terminal arrangement.
- After making the connection, use a tester, etc., to confirm the connection.

For the terminal arrangement, see Para. 16.1.9.

! CAUTION

- Unless the main power is normally supplied to the controller due to phase defect or voltage drop, an error of "8–027 Slow Charge error" occurs. When this happens, make sure that the maser power voltage at the controller power connector satisfies the specified input power of the controller, and that the same voltage is stabilized.
- For details of the 8–027 error, see Para. 13.7 of the "Instruction Manual: Operations".

16.1.3 Connecting the Motor Cable "ROBOT" (② of Fig. 16.1) (THL300 to THL1200)

The motor cable is integrated with the encoder cable, robot control signal cable, and brake signal cable. This integrated cable is further integrated with the robot. The motor cable connects the controller and robot, and supplies the power required to rotate the motor from the controller servo driver to each axis feed motor of the robot.

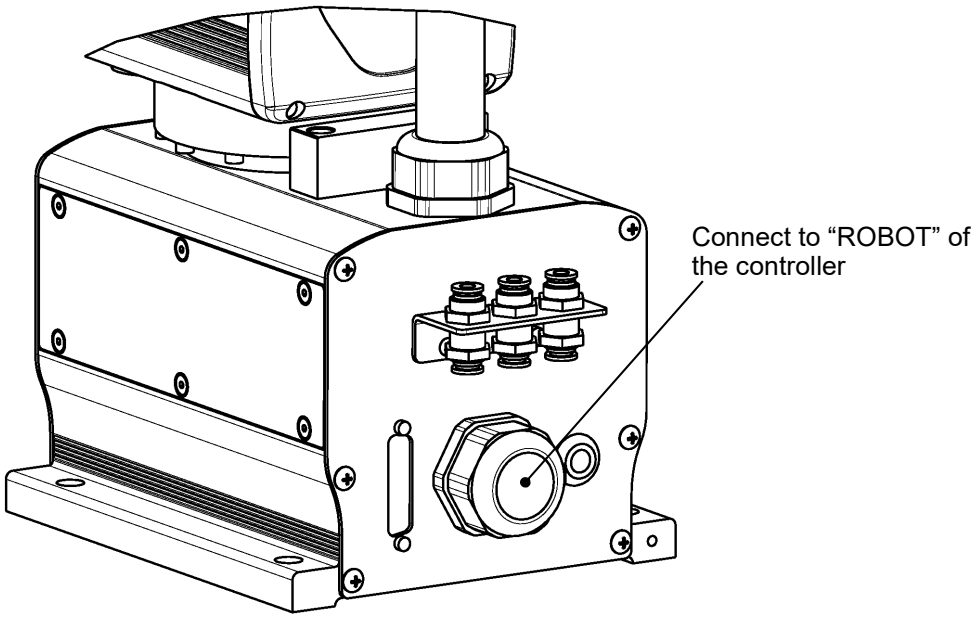


Fig. 16.2 Robot side connector arrangement (TSL3000)

16.1.4 Connecting the Encoder Cable "ROBOT" (② of Fig. 16.1)

The encoder cable is integrated with the motor cable, robot control signal cable, and brake signal cable. This integrated cable is further integrated with the robot. The encoder cable is a signal line used to transmit a signal from the rotation angle detecting encoder of each robot axis to the controller.

16.1.5 Connecting the Robot Control Signal Cable "ROBOT" (② of Fig. 16.1)

The robot control signal cable is integrated with the motor cable, encoder cable, and brake signal cable. This integrated cable is further integrated with the robot. The robot control signal cable is used for motor brake ON and OFF, and input and output of robot control signals such as hand operation signal.

16.1.6 Connecting the Brake Signal Cable "ROBOT" (② of Fig. 16.1)

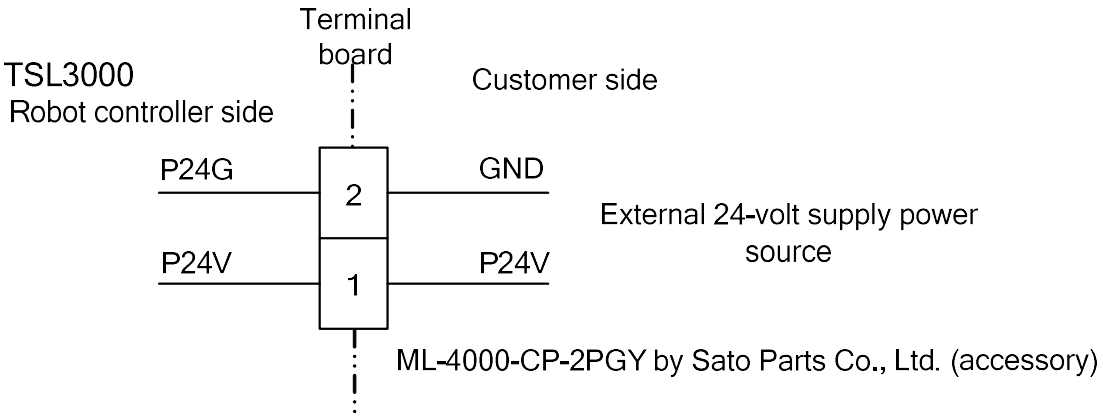
The brake signal cable is integrated with the motor cable, encoder cable, and robot control signal cable. This integrated cable is further integrated with the robot. The brake signal cable is used to turn ON and OFF the brake for fixing the motor axis.

16.1.7 Connecting the External Input/Output Power Cable "GNDP24V" (③ of Fig. 16.1)

An attached connector (ML-4000-CP-2PGY) is used to connect the external input/output power cable. The P24V power is supplied from the front of the controller.

The following shows the I/O using the external supply power (24 VDC). Be sure to supply power (24 VDC) from the outside.

- External input/output
- Externally operated input/output
- Extended input/output
- Manual input/output



The connector compatible cables are "AWG24 through 16"
Select the optimum external power source in conformity to the customer's system specifications (power capacity).
To connect and disconnect the external input/output power cable, see the description of "Interface Part".

CAUTION

Be sure to supply external power (24 VDC). If not, safety signals are not enabled and the controller servo power cannot be turned on.

16.1.8 Connecting and Disconnecting Cables

CAUTION

- Before connecting or disconnecting any controller cable, be sure to turn off the main power.
- When disconnecting a cable, be sure to pull the connector and not the cord. Otherwise, you may damage the cable.
- When disconnecting a cable, pull out the plug while holding down the controller. If failed to hold down the controller, the controller may fall down when pulling out the connector.

a) ROBOT connector: ROBOT

When connecting the connector, completely insert it into the connector on the controller body side while holding down the buttons located at the top and bottom of the connector on the cable side. If connection is loose, it may cause accidents due to connector contact failure, so be sure to check that the connector is completely inserted.

To remove the connector, reverse the process: Pull out the connector on the cable side while holding down the buttons located at the top and bottom of the connector.

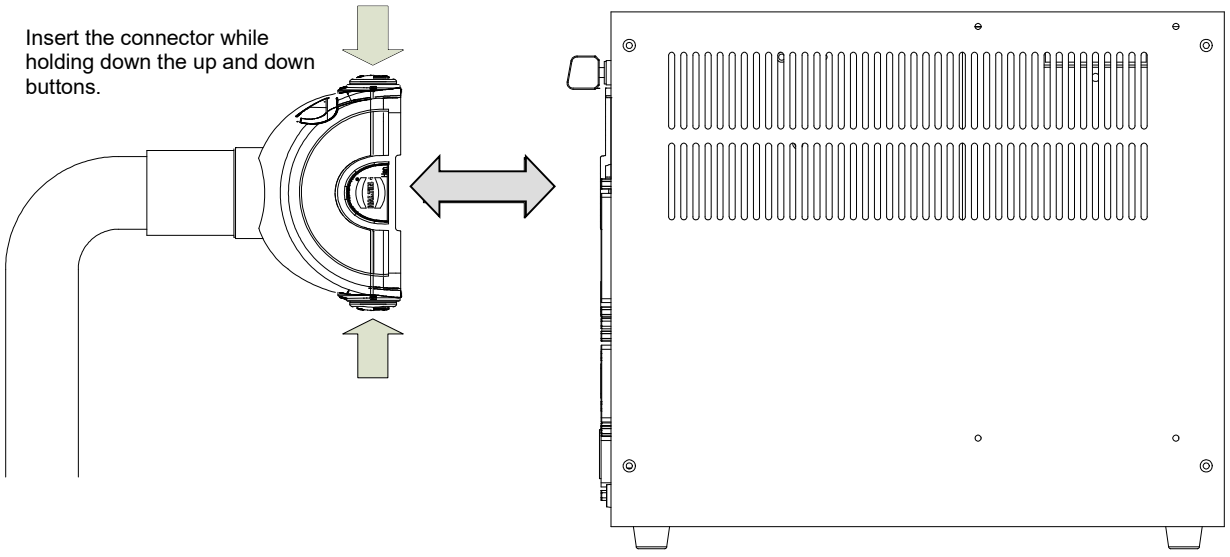


Fig. 16.3 Connecting and disconnecting the ROBOT connector (TSL3000)

b) Square connectors: INPUT, OUTPUT, TP, HOST/TCPRG, COM1
Firstly, completely insert the cable side connector into the controller connector. Then tighten the lock screws on both ends of the cable side connector with a Phillips head screwdriver (No. 1). A loose screw can cause a contact failure or other accident. To avoid this, make sure that the screws are clamped completely. To disconnect the connectors, first loosen the lock screws, and then pull out the cable side connector.

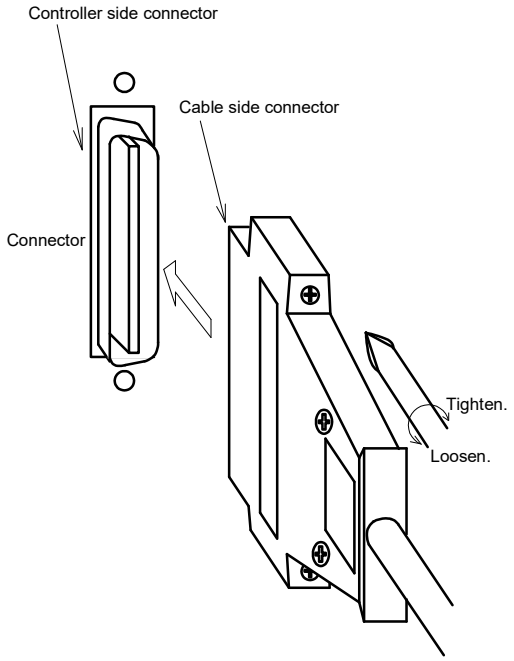
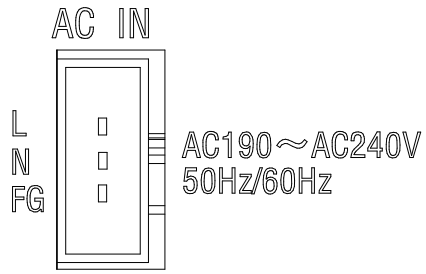


Fig. 16.4 Connecting and disconnecting a square connector (TSL3000)

16.1.9 Examples of Connector Terminal Arrangement (Controller side)

a) Power cable connector "ACIN"



Connects to controller.

Type: 03JFAT-SAYGF-I

Manufacturer: J.S.T. Mfg. Co., Ltd.

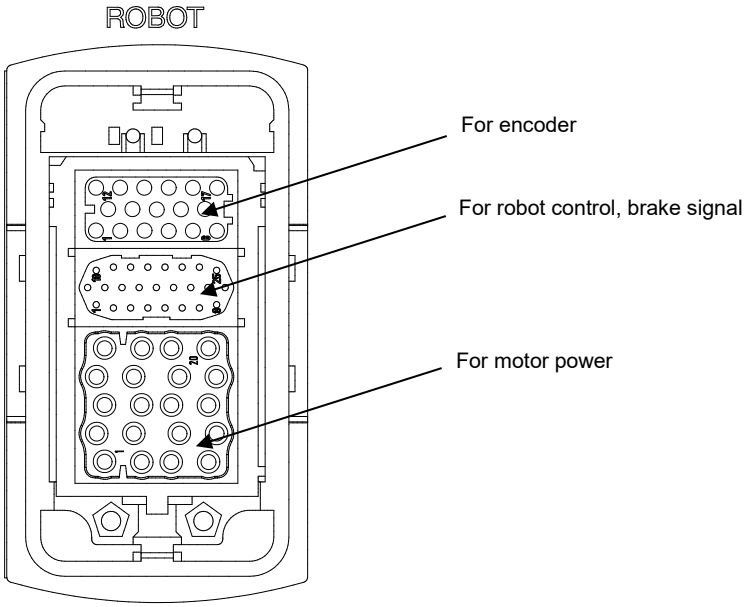
1(L) ┐
2(N) ┘ Single phase, 190 ~ 240 VAC, 50/60 Hz

3(FG) Grounding (JIS Class D or higher)

! DANGER

- Completely connect the grounding cable. Otherwise, an electric shock or fire may be caused if a fault or electric leak occurs, or misoperation may be caused by noise.

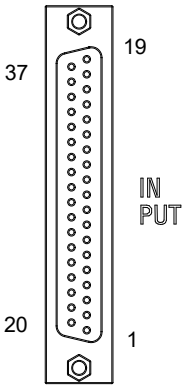
b) Connector for motor, encoder, robot control signal cable, and brake signal cable "ROBOT"



Connects to controller.

Manufacturer: Harting Co., Ltd.

c) Universal input signal cable connector "INPUT"



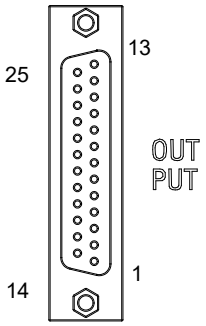
Connects to controller.

Type: XM3B-3722-112

Manufacturer: OMRON Corporation

The mating components are attached to the controller.
For details, refer to the "Instruction Manual: Interface".

d) Universal output signal cable connector "OUTPUT"



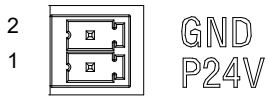
Connects to controller.

Type: XM3B-2522-112

Manufacturer: OMRON Corporation

The mating components are attached to the controller.
For details, refer to the "Instruction Manual: Interface".

e) Connector for supplying power to I/O signals



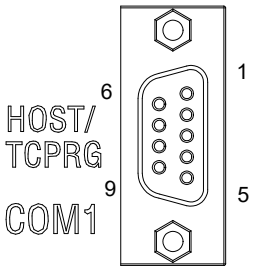
Connects to controller.

Type: ML-4000-CWJH 02PGY

Manufacturer: Sato Parts Co., Ltd.

The mating components are attached to the controller.
For details, refer to the "Instruction Manual: Interface".

f) Communication connector HOST/TCPRG, COM1

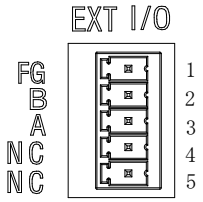


Connects to controller.

Type: XM2C-0942-132L

Manufacturer: OMRON Corporation

g) Connector for distribution I/O cable connection EXT I/O (Back of controller)



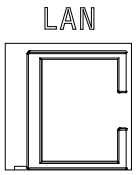
Connects to controller.

Type: ML-4000-CWJH 05PGY

Manufacturer: Sato Parts Co., Ltd.

The mating components are attached to the controller.
For details, refer to the "Instruction Manual: Interface".

h) Connector for ethernet LAN (Back of controller)



Connects to controller.

Type: J0026D21BNL

Manufacturer: PULSE Electronics

16.2 Controller Connector Signals

For information on TSL3000E, refer to Section 17. For information on TS3000 and TS3000E, refer to Section 18.

16.2.1 Connector Signal Connection Diagrams

Diagrams showing which signals correspond to which terminals are shown in Section 2 of the TSL3000 Interface Manual STE85364.

16.2.2 Jumpers for Safety Related Signals

The following system input signals are provided to serve for the safety purpose.

System input signals	INPUT-12	(STOP)	
	INPUT-14	(SVOFF)	
	INPUT-32	(BREAK)	
	INPUT-18, 19	(EMS1B–EMS1C)	
	INPUT-36, 37	(EMS2B–EMS2C)	
	INPUT-17, 35	(INCOM–P24V)	Presuming standard P24V(+) common

These signals are already jumpered for the connectors provided for the TSL3100 robot controller. If you wish to use or change them, therefore, you should remove the jumpers and rewire as appropriate. If you plan to use the robot without using system input signals, be sure to connect the attached connectors to the controller side INPUT connector.

Unless the following signals are used as the system signals, jumper them also.

INPUT-13 (LOW_SPD)
INPUT-31 (CYCLE)

Connector jumpers

INPUT			
12-16	14-16	32-16	32-37
(13-16)	(31-16)	18-19	17-35

! CAUTION

- Unless the signals of SVOFF and emergency stop contacts 1, 2 are jumpered, the controller servo power cannot be turned on.
- Unless the CYCLE signal is jumpered, the controller enters the cycle operation mode.
- Unless the LOW_SPD signal is jumpered, the robot is operated at low speed during automatic operation.
- Unless the STOP signal is jumpered, automatic operation of the robot is not possible.
- Unless the BREAK signal is jumpered, automatic operation of the robot is not possible.

17. System Connections (TSL3000E)

17.1 Cable Wiring

This section describes the various types of cables and connectors and explains how these are to be connected.

17.1.1 Connector Arrangement on the Controller

The cables connected to the robot controller are shown in Fig. 17.1.

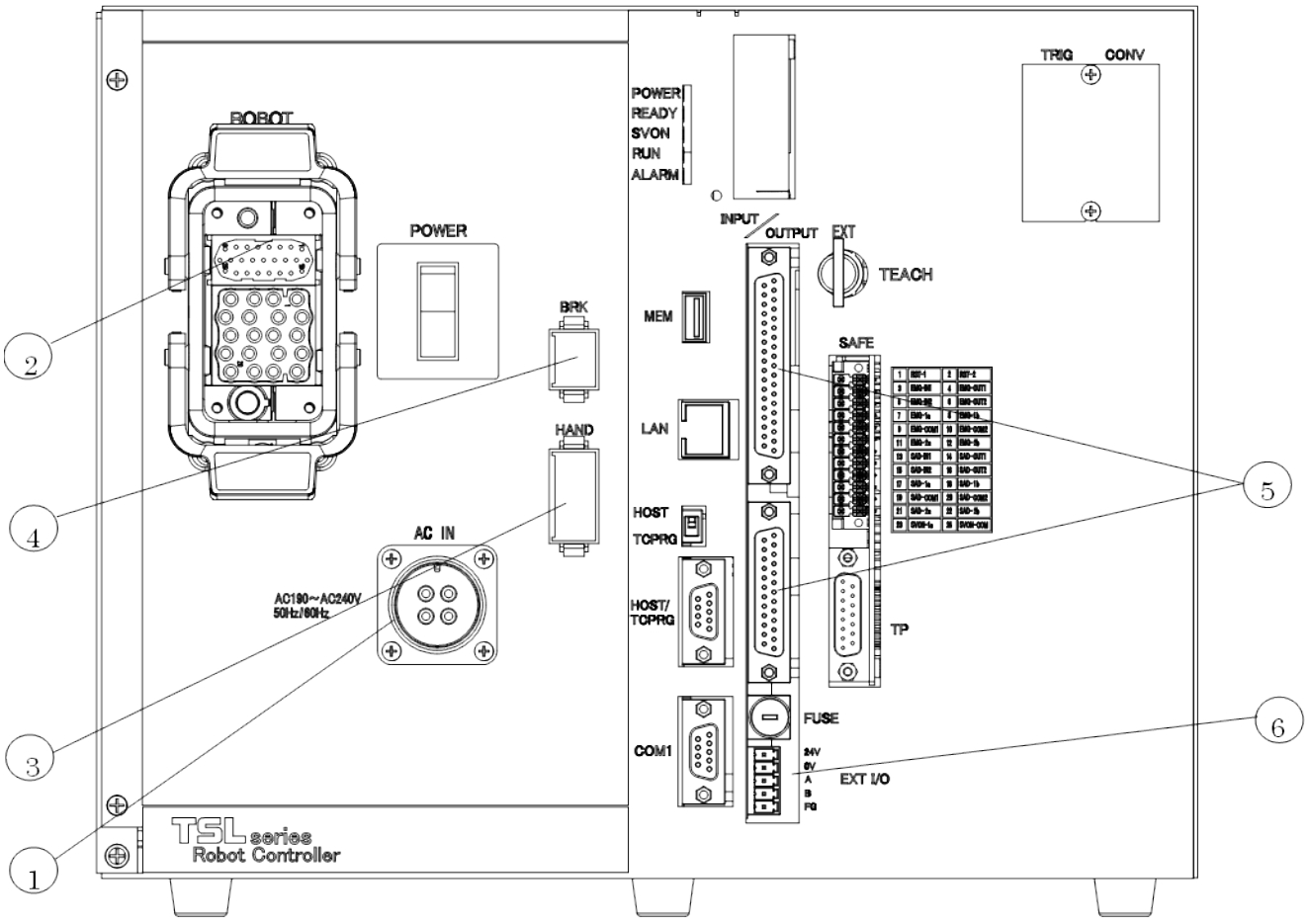


Fig. 17.1 Robot controller connector arrangement (TSL3000E)

① Power cable (ACIN)
② Motor cable, encoder cable (ROBOT)
③ Robot control signal cable (HAND)
④ Brake signal cable (BRK)
⑤ External operation input signal cables (INPUT/OUTPUT)
⑥ Power supply cable for external input/output (EXT I/O)

In the subsequent paragraphs, we explain how to connect cables ① to ④. For information on how to connect other cables, refer to the Interface Manual.

17.1.2 Connecting the Power Cable

The power cable (“ACIN” (① of Fig. 17.1)) is used to supply the main AC power to the controller.

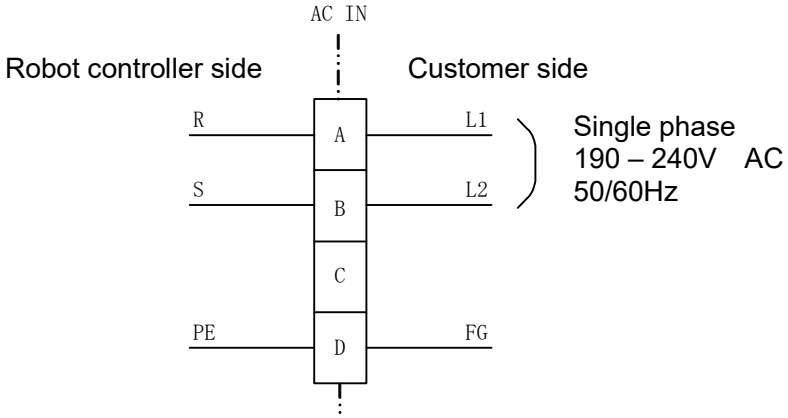


Table 17.1 Power supply specifications (TSL3000E)

Power supply	Single phase, 190 - 240 V AC, 50/60 Hz ±1 Hz
Instantaneous power failure	Within 40 msec
Grounding	JIS class D

The connector is a standard accessory.

ACIN plug connector	Type: JL04V-6A18-10SE-EB	Maker: Japan Aviation Electronics Industry, Ltd
	Type: JL04V-18CK(13)	Maker: Japan Aviation Electronics Industry, Ltd
Wire	2.0mm2 (AWG#14)	

As the cable is not an accessory, use the attached plug connector connected to ACIN on the controller side to manufacture a cable.
For connection between the connector and cable, place the conductor in-between.

For the terminal arrangement, see Para. 17.1.7.

! DANGER

- Be sure to use the designated wire. Failure to do so could lead to fires or faults.
- When connecting the connector and wires, make sure not to mistake the terminal arrangement.
- After making the connection, use a tester, etc., to confirm the connection.

! CAUTION

- Unless the main power is normally supplied to the controller due to phase defect or voltage drop, an error of "8–027 Slow Charge error" occurs. When this happens, make sure that the maser power voltage at the controller power connector satisfies the specified input power of the controller, and that the same voltage is stabilized.
- For details on "8-027 Slow Charge error," see the "Instruction Manual: Alarms" provided as a separate volume.

17.1.3 Connecting the Motor Cable/Encorder Cable

The motor cable connects the controller and robot, and supplies the power required to rotate the motor from the controller servo driver to each axis feed motor of the robot.
The encoder cable is a signal line used to transmit a signal from the rotation angle detecting encoder of each robot axis to the controller.
The motor cable connector and encoder connector are for ROBOT (② in Fig. 17.1).
The motor cable and encoder cable on the robot side are located at Fig. 17.2.
The cables and connectors for the motor cable and encoder cable are standard accessories.

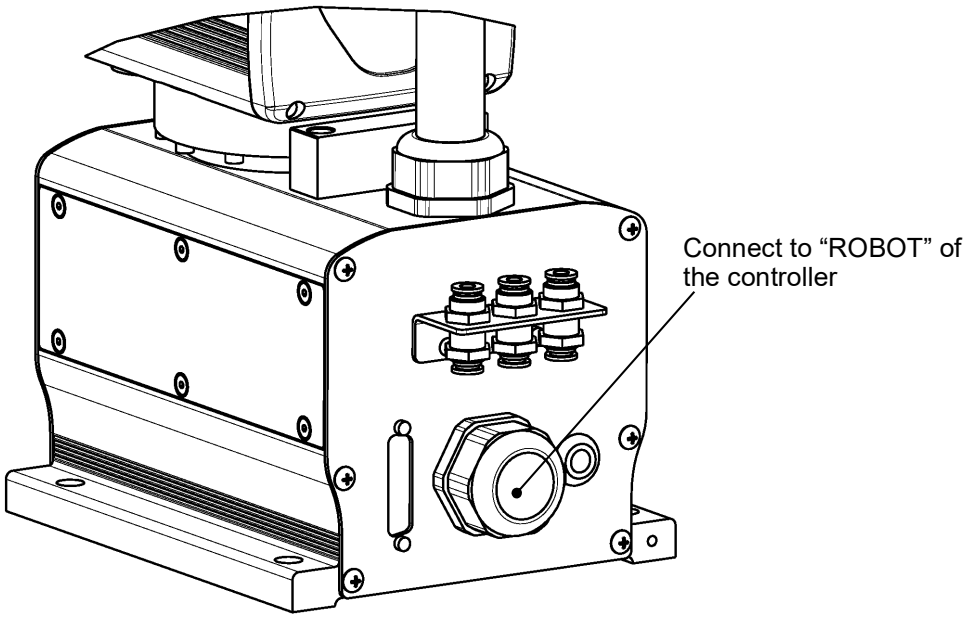


Fig. 17.2 Robot side connector arrangement (TSL3000E)

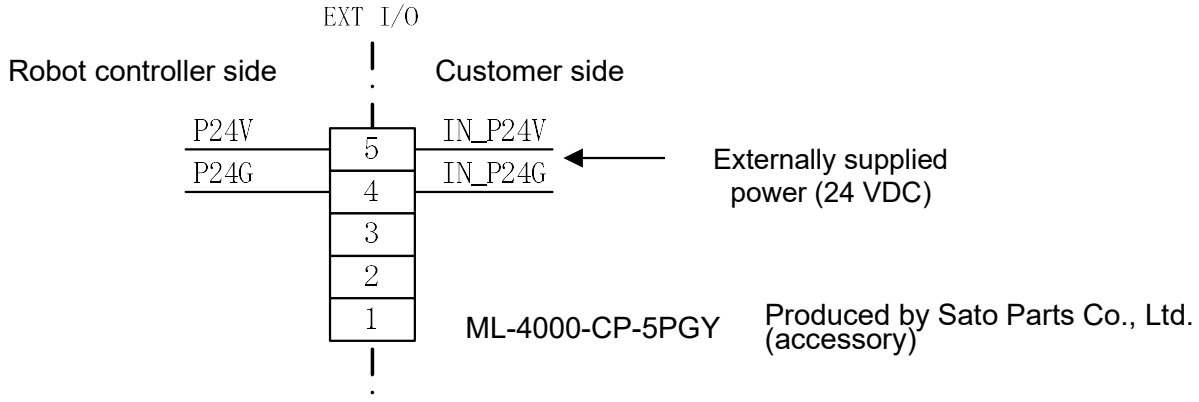
17.1.4 Connecting the Brake Signal Cable and the Robot Control Signal Cable

The brake cable is used to input and output an ON/OFF signal for the brake to lock the motor shaft.
The brake signal cable used to lock the motor shaft is connected to the connector BRK (④ in Fig. 17.1).
The robot control signal cable is used for input and output of robot control signals such as hand operation signal.
The robot control signal cable is connected to the connector "HAND" (③ of Fig. 17.1).
The brake cable and robot control signal cable on the robot side are located at Fig. 17.2.
The cables and connectors for the brake cable and robot control signal cable are standard accessories.

17.1.5 Connecting the External Input/Output Power Cable

Use the attached connector [ML-4000-CP-5PGY] to connect the power supply cable for external input/output ("EXT I/O" (⑥ in Fig. 17.1)). The P24V power is supplied from the front of the controller.
Shown below are the input/output signals that use the externally supplied power (24 VDC). Be sure to supply external power (24 VDC).

- External input/output
- External operation input/output
- Extended input/output
- Hand input/output



The wires for the connector are AWG24 to AWG16.
Choose the most suitable external power supply for your system specification (ampacity).
See the "Instruction Manual: Interface" for how to connect/disconnect the power supply cable for external input/output.

CAUTION

Be sure to supply external power (24 DVC). Otherwise, the safety signal will not be enabled and the controller servo power cannot be turned on.

17.1.6 Connecting and Disconnecting Cables

CAUTION

- Before connecting or disconnecting any controller cable, be sure to turn off the main power.
- When disconnecting a cable, be sure to pull the connector and not the cord. Otherwise, you may damage the cable.
- When removing the cable, pull off the plug while holding the controller. If you do not hold the controller, the controller may be overturned when removing the plug.

a) ROBOT connector: ROBOT

To connect the connector, firmly insert the connector on the cable side into the connector on the side of the controller body. Raise the levers located on the upper and lower portions on the side of the controller body, and lock it in position.

To remove it, reverse the above-mentioned procedure.

Tilt the levers located on the upper and lower portions. After that, pull off the connector on the cable side.

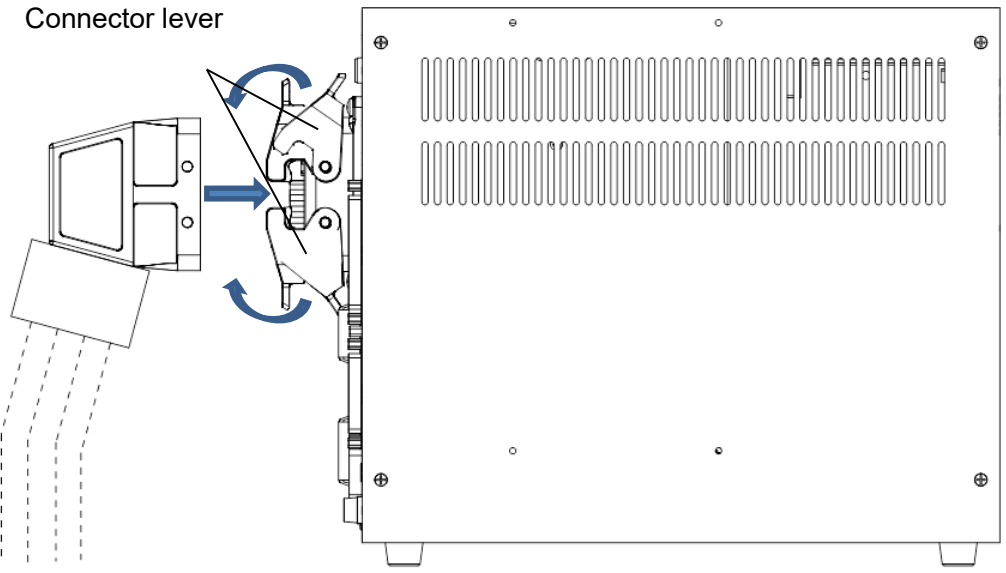


Fig. 17.3 Connecting and disconnecting a ROBOT connector (TSL3000E)

b) Square connectors: INPUT, OUTPUT, TP, HOST/TCPRG, COM1, BRK, HAND
Firstly, completely insert the cable side connector into the controller connector. Then tighten the lock screws on both ends of the cable side connector with a screwdriver. A loose screw can cause a contact failure or other accident. To avoid this, make sure that the screws are clamped completely
To disconnect the connectors, first loosen the lock screws, then pull out the cable side connector. BRK and HAND are quick-operated lock type connectors, however.

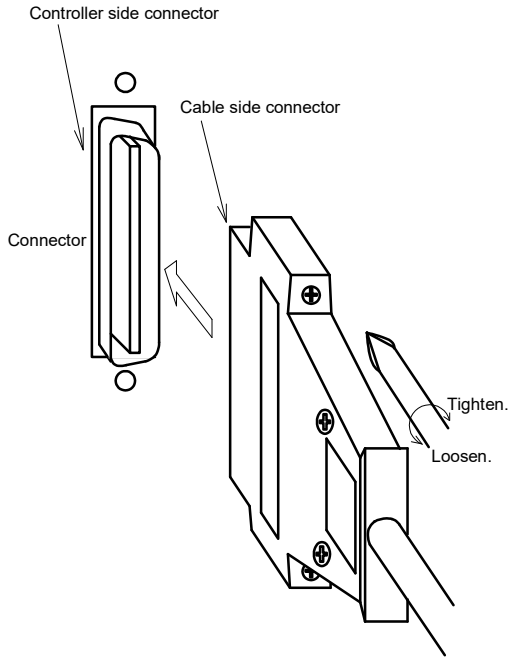


Fig. 17.4 Connecting and disconnecting a square connector (TSL3000E)

c) Circular connector: ACIN

To connect the connector, align the key position, securely insert the connector on the cable side into the connector on the robot controller body side, and then tighten by rotating the lock screw on the cable side clockwise. If the connector is loose, it may cause accidents due to connector contact failure, so be sure to check that the connector is securely inserted.

To disconnect the connector, rotate the lock screw counterclockwise, in the opposite direction from connecting, to loosen the connector, and then pull out the connector on the cable side.

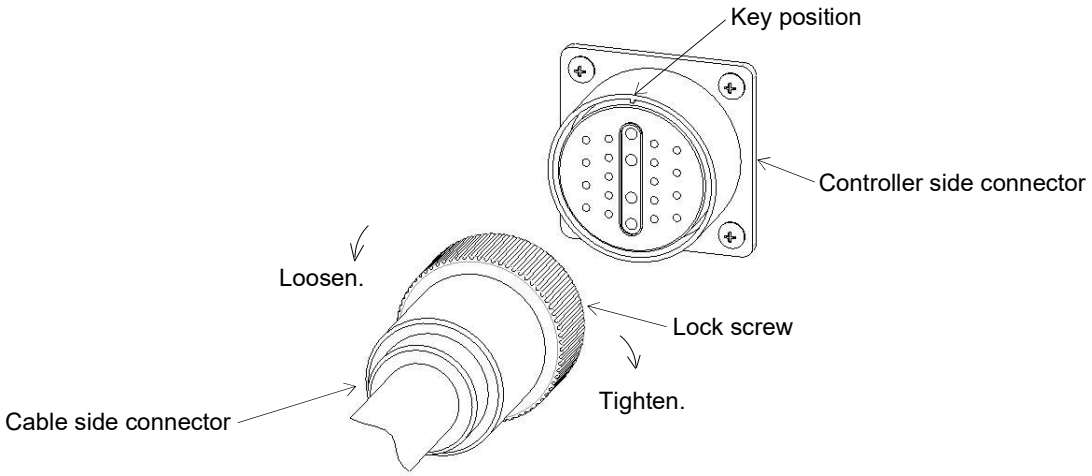
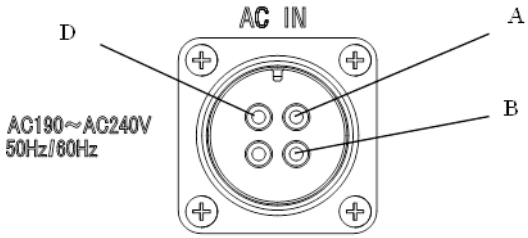


Fig. 17.5 Connectin and disconnecting a circular connector (TSL3000E)

17.1.7 Examples of Connector Terminal Arrangement (for the TSL3000E)

a) TSL3000E Power cable connector "ACIN"



Connects to controller.

Type: JL04V-2E18-10PE-B

Manufacturer: Japan Aviation Electronics Industry, Ltd

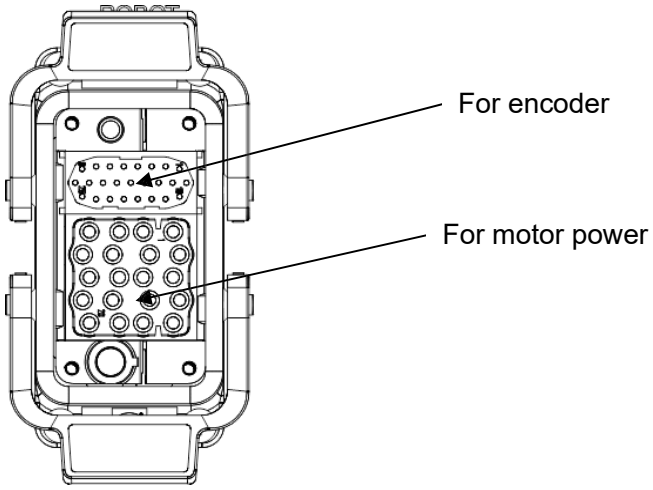
A
B — Single phase, 190 ~ 240 VAC, 50/60 Hz

D — Grounding (Class D or higher level)

⚠ DANGER

- Completely connect the grounding cable. Otherwise, an electric shock or fire may be caused if a fault or electric leak occurred. Or mis-operation may be caused by noise.

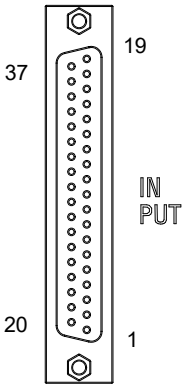
b) Motor/Encoder cable connector "ROBOT"



Connects to controller.

Manufacturer: Harting Co., Ltd.

c) Universal input signal cable connector "INPUT"



Connects to controller.

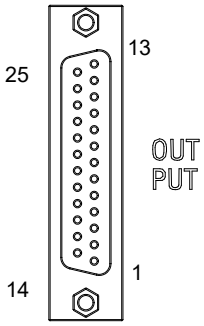
Type: XM3B-3722-112

Manufacturer: Omron Corporation

The counterpart is attached to the controller.

For details, see Part STE85364 TSL3000 Interface or Part STE85457 TSL3000E Interface.

d) Universal output signal cable connector "OUTPUT"



Connects to controller.

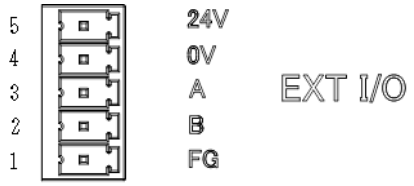
Type: XM3B-2522-112

Manufacturer: Omron Corporation

The counterpart is attached to the controller.

For details, see Part STE85364 TSL3000 Interface or Part STE85457 TSL3000E Interface.

e) TSL3000E Input/output signal power supply and Distributed I/O connector EXT I/O connector



Connects to controller.

Type: ML-4000-CWJH 05PGY

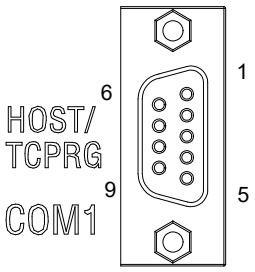
Manufacturer: Sato Parts Co., Ltd.

The counterpart is attached to the controller.

For details, see Part STE85364 TSL3000 Interface or Part STE85457 TSL3000E Interface.

Manufacturer: Sato Parts Co., Ltd.

f) Communication connector "HOST/TCPRG, COM1"

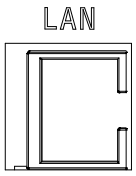


Connects to controller.

Type: XM2C-0942-132L

Manufacturer: Omron Corporation

g) Connector for etherne "LAN"

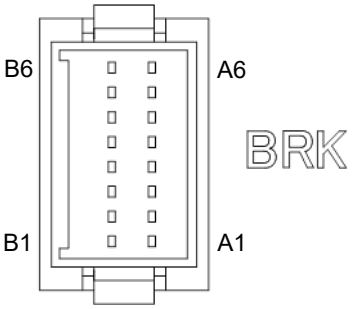


Connects to controller.

Type: J0026D21BNL

Manufacturer: PULSE Electronics

h) Brake signal cable connector "BRK"

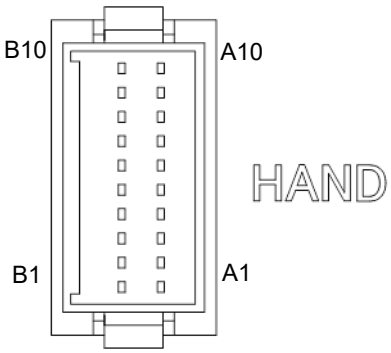


Connects to controller.

Type: J21DPM-12V-KX

Manufacturer: J.S.T. Mfg. Co., Ltd.

i) Robot control signal cable connector “HAND”



Connects to controller.

Type: J21DPM-20V-KX

Manufacturer: J.S.T. Mfg. Co., Ltd.

17.2 Controller Connector Signals

17.2.1 Connector Signal Connection Diagrams

Diagrams showing which signals correspond to which terminals are shown in Section 2 of the TSL3000 Interface Manual STE85364 or the TSL3000E Interface Manual STE85457.

17.2.2 Jumpers for Safety Related Signals

The following system input signals are provided to serve for the safety purpose.

System input signals	INPUT-12	(STOP)	
	INPUT-14	(SVOFF)	
	INPUT-32	(BREAK)	
	INPUT-17, 35	(INCOM–P24V)	Presuming standard P24V(+) common

These signals are already jumpered for the connectors provided for the TSL3100E robot controller. If you wish to use or change them, therefore, you should remove the jumpers and rewire as appropriate. If you plan to use the robot without using system input signals, be sure to connect the attached connectors to the controller side INPUT connector.

Unless the following signals are used as the system signals, jumper them also.

INPUT-13	(LOW_SPD)
INPUT-31	(CYCLE)

Connector jumpers (TSL3000E)

INPUT		
12-16	14-16	32-16
17-35	(13-16)	(31-16)

! CAUTION

- Unless the signals of SVOFF and emergency stop contacts 1, 2 are jumpered, the controller servo power cannot be turned on.
- Unless the CYCLE signal is jumpered, the controller enters the cycle operation mode.
- Unless the LOW_SPD signal is jumpered, the robot is operated at low speed during automatic operation.
- Unless the STOP signal is jumpered, automatic operation of the robot is not possible.
- Unless the BREAK signal is jumpered, automatic operation of the robot is not possible.

18. System Connections (TS3000 and TS3000E)

18.1 Cable Wiring

For information on TSL3000, refer to Section 16. For information on TSL3000E, refer to Section 17. This section describes the various types of cables and connectors and explains how these are to be connected.

18.1.1 Connector Arrangement on the Controller

The cables connected to the robot controller are shown in Fig. 18.1.

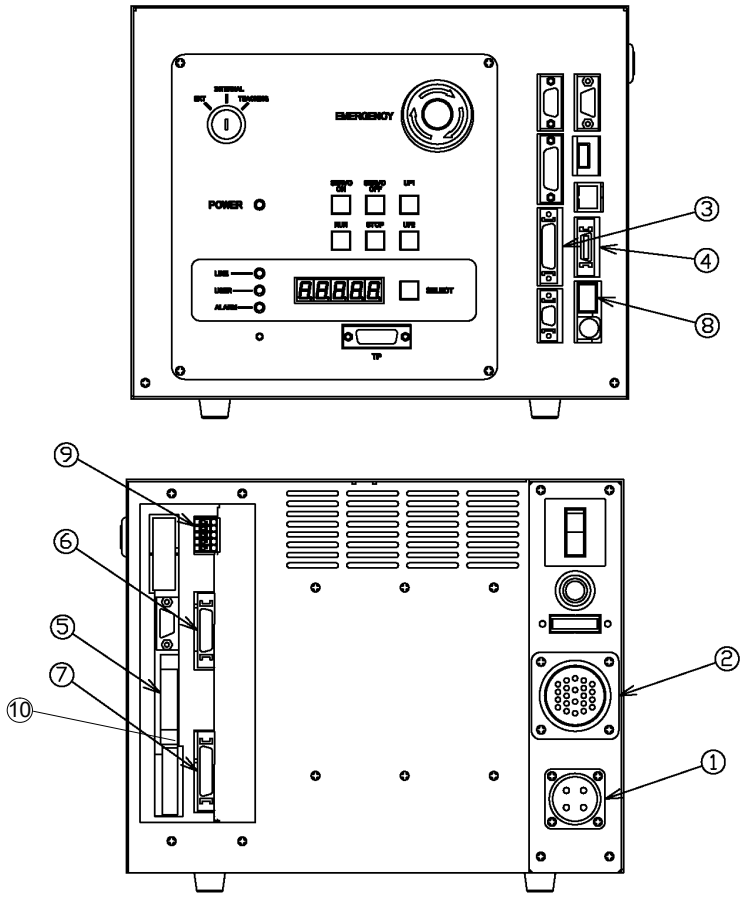


Fig. 18.1 Robot controller connector arrangement (TS3000 and TS3000E)

① Power cable (ACIN)
② Motor cable (MOTOR)
③ Encoder cable (ENC)
④ Robot control signal cable (HAND)
⑤ External operation input signal cables (SYSTEM)
⑥ External operation input signal cables (INPUT)
⑦ External operation input signal cables (OUTPUT)

⑧ Brake signal cable (BRK)
⑨ Distribution I/O cable (EXT-I/O)
⑩ External input/output power cable "EMS"

In the subsequent paragraphs, we explain how to connect cables ① to ④ and ⑩ inclusive. For information on how to connect cables ⑤ to ⑨, refer to the TS3000 Interface Manual STE80729.

18.1.2 Connecting the Power Cable "ACIN" (① of Fig. 18.1) (plug connector attached)

The power cable is used to supply the main AC power to the controller.

Table 18.1 Power supply specifications (TS3000 and TS3000E)

Power supply	Single phase, 200 - 240 VAC, 50/60 Hz ±1 Hz
Power capacity	0.7 kVA (THL300 and THL400) 1.4 kVA (THL500 to THL1000)
Instantaneous power failure	With 50 Hz: Within 40 msec With 60 Hz: Within 32 msec
Grounding	JIS class D

The connector is ACIN (① of Fig. 18.1).

ACIN plug connector	Type:JL04V–6A18–10SE–EB	Maker: Japan Aviation Electronics Industry, Ltd
ACIN cable clamp	Type: JL04–18CK	Maker: Japan Aviation Electronics Industry, Ltd
Wire	3.5 mm^2 - 5.5 mm^2	

As the cable is not an accessory, use the attached plug connector connected to ACIN on the controller side to manufacture a cable.
Wires are to be soldered to the connector.

! DANGER

- Be sure to use the designated wire. Failure to do so could lead to fires or faults.
- When connecting the connector and wires, make sure not to mistake the terminal arrangement.
- After making the connection, use a tester, etc., to confirm the connection.

For the terminal arrangement, see Para. 18.1.9.

! CAUTION

- Unless the main power is normally supplied to the controller due to phase defect or voltage drop, an error of "8–027 Slow Charge error" occurs. When this happens, make sure that the maser power voltage at the controller power connector satisfies the specified input power of the controller, and that the same voltage is stabilized.
- For details of the 8–027 error, see Para. 13.7 of the Operator's Manual.

18.1.3 Connecting the Motor Cable "MOTOR" (② of Fig. 18.1) (THL300 to THL1200)

The motor cable is integrated with the encoder cable, robot control signal cable, and brake signal cable. This integrated cable is further integrated with the robot. The motor cable connects the controller and robot, and supplies the power required to rotate the motor from the controller servo driver to each axis feed motor of the robot.

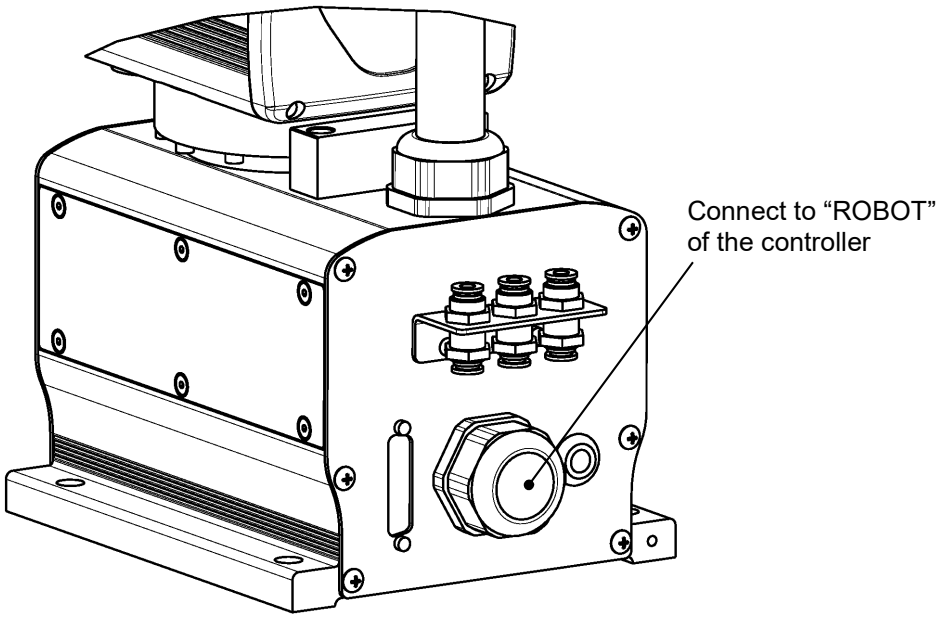


Fig. 18.2 Robot side connector arrangement (TS3000 and TS3000E)

18.1.4 Connecting the Encoder Cable "ENC" (③ of Fig. 18.1)

The encoder cable is integrated with the motor cable, robot control signal cable, and brake signal cable. This integrated cable is further integrated with the robot. The encoder cable is a signal line used to transmit a signal from the rotation angle detecting encoder of each robot axis to the controller.
The connector for connecting the encoder cable is ENC (③ of Fig. 18.1).

18.1.5 Connecting the Robot Control Signal Cable "HAND" (④ of Fig. 18.1)

The robot control signal cable is integrated with the motor cable, encoder cable, and brake signal cable. This integrated cable is further integrated with the robot. The robot control signal cable is used for input and output of robot control signals such as hand operation signal.
The connector for connecting the robot control signal cable is HAND (④ of Fig. 18.1).

18.1.6 Connecting the Brake Signal Cable "BRK" (⑧ of Fig. 18.1)

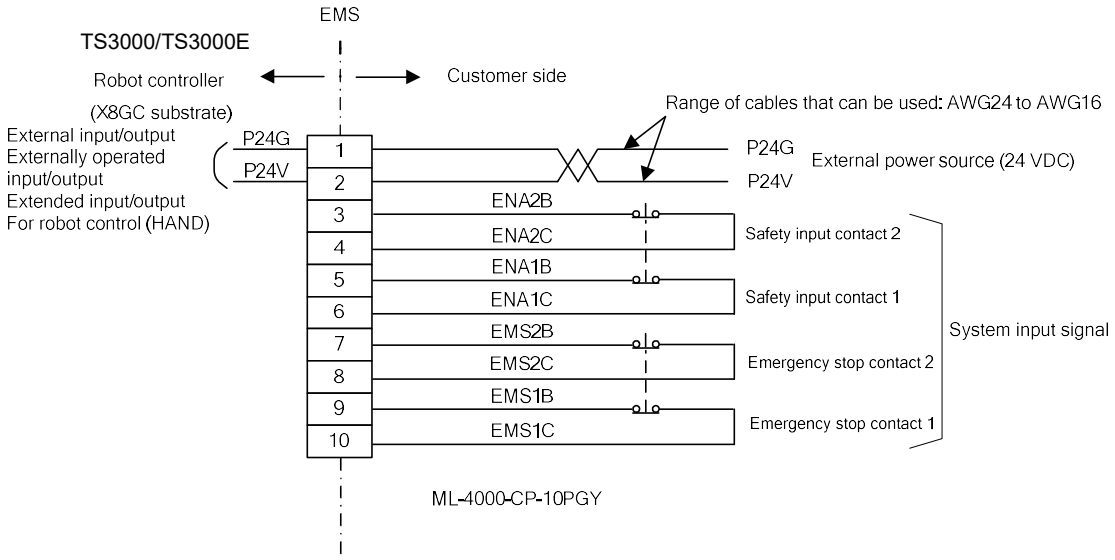
The brake signal cable is integrated with the motor cable, encoder cable, and robot control signal cable. This integrated cable is further integrated with the robot. The brake signal cable is used for motor brake ON and OFF.
The connector for connecting the brake signal cable is BRK (⑧ of Fig. 18.1).

18.1.7 Connecting the External Input/Output Power Cable "EMS" (⑩ of Fig. 18.1)

Use the attached connector (ML-4000-CP-10PGYT) to connect the external input/output power cable, similarly to the case of connecting the input signal cable for safety. The P24V power is supplied from the connector "EMS" on the rear of the controller.
The following shows the I/O using the external supply power (24 VDC). Be sure to supply power (24 VDC) from the outside.

- External input/output
- Externally operated input/output
- Extended input/output
- Manual input/output

It should be noted that the controller internal power (24 VDC) is used for the trigger input and brake output.



The connector EMS compatible cables are "AWG24 through 16").
Select the optimum external power source in conformity to the customer's system specifications (power capacity).
To connect and disconnect the connector EMS, see the description of "Interface Part".

CAUTION

Be sure to supply external power (24 VDC). If not, power cannot be turned on, independently of safety measures.

18.1.8 Connecting and Disconnecting Cables

CAUTION

- Before connecting or disconnecting any controller cable, be sure to turn off the main power ("POWER") switch.
- When disconnecting a cable, be sure to pull the plug and not the cord. Otherwise, you may damage the cable.

a) Circular connectors: ACIN, MOTOR

Firstly align the key position, and completely insert the connector on the cable side into the controller connector. Then turn the cable side lock screw to the right to clamp the cable. A loose screw can cause a contact failure or other accident. To avoid this, make sure that the screw is clamped completely.

To disconnect the connectors, turn the lock screw to the left and pull out the cable side connector.

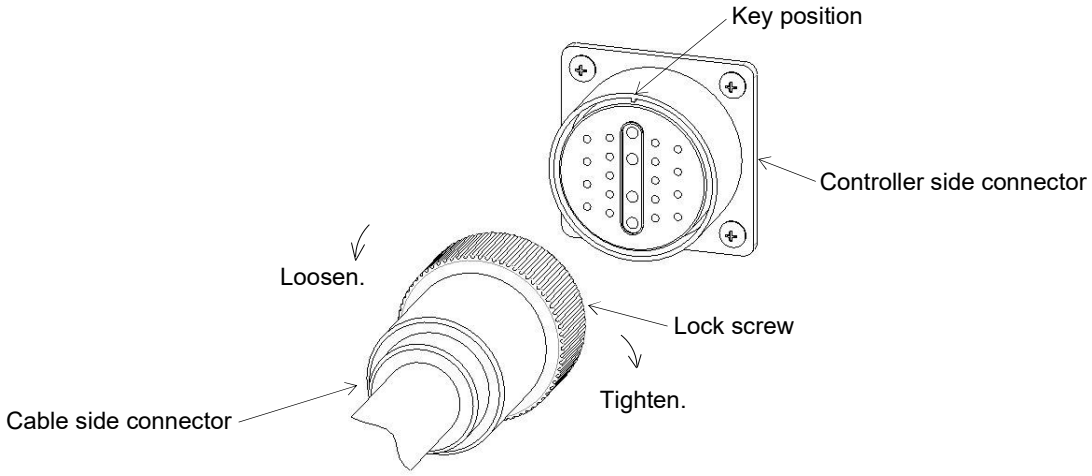


Fig. 18.3 Connecting and disconnecting a circular connector

b) Square connectors: ENC, HAND, SYSTEM, INPUT, OUTPUT, TRIG, CONV
Firstly, completely insert the cable side connector into the controller connector. Then tighten the lock screws on both ends of the cable side connector with a screwdriver. A loose screw can cause a contact failure or other accident. To avoid this, make sure that the screws are clamped completely.
To disconnect the connectors, first loosen the lock screws, then pull out the cable side connector. INPUT, OUTPUT is a quick-operated lock type connector, however.

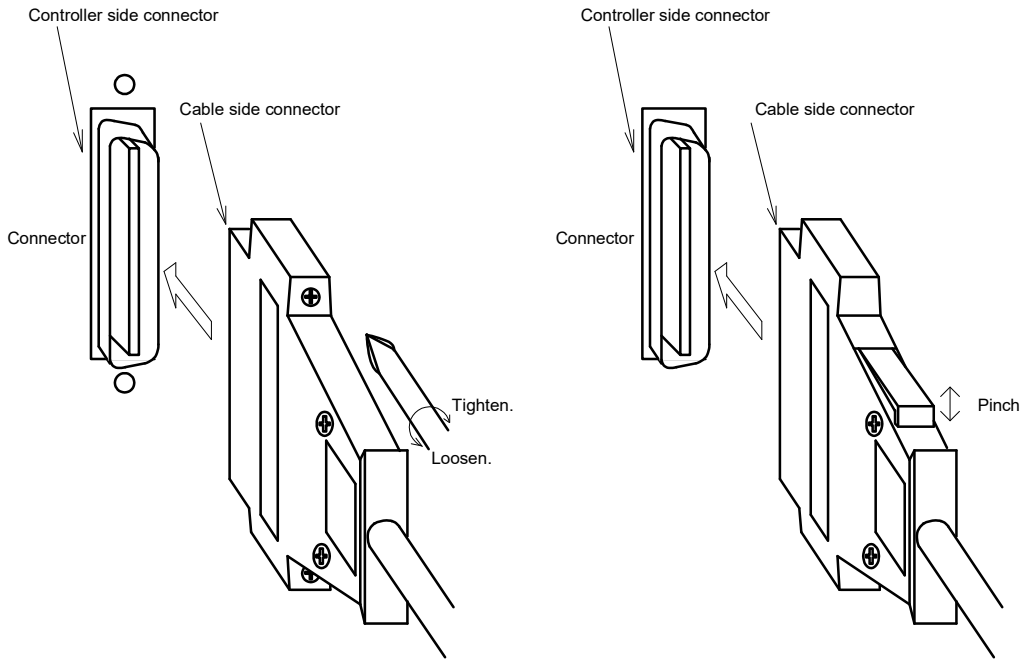
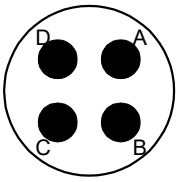


Fig. 18.4 Connecting and disconnecting a square connector

18.1.9 Examples of Connector Terminal Arrangement (Controller Side)

a) Power cable connector ACIN



Connects to controller.
Type: JL04V–2E18–10PE–B
Manufacturer:
Japan Aviation Electronics Industry, Ltd

A
B
C
D

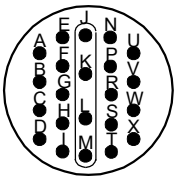
Single phase, 200 - 240 VAC, 50/60 Hz

Grounding (Class D)

DANGER

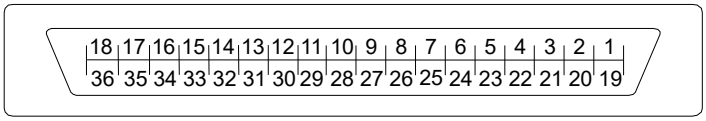
- Completely connect the grounding cable.
  Otherwise, an electric shock or fire may be caused if a fault or electric leak occurred. Or misoperation may be caused by noise.

b) Motor cable connector MOTOR



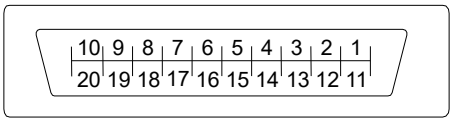
Connects to controller.
Type: JL04V-2A28-11SE
Manufacturer:
Japan Aviation Electronics Industry, Ltd

c) Encoder cable connector ENC



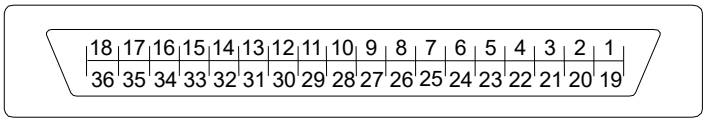
Connects to controller.
Type: 52986-3659
Manufacturer: MOLEX

d) Robot control signal cable connector HAND



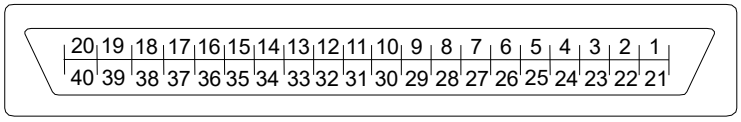
Connects to controller.
Type: 52986-2079
Manufacturer: MOLEX

e) General-purpose input signal cable connector INPUT



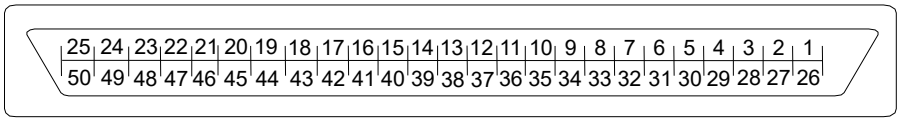
Connects to controller.
Type: DHA-RC36-R132N-FA
Manufacturer: DDK

f) General-purpose output signal cable connector OUTPUT



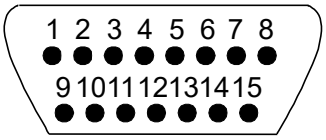
Connects to controller.
Type: DHA-RC40-R132N-FA
Manufacturer: DDK

g) System input/output signal cable connector SYSTEM



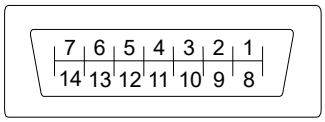
Connects to controller.
Type: 52986-5079
Manufacturer: MOLEX

h) Trigger input connector TRIG



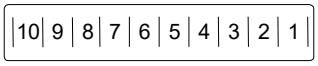
Connects to controller.
Type: XM2C-1542-112L
Manufacturer: OMRON

i) Encoder cable connector CONV



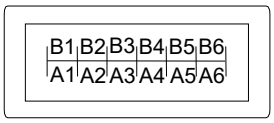
Connects to controller.
Type: 52986-1479
Manufacturer: MOLEX

j) Emergency stop, safety input, and external P24V supply connector EMS



Connects to controller.
Type: ML-4000CWJH-10PGY
Manufacturer: SATO PARTS

k) Brake connector BRK



Connects to controller.
Type: 1-1827876-6
Manufacturer:
Tyco Electronics AMP

18.2 Controller Connector Signals

For information on TSL3000, refer to Section 16. For information on TSL3000E, refer to Section 17.

18.2.1 Connector Signal Connection Diagrams

Diagrams showing which signals correspond to which terminals are shown in Section 2 of the TS3000 Interface Manual STE80729.

18.2.2 Jumpers for Safety Related Signals

The following system input signals are provided to serve for the safety purpose.

System input signals	SYSTEM-12	(STOP)
	SYSTEM-16	(SVOFF)
	SYSTEM-14	(BREAK)
	EMS-7, 8	(EMS2B - EMS2C)
	EMS-9, 10	(EMS1B - EMS1C)
	EMS-3, 4	(ENA2B - ENA2C)
	EMA-5, 6	(ENA1B - ENA1C)

These signals are already jumpered for the connectors provided for the TS3000 and TS3000E robot controller. If you wish to use or change them, therefore, you should remove the jumpers and rewire as appropriate. If you plan to use the robot without using system input signals, be sure to connect the attached connectors to the controller side SYSTEM, EMS connectors.
Unless the following signals are used as the system signals, jumper them also.

SYSTEM-15	(LOW_SPD)
SYSTEM-13	(CYCLE)

Connector jumpers

SYSTEM		EMS	
12-17 (18)	14-17 (18)	3-4	5-6
16-17 (18)	(13-17 (18))	7-8	9-10
(15-17 (18))	-		

! CAUTION

- Unless the signals of SVOFF and emergency stop contacts 1, 2 are jumpered, the controller servo power cannot be turned on.
- Unless the CYCLE signal is jumpered, the controller enters the cycle operation mode.
- Unless the LOW_SPD signal is jumpered, the robot is operated at low speed during automatic operation.
- Unless the STOP signal is jumpered, automatic operation of the robot is not possible.
- Unless the BREAK signal is jumpered, automatic operation of the robot is not possible.

18.3 Separating Control Panel from Controller

Separate control panel option is not available for TSL3000 and TS3000E.

18.3.1 Removing Control Panel

Remove the control panel in the following manner.

a) Loosen the four (4) screws at the four (4) corners, which secure the control panel.

b) Remove these screws, then carefully draw out the control panel toward your side.

Caution: Be careful of the cable connected on the rear side.

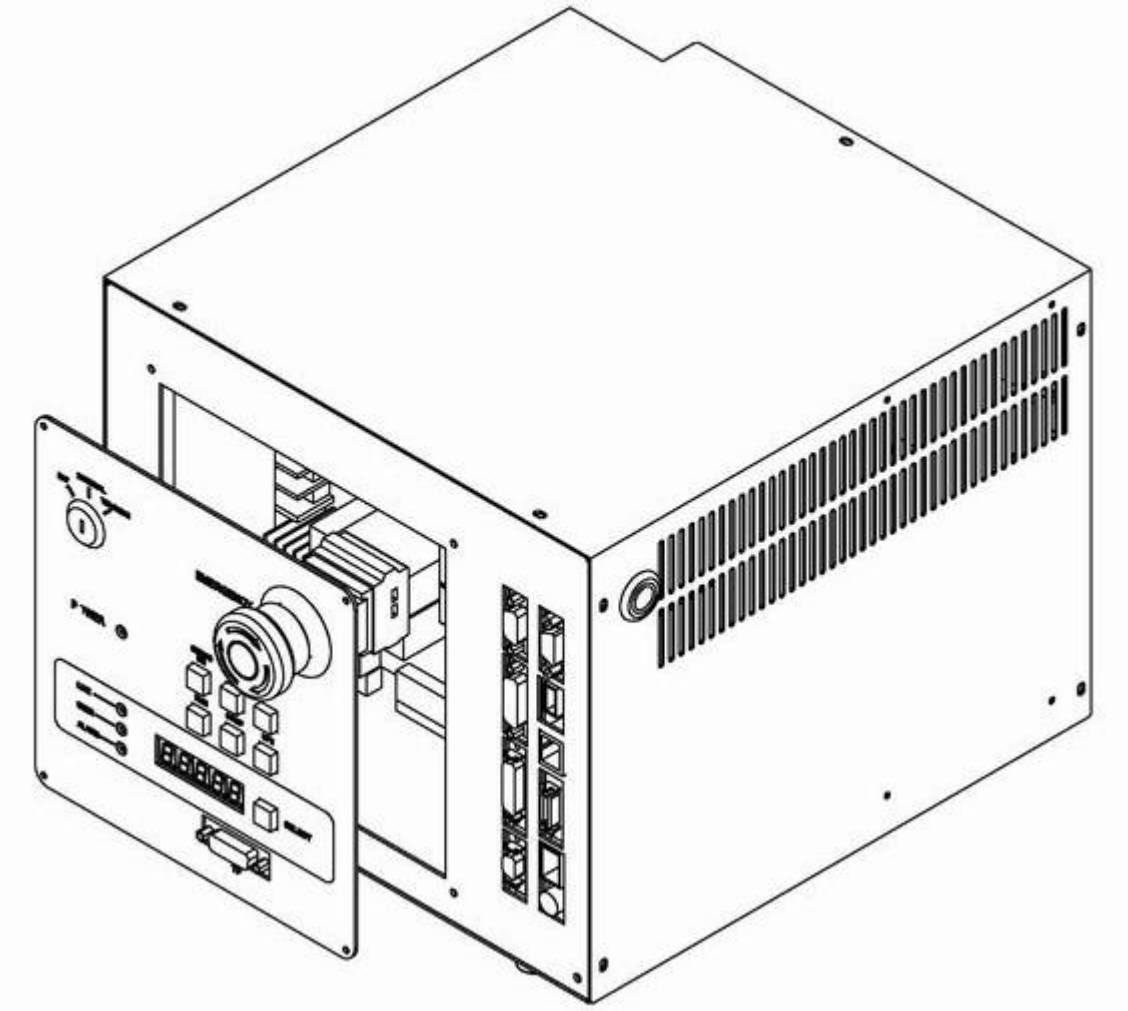


Fig. 18.5 Removing control panel

18.3.2 Cable between Controller and Control Panel (option)

The cable required to connect the control panel and controller when they are installed separately can be provided optionally.

18.3.3 Control Panel Mounting Dimensions

The dimensions of mounting the control panel are shown in Fig. 18.6.
Cross truss head screws (ϕ3 x 6, ZN3–B) are used.

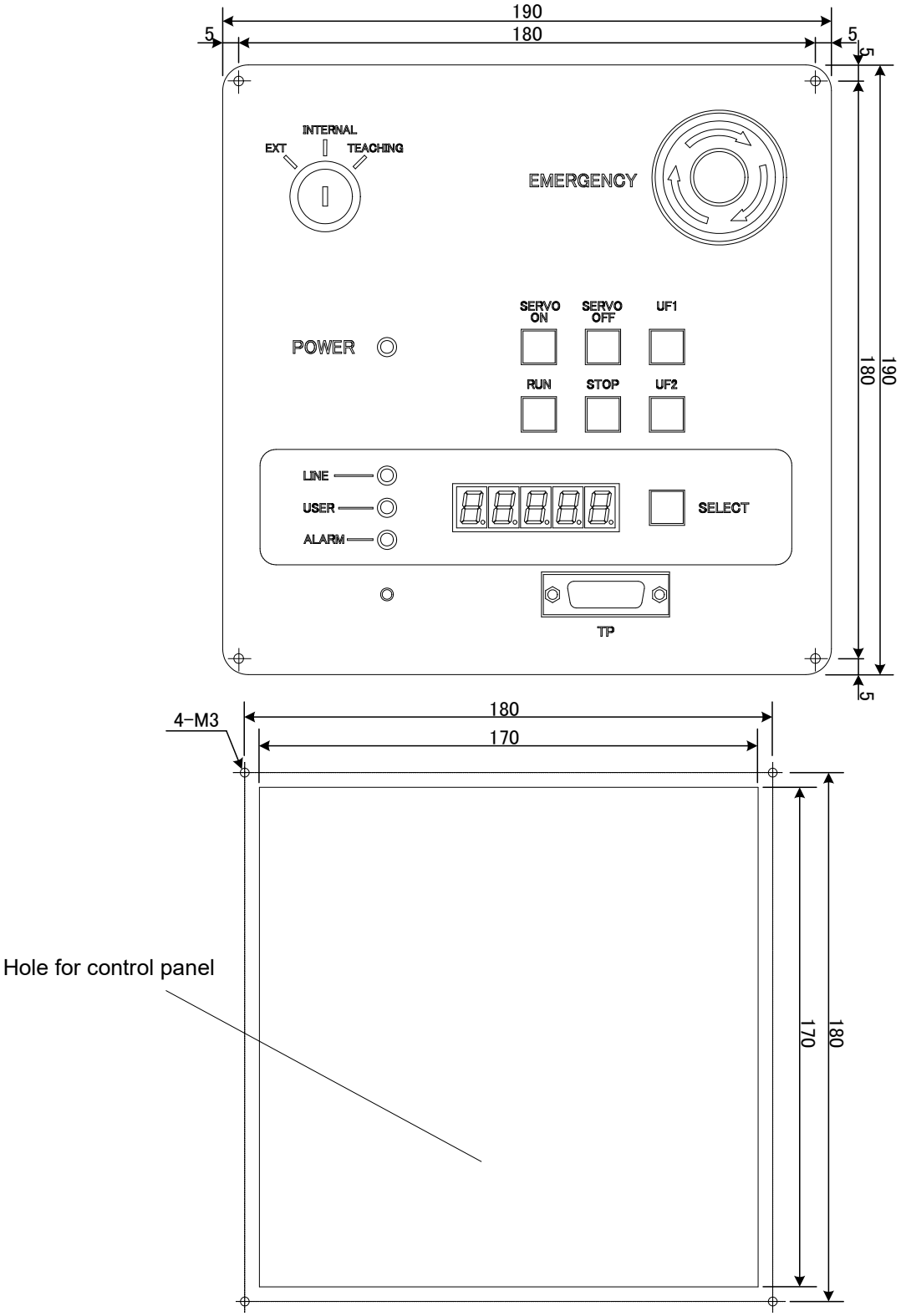


Fig. 18.6 Control panel mounting dimensions

18.3.4 Mounting Dummy Panel on Controller (option)

When the control panel has been disengaged from the controller, mount a dummy panel on the place where the control panel was set before, as shown in Fig. 18.7. The dummy panel, mounting parts, etc. are provided optionally.

a) Connect the cable connector which was disconnected when separating the controller from the control panel, to the rear side of the dummy panel, then screw both ends of the connector. When mounting the connector, use the cross truss screws (ϕ3 x 6, ZN3–B).

b) Screw the dummy panel into the controller.

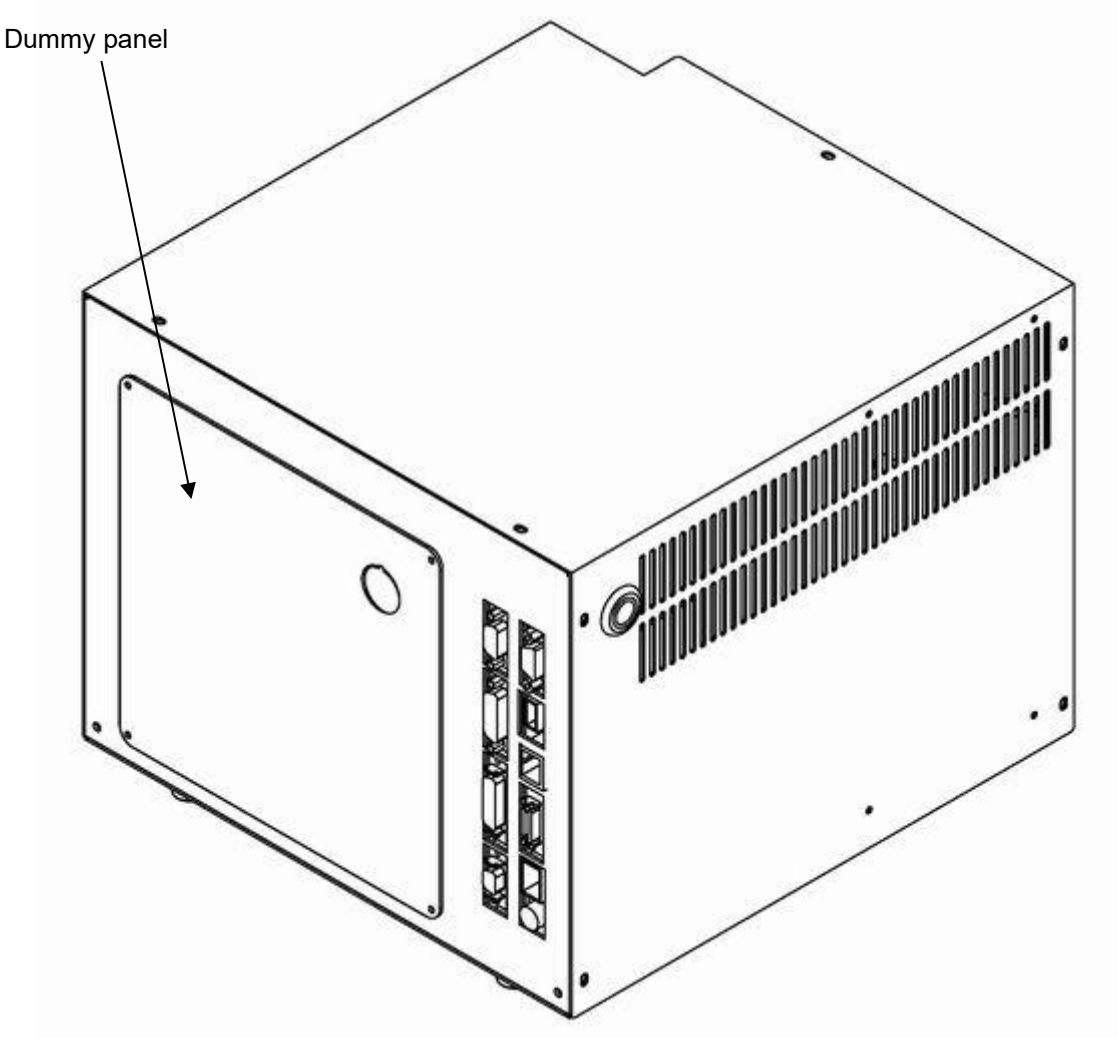


Fig. 18.7 Mounting dummy panel

18.3.5 Dimensions when Separating Control Panel

Fig. 18.8 shows the connections of the control panel and dummy panel. Provide a clearance of 50 mm or more (with cover, 60 mm or more) on the rear side of the separated control panel.
When the cable is connected to the dummy panel of the controller, provide a clearance of 80 mm or so in front of the controller as the cable connector sticks out of the panel surface.

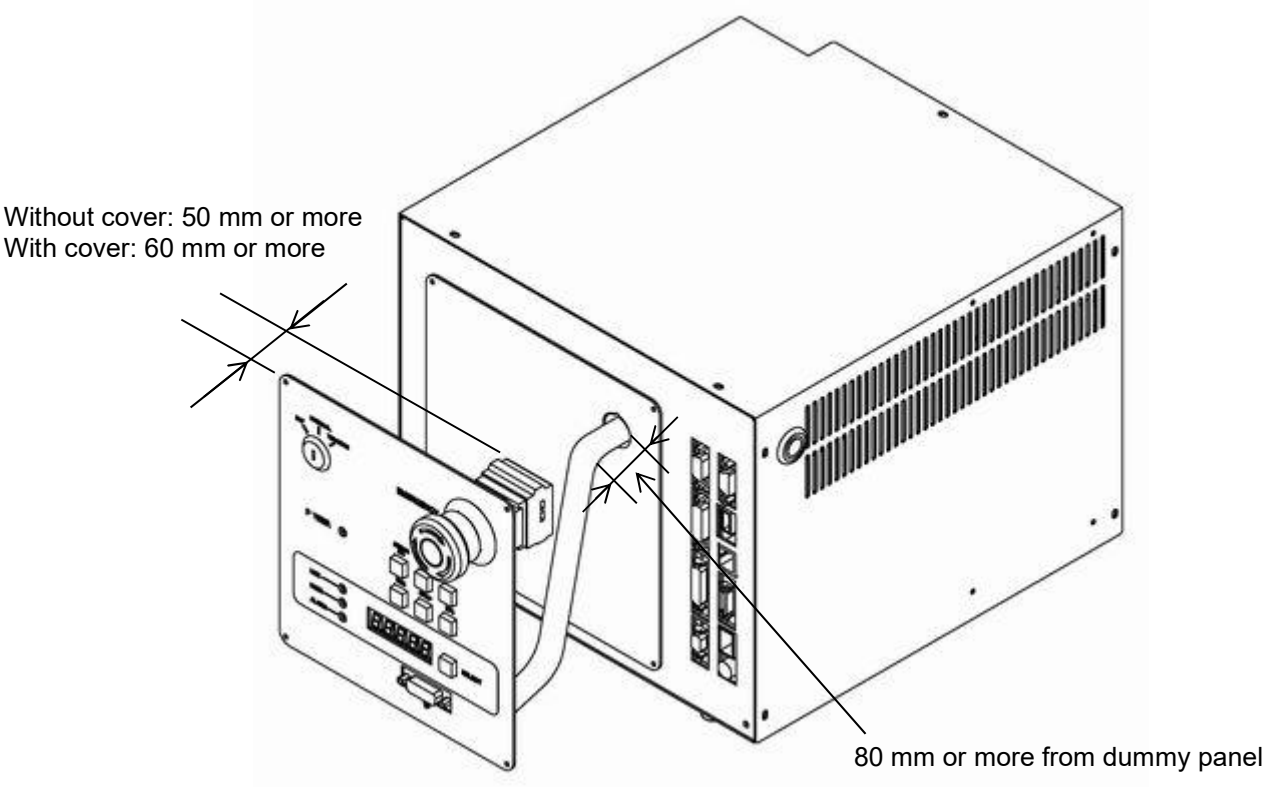


Fig. 18.8 Cable connections of dummy panel and control panel

19. Tool Interface (THL300 and THL400)

19.1 Mounting Tool

The tool is mounted on the end of the tool shaft. Dimensions of the tool shaft section are shown in Fig. 19.1. As shown in Fig. 19.1, the tool is centered with the ϕ12H7 mating section. The tool direction is adjusted by means of the 4 x 4 keys and secured with four (4) M4 bolts.
The tool flange is optional.

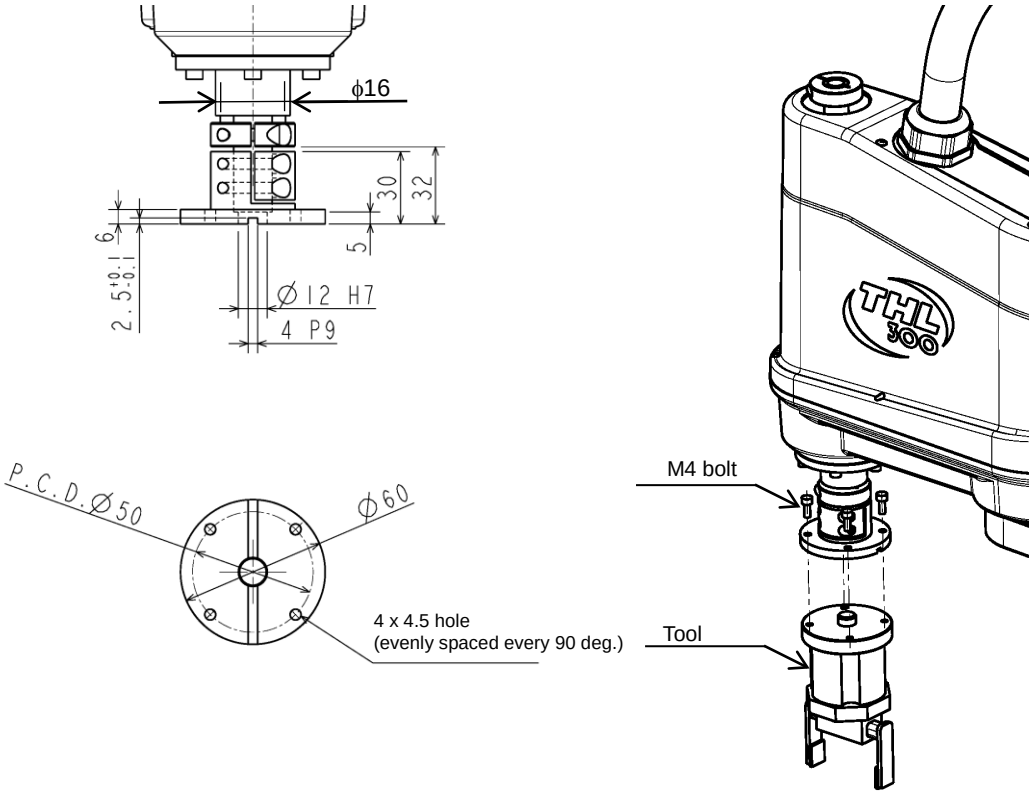


Fig. 19.1 Tool mounting dimensions (THL300 and THL400)

19.2 Tool Air Piping

The customer is requested to install the tool air piping using the following accessories.

- Air tube x3 (red, white, blue)

Fig. 19.2 shows an installation example of tool air piping for your reference.

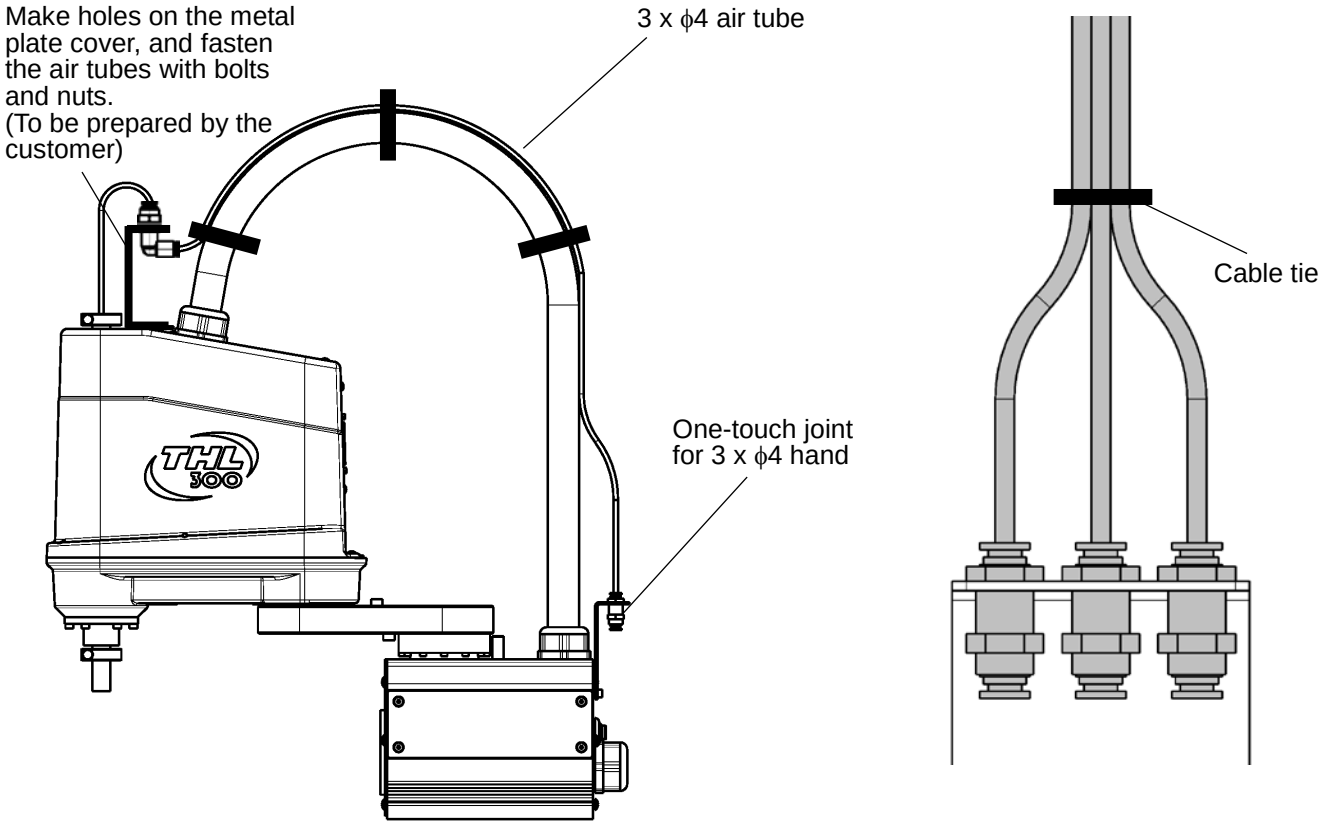


Fig. 19.2 Tool air piping installation example (THL300 and THL400)

! CAUTION

- Air tubes are consumable supplies. Check the air tubes during periodic inspection, and change them if any damage is found.
- The customer is requested to prepare solenoid valve air.
- Fig. 19.2 is a reference example. Tool air piping is not guaranteed. The warranty does not cover breakdowns of the harness, the air tubes or their accessories due to piping by the customer.

19.3 Permissible Load Conditions and Program Setting

This paragraph describes the permissible load conditions of the robot and how to set up the program according to the load.

19.3.1 Permissible Load Conditions

The robot load conditions are defined by the tool mass, moment of inertia and offset value of tool gravity center from the center of the tool shaft, as shown in Fig. 19.3. The permissible load conditions are shown in Table 19.1.

Table 19.1 Permissible load conditions (THL300 and THL400)

Conditions	Permissible values
Mass	Max. 5 kg
Load inertia	Max 0.05 kg·m^2
Offset value of load gravity center	Max. 100 mm

! CAUTION

- NEVER operate the robot under the load conditions exceeding the permissible values. Otherwise, the robot life and safety cannot be guaranteed.

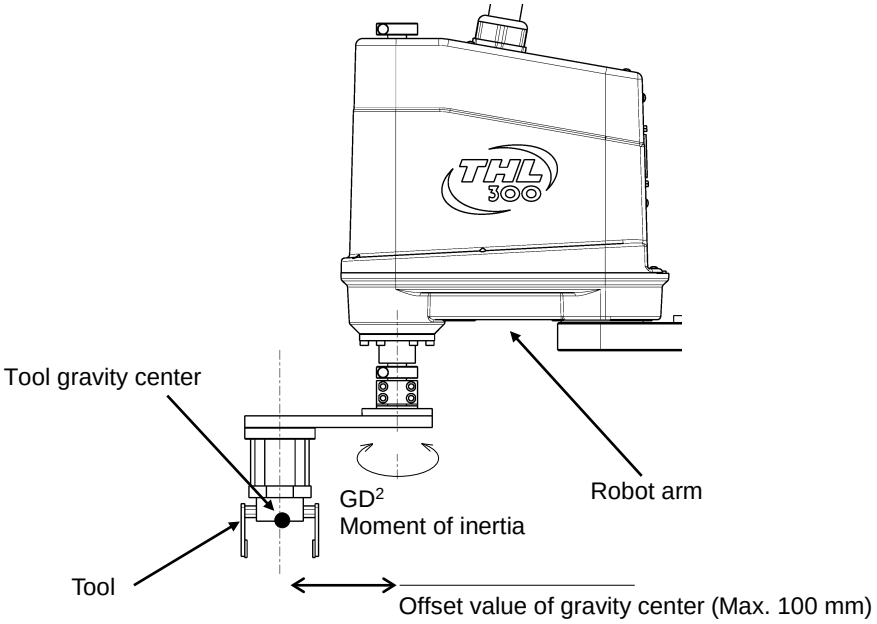


Fig. 19.3 Robot tool

19.3.2 Load Conditions and Program Setting

This robot can automatically change the maximum speed, acceleration/deceleration and servo gain by using the PAYLOAD command in the program according to the load conditions.

Be sure to use the PAYLOAD command.

The specific method for using this function is explained below.

a) PAYLOAD command format

The PAYLOAD command format is written as shown below if the tool mass is M kg and the gravity center offset is L mm.

PAYLOAD = {M, L}

M	:	Load mass	(unit: kg)
L	:	Offset value of gravity center	(unit: mm)

The PAYLOAD command has the following functions.

- The maximum speed and acceleration/deceleration of each robot axis are automatically changed according to the set load conditions.
- The servo gain of each robot axis is automatically changed according to the set load conditions.

b) Program examples

Basic program examples using the PAYLOAD command are shown below. For further information, see the Robot Language Manual.

(Program example 1)

The robot is moved under the load conditions of 5 kg mass and 100 mm gravity center offset.

```
PROGRAM SAMPLE
   SPEED=100
   PAYLOAD={5,100}
   MOVE P1
   MOVE P2
   STOP
END
```

(Program example 2)

When the hand mass is 3 kg and the gravity center offset is 30 mm, and the mass is 5 kg and gravity center offset is 50 mm when the workpiece is grasped.

Pick-and-place operation is executed under the above conditions.

```
PROGRAM SAMPLE
    PAYLOAD={3,30}
    ACCUR=COARSE
    ENABLE NOWAIT
    RESET DOUT
    MOVE P0
    DOUT(1)
    WAIT DIN (1)
LOOP:
    MOVE P1+POINT(0,0,100)
    IF DIN (-1)THEN GOTO FIN
    MOVE P1
    WAIT MOTION>=100
    DOUT(213)
    DELAY 1
    PAYLOAD={5,50}
    MOVE P1+POINT(0,0,100)
    MOVE P2+POINT(0,0,100)
    MOVE P2
    WAIT MOTION>=100
    DOUT(-213)
    DELAY 1
    PAYLOAD={3,30}
    MOVE P2+POINT(0,0,100)
    GOTO LOOP
FIN:
    MOVE P0
    DOUT(1)
    STOP
END
```

c) Setting of PAYLOAD command

In the default state, or when the PAYLOAD command is not used, the maximum speed and acceleration/deceleration are set to 100 % and the servo gain is set to the value under the minimum load. See "Setting Maximum Speed and Robot Acceleration/Deceleration for Load Conditions" of 19.3.3, 20.3.3, 21.3.3 and 22.3.3.

! CAUTION

- Be sure to use the PAYLOAD command.
- Failure to use the PAYLOAD command will cause malfunctioning of the robot or shortening of the life of the mechanisms. In the worst case, the mechanism will be damaged.
- Even when the PAYLOAD command is used, regulate the speed by using the SPEED or DECEL command while confirming the workpiece behavior subject to handling.
- Micro vibration may occur depending on the robot posture. If micro vibration occurs, reduce the acceleration speed to use the robot.

! CAUTION

- The load moment of inertia should be within the tolerances given in Tables 19.1, 20.1, 21.1 and 22.1.
- Even if there is no offset of load gravity center, if the moment of inertia is large, the robot may vibrate. When this happens, figure out the virtual gravity center offset (L mm) from the following equation, using the moment of inertia (J kg·m^2) and mass (M kg).

  $L = \sqrt{(J \times 10^6 / M)}$

  Then, designate the following command.

  PAYLOAD = {M, L}

! CAUTION

- When guiding manually, the robot may vibrate if the load mass or gravity center offset is large. This is because the servo gain is not appropriate. When this happens, perform the following operation while setting the load conditions in the test run mode.

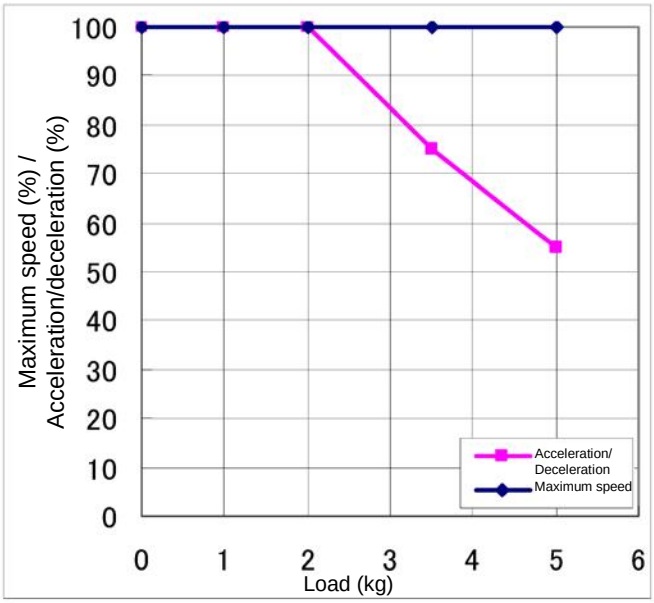
  DO PAYLOAD = {M, L} M: Mass [kg] L: Offset [mm]
  EXE

  The servo gain is changed then to the value which meets the load conditions.

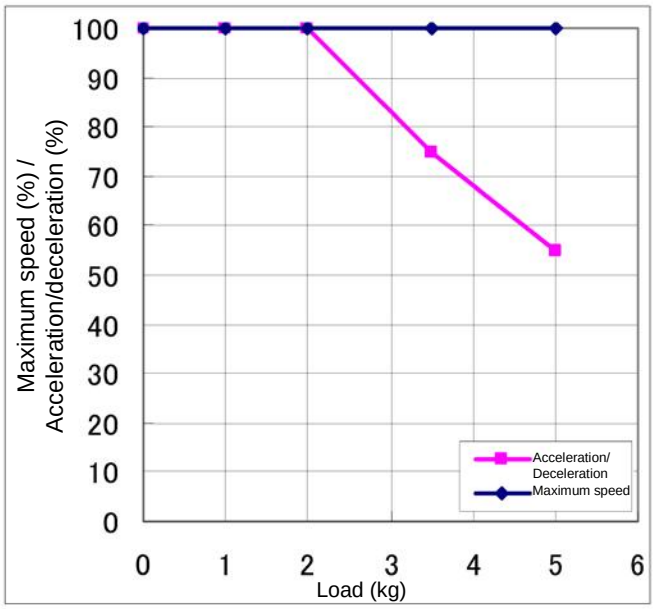
19.3.3 Setting Maximum Speed and Robot Acceleration/Deceleration for Load Conditions

When the PAYLOAD command is used, the maximum speed and acceleration/deceleration of the robot are automatically changed according to the load conditions.

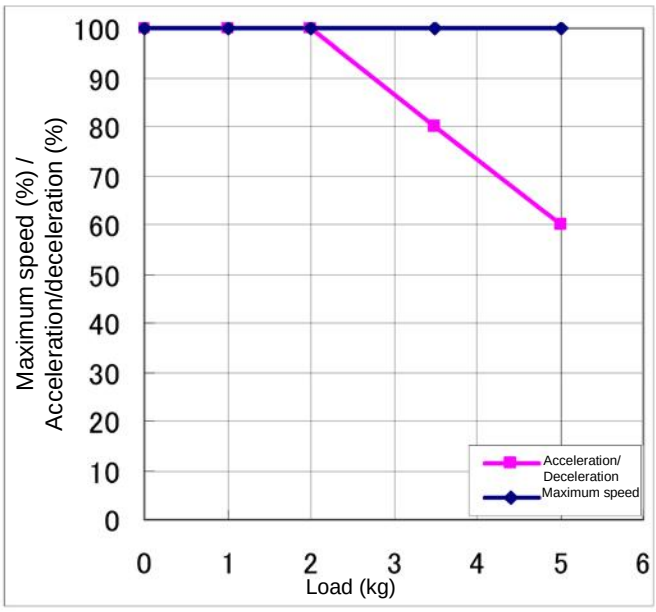
The maximum speed and acceleration/deceleration change with the load mass, as shown in Figs. 19.4 to 19.4.



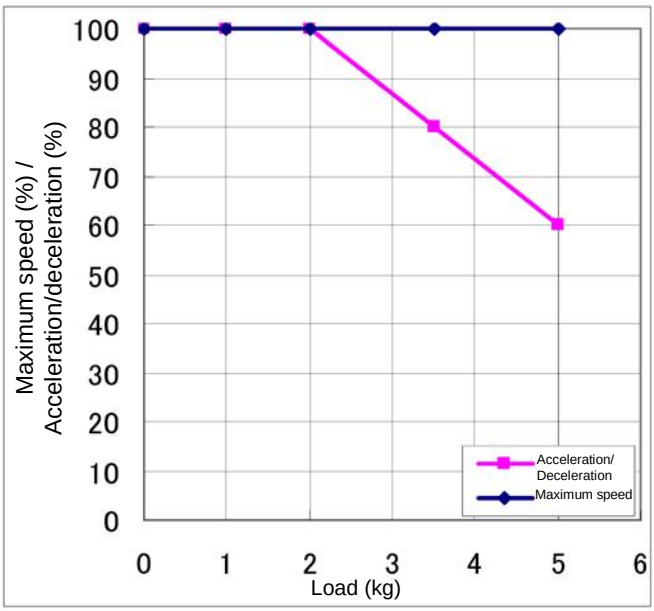
Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 1)



Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 2)

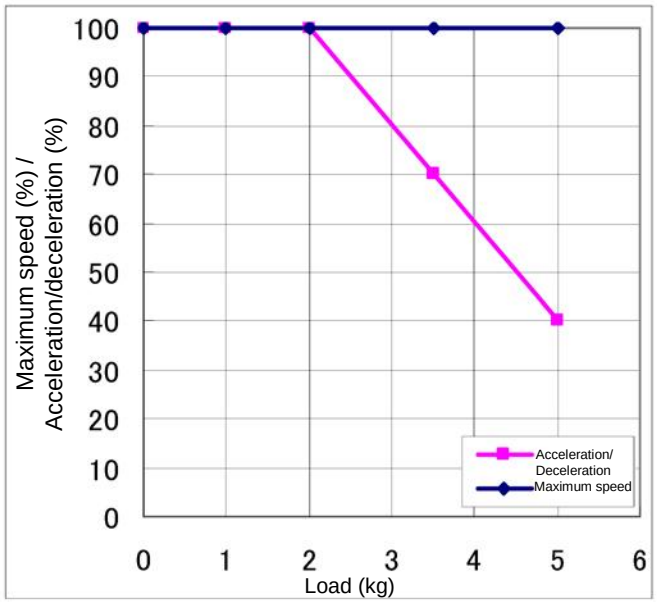


Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 3)

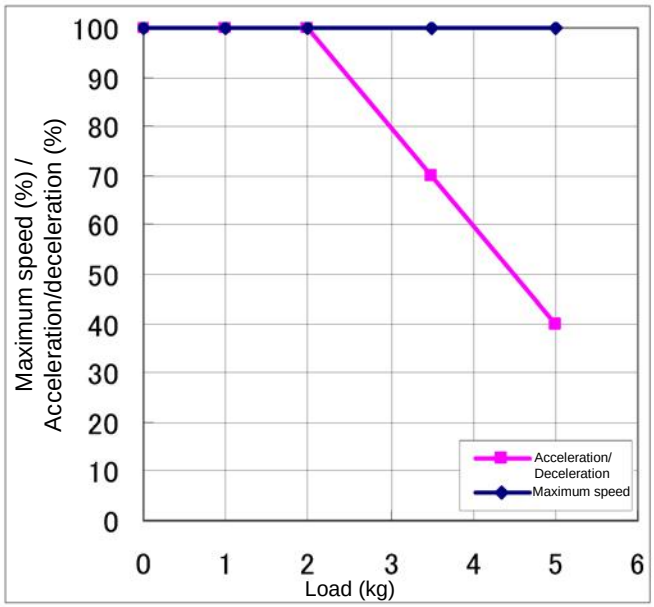


Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 4)

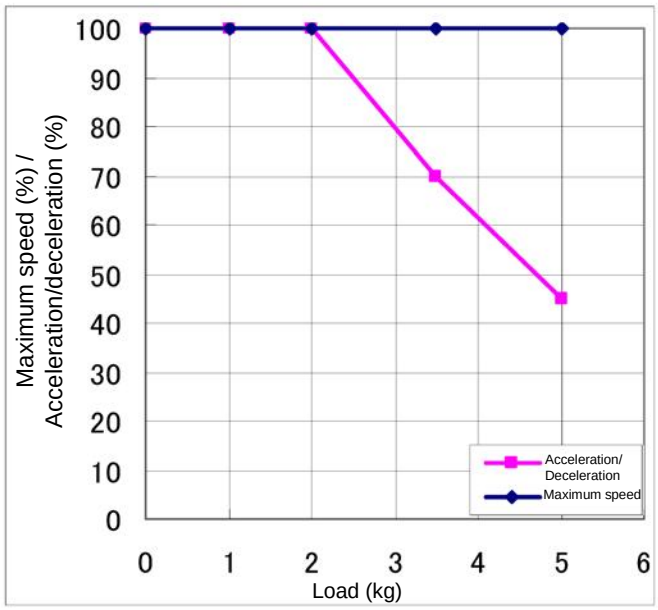
Fig. 19.4 Setting of maximum speed and acceleration/deceleration in relation to load mass (THL300)



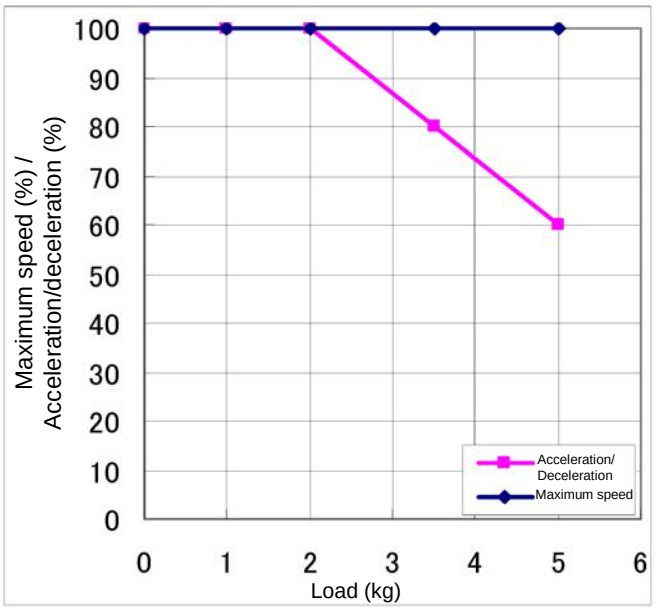
Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 1)



Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 2)



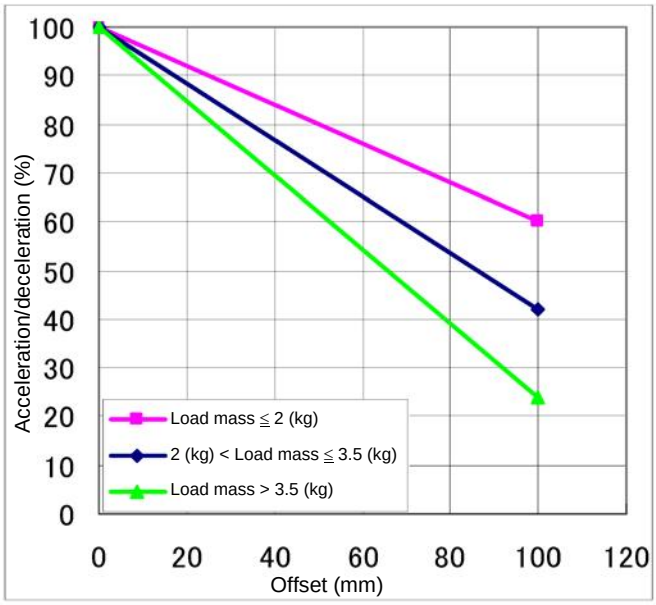
Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 3)



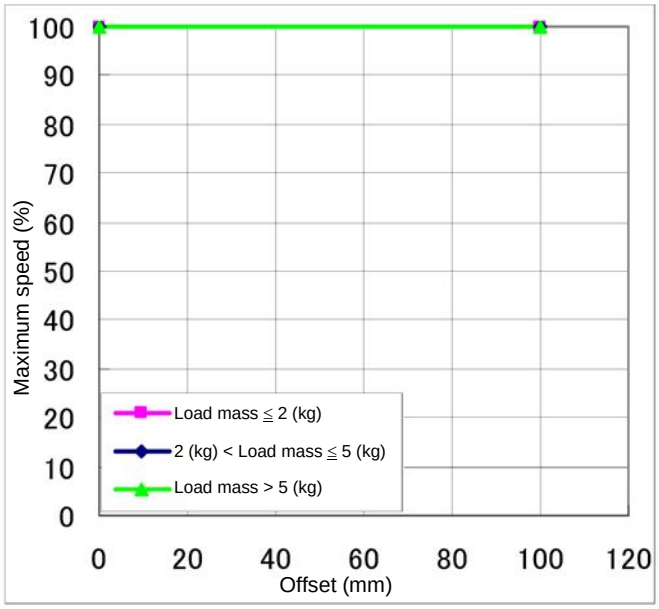
Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 4)

Fig. 19.5 Setting of maximum speed and acceleration/deceleration in relation to load mass (THL400)

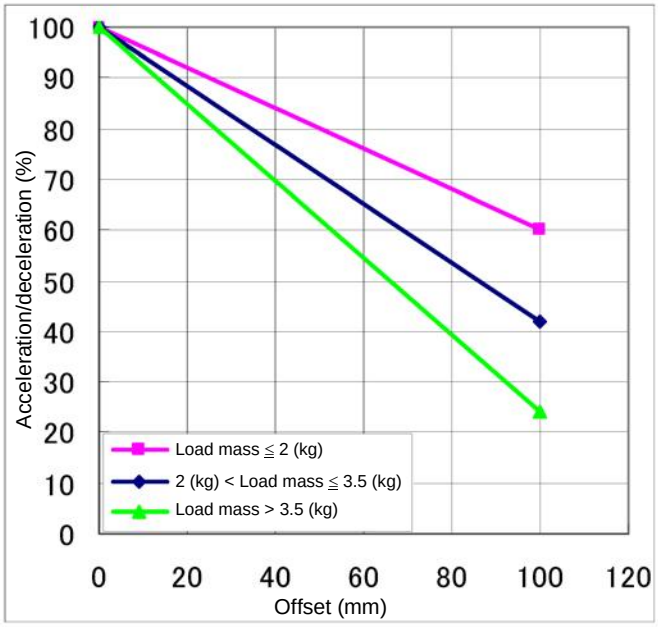
Additionally, if there is an offset of load gravity center, the maximum speed and acceleration/deceleration change as shown in Fig. 19.6 to Fig. 19.9.



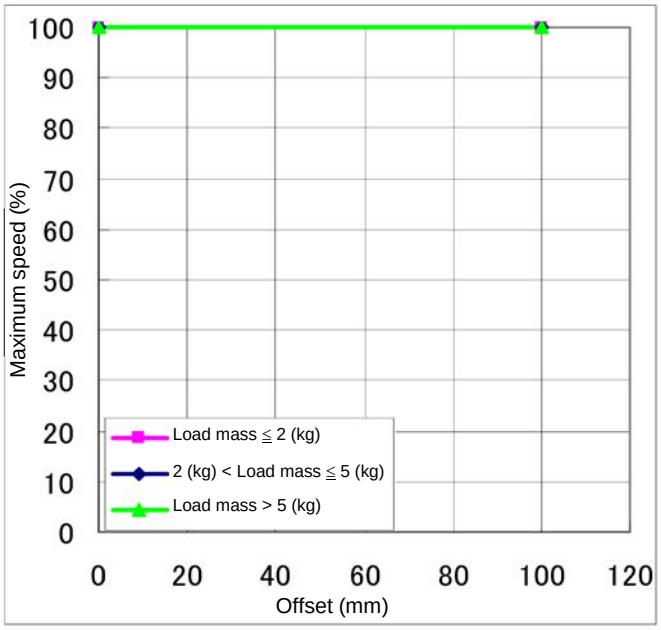
Setting of acceleration/deceleration in relation to offset (Axis 1)



Setting of maximum speed in relation to offset (Axis 1)

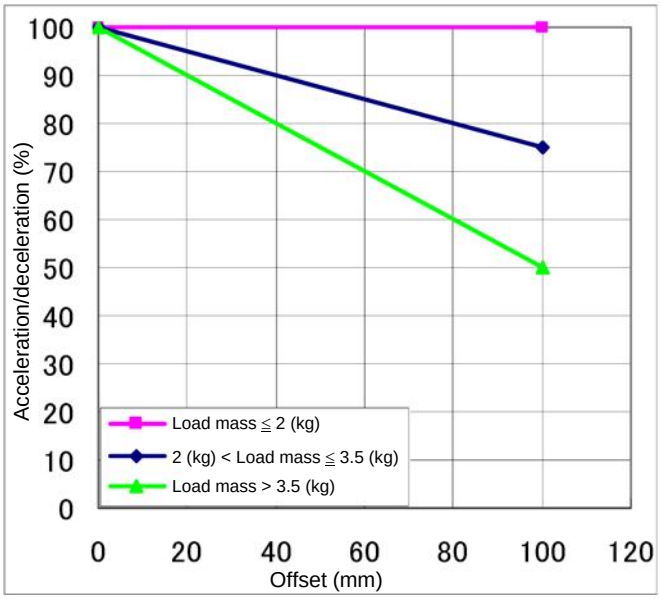


Setting of acceleration/deceleration in relation to offset (Axis 2)

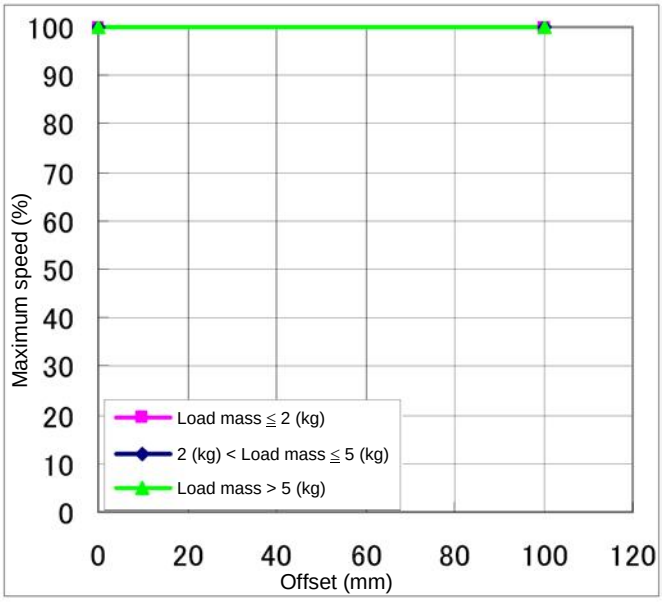


Setting of maximum speed in relation to offset (Axis 2)

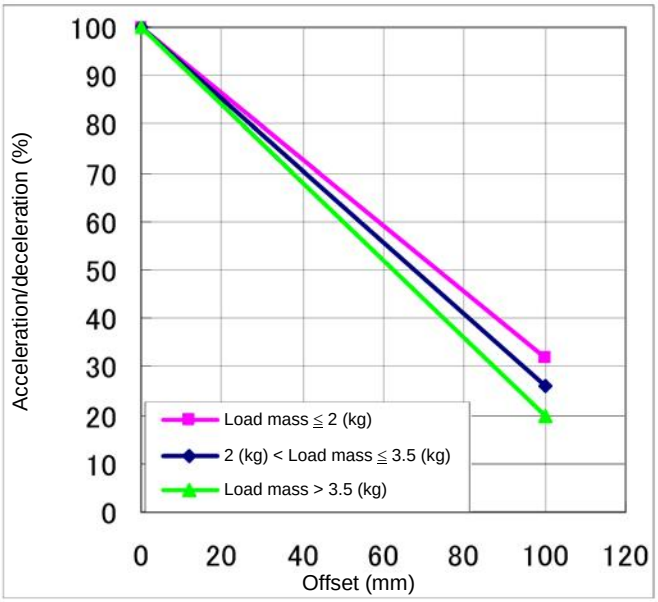
Fig. 19.6 Setting of maximum speed and acceleration/deceleration in relation to gravity center offset (Axes 1 and 2) (THL300)



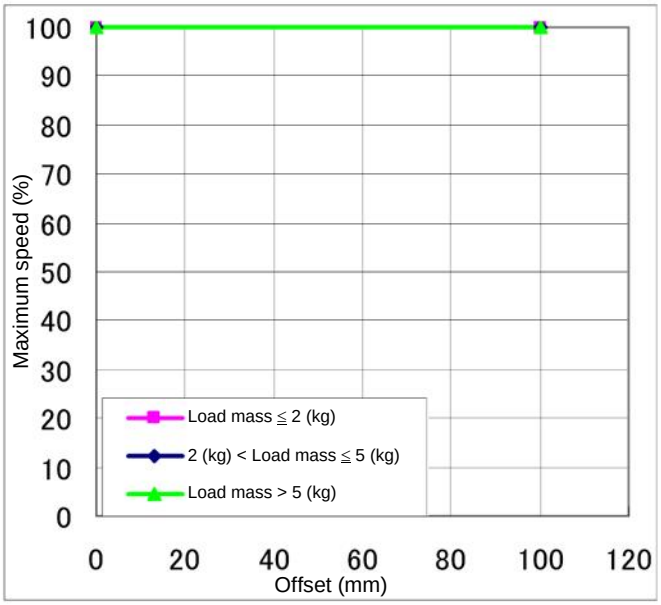
Setting of acceleration/deceleration in relation to offset (Axis 3)



Setting of maximum speed in relation to offset (Axis 3)

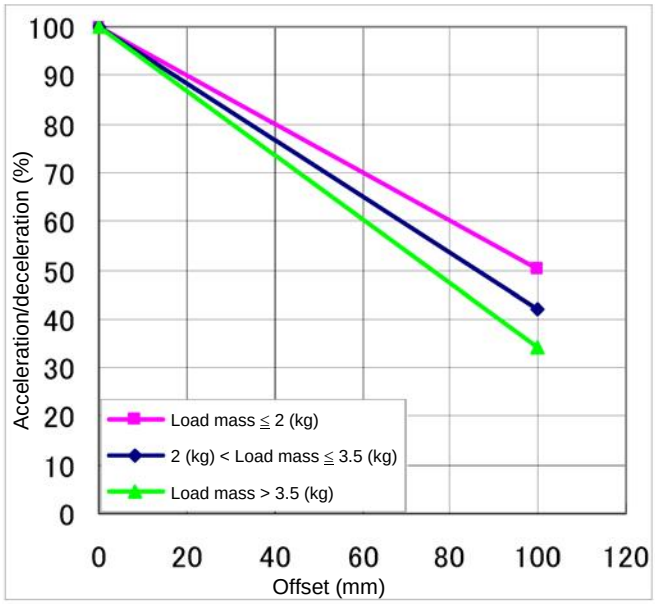


Setting of acceleration/deceleration in relation to offset (Axis 4)

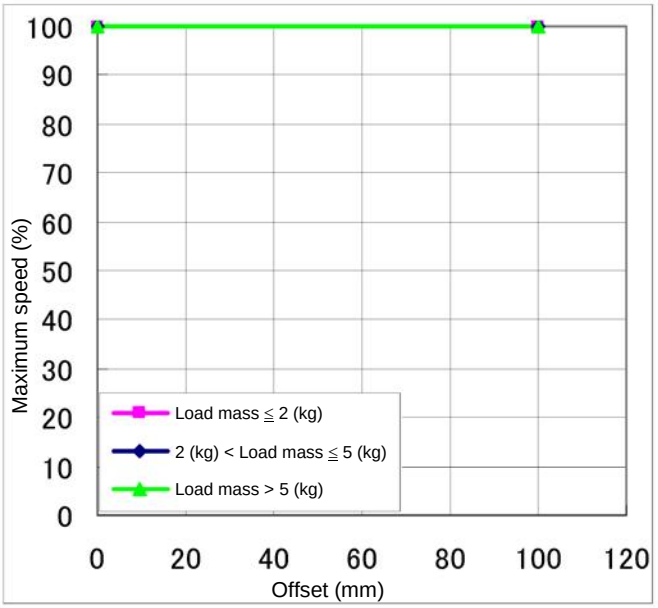


Setting of maximum speed in relation to offset (Axis 4)

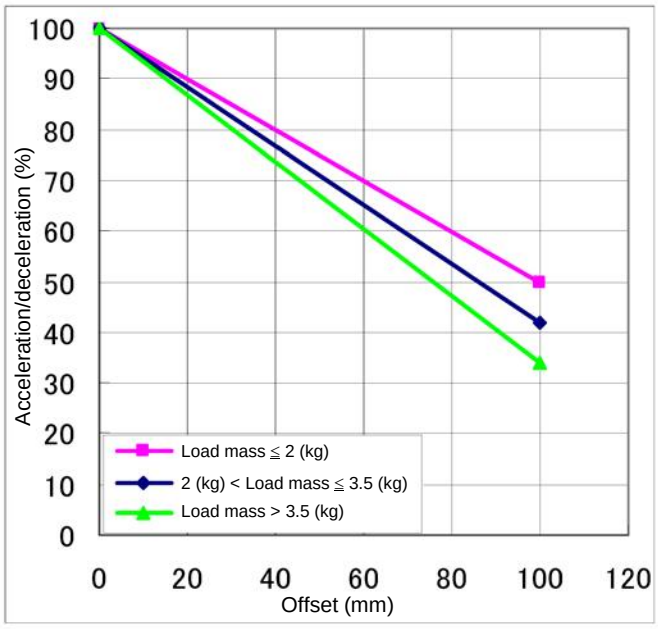
Fig. 19.7 Setting of maximum speed and acceleration/deceleration in relation to gravity center offset (Axes 3 and 4) (THL300)



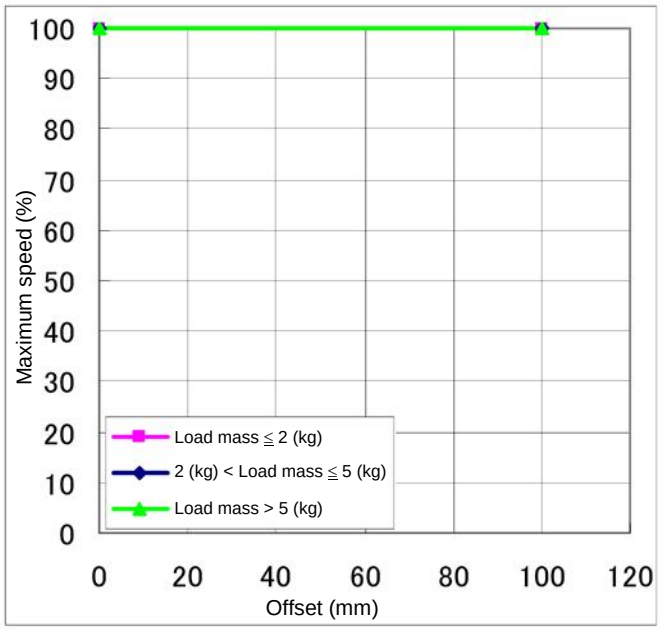
Setting of acceleration/deceleration in relation to offset (Axis 1)



Setting of maximum speed in relation to offset (Axis 1)

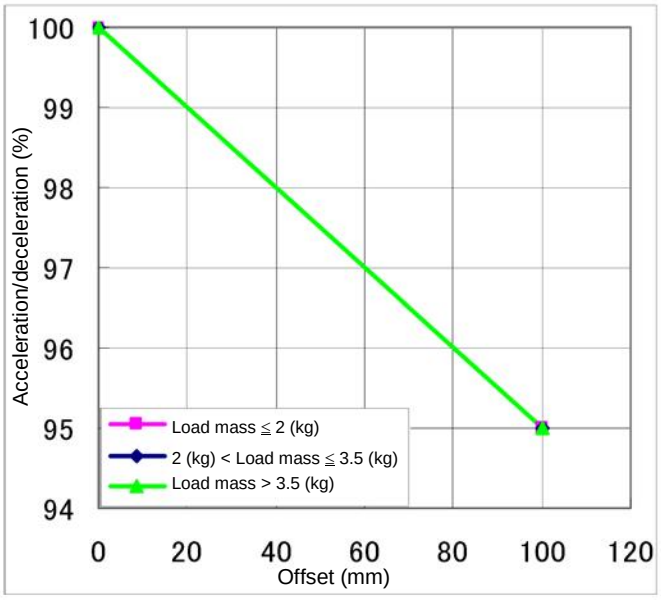


Setting of acceleration/deceleration in relation to offset (Axis 2)

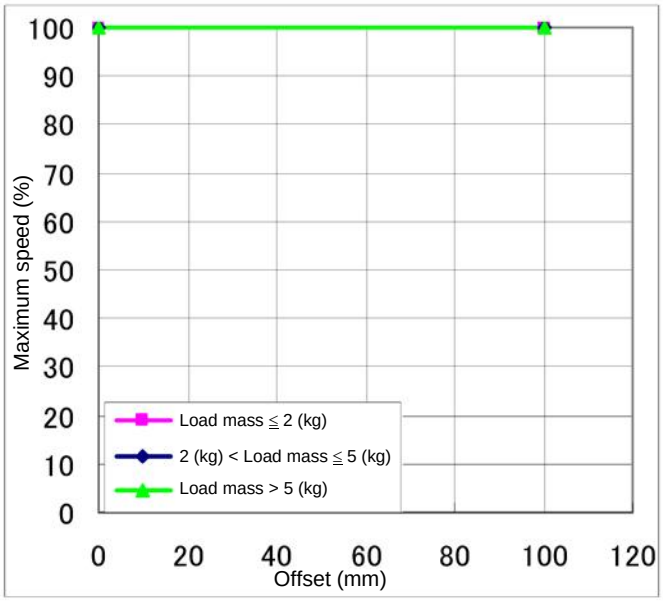


Setting of maximum speed in relation to offset (Axis 2)

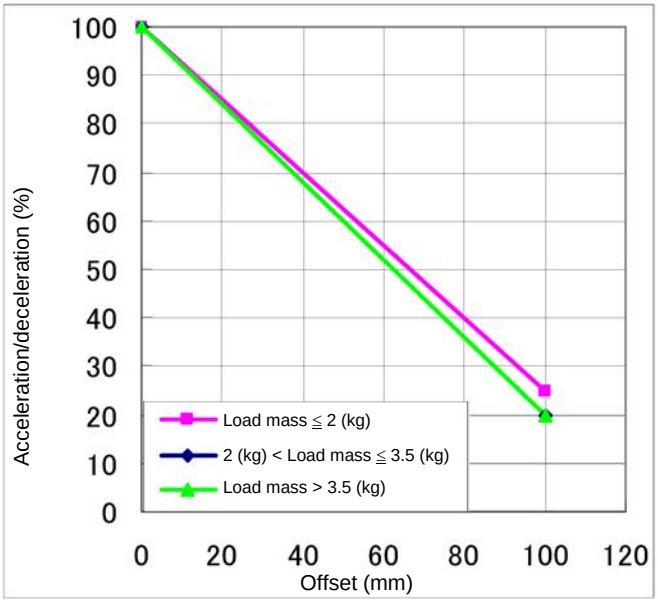
Fig. 19.8 Setting of maximum speed and acceleration/deceleration in relation to gravity center offset (Axes 1 and 2) (THL400)



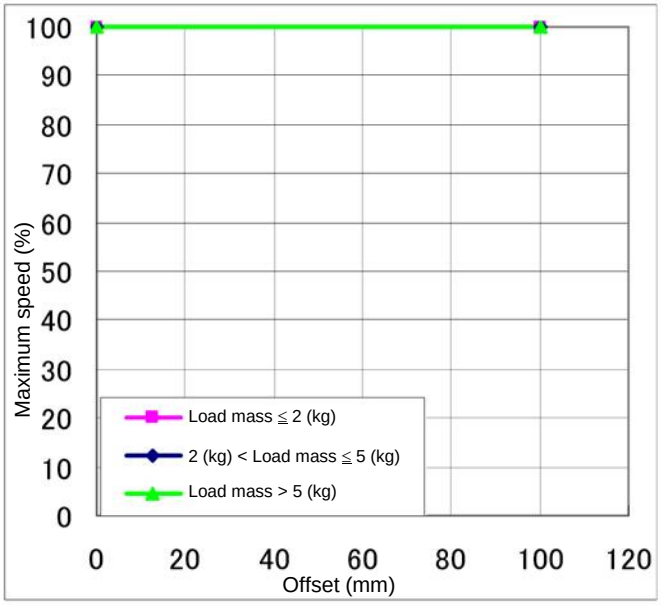
Setting of acceleration/deceleration in relation to offset (Axis 3)



Setting of maximum speed in relation to offset (Axis 3)



Setting of acceleration/deceleration in relation to offset (Axis 4)



Setting of maximum speed in relation to offset (Axis 4)

Fig. 19.9 Setting of maximum speed and acceleration/deceleration in relation to gravity center offset (Axes 3 and 4) (THL400)

20. Tool Interface (THL500, THL600 and THL700)

20.1 Mounting Tool

The tool is mounted on the end of the tool shaft. Dimensions of the tool shaft section are shown in Fig. 20.1. As shown in Fig. 20.1, the tool is centered with the ϕ12H7 mating section. The tool direction is adjusted by means of the 4 x 4 keys and secured with four (4) M4 bolts.
The tool flange is optional.

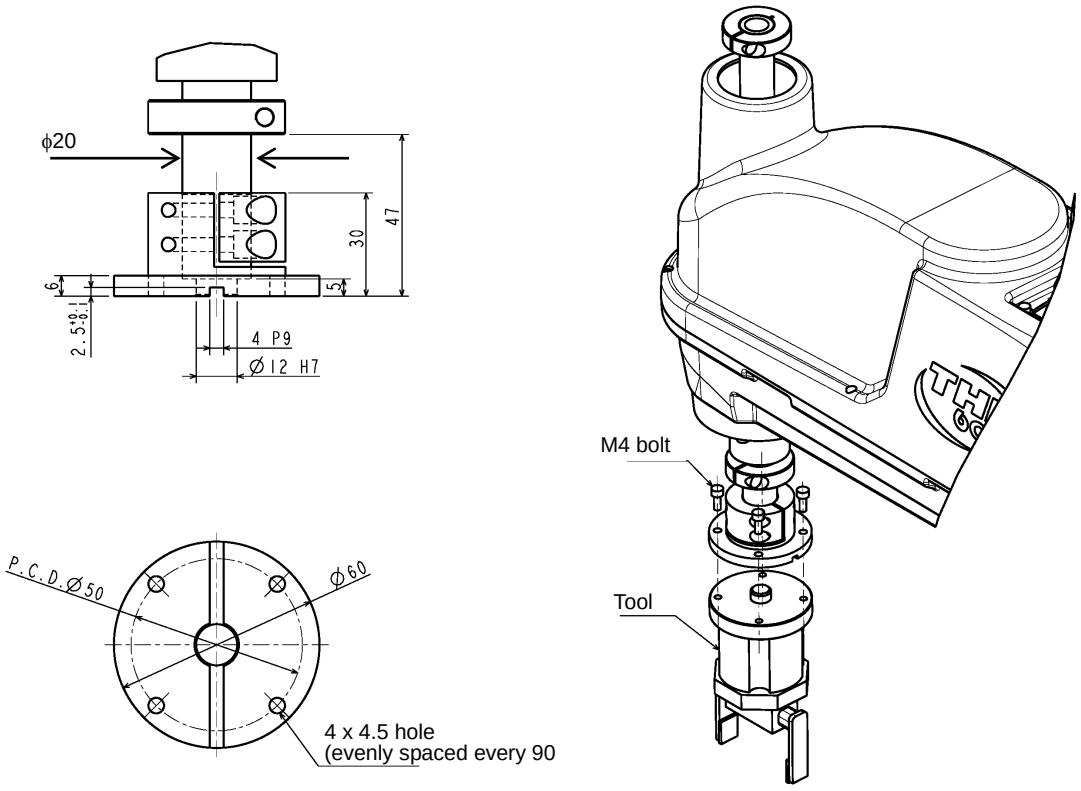


Fig. 20.1 Tool mounting dimensions (THL500, THL600 and THL700)

20.2 Tool Air Piping

The customer is requested to install the tool air piping using the following accessories.

- Air tube x3 (red, white, blue)

Fig. 20.2 shows an installation example of tool air piping for your reference.

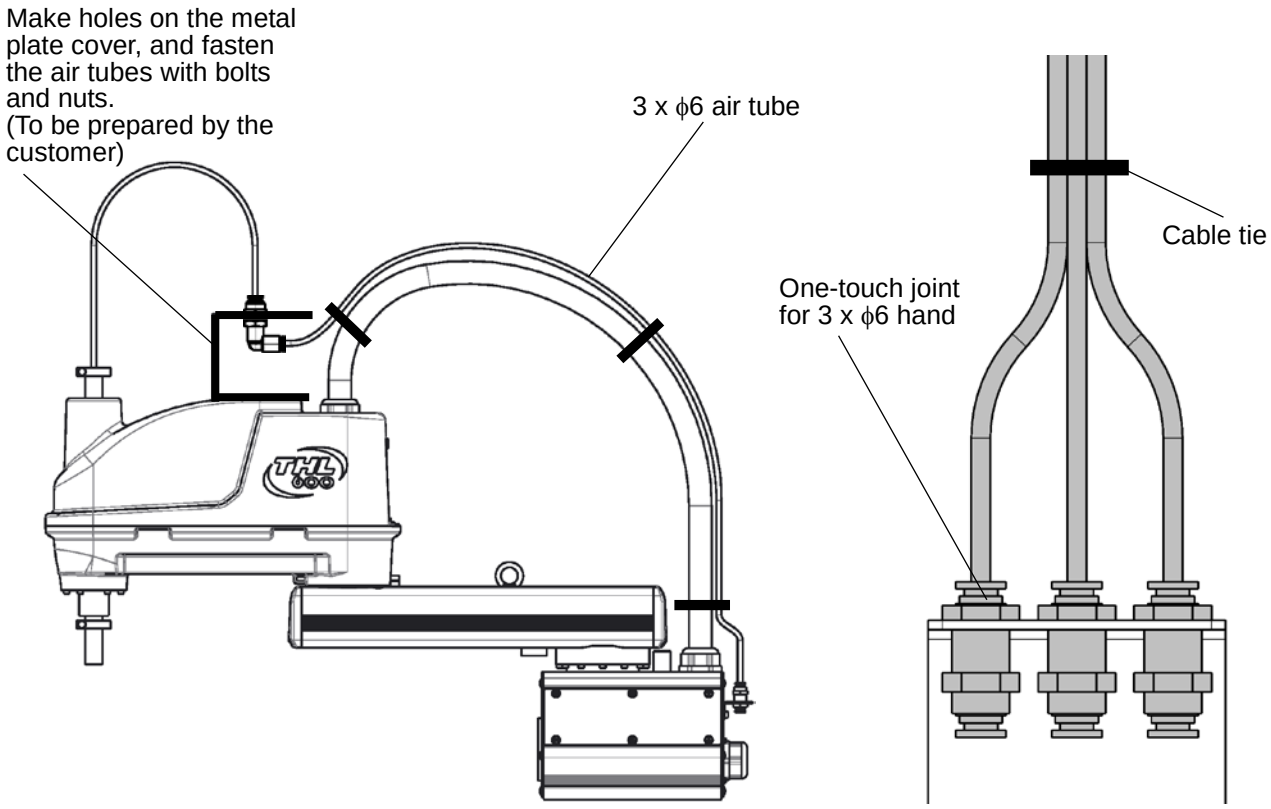


Fig. 20.2 Tool air piping installation example (THL500, THL600 and THL700)

! CAUTION

- Air tubes are consumable supplies. Check the air tubes during periodic inspection, and change them if any damage is found.
- The customer is requested to prepare solenoid valve air.
- Fig. 20.2 is a reference example. Tool air piping is not guaranteed. The warranty does not cover breakdowns of the harness, the air tubes or their accessories due to piping by the customer.

20.3 Permissible Load Conditions and Program Setting

This paragraph describes the permissible load conditions of the robot and how to set up the program according to the load.

20.3.1 Permissible Load Conditions

The robot load conditions are defined by the tool mass, moment of inertia and offset value of tool gravity center from the center of the tool shaft, as shown in Fig. 20.3. The permissible load conditions are shown in Table 20.1.

Table 20.1 Permissible load conditions (THL500, THL600 and THL700)

Conditions	Permissible values
Mass	Max. 10 kg
Load inertia	Max 0.20 kg·m^2
Offset value of load gravity center	Max. 100 mm

! CAUTION

- NEVER operate the robot under the load conditions exceeding the permissible values. Otherwise, the robot life and safety cannot be guaranteed.

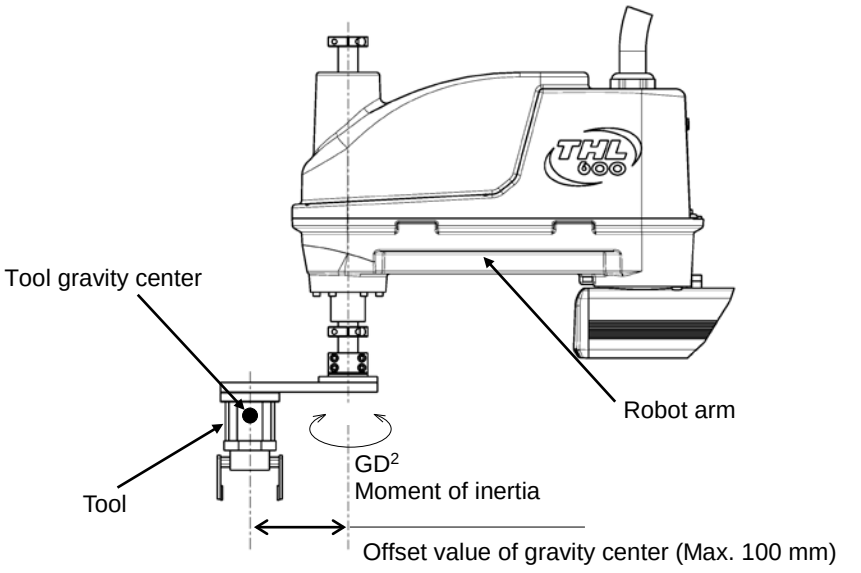


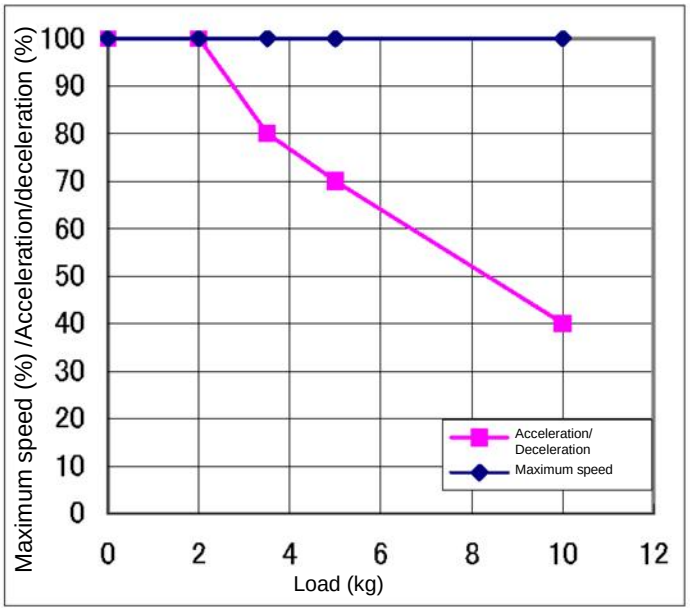
Fig. 20.4 Robot tool

20.4.2 Load Conditions and Program Setting

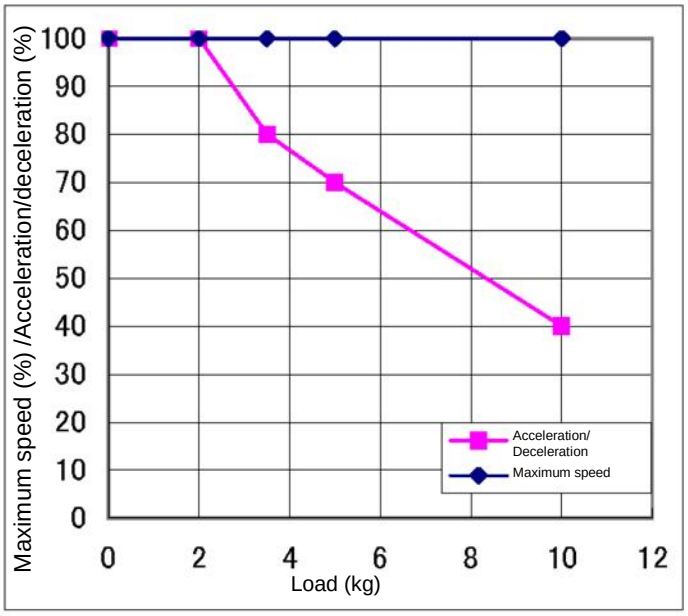
For load conditions and program setting, see "20.3.2 Load Conditions and Program Setting".

20.4.3 Setting Maximum Speed and Robot Acceleration/Deceleration for Load Conditions

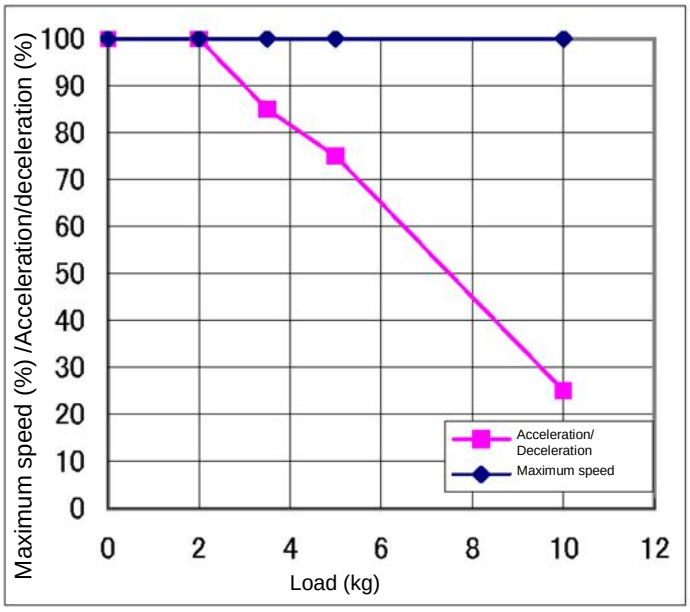
When the PAYLOAD command is used, the maximum speed and acceleration/ deceleration of the robot are automatically changed according to the load conditions. The maximum speed and acceleration/deceleration change with the load mass, as shown in Figs. 20.4 to 20.6.



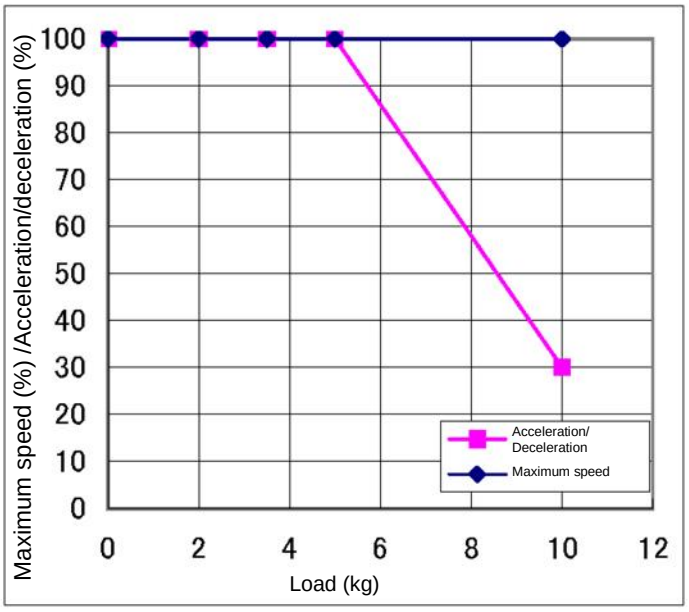
Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 1)



Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 2)

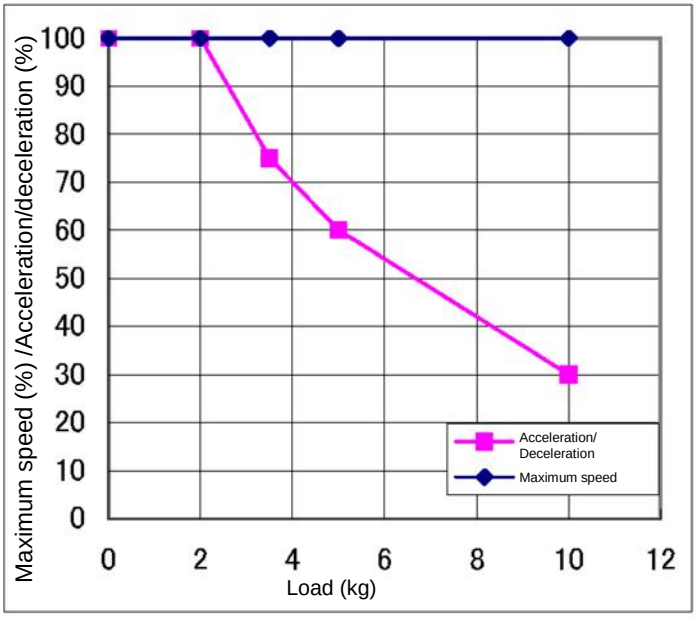


Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 3)

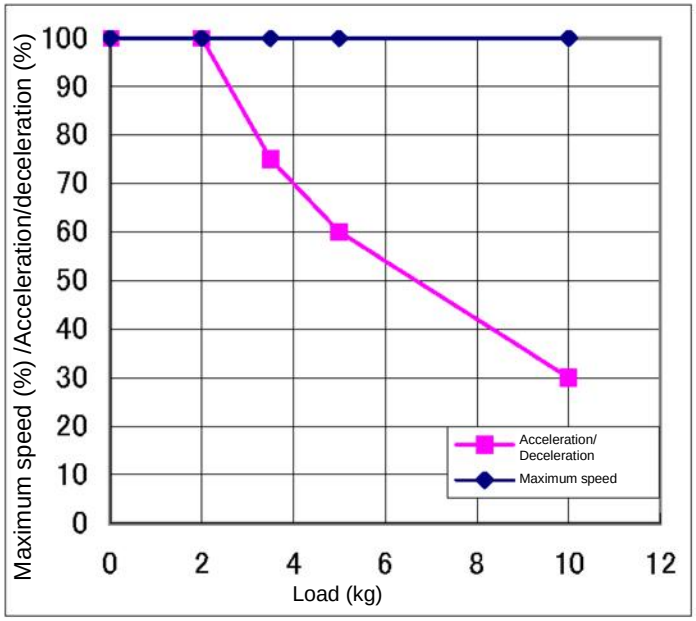


Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 4)

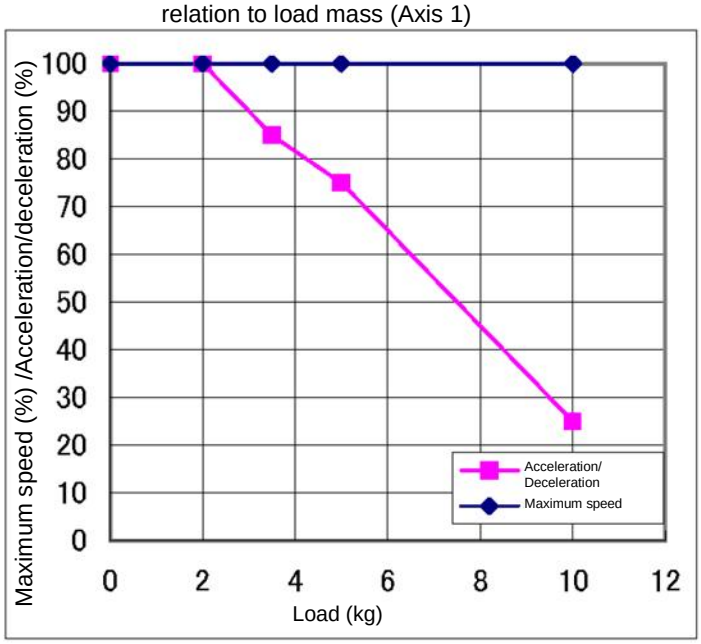
Fig. 20.4 Setting of maximum speed and acceleration/deceleration in relation to load mass (THL500)



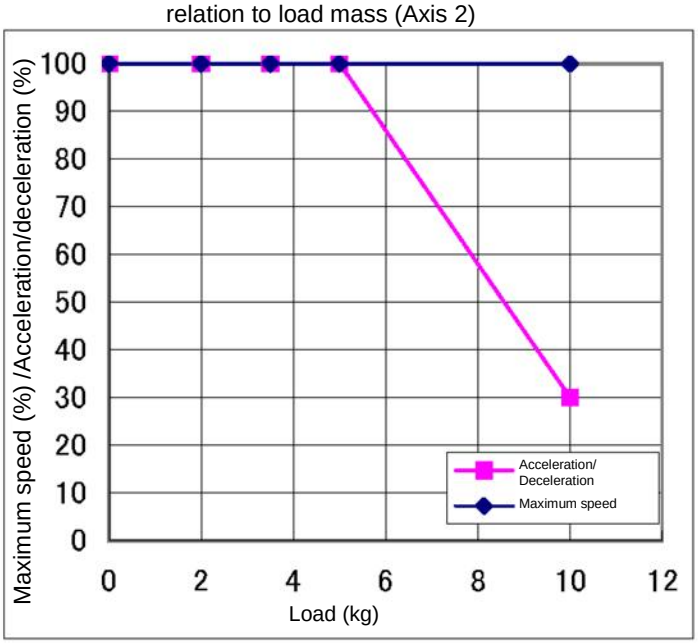
Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 1)



Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 2)

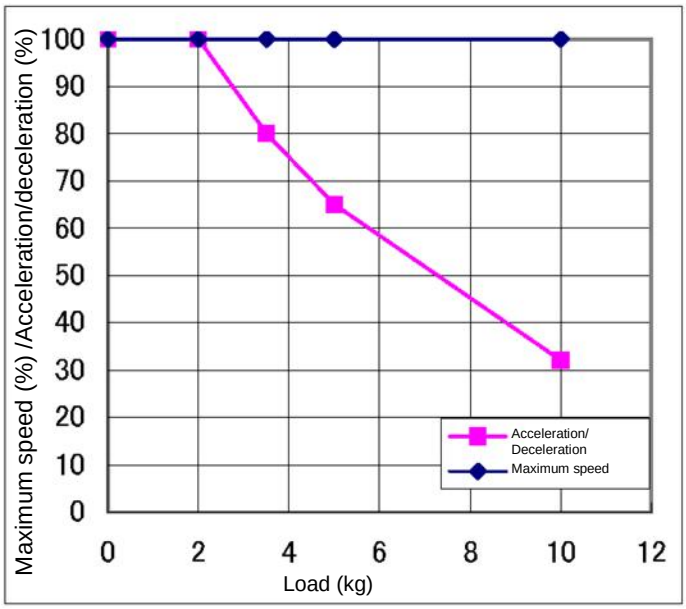


Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 3)

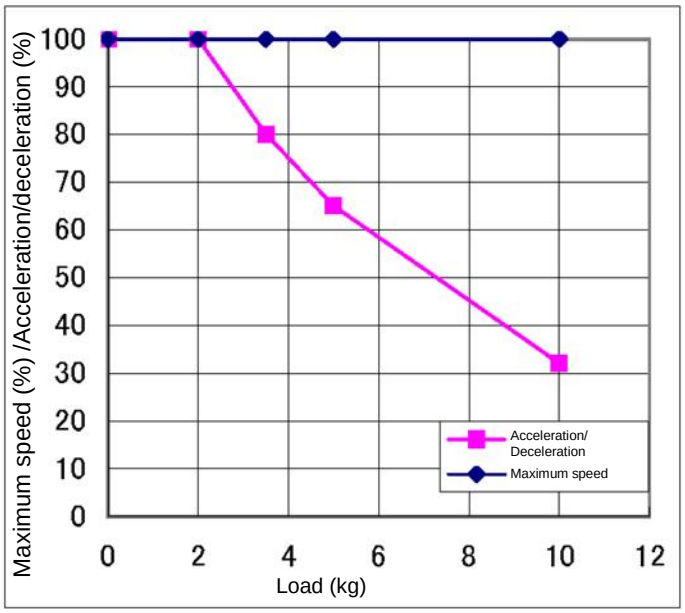


Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 4)

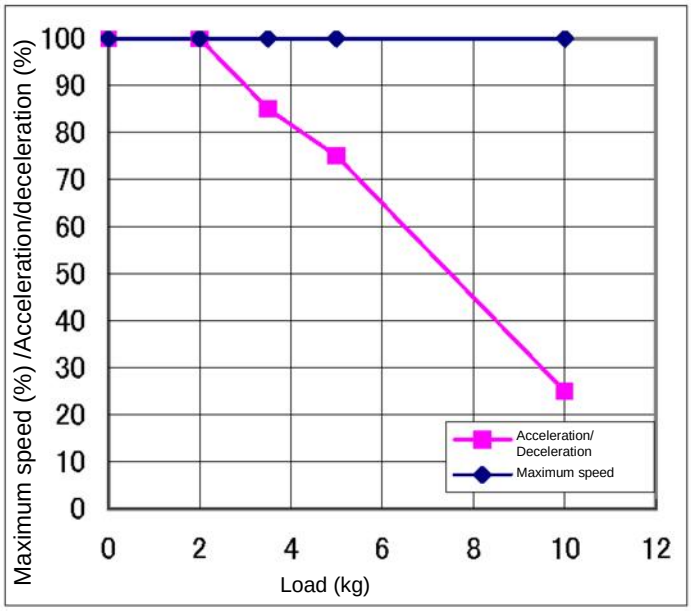
Fig. 20.5 Setting of maximum speed and acceleration/deceleration in relation to load mass (THL600)



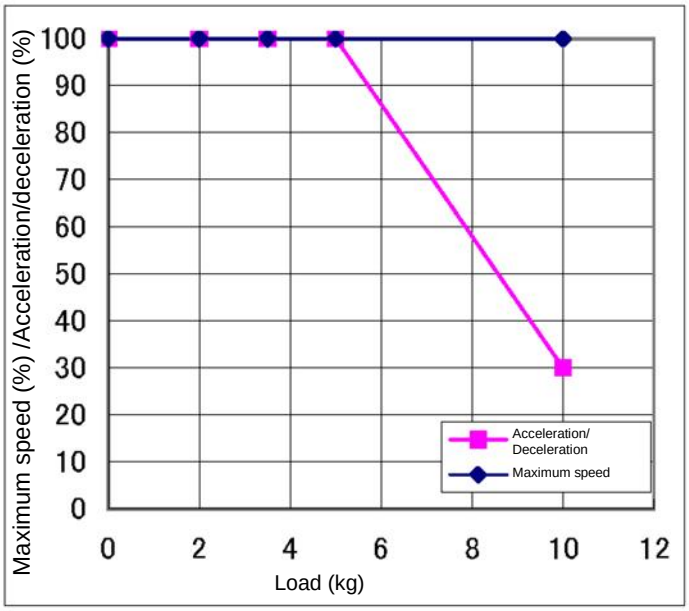
Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 1)



Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 2)



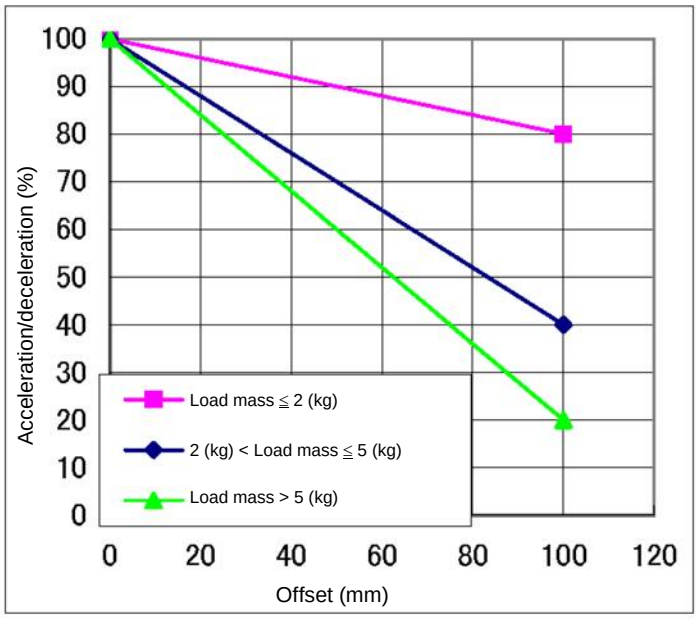
Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 3)



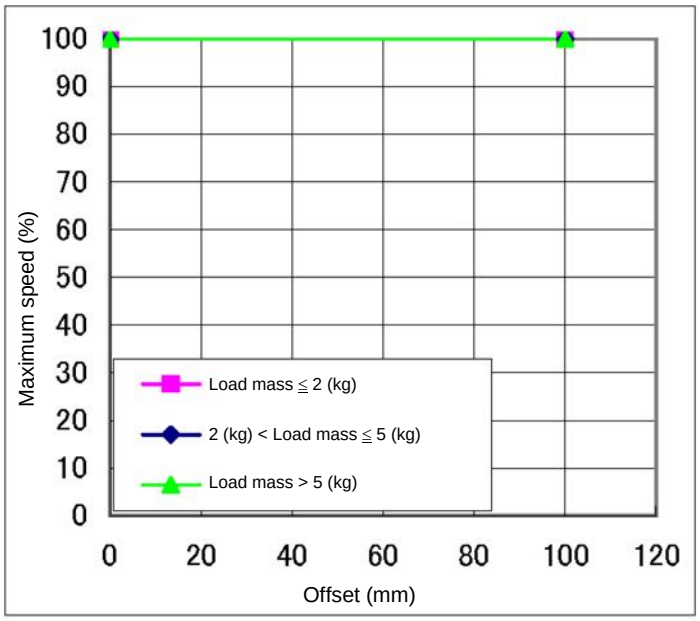
Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 4)

Fig. 20.6 Setting of maximum speed and acceleration/deceleration in relation to load mass (THL700)

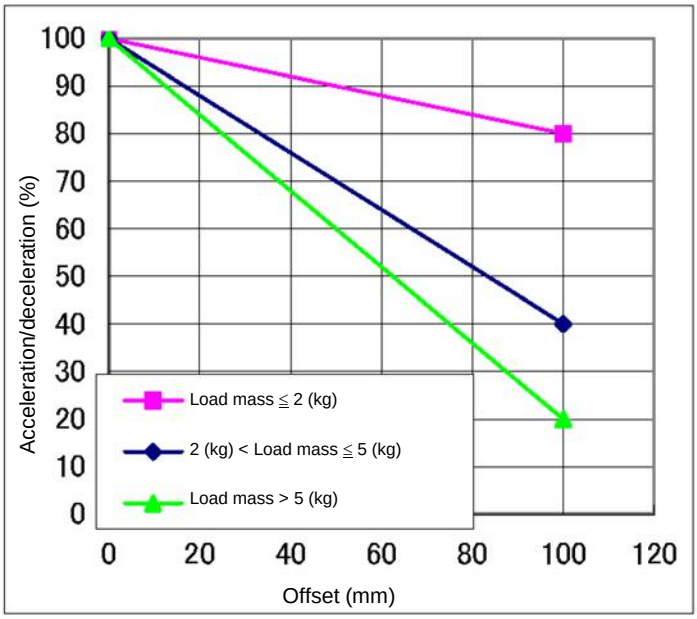
Additionally, if there is an offset of load gravity center, the maximum speed and acceleration/deceleration change as shown in Fig. 20.7 to Fig. 20.12.



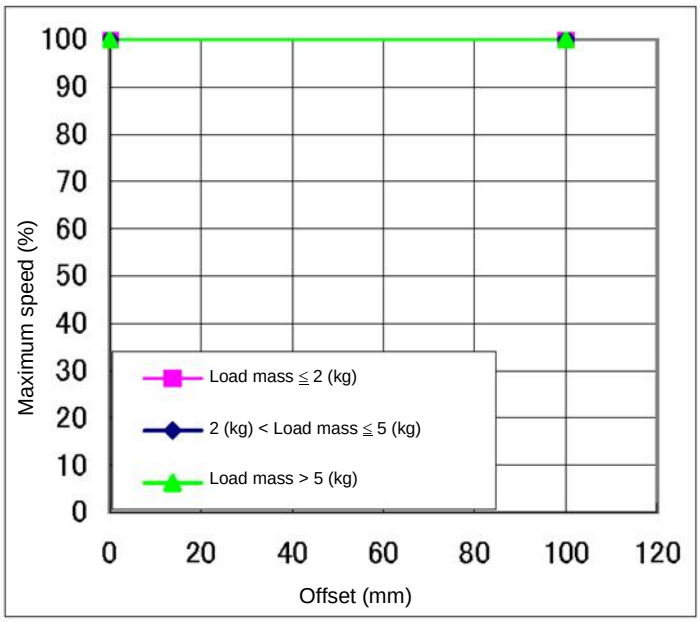
Setting of acceleration/deceleration in relation to offset (Axis 1)



Setting of maximum speed in relation to offset (Axis 1)

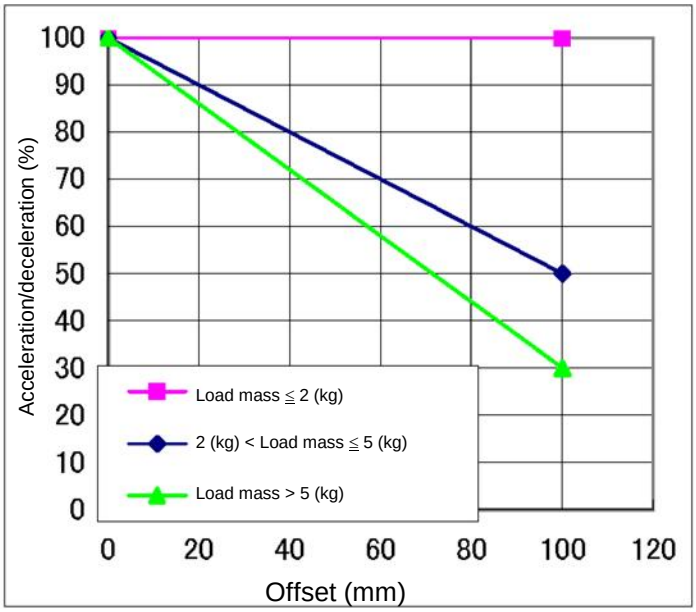


Setting of acceleration/deceleration in relation to offset (Axis 2)

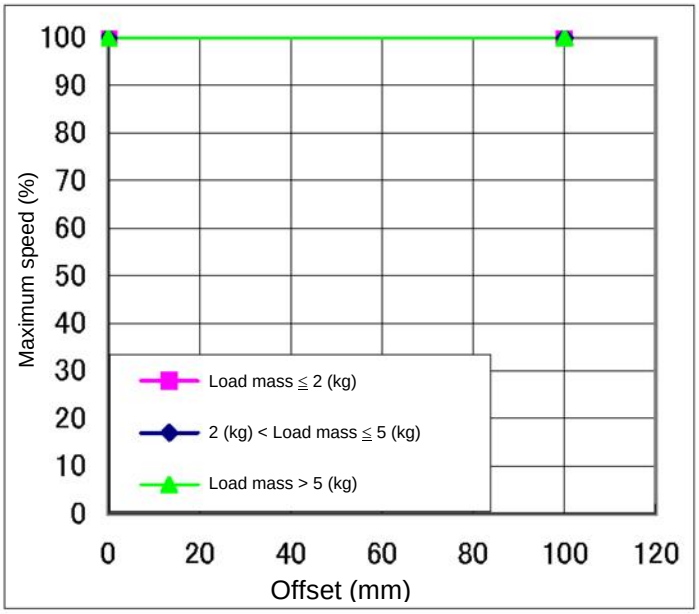


Setting of maximum speed in relation to offset (Axis 2)

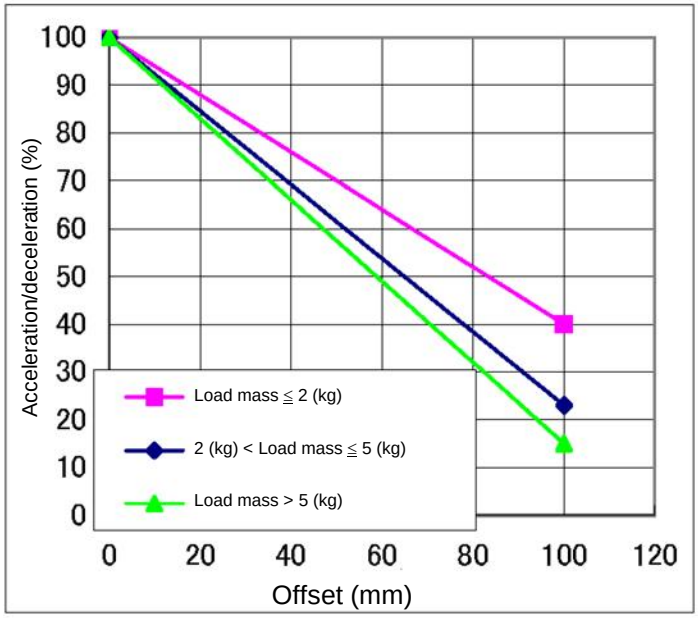
Fig. 20.7 Setting of maximum speed and acceleration/deceleration in relation to gravity center offset (Axes 1 and 2) (THL500)



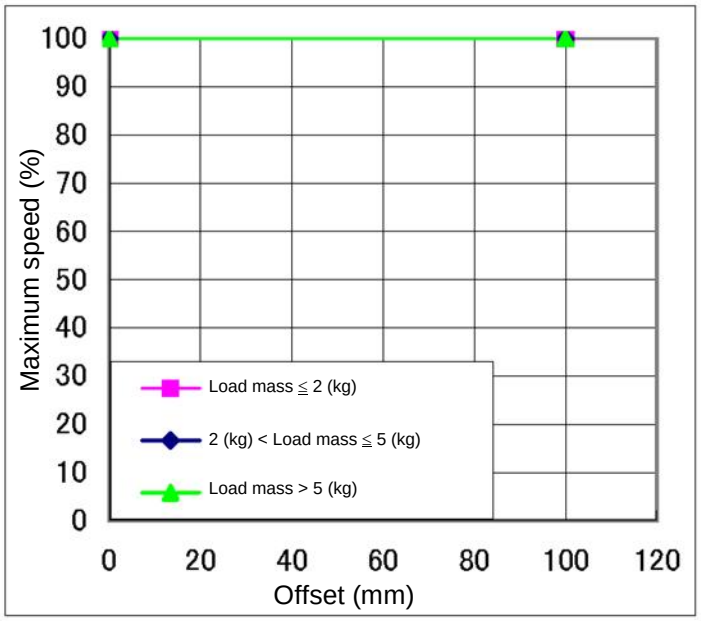
Setting of acceleration/deceleration in relation to offset (Axis 3)



Setting of maximum speed in relation to offset (Axis 3)

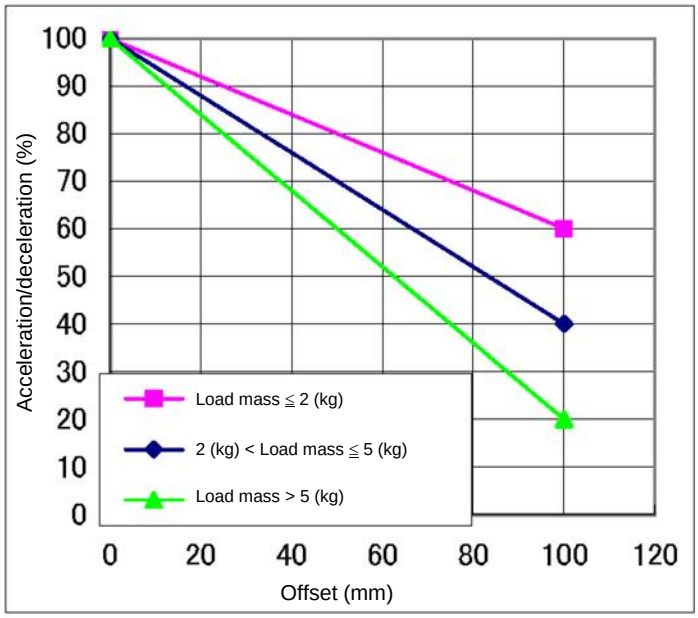


Setting of acceleration/deceleration in relation to offset (Axis 4)

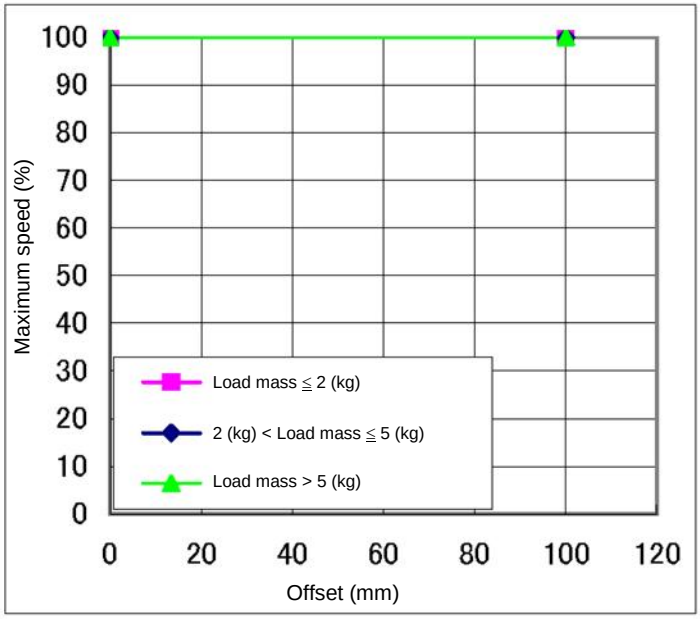


Setting of maximum speed in relation to offset (Axis 4)

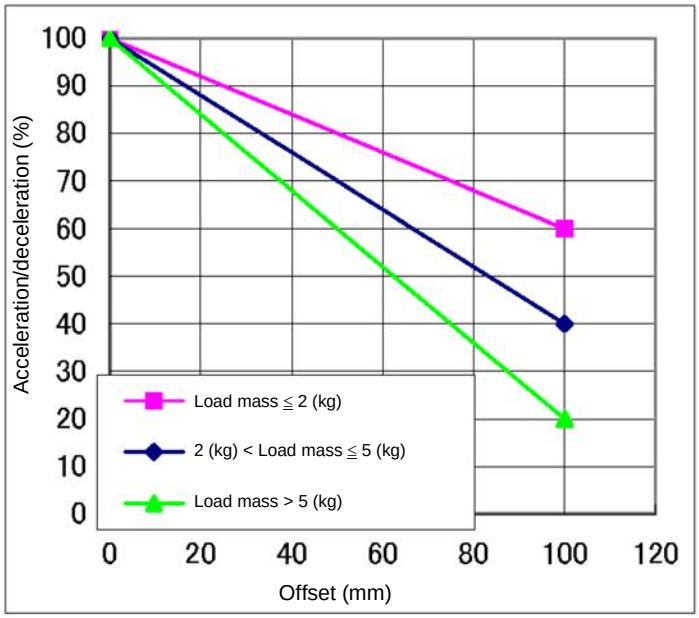
Fig. 20.8 Setting of maximum speed and acceleration/deceleration in relation to gravity center offset (Axes 3 and 4) (THL500)



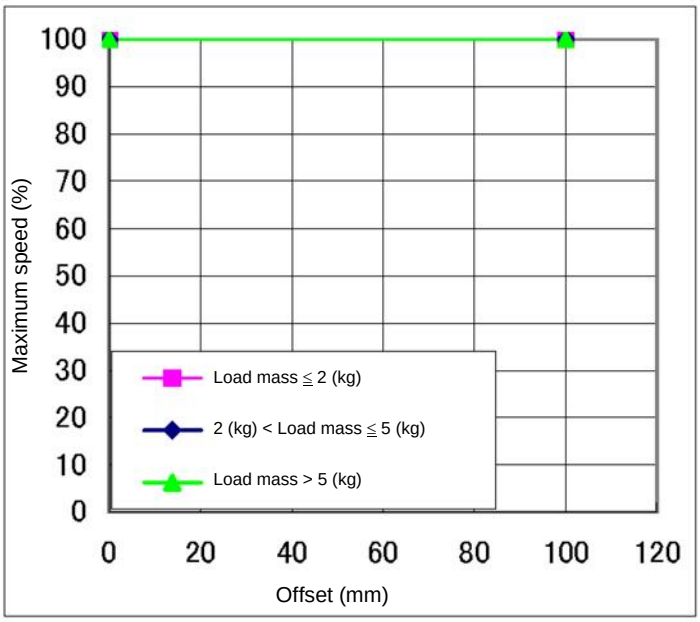
Setting of acceleration/deceleration in relation to offset (Axis 1)



Setting of maximum speed in relation to offset (Axis 1)

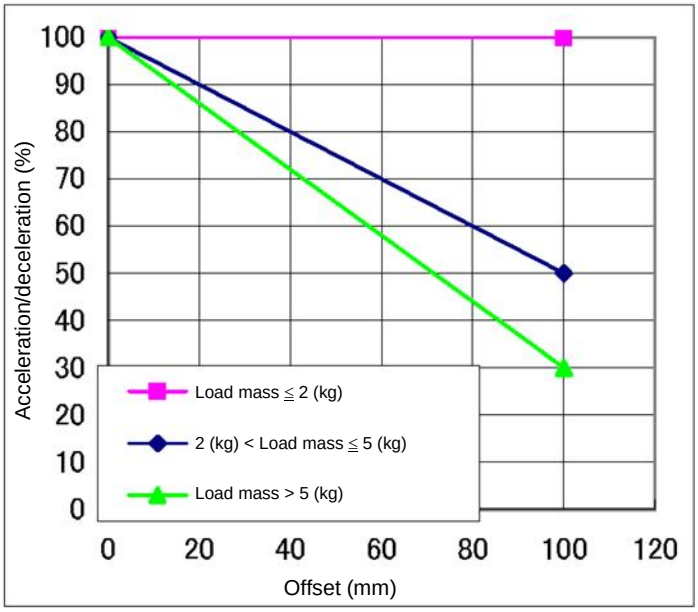


Setting of acceleration/deceleration in relation to offset (Axis 2)

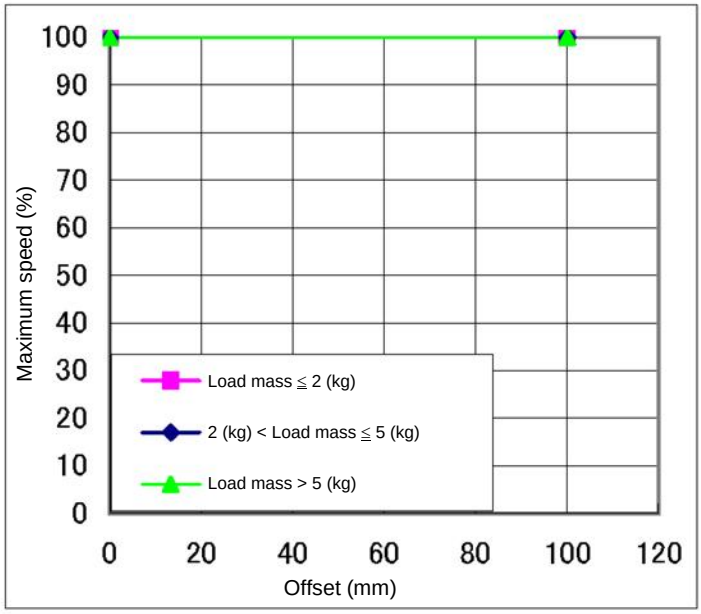


Setting of maximum speed in relation to offset (Axis 2)

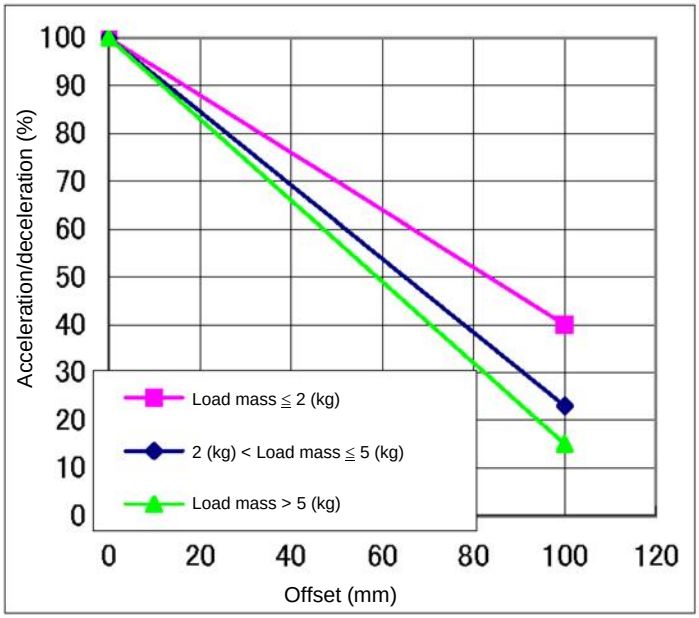
Fig. 20.9 Setting of maximum speed and acceleration/deceleration in relation to gravity center offset (Axes 1 and 2) (THL600)



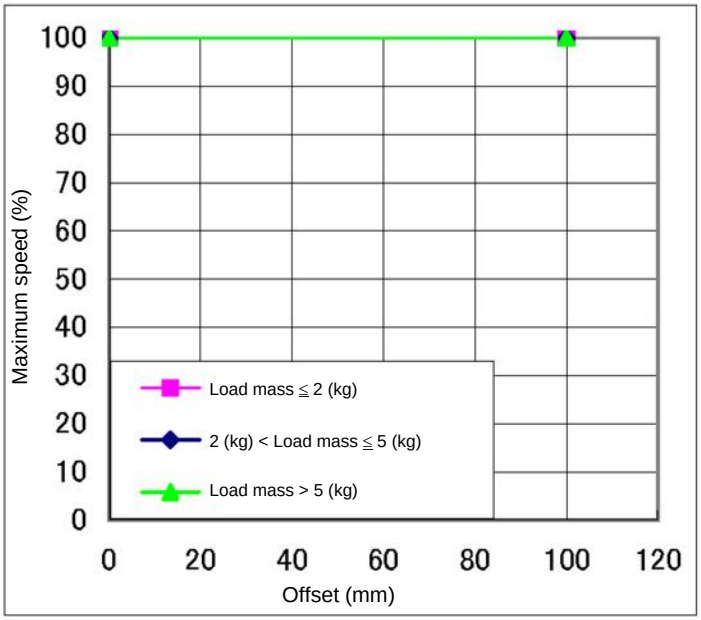
Setting of acceleration/deceleration in relation to offset (Axis 3)



Setting of maximum speed in relation to offset (Axis 3)

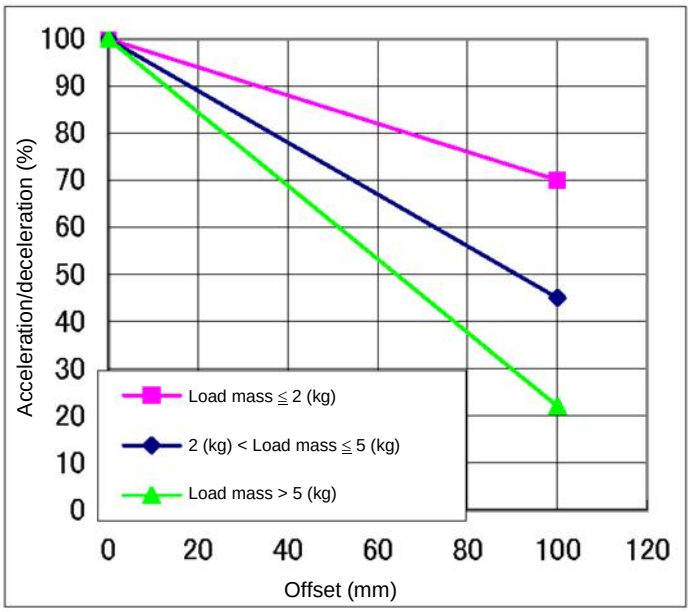


Setting of acceleration/deceleration in relation to offset (Axis 4)

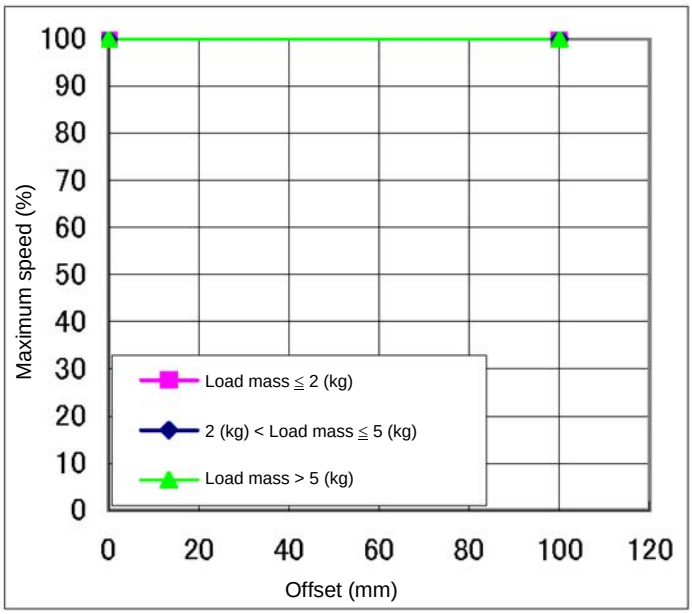


Setting of maximum speed in relation to offset (Axis 4)

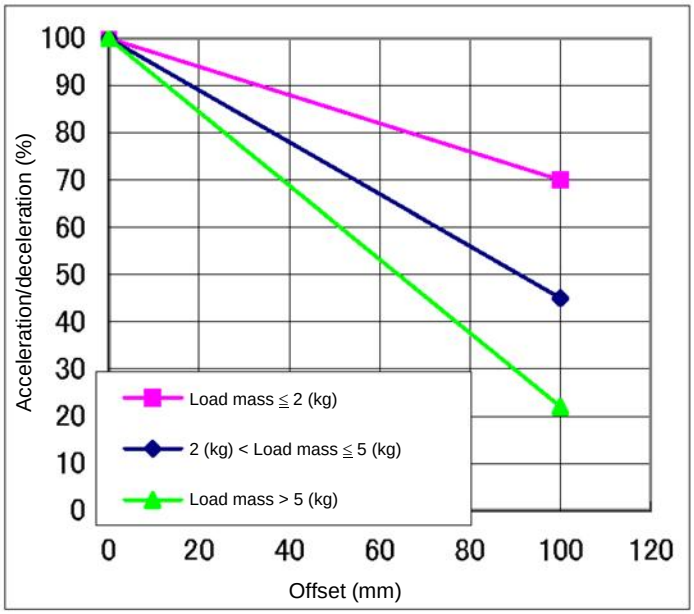
Fig. 20.10 Setting of maximum speed and acceleration/deceleration in relation to gravity center offset (Axes 3 and 4) (THL600)



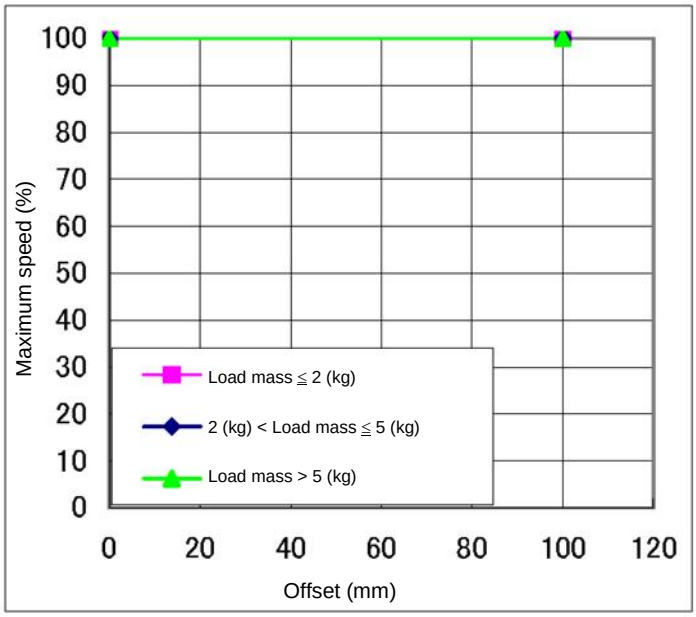
Setting of acceleration/deceleration in relation to offset (Axis 1)



Setting of maximum speed in relation to offset (Axis 1)

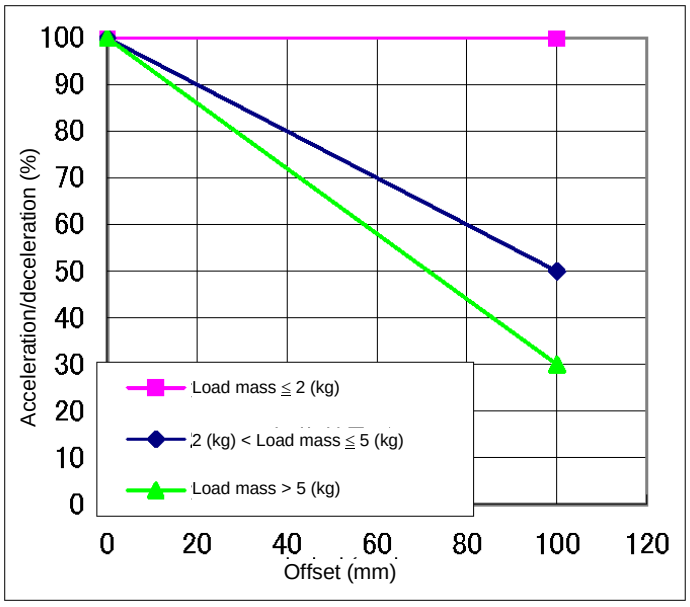


Setting of acceleration/deceleration in relation to offset (Axis 2)

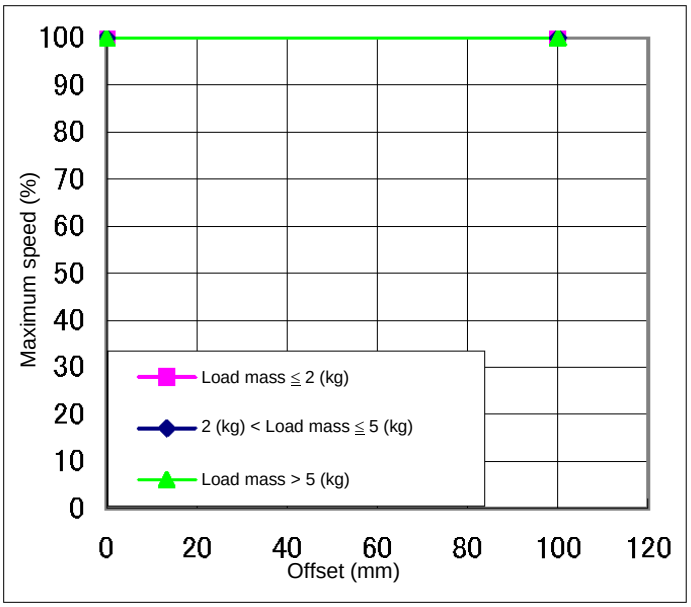


Setting of maximum speed in relation to offset (Axis 2)

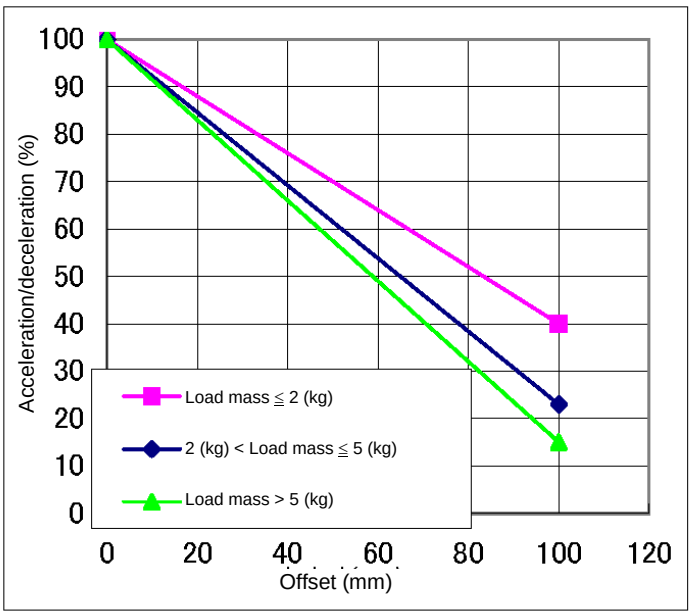
Fig. 20.11 Setting of maximum speed and acceleration/deceleration in relation to gravity center offset (Axes 1 and 2) (THL700)



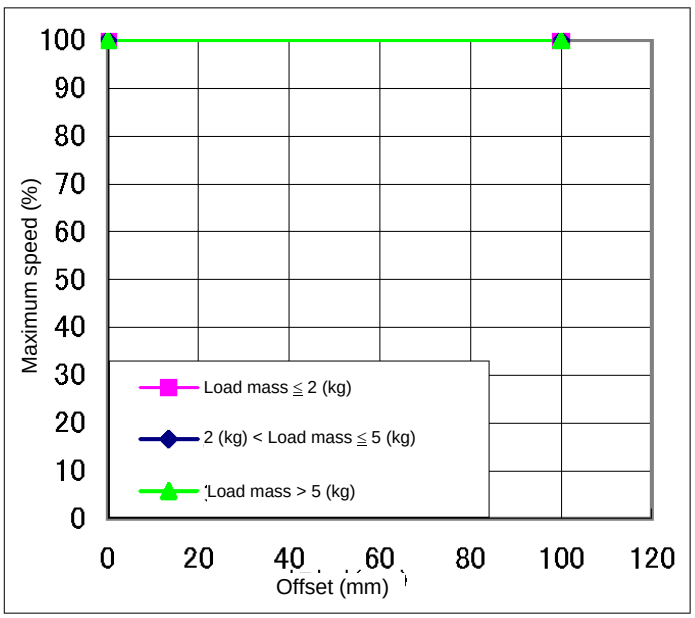
Setting of acceleration/deceleration in relation to offset (Axis 3)



Setting of maximum speed in relation to offset (Axis 3)



Setting of acceleration/deceleration in relation to offset (Axis 4)



Setting of maximum speed in relation to offset (Axis 4)

Fig. 20.12 Setting of maximum speed and acceleration/deceleration in relation to gravity center offset (Axes 3 and 4) (THL700)

21. Tool Interface (THL800, THL900 and THL1000)

21.1 Mounting Tool

The tool is mounted on the end of the tool shaft. Dimensions of the tool shaft section are shown in Fig. 21.1. As shown in Fig. 21.1, the tool is centered with the ϕ12H7 mating section. The tool direction is adjusted by means of the 4 x 4 keys and secured with four (4) M4 bolts.
The tool flange is optional.

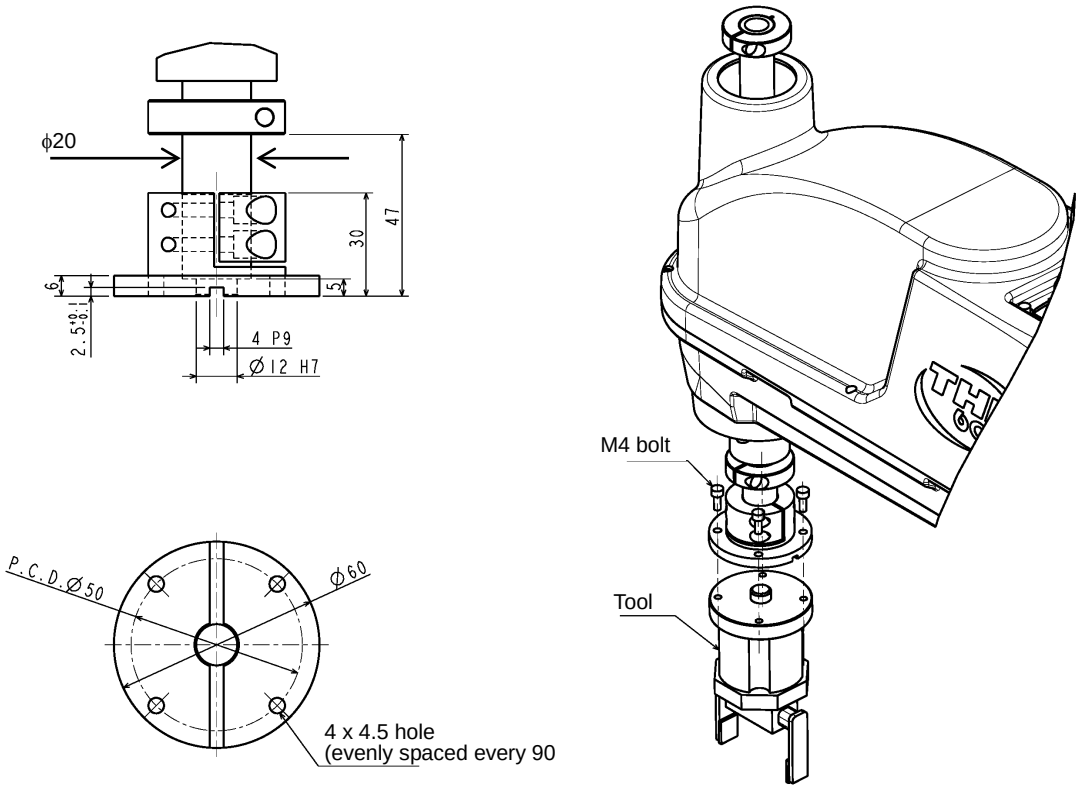


Fig. 21.1 Tool mounting dimensions (THL800, THL900 and THL1000)

21.2 Tool Air Piping

The customer is requested to install the tool air piping using the following accessories.

- Air tube x3 (red, white, blue)

Fig. 21.2 shows an installation example of tool air piping for your reference.

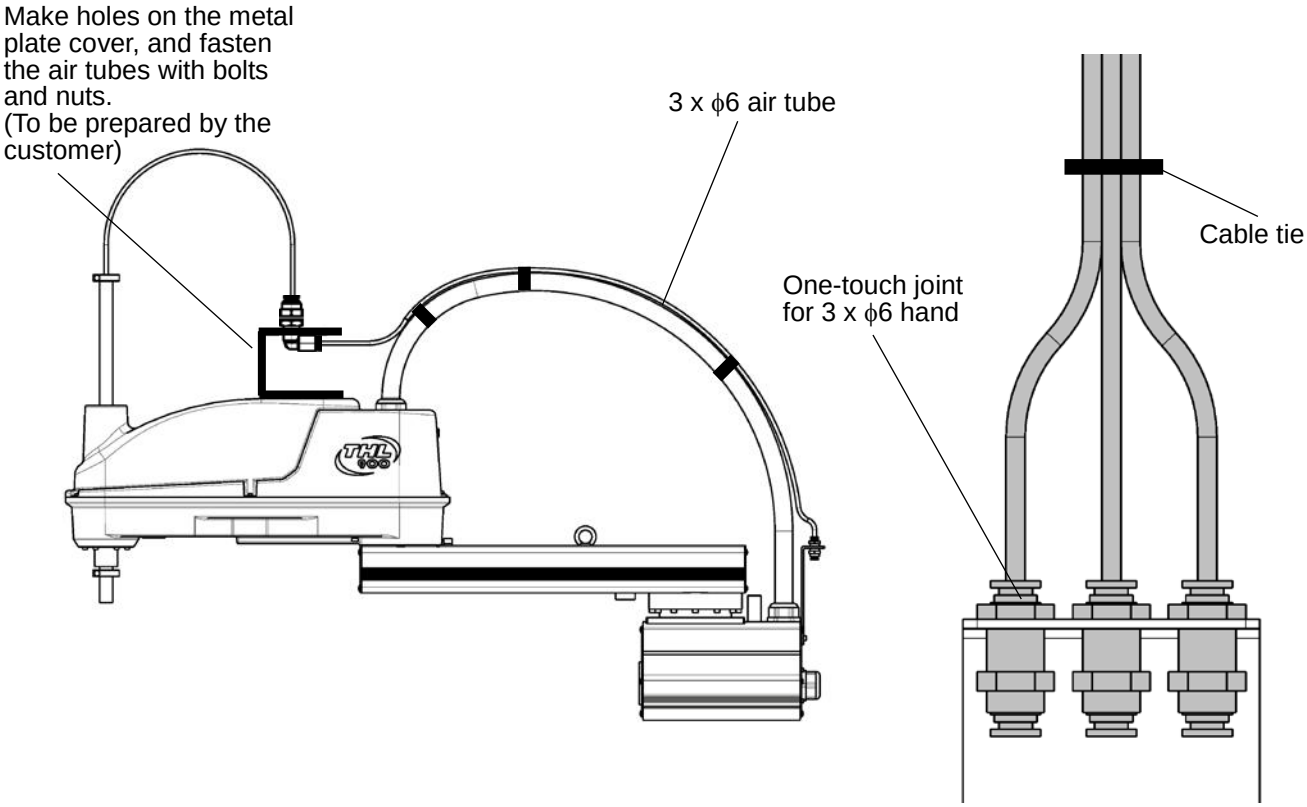


Fig. 21.2 Tool air piping installation example (THL800, THL900 and THL1000)

! CAUTION

- Air tubes are consumable supplies. Check the air tubes during periodic inspection, and change them if any damage is found.
- The customer is requested to prepare solenoid valve air.
- Fig. 21.2 is a reference example. Tool air piping is not guaranteed. The warranty does not cover breakdowns of the harness, the air tubes or their accessories due to piping by the customer.

21.3 Permissible Load Conditions and Program Setting

This paragraph describes the permissible load conditions of the robot and how to set up the program according to the load.

21.3.1 Permissible Load Conditions

The robot load conditions are defined by the tool mass, moment of inertia and offset value of tool gravity center from the center of the tool shaft, as shown in Fig. 21.3. The permissible load conditions are shown in Table 21.1.

Table 21.1 Permissible load conditions (THL800, THL900 and THL1000)

Conditions	Permissible values
Mass	Max. 10 kg
Load inertia	Max 0.20 kg·m^2
Offset value of load gravity center	Max. 100 mm

! CAUTION

- NEVER operate the robot under the load conditions exceeding the permissible values. Otherwise, the robot life and safety cannot be guaranteed.

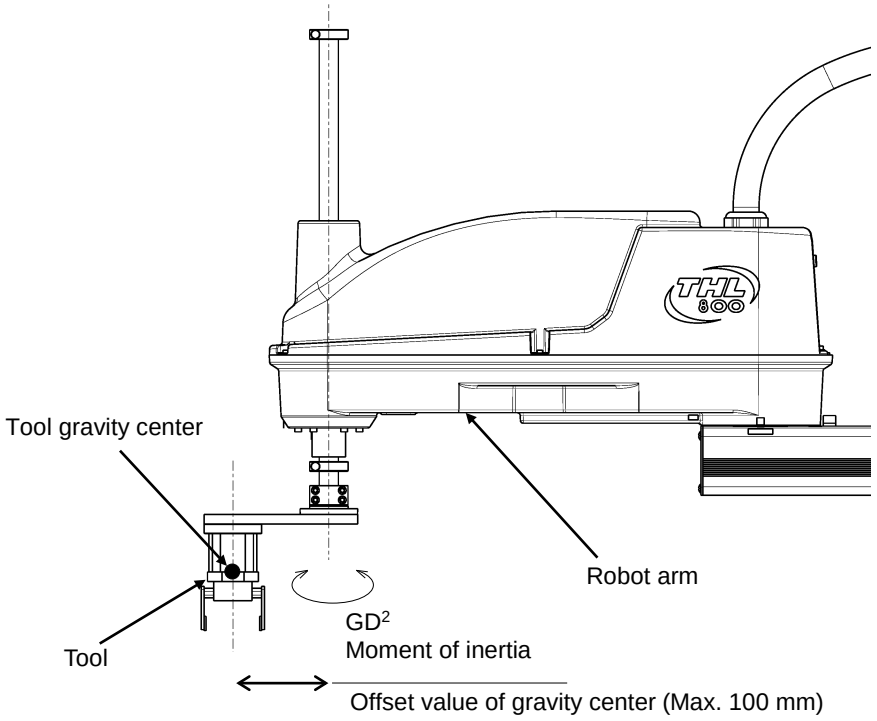


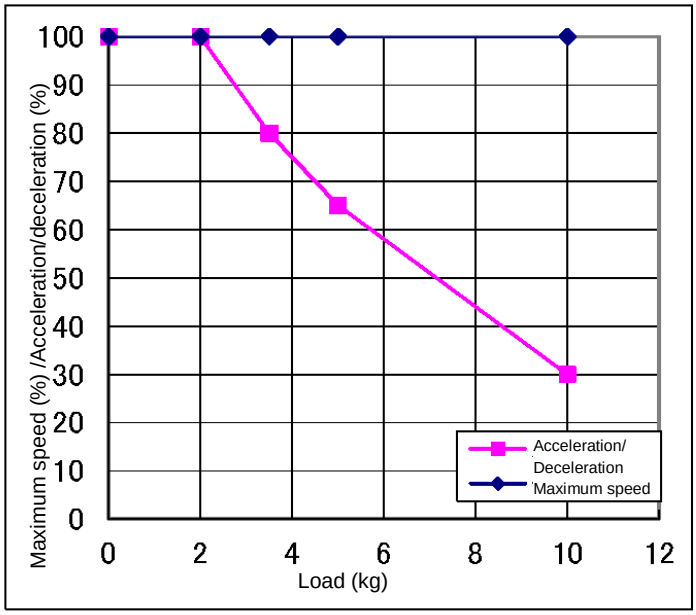
Fig. 21.3 Robot tool (THL800, THL900 and THL1000)

21.3.2 Load Conditions and Program Setting

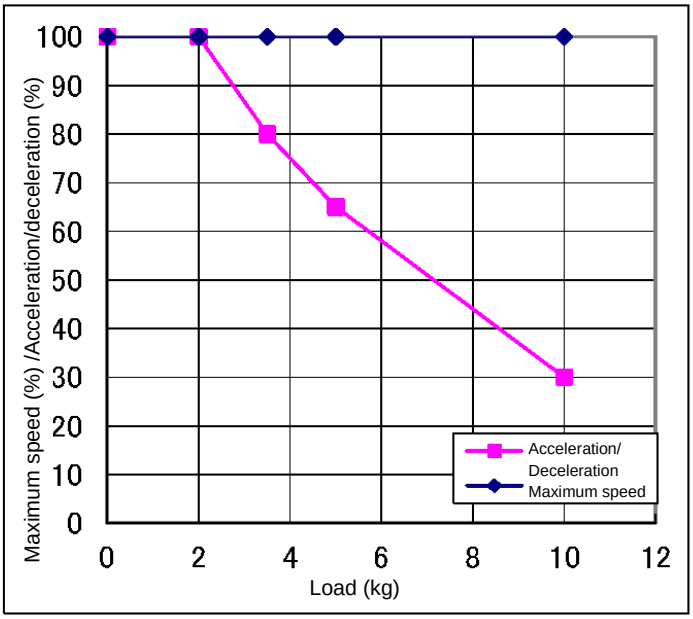
For load conditions and program setting, see "19.3.2 Load Conditions and Program Setting".

21.3.3 Setting Maximum Speed and Robot Acceleration/Deceleration for Load Conditions

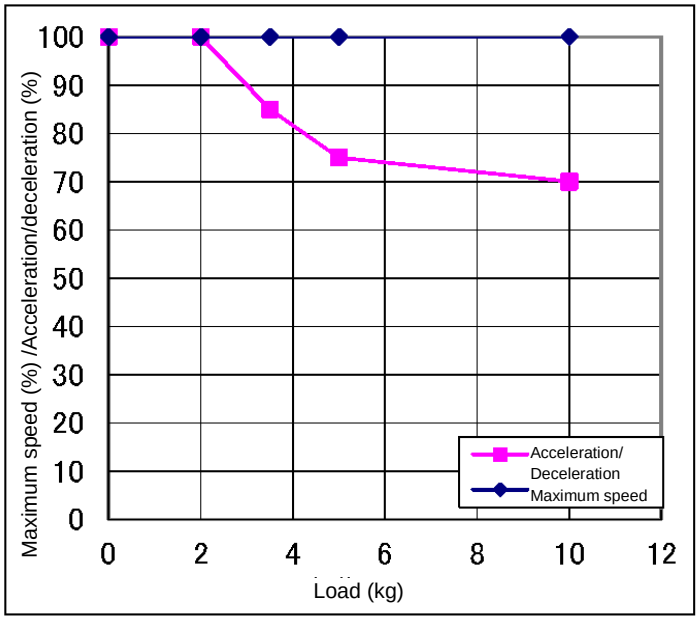
When the PAYLOAD command is used, the maximum speed and acceleration/ deceleration of the robot are automatically changed according to the load conditions. The maximum speed and acceleration/deceleration change with the load mass, as shown in Figs. 21.4 to 21.6.



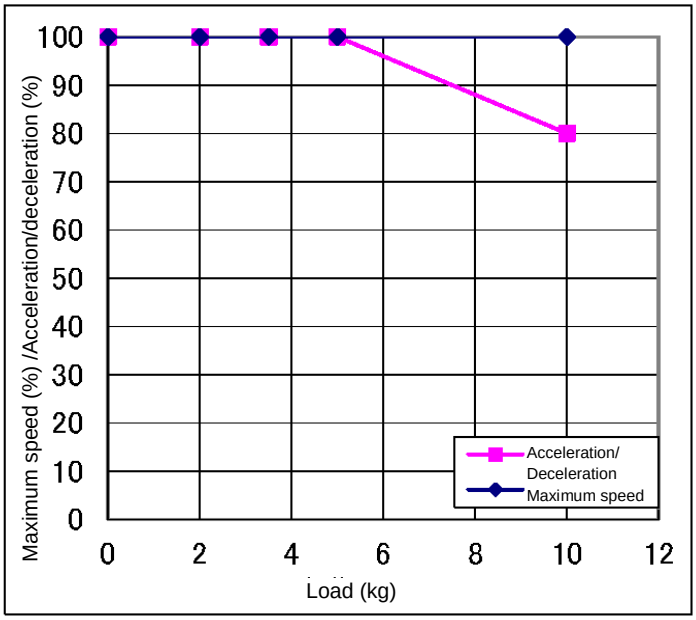
Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 1)



Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 2)

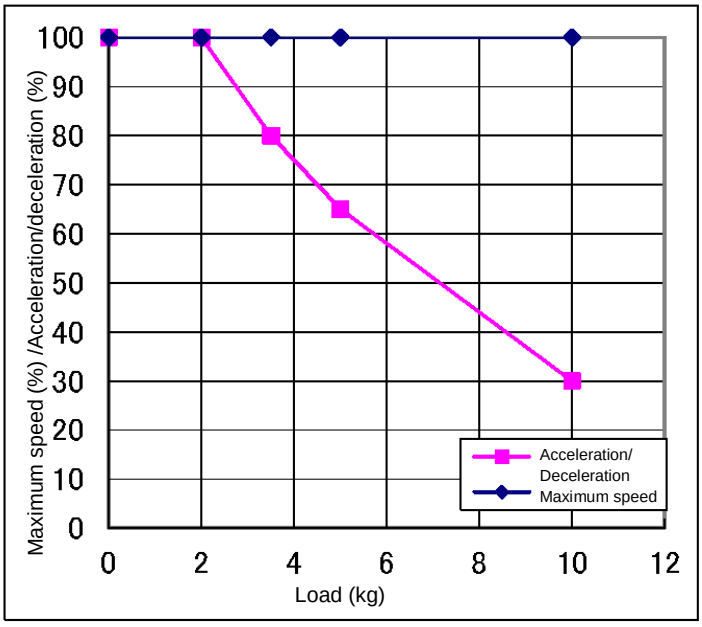


Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 3)

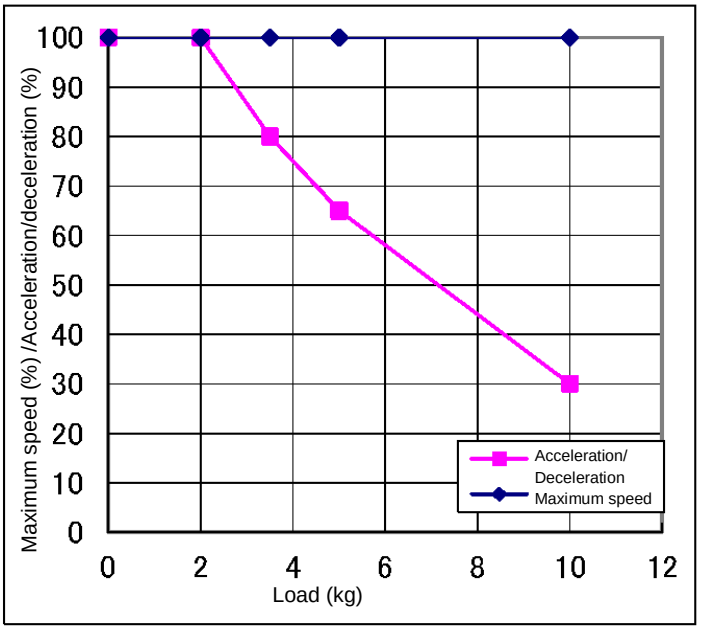


Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 4)

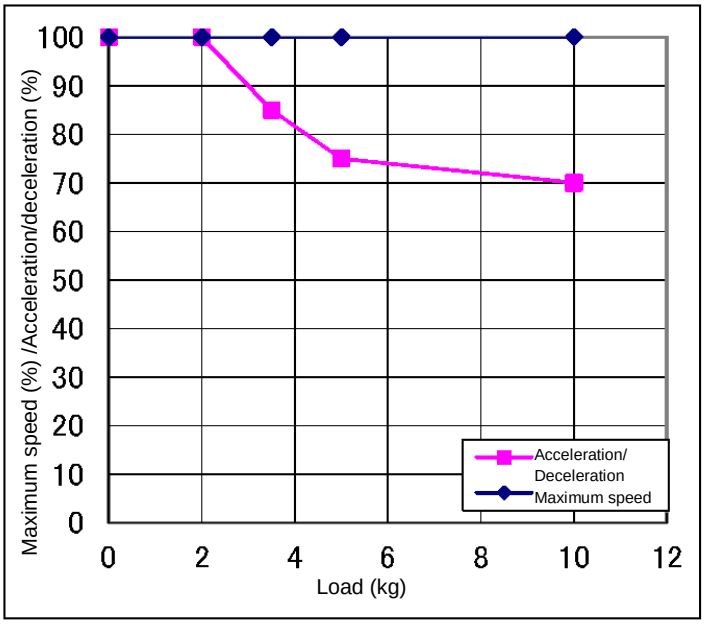
Fig. 21.4 Setting of maximum speed and acceleration/deceleration in relation to load mass (THL800)



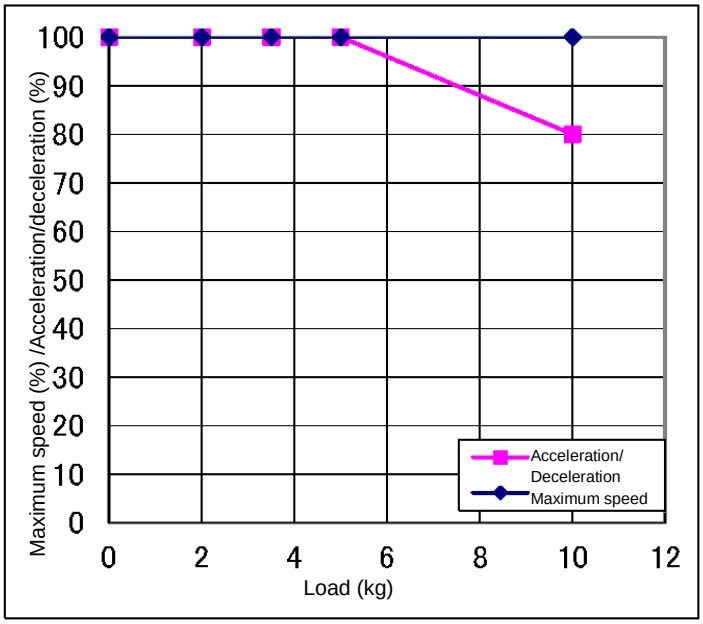
Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 1)



Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 2)

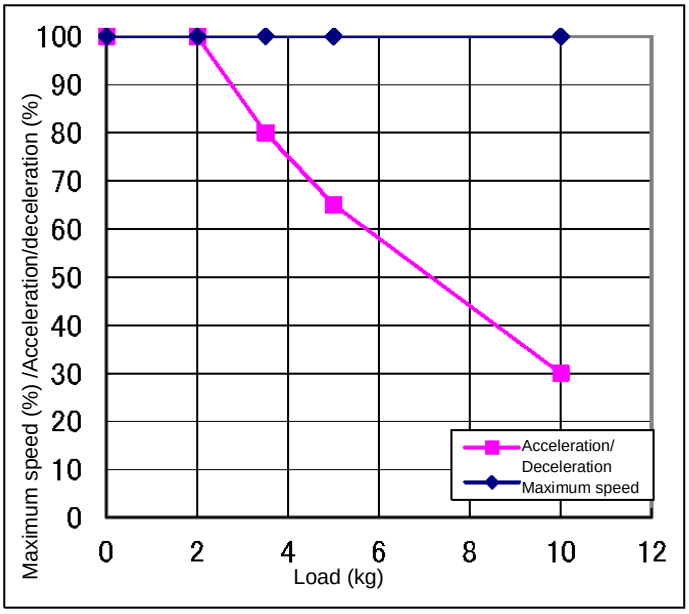


Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 3)

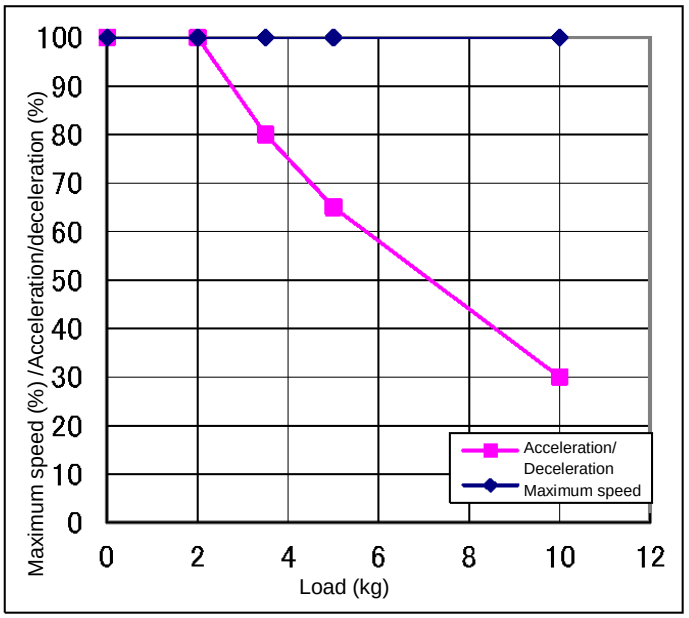


Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 4)

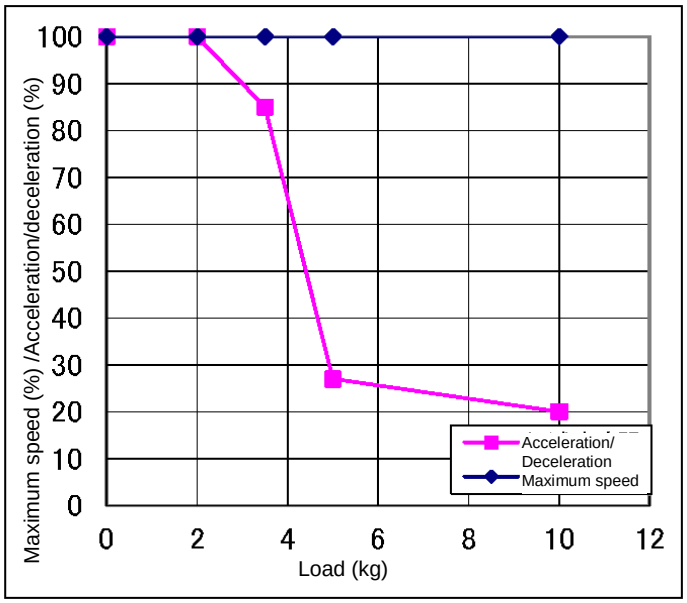
Fig. 21.5 Setting of maximum speed and acceleration/deceleration in relation to load mass (THL900)



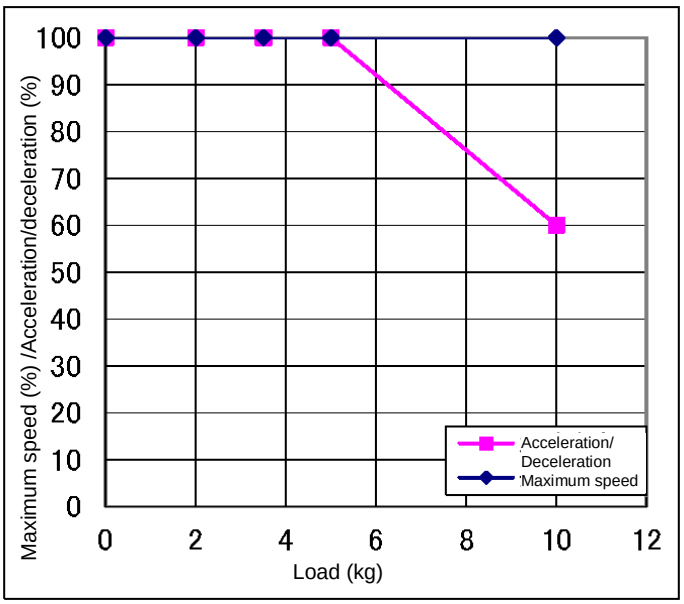
Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 1)



Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 2)



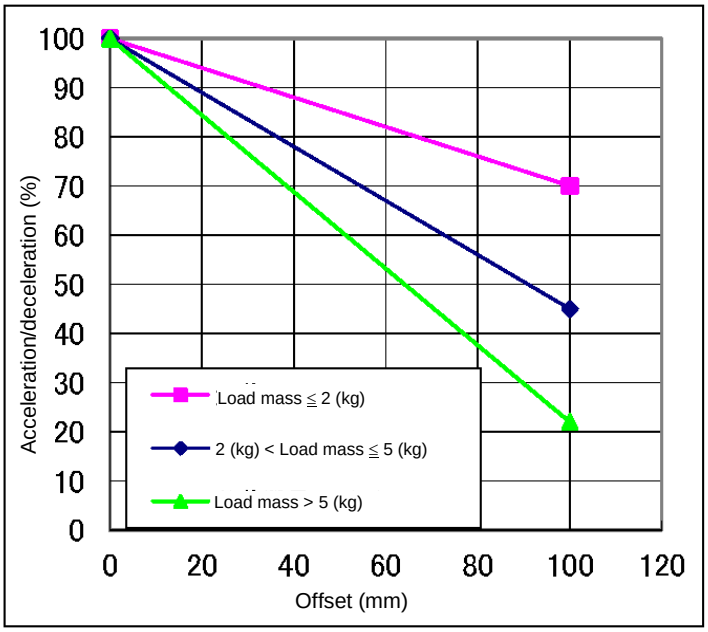
Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 3)



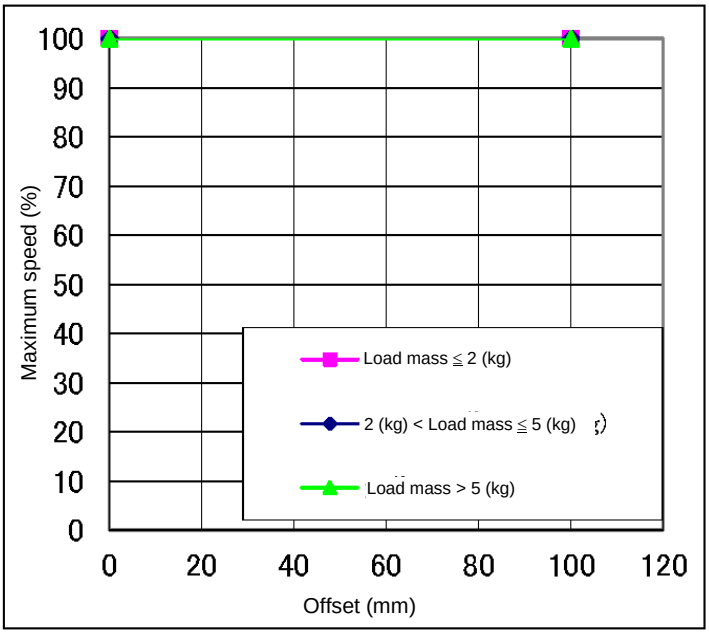
Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 4)

Fig. 21.6 Setting of maximum speed and acceleration/deceleration in relation to load mass (THL1000)

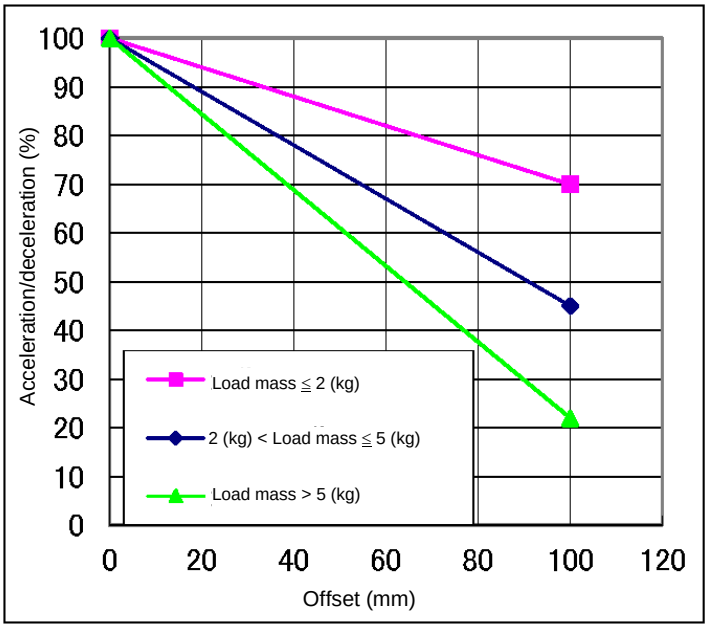
Additionally, if there is an offset of load gravity center, the maximum speed and acceleration/deceleration change as shown in Fig. 21.7 to Fig. 21.12.



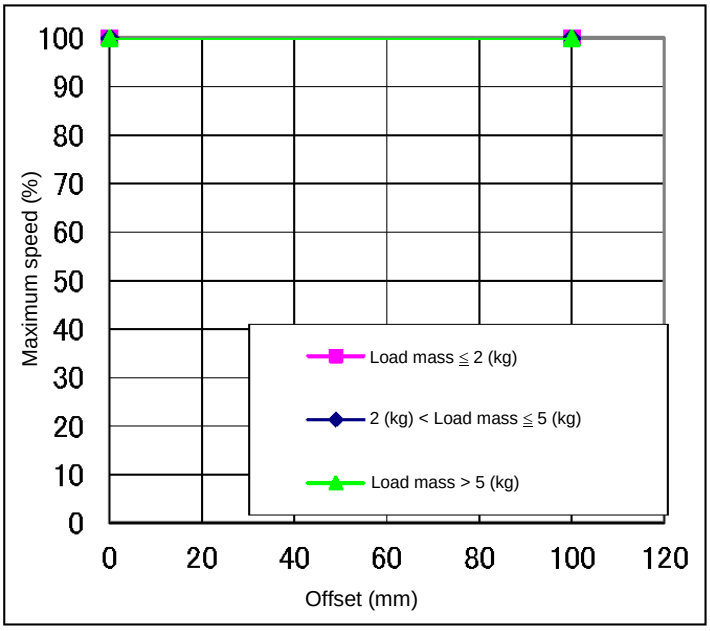
Setting of acceleration/deceleration in relation to offset (Axis 1)



Setting of maximum speed in relation to offset (Axis 1)

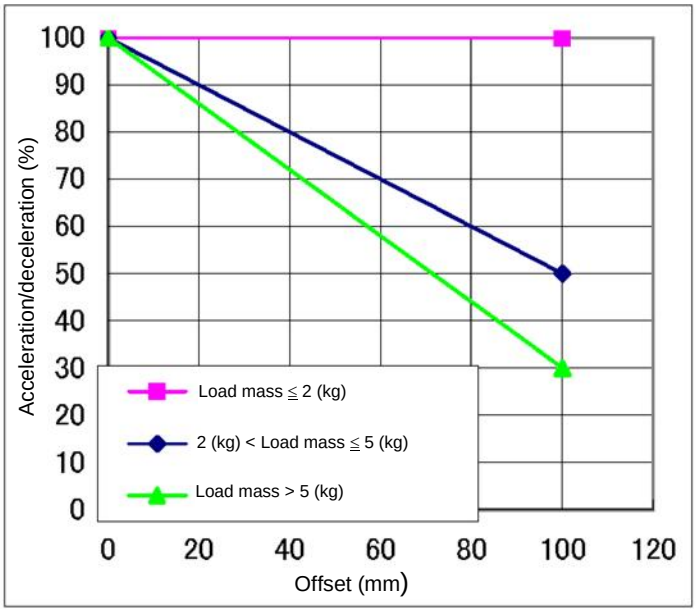


Setting of acceleration/deceleration in relation to offset (Axis 2)

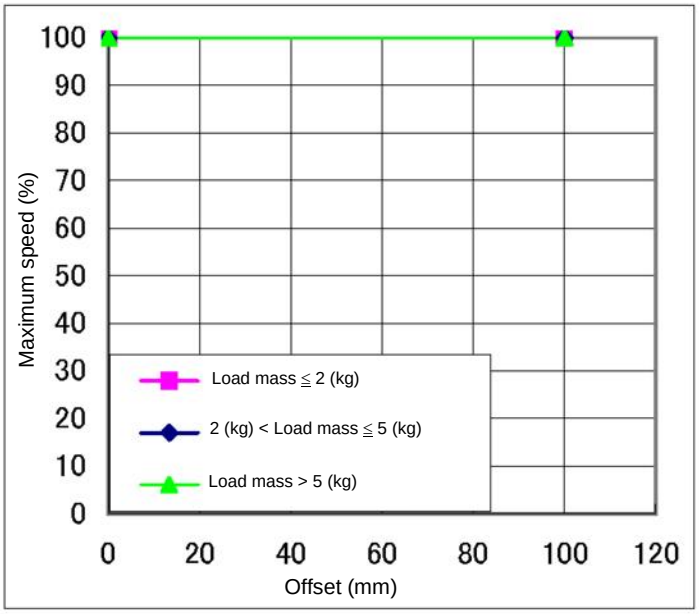


Setting of maximum speed in relation to offset (Axis 2)

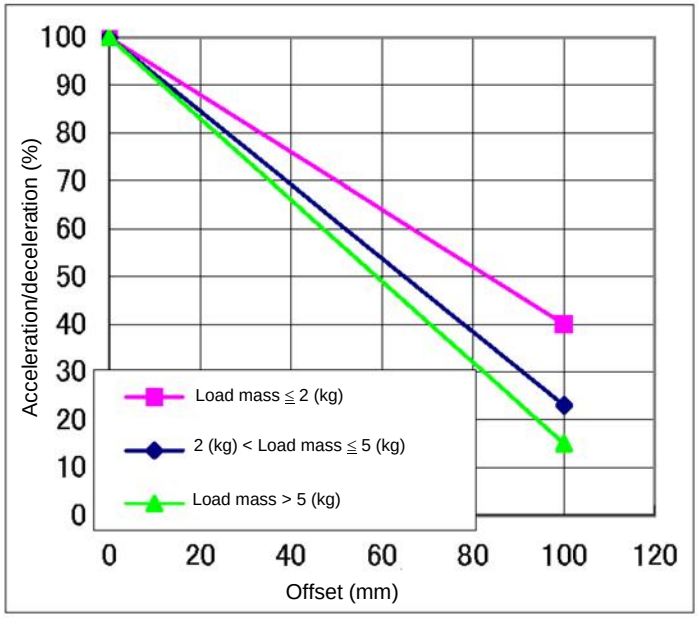
Fig. 21.7 Setting of maximum speed and acceleration/deceleration in relation to gravity center offset (Axes 1 and 2) (THL800)



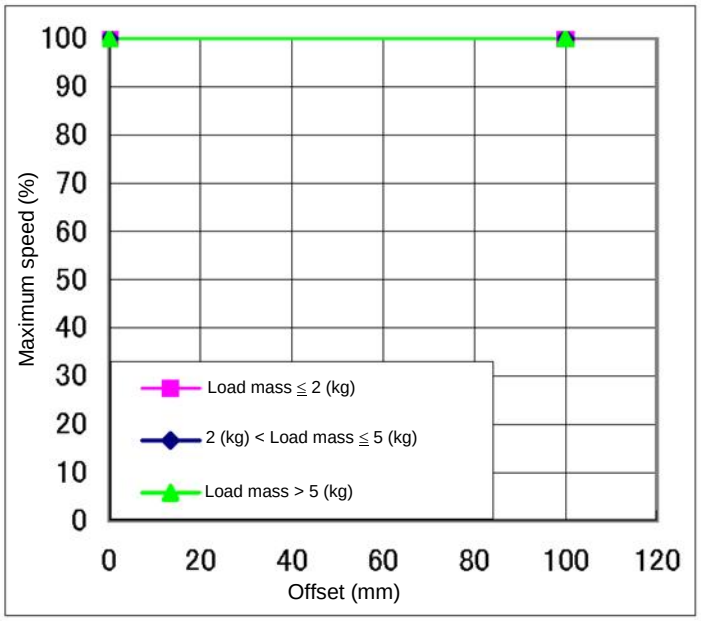
Setting of acceleration/deceleration in relation to offset (Axis 3)



Setting of maximum speed in relation to offset (Axis 3)

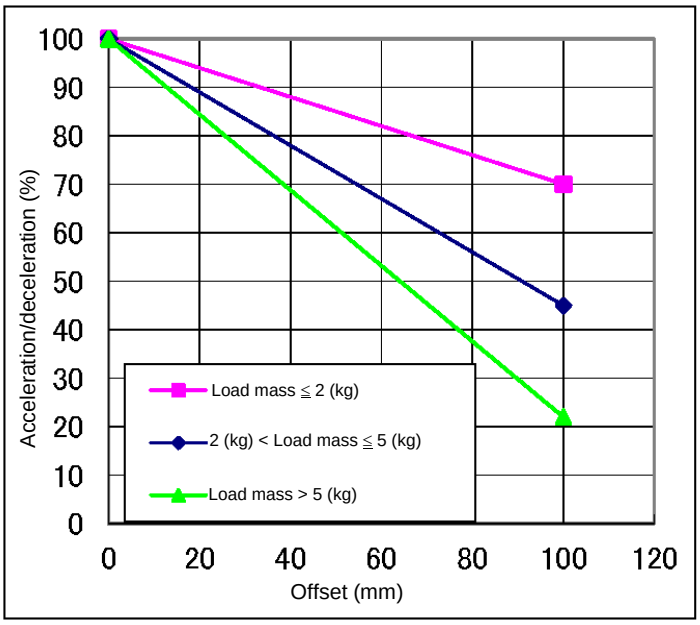


Setting of acceleration/deceleration in relation to offset (Axis 4)

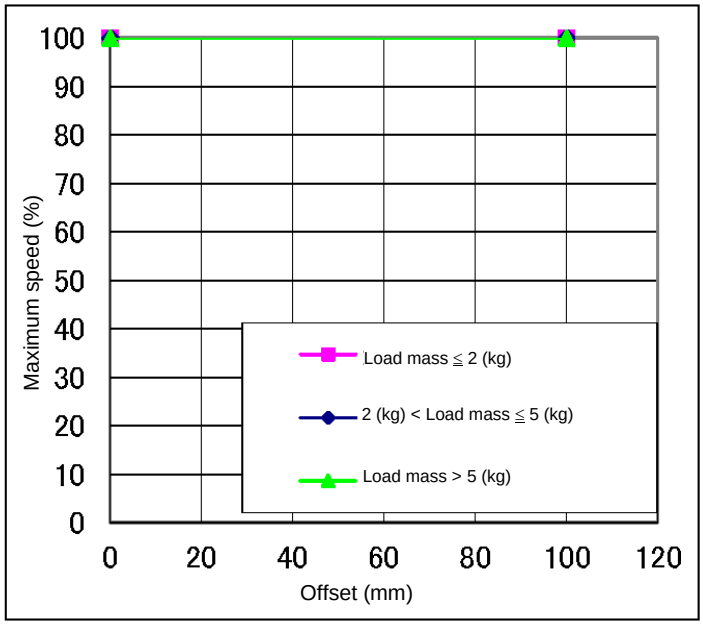


Setting of maximum speed in relation to offset (Axis 4)

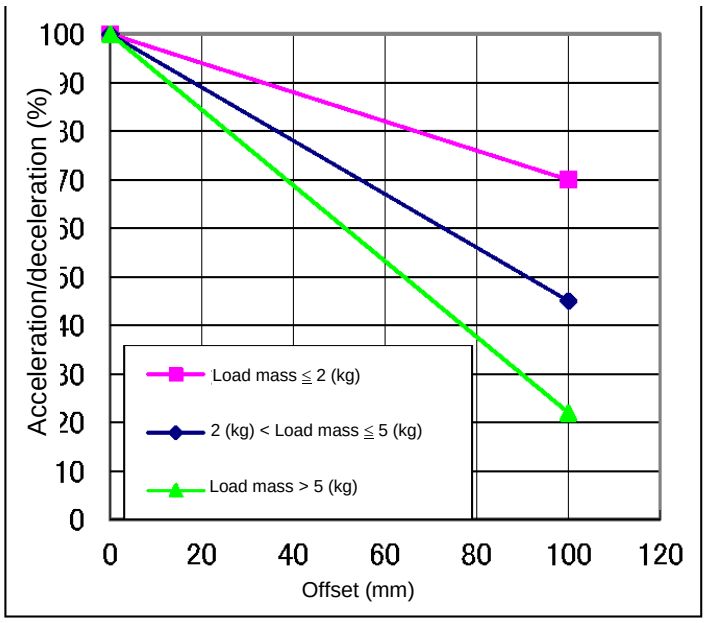
Fig. 21.8 Setting of maximum speed and acceleration/deceleration in relation to gravity center offset (Axes 3 and 4) (THL800)



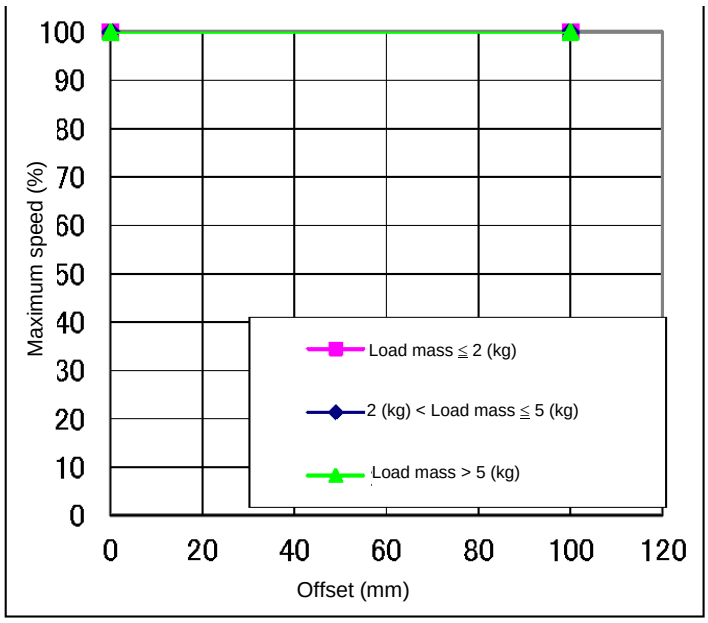
Setting of acceleration/deceleration in relation to offset (Axis 1)



Setting of maximum speed in relation to offset (Axis 1)

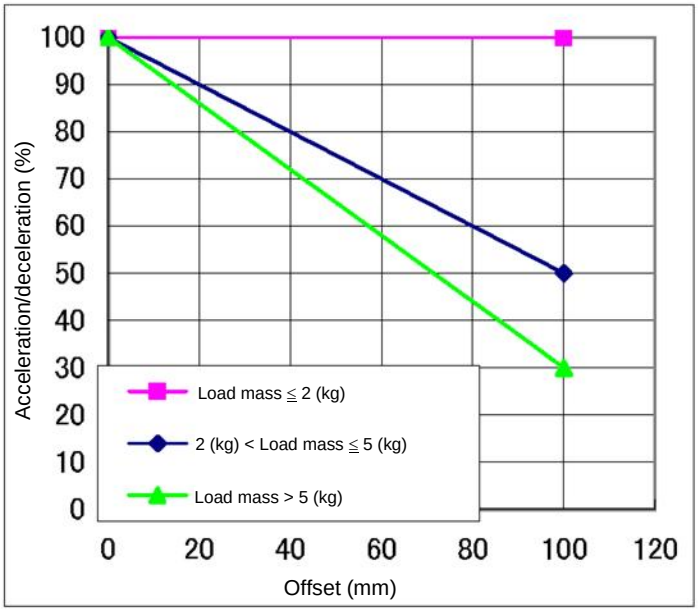


Setting of acceleration/deceleration in relation to offset (Axis 2)

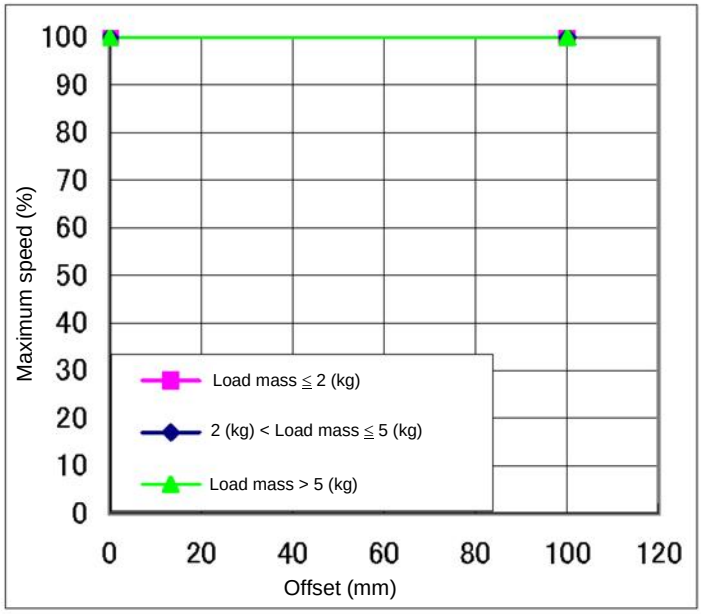


Setting of maximum speed in relation to offset (Axis 2)

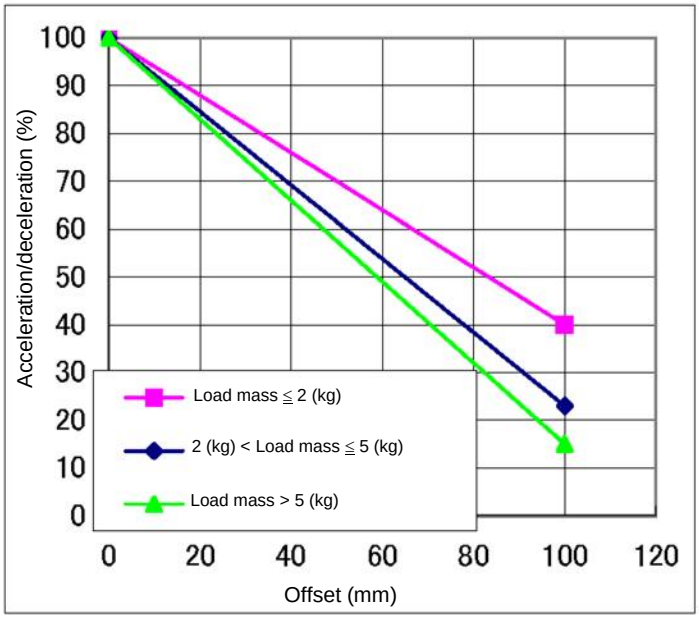
Fig. 21.9 Setting of maximum speed and acceleration/deceleration in relation to gravity center offset (Axes 1 and 2) (THL900)



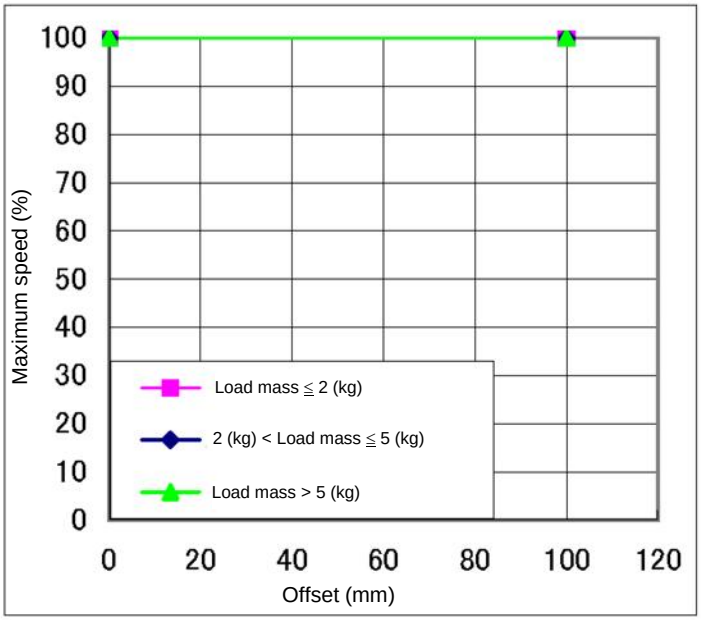
Setting of acceleration/deceleration in relation to offset (Axis 3)



Setting of maximum speed in relation to offset (Axis 3)

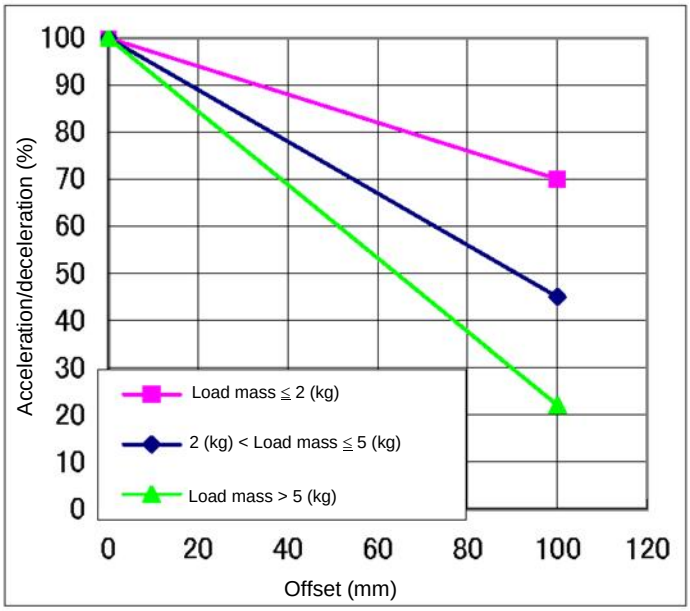


Setting of acceleration/deceleration in relation to offset (Axis 4)

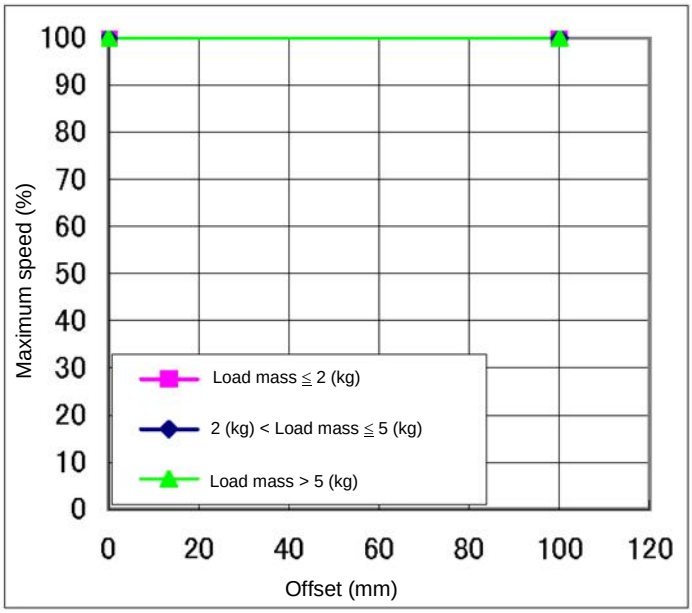


Setting of maximum speed in relation to offset (Axis 4)

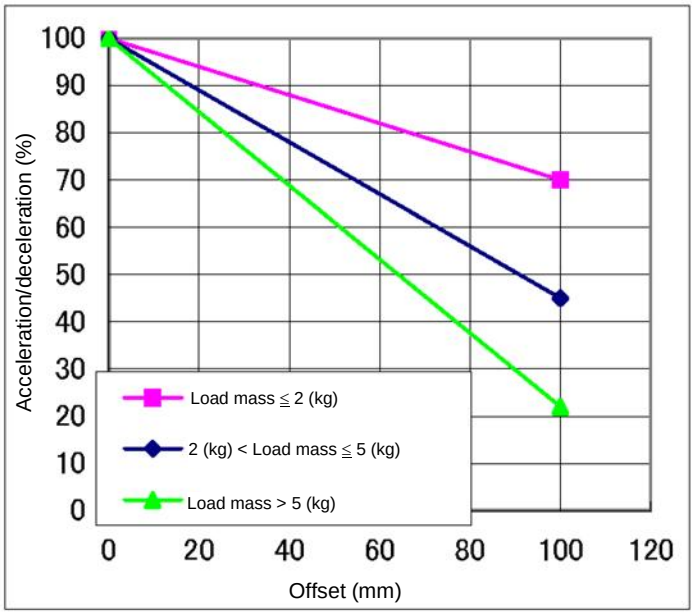
Fig. 21.10 Setting of maximum speed and acceleration/deceleration in relation to gravity center offset (Axes 3 and 4) (THL900)



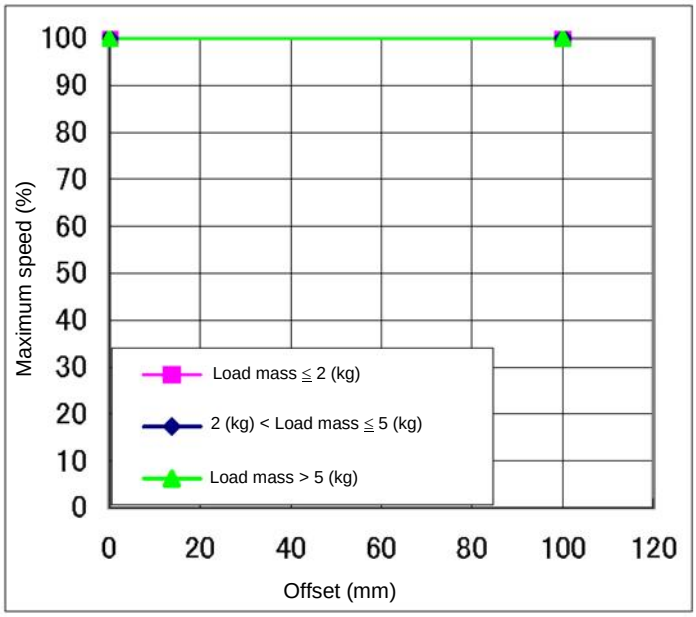
Setting of acceleration/deceleration in relation to offset (Axis 1)



Setting of maximum speed in relation to offset (Axis 1)

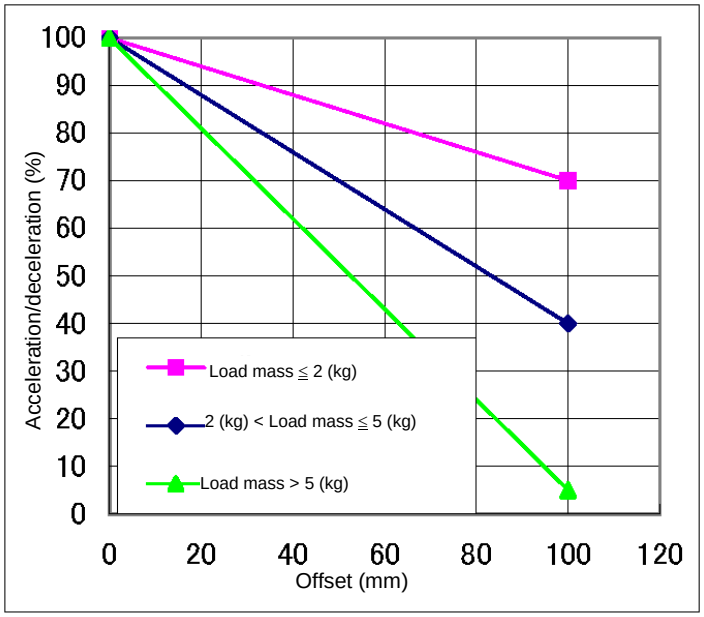


Setting of acceleration/deceleration in relation to offset (Axis 2)

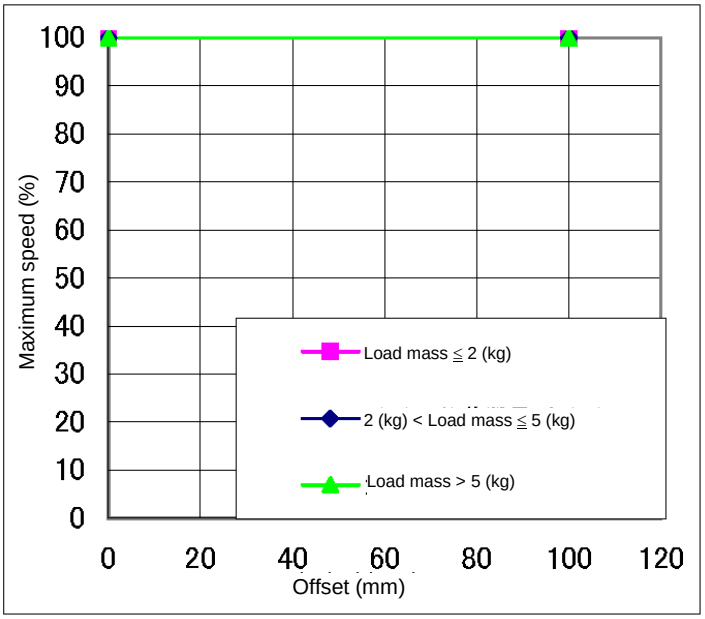


Setting of maximum speed in relation to offset (Axis 2)

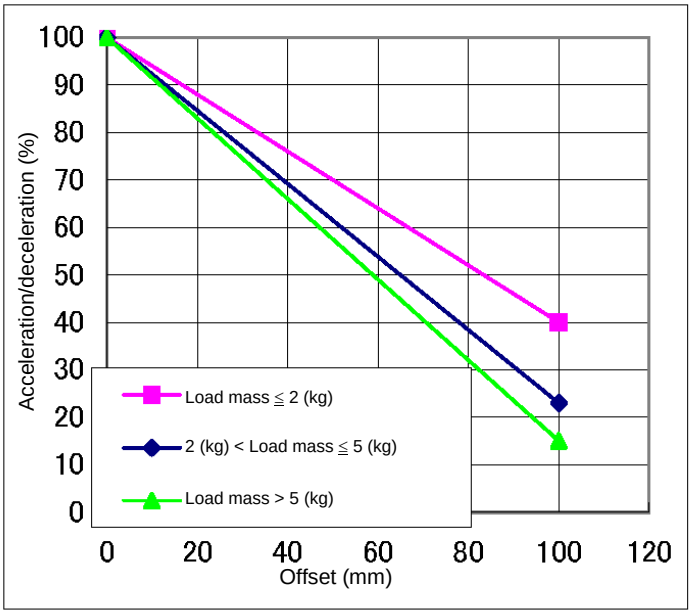
Fig. 21.11 Setting of maximum speed and acceleration/deceleration in relation to gravity center offset (Axes 1 and 2) (THL1000)



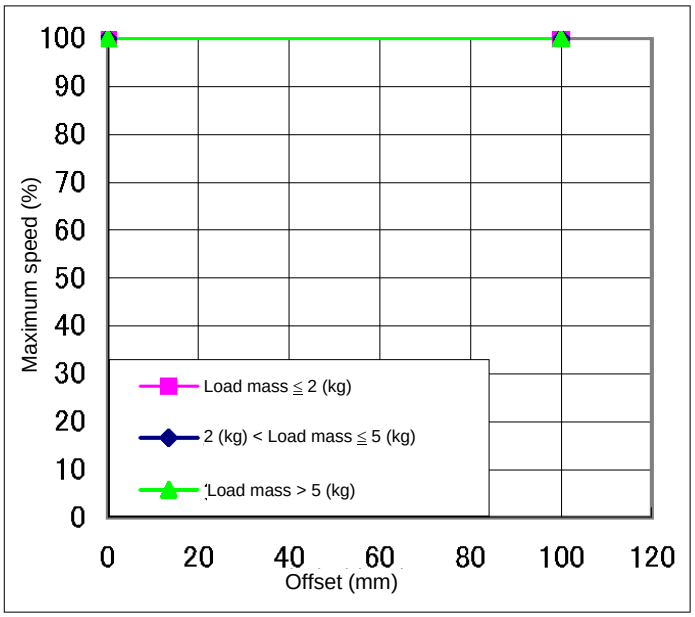
Setting of acceleration/deceleration in relation to offset (Axis 3)



Setting of maximum speed in relation to offset (Axis 3)



Setting of acceleration/deceleration in relation to offset (Axis 4)



Setting of maximum speed in relation to offset (Axis 4)

Fig. 21.12 Setting of maximum speed and acceleration/deceleration in relation to gravity center offset (Axes 3 and 4) (THL1000)

22. Tool Interface (THL1200)

22.1 Mounting Tool

The tool is mounted on the end of the tool shaft. Dimensions of the tool shaft section are shown in Fig. 22.1. As shown in Fig. 22.1, the tool is centered with the ϕ12H7 mating section. The tool direction is adjusted by means of the 4 x 4 keys and secured with four (4) M4 bolts.
The tool flange is optional.

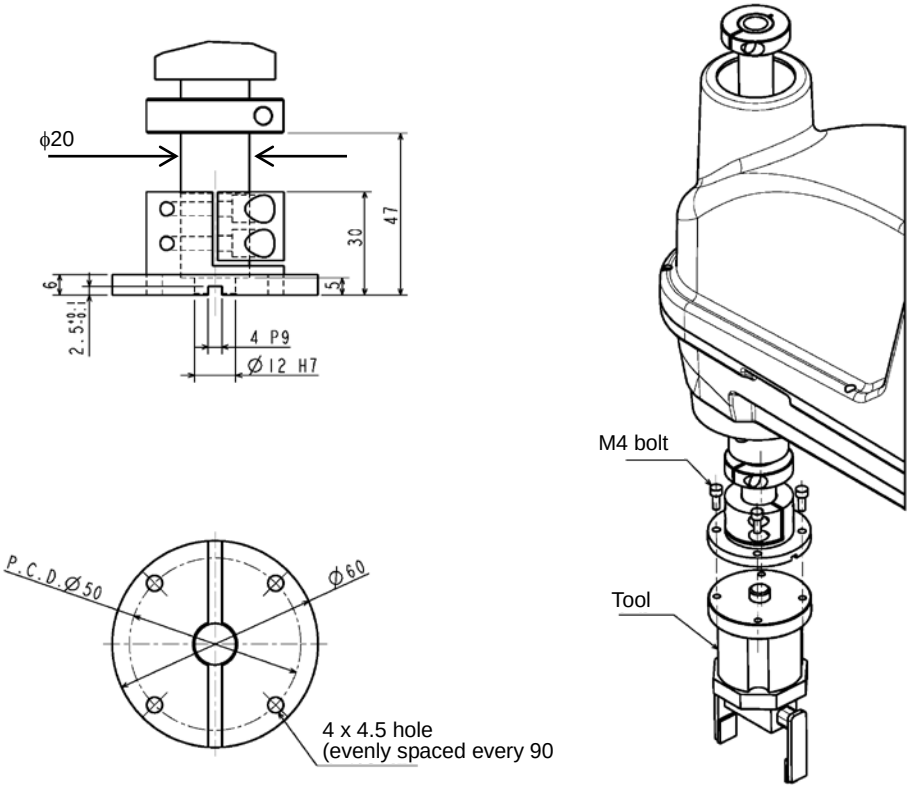


Fig. 22.1 Tool mounting dimensions (THL1200)

22.2 Tool Air Piping

The customer is requested to install the tool air piping using the following accessories. Fig. 22.2 shows an installation example of tool air piping for your reference.

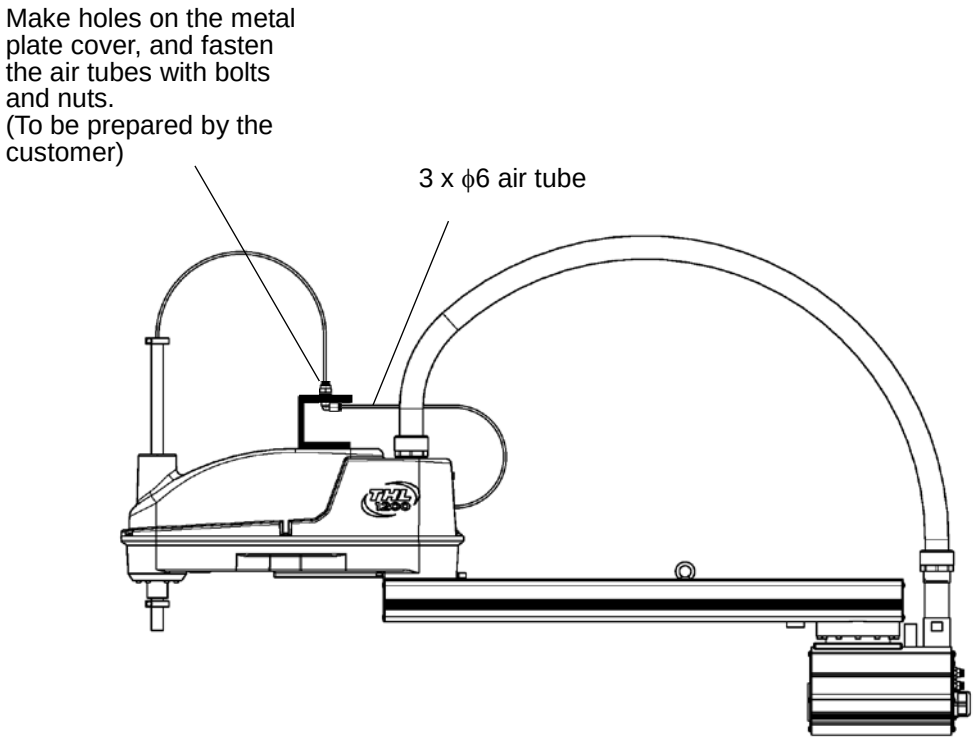


Fig. 22.2 Tool air piping installation example (THL1200)

! CAUTION

- Air tubes are consumable supplies. Check the air tubes during periodic inspection, and change them if any damage is found.
- The customer is requested to prepare solenoid valve air.
- Fig. 22.2 is a reference example. Tool air piping is not guaranteed. The warranty does not cover breakdowns of the harness, the air tubes or their accessories due to piping by the customer.

22.3 Permissible Load Conditions and Program Setting

This paragraph describes the permissible load conditions of the robot and how to set up the program according to the load.

22.3.1 Permissible Load Conditions

The robot load conditions are defined by the tool mass, moment of inertia and offset value of tool gravity center from the center of the tool shaft, as shown in Fig. 22.3. The permissible load conditions are shown in Table 22.1.

Table 22.1 Permissible load conditions (THL1200)

Conditions	Permissible values
Mass	Max. 10 kg
Load inertia	Max 0.20 kg·m^2
Offset value of load gravity center	Max. 100 mm

! CAUTION

- NEVER operate the robot under the load conditions exceeding the permissible values. Otherwise, the robot life and safety cannot be guaranteed.

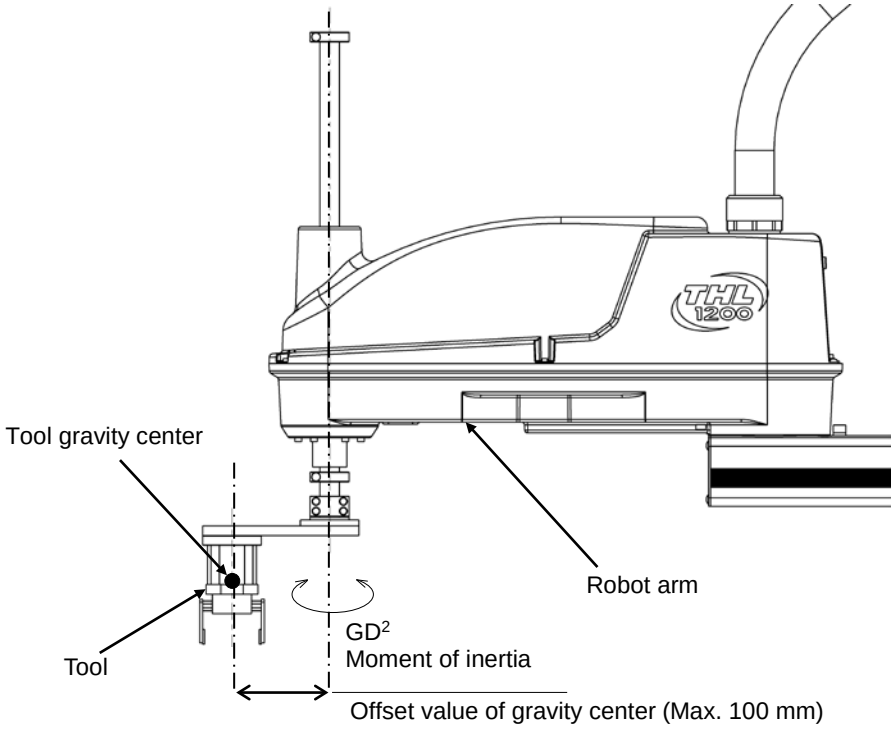


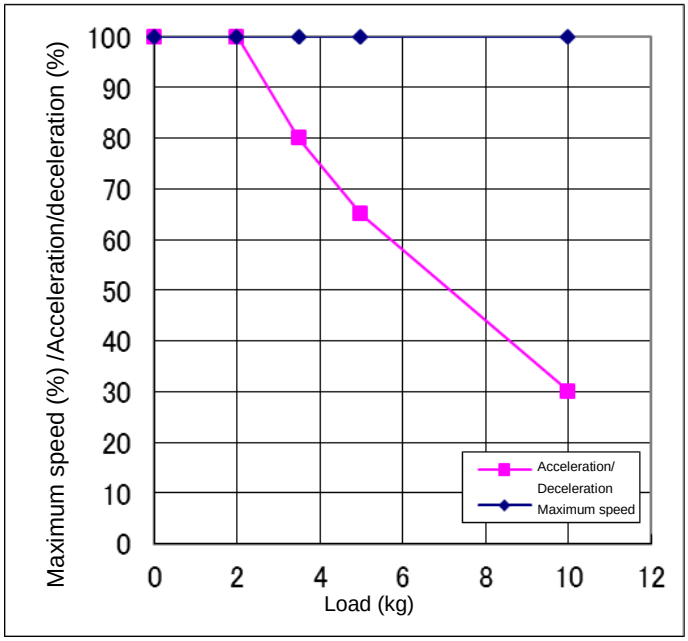
Fig. 22.4 Robot tool (THL1200)

22.4.2 Load Conditions and Program Setting

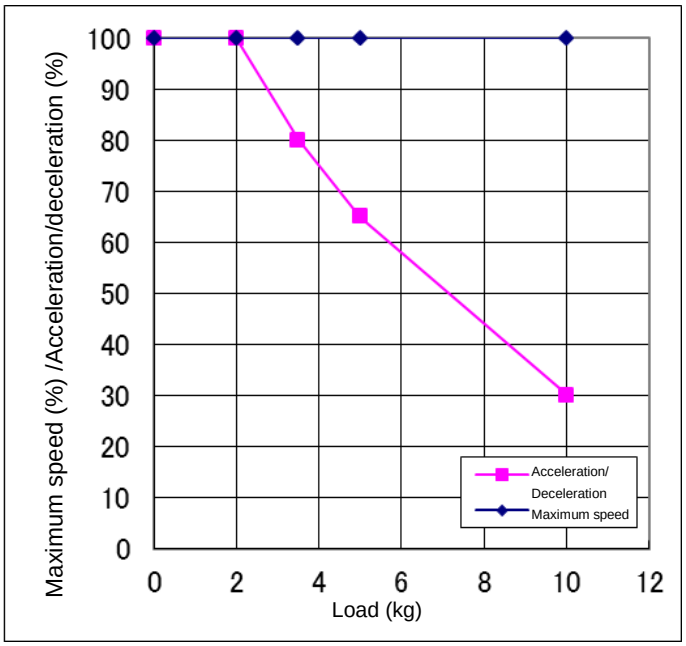
For load conditions and program setting, see "19.3.2 Load Conditions and Program Setting".

22.3.3 Setting Maximum Speed and Robot Acceleration/Deceleration for Load Conditions

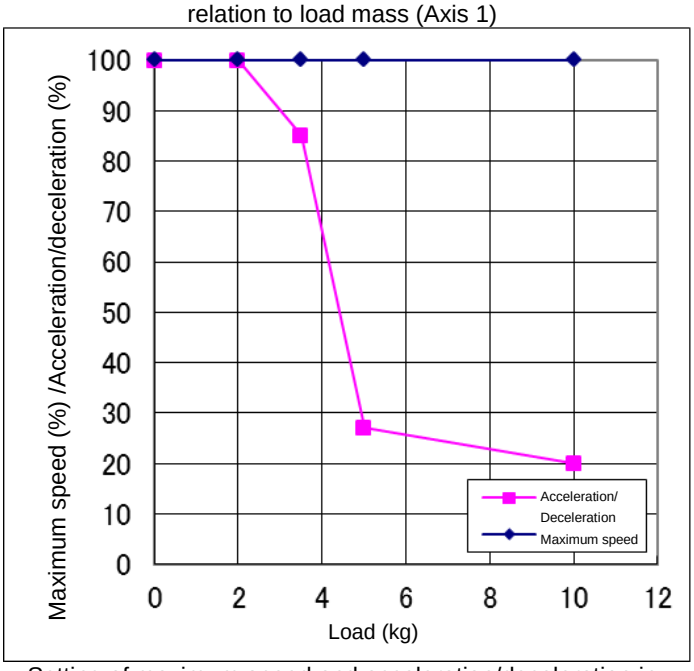
When the PAYLOAD command is used, the maximum speed and acceleration/ deceleration of the robot are automatically changed according to the load conditions. The maximum speed and acceleration/deceleration change with the load mass, as shown in Figs. 22.4.



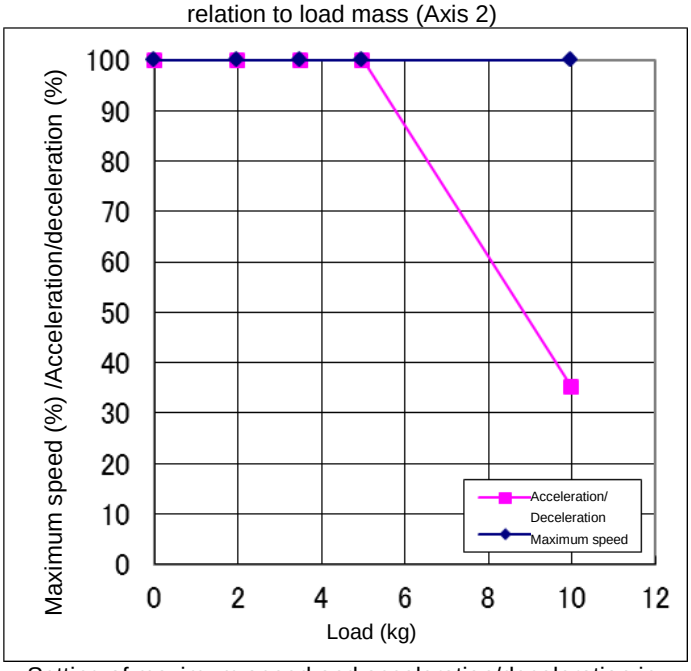
Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 1)



Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 2)



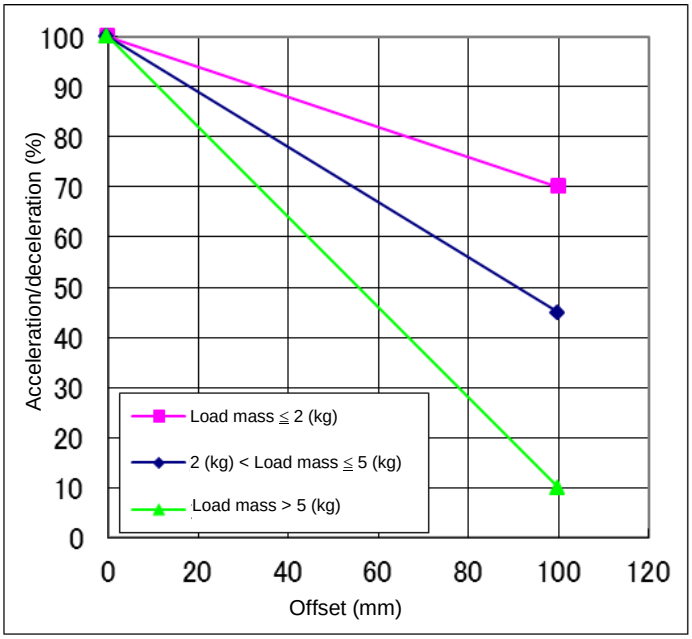
Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 3)



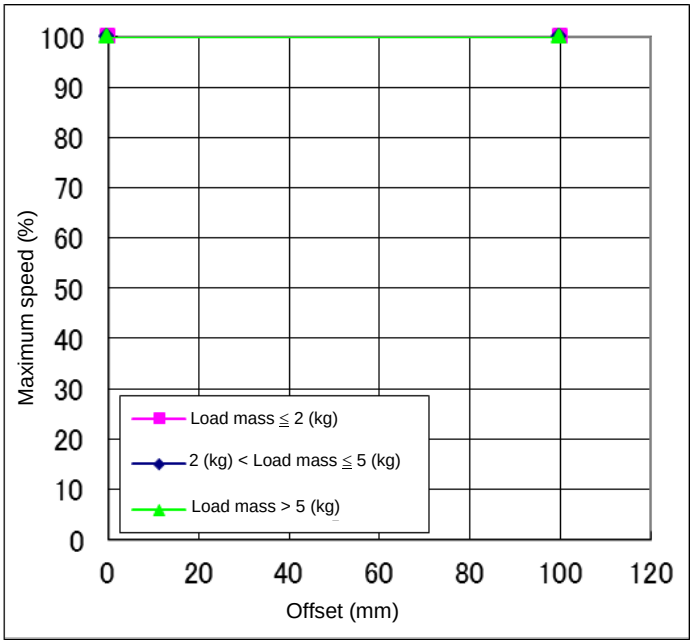
Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 4)

Fig. 22.4 Setting of maximum speed and acceleration/deceleration in relation to load mass (THL1200)

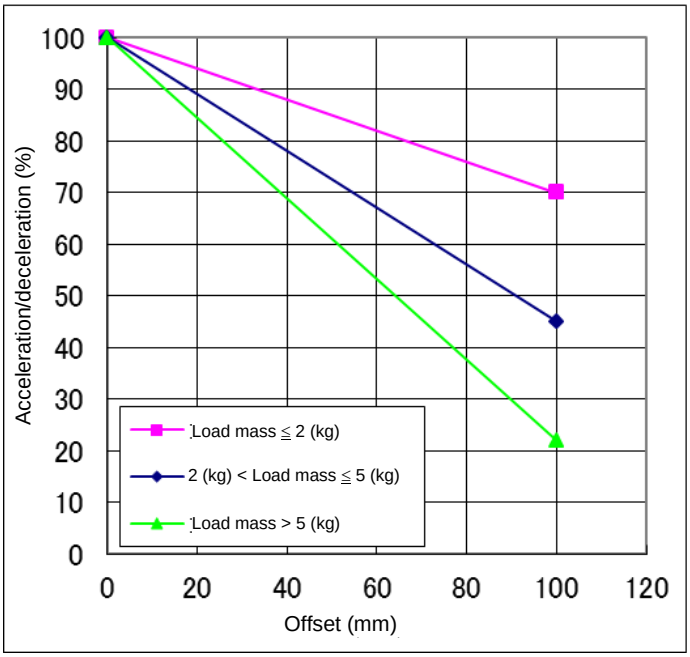
Additionally, if there is an offset of load gravity center, the maximum speed and acceleration/deceleration change as shown in Fig. 22.5 to 22.6.



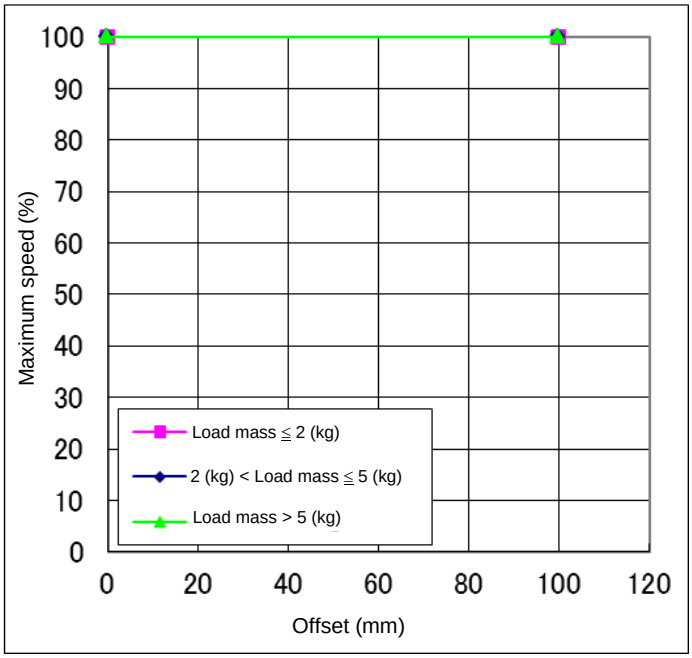
Setting of acceleration/deceleration in relation to offset (Axis 1)



Setting of maximum speed in relation to offset (Axis 1)

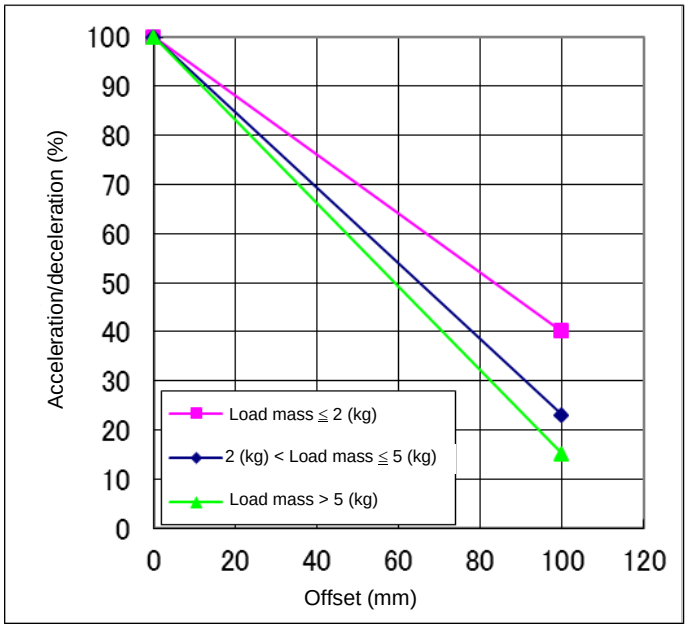


Setting of acceleration/deceleration in relation to offset (Axis 2)

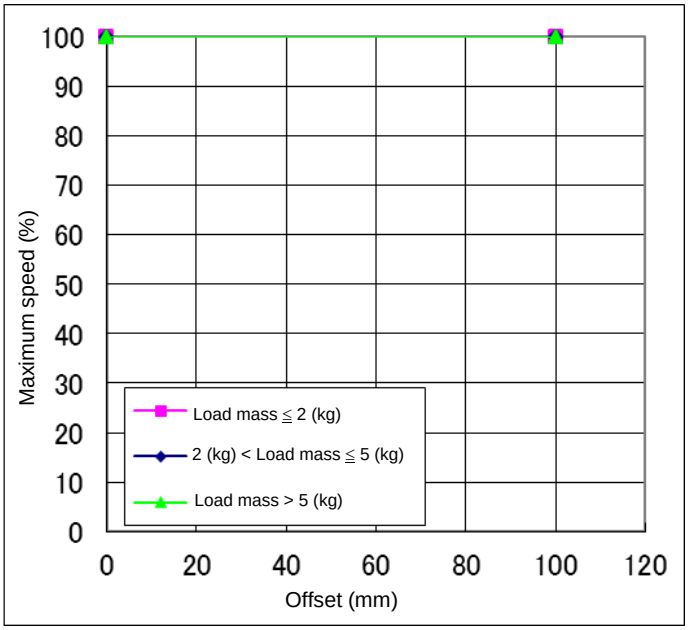


Setting of maximum speed in relation to offset (Axis 2)

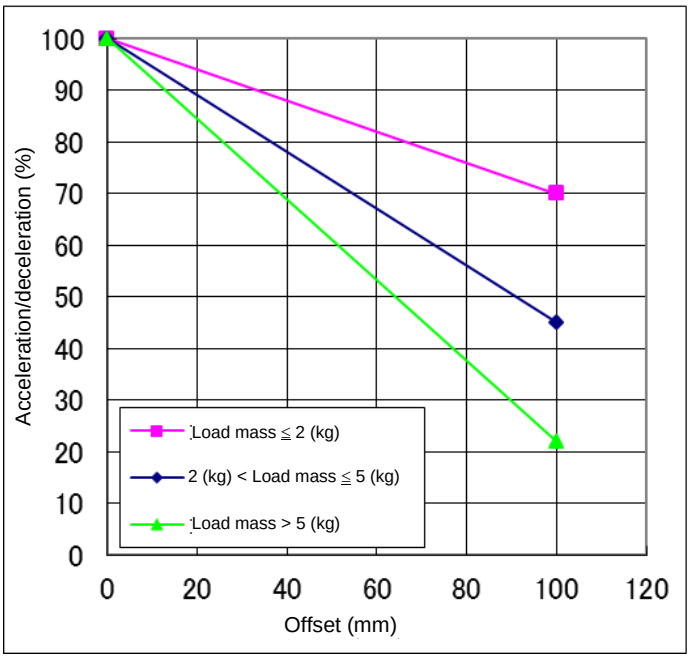
Fig. 22.5 Setting of maximum speed and acceleration/deceleration in relation to gravity center offset (Axes 1 and 2) (THL1200)



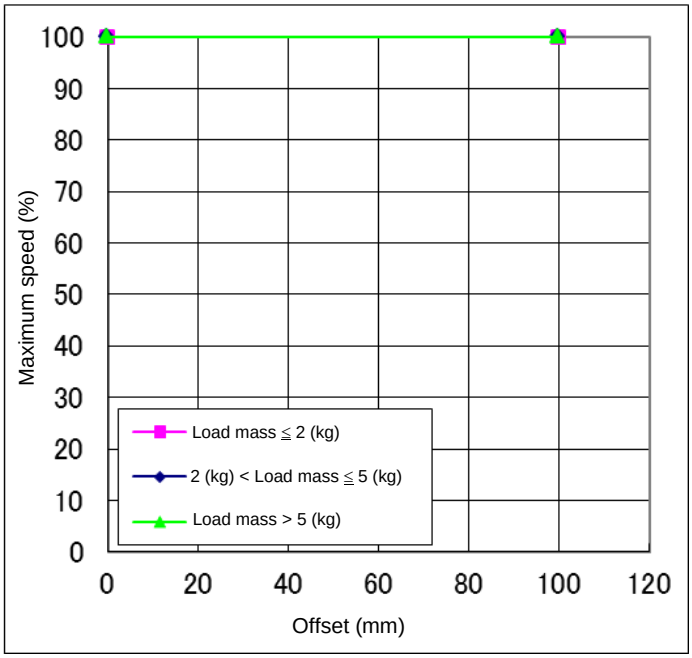
Setting of acceleration/deceleration in relation to offset (Axis 3)



Setting of maximum speed in relation to offset (Axis 3)



Setting of acceleration/deceleration in relation to offset (Axis 4)



Setting of maximum speed in relation to offset (Axis 4)

Fig. 22.6 Setting of maximum speed and acceleration/deceleration in relation to gravity center offset (Axes 3 and 4) (THL1200)

23. Tool Interface (TSL3000)

23.1 Tool Wiring

Eight (8) input signals are provided for sensors, etc. and eight (8) control signals for solenoid valves, etc. A supply power signal of 24 VDC is also provided. They are connected to the controller. The wiring arrangement for these cables is shown in Fig. 23.9. The wires are connected to the connectors on the upper side of Arm 2. The user should provide the following connectors to connect the cables.

D-SUB connector (standard) Type: <Hood> XM2S–2511 (Maker: OMRON)
<Basic body> XM3A–2521 (Maker: OMRON)

Adaptive cable Conductive cross section area: 0.2 mm^2 to 0.5 mm^2

Each connector and cable are connected by soldering.

! DANGER

- Be sure to use the designated wire. Otherwise, fires or faults may be caused.
- When connecting the connector and wires, make sure not to mistake the terminal arrangement.
- After making the connection, use a tester, etc., to confirm the connection.

When controlling the robot from the sequencer (i.e., programmable ladder controller: PLC), etc. installed separately, remove the base rear cover from the base section, remove connectors JOES and JOFS on the rear side, then connect the cables running from the PLC, etc. through the cable inlets provided on the base rear cover. Also, the D-SUB 25 pins can be used separately for THL500 to THL1200. (See Figs. 23.1, 23.2 ,23.3 and 23.4.) For ahead of the JOES and JOFS connectors, the user should prepare the following plug connectors and connect the cables. The current is 1 A or less per cable.

Type of connector: JOES SMP–10V–BC (Maker: J.S.T. Mfg.)
JOFS SMP–11V–BC (Maker: J.S.T. Mfg.)

Type of contact: BHF–001T–0.8SS (Maker: J.S.T. Mfg.)

Adaptive cable Conductive cross section area: 0.2 mm^2 to 0.3 mm^2

Opposite connector type (THL500 to THL1200)

Type of connector: JOEP SMR–10V–B (Maker: J.S.T. Mfg.)
JOFP SMR–11V–B (Maker: J.S.T. Mfg.)

Type of contact: BYM–001T–0.6 (Maker: J.S.T. Mfg.)

D-SUB 25 pin connector types

Connector type (male)	<Hood> XM2S-2511 (maker: OMRON)
	<Body> XM3A-2521 (maker: OMRON)
Connector type (female)	<Body> XM3D-2521 (maker: OMRON)

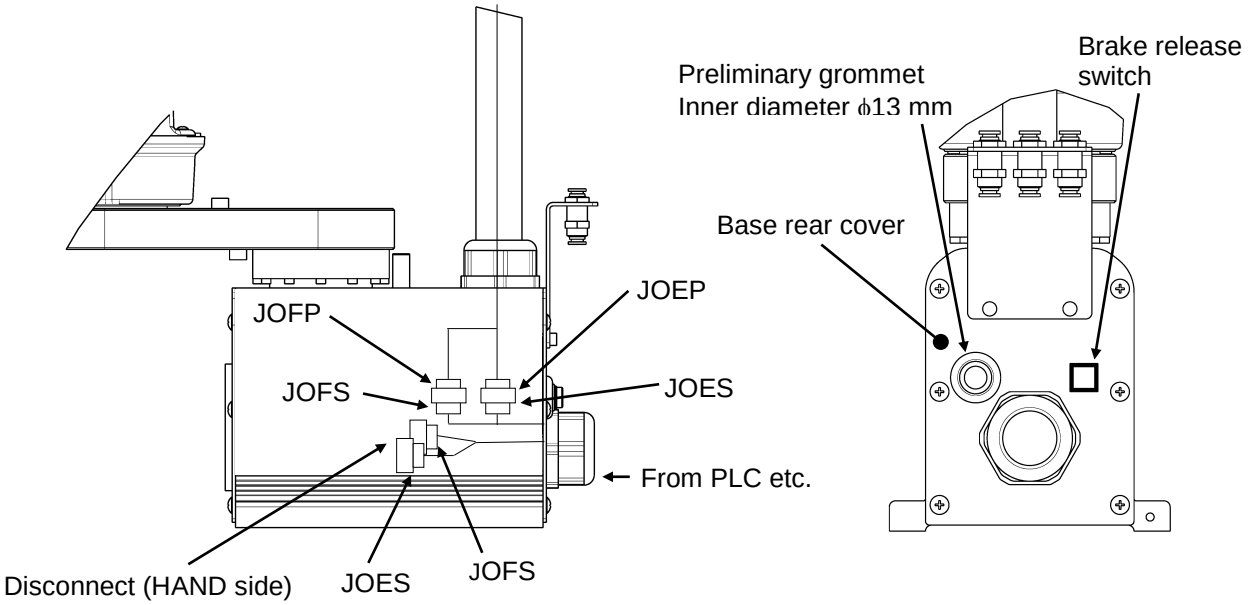


Fig. 23.1 Wiring to PLC, etc. (THL300, THL400 and TSL3000)

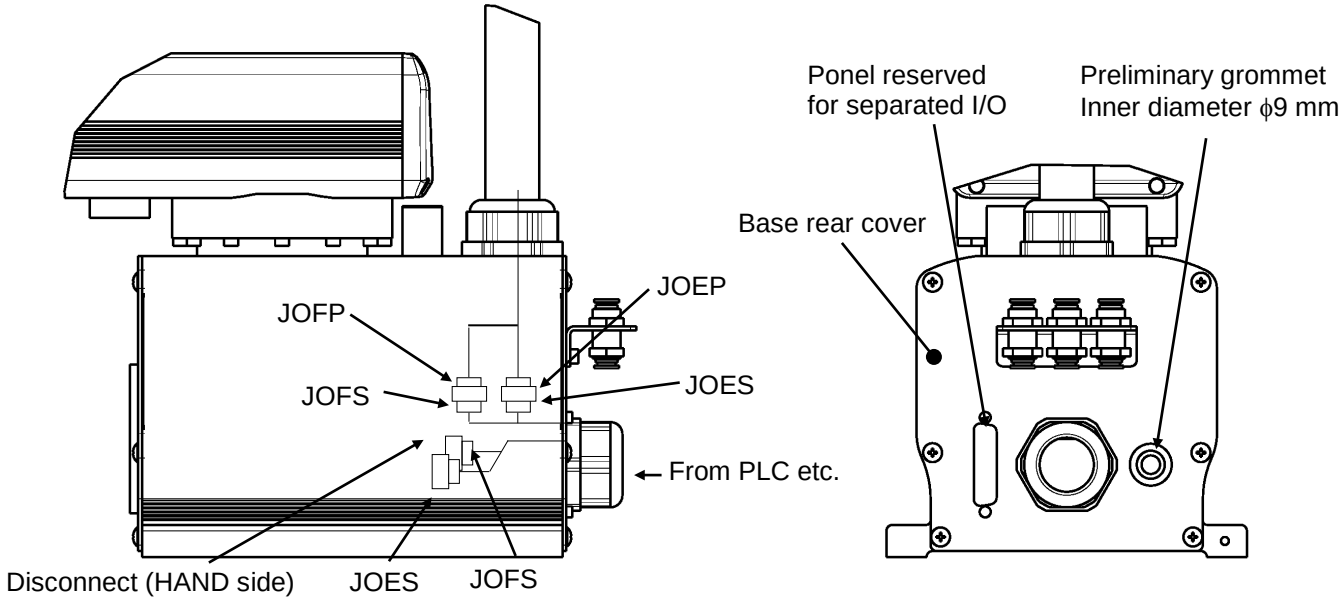


Fig. 23.2 Wiring to PLC, etc. (THL500, THL600, THL700 and TSL3000)

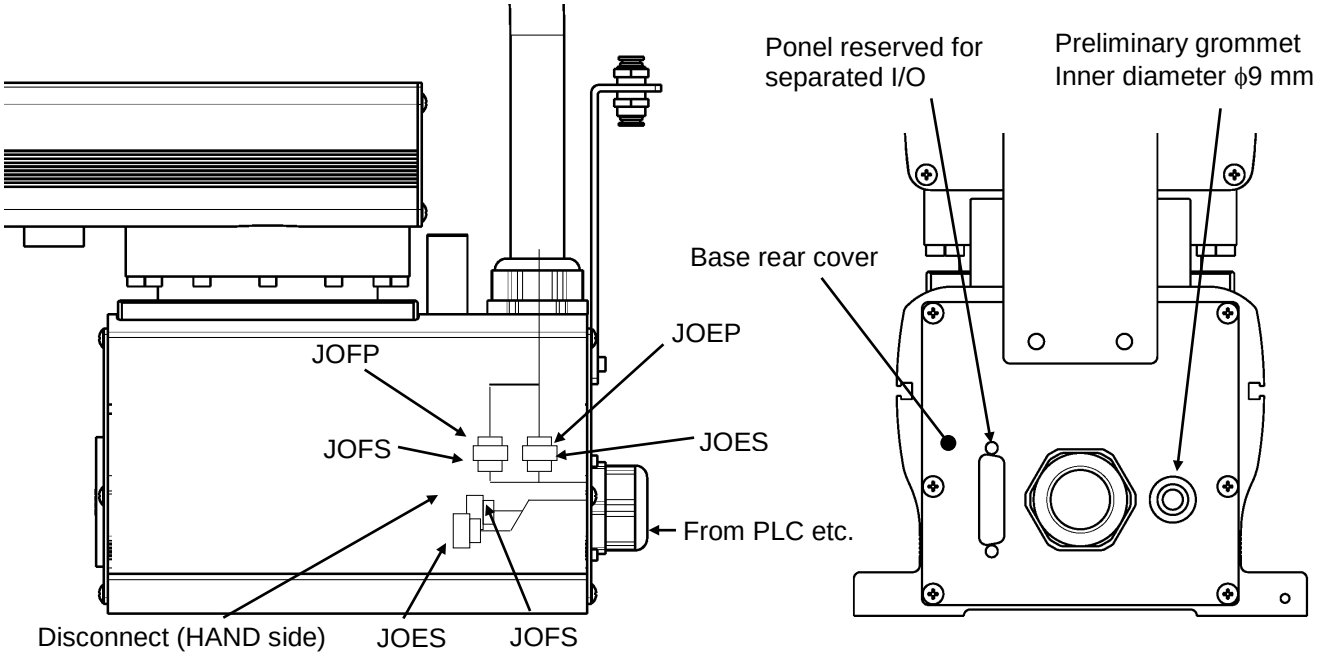


Fig. 23.3 Wiring to PLC, etc. (THL800, THL900, THL1000 and TSL3000)

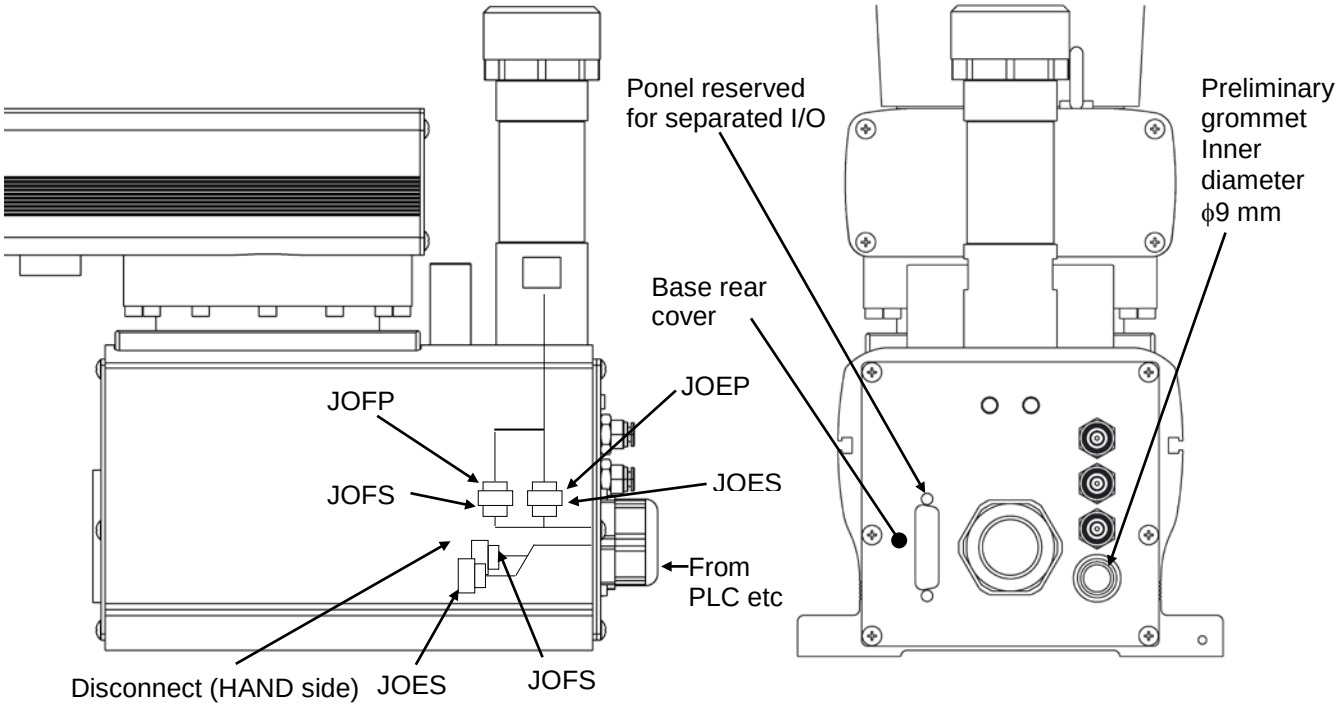
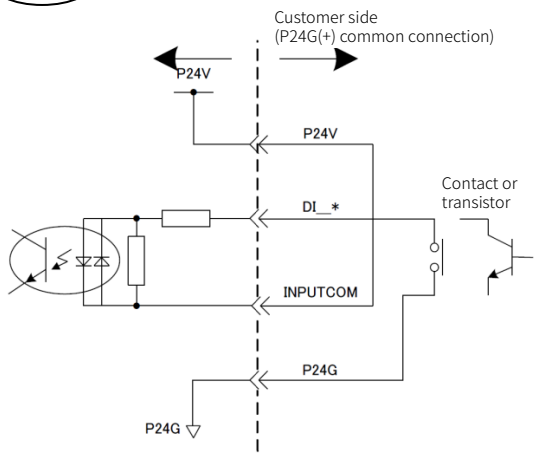


Fig. 23.4 Wiring to PLC, etc. (THL1200 and TSL3000)

Table 23.1 Input/output signal connector CN0 (TSL3000, Type-N)

Pin (D-SUB)	Signal name		Signal No.	Input/output circuit and example of connections
1	D-IN1	Input signal 1	201	Input
2	D-IN2	Input signal 2	202	
3	D-IN3	Input signal 3	203	
4	D-IN4	Input signal 4	204	
5	D-IN5	Input signal 5	205	
6	D-IN6	Input signal 6	206	
7	D-IN7	Input signal 7	207	
8	D-IN8	Input signal 8	208	
9	DC 24VGND (P24G)			
10	Shield (FG)			
11	D-OUT1	Output signal 1	201	Output
12	D-OUT2	Output signal 2	202	
13	D-OUT3	Output signal 3	203	
14	D-OUT4	Output signal 4	204	
15	D-OUT5	Output signal 5	205	
16	D-OUT6	Output signal 6	206	
17	D-OUT7	Output signal 7	207	
18	D-OUT8	Output signal 8	208	
19	DC 24V (P24V)			



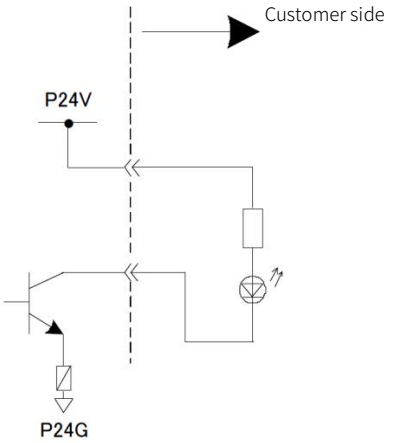
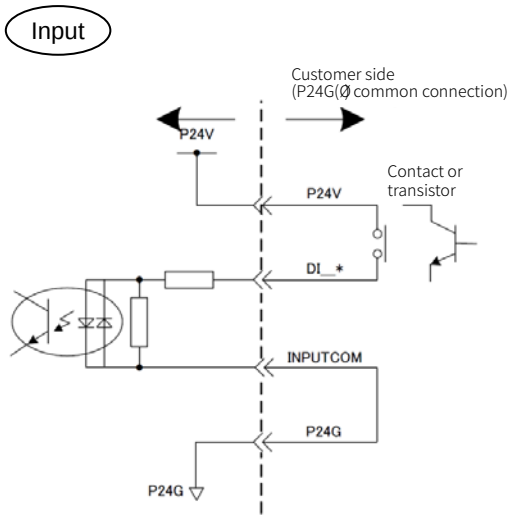
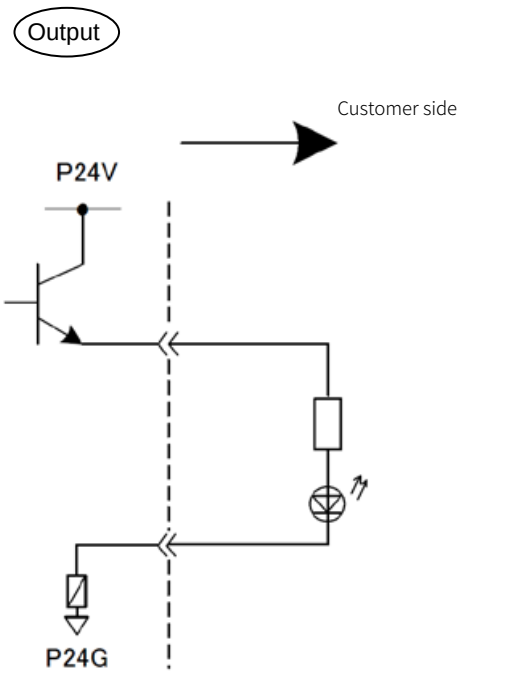


Table 23.2 Input/output signal connector CN0 (TSL3000, Type-P)

Pin (D-SUB)	Signal name		Signal No.	Input/output circuit and example of connections
1	D-IN1	Input signal 1	201	
2	D-IN2	Input signal 2	202	
3	D-IN3	Input signal 3	203	
4	D-IN4	Input signal 4	204	
5	D-IN5	Input signal 5	205	
6	D-IN6	Input signal 6	206	
7	D-IN7	Input signal 7	207	
8	D-IN8	Input signal 8	208	
9	DC 24VGND (P24G)			
10	Shield (FG)			
11	D-OUT1	Output signal 1	201	
12	D-OUT2	Output signal 2	202	
13	D-OUT3	Output signal 3	203	
14	D-OUT4	Output signal 4	204	
15	D-OUT5	Output signal 5	205	
16	D-OUT6	Output signal 6	206	
17	D-OUT7	Output signal 7	207	
18	D-OUT8	Output signal 8	208	
19	DC 24VGND (P24G)			





As input signals, no-voltage contacts or transistor open collector inputs are used.

No-voltage contact specifications:

Contact rating: 24 VDC, 10 mA or over (circuit current: approx. 7 mA)

Minimum contact current: 24 VDC, 1 mA

Contact impedance: 100 Ω or less

Transistor specifications:

Withhold voltage between collector and emitter: 30 V or over

Current between collector and emitter: 10 mA or over (circuit current: approx. 7 mA)

Leak current between collector and emitter: 100 μA or less

By using P24V power of the controller, a relay, solenoid valve, etc., can be driven. When the external power is used, GND of the external power should be common to GND (P24G) of the robot controller.

Output specifications:

Rated voltage : 24 VDC (max. 30 VDC)

Rated current : 100 mA

- If the P24V power supplied from the robot controller is used, the total current should be 2 A or less.
- When the external power is used, the total current should also be 2 A or less.
- When a relay or solenoid valve, etc., is connected, it is necessary to use a surge killer or diode to absorb the surge voltage.

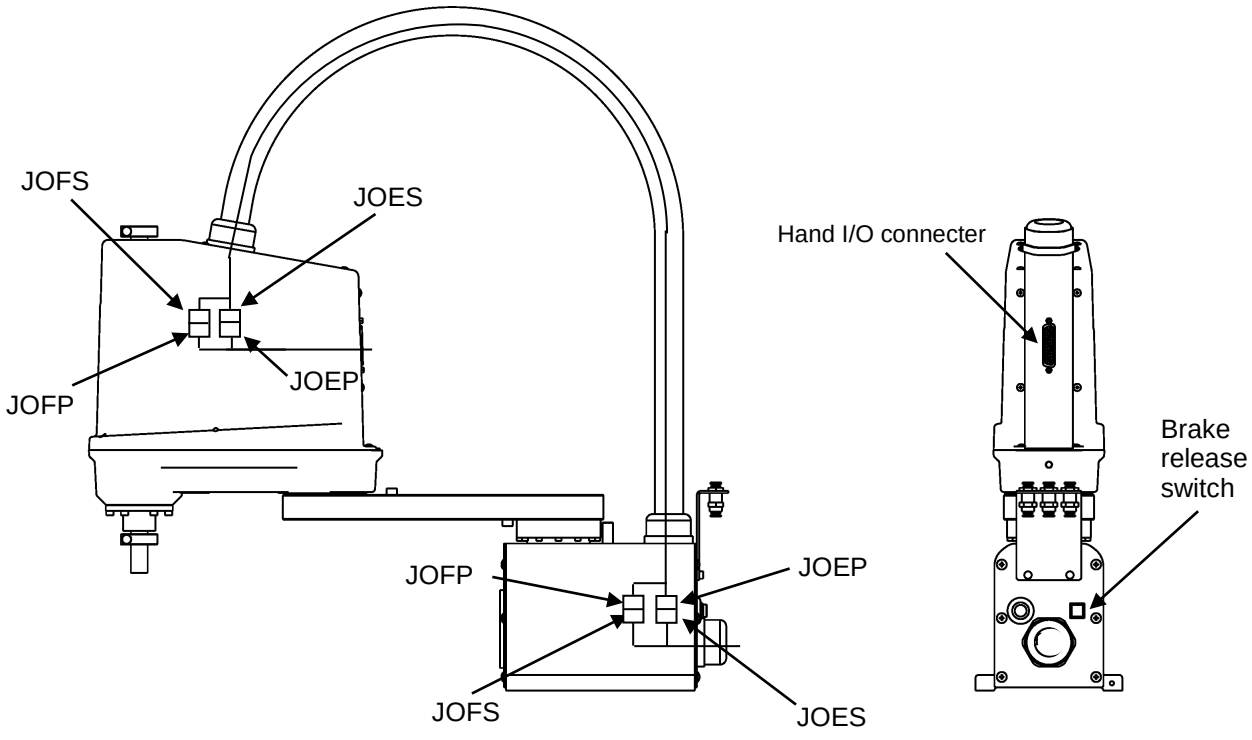


Fig. 23.5 Tool wiring (THL300 and THL400)

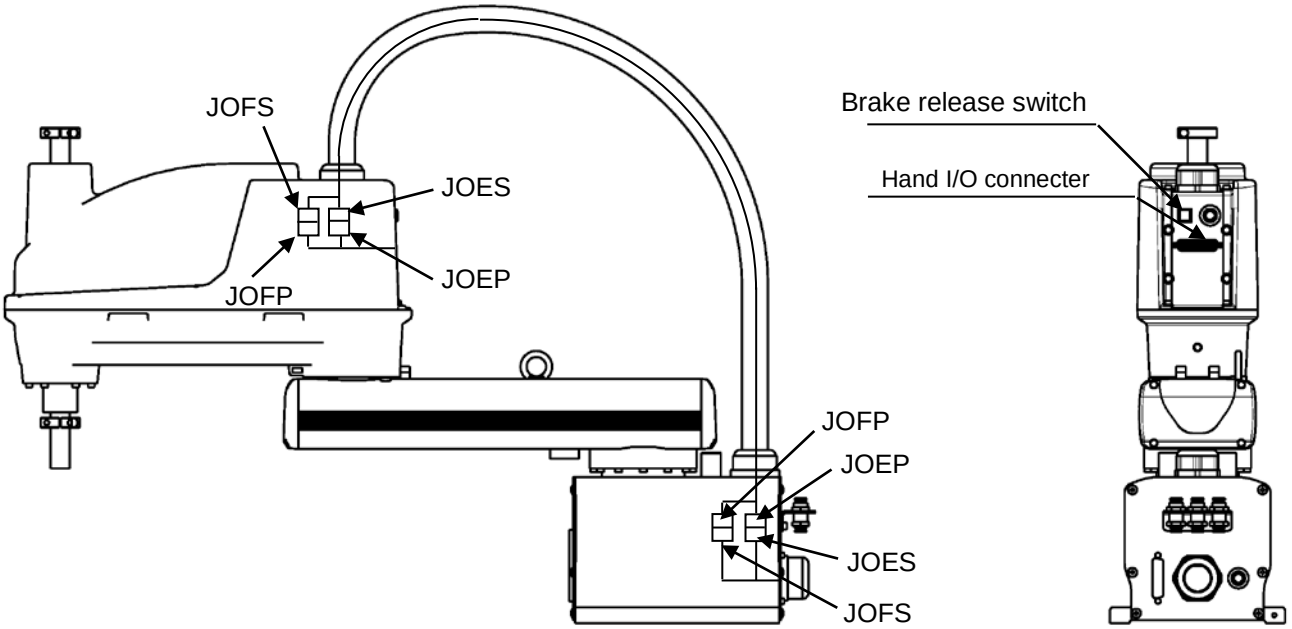


Fig. 23.6 Tool wiring (THL500, THL600 and THL700)

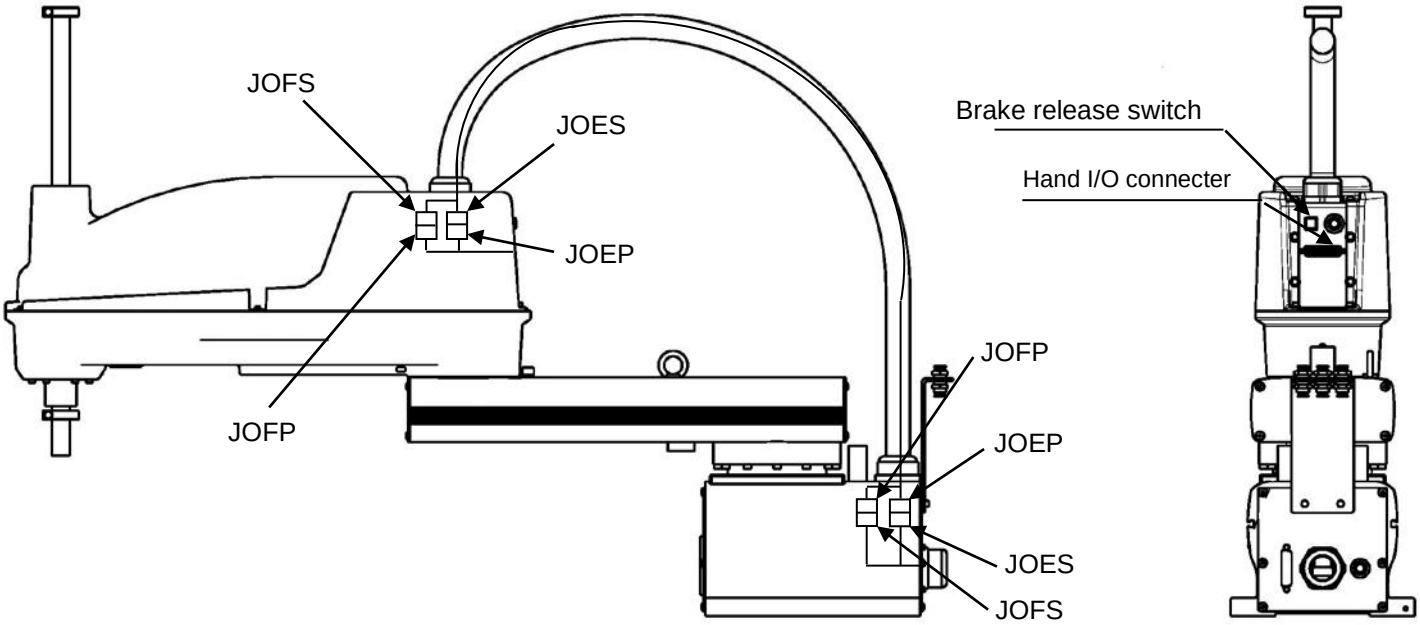


Fig. 23.7 Tool wiring (THL800, THL900 and THL1000)

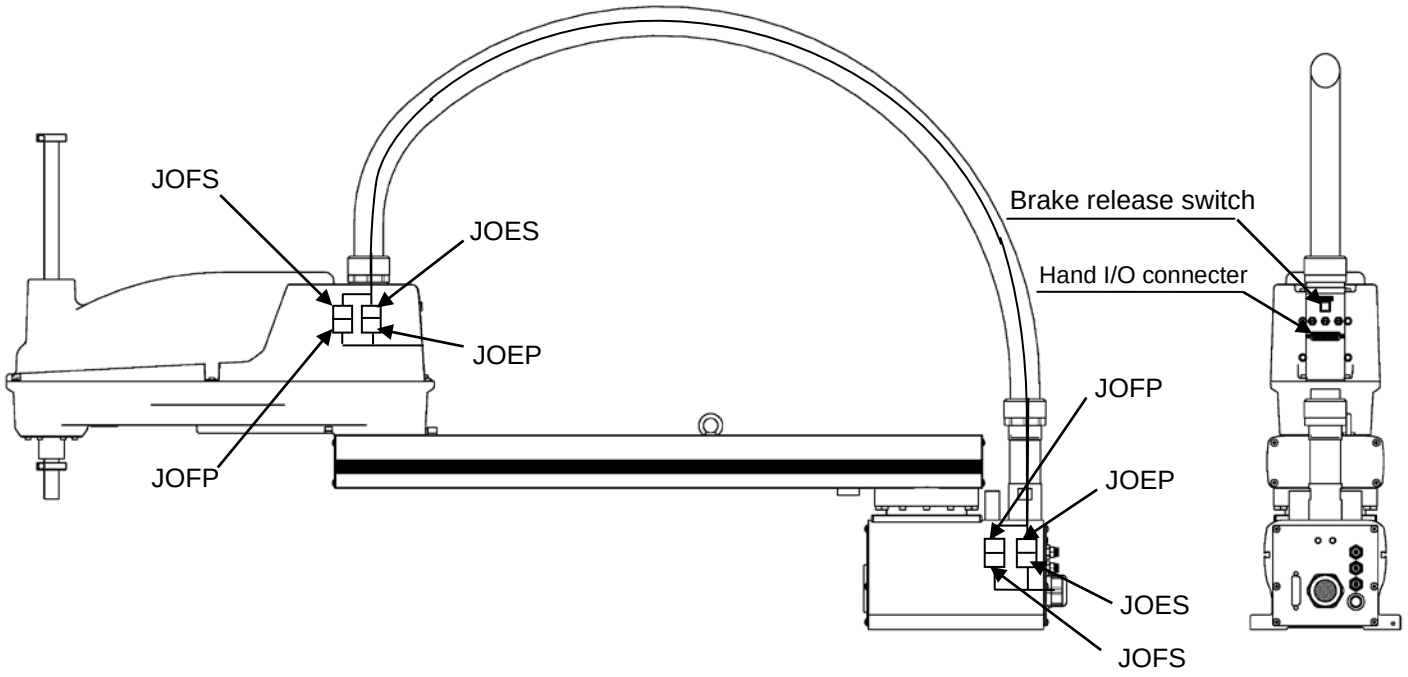


Fig. 23.8 Tool wiring (THL1200)

! CAUTION

- Be sure to connect all connectors properly. Otherwise, the robot may malfunction.

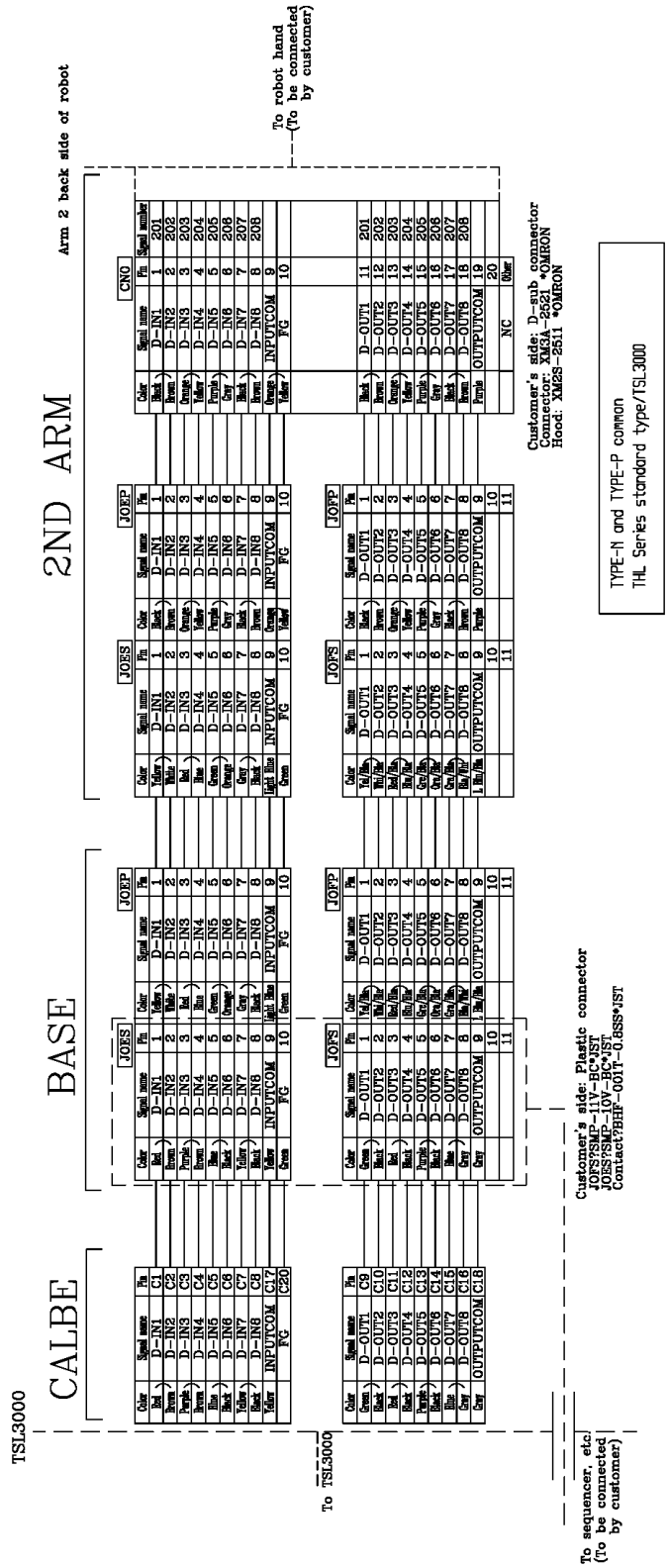


Fig. 23.9 Tool wiring (Common to Type-N, Type-P and TSL3000)

25.3 Tool Cable wiring

The customer is requested to install the cable wiring of the tool (the electric chuck, etc) which is prepared by the customer.

Fig. 23.10 to 23.13 shows an installation example of tool cable wiring using the connector (D-SUB) for the reference. Parts used for examples are not accessories. Please design and prepare such parts.

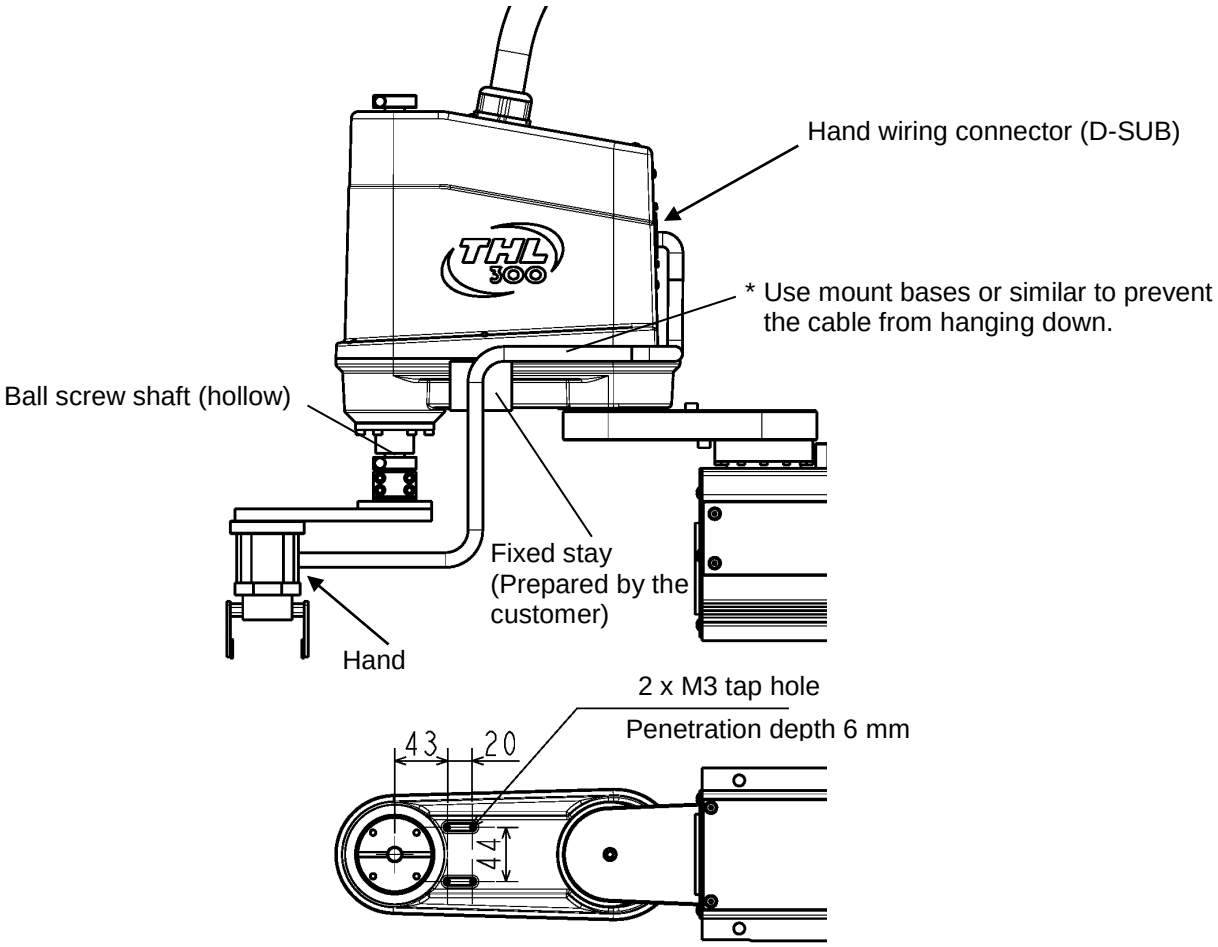


Fig. 23.10 Tool cable wiring installation example using the connector (D-SUB) (THL300 and THL400)

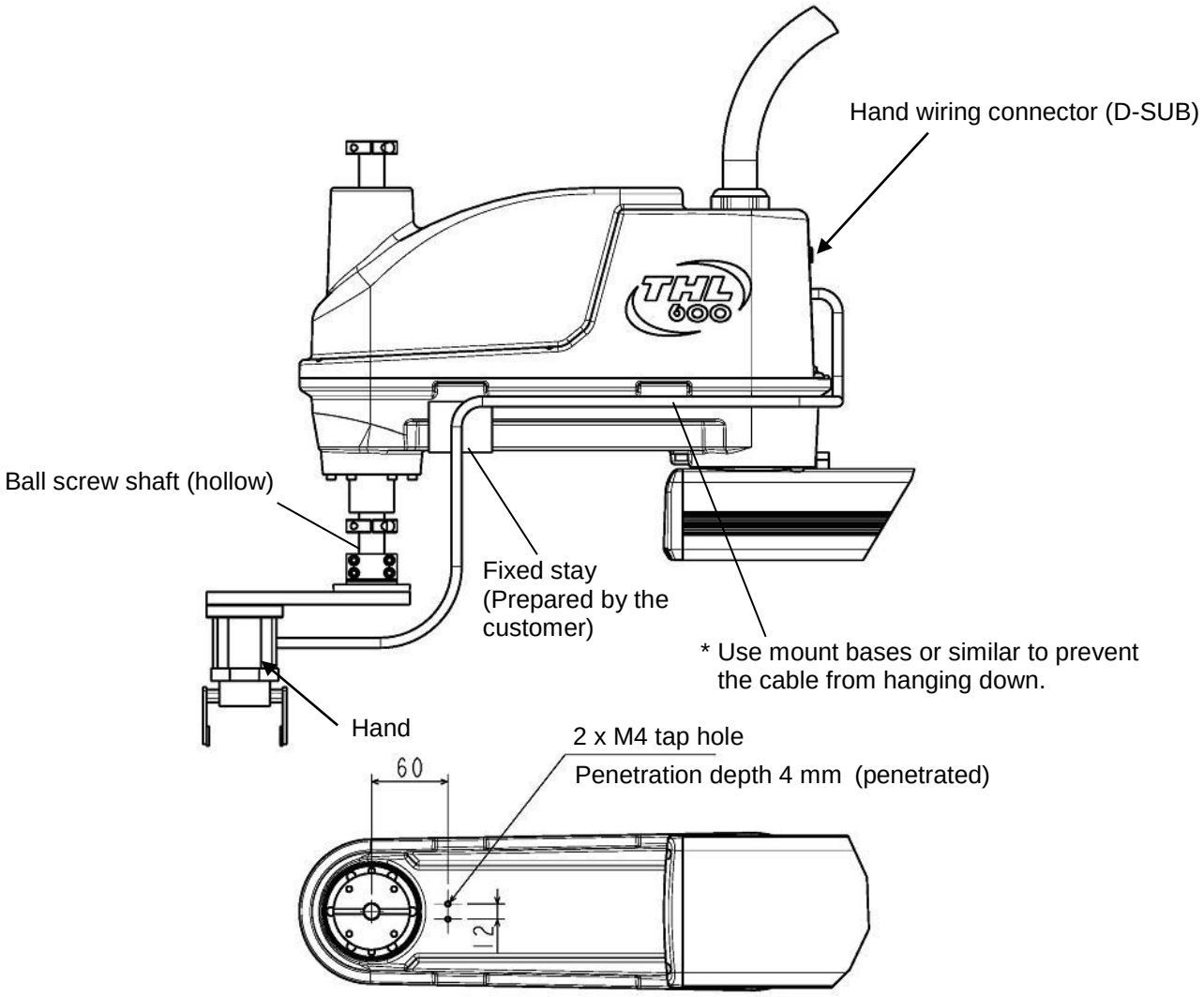


Fig. 23.11 Tool cable wiring installation example using the connector (D-SUB)
(THL500, THL600 and THL700)

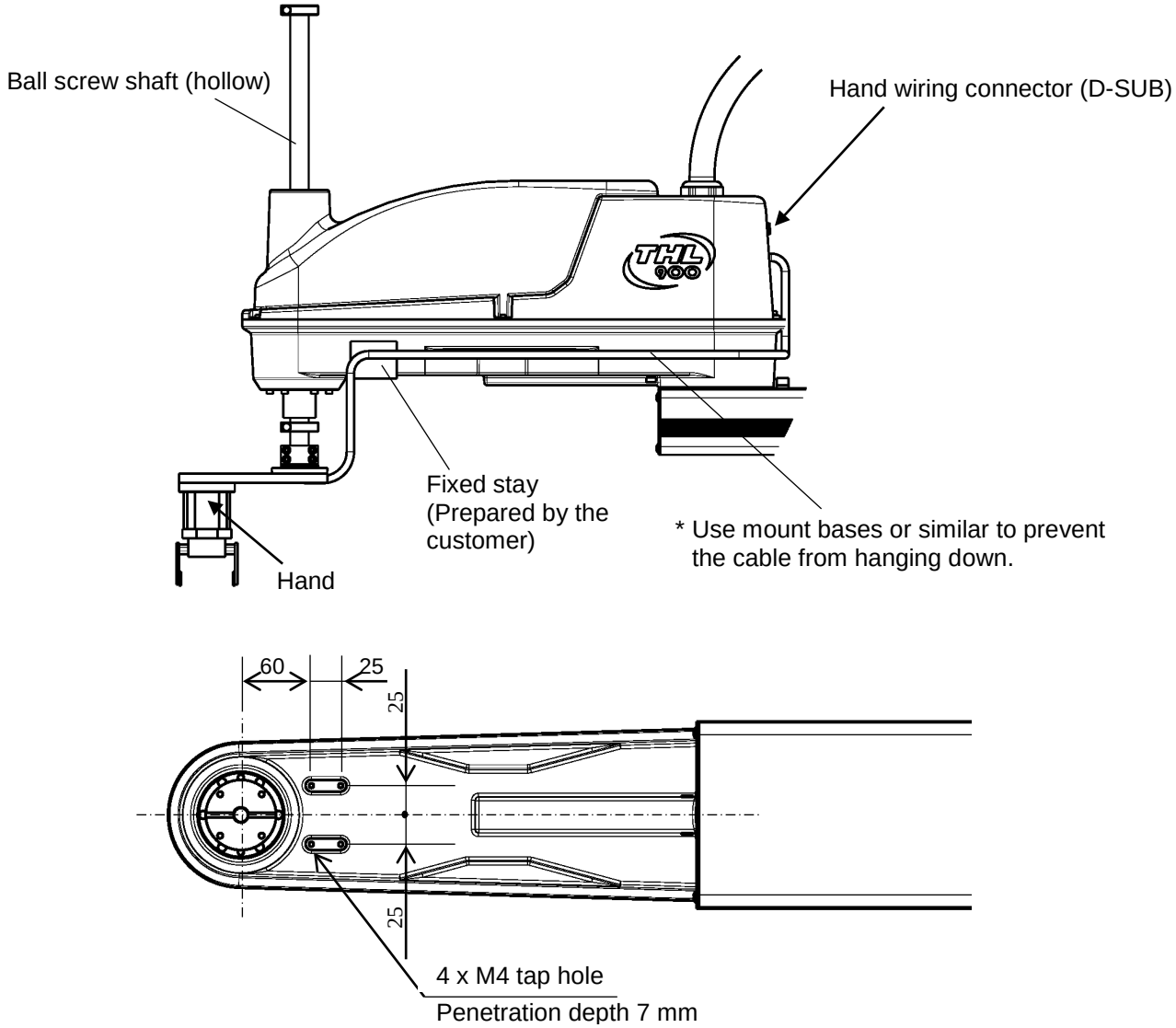


Fig. 23.12 Tool cable wiring installation example using the connector (D-SUB)
(THL800, THL900 and THL1000)

Secure the fixing stay and others, using the tap hole on the bottom of Arm 2.

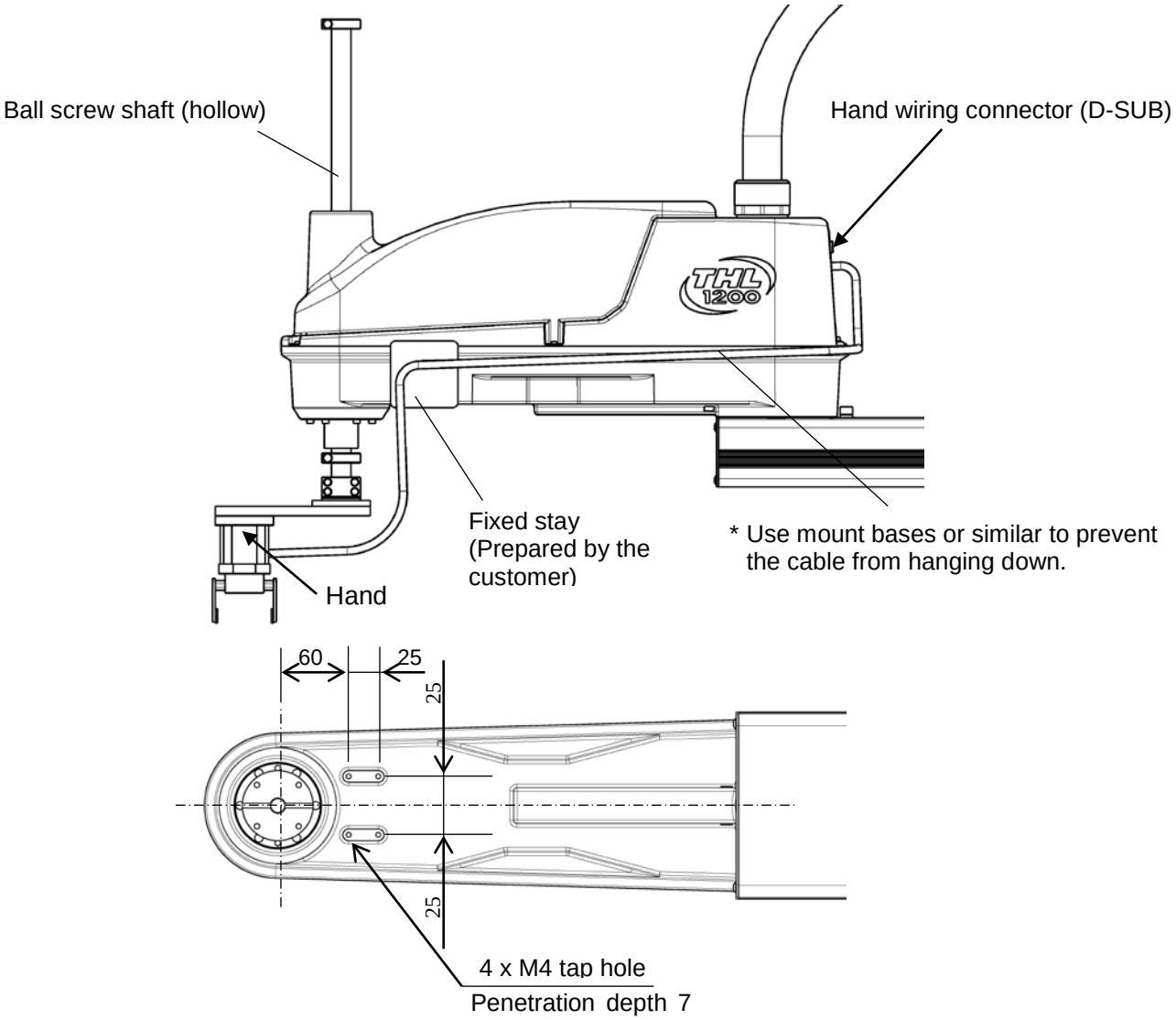


Fig. 23.13 Tool cable wiring installation example using the connector (D-SUB) (THL1200)

Secure the fixing stay and others, using the tap hole on the bottom of Arm 2.

! CAUTION

- Be sure to use a highly flexible robot cable, which should be secured below the arm with a cable clamp, etc. Unless a robot cable is used, the wire may be broken.
- When performing tool wiring and piping, take all necessary measures against breakage due to rub, etc.
- At robot operation, take careful precautions not to exert a load on respective connectors.
- In operation, confirm that the fixed stay does not interfere with the arm 1, etc.

When wiring is insufficient for the connector (D-SUB), if the cable wiring and robot's wire harness cere bundled, robot's wire harness may be broken and disconnected, due to unexpected fluctuations or un balanced loads on the robot's wire harness during robot operation depending on the weight of the cable wiring the operating speed and range of the robot.If the cable wiring is routed to the tip of the robot, do not bundle it to the robot's wire harness. Prepare a pillar, for example, in order to route the cable wiring.

Fig. 23.14 to 23.17 shows an installation example of tool cable wiring for your reference. Parts used for examples are not accessories. Please design and prepare such parts.

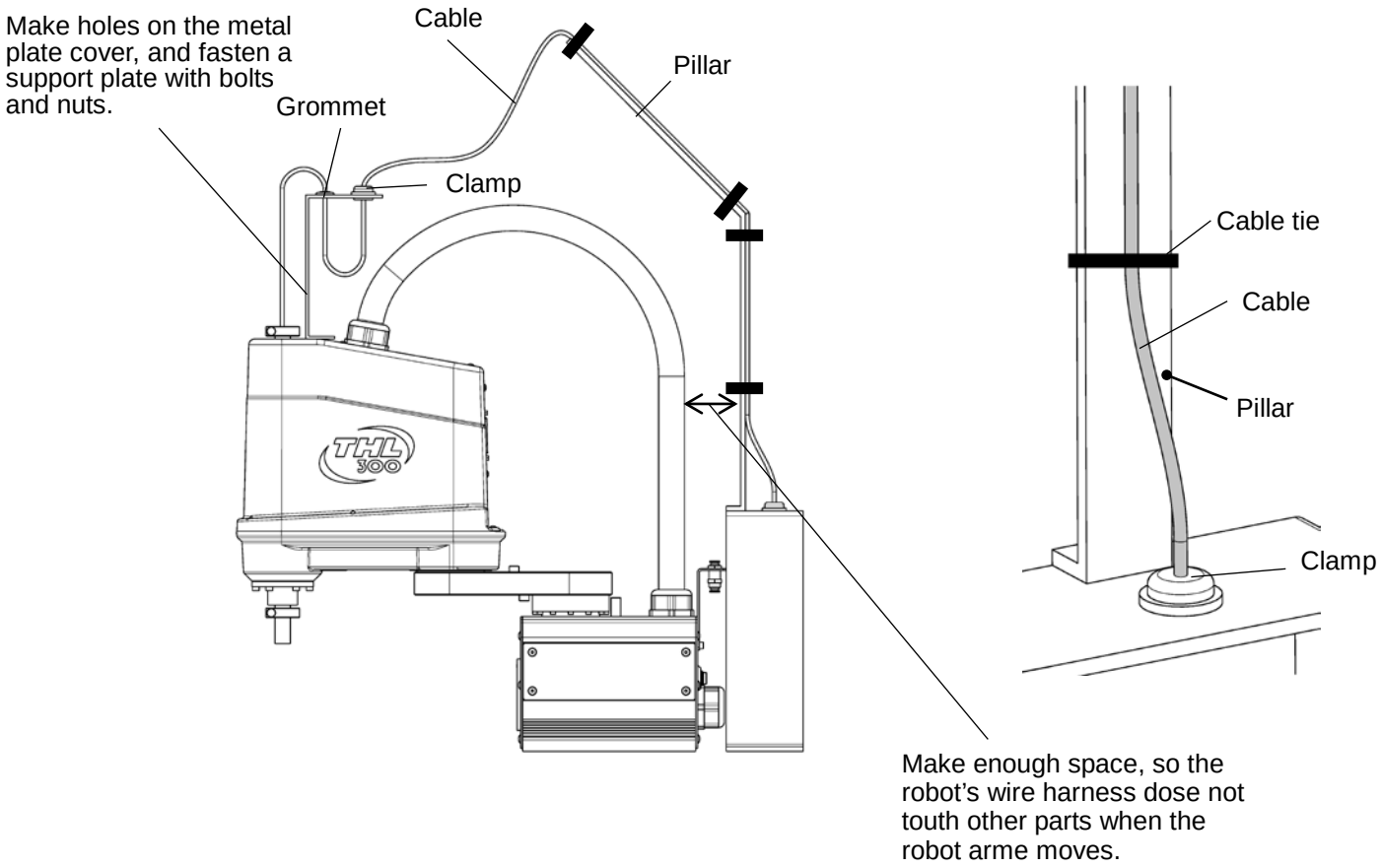


Fig. 23.14 Tool cable wiring installation example (THL300/THL400)

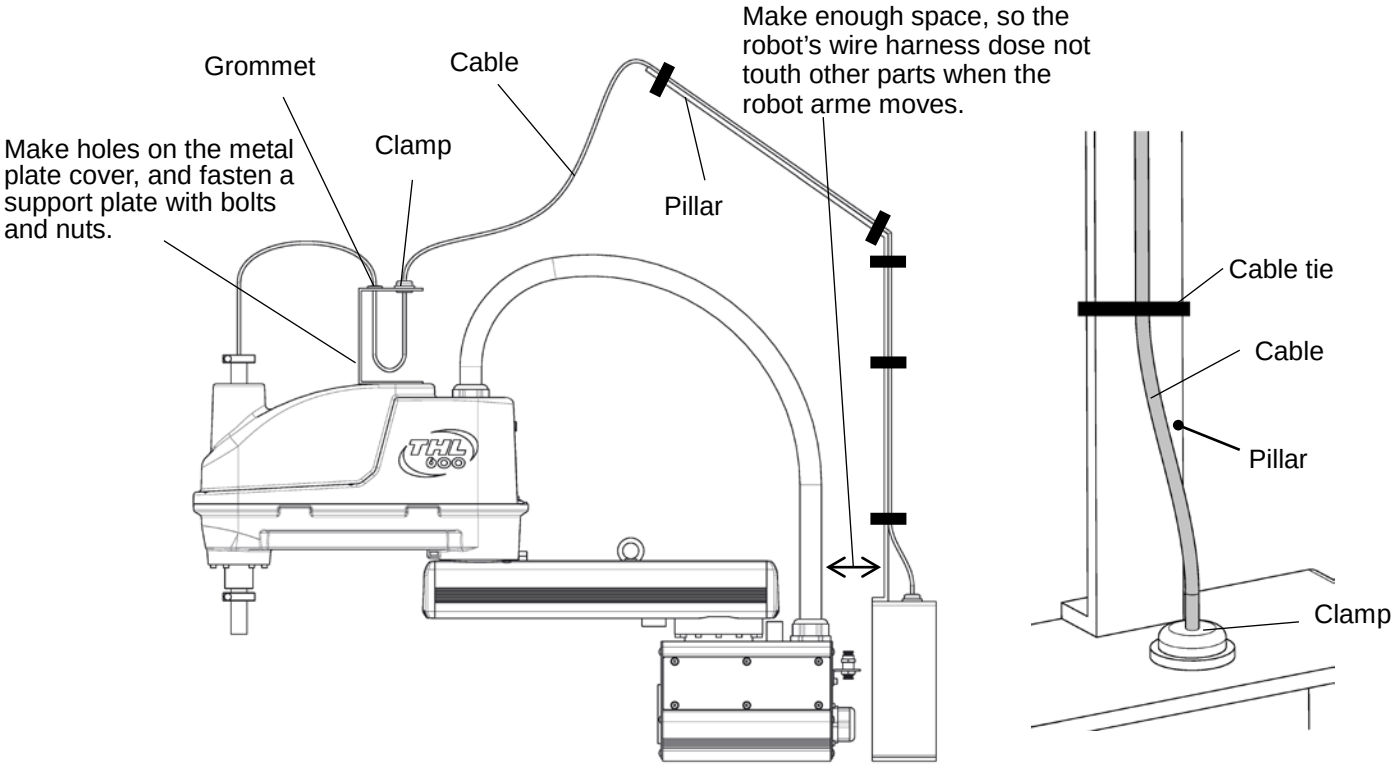


Fig. 23.15 Tool cable wiring installation example (THL500/THL600/THL700)

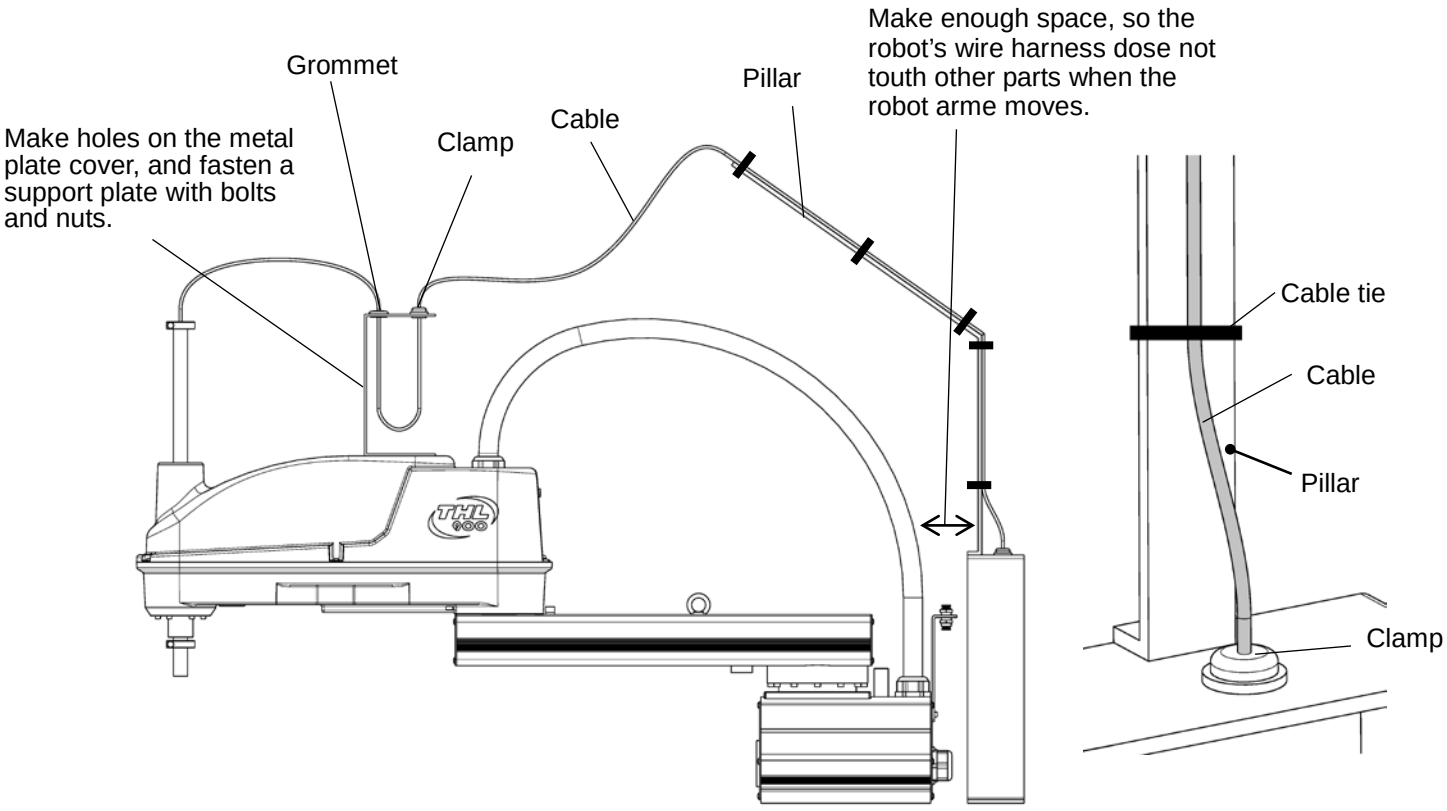


Fig. 23.16 Tool cable wiring installation example (THL800/THL900/THL1000)

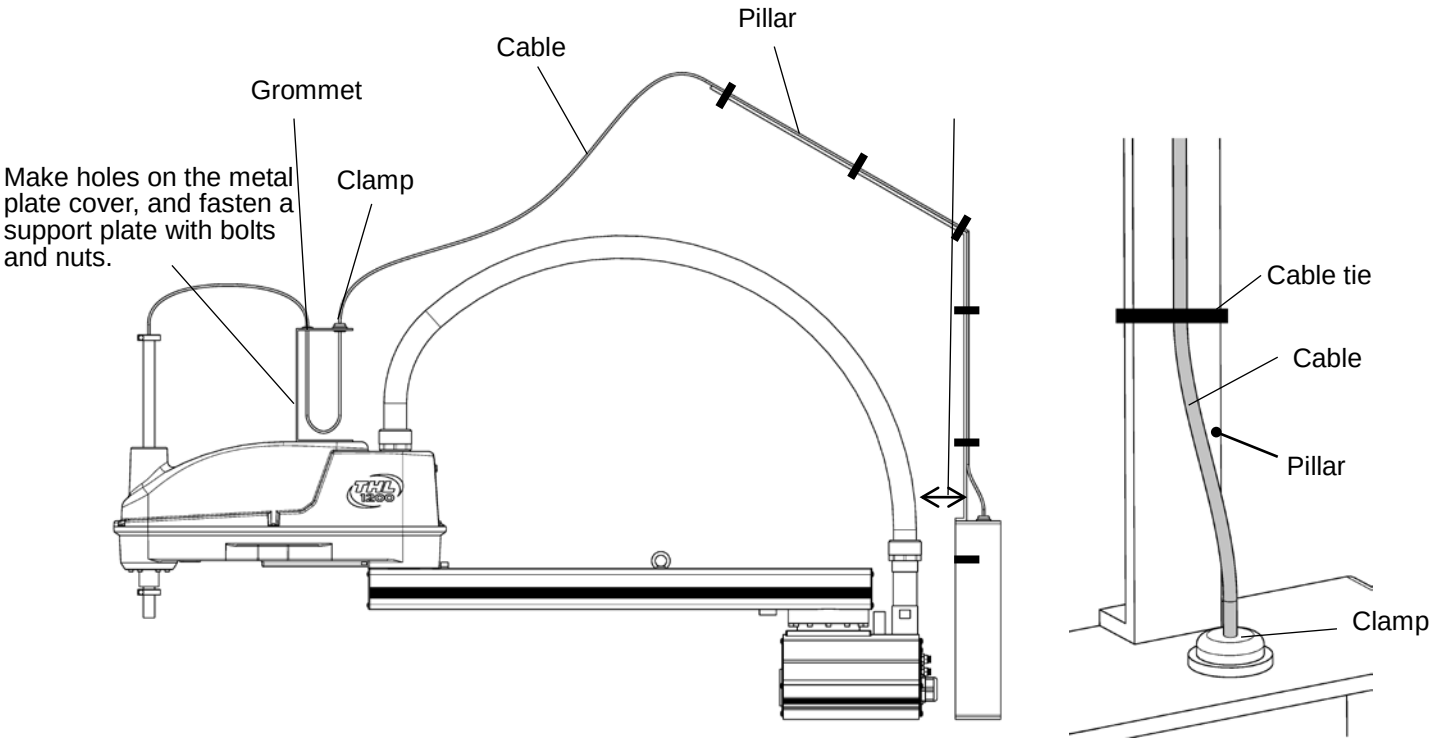


Fig. 23.17 Tool cable wiring installation example (THL1200)

CAUTION

- Fig. 23.10 to 23.17 is a reference example and it dose not guarantee the robot's wire harness. The warranty does not cover breakdowns of the robot's wire harness due to wiring by the customer.

23.2 The wiring arrangement for the hand I/O cable (Option)

The wiring arrangement for the hand I/O cable (Option) is shown in Fig. 23.18～23.19. And use parts list of the hand I/O cable (Option) is shown in Table 23.3.

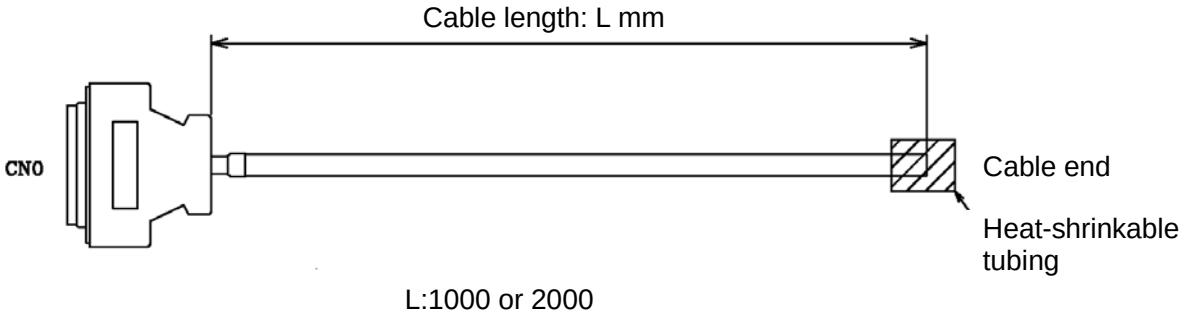


Fig. 23.18 Outline drawing for the hand I/O cable (Option)

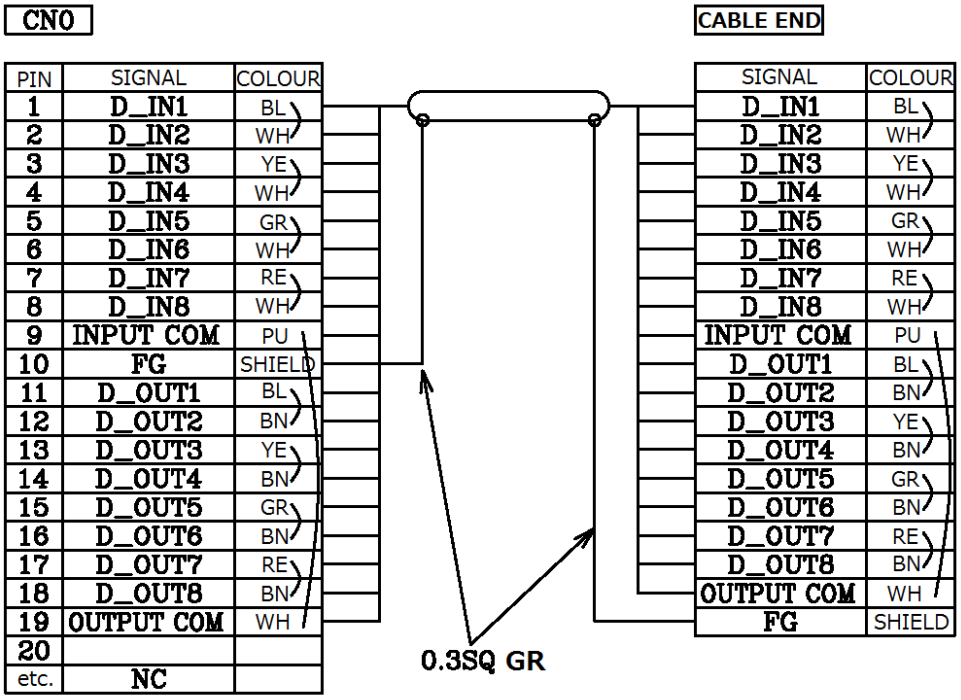


Fig. 23.19 The wiring arrangement for the hand I/O cable (Option)

Table 23.3 Use parts list of the hand I/O cable (Option)

No.	Part name	Type	Maker	Q'ty
1	Cable	RMFEV-SB 0.2SQ-10P	Dyden	1
2	Plug with solder-cup terminal	XM3A-2521	Omron	1
3	D-sub connector	XM2S-2511		1

24. Tool Interface (TSL3000E)

24.1 Tool Wiring

Eight (8) input signals are provided for sensors, etc. and eight (8) control signals for solenoid valves, etc. A supply power signal of 24 VDC is also provided. They are connected to the controller. The wiring arrangement for these cables is shown in Fig. 24.9. The wires are connected to the connectors on the upper side of Arm 2. The user should provide the following connectors to connect the cables.

D-SUB connector (standard) Type: <Hood> XM2S–2511 (Maker: OMRON)
<Basic body> XM3A–2521 (Maker: OMRON)

Adaptive cable Conductive cross section area: 0.2 mm^2 to 0.5 mm^2

Each connector and cable are connected by soldering.

! DANGER

- Be sure to use the designated wire. Otherwise, fires or faults may be caused.
- When connecting the connector and wires, make sure not to mistake the terminal arrangement.
- After making the connection, use a tester, etc., to confirm the connection.

When controlling the robot from the sequencer (i.e., programmable ladder controller: PLC), etc. installed separately, remove the base rear cover from the base section, remove connectors JOES and JOFS on the rear side, then connect the cables running from the PLC, etc. through the cable inlets provided on the base rear cover. Also, the D-SUB 25 pins can be used separately for THL500 to THL1200. (See Figs. 24.1, 24.2 and 24.3.) For ahead of the JOES and JOFS connectors, the user should prepare the following plug connectors and connect the cables. The current is 1 A or less per cable.

Type of connector: JOES SMP–10V–BC (Maker: J.S.T. Mfg.)
JOFS SMP–11V–BC (Maker: J.S.T. Mfg.)

Type of contact: BHF–001T–0.8SS (Maker: J.S.T. Mfg.)

Adaptive cable Conductive cross section area: 0.2 mm^2 to 0.3 mm^2

Opposite connector type (THL500 to THL1000)

Type of connector: JOEP SMR–10V–B (Maker: J.S.T. Mfg.)
JOFP SMR–11V–B (Maker: J.S.T. Mfg.)

Type of contact: BYM–001T–0.6 (Maker: J.S.T. Mfg.)

D-SUB 25 pin connector types

Connector type (male)	<Hood> XM2S-2511 (maker: OMRON)
	<Body> XM3A-2521 (maker: OMRON)
Connector type (female)	<Body> XM3D-2521 (maker: OMRON)

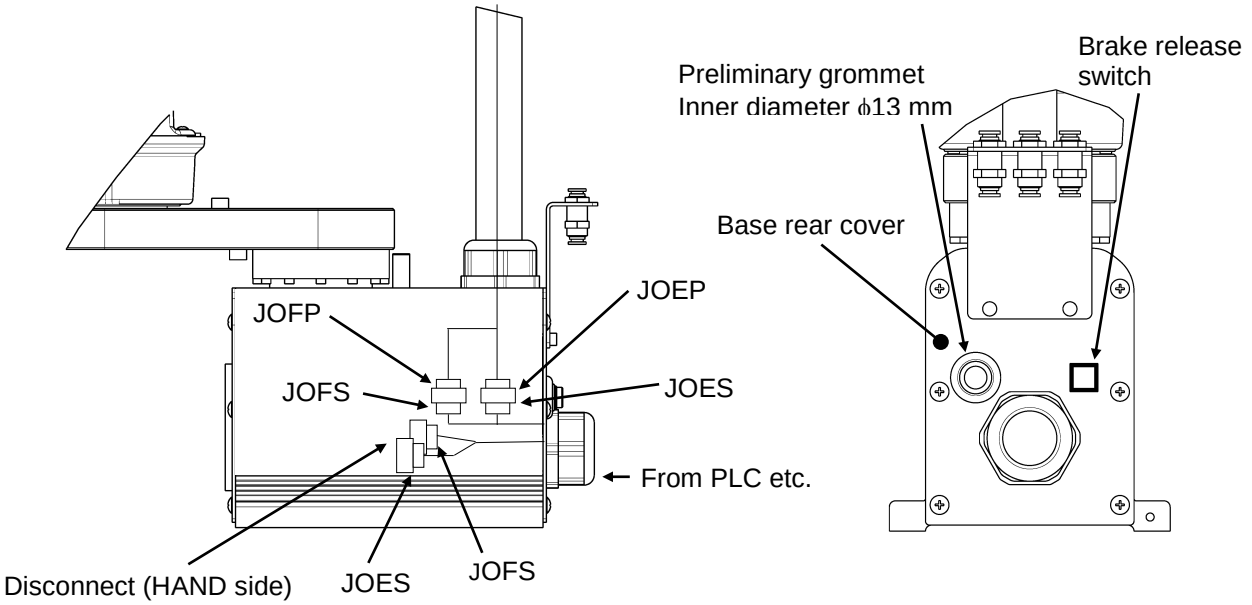


Fig. 24.1 Wiring to PLC, etc. (THL300, THL400 and TSL3000E)

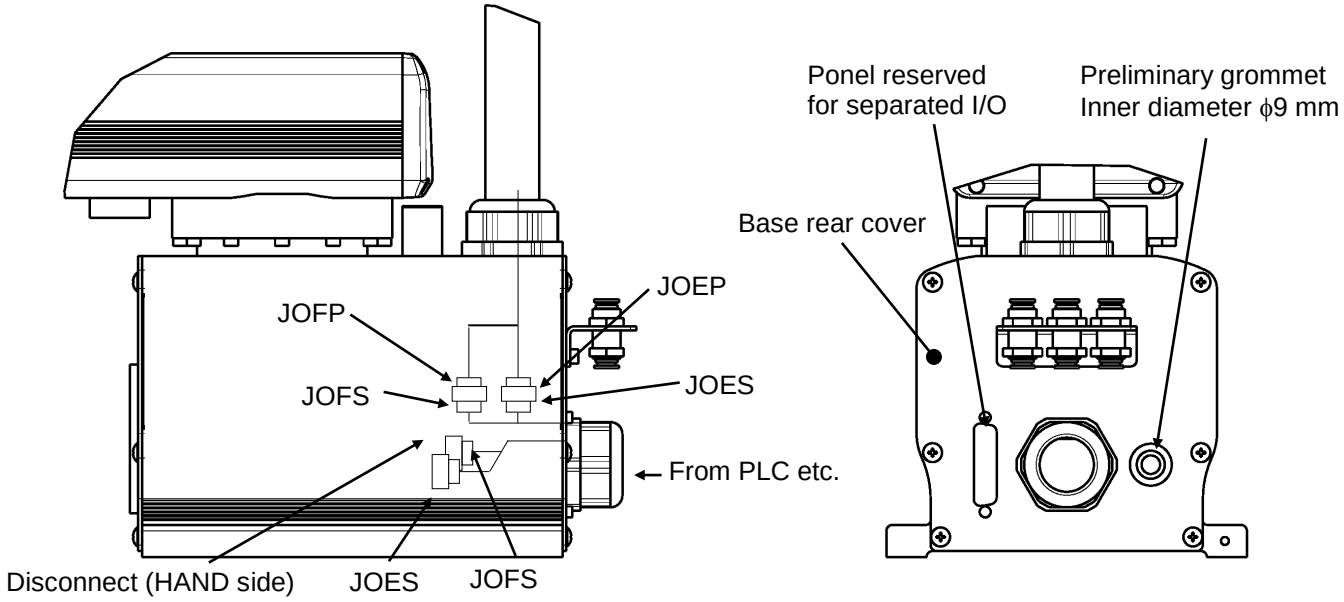


Fig. 24.2 Wiring to PLC, etc. (THL500, THL600, THL700 and TSL3000E)

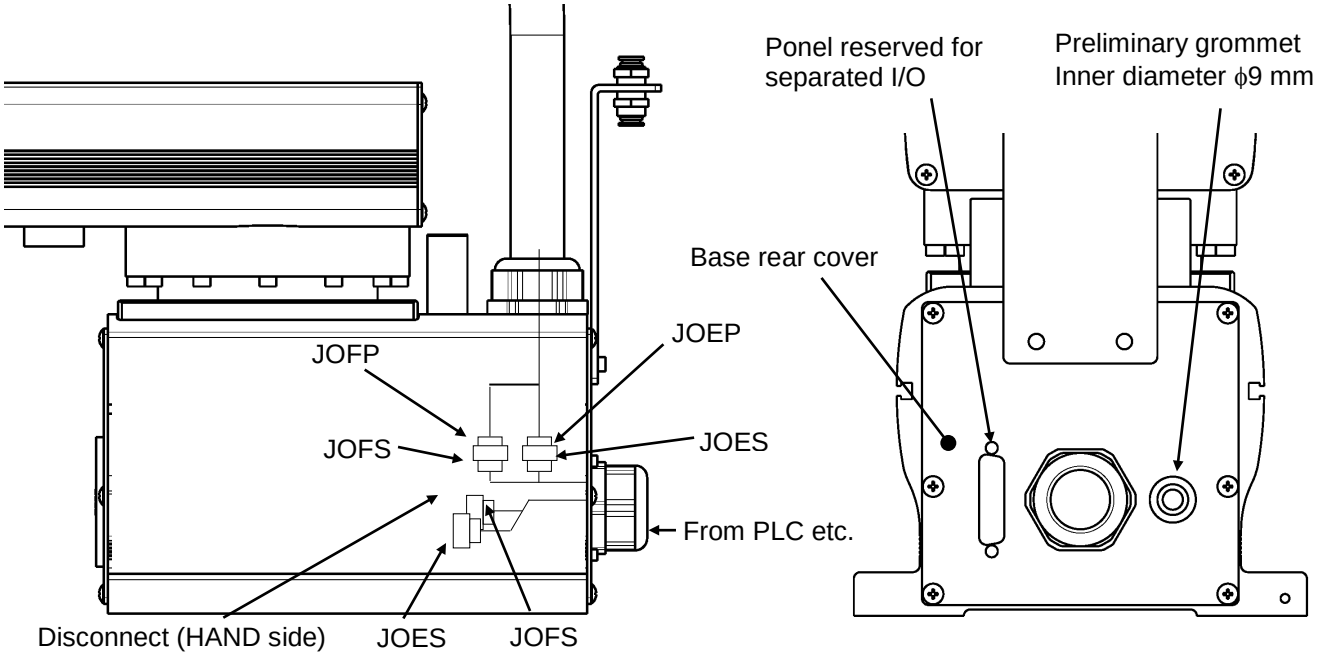


Fig. 24.3 Wiring to PLC, etc. (THL800, THL900, THL1000 and TSL3000E)

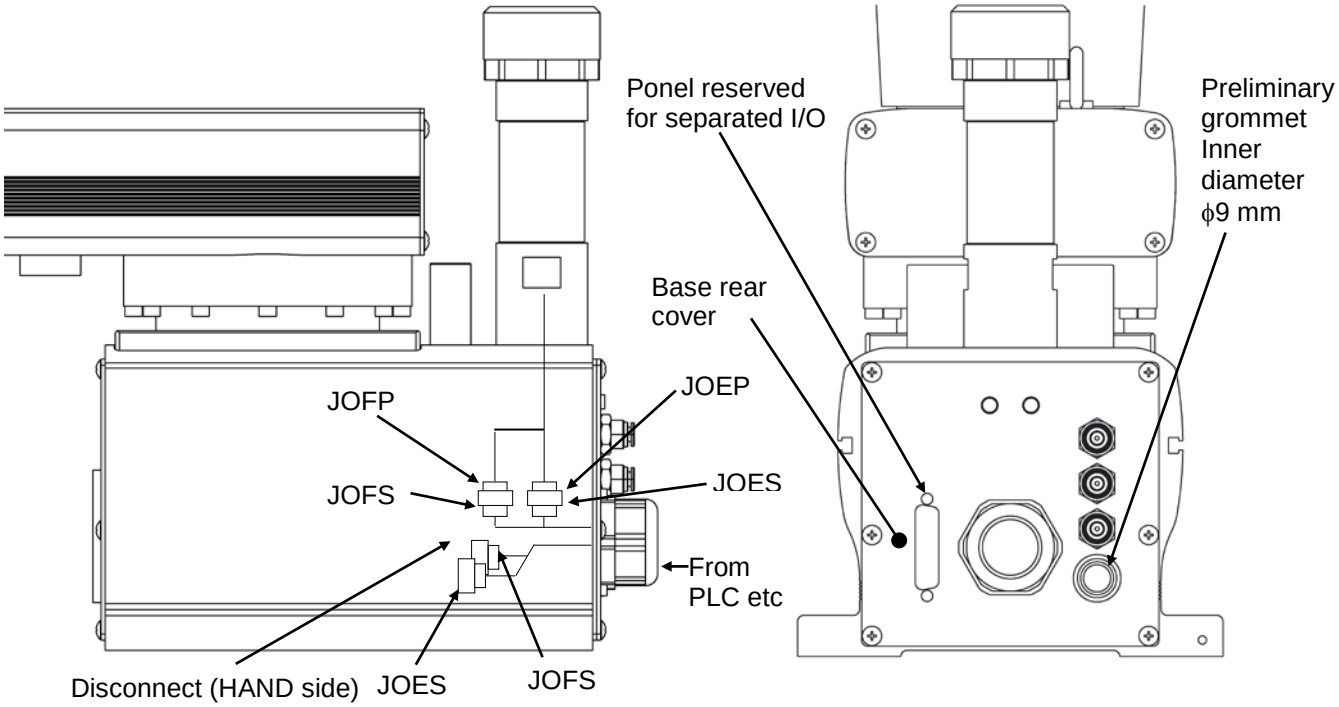
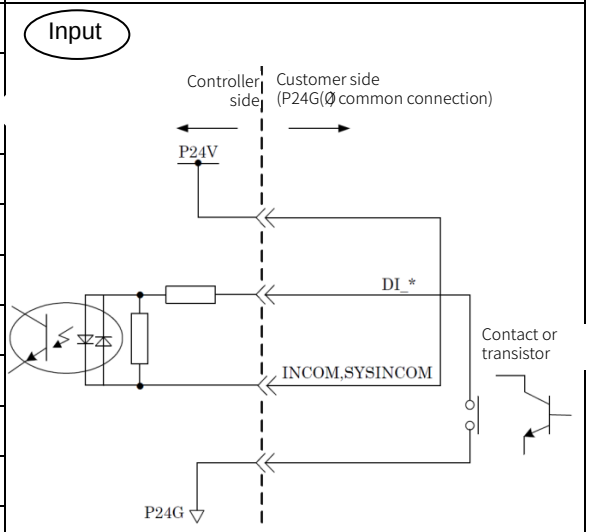


Fig. 24.4 Wiring to PLC, etc. (THL1200 and TSL3000E)

Table 24.1 Input/output signal connector CN0 (TSL3000E, Type-N)

Pin (D-SUB)	Signal name		Signal No.	Input/output circuit and example of connections
1	D-IN1	Input signal 1	201	
2	D-IN2	Input signal 2	202	
3	D-IN3	Input signal 3	203	
4	D-IN4	Input signal 4	204	
5	D-IN5	Input signal 5	205	
6	D-IN6	Input signal 6	206	
7	D-IN7	Input signal 7	207	
8	D-IN8	Input signal 8	208	
9	DC 24VGND (P24G)			
10	Shield (FG)			
11	D-OUT1	Output signal 1	201	
12	D-OUT2	Output signal 2	202	
13	D-OUT3	Output signal 3	203	
14	D-OUT4	Output signal 4	204	
15	D-OUT5	Output signal 5	205	
16	D-OUT6	Output signal 6	206	
17	D-OUT7	Output signal 7	207	
18	D-OUT8	Output signal 8	208	
19	DC 24V (P24V)			



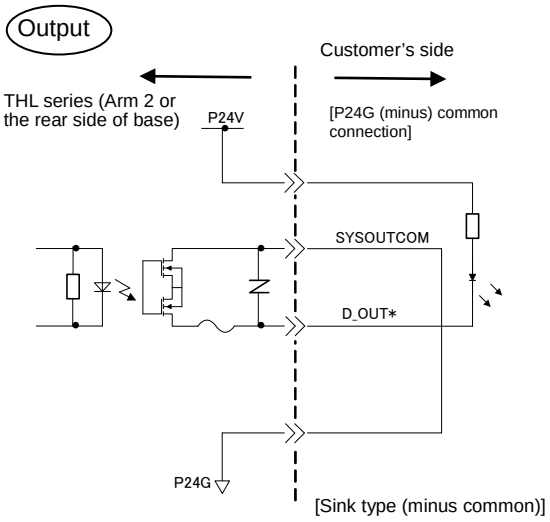
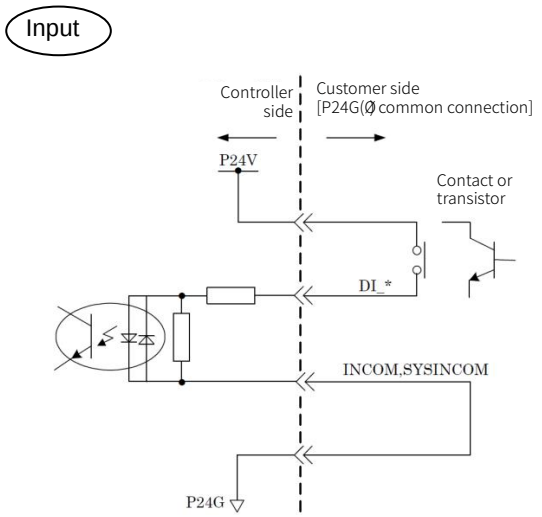
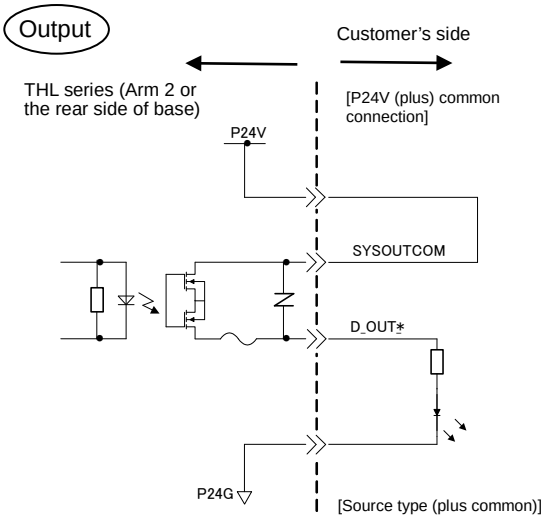


Table 24.2 Input/output signal connector CN0 (TSL3000E, Type-P)

Pin (D-SUB)	Signal name		Signal No.	Input/output circuit and example of connections
1	D-IN1	Input signal 1	201	
2	D-IN2	Input signal 2	202	
3	D-IN3	Input signal 3	203	
4	D-IN4	Input signal 4	204	
5	D-IN5	Input signal 5	205	
6	D-IN6	Input signal 6	206	
7	D-IN7	Input signal 7	207	
8	D-IN8	Input signal 8	208	
9	DC 24VGND (P24G)			
10	Shield (FG)			
11	D-OUT1	Output signal 1	201	
12	D-OUT2	Output signal 2	202	
13	D-OUT3	Output signal 3	203	
14	D-OUT4	Output signal 4	204	
15	D-OUT5	Output signal 5	205	
16	D-OUT6	Output signal 6	206	
17	D-OUT7	Output signal 7	207	
18	D-OUT8	Output signal 8	208	
19	DC 24VGND (P24G)			





As input signals, no-voltage contacts or transistor open collector inputs are used.

No-voltage contact specifications:

Contact rating: 24 VDC, 10 mA or over (circuit current: approx. 7 mA)

Minimum contact current: 24 VDC, 1 mA

Contact impedance: 100 Ω or less

Transistor specifications:

Withhold voltage between collector and emitter: 30 V or over

Current between collector and emitter: 10 mA or over (circuit current: approx. 7 mA)

Leak current between collector and emitter: 100 μA or less

By using P24V power of the controller, a relay, solenoid valve, etc., can be driven.

When the external power is used, GND of the external power should be common to GND (P24G) of the robot controller.

Output specifications:

Rated voltage : 24 VDC (max. 30 VDC)

Rated current : 100 mA

- If the P24V power supplied from the robot controller is used, the total current should be 2 A or less.
- When the external power is used, the total current should also be 2 A or less.
- When a relay or solenoid valve, etc., is connected, it is necessary to use a surge killer or diode to absorb the surge voltage.

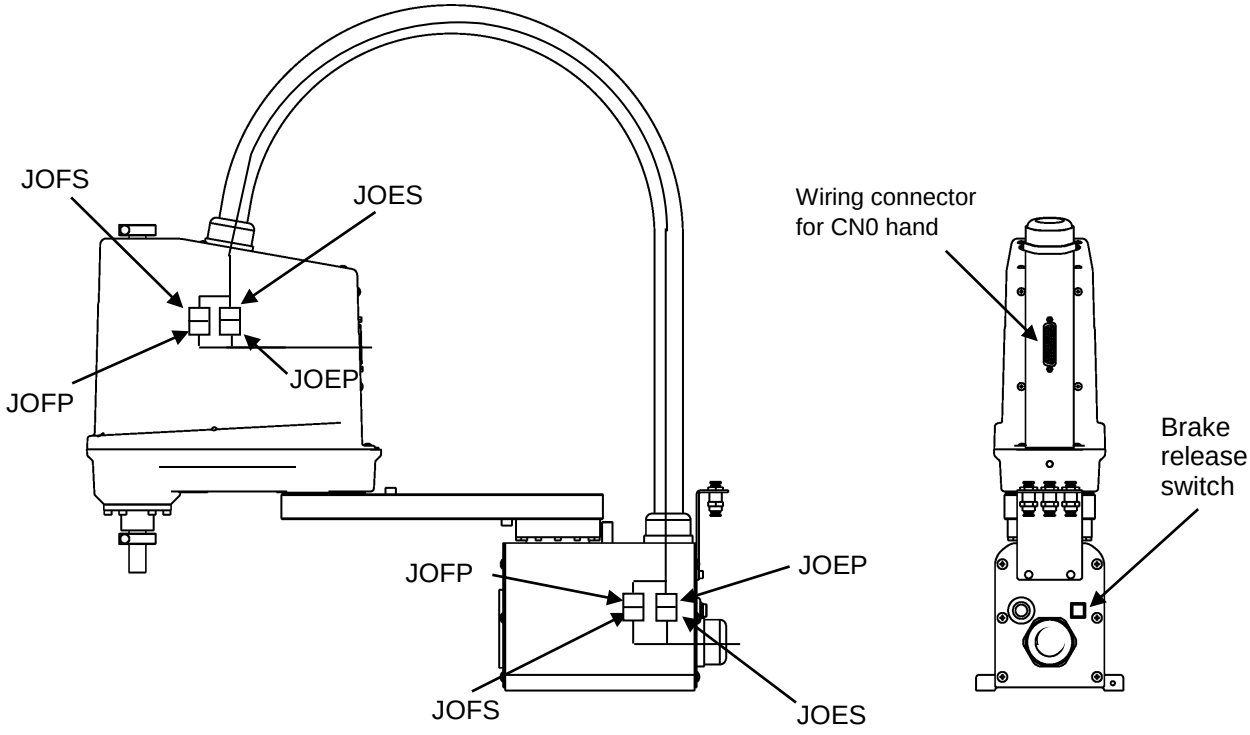


Fig. 24.5 Tool wiring (THL300 and THL400)

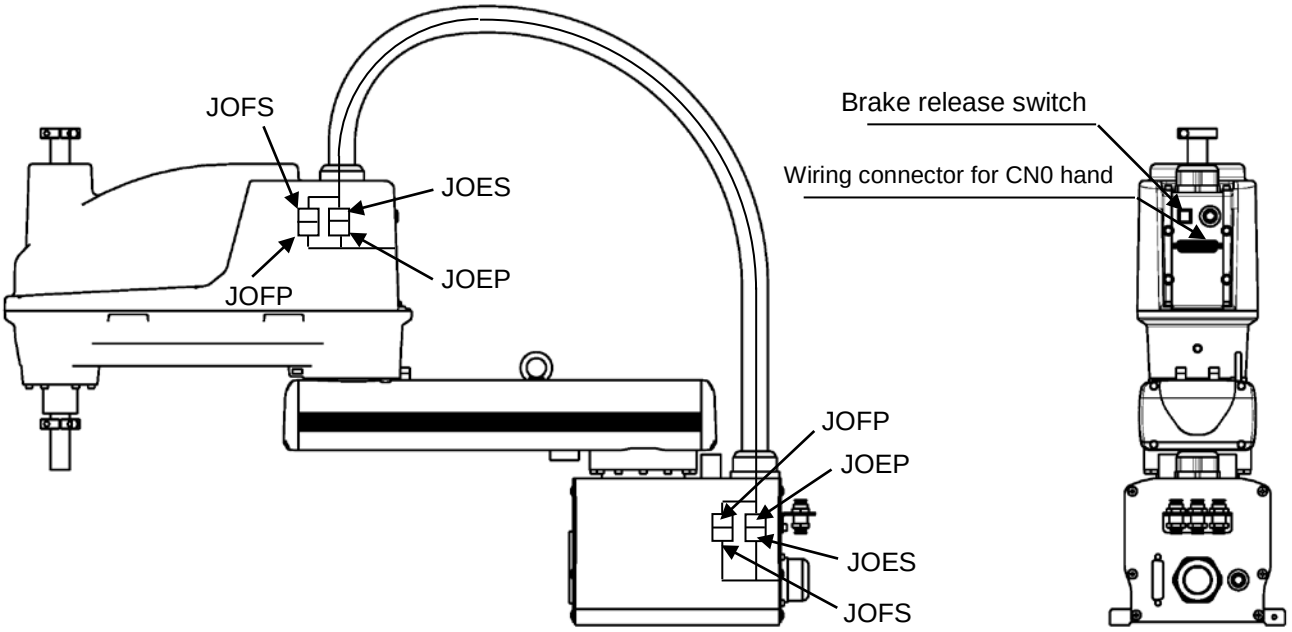


Fig. 24.6 Tool wiring (THL500, THL600 and THL700)

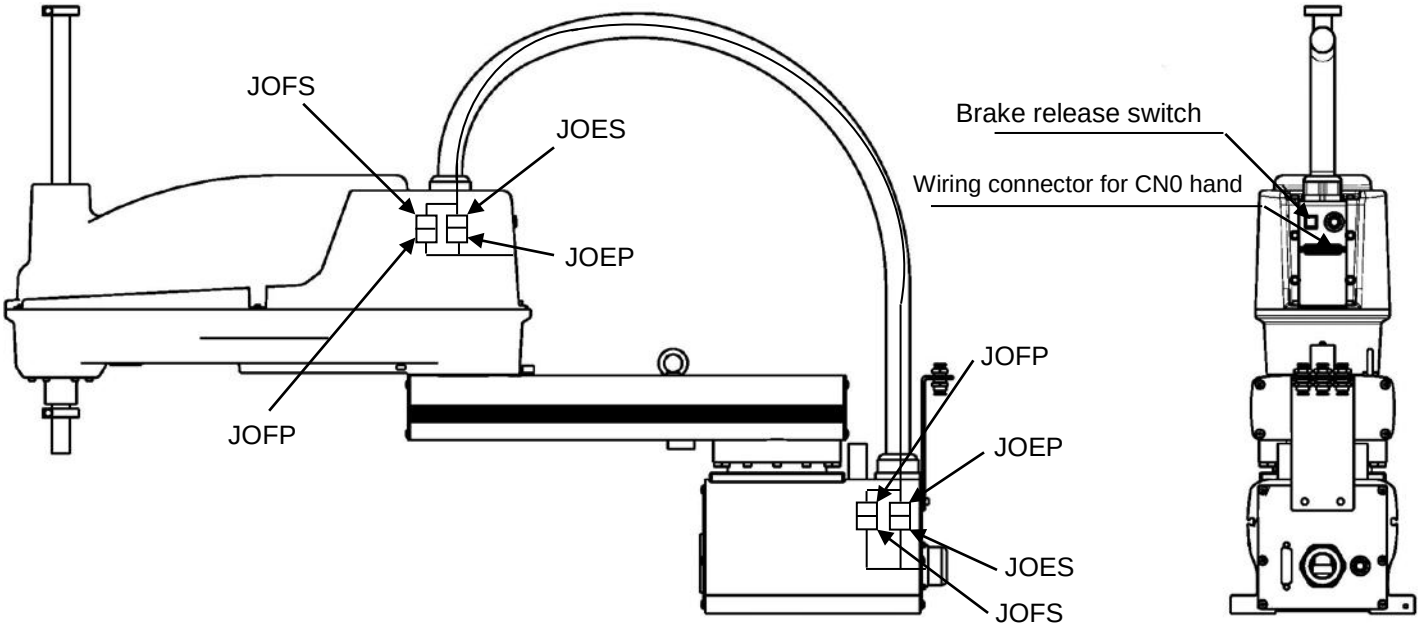


Fig. 24.7 Tool wiring (THL800, THL900 and THL1000)

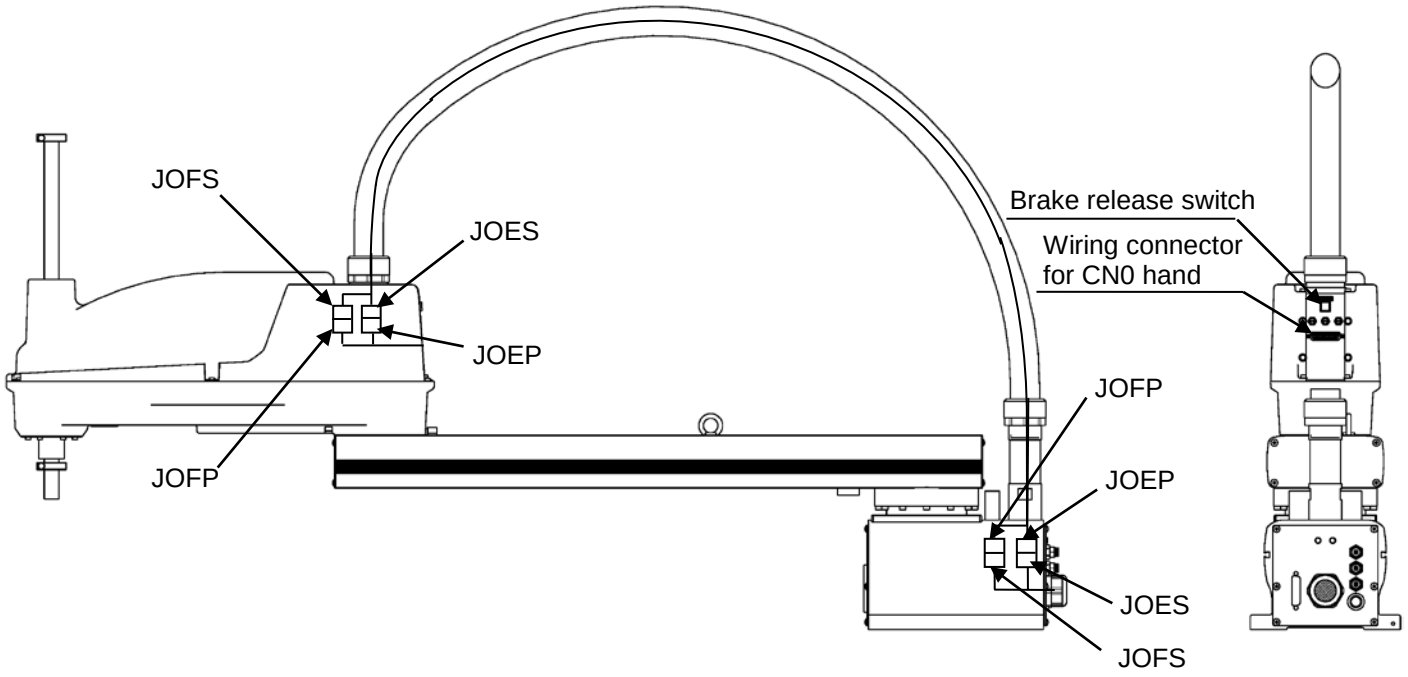


Fig. 24.8 Tool wiring (THL1200)

! CAUTION

- Be sure to connect all connectors properly. Otherwise, the robot may malfunction.

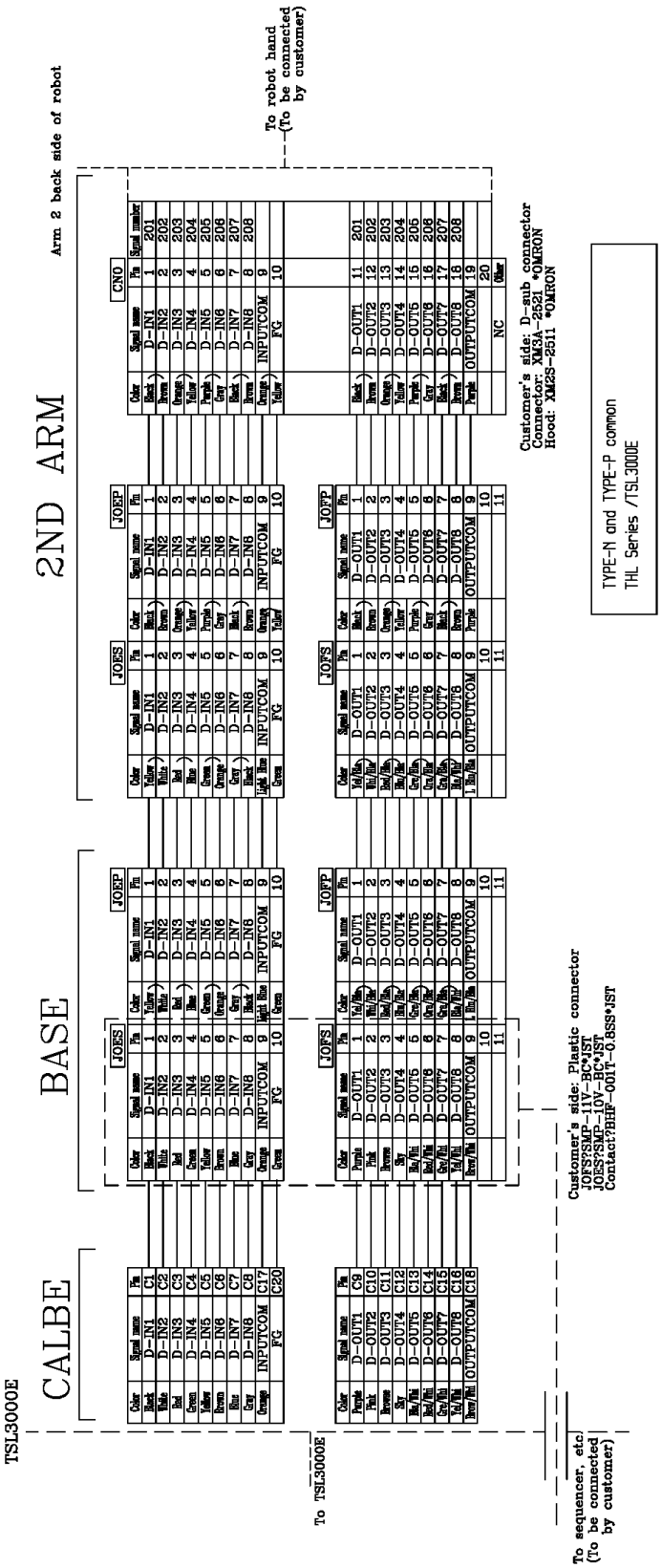


Fig. 24.9 Tool wiring (Common to Type-N, Type-P and TSL3000E)

24.2 Tool Cable wiring

Tool Cable wiring is shown in 23.2

24.3 The wiring arrangement for the hand I/O cable (Option)

The wiring arrangement for the hand I/O cable (Option) is shown in 23.3

25. Tool Interface (TS3000 and TS3000E)

25.1 Tool Wiring

Eight (8) input signals are provided for sensors, etc. and eight (8) control signals for solenoid valves, etc. A supply power signal of 24 VDC is also provided. They are connected to the controller. The wiring arrangement for these cables is shown in Figs. 25.9 and 25.10. The wires are connected to the connectors on the upper side of Arm 2. The user should provide the following connectors to connect the cables.

D-SUB connector (standard) Type: <Hood> XM2S–2511 (Maker: OMRON)
<Basic body> XM3A–2521 (Maker: OMRON)

Adaptive cable Conductive cross section area: 0.2 mm^2 to 0.5 mm^2

Each connector and cable are connected by soldering.

! DANGER

- Be sure to use the designated wire. Otherwise, fires or faults may be caused.
- When connecting the connector and wires, make sure not to mistake the terminal arrangement.
- After making the connection, use a tester, etc., to confirm the connection.

When controlling the robot from the sequencer (i.e., programmable ladder controller: PLC), etc. installed separately, remove the base rear cover from the base section, remove connectors JOES and JOFS on the rear side, then connect the cables running from the PLC, etc. through the cable inlets provided on the base rear cover. Also, the D-SUB 25 pins can be used separately for THL500 to THL1200. (See Figs. 25.1, 25.2 and 25.3.) For ahead of the JOES and JOFS connectors, the user should prepare the following plug connectors and connect the cables. The current is 1 A or less per cable.

Type of connector: JOES SMP–10V–BC (Maker: J.S.T. Mfg.)
JOFS SMP–11V–BC (Maker: J.S.T. Mfg.)

Type of contact: BHF–001T–0.8SS (Maker: J.S.T. Mfg.)

Adaptive cable Conductive cross section area: 0.2 mm^2 to 0.3 mm^2

Opposite connector type (THL500 to THL1200)

Type of connector: JOEP SMR–10V–B (Maker: J.S.T. Mfg.)
JOFP SMR–11V–B (Maker: J.S.T. Mfg.)

Type of contact: BYM–001T–0.6 (Maker: J.S.T. Mfg.)

D-SUB 25 pin connector types (THL500 to THL1200)

Connector type (male) <Hood> XM2S-2511 (maker: OMRON)

<Body> XM3A-2521 (maker: OMRON)

Connector type (female) <Body> XM3D-2521 (maker: OMRON)

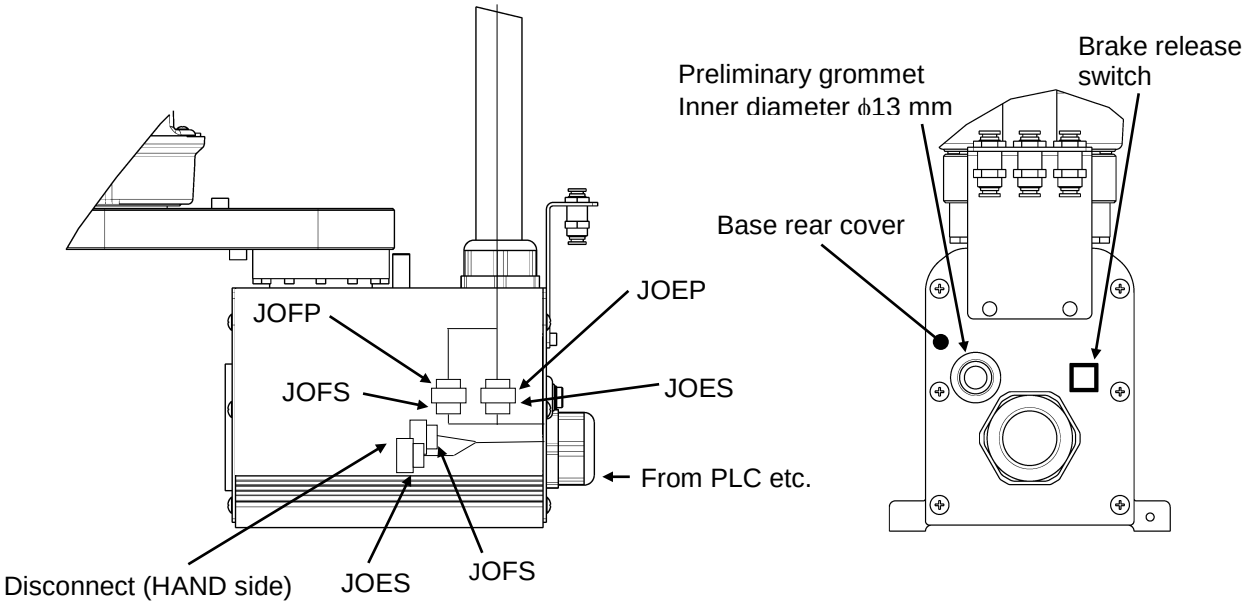


Fig. 25.1 Wiring to PLC, etc. (THL300, THL400 and TS3000/TS3000E)

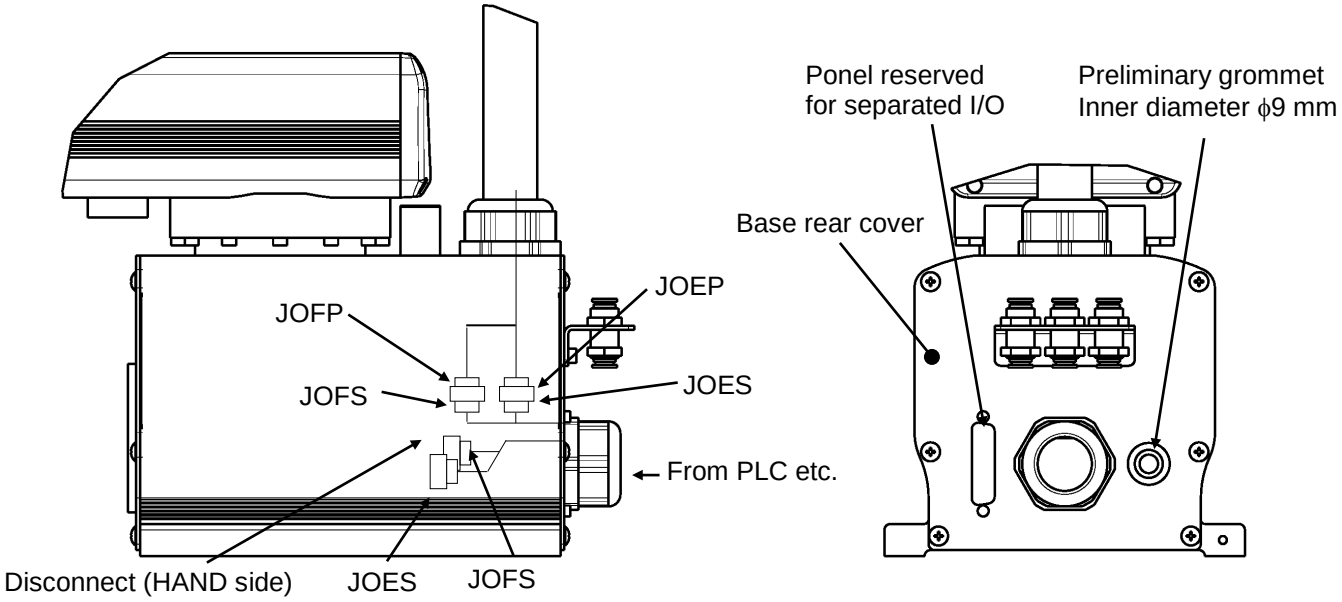


Fig. 25.2 Wiring to PLC, etc. (THL500, THL600, THL700 and TS3000/TS3000E)

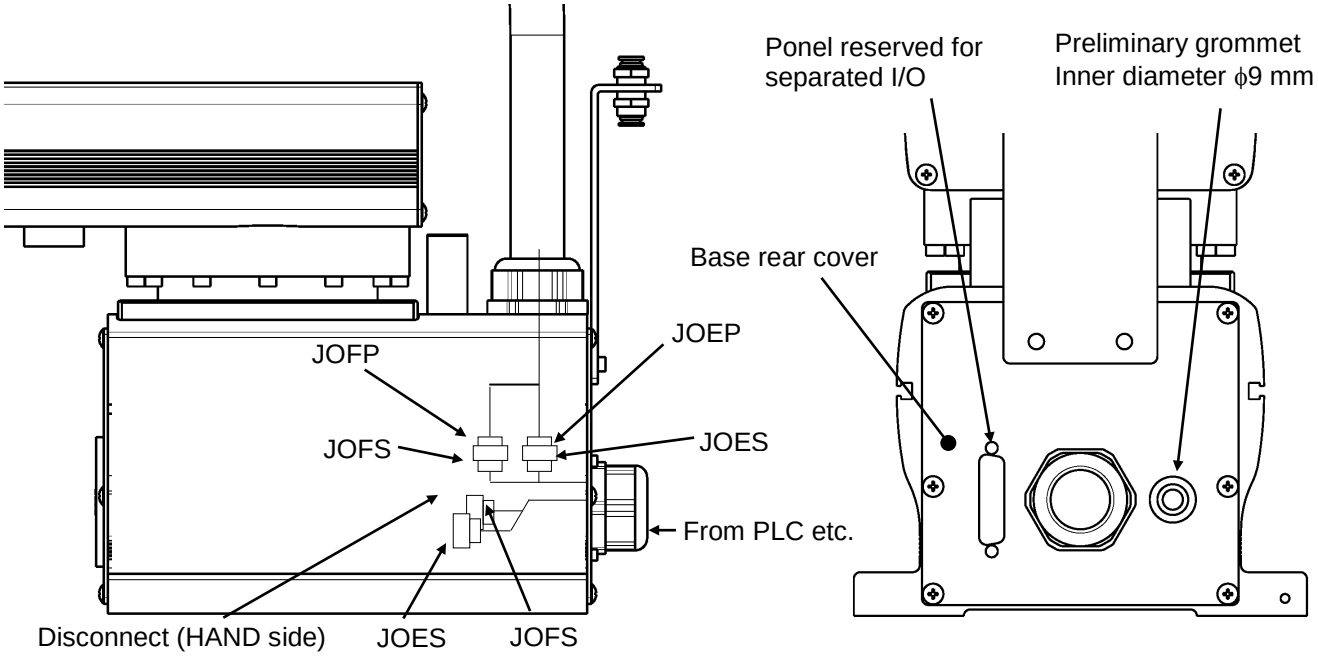


Fig. 25.3 Wiring to PLC, etc. (THL800, THL900, THL1000 and TS3000/TS3000E)

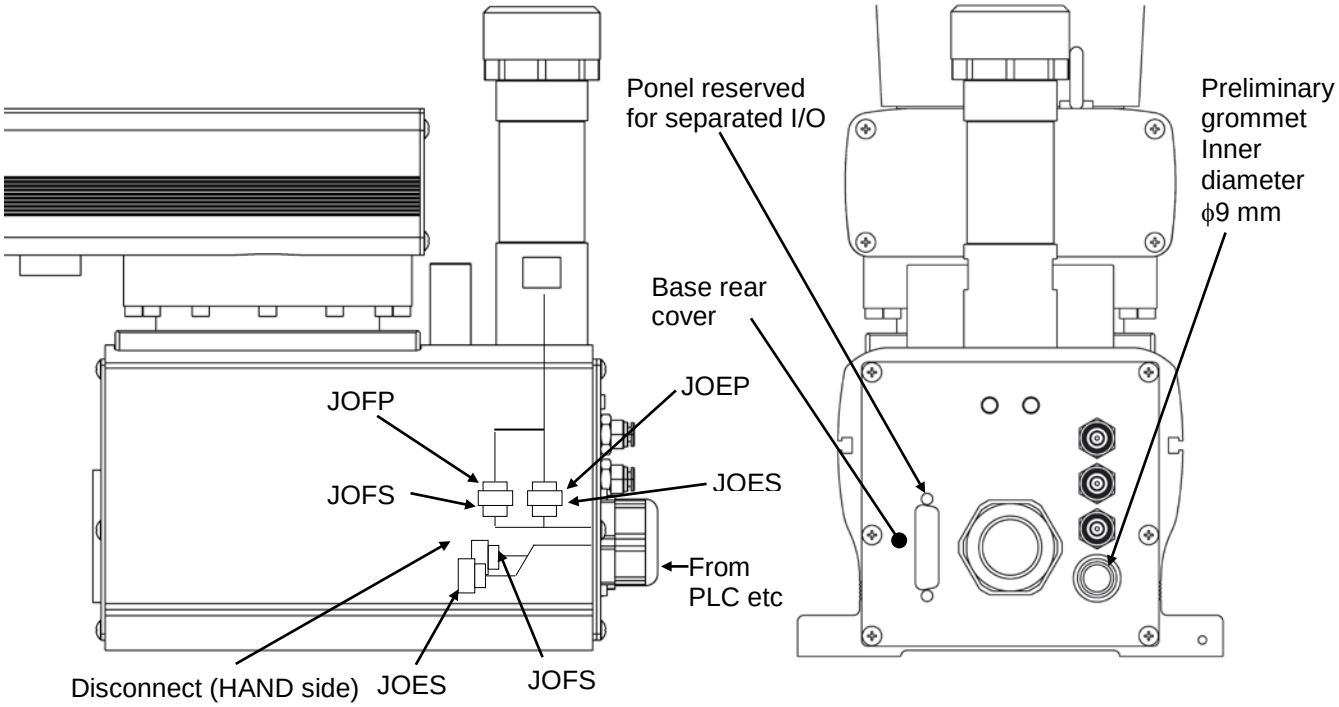
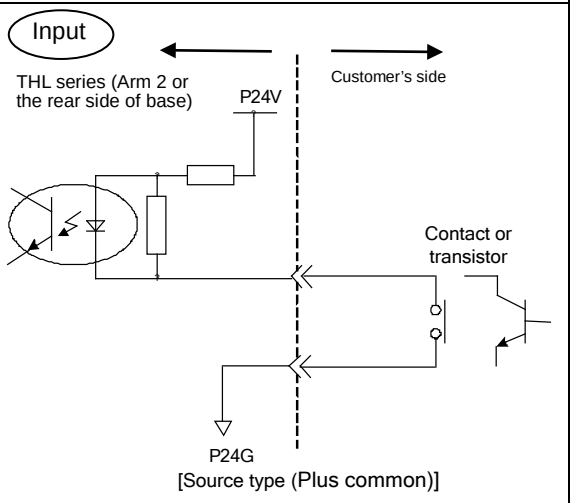


Fig. 25.4 Wiring to PLC, etc. (THL1200 and TS3000/TS3000E)

Table 25.1 Input/output signal connector CN0 (TS3000 and TS3000E, Type-N)

Pin (D-SUB)	Signal name		Signal No.	Input/output circuit and example of connections
1	D-IN1	Input signal 1	201	
2	D-IN2	Input signal 2	202	
3	D-IN3	Input signal 3	203	
4	D-IN4	Input signal 4	204	
5	D-IN5	Input signal 5	205	
6	D-IN6	Input signal 6	206	
7	D-IN7	Input signal 7	207	
8	D-IN8	Input signal 8	208	
9	DC 24VGND (P24G)			
10	Shield (FG)			
11	D-OUT1	Output signal 1	201	
12	D-OUT2	Output signal 2	202	
13	D-OUT3	Output signal 3	203	
14	D-OUT4	Output signal 4	204	
15	D-OUT5	Output signal 5	205	
16	D-OUT6	Output signal 6	206	
17	D-OUT7	Output signal 7	207	
18	D-OUT8	Output signal 8	208	
19	DC 24V (P24V)			



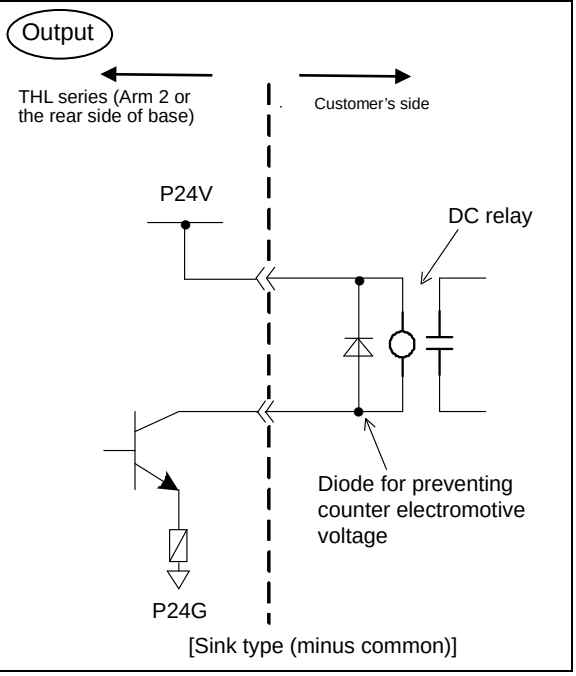
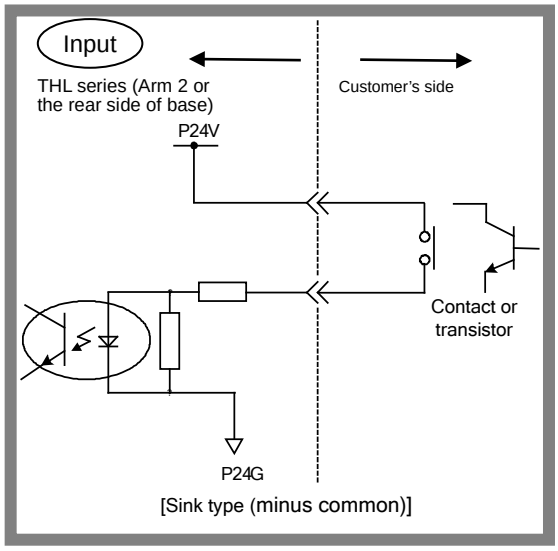
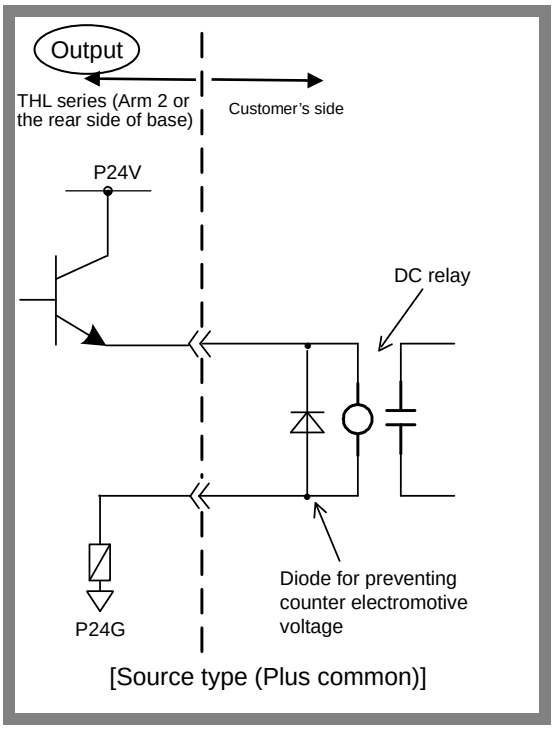


Table 25.2 Input/output signal connector CN0 (TS3000 and TS3000E, Type-P)

Pin (D-SUB)	Signal name		Signal No.	Input/output circuit and example of connections
1	D-IN1	Input signal 1	201	
2	D-IN2	Input signal 2	202	
3	D-IN3	Input signal 3	203	
4	D-IN4	Input signal 4	204	
5	D-IN5	Input signal 5	205	
6	D-IN6	Input signal 6	206	
7	D-IN7	Input signal 7	207	
8	D-IN8	Input signal 8	208	
9	DC 24VGND (P24G)			
10	Shield (FG)			
11	D-OUT1	Output signal 1	201	
12	D-OUT2	Output signal 2	202	
13	D-OUT3	Output signal 3	203	
14	D-OUT4	Output signal 4	204	
15	D-OUT5	Output signal 5	205	
16	D-OUT6	Output signal 6	206	
17	D-OUT7	Output signal 7	207	
18	D-OUT8	Output signal 8	208	
19	DC 24VGND (P24G)			





As input signals, no-voltage contacts or transistor open collector inputs are used.

No-voltage contact specifications:

Contact rating: 24 VDC, 10 mA or over (circuit current: approx. 7 mA)

Minimum contact current: 24 VDC, 1 mA

Contact impedance: 100 Ω or less

Transistor specifications:

Withhold voltage between collector and emitter: 30 V or over

Current between collector and emitter: 10 mA or over (circuit current: approx. 7 mA)

Leak current between collector and emitter: 100 μA or less

By using P24V power of the controller, a relay, solenoid valve, etc., can be driven. When the external power is used, GND of the external power should be common to GND (P24G) of the robot controller.

Output specifications:

Rated voltage : 24 VDC (max. 30 VDC)

Rated current : 100 mA

- If the P24V power supplied from the robot controller is used, the total current should be 2 A or less.
- When the external power is used, the total current should also be 2 A or less.
- When a relay or solenoid valve, etc., is connected, it is necessary to use a surge killer or diode to absorb the surge voltage.

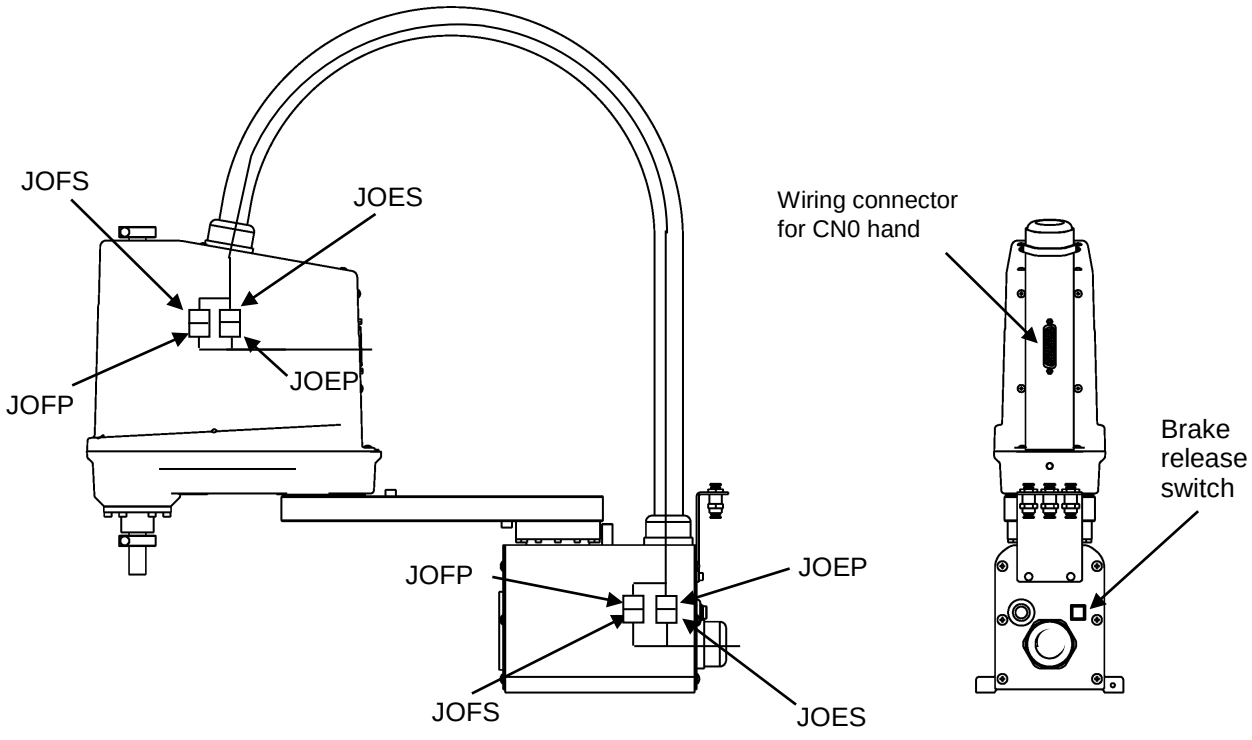


Fig. 25.5 Tool wiring (THL300 and THL400)

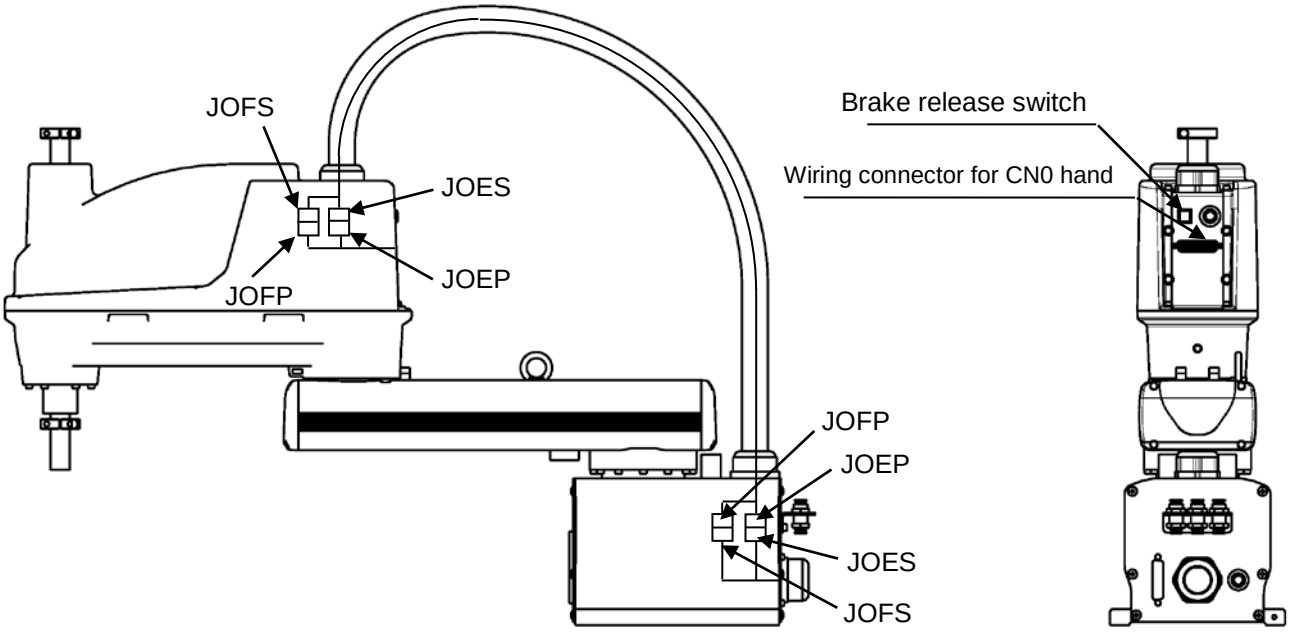


Fig. 25.6 Tool wiring (THL500, THL600 and THL700)

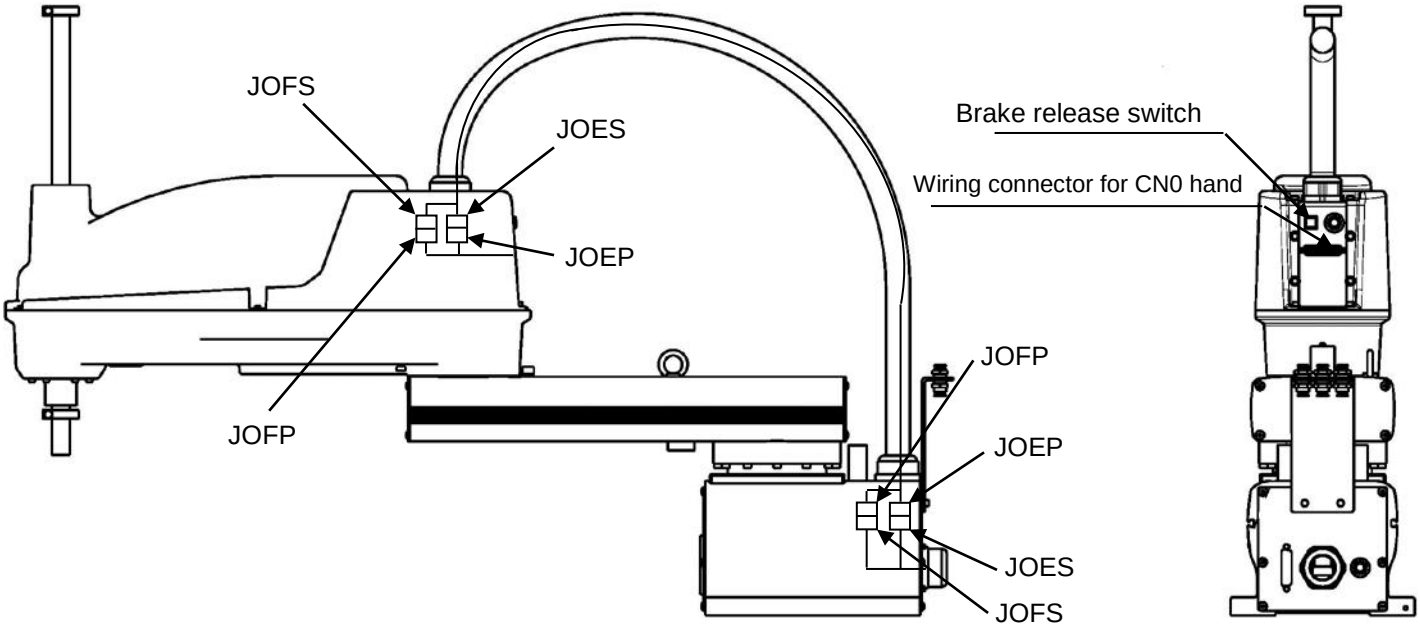


Fig. 25.7 Tool wiring (THL800, THL900 and THL1000)

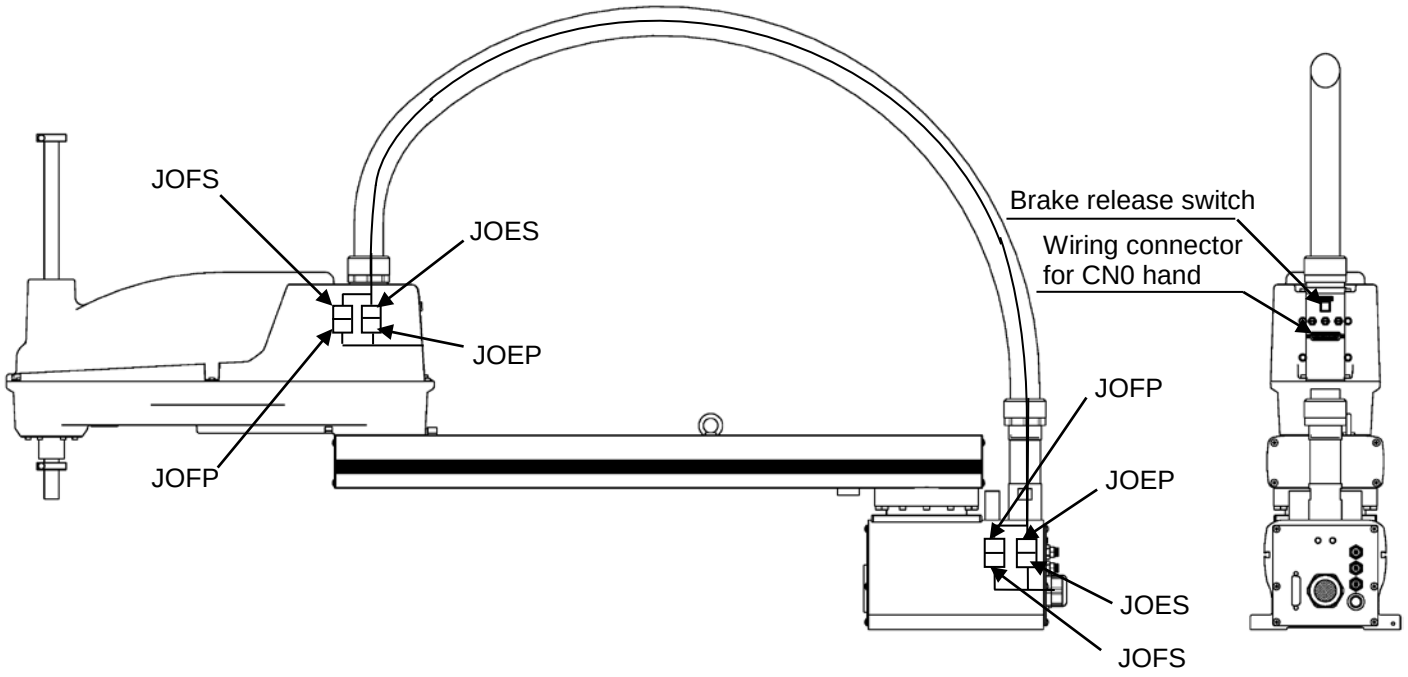


Fig. 25.8 Tool wiring (THL1200)

! CAUTION

- Be sure to connect all connectors properly. Otherwise, the robot may malfunction.

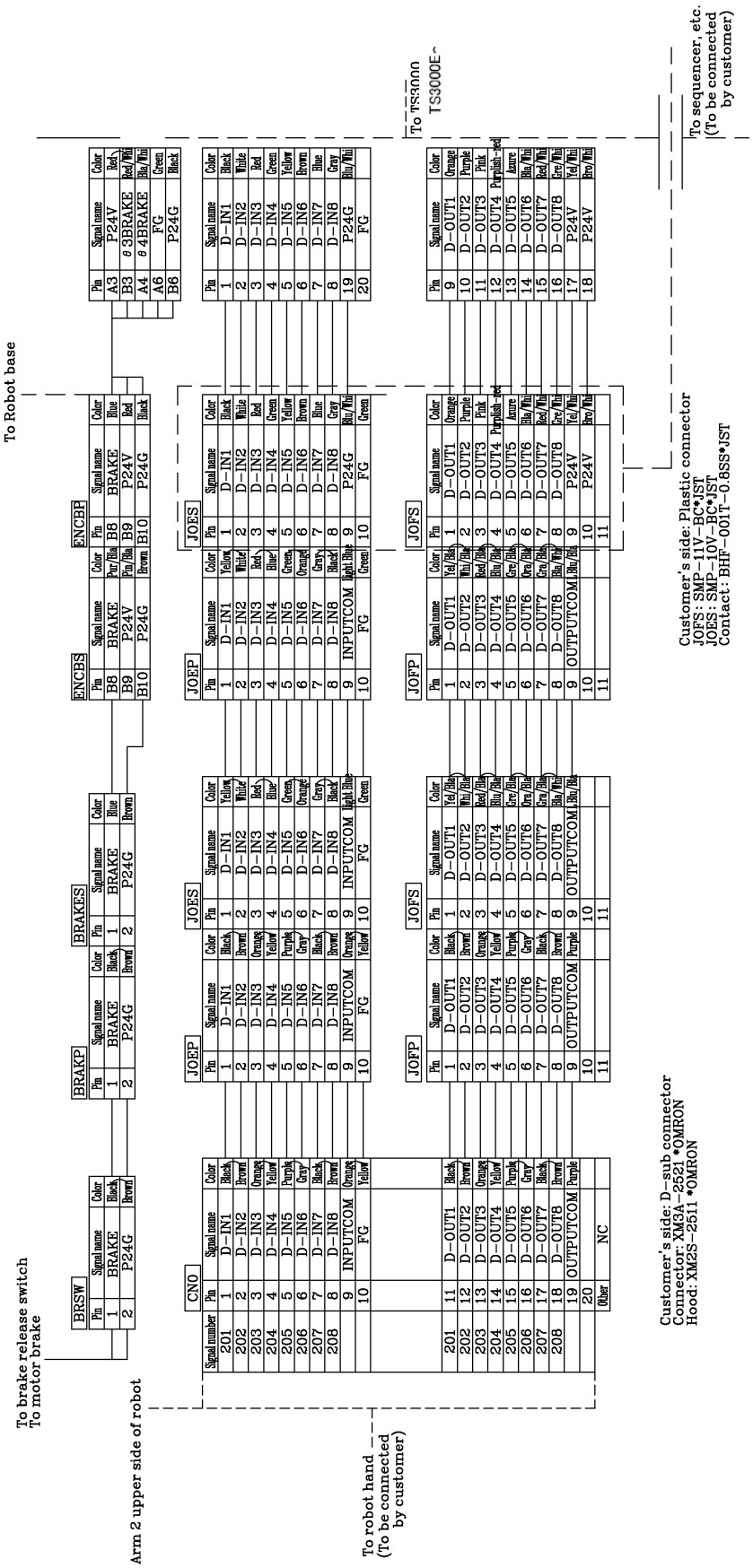


Fig. 25.9 Tool wiring (TS3000 and TS3000E, Type-N)

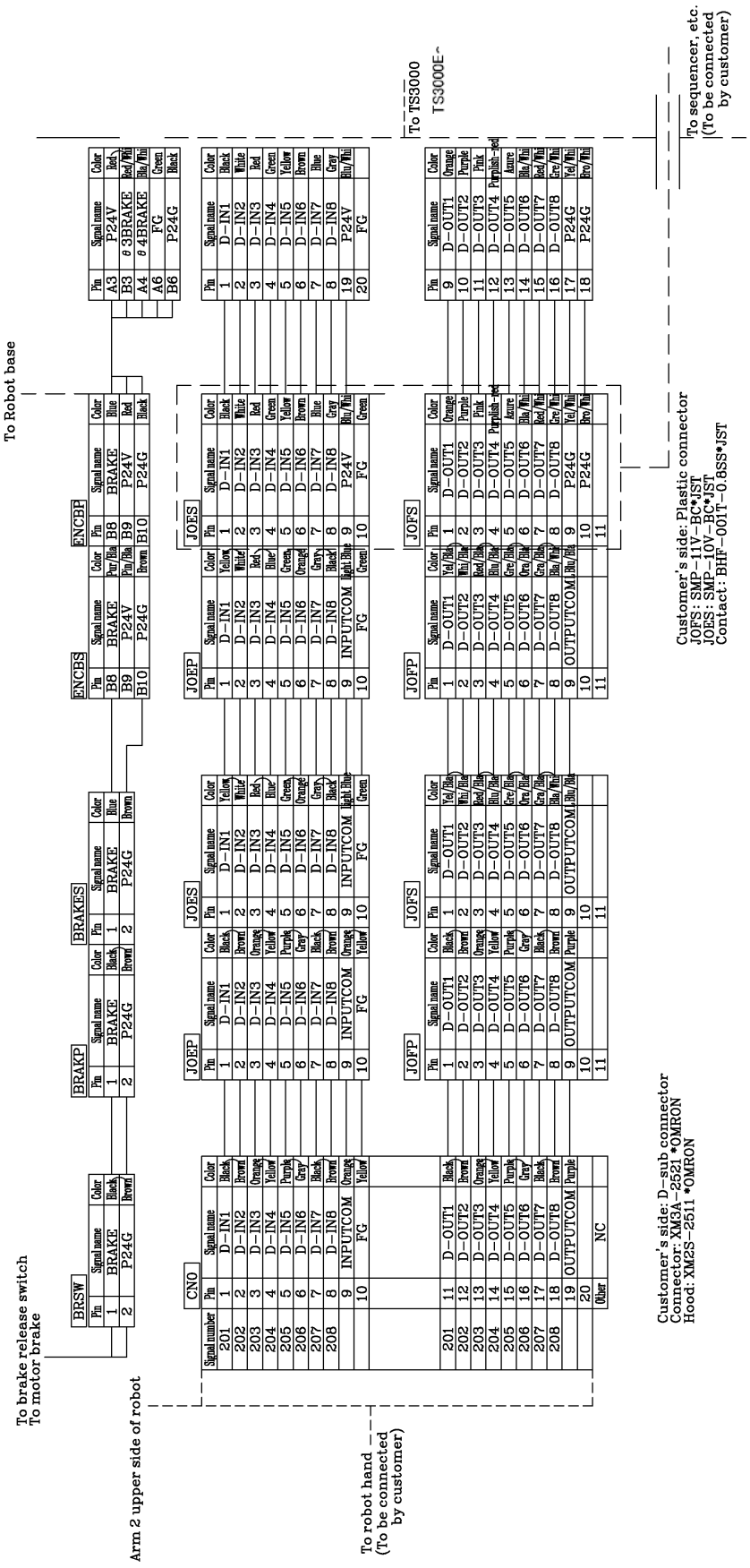


Fig. 25.10 Tool wiring (TS3000 and TS3000E, Type-P)

25.2 Tool Cable wiring

Tool Cable wiring is shown in 23.2

25.3 The wiring arrangement for the hand I/O cable (Option)

The wiring arrangement for the hand I/O cable (Option) is shown in 23.3

25.4 Tool Signals (Controller Side)

The controller is provided with tool signals (i.e., eight (8) input signals for sensor, etc., eight (8) control signals for solenoid valve, etc., 24 VDC(P24V) and 24 VDC GND (P24G) signals), which can be connected also from the controller side. The relevant signals are described below.

Table 25.3 Input signal connector HAND (TS3000 and TS3000E, Type N)

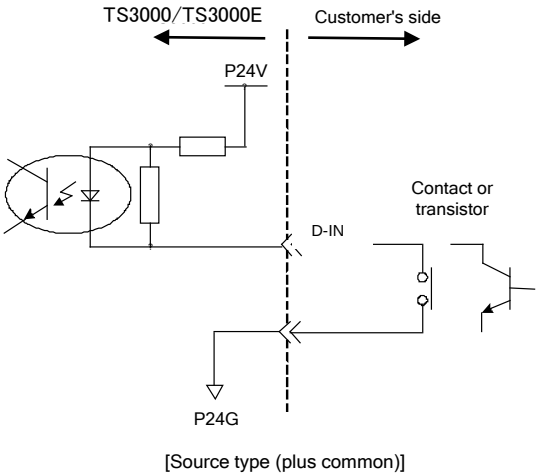
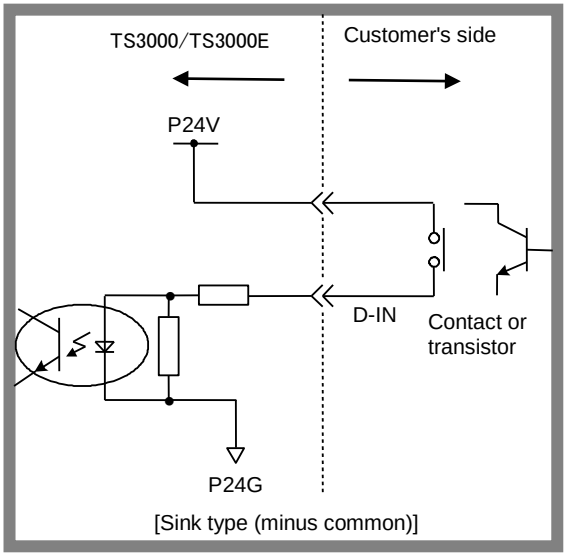
Pin	Signal name		Signal No.	Input circuit and example of connections
1	D-IN1	Input signal 0	201	
2	D-IN2	Input signal 1	202	
3	D-IN3	Input signal 2	203	
4	D-IN4	Input signal 3	204	
5	D-IN5	Input signal 4	205	
6	D-IN6	Input signal 5	206	
7	D-IN7	Input signal 6	207	
8	D-IN8	Input signal 7	208	
19	DC 24VGND (P24G)			
20	FG			

Table 25.4 Input signal connector HAND (TS3000 and TS3000E, Type P)

Pin	Signal name		Signal No.	Input circuit and example of connections
1	D-IN1	Input signal 0	201	
2	D-IN2	Input signal 1	202	
3	D-IN3	Input signal 2	203	
4	D-IN4	Input signal 3	204	
5	D-IN5	Input signal 4	205	
6	D-IN6	Input signal 5	206	
7	D-IN7	Input signal 6	207	
8	D-IN8	Input signal 7	208	
19	DC 24VGND (P24V)			
20	FG			



As input signals, no-voltage contacts or transistor open collector inputs are used.

No-voltage contact specifications:

Contact rating: 24 VDC, 10 mA or over (circuit current: approx. 7 mA)

Minimum contact current: 24 VDC, 1 mA

Contact impedance: 100 Ω or less

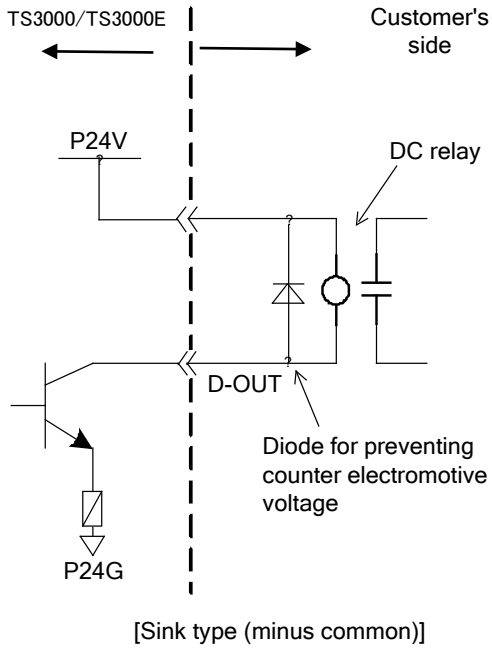
Transistor specifications:

Withhold voltage between collector and emitter: 30 V or over

Current between collector and emitter: 10 mA or over (circuit current: approx. 7 mA)

Leak current between collector and emitter: 100 μA or less

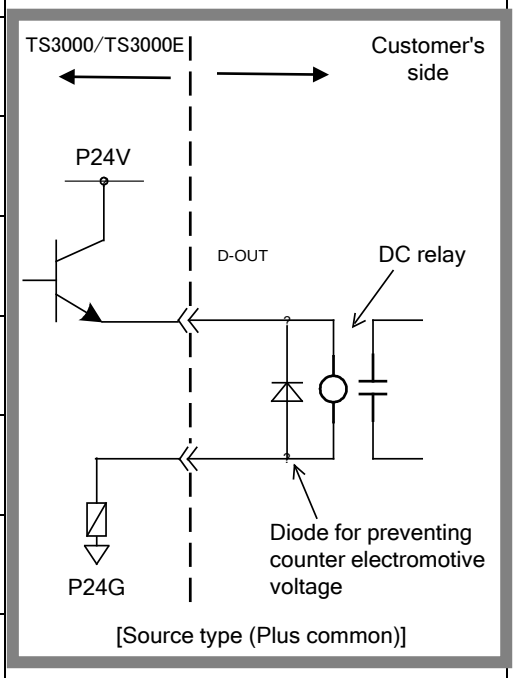
Table 25.5 Output signal connector HAND (TS3000 and TS3000E, Type N)

<table>
<tr><th>Pin</th><th colspan="2">Signal name</th><th>Signal No.</th><th>Output circuit and example of connections</th></tr>
<tr><td>9</td><td>D-OUT1</td><td>Output signal 0</td><td>201</td><td rowspan="9"> 

[Sink type (minus common)]</td></tr>
<tr><td>10</td><td>D-OUT2</td><td>Output signal 1</td><td>202</td></tr>
<tr><td>11</td><td>D-OUT3</td><td>Output signal 2</td><td>203</td></tr>
<tr><td>12</td><td>D-OUT4</td><td>Output signal 3</td><td>204</td></tr>
<tr><td>13</td><td>D-OUT5</td><td>Output signal 4</td><td>205</td></tr>
<tr><td>14</td><td>D-OUT6</td><td>Output signal 5</td><td>206</td></tr>
<tr><td>15</td><td>D-OUT7</td><td>Output signal 6</td><td>207</td></tr>
<tr><td>16</td><td>D-OUT8</td><td>Output signal 7</td><td>208</td></tr>
<tr><td>17</td><td colspan="2" rowspan="2">DC 24V (P24V)</td><td rowspan="2"></td><td rowspan="2"></td></tr>
<tr><td>18</td></tr>
</table>

Table 25.6 Output signal connector HAND (TS3000 and TS3000E, Type P)

Pin	Signal name		Signal No.	Output circuit and example of connections
9	D-OUT1	Output signal 0	201	
10	D-OUT2	Output signal 1	202	
11	D-OUT3	Output signal 2	203	
12	D-OUT4	Output signal 3	204	
13	D-OUT5	Output signal 4	205	
14	D-OUT6	Output signal 5	206	
15	D-OUT7	Output signal 6	207	
16	D-OUT8	Output signal 7	208	
17	DC 24VGND (P24G)			
18				



By using the P24 V power of the controller, a relay, solenoid valve, etc., can be driven. When the external power is used, GND of the external power should be common to GND (P24G) of the robot controller.

Output specifications:

Rated voltage : 24 VDC (P24V)
Rated current : 100 mA

- When a relay or solenoid valve, etc., is connected, it is necessary to use a surge killer or diode to absorb the surge voltage.
- When a double solenoid is used, HO_1 and HO_2, and HO_3 and HO_4 are used as pairs.

The right figure shows the DC relay circuit when the external power is used.